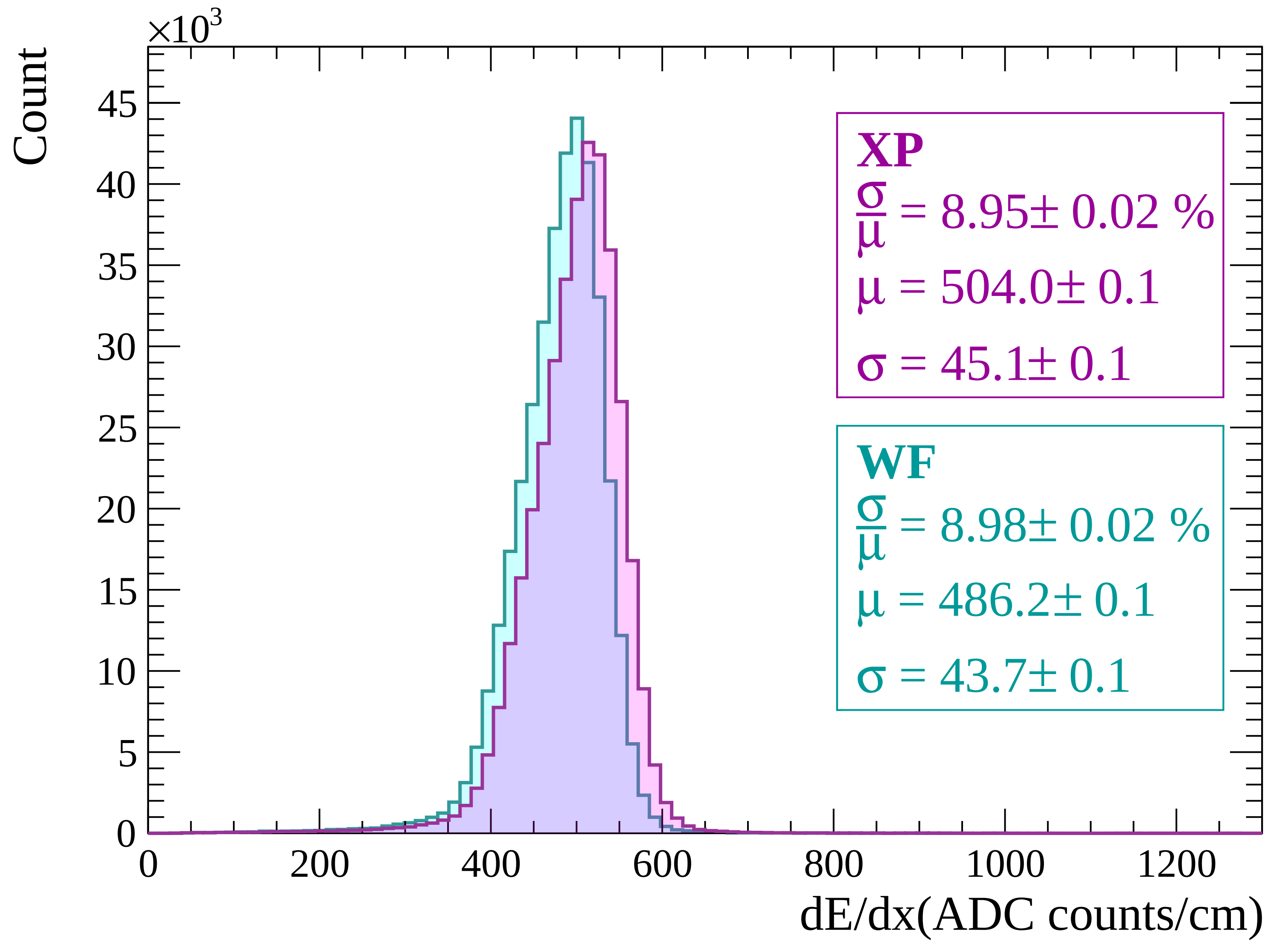
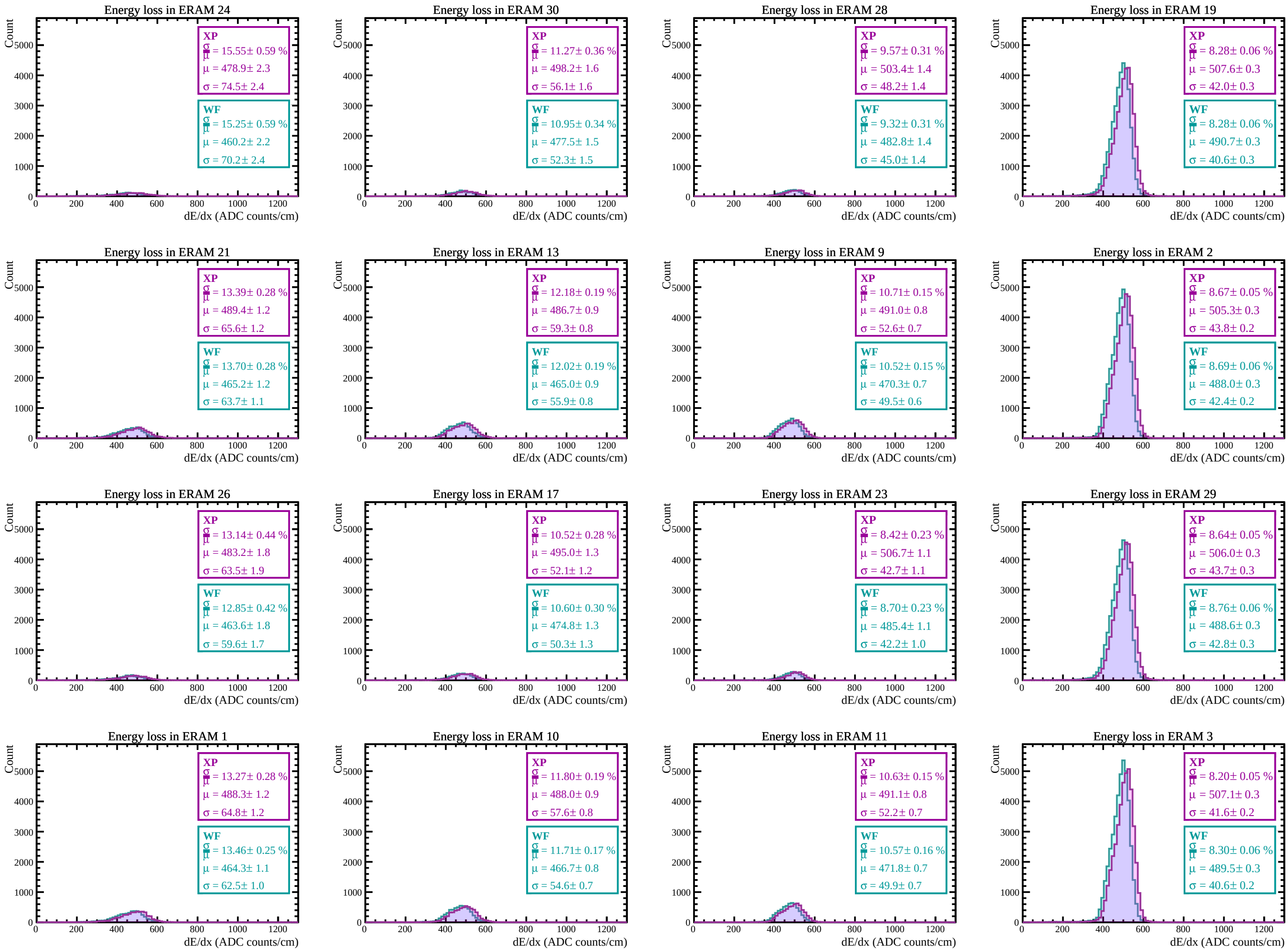
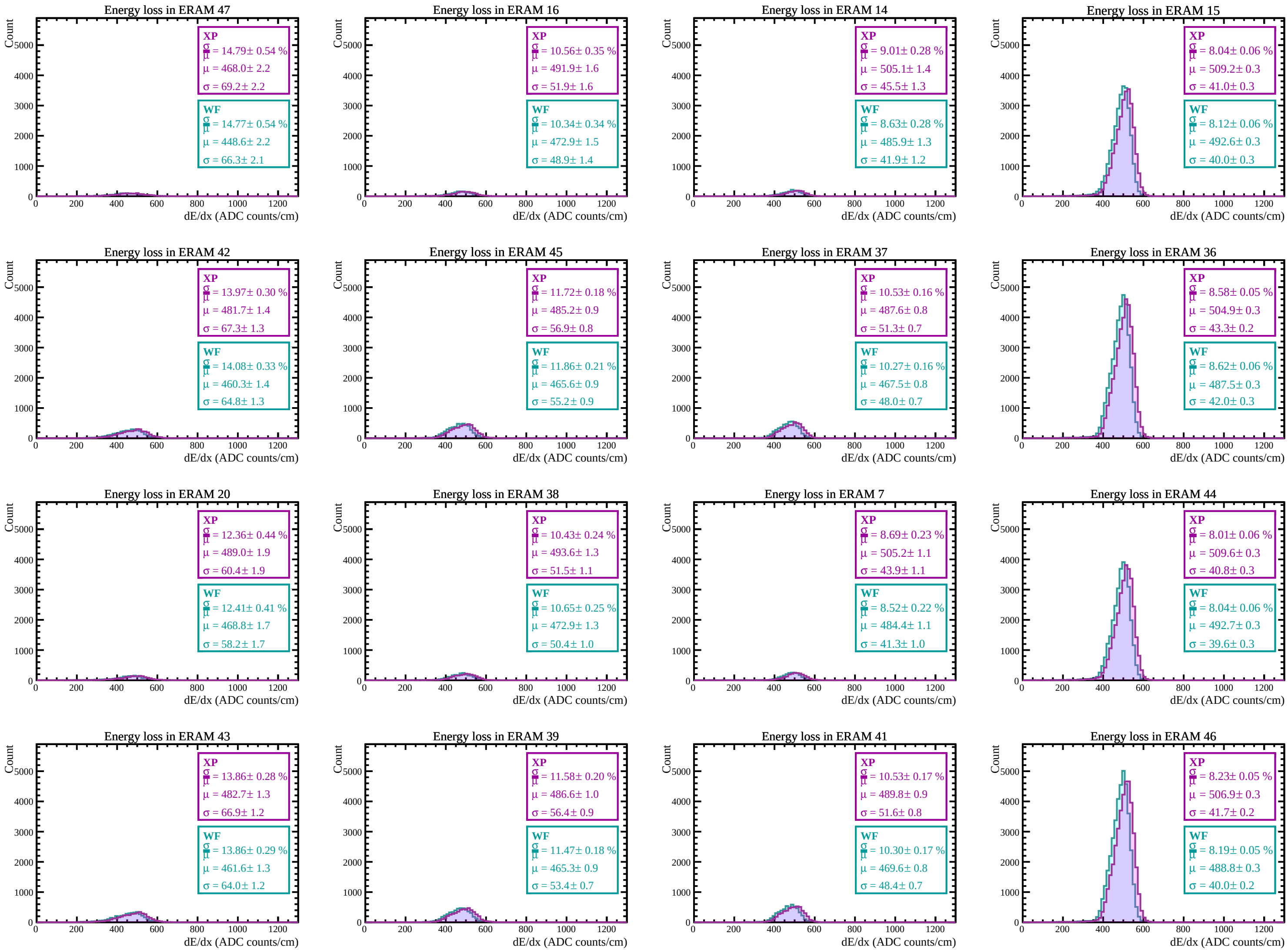
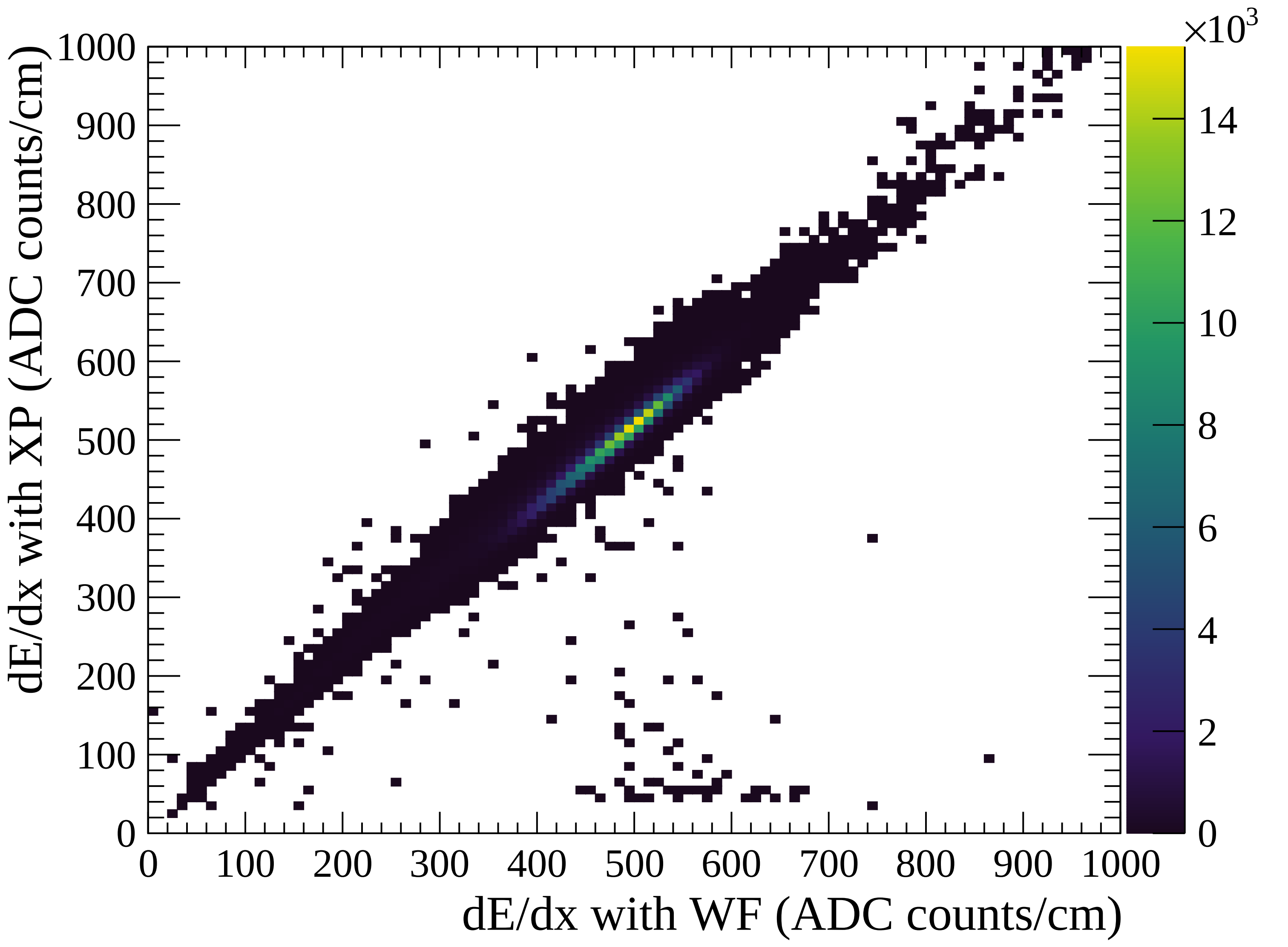


×10^3
Count
45
40
35
30
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx(ADC counts/cm)
XP
σ/μ = 8.95± 0.02 %
μ = 504.0± 0.1
σ = 45.1± 0.1
WF
σ/μ = 8.98± 0.02 %
μ = 486.2± 0.1
σ = 43.7± 0.1

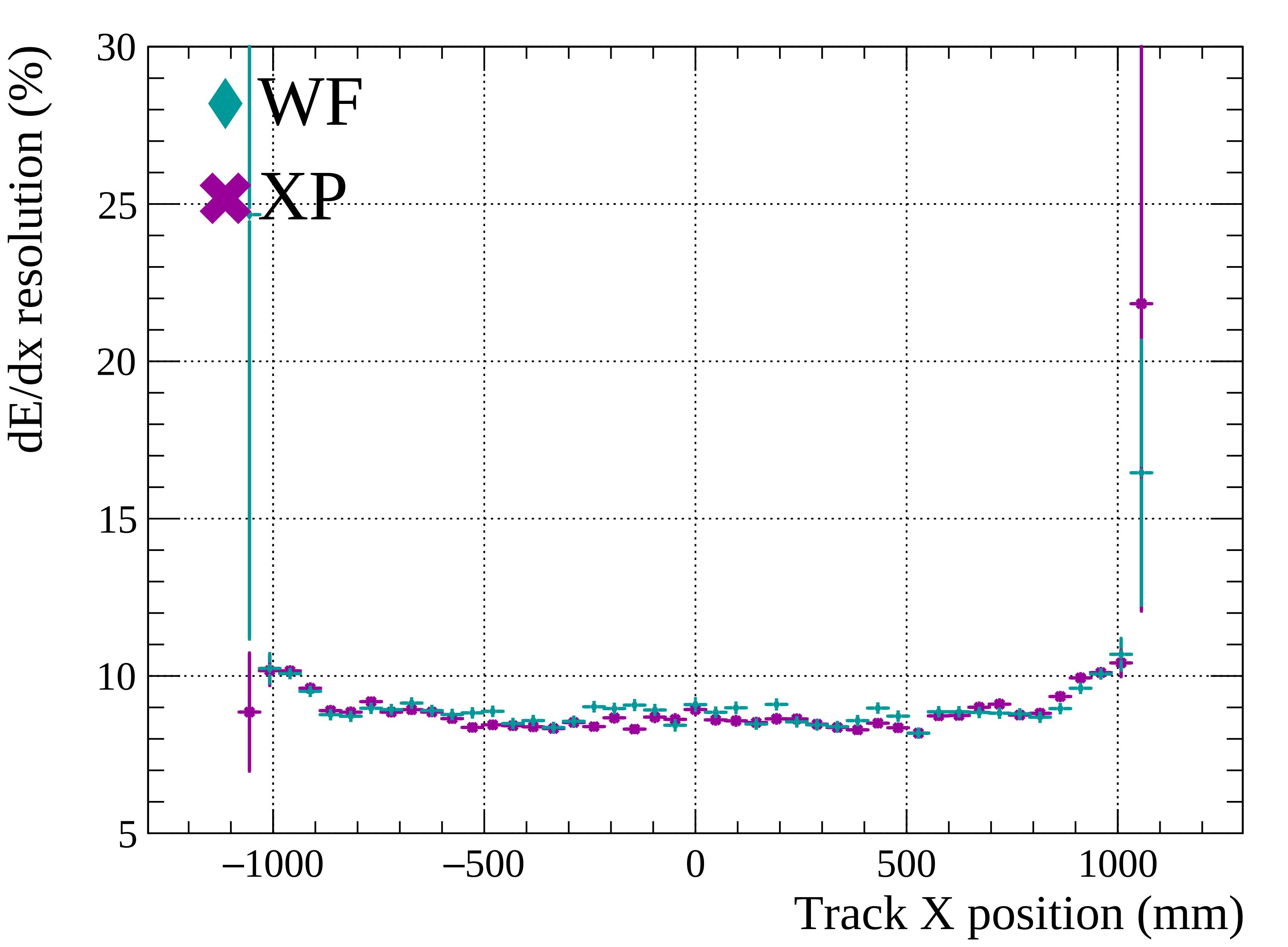

Energy loss in ERAM 24
XP σ/μ = 15.55± 0.59 % μ = 478.9± 2.3 σ = 74.5± 2.4
WF σ/μ = 15.25± 0.59 % μ = 460.2± 2.2 σ = 70.2± 2.4
Energy loss in ERAM 30
XP σ/μ = 11.27± 0.36 % μ = 498.2± 1.6 σ = 56.1± 1.6
WF σ/μ = 10.95± 0.34 % μ = 477.5± 1.5 σ = 52.3± 1.5
Energy loss in ERAM 28
XP σ/μ = 9.57± 0.31 % μ = 503.4± 1.4 σ = 48.2± 1.4
WF σ/μ = 9.32± 0.31 % μ = 482.8± 1.4 σ = 45.0± 1.4
Energy loss in ERAM 19
XP σ/μ = 8.28± 0.06 % μ = 507.6± 0.3 σ = 42.0± 0.3
WF σ/μ = 8.28± 0.06 % μ = 490.7± 0.3 σ = 40.6± 0.3
Energy loss in ERAM 21
XP σ/μ = 13.39± 0.28 % μ = 489.4± 1.2 σ = 65.6± 1.2
WF σ/μ = 13.70± 0.28 % μ = 465.2± 1.2 σ = 63.7± 1.1
Energy loss in ERAM 13
XP σ/μ = 12.18± 0.19 % μ = 486.7± 0.9 σ = 59.3± 0.8
WF σ/μ = 12.02± 0.19 % μ = 465.0± 0.9 σ = 55.9± 0.8
Energy loss in ERAM 9
XP σ/μ = 10.71± 0.15 % μ = 491.0± 0.8 σ = 52.6± 0.7
WF σ/μ = 10.52± 0.15 % μ = 470.3± 0.7 σ = 49.5± 0.6
Energy loss in ERAM 2
XP σ/μ = 8.67± 0.05 % μ = 505.3± 0.3 σ = 43.8± 0.2
WF σ/μ = 8.69± 0.06 % μ = 488.0± 0.3 σ = 42.4± 0.2
Energy loss in ERAM 26
XP σ/μ = 13.14± 0.44 % μ = 483.2± 1.8 σ = 63.5± 1.9
WF σ/μ = 12.85± 0.42 % μ = 463.6± 1.8 σ = 59.6± 1.7
Energy loss in ERAM 17
XP σ/μ = 10.52± 0.28 % μ = 495.0± 1.3 σ = 52.1± 1.2
WF σ/μ = 10.60± 0.30 % μ = 474.8± 1.3 σ = 50.3± 1.3
Energy loss in ERAM 23
XP σ/μ = 8.42± 0.23 % μ = 506.7± 1.1 σ = 42.7± 1.1
WF σ/μ = 8.70± 0.23 % μ = 485.4± 1.1 σ = 42.2± 1.0
Energy loss in ERAM 29
XP σ/μ = 8.64± 0.05 % μ = 506.0± 0.3 σ = 43.7± 0.3
WF σ/μ = 8.76± 0.06 % μ = 488.6± 0.3 σ = 42.8± 0.3
Energy loss in ERAM 1
XP σ/μ = 13.27± 0.28 % μ = 488.3± 1.2 σ = 64.8± 1.2
WF σ/μ = 13.46± 0.25 % μ = 464.3± 1.1 σ = 62.5± 1.0
Energy loss in ERAM 10
XP σ/μ = 11.80± 0.19 % μ = 488.0± 0.9 σ = 57.6± 0.8
WF σ/μ = 11.71± 0.17 % μ = 466.7± 0.8 σ = 54.6± 0.7
Energy loss in ERAM 11
XP σ/μ = 10.63± 0.15 % μ = 491.1± 0.8 σ = 52.2± 0.7
WF σ/μ = 10.57± 0.16 % μ = 471.8± 0.7 σ = 49.9± 0.7
Energy loss in ERAM 3
XP σ/μ = 8.20± 0.05 % μ = 507.1± 0.3 σ = 41.6± 0.2
WF σ/μ = 8.30± 0.06 % μ = 489.5± 0.3 σ = 40.6± 0.2
Count
dE/dx (ADC counts/cm)

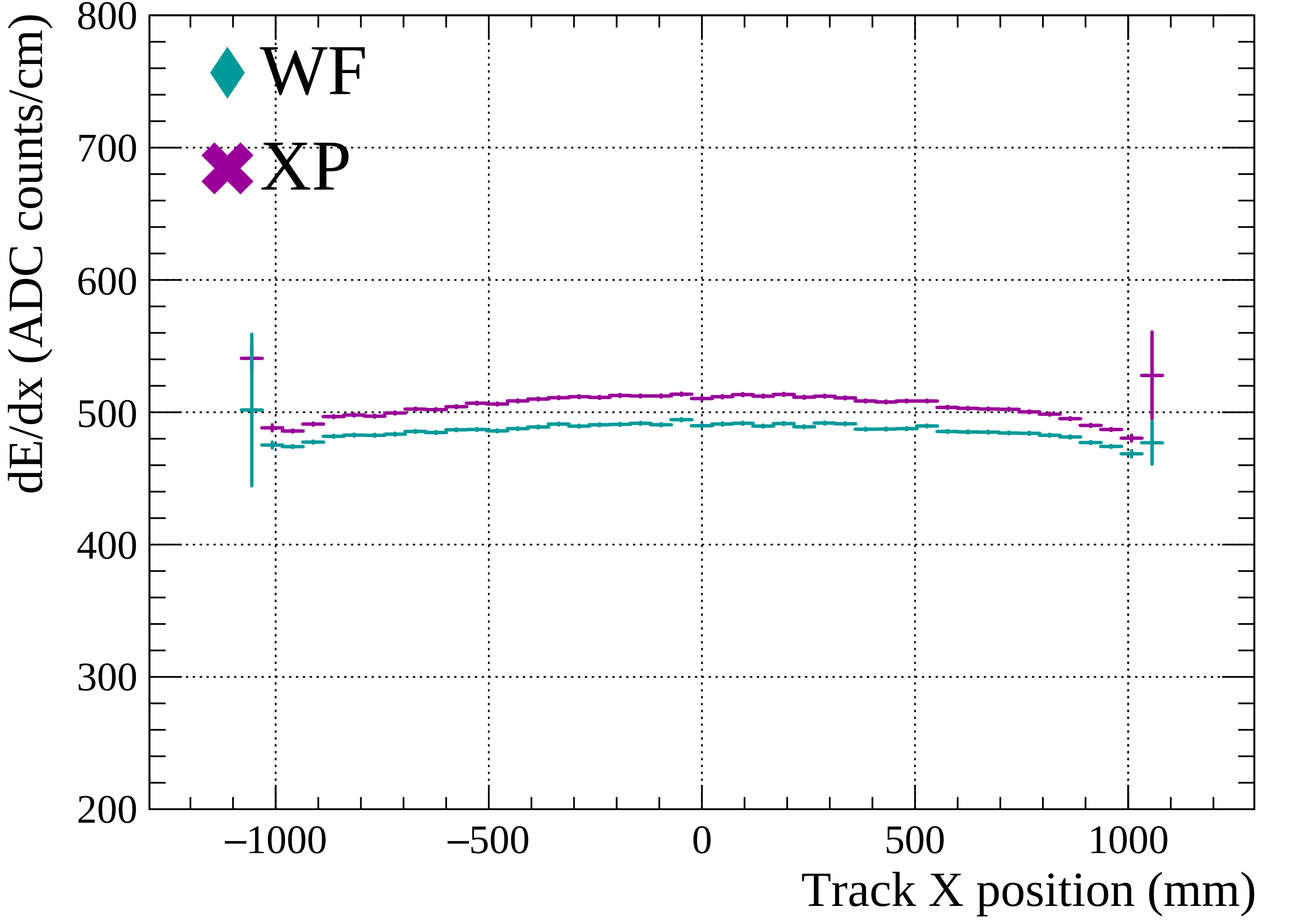


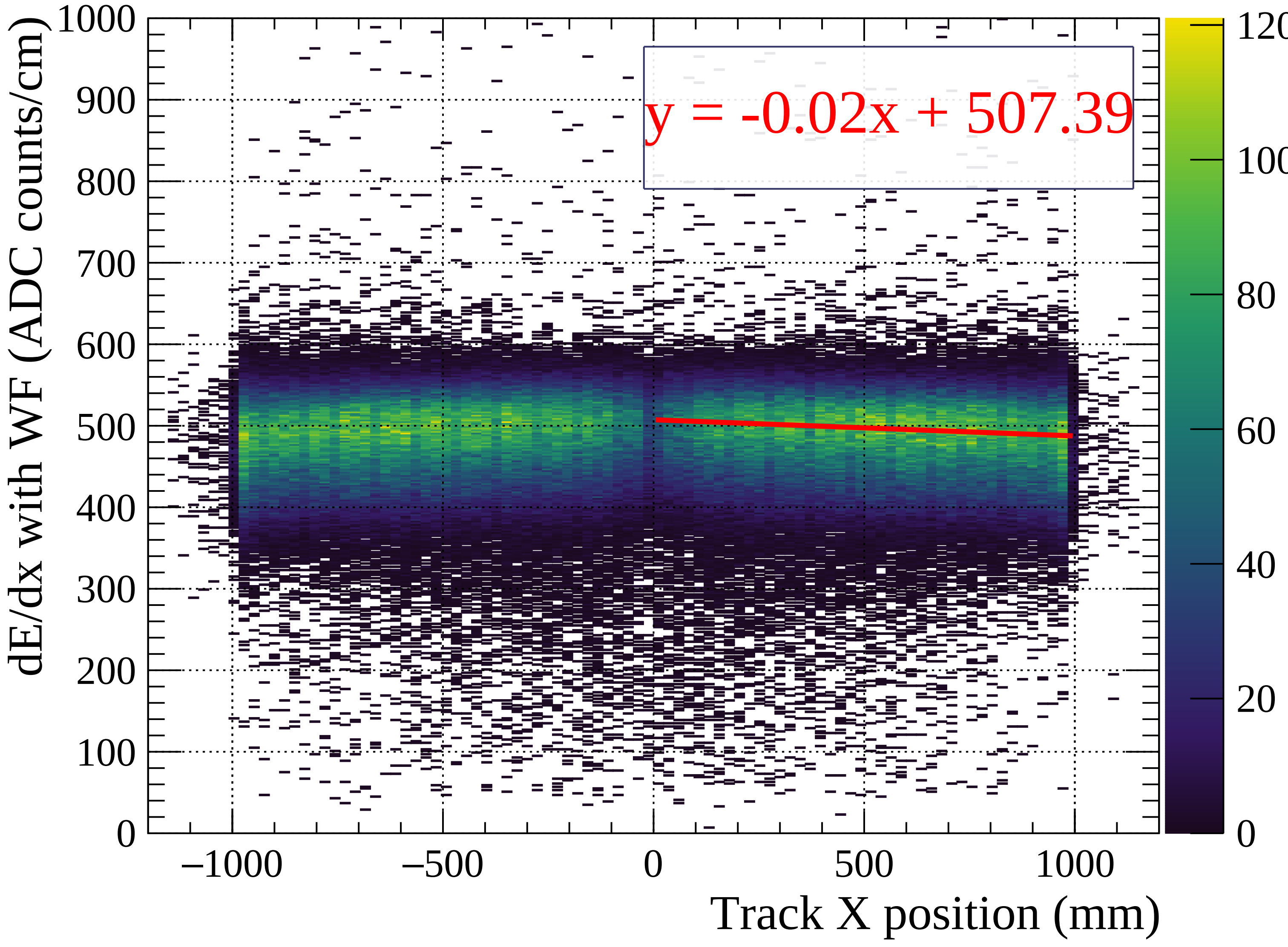
Energy loss in ERAM 47
XP
σ/μ = 14.79± 0.54 %
μ = 468.0± 2.2
σ = 69.2± 2.2
WF
σ/μ = 14.77± 0.54 %
μ = 448.6± 2.2
σ = 66.3± 2.1
Energy loss in ERAM 16
XP
σ/μ = 10.56± 0.35 %
μ = 491.9± 1.6
σ = 51.9± 1.6
WF
σ/μ = 10.34± 0.34 %
μ = 472.9± 1.5
σ = 48.9± 1.4
Energy loss in ERAM 14
XP
σ/μ = 9.01± 0.28 %
μ = 505.1± 1.4
σ = 45.5± 1.3
WF
σ/μ = 8.63± 0.28 %
μ = 485.9± 1.3
σ = 41.9± 1.2
Energy loss in ERAM 15
XP
σ/μ = 8.04± 0.06 %
μ = 509.2± 0.3
σ = 41.0± 0.3
WF
σ/μ = 8.12± 0.06 %
μ = 492.6± 0.3
σ = 40.0± 0.3
Energy loss in ERAM 42
XP
σ/μ = 13.97± 0.30 %
μ = 481.7± 1.4
σ = 67.3± 1.3
WF
σ/μ = 14.08± 0.33 %
μ = 460.3± 1.4
σ = 64.8± 1.3
Energy loss in ERAM 45
XP
σ/μ = 11.72± 0.18 %
μ = 485.2± 0.9
σ = 56.9± 0.8
WF
σ/μ = 11.86± 0.21 %
μ = 465.6± 0.9
σ = 55.2± 0.9
Energy loss in ERAM 37
XP
σ/μ = 10.53± 0.16 %
μ = 487.6± 0.8
σ = 51.3± 0.7
WF
σ/μ = 10.27± 0.16 %
μ = 467.5± 0.8
σ = 48.0± 0.7
Energy loss in ERAM 36
XP
σ/μ = 8.58± 0.05 %
μ = 504.9± 0.3
σ = 43.3± 0.2
WF
σ/μ = 8.62± 0.06 %
μ = 487.5± 0.3
σ = 42.0± 0.3
Energy loss in ERAM 20
XP
σ/μ = 12.36± 0.44 %
μ = 489.0± 1.9
σ = 60.4± 1.9
WF
σ/μ = 12.41± 0.41 %
μ = 468.8± 1.7
σ = 58.2± 1.7
Energy loss in ERAM 38
XP
σ/μ = 10.43± 0.24 %
μ = 493.6± 1.3
σ = 51.5± 1.1
WF
σ/μ = 10.65± 0.25 %
μ = 472.9± 1.3
σ = 50.4± 1.0
Energy loss in ERAM 7
XP
σ/μ = 8.69± 0.23 %
μ = 505.2± 1.1
σ = 43.9± 1.1
WF
σ/μ = 8.52± 0.22 %
μ = 484.4± 1.1
σ = 41.3± 1.0
Energy loss in ERAM 44
XP
σ/μ = 8.01± 0.06 %
μ = 509.6± 0.3
σ = 40.8± 0.3
WF
σ/μ = 8.04± 0.06 %
μ = 492.7± 0.3
σ = 39.6± 0.3
Energy loss in ERAM 43
XP
σ/μ = 13.86± 0.28 %
μ = 482.7± 1.3
σ = 66.9± 1.2
WF
σ/μ = 13.86± 0.29 %
μ = 461.6± 1.3
σ = 64.0± 1.2
Energy loss in ERAM 39
XP
σ/μ = 11.58± 0.20 %
μ = 486.6± 1.0
σ = 56.4± 0.9
WF
σ/μ = 11.47± 0.18 %
μ = 465.3± 0.9
σ = 53.4± 0.7
Energy loss in ERAM 41
XP
σ/μ = 10.53± 0.17 %
μ = 489.8± 0.9
σ = 51.6± 0.8
WF
σ/μ = 10.30± 0.17 %
μ = 469.6± 0.8
σ = 48.4± 0.7
Energy loss in ERAM 46
XP
σ/μ = 8.23± 0.05 %
μ = 506.9± 0.3
σ = 41.7± 0.2
WF
σ/μ = 8.19± 0.05 %
μ = 488.8± 0.3
σ = 40.0± 0.2
Count
dE/dx (ADC counts/cm)



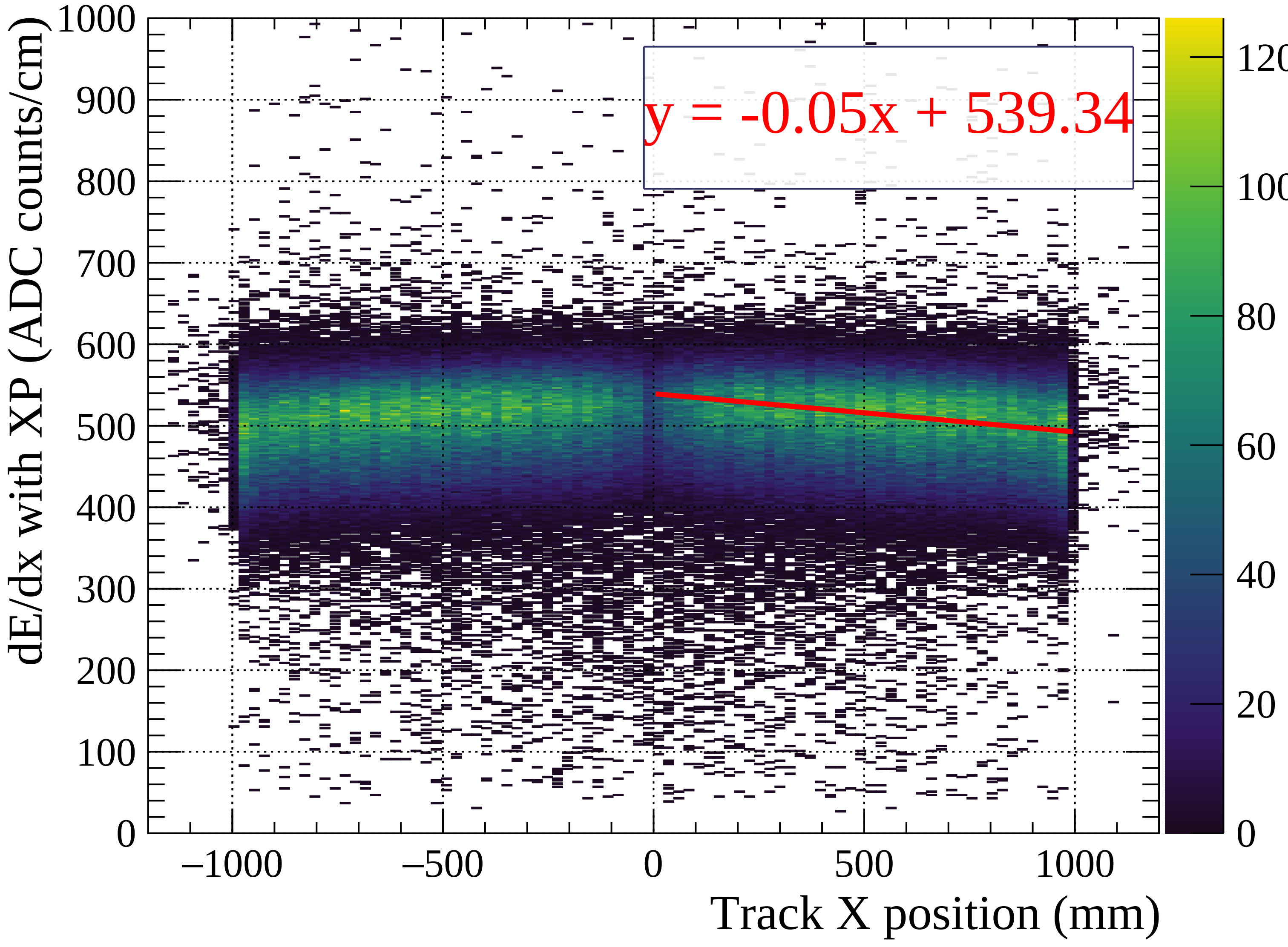
×10^3
14
12
10
8
6
4
2
0
1000
900
800
700
600
500
400
300
200
100
0
0
100
200
300
400
500
600
700
800
900
1000
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)


WF
XP
dE/dx resolution (%)
30
25
20
15
10
5
−1000
−500
0
500
1000
Track X position (mm)

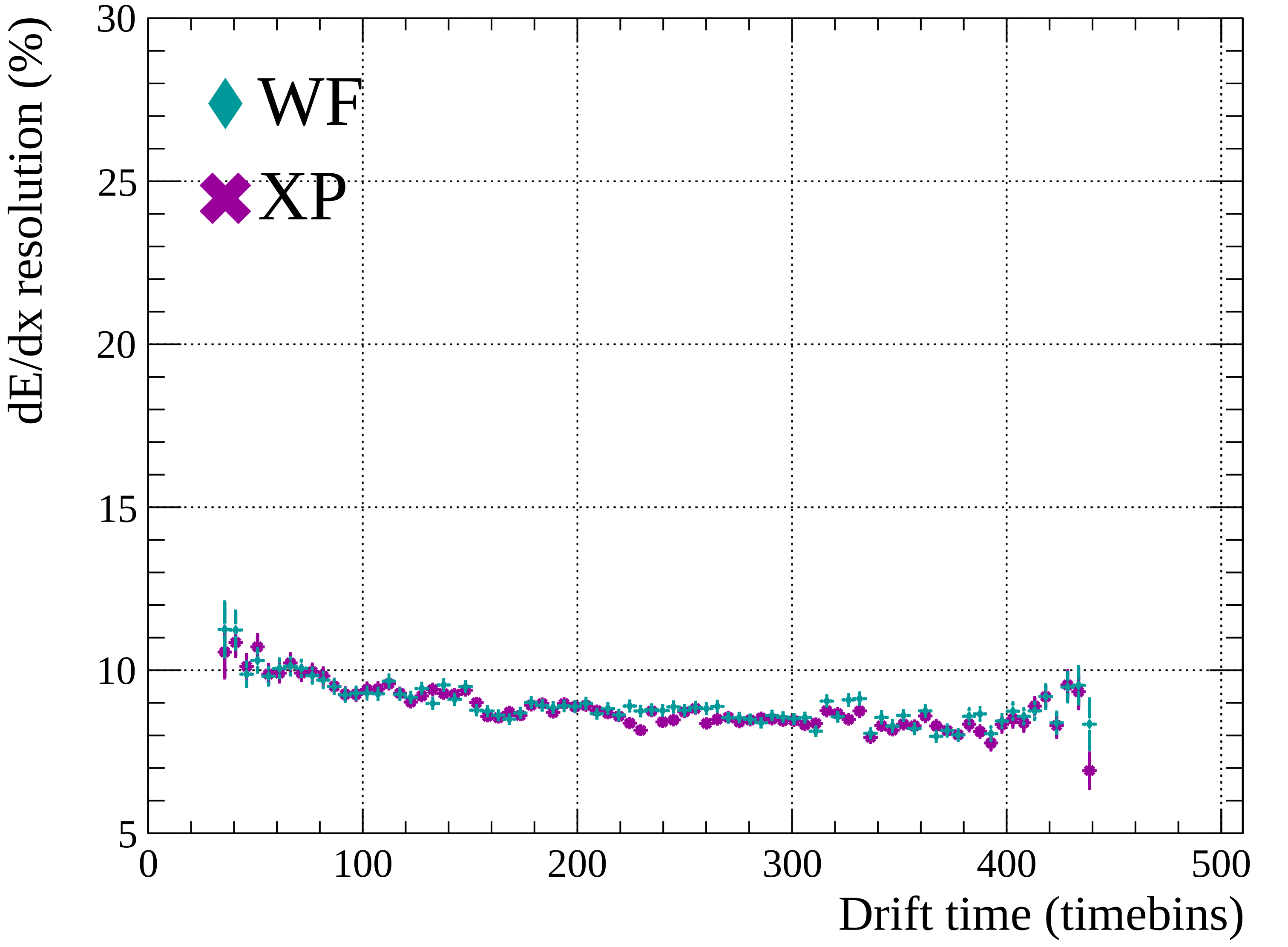

WF
XP
800
700
600
500
400
300
200
−1000
−500
0
500
1000
dE/dx (ADC counts/cm)
Track X position (mm)



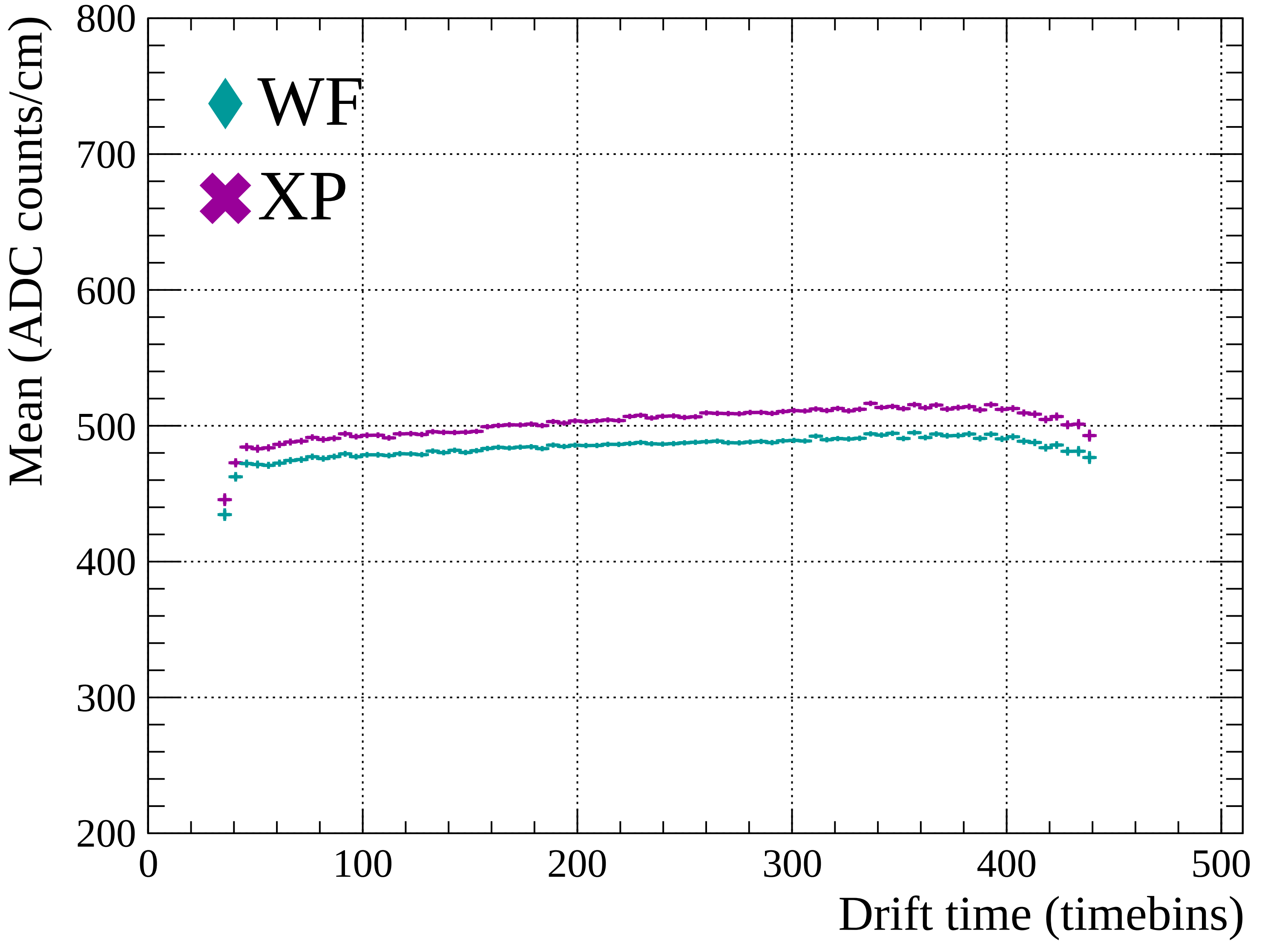
dE/dx with WF (ADC counts/cm)
Track X position (mm)
y = -0.02x + 507.39
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000
0
20
40
60
80
100
120

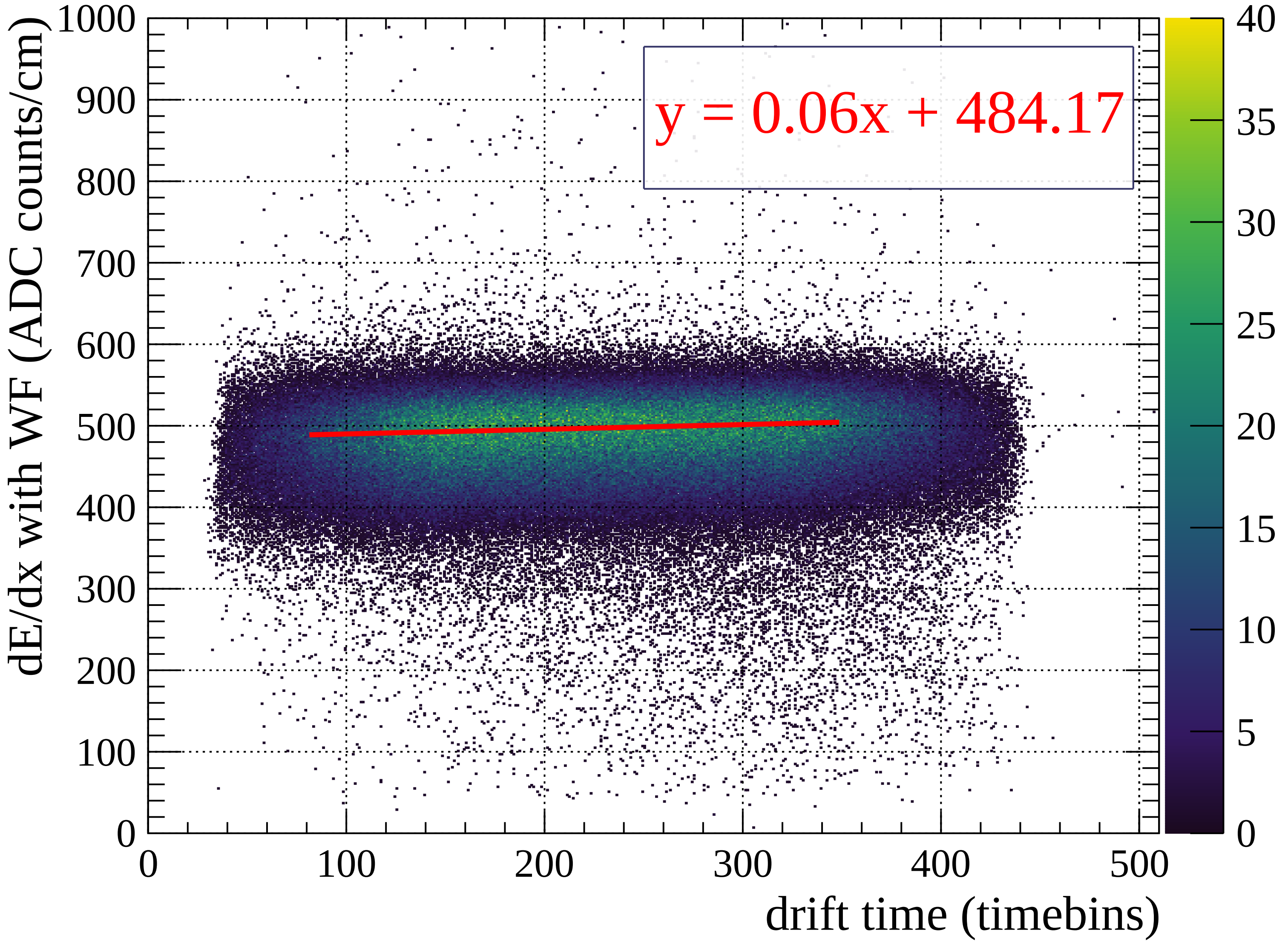


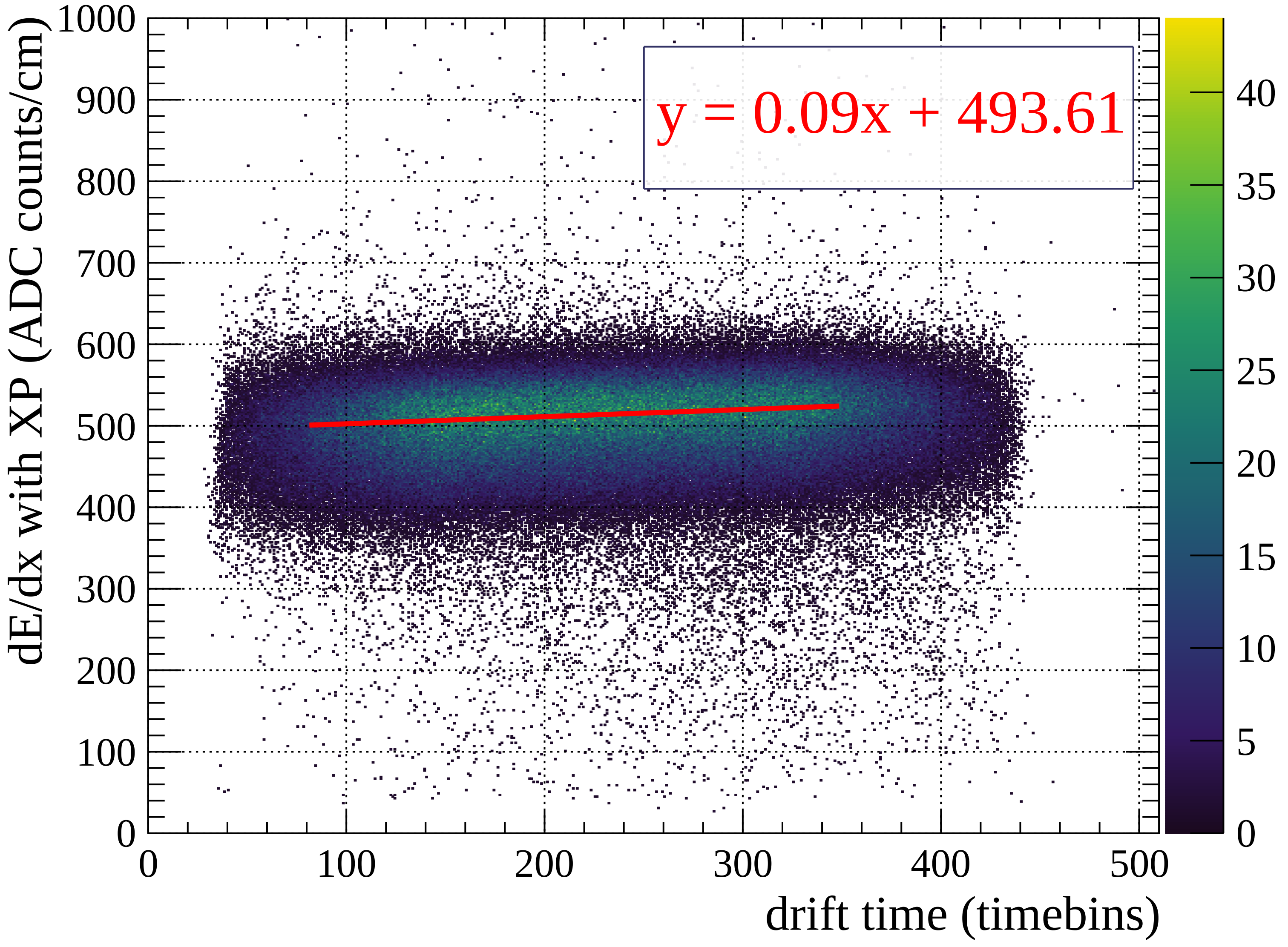
dE/dx with XP (ADC counts/cm)
Track X position (mm)
y = -0.05x + 539.34
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000
20
40
60
80
100
120



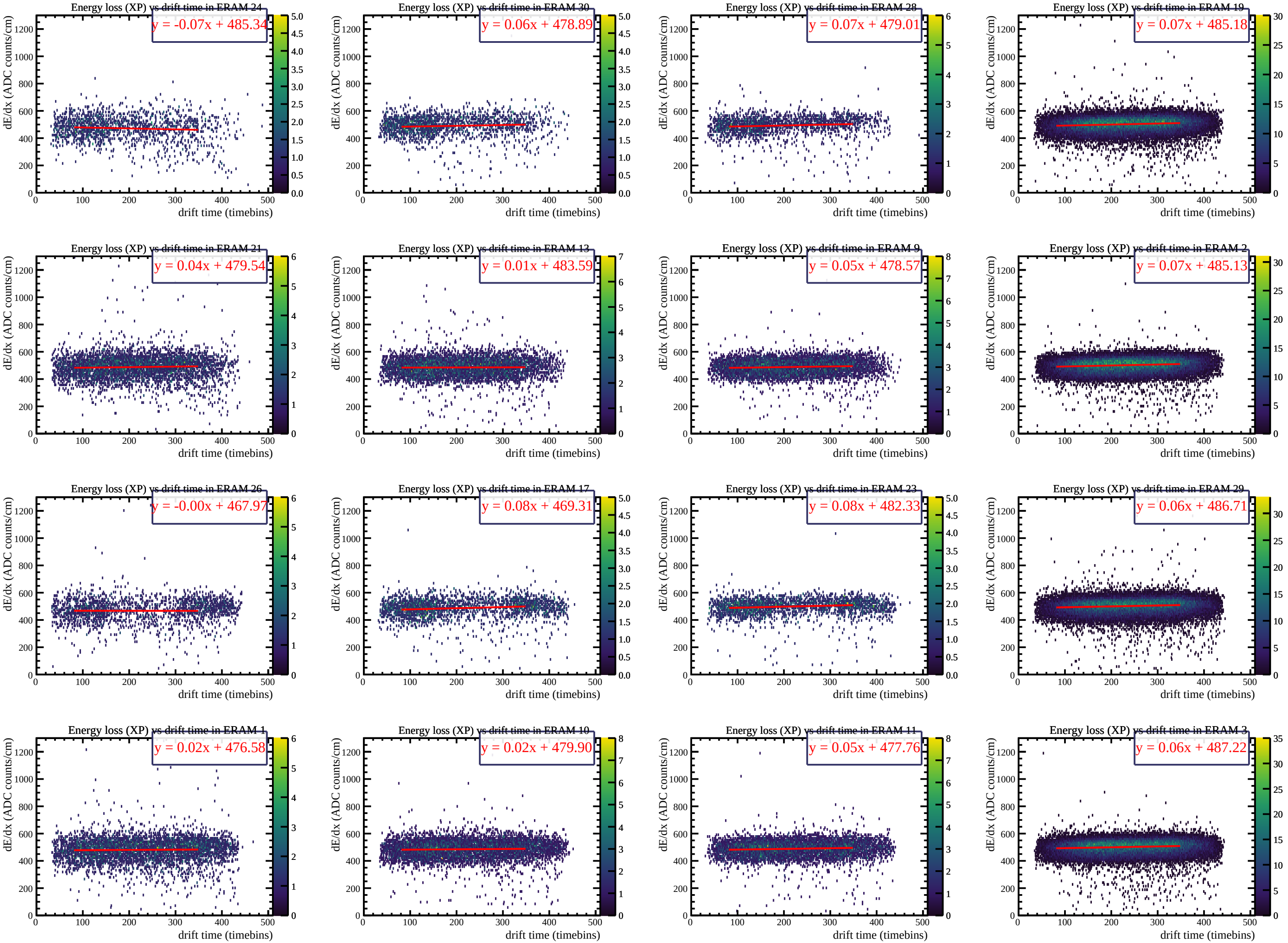
dE/dx resolution (%)
30
25
20
15
10
5
WF
XP
0
100
200
300
400
500
Drift time (timebins)

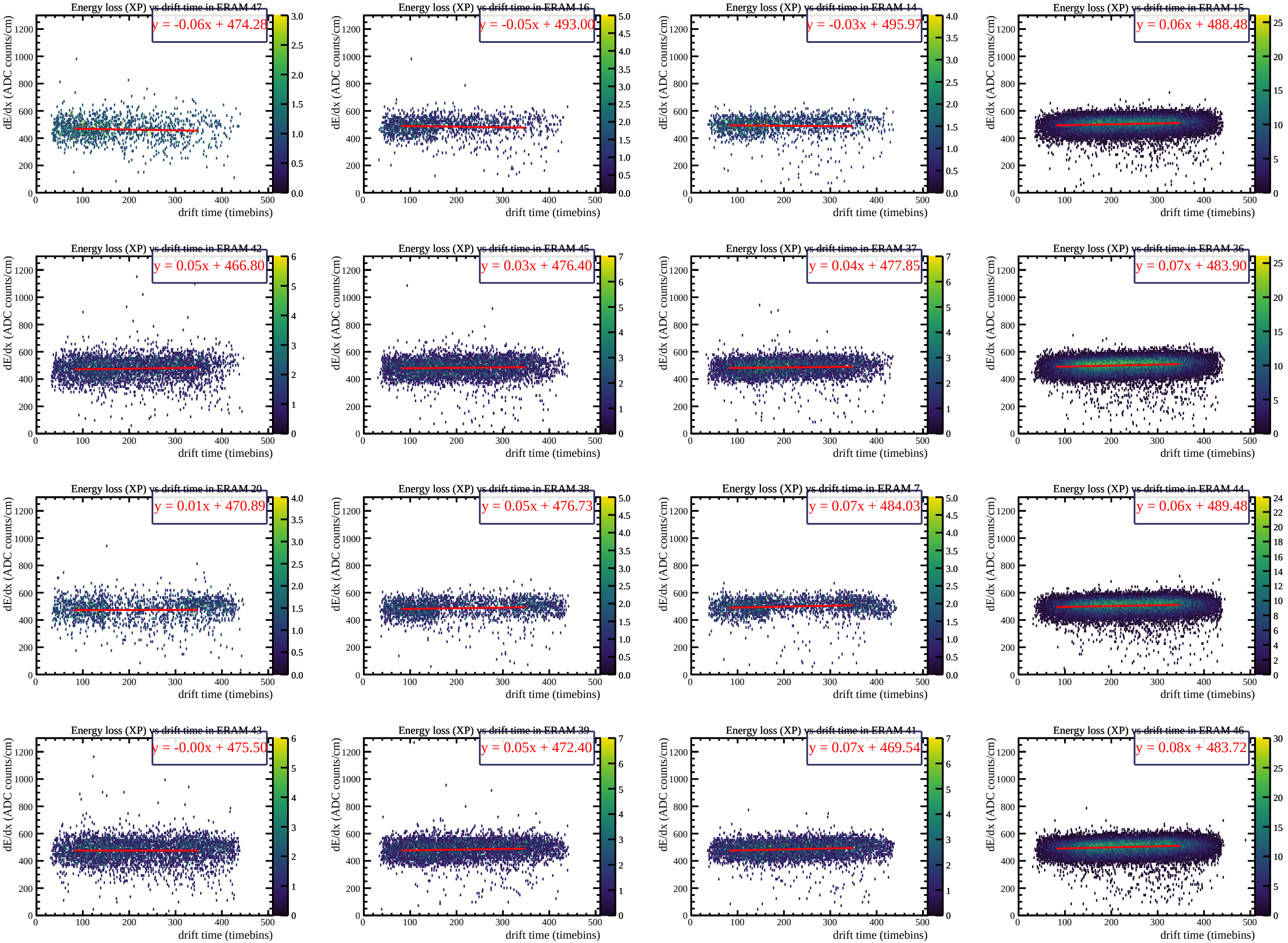

WF
XP
Mean (ADC counts/cm)
Drift time (timebins)
800
700
600
500
400
300
200
0
100
200
300
400
500

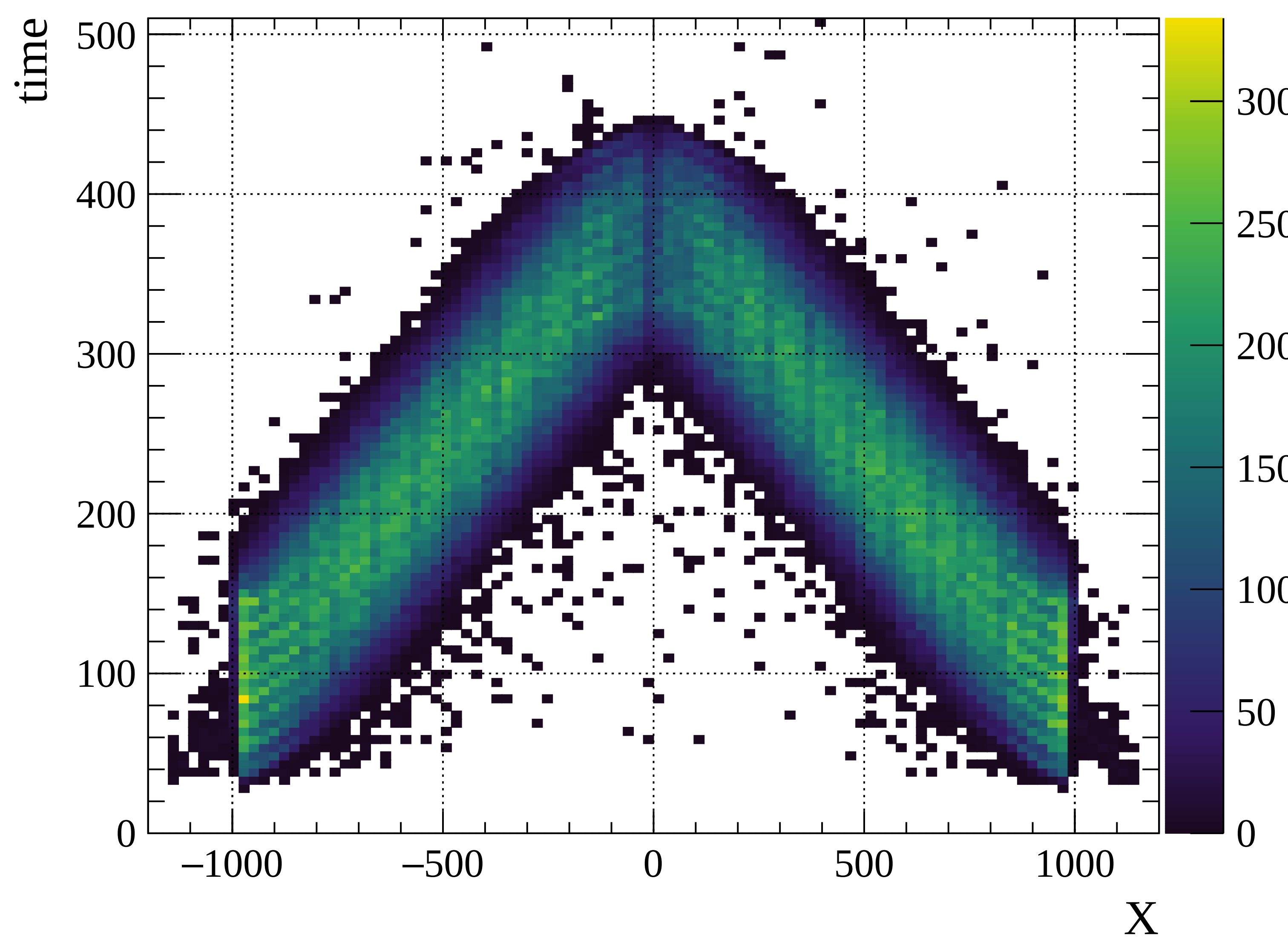

y = 0.06x + 484.17
dE/dx with WF (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
5
10
15
20
25
30
35
40



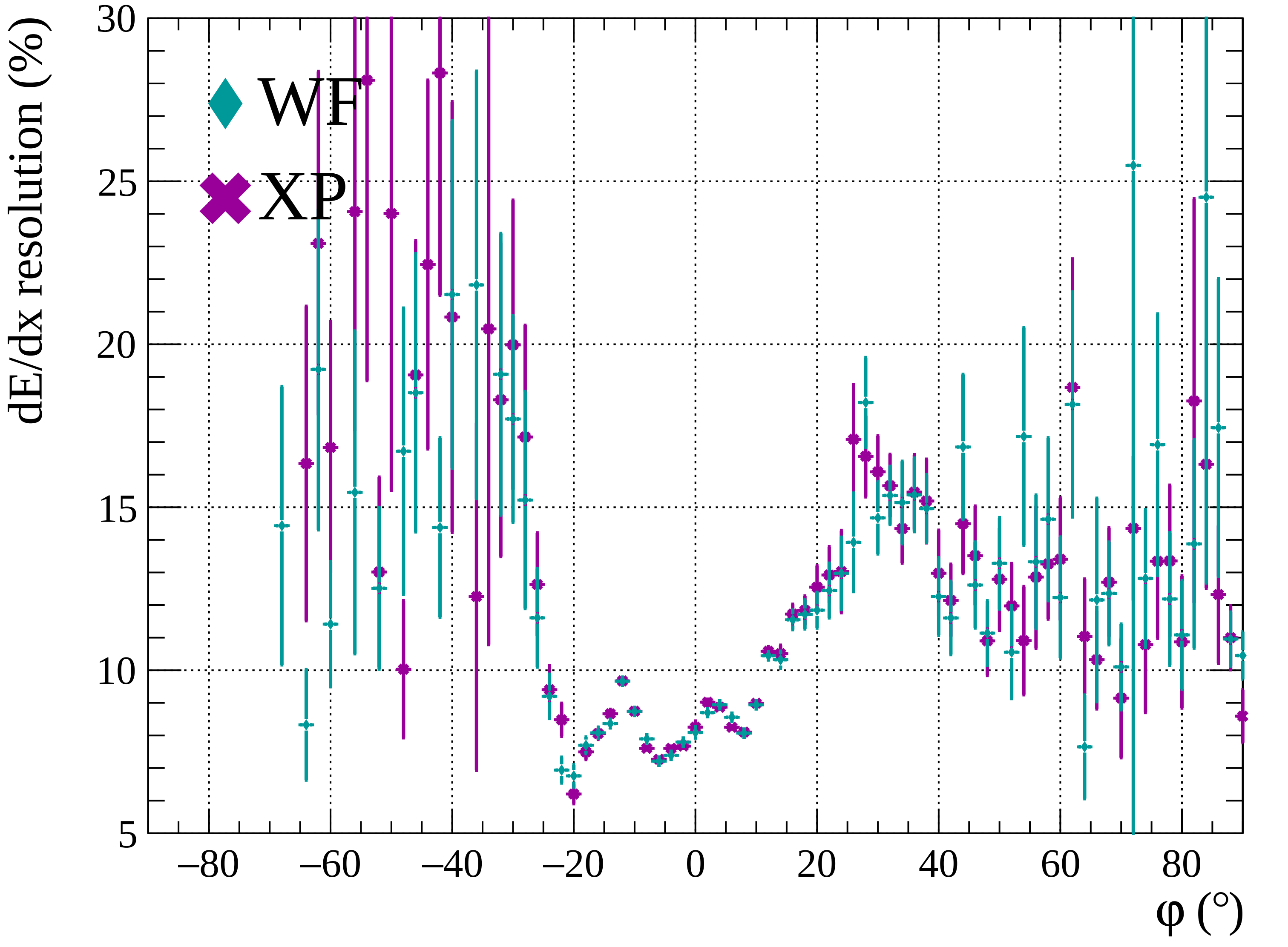
y = 0.09x + 493.61
dE/dx with XP (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
5
10
15
20
25
30
35
40


Energy loss (XP) vs drift time in ERAM 24
y = -0.07x + 485.34
Energy loss (XP) vs drift time in ERAM 30
y = 0.06x + 478.89
Energy loss (XP) vs drift time in ERAM 28
y = 0.07x + 479.01
Energy loss (XP) vs drift time in ERAM 19
y = 0.07x + 485.18
Energy loss (XP) vs drift time in ERAM 21
y = 0.04x + 479.54
Energy loss (XP) vs drift time in ERAM 13
y = 0.01x + 483.59
Energy loss (XP) vs drift time in ERAM 9
y = 0.05x + 478.57
Energy loss (XP) vs drift time in ERAM 2
y = 0.07x + 485.13
Energy loss (XP) vs drift time in ERAM 26
y = -0.00x + 467.97
Energy loss (XP) vs drift time in ERAM 17
y = 0.08x + 469.31
Energy loss (XP) vs drift time in ERAM 23
y = 0.08x + 482.33
Energy loss (XP) vs drift time in ERAM 29
y = 0.06x + 486.71
Energy loss (XP) vs drift time in ERAM 1
y = 0.02x + 476.58
Energy loss (XP) vs drift time in ERAM 10
y = 0.02x + 479.90
Energy loss (XP) vs drift time in ERAM 11
y = 0.05x + 477.76
Energy loss (XP) vs drift time in ERAM 3
y = 0.06x + 487.22
dE/dx (ADC counts/cm)
drift time (timebins)

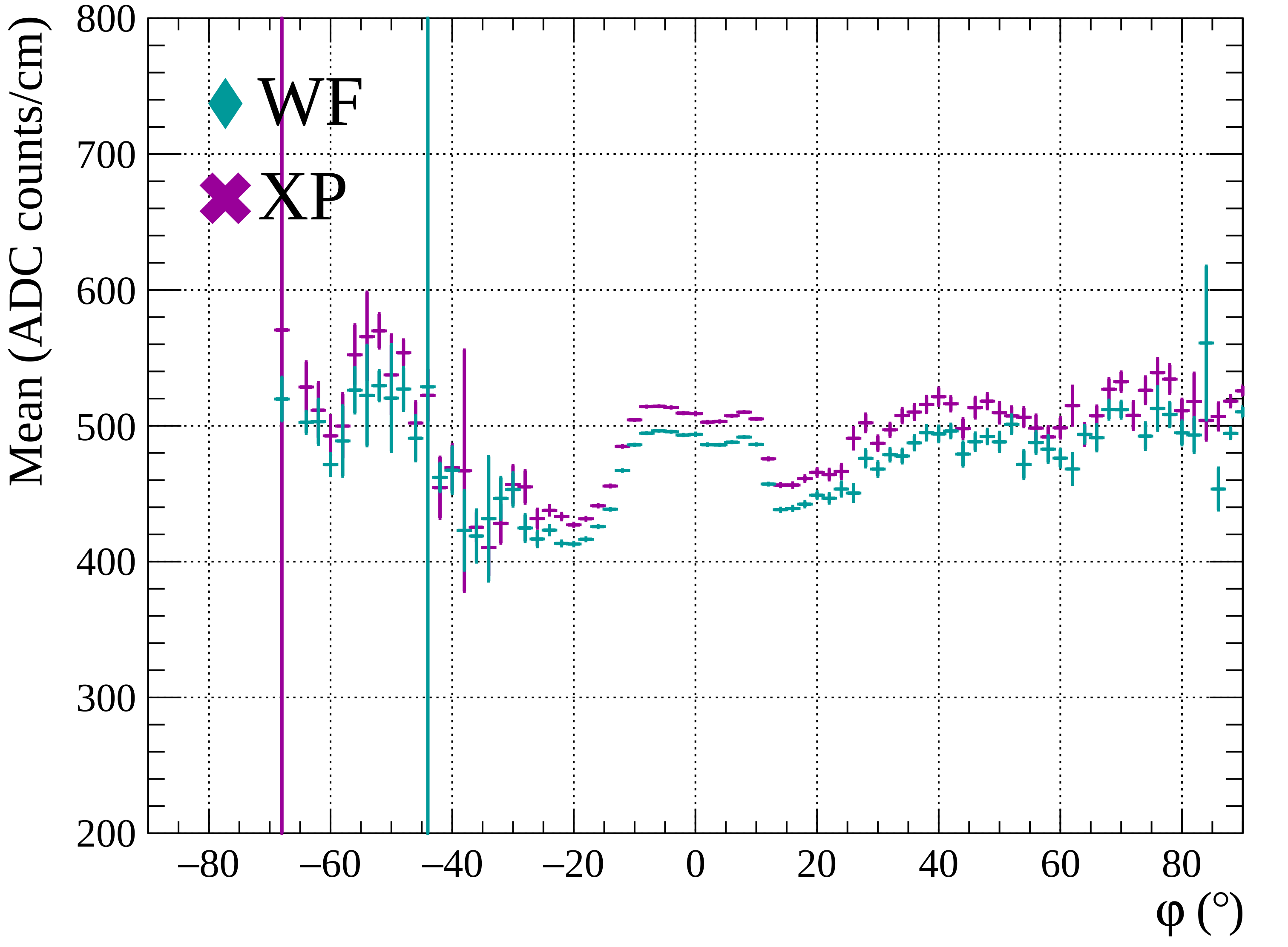

Energy loss (XP) vs drift time in ERAM 47
y = -0.06x + 474.28
Energy loss (XP) vs drift time in ERAM 16
y = -0.05x + 493.00
Energy loss (XP) vs drift time in ERAM 14
y = -0.03x + 495.97
Energy loss (XP) vs drift time in ERAM 15
y = 0.06x + 488.48
Energy loss (XP) vs drift time in ERAM 42
y = 0.05x + 466.80
Energy loss (XP) vs drift time in ERAM 45
y = 0.03x + 476.40
Energy loss (XP) vs drift time in ERAM 37
y = 0.04x + 477.85
Energy loss (XP) vs drift time in ERAM 36
y = 0.07x + 483.90
Energy loss (XP) vs drift time in ERAM 20
y = 0.01x + 470.89
Energy loss (XP) vs drift time in ERAM 38
y = 0.05x + 476.73
Energy loss (XP) vs drift time in ERAM 7
y = 0.07x + 484.03
Energy loss (XP) vs drift time in ERAM 44
y = 0.06x + 489.48
Energy loss (XP) vs drift time in ERAM 43
y = -0.00x + 475.50
Energy loss (XP) vs drift time in ERAM 39
y = 0.05x + 472.40
Energy loss (XP) vs drift time in ERAM 41
y = 0.07x + 469.54
Energy loss (XP) vs drift time in ERAM 46
y = 0.08x + 483.72
dE/dx (ADC counts/cm)
drift time (timebins)



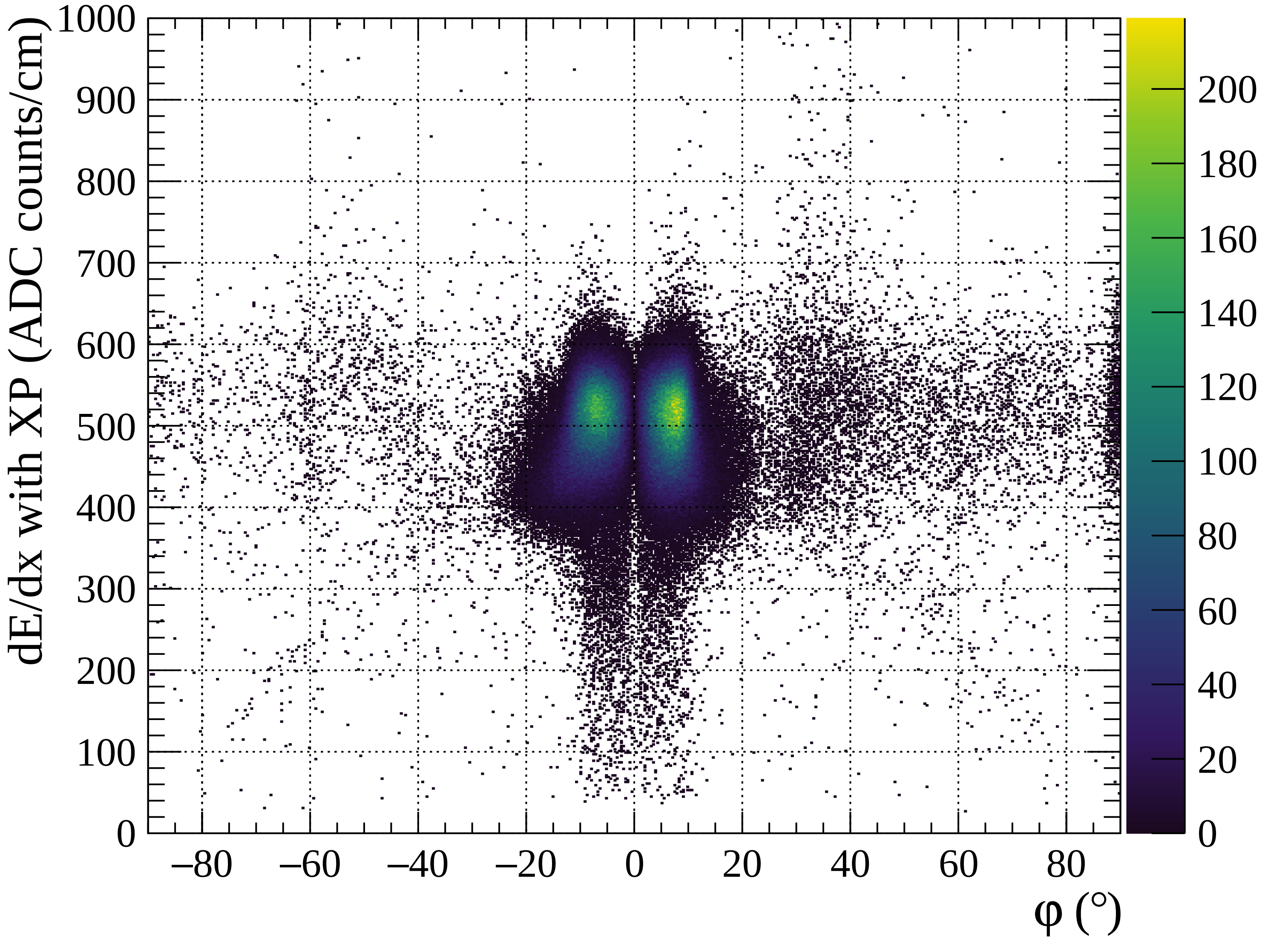
time
500
400
300
200
100
0
−1000
−500
0
500
1000
X
300
250
200
150
100
50
0

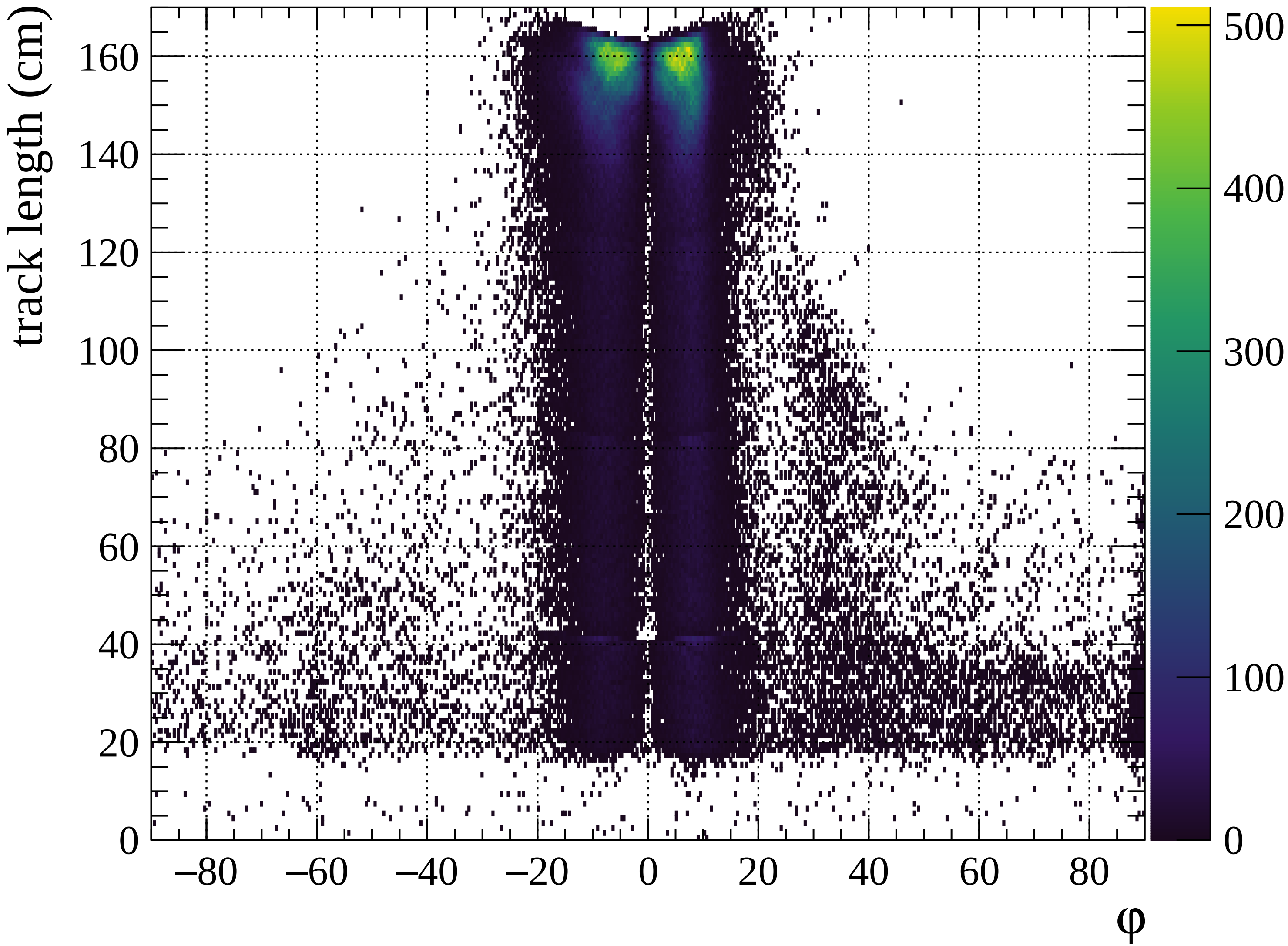


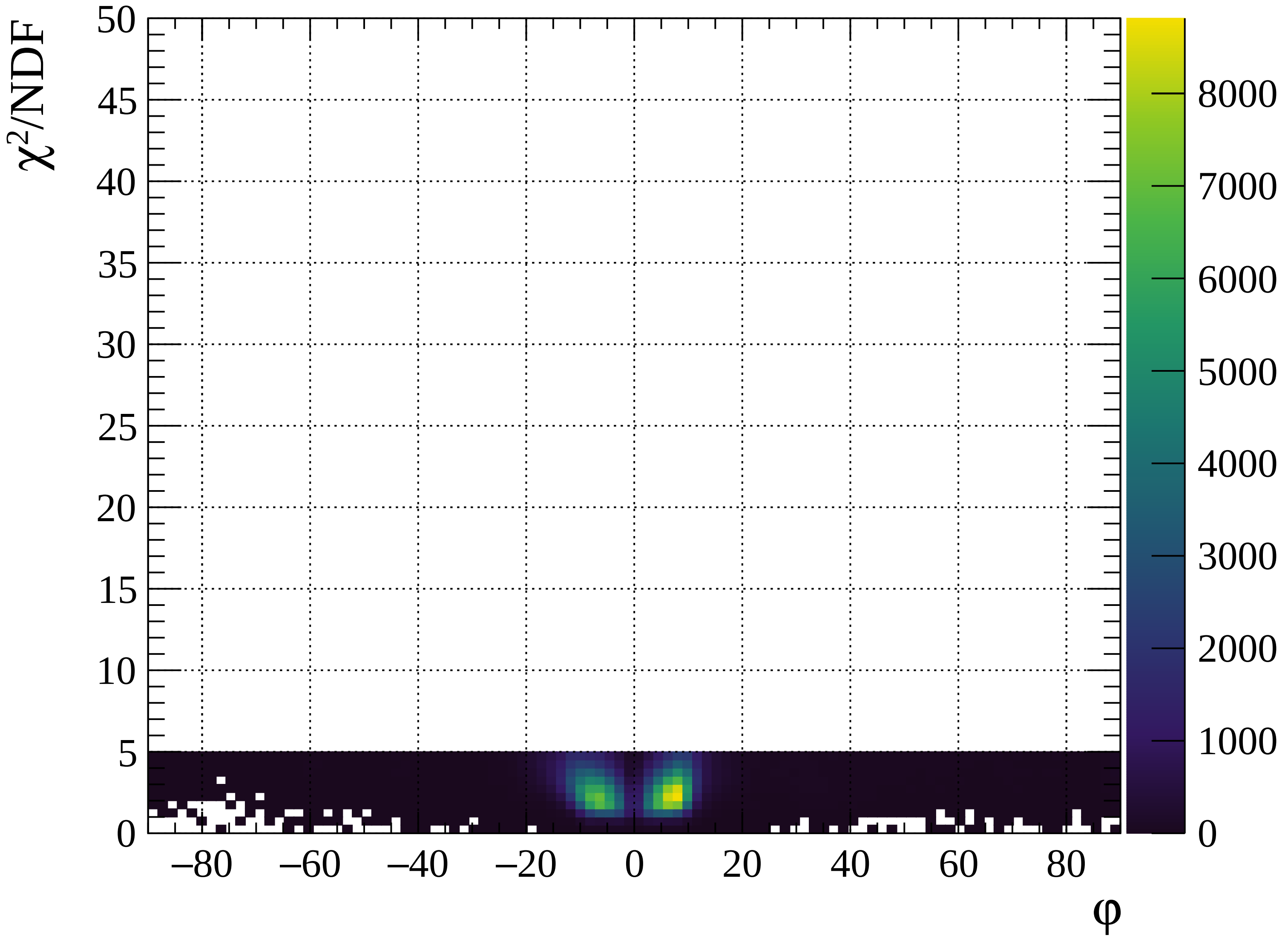
WF
XP
dE/dx resolution (%)
φ (°)
30
25
20
15
10
5
−80
−60
−40
−20
0
20
40
60
80



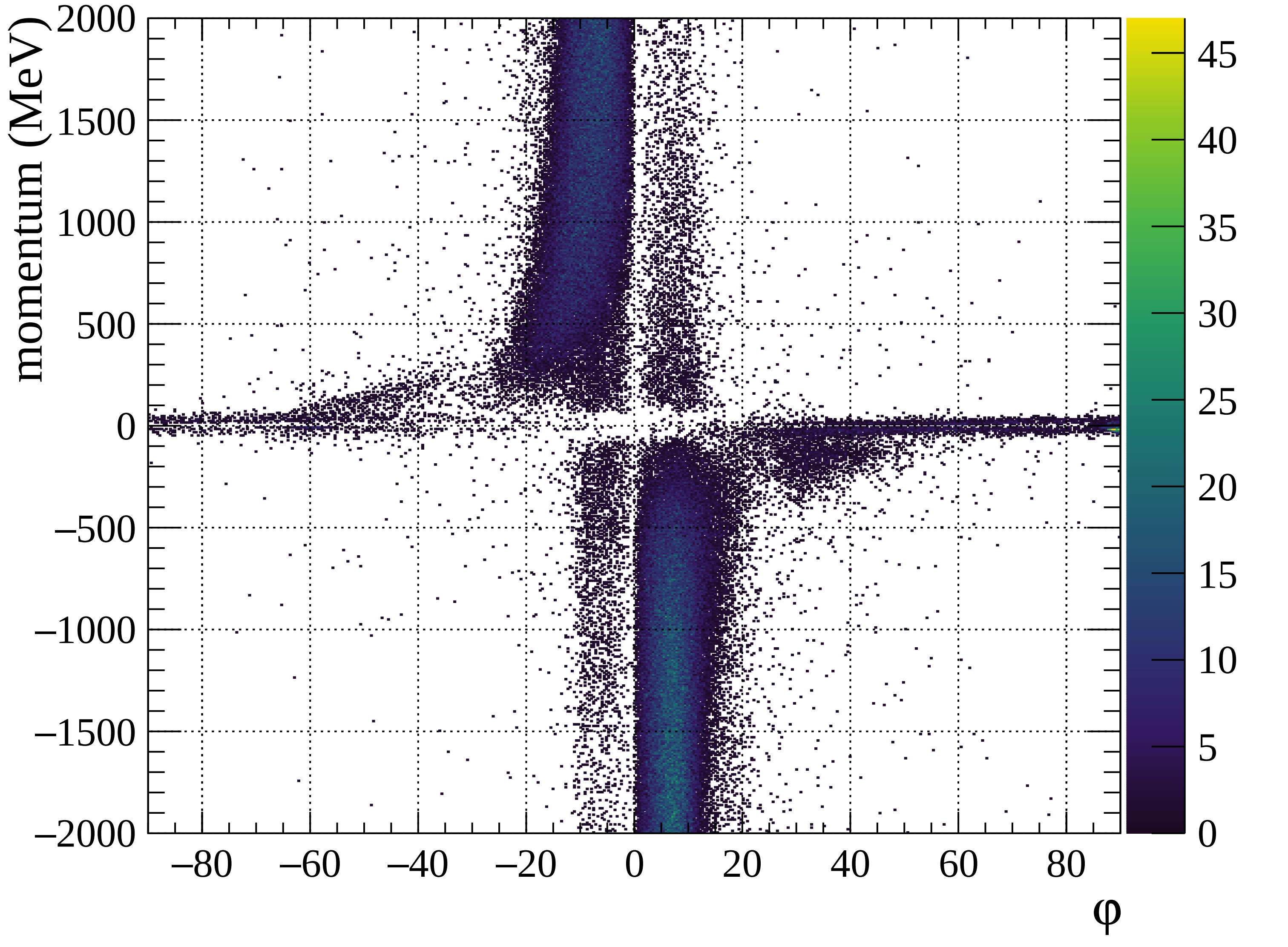
Mean (ADC counts/cm)
800
700
600
500
400
300
200
WF
XP
−80
−60
−40
−20
0
20
40
60
80
φ (°)


dE/dx with XP (ADC counts/cm)
φ (°)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
200
180
160
140
120
100
80
60
40
20
0

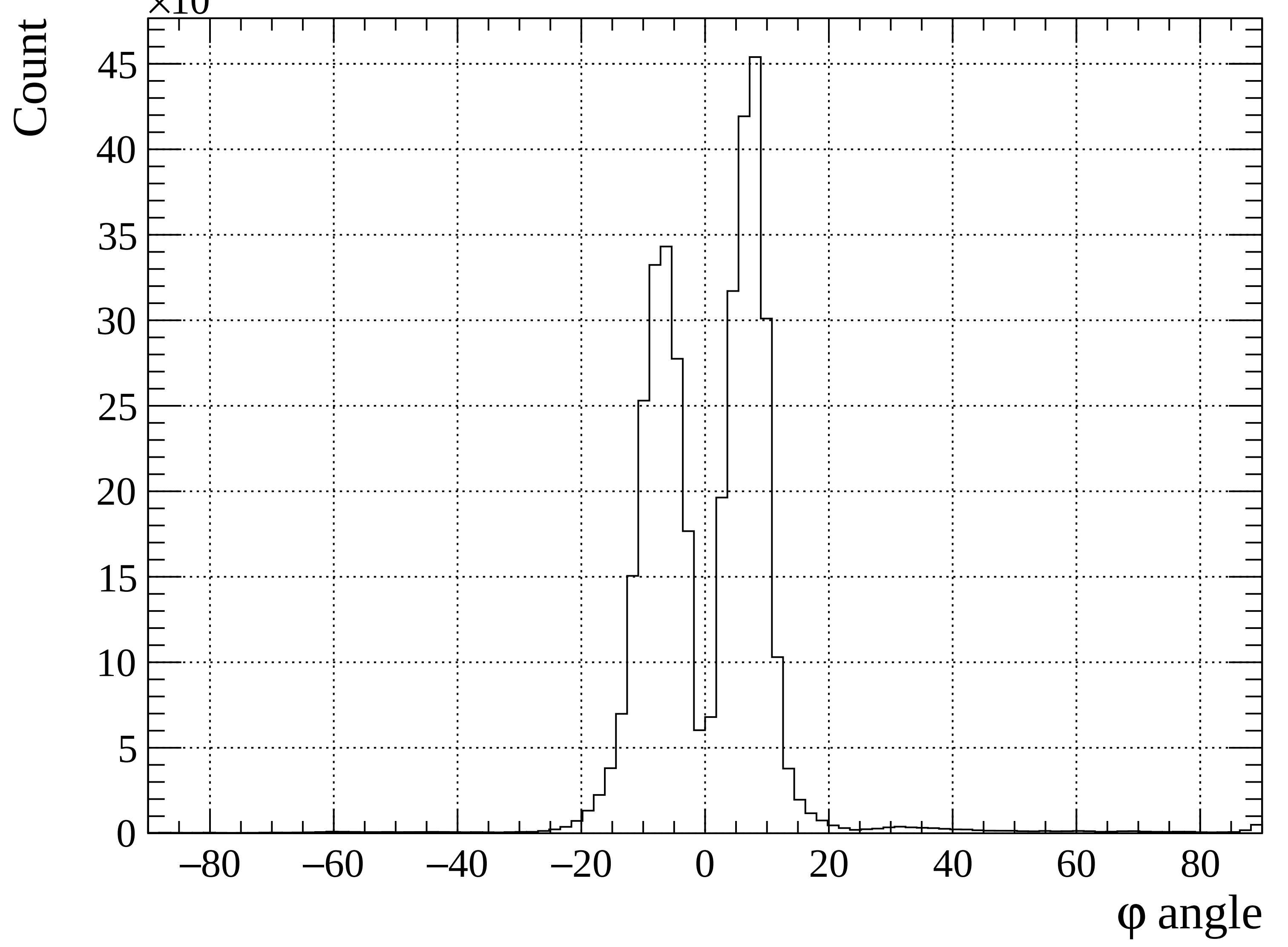

track length (cm)
160
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
φ
500
400
300
200
100
0



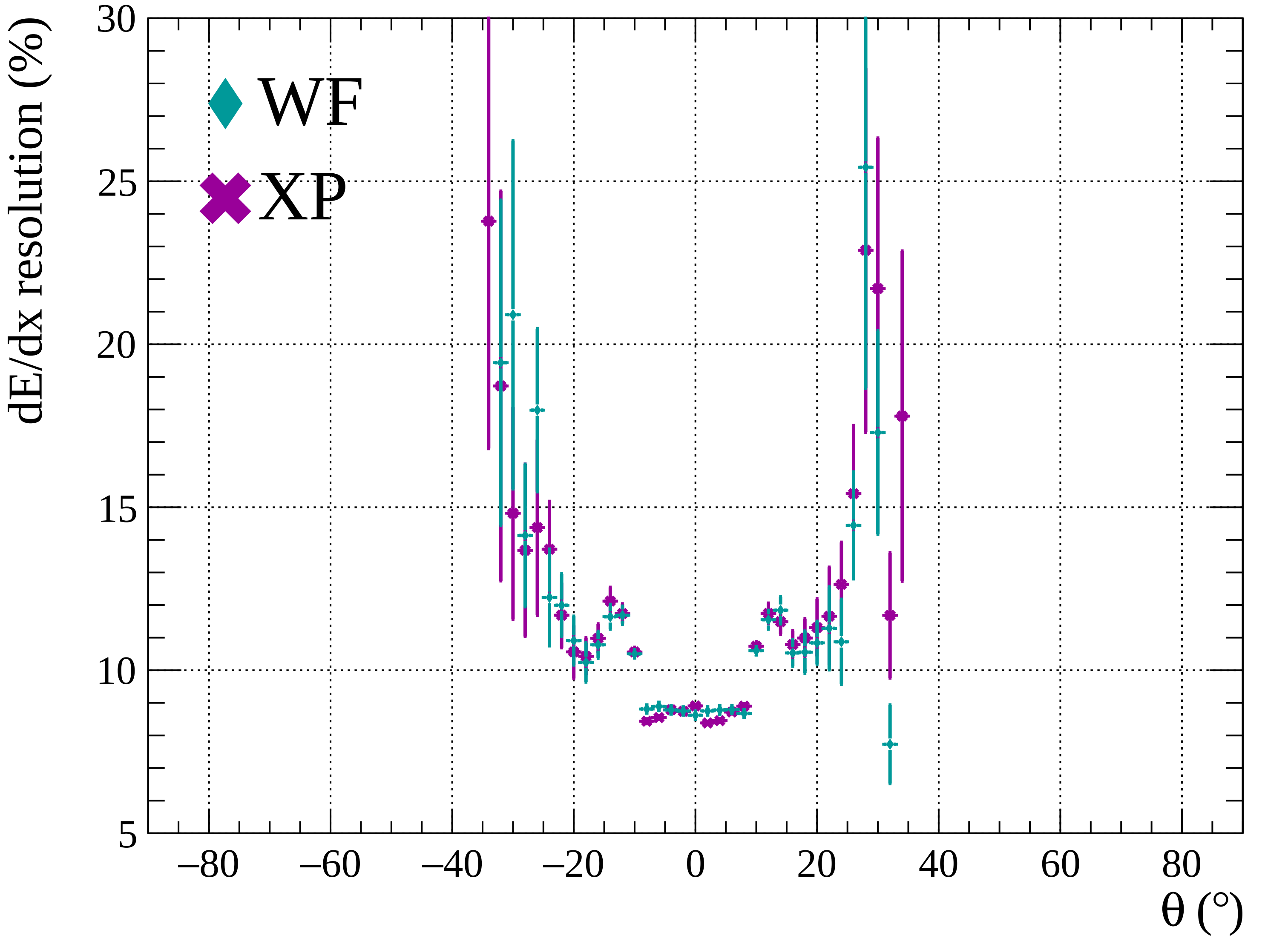
χ^2/NDF
φ
0
5
10
15
20
25
30
35
40
45
50
−80
−60
−40
−20
0
20
40
60
80
0
1000
2000
3000
4000
5000
6000
7000
8000

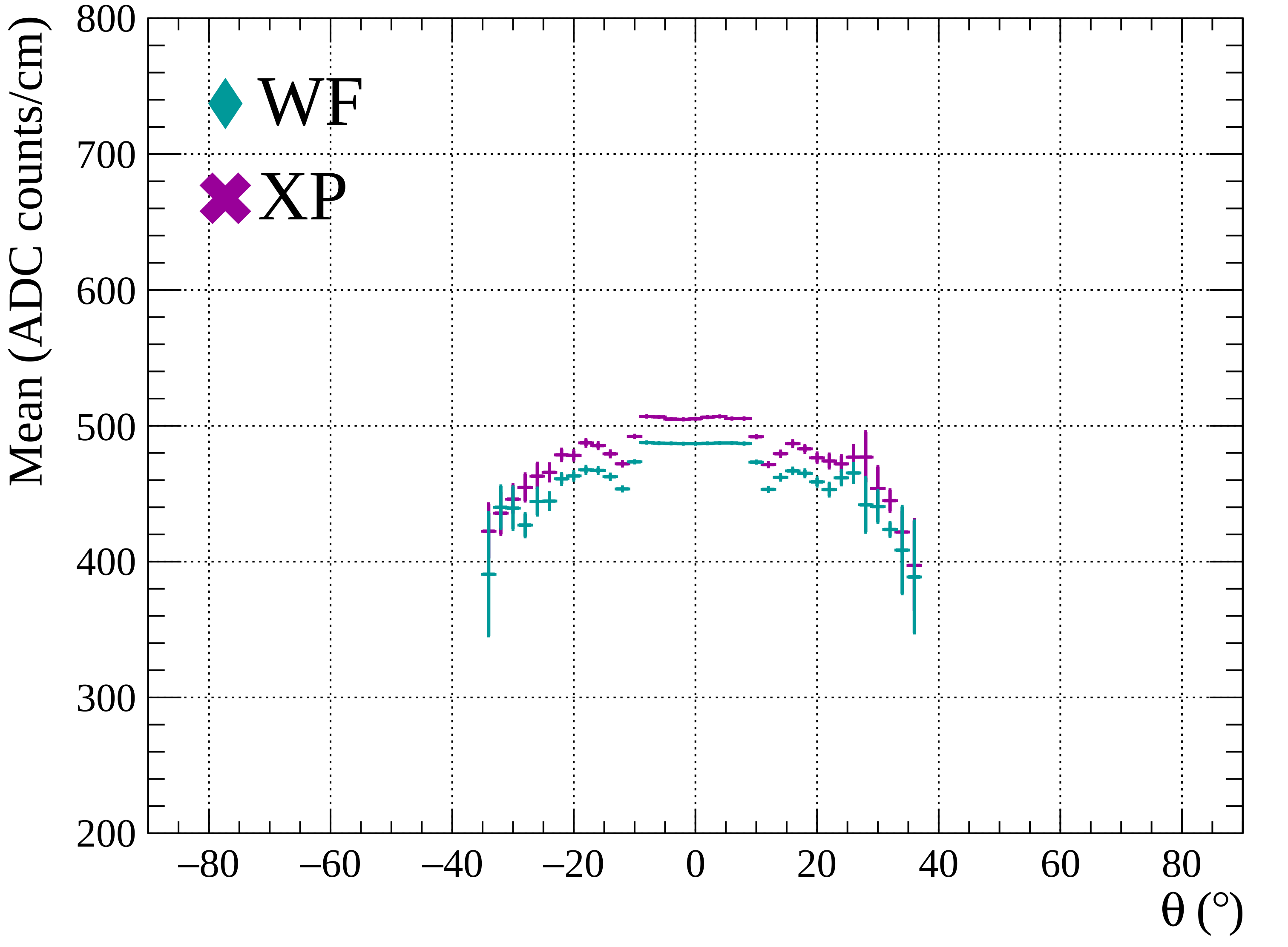


momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
φ
45
40
35
30
25
20
15
10
5
0

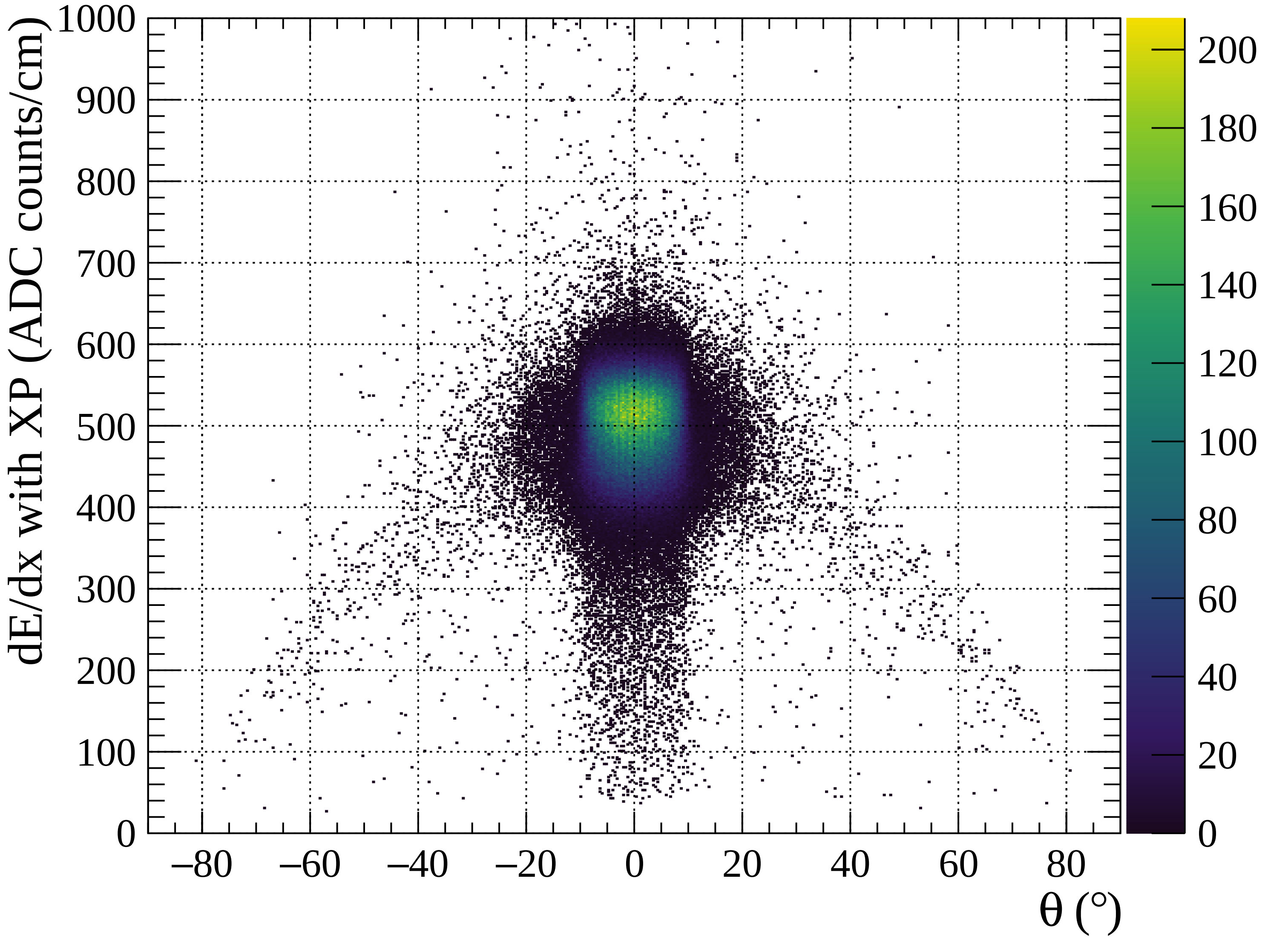


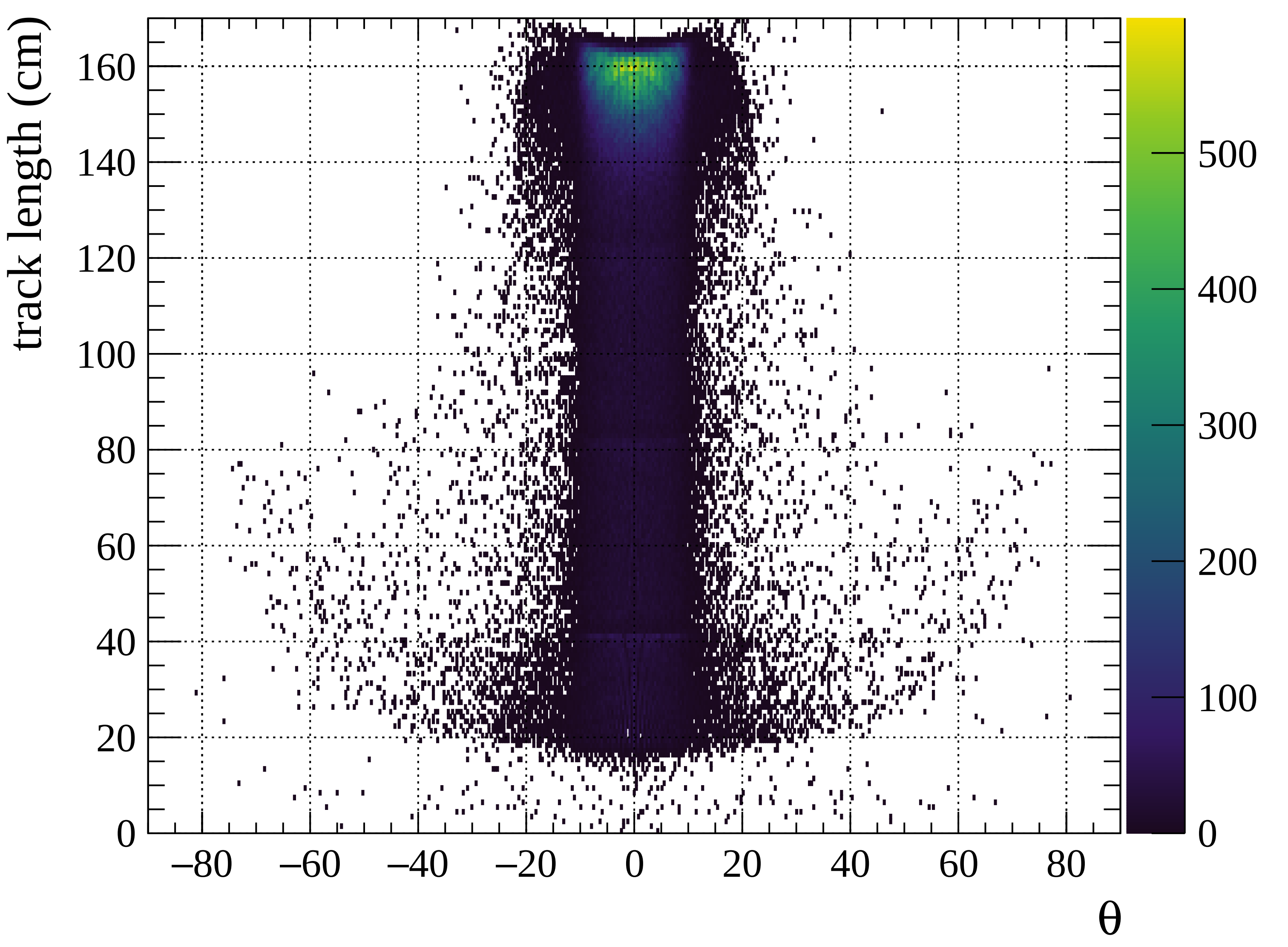
Count
×10
φ angle

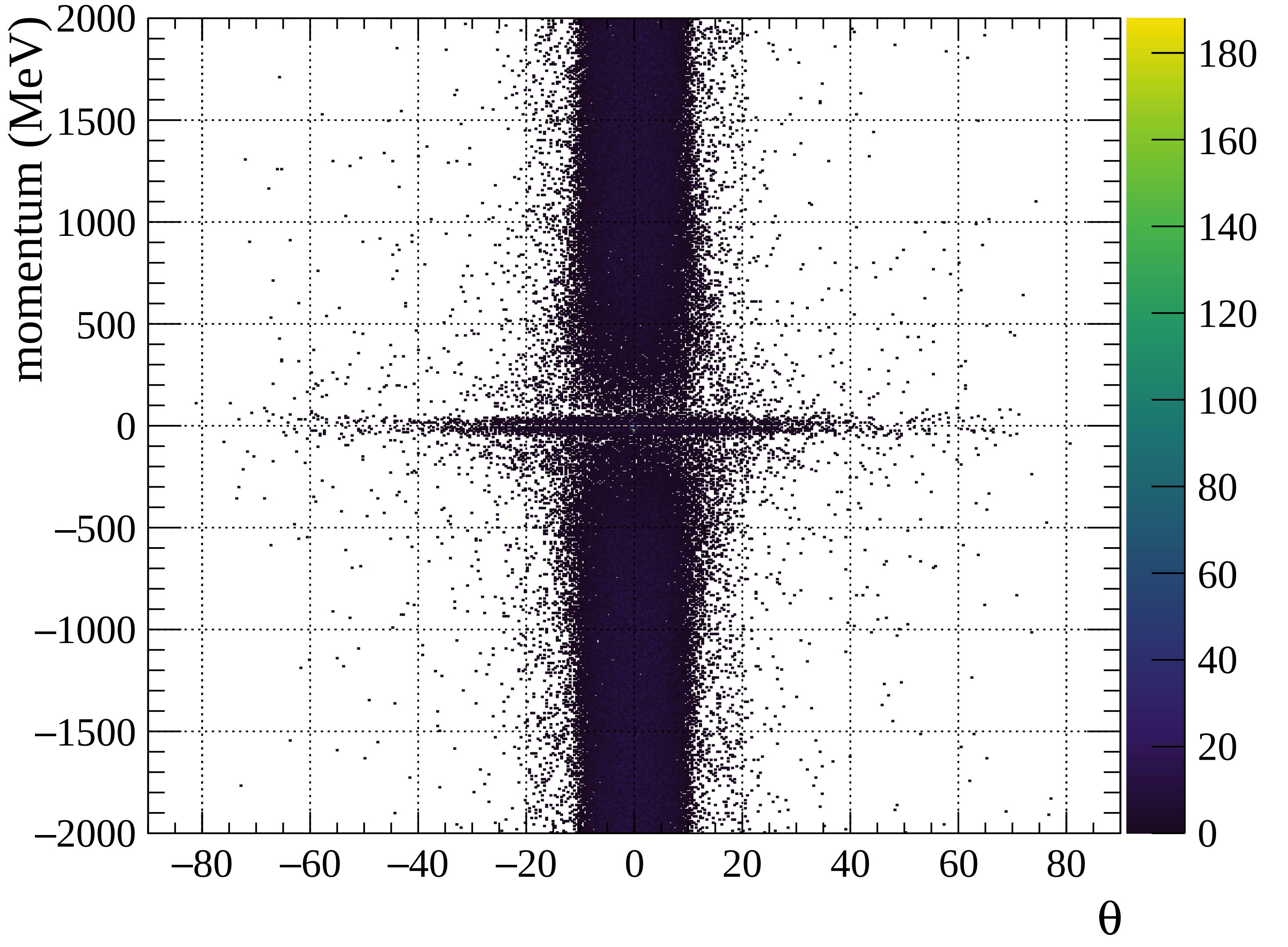

WF
XP
dE/dx resolution (%)
θ (°)
5
10
15
20
25
30
−80
−60
−40
−20
0
20
40
60
80

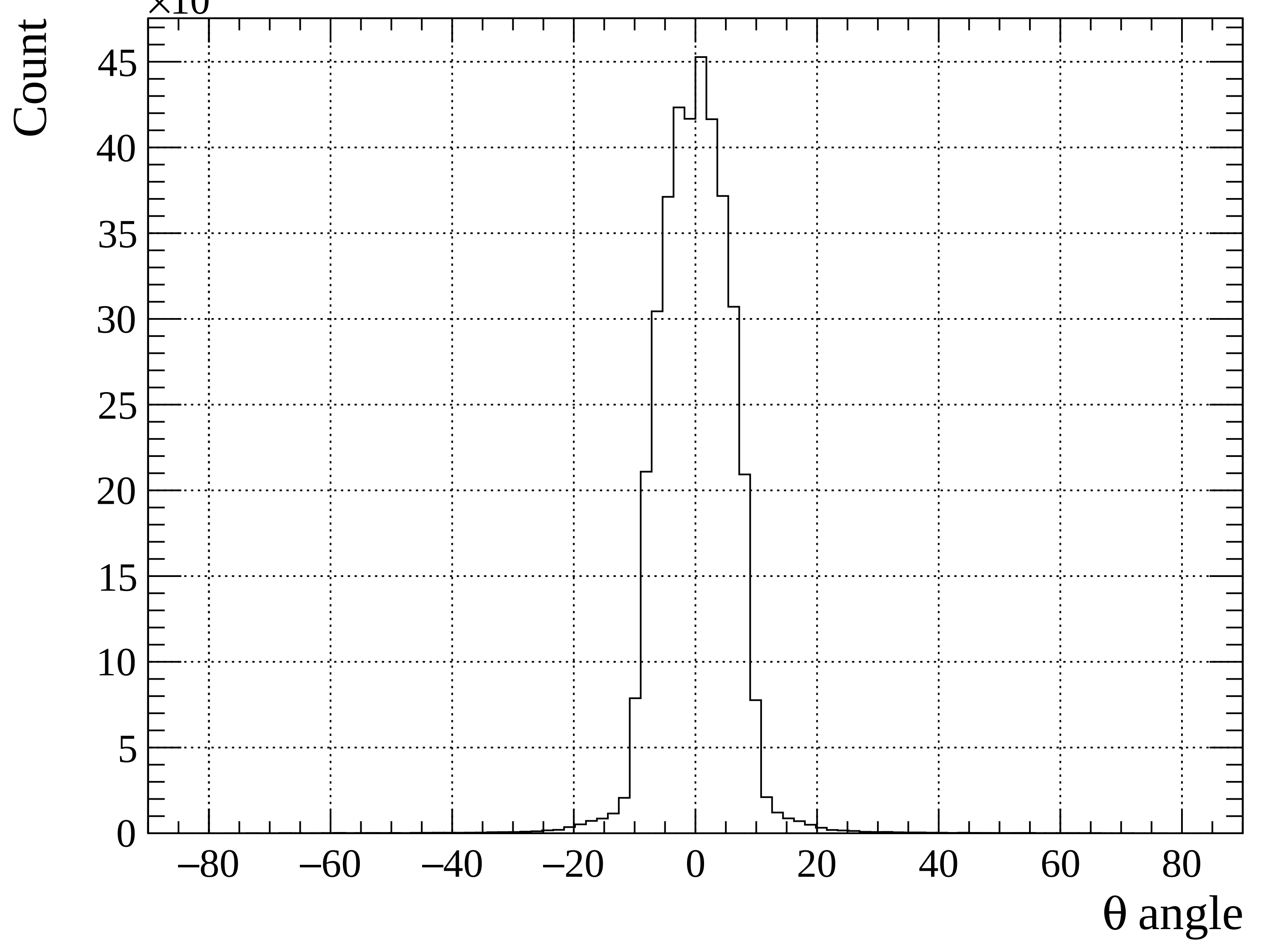


Mean (ADC counts/cm)
800
700
600
500
400
300
200
WF
XP
−80
−60
−40
−20
0
20
40
60
80
θ (°)

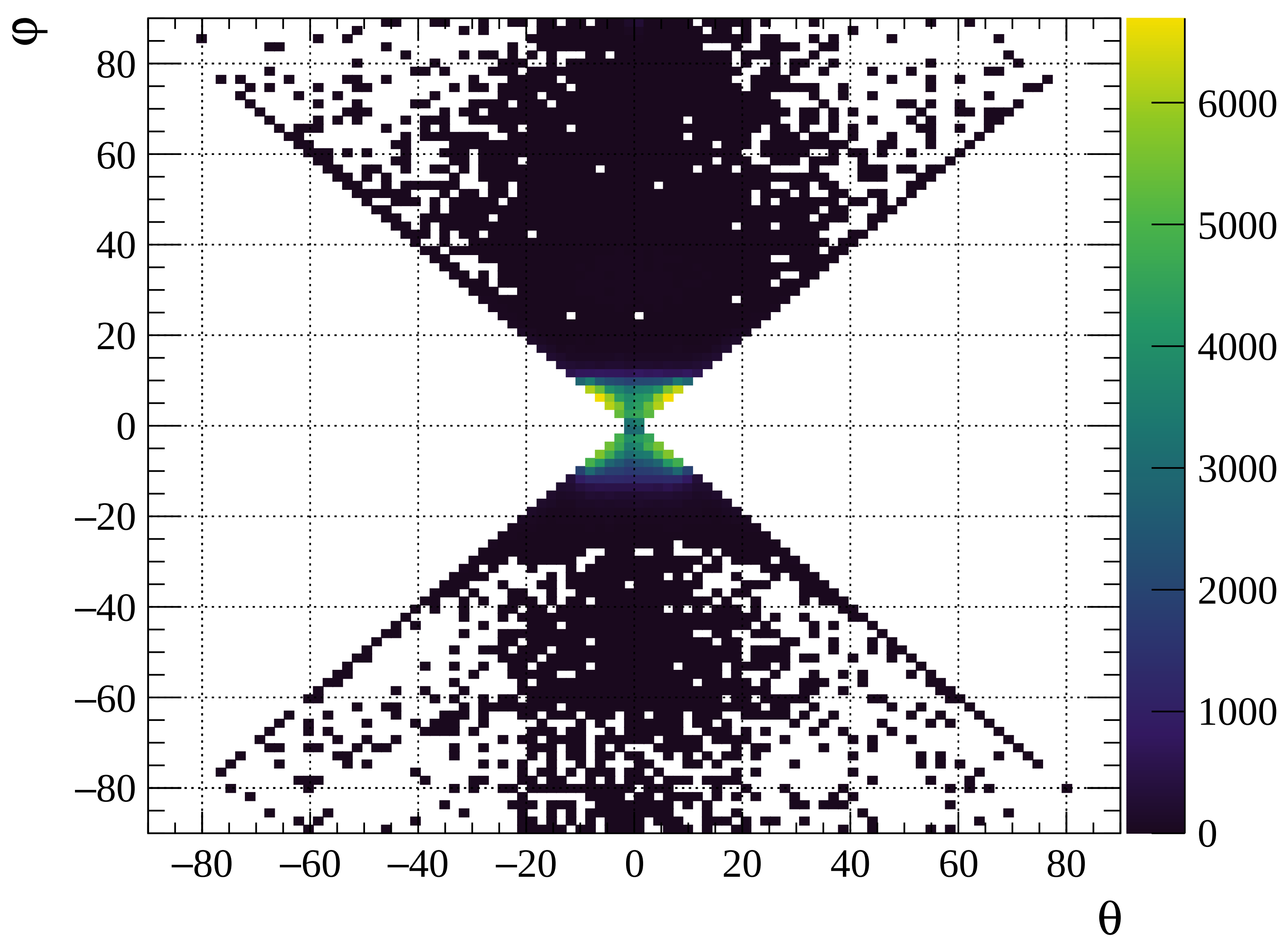

dE/dx with XP (ADC counts/cm)
θ (°)
0
100
200
300
400
500
600
700
800
900
1000
−80
−60
−40
−20
0
20
40
60
80
0
20
40
60
80
100
120
140
160
180
200

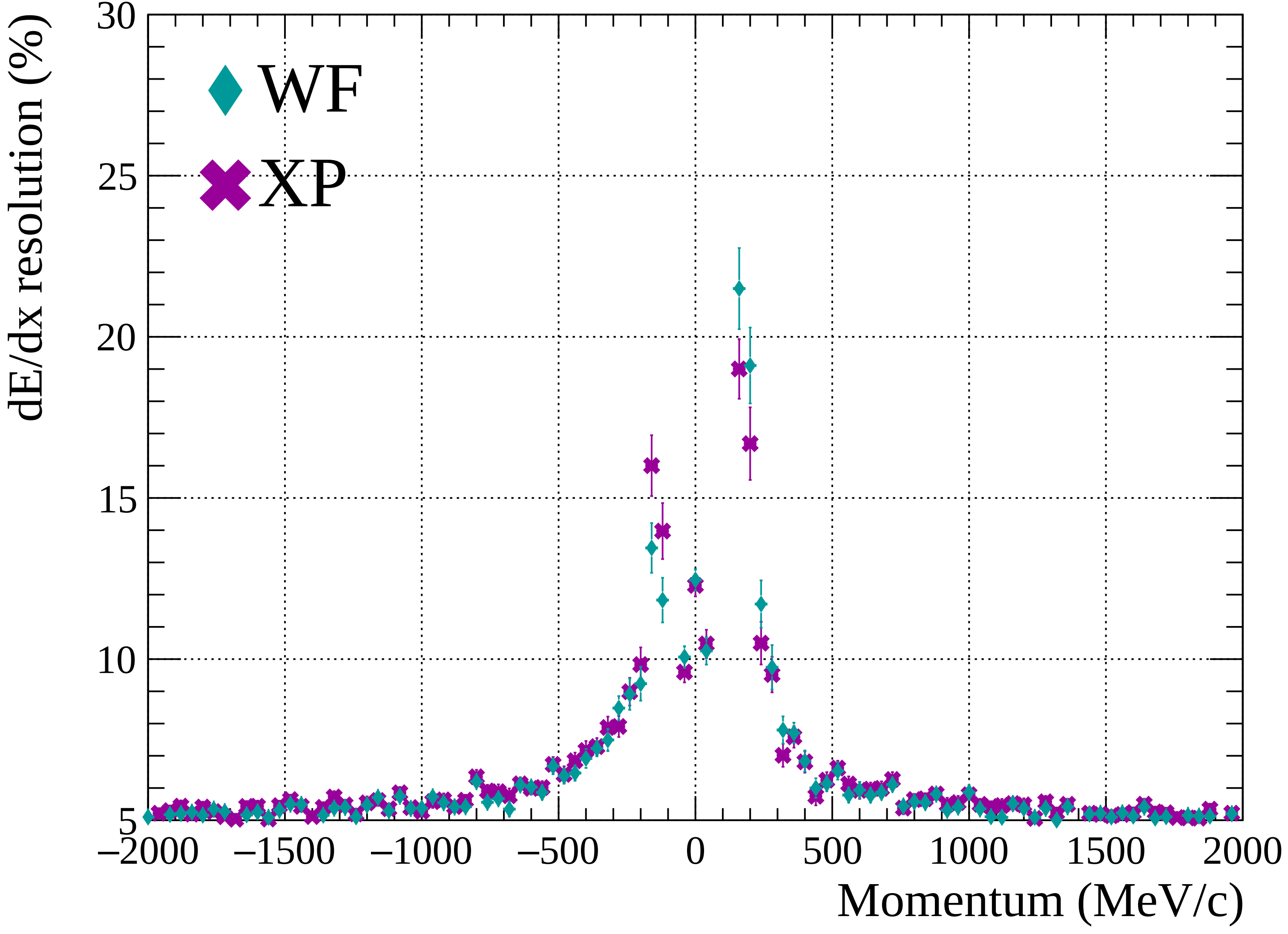

track length (cm)
160
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
θ
500
400
300
200
100
0


momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
θ
180
160
140
120
100
80
60
40
20
0

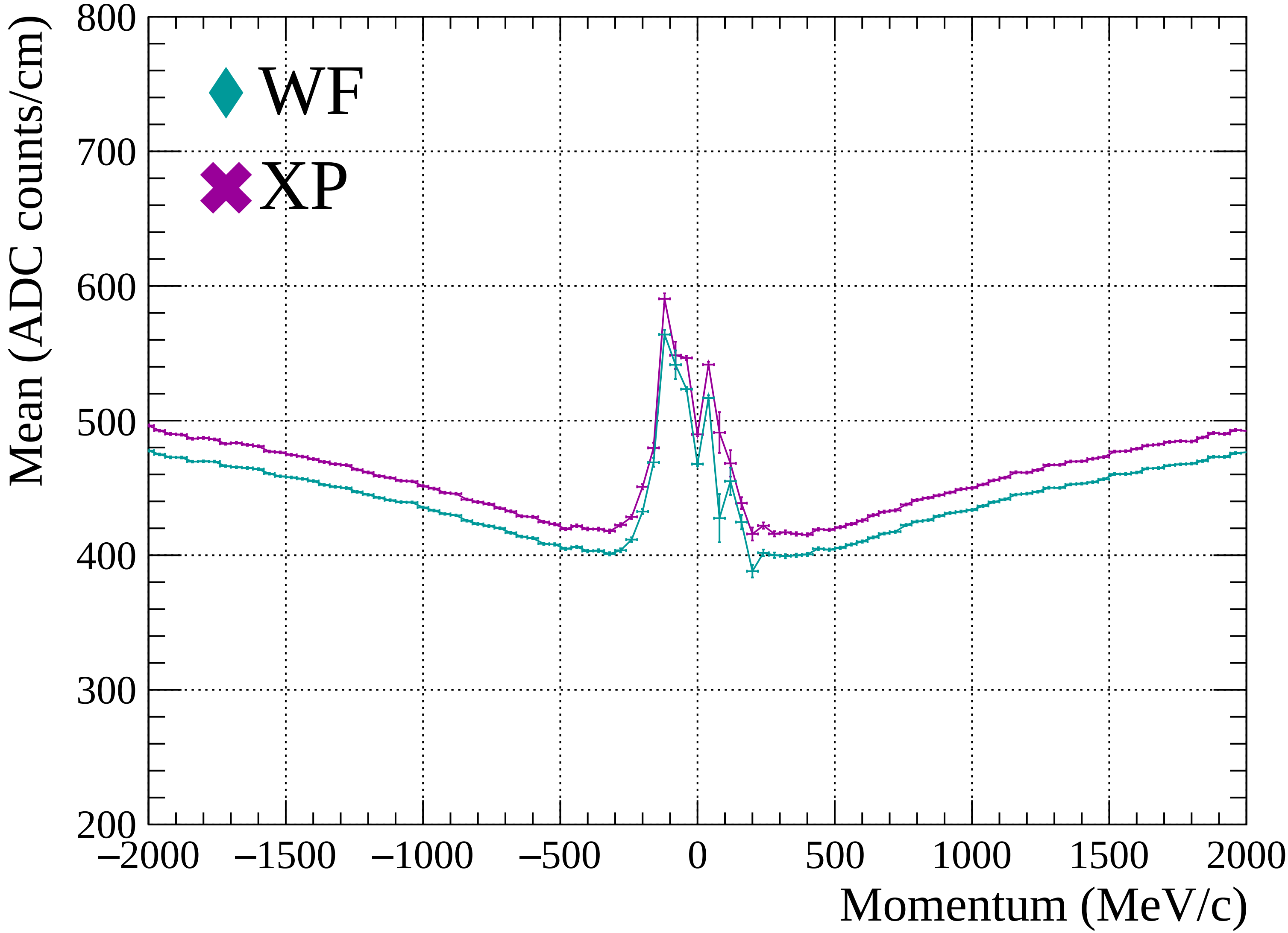


×10
Count
45
40
35
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
θ angle

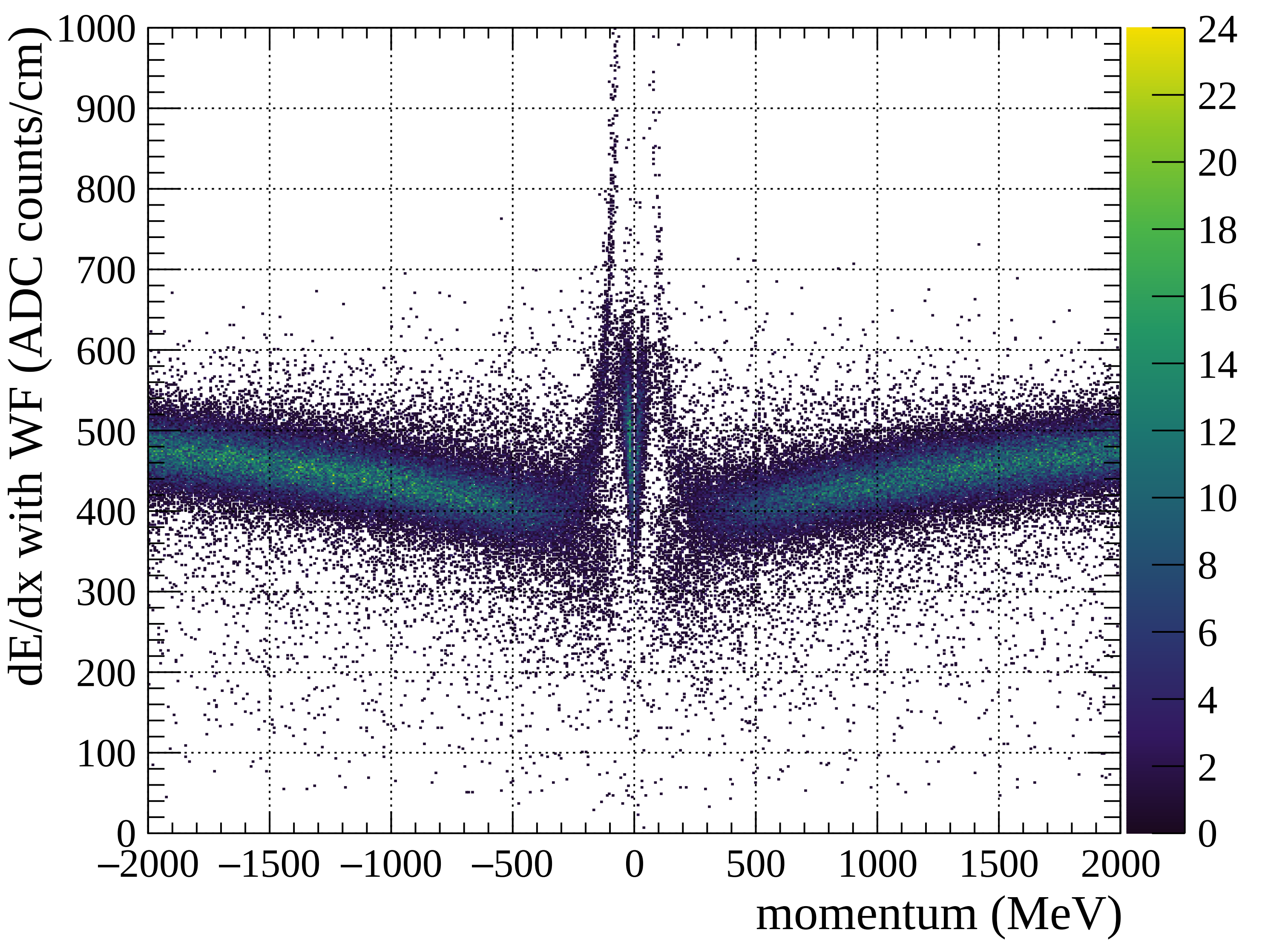

φ
80
60
40
20
0
−20
−40
−60
−80
−80
−60
−40
−20
0
20
40
60
80
θ
6000
5000
4000
3000
2000
1000
0

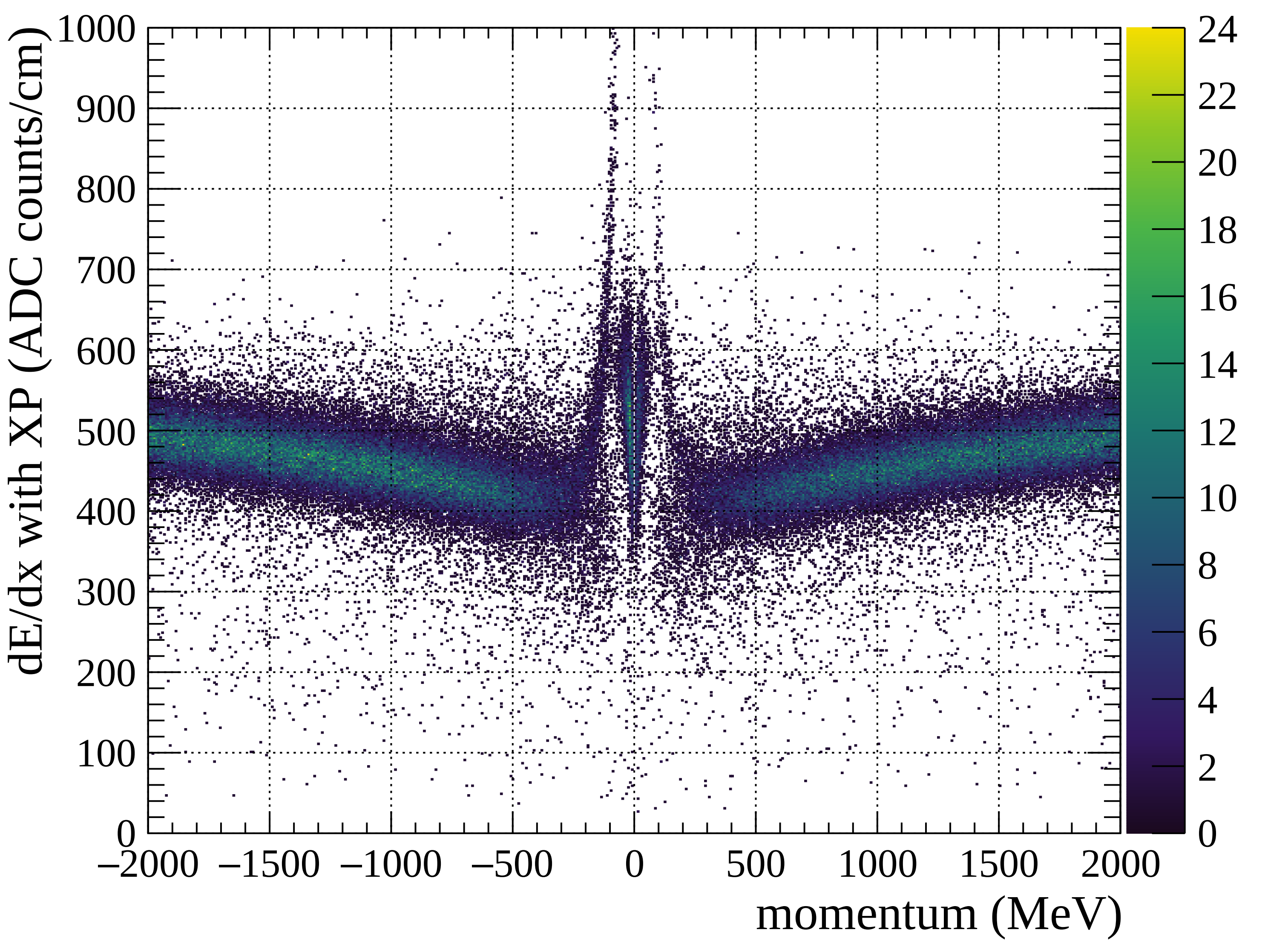


WF
XP
dE/dx resolution (%)
30
25
20
15
10
5
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)

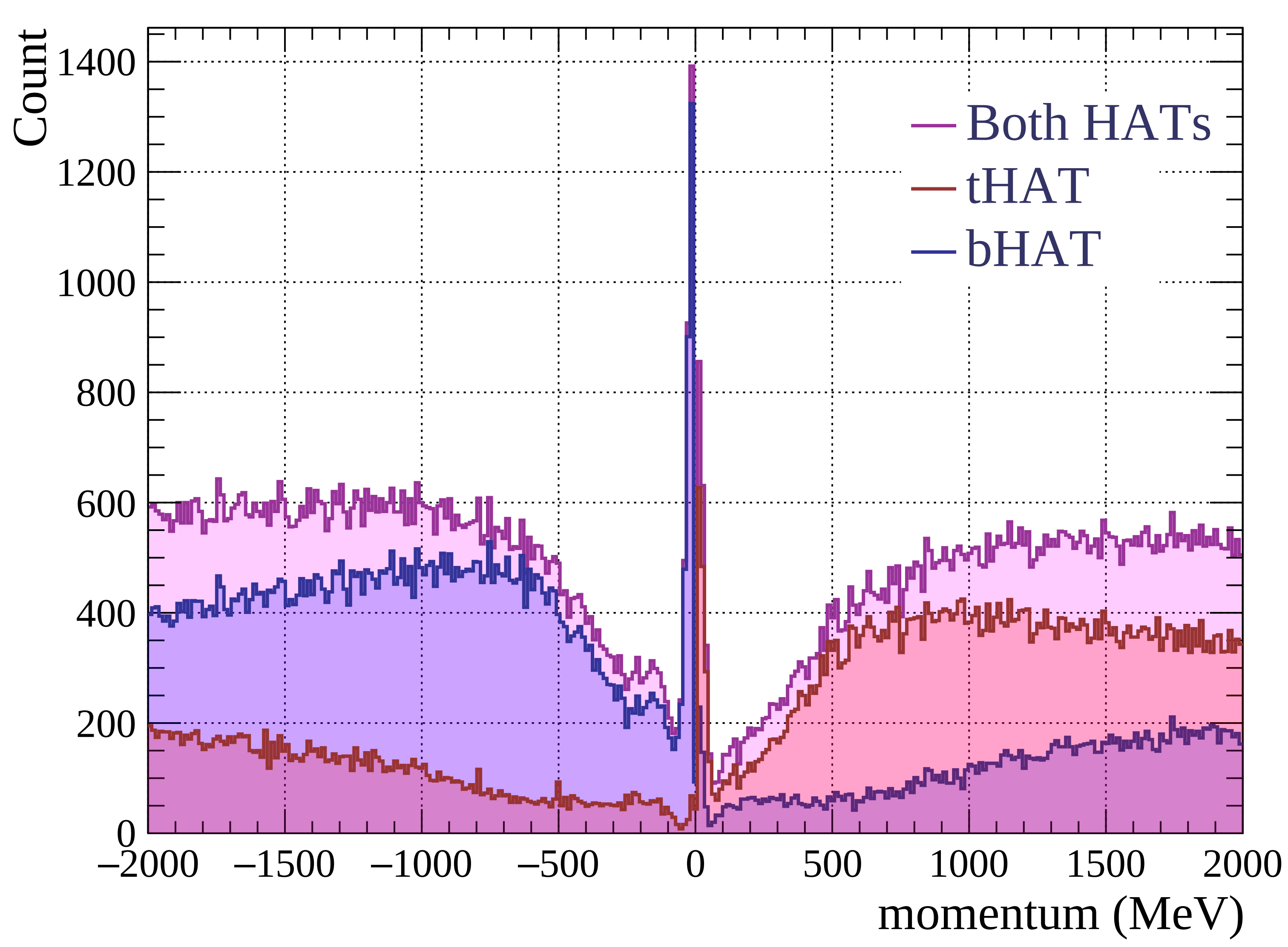


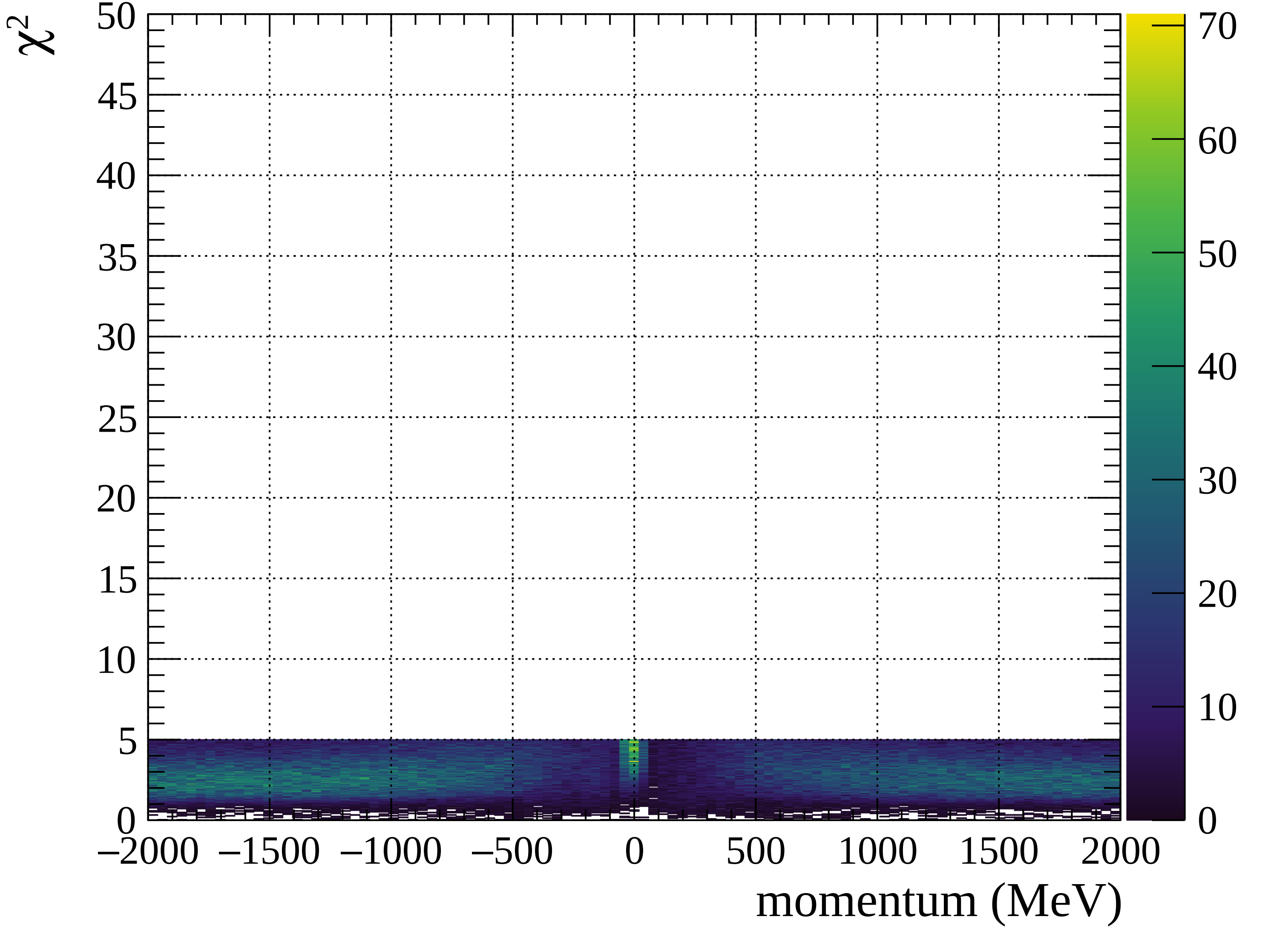
WF
XP
Mean (ADC counts/cm)
800
700
600
500
400
300
200
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)


dE/dx with WF (ADC counts/cm)
momentum (MeV)
0
100
200
300
400
500
600
700
800
900
1000
−2000
−1500
−1000
−500
0
500
1000
1500
2000
0
2
4
6
8
10
12
14
16
18
20
22
24

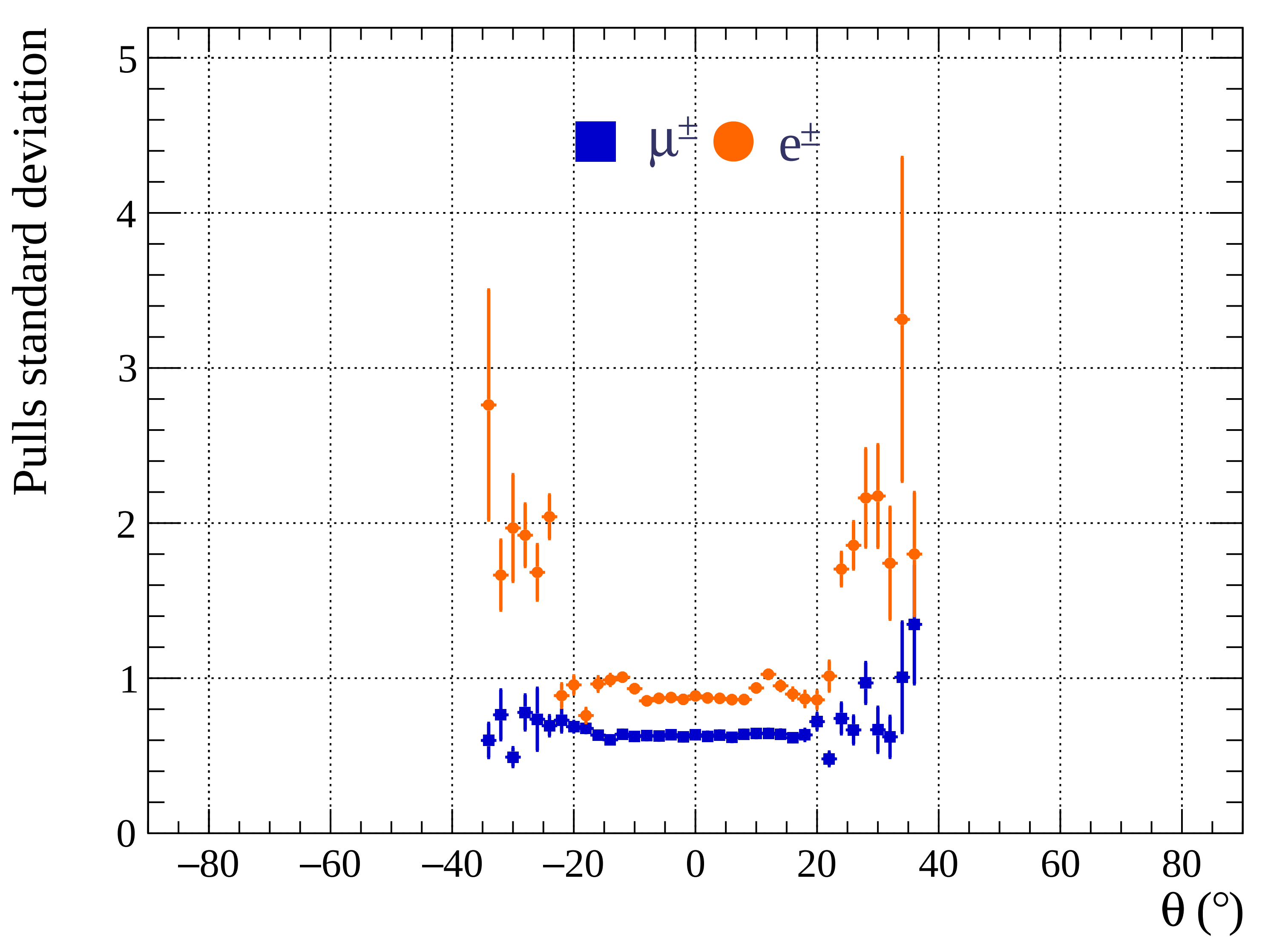


dE/dx with XP (ADC counts/cm)
momentum (MeV)
0
100
200
300
400
500
600
700
800
900
1000
−2000
−1500
−1000
−500
0
500
1000
1500
2000
0
2
4
6
8
10
12
14
16
18
20
22
24

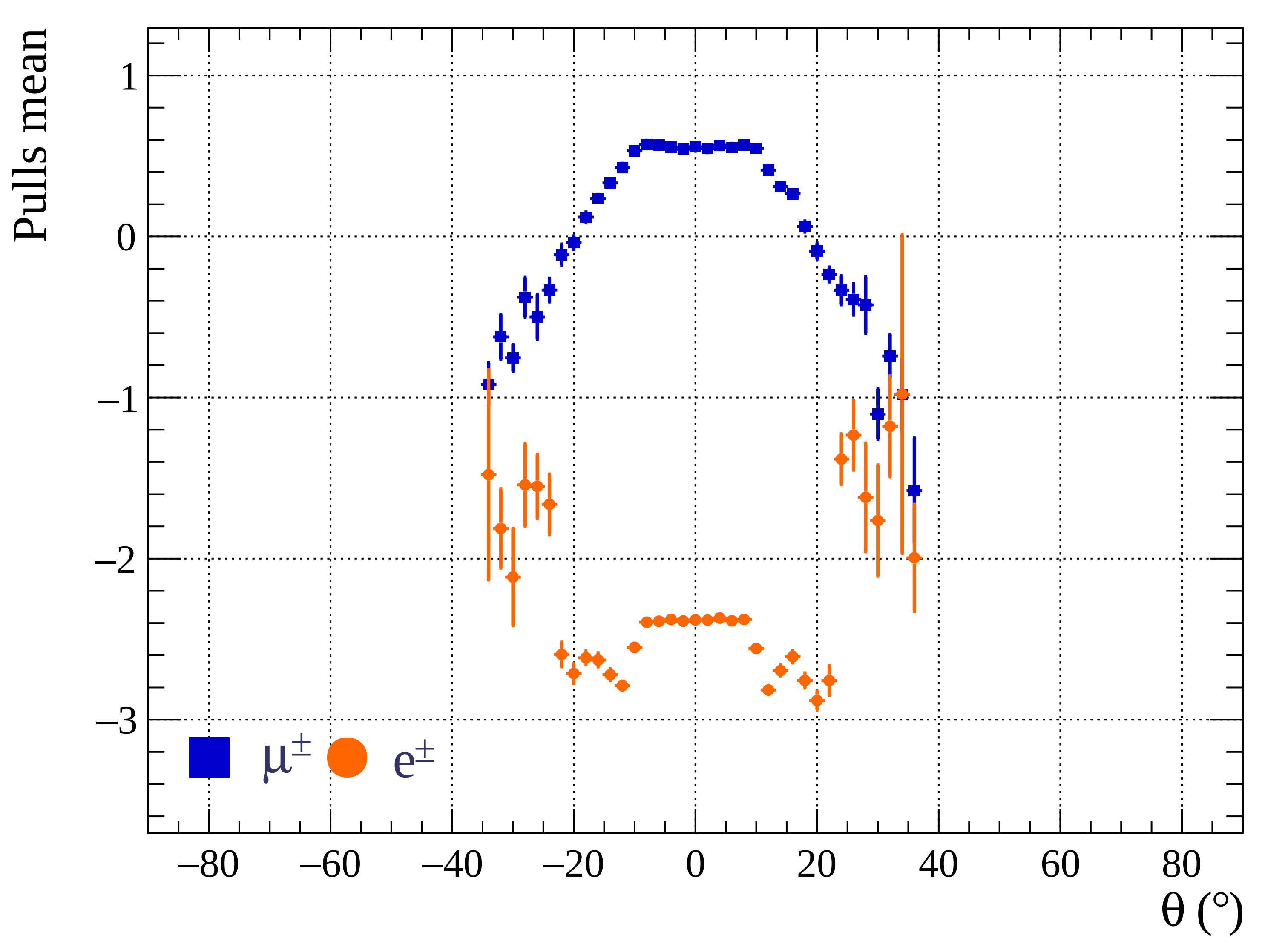

Count
1400
1200
1000
800
600
400
200
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
Both HATs
tHAT
bHAT



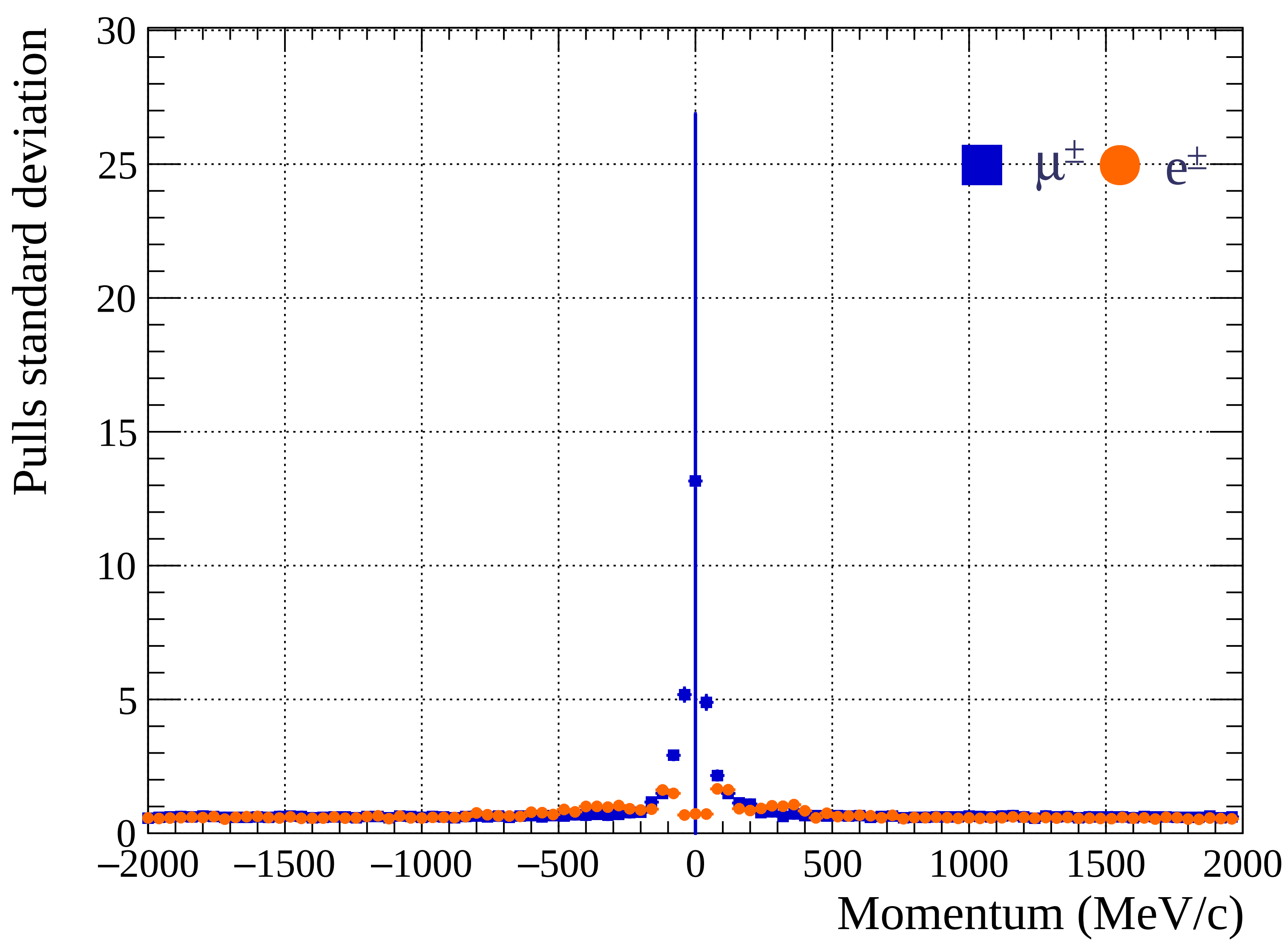
χ^2
50
45
40
35
30
25
20
15
10
5
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
70
60
50
40
30
20
10
0

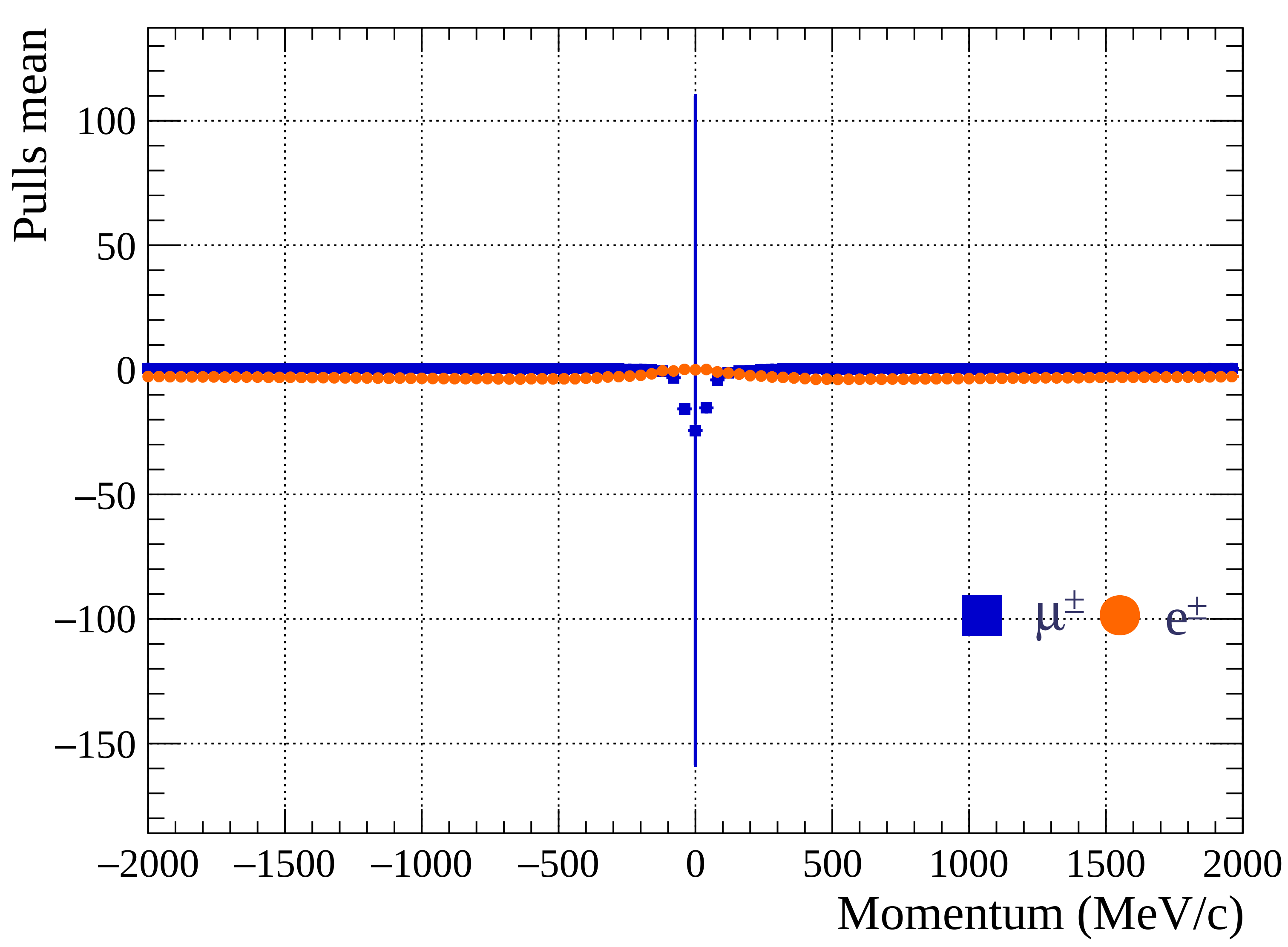


Pulls standard deviation
θ (°)
μ±
e±
0
1
2
3
4
5
−80
−60
−40
−20
0
20
40
60
80

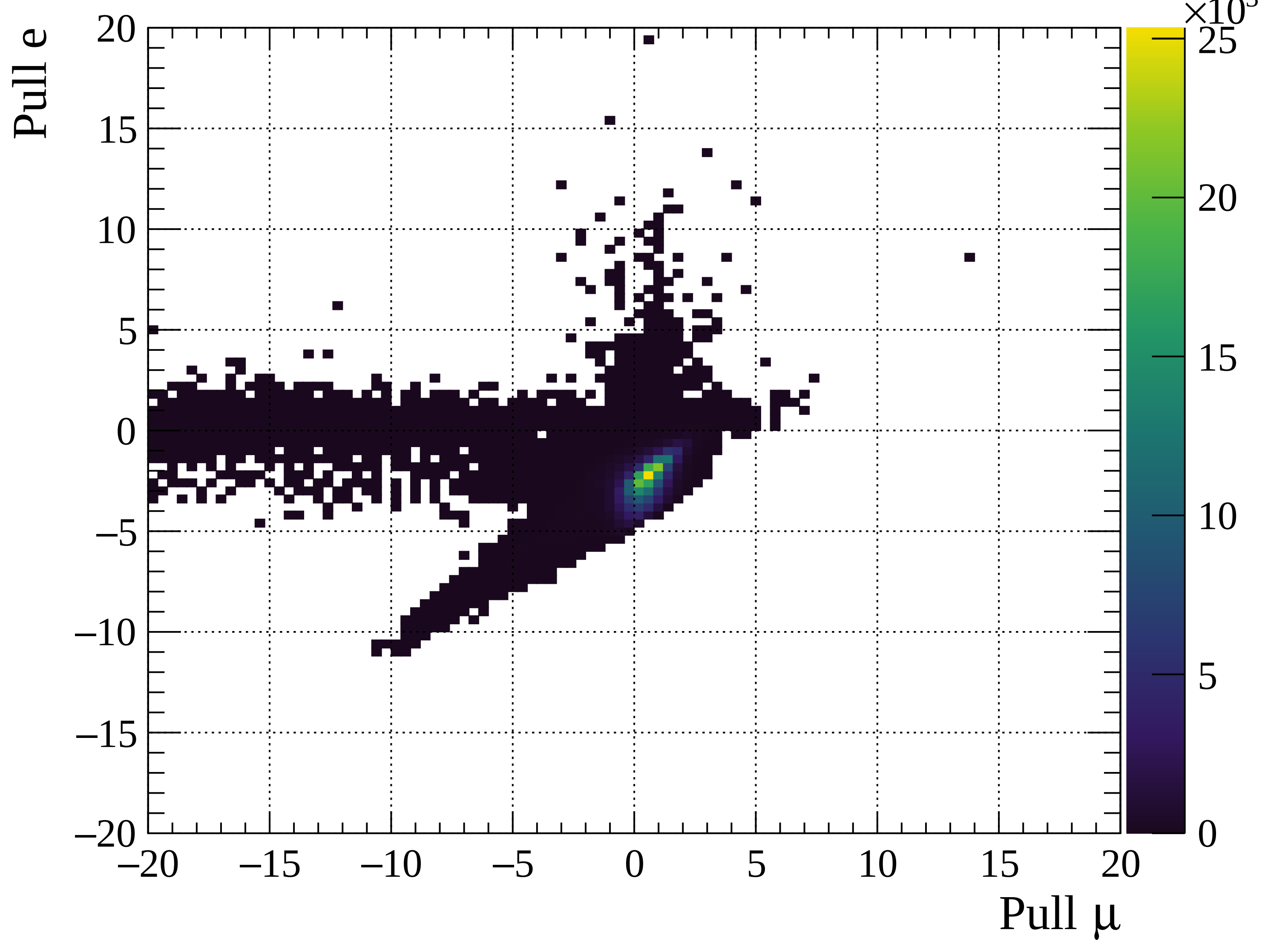


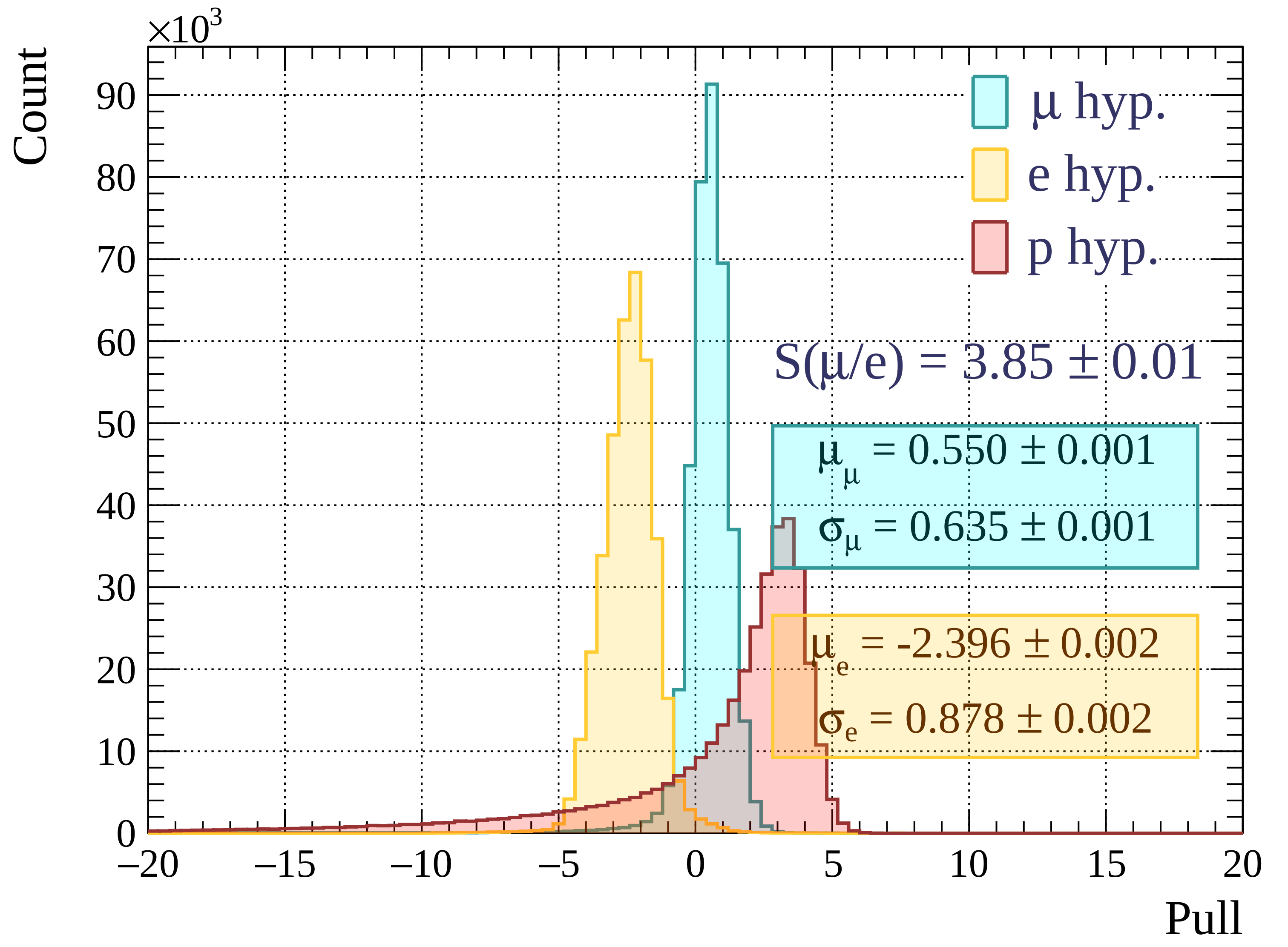
Pulls mean
1
0
−1
−2
−3
−80
−60
−40
−20
0
20
40
60
80
θ (°)
μ±
e±


Pulls standard deviation
30
25
20
15
10
5
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
$\mu^{\pm}$
$e^{\pm}$

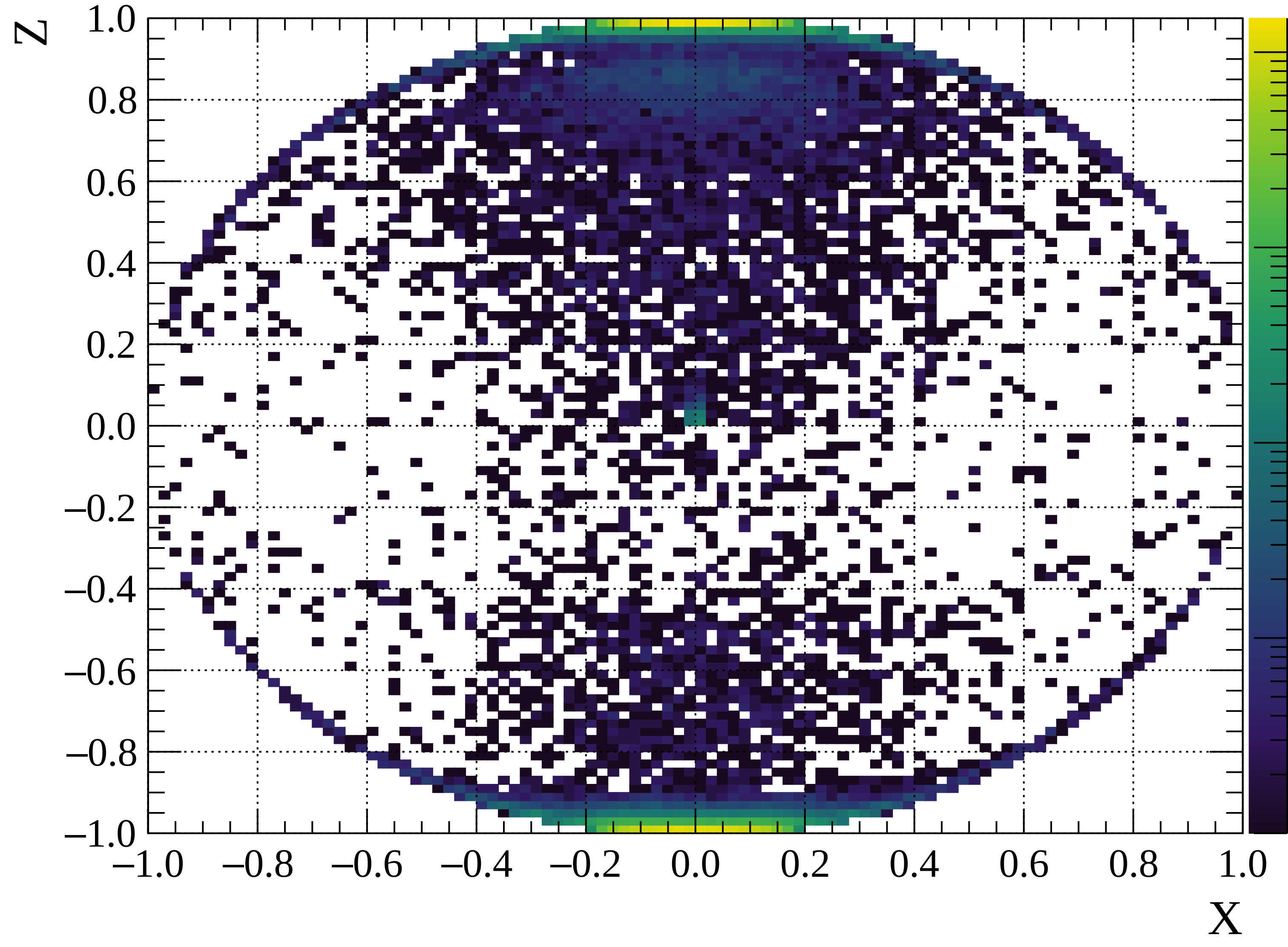


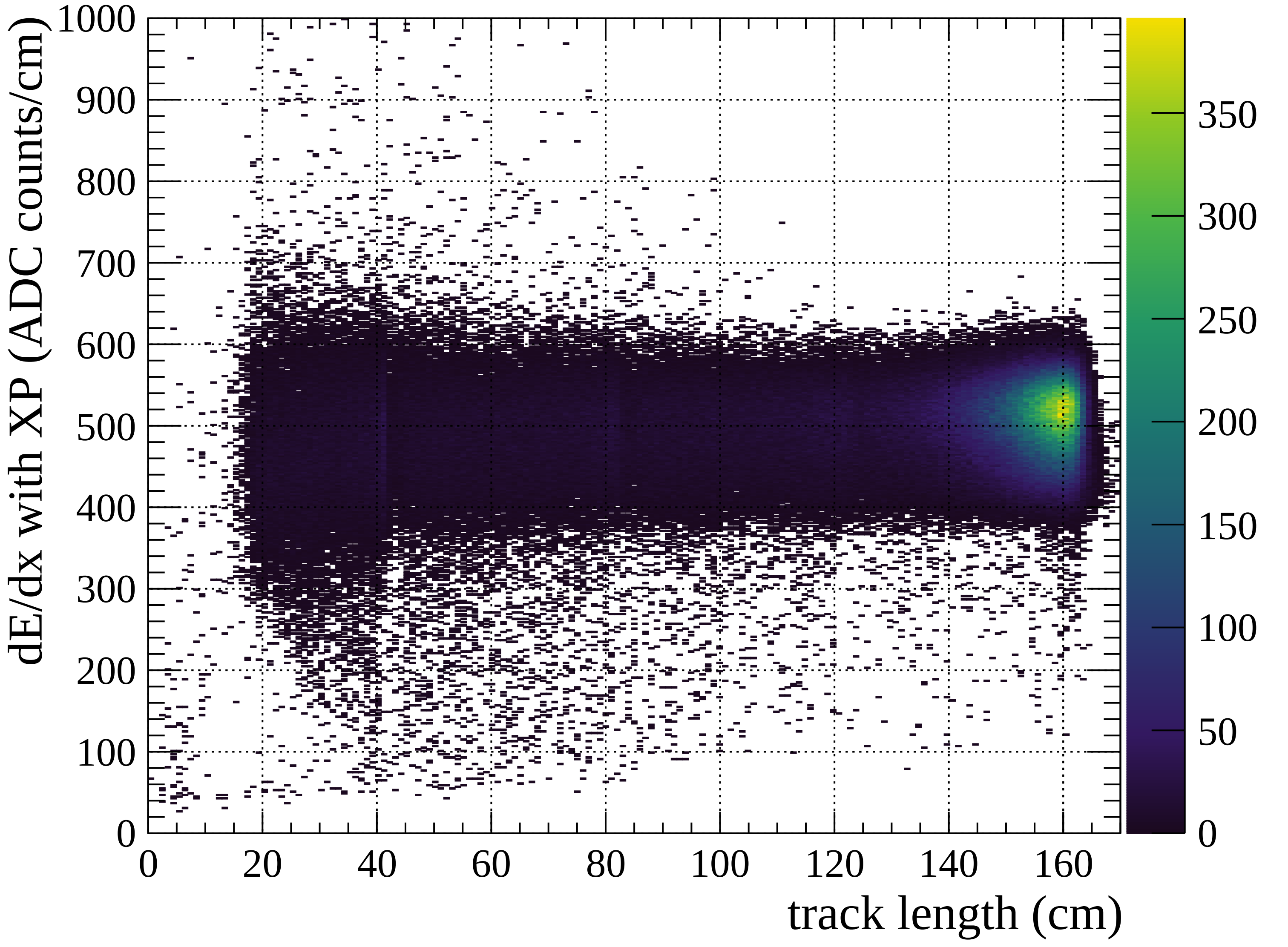
Pulls mean
100
50
0
−50
−100
−150
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
$\mu^{\pm}$
$e^{\pm}$

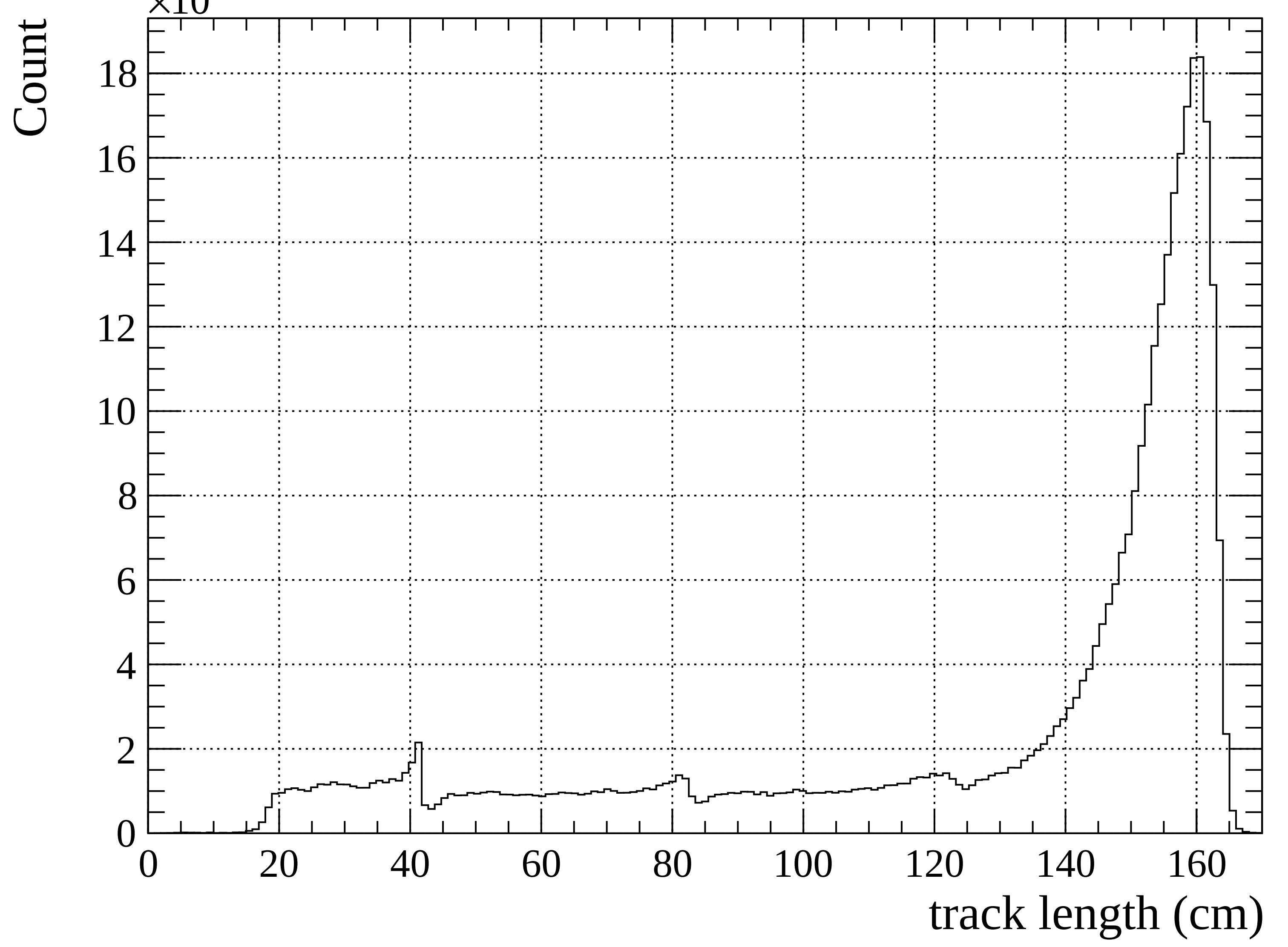

Pull e
Pull μ
×10³
20
15
10
5
0
−5
−10
−15
−20
−20
−15
−10
−5
0
5
10
15
20
25
20
15
10
5
0



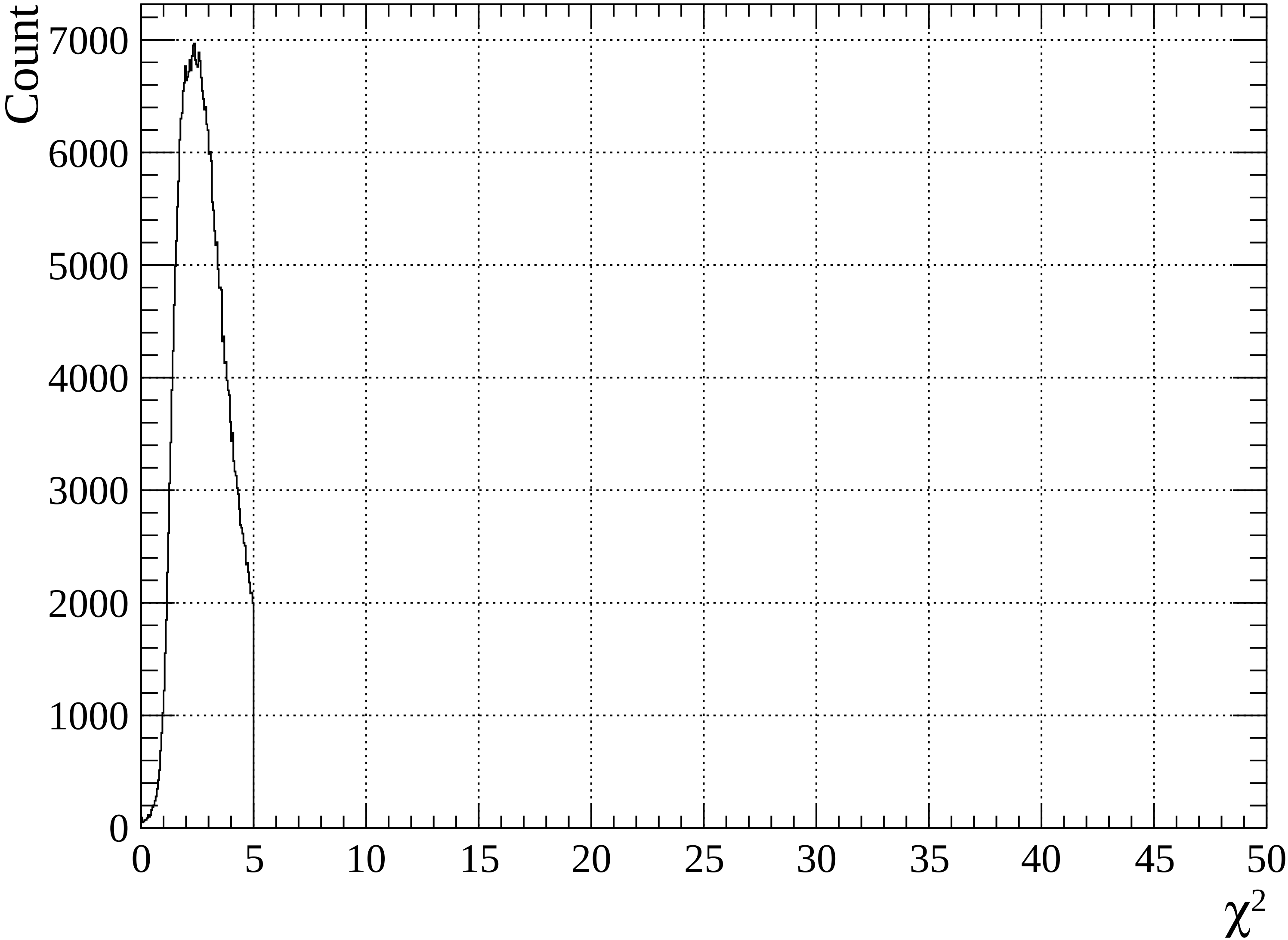
×10^3
Count
90
80
70
60
50
40
30
20
10
0
−20
−15
−10
−5
0
5
10
15
20
Pull
μ hyp.
e hyp.
p hyp.
S(μ/e) = 3.85 ± 0.01
μ_μ = 0.550 ± 0.001
σ_μ = 0.635 ± 0.001
μ_e = -2.396 ± 0.002
σ_e = 0.878 ± 0.002


Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X

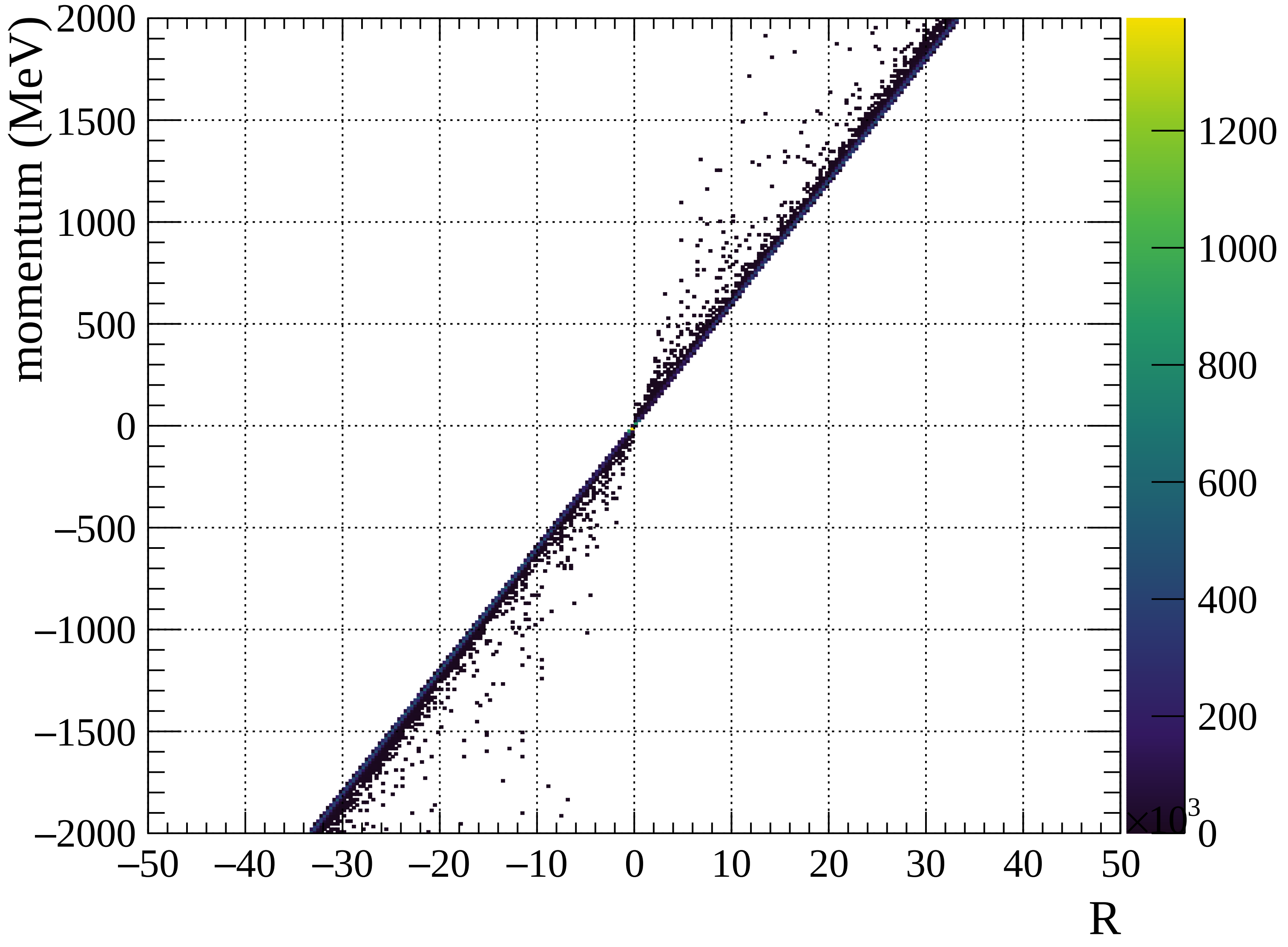

dE/dx with XP (ADC counts/cm)
track length (cm)
0
100
200
300
400
500
600
700
800
900
1000
0
20
40
60
80
100
120
140
160
0
50
100
150
200
250
300
350



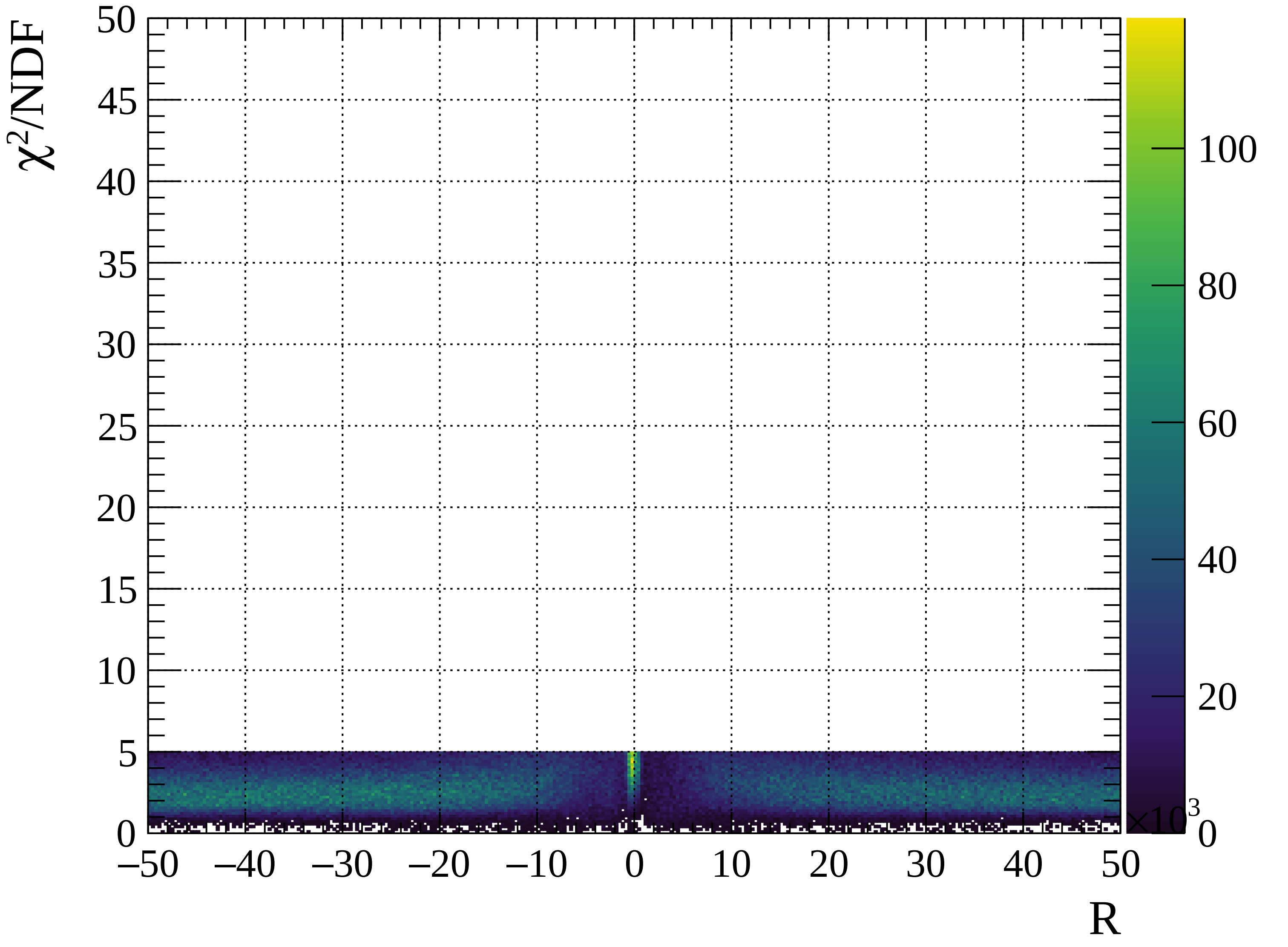
×10
Count
0
2
4
6
8
10
12
14
16
18
0
20
40
60
80
100
120
140
160
track length (cm)



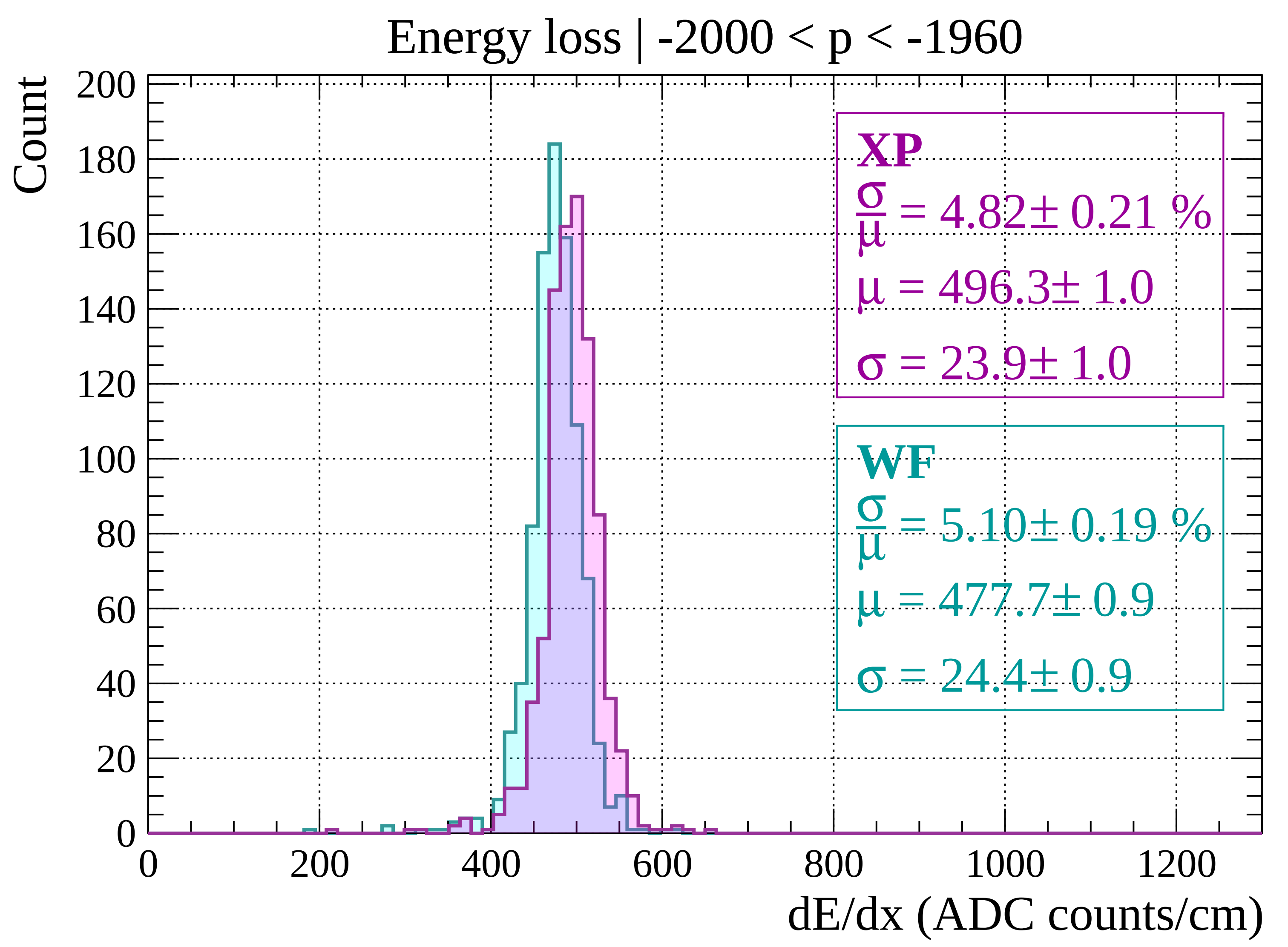
Count
7000
6000
5000
4000
3000
2000
1000
0
0
5
10
15
20
25
30
35
40
45
50
χ^2

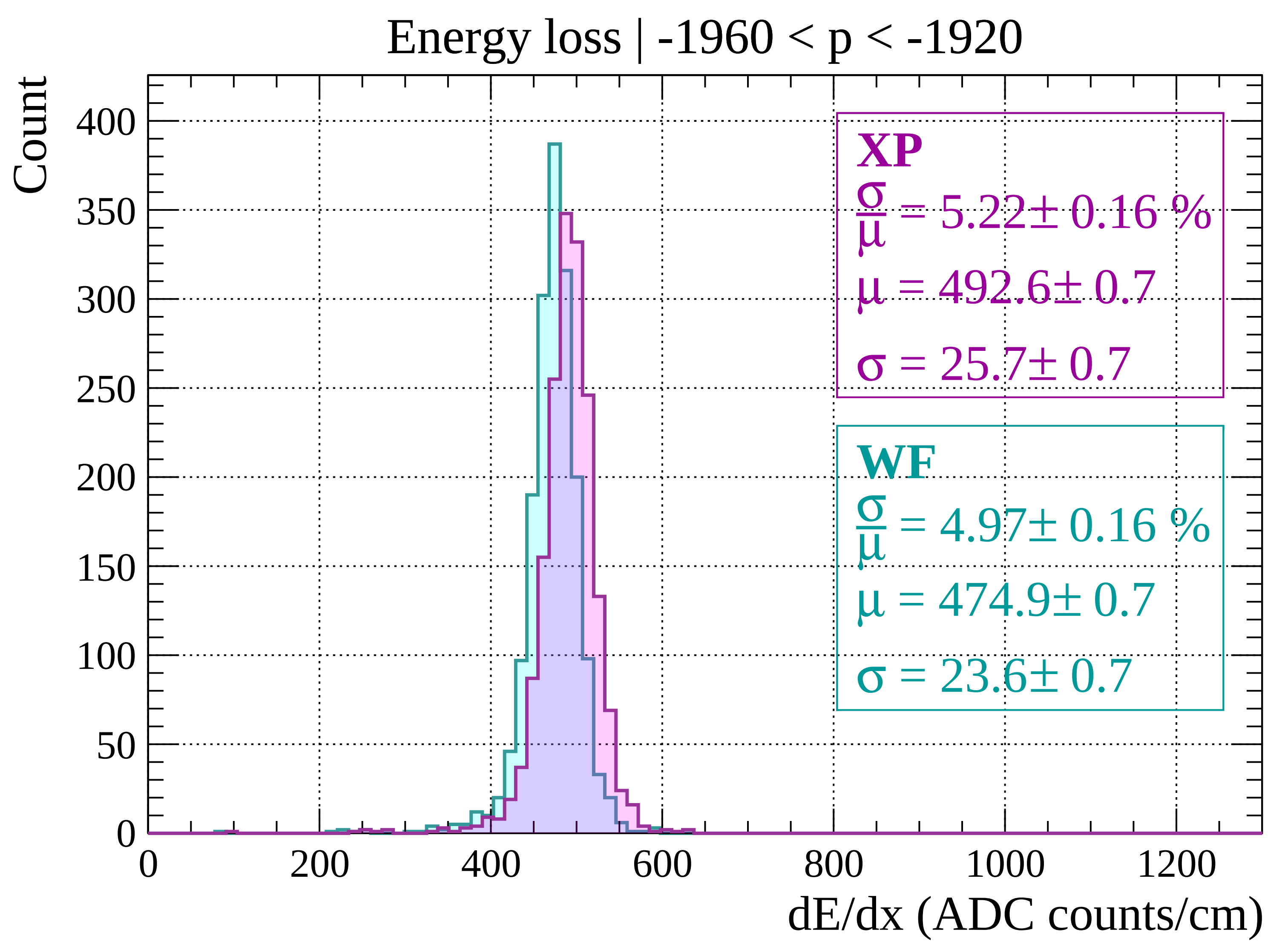


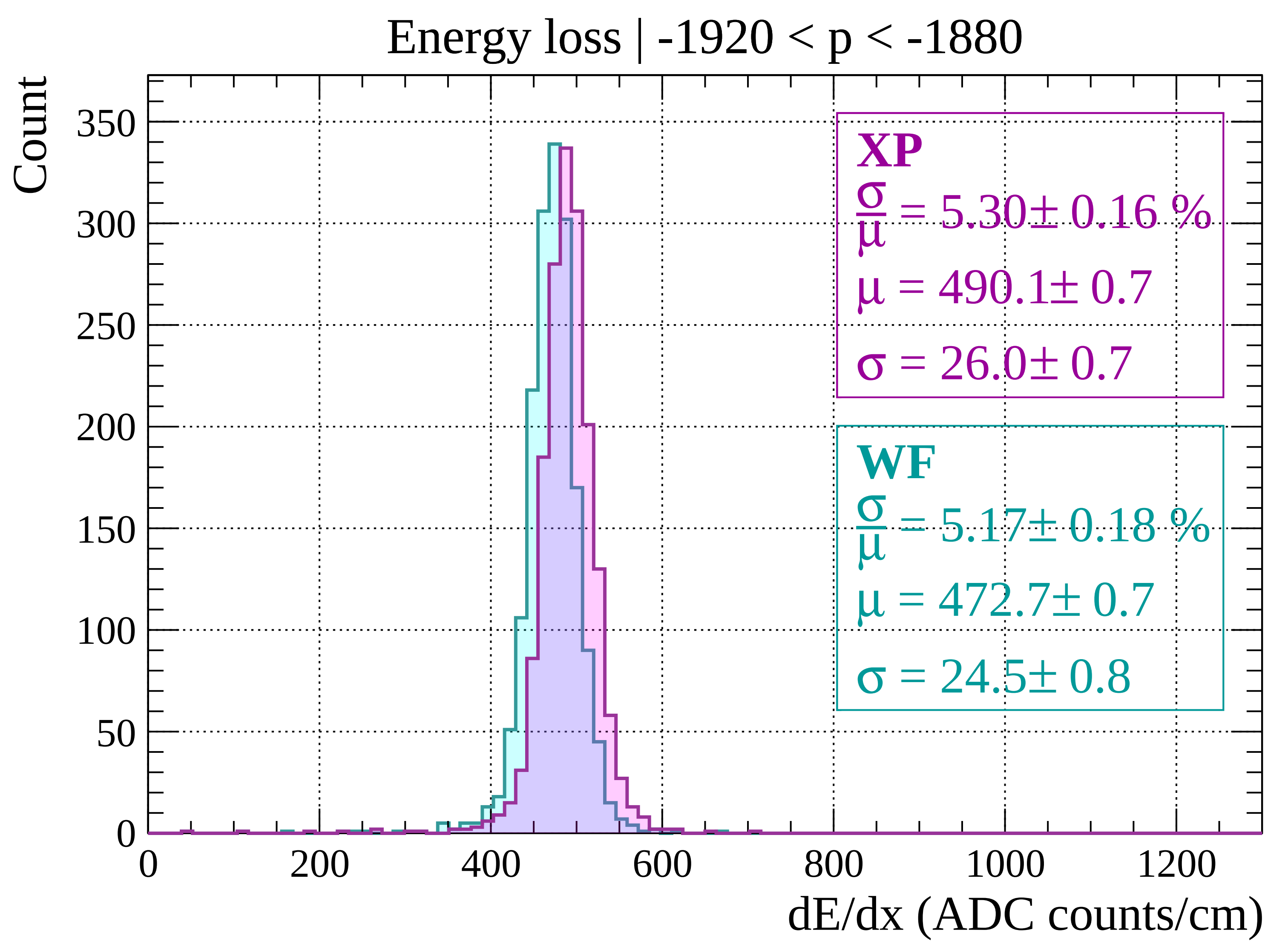
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
1200
1000
800
600
400
200
0
$\times10^{3}$



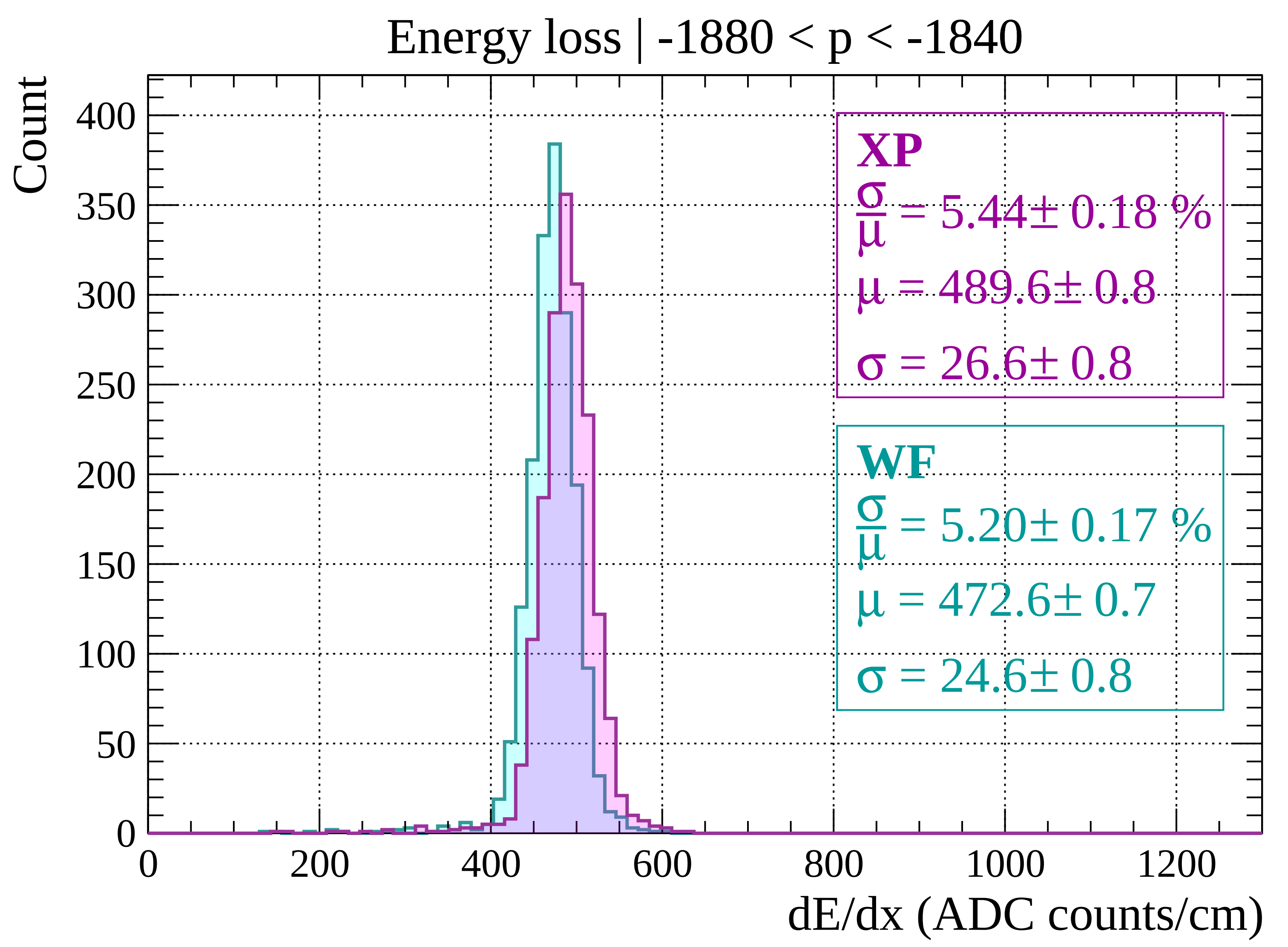
χ^2/NDF
R
$\times 10^3$

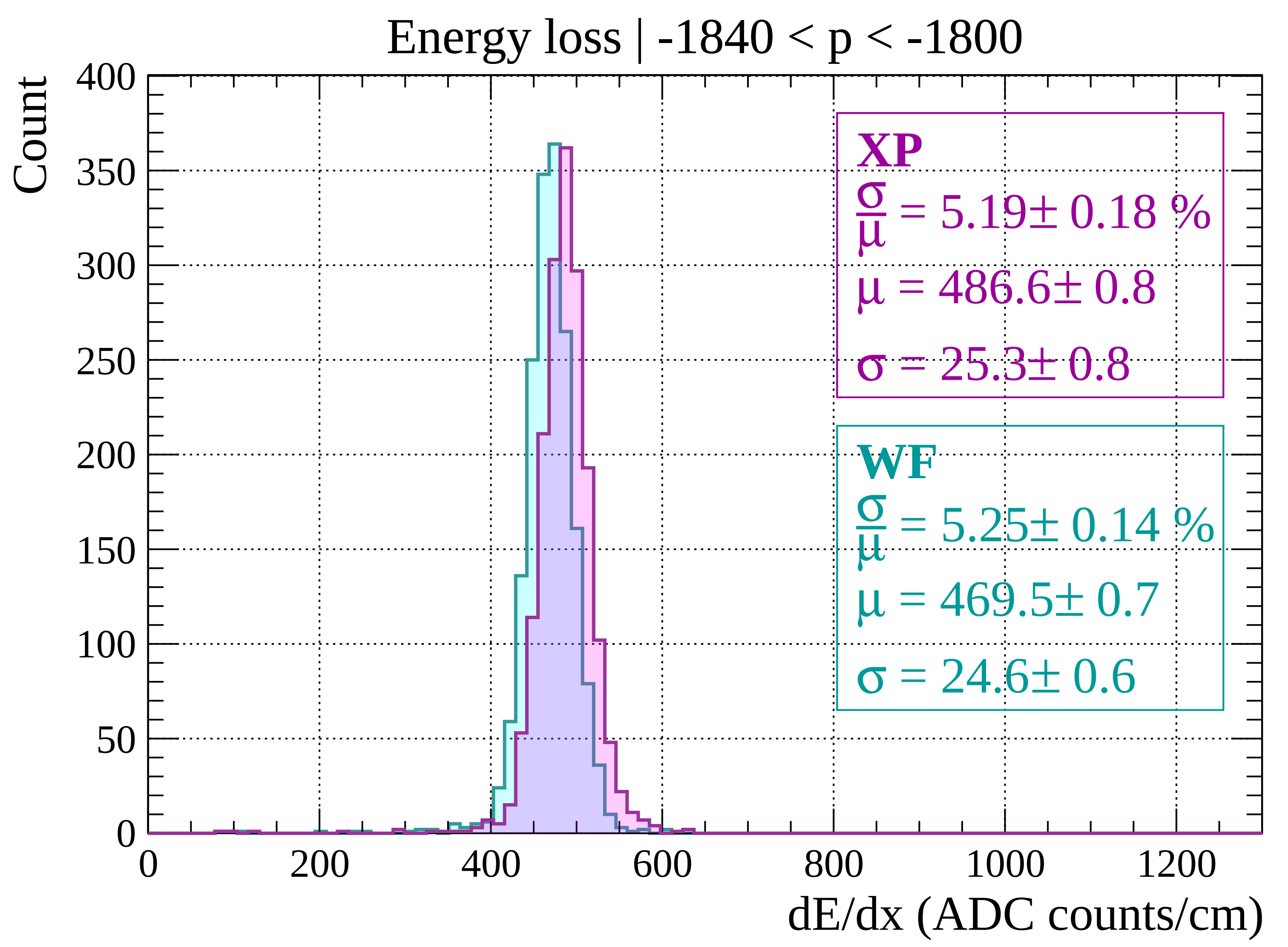

Energy loss | -2000 < p < -1960
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
XP
σ/μ = 4.82± 0.21 %
μ = 496.3± 1.0
σ = 23.9± 1.0
WF
σ/μ = 5.10± 0.19 %
μ = 477.7± 0.9
σ = 24.4± 0.9

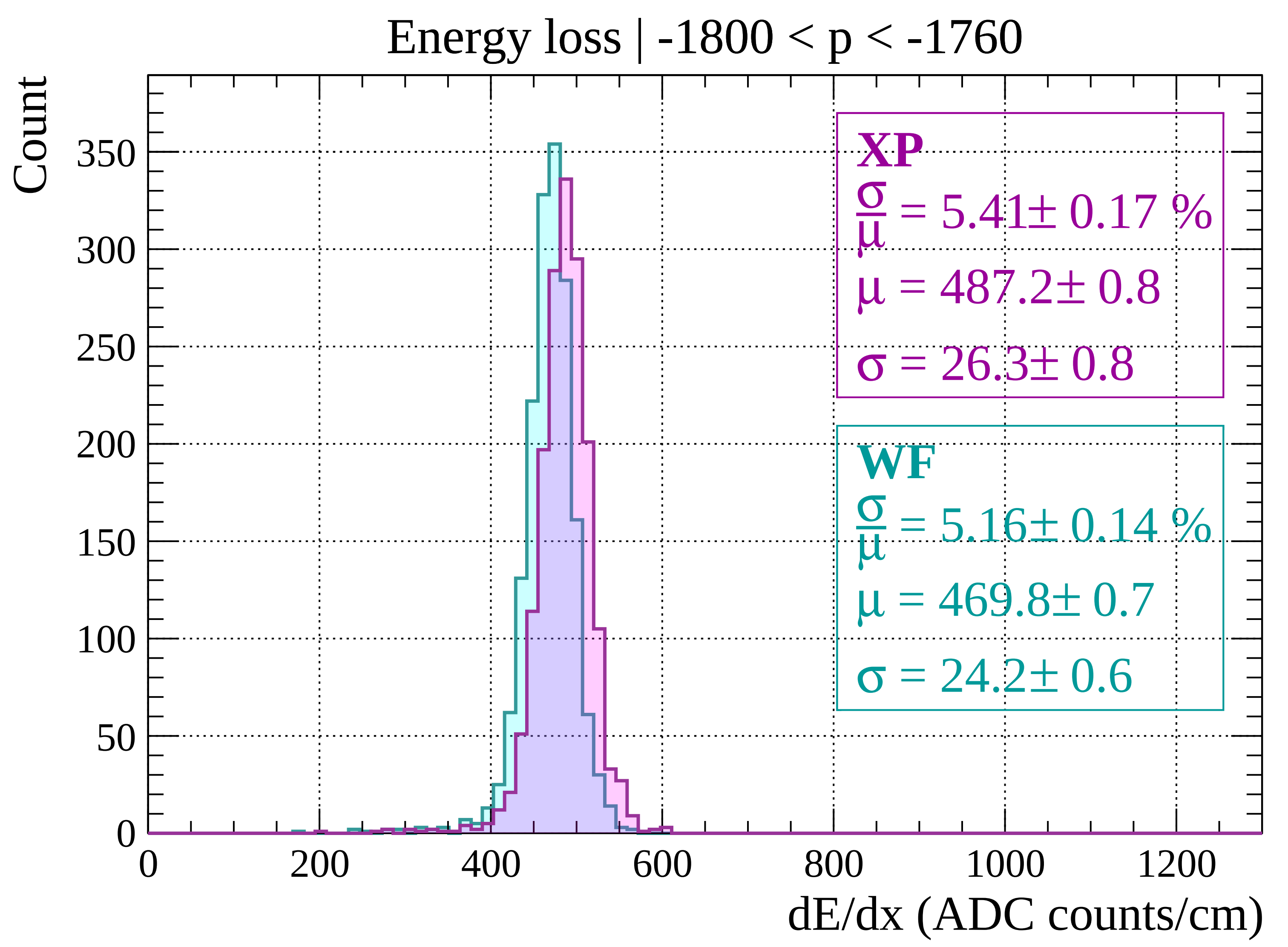

Energy loss | -1960 < p < -1920
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 5.22 ± 0.16 %
μ = 492.6 ± 0.7
σ = 25.7 ± 0.7
WF
σ/μ = 4.97 ± 0.16 %
μ = 474.9 ± 0.7
σ = 23.6 ± 0.7



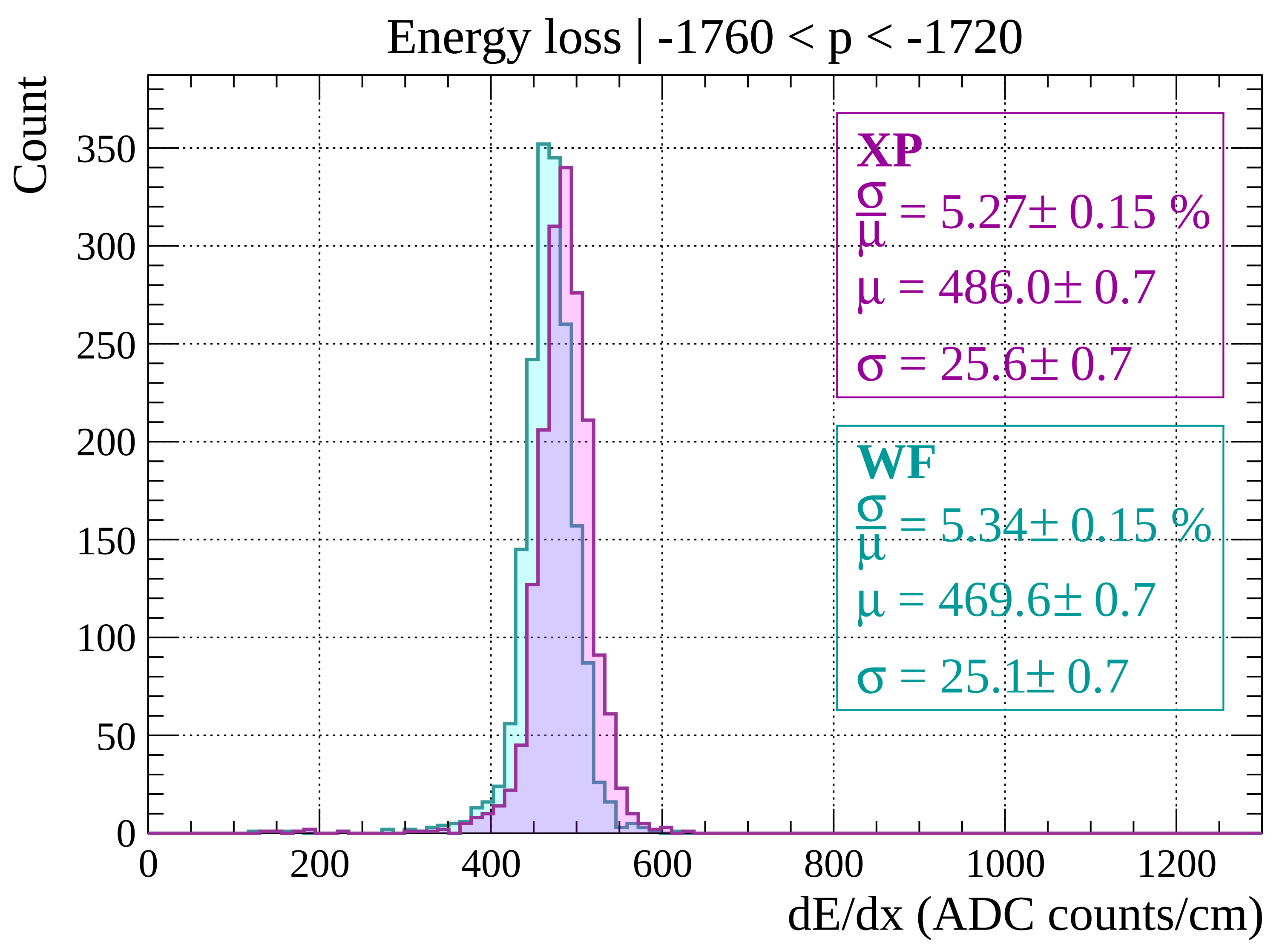
Energy loss | -1920 < p < -1880
Count
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 5.30± 0.16 %
μ = 490.1± 0.7
σ = 26.0± 0.7
WF
σ/μ = 5.17± 0.18 %
μ = 472.7± 0.7
σ = 24.5± 0.8


Energy loss | -1880 < p < -1840
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
σ/μ = 5.44± 0.18 %
μ = 489.6± 0.8
σ = 26.6± 0.8
WF
σ/μ = 5.20± 0.17 %
μ = 472.6± 0.7
σ = 24.6± 0.8

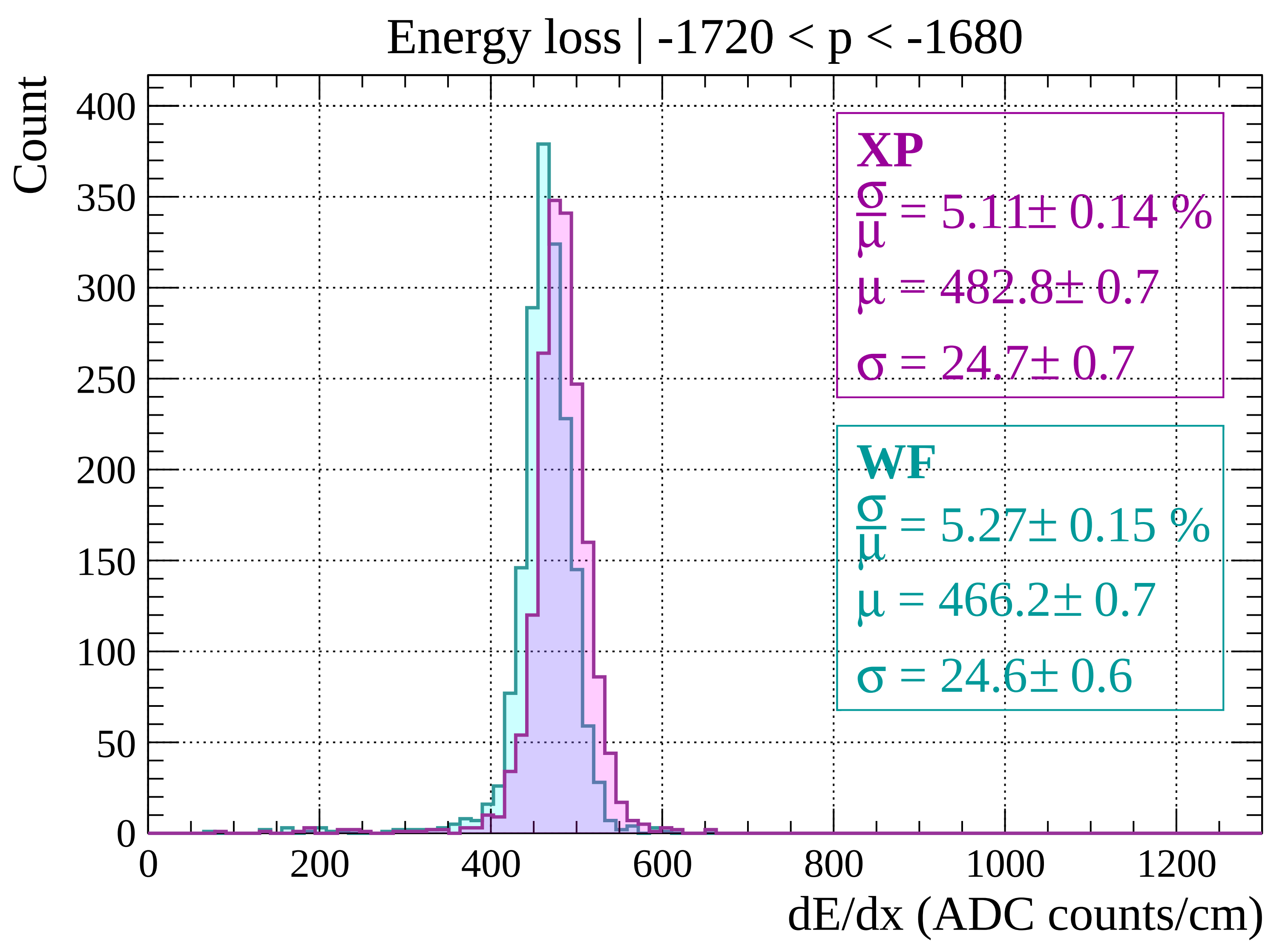

Energy loss | -1840 < p < -1800
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
350
400
XP
$\frac{\sigma}{\mu}$ = 5.19± 0.18 %
μ = 486.6± 0.8
σ = 25.3± 0.8
WF
$\frac{\sigma}{\mu}$ = 5.25± 0.14 %
μ = 469.5± 0.7
σ = 24.6± 0.6

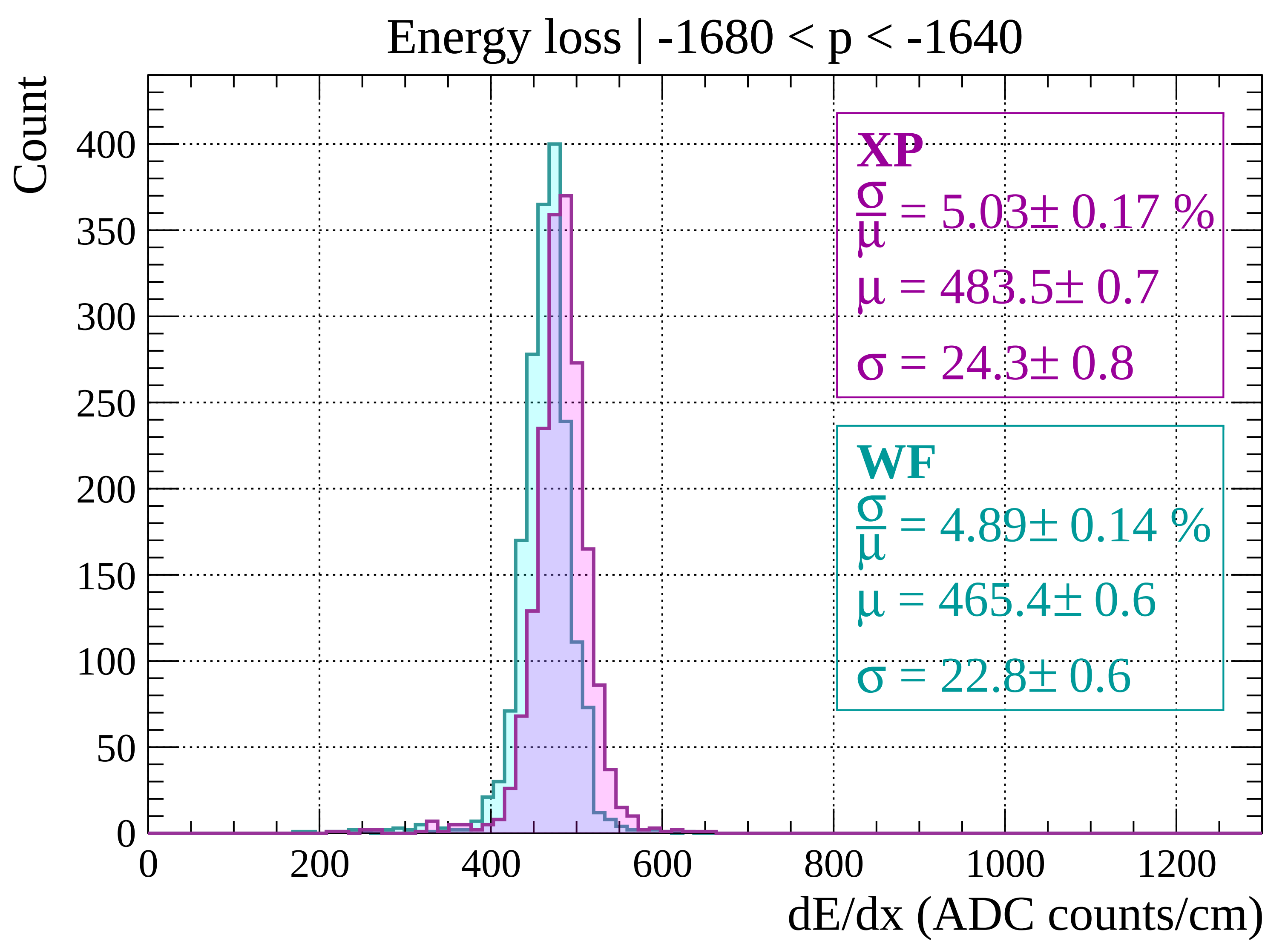


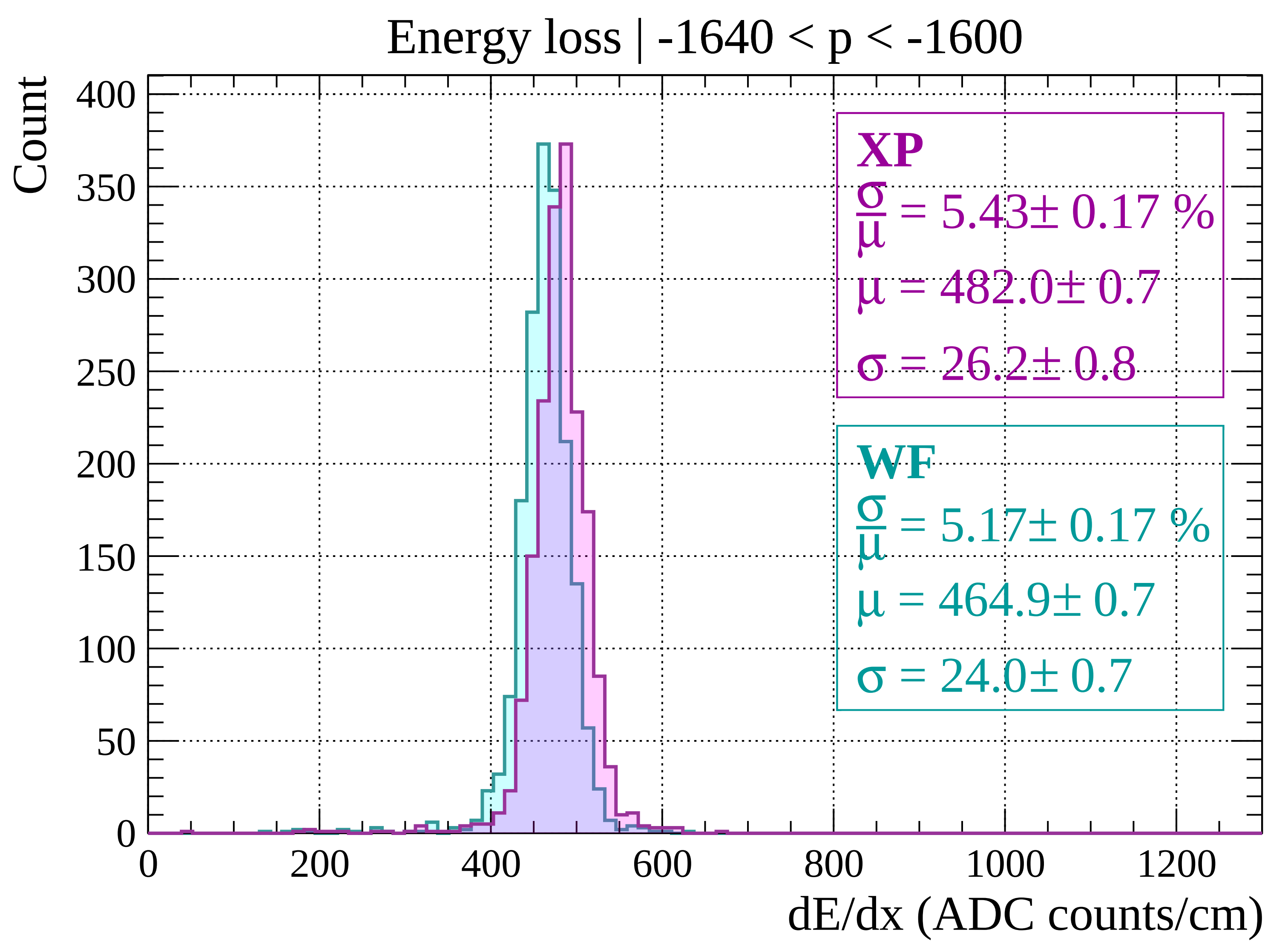
Energy loss | -1800 < p < -1760
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
350
XP
$\frac{\sigma}{\mu}$ = 5.41± 0.17 %
μ = 487.2± 0.8
σ = 26.3± 0.8
WF
$\frac{\sigma}{\mu}$ = 5.16± 0.14 %
μ = 469.8± 0.7
σ = 24.2± 0.6



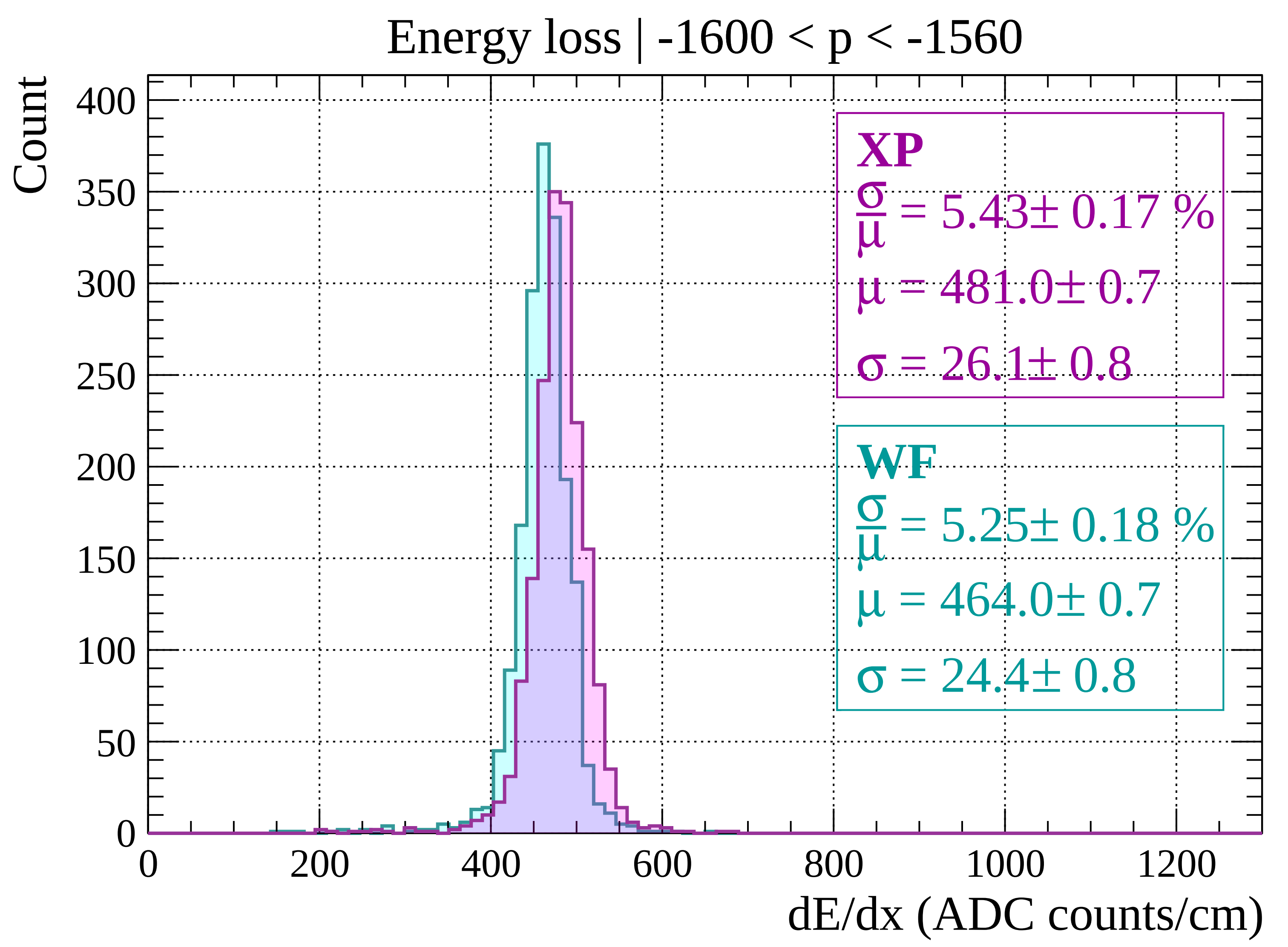
Energy loss | -1760 < p < -1720
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 5.27± 0.15 %
μ = 486.0± 0.7
σ = 25.6± 0.7
WF
σ/μ = 5.34± 0.15 %
μ = 469.6± 0.7
σ = 25.1± 0.7

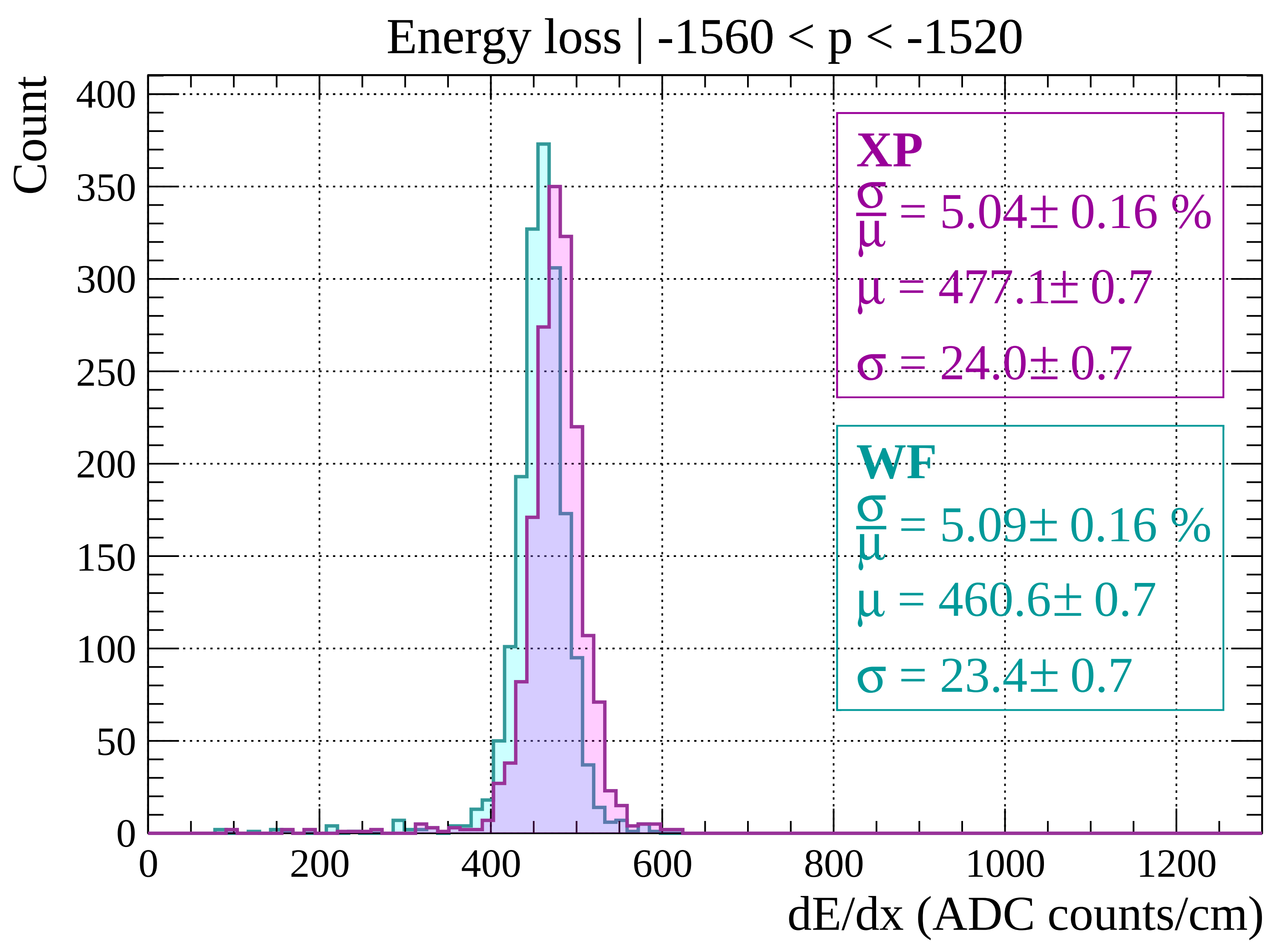

Energy loss | -1720 < p < -1680
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 5.11± 0.14 %
μ = 482.8± 0.7
σ = 24.7± 0.7
WF
$\frac{\sigma}{\mu}$ = 5.27± 0.15 %
μ = 466.2± 0.7
σ = 24.6± 0.6


Energy loss | -1680 < p < -1640
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 5.03± 0.17 %
μ = 483.5± 0.7
σ = 24.3± 0.8
WF
$\frac{\sigma}{\mu}$ = 4.89± 0.14 %
μ = 465.4± 0.6
σ = 22.8± 0.6

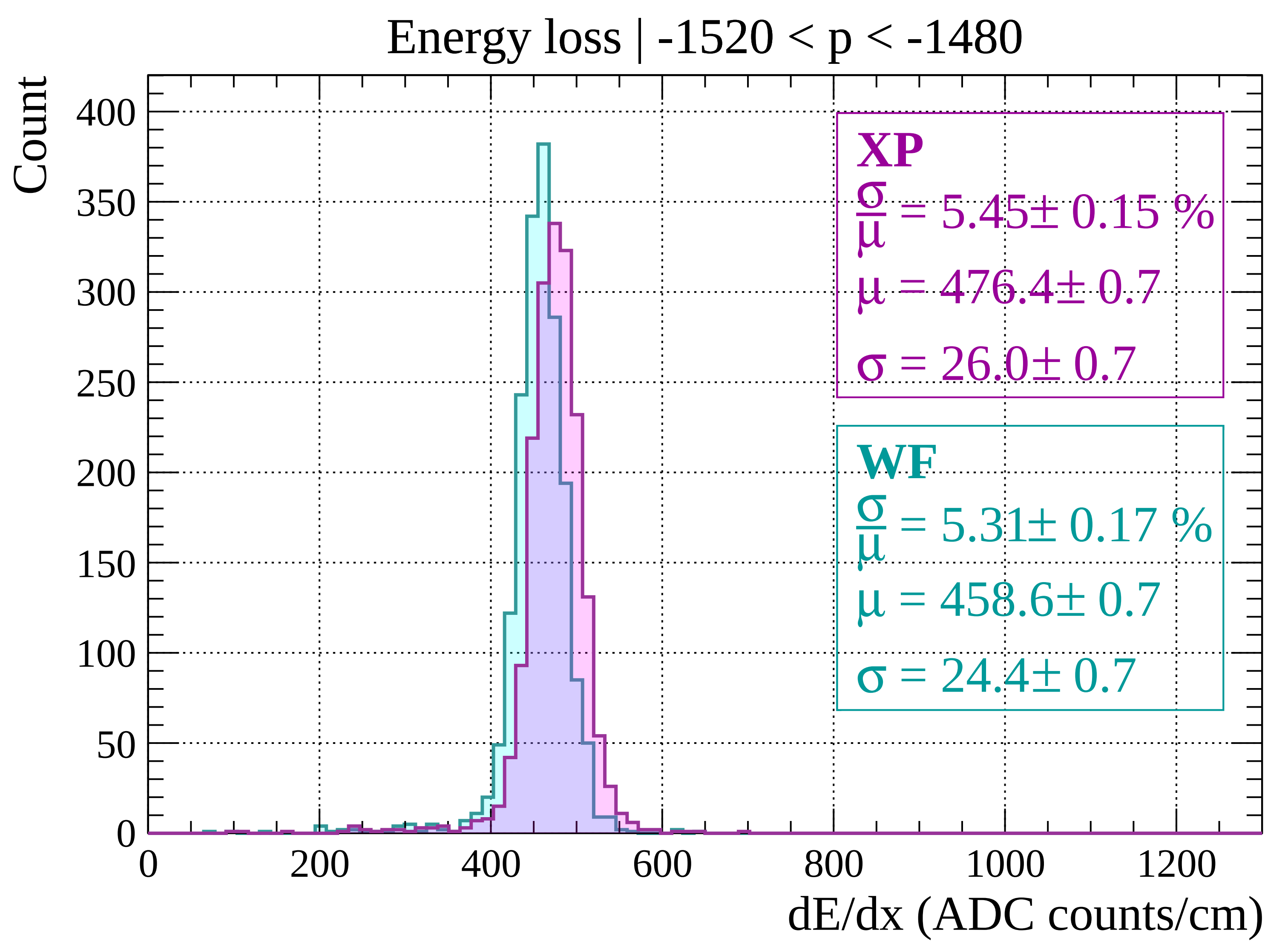


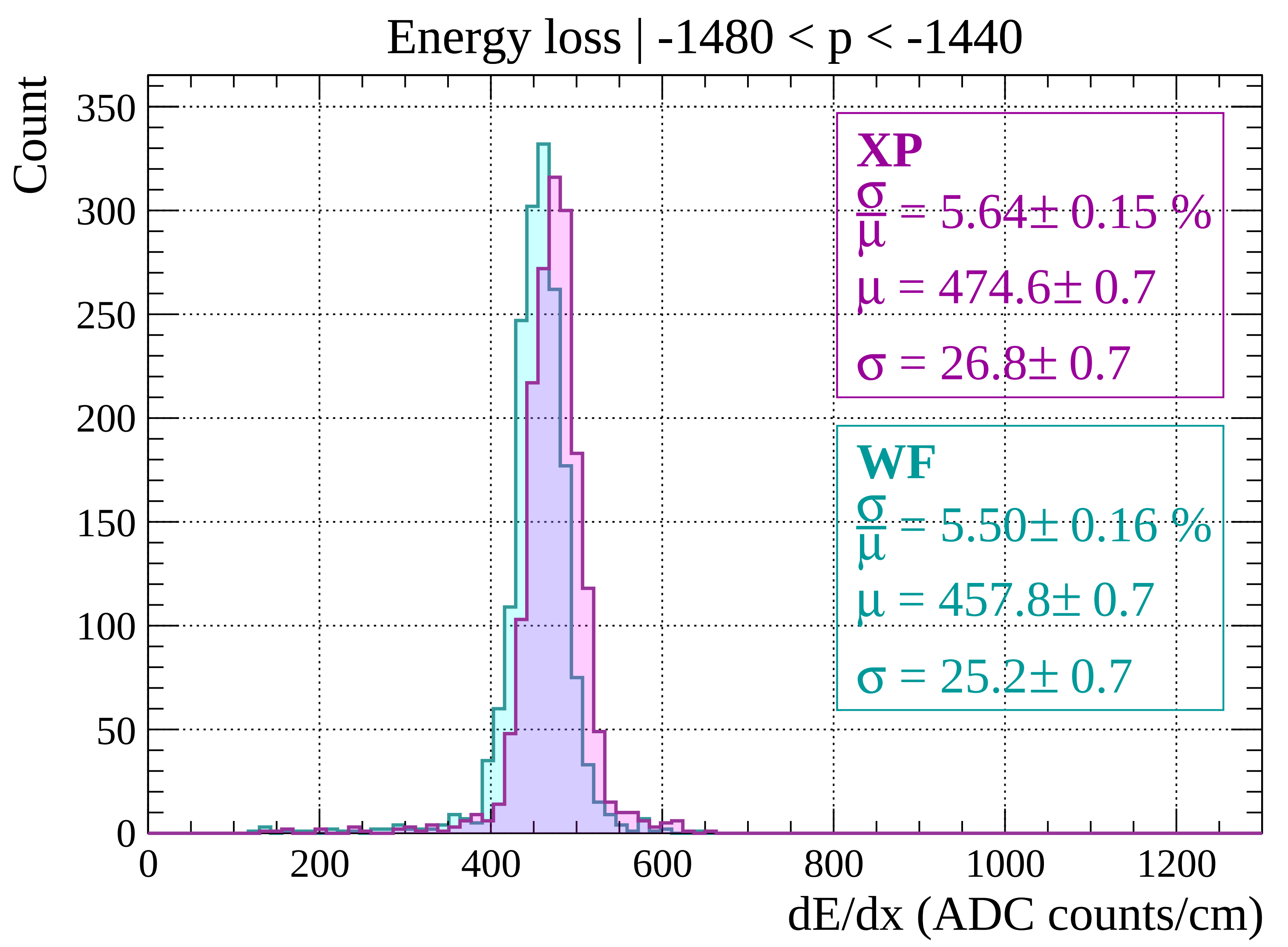
Energy loss | -1640 < p < -1600
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 5.43± 0.17 %
μ = 482.0± 0.7
σ = 26.2± 0.8
WF
$\frac{\sigma}{\mu}$ = 5.17± 0.17 %
μ = 464.9± 0.7
σ = 24.0± 0.7


Energy loss | -1600 < p < -1560
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
350
400
XP
$\frac{\sigma}{\mu}$ = 5.43± 0.17 %
μ = 481.0± 0.7
σ = 26.1± 0.8
WF
$\frac{\sigma}{\mu}$ = 5.25± 0.18 %
μ = 464.0± 0.7
σ = 24.4± 0.8

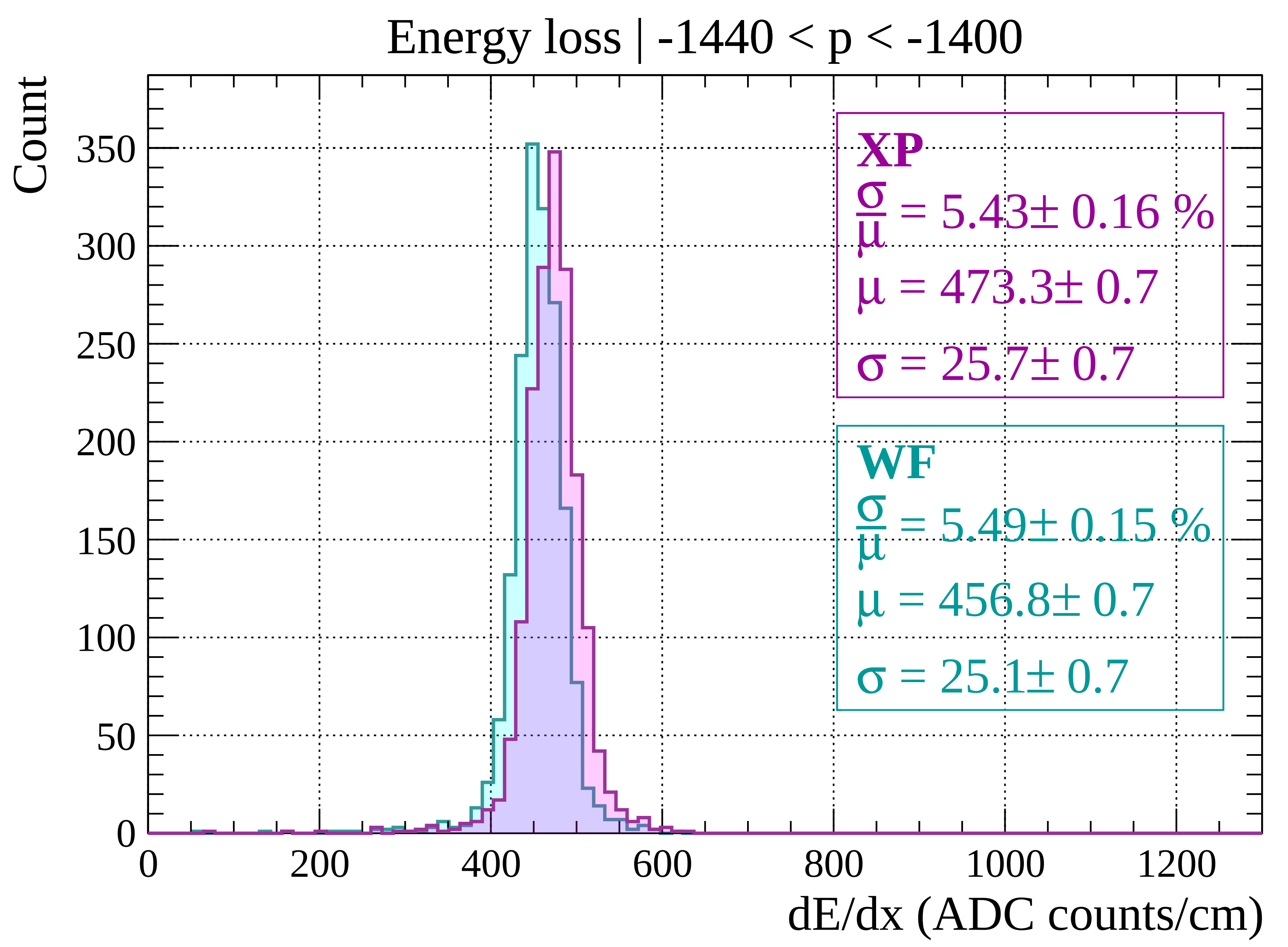


Energy loss | -1560 < p < -1520
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 5.04± 0.16 %
μ = 477.1± 0.7
σ = 24.0± 0.7
WF
$\frac{\sigma}{\mu}$ = 5.09± 0.16 %
μ = 460.6± 0.7
σ = 23.4± 0.7

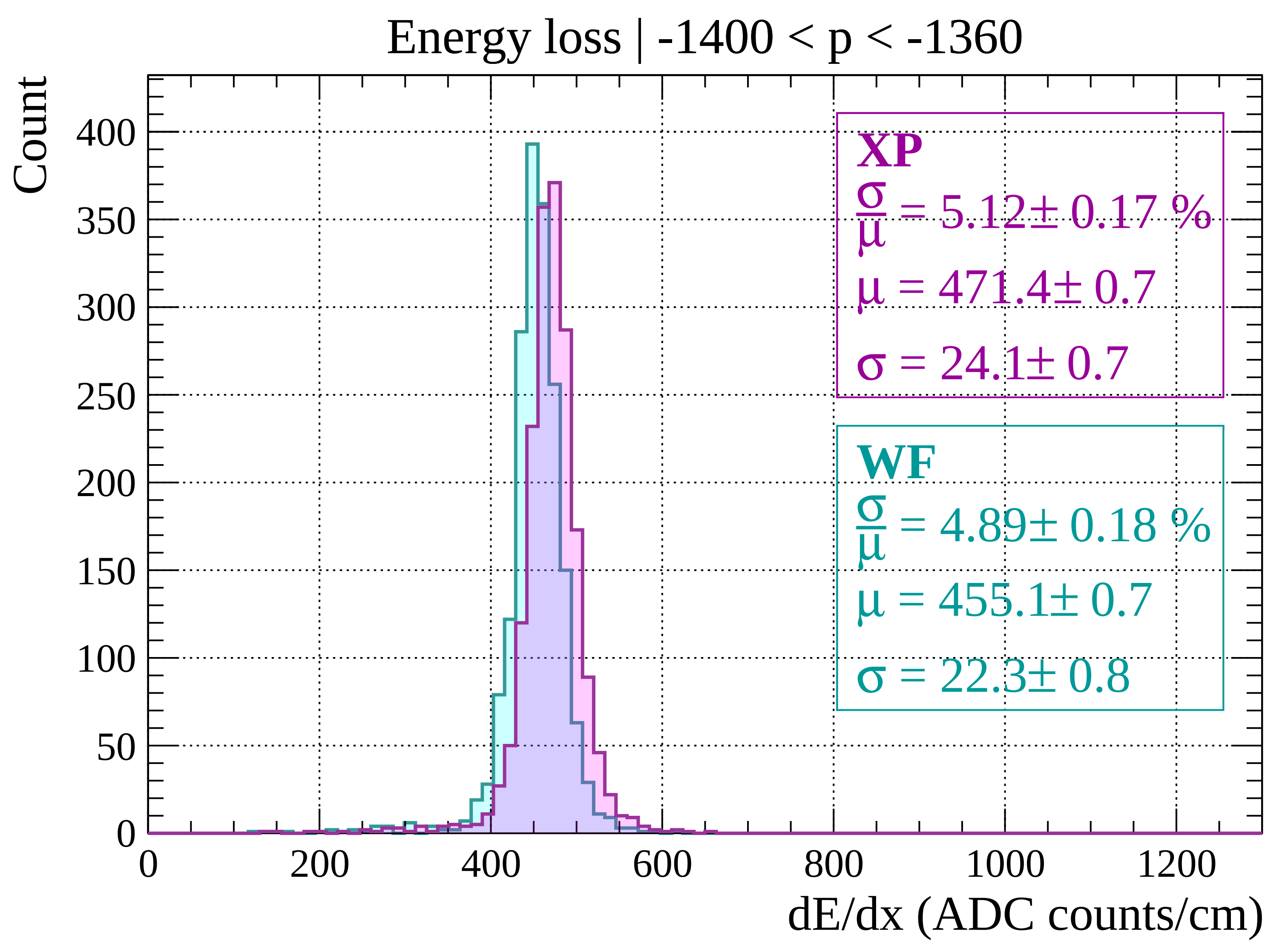

Energy loss | -1520 < p < -1480
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
σ/μ = 5.45± 0.15 %
μ = 476.4± 0.7
σ = 26.0± 0.7
WF
σ/μ = 5.31± 0.17 %
μ = 458.6± 0.7
σ = 24.4± 0.7



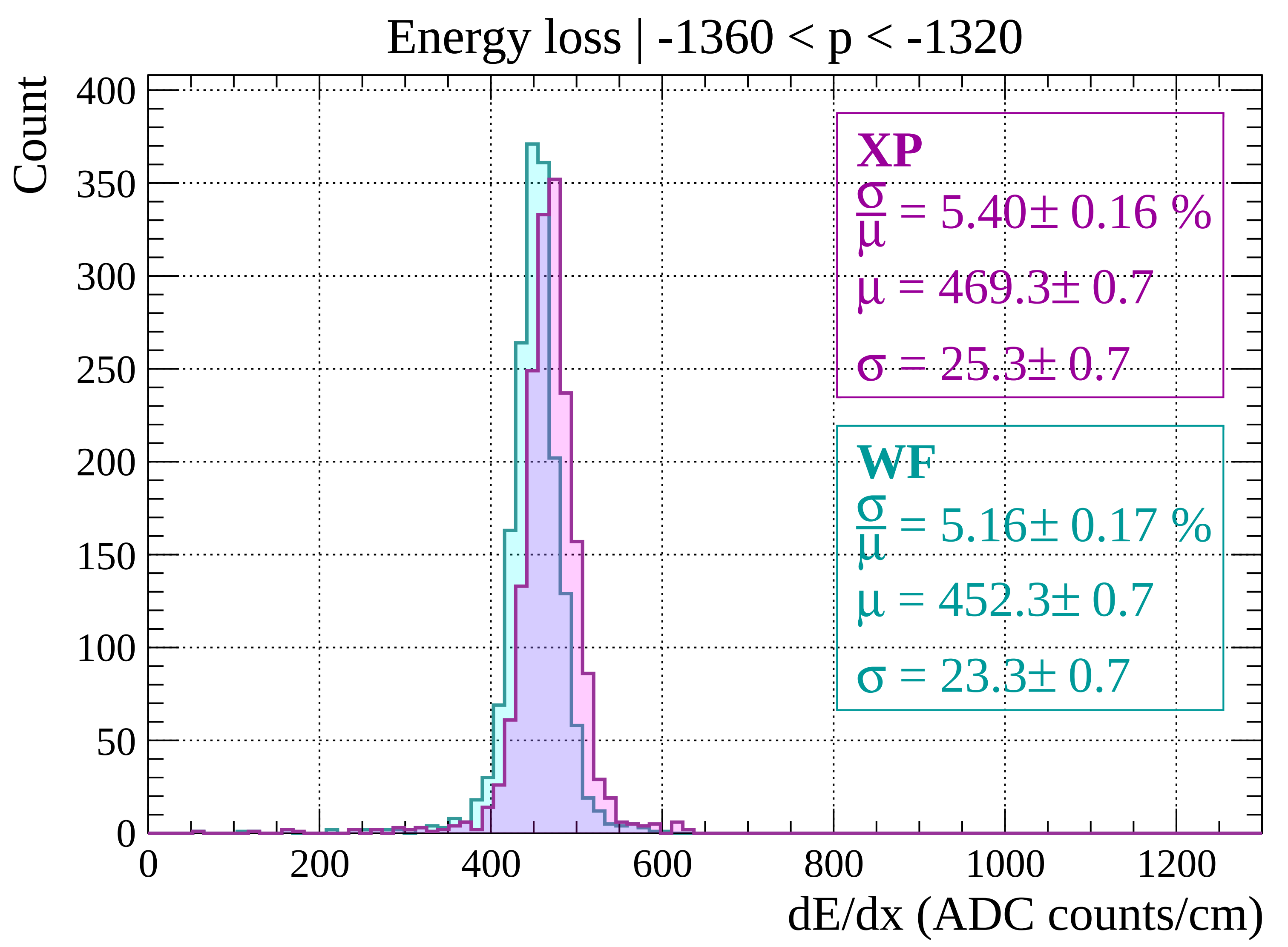
Energy loss | -1480 < p < -1440
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
σ/μ = 5.64 ± 0.15 %
μ = 474.6 ± 0.7
σ = 26.8 ± 0.7
WF
σ/μ = 5.50 ± 0.16 %
μ = 457.8 ± 0.7
σ = 25.2 ± 0.7



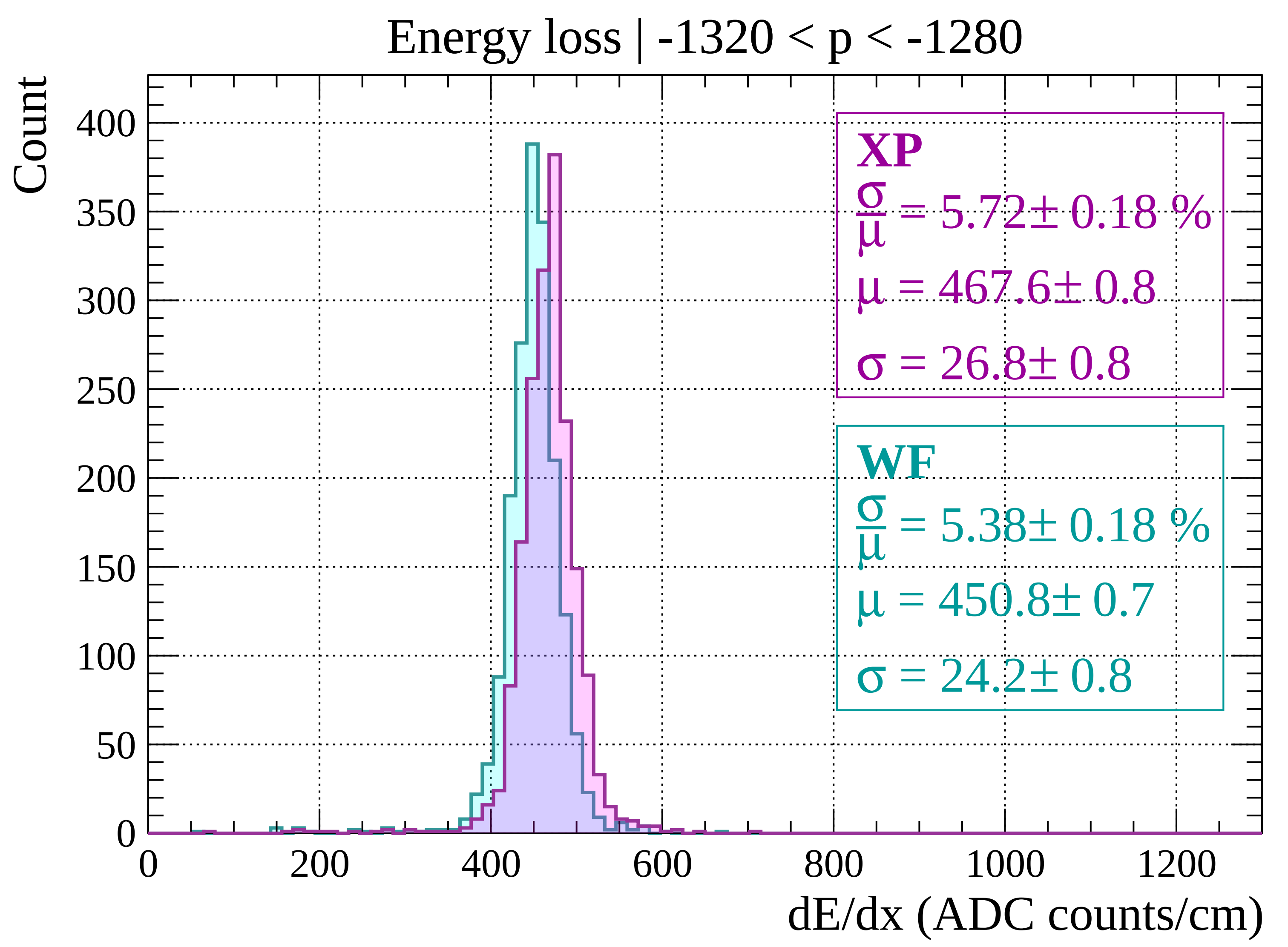
Energy loss | -1440 < p < -1400
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
σ/μ = 5.43± 0.16 %
μ = 473.3± 0.7
σ = 25.7± 0.7
WF
σ/μ = 5.49± 0.15 %
μ = 456.8± 0.7
σ = 25.1± 0.7

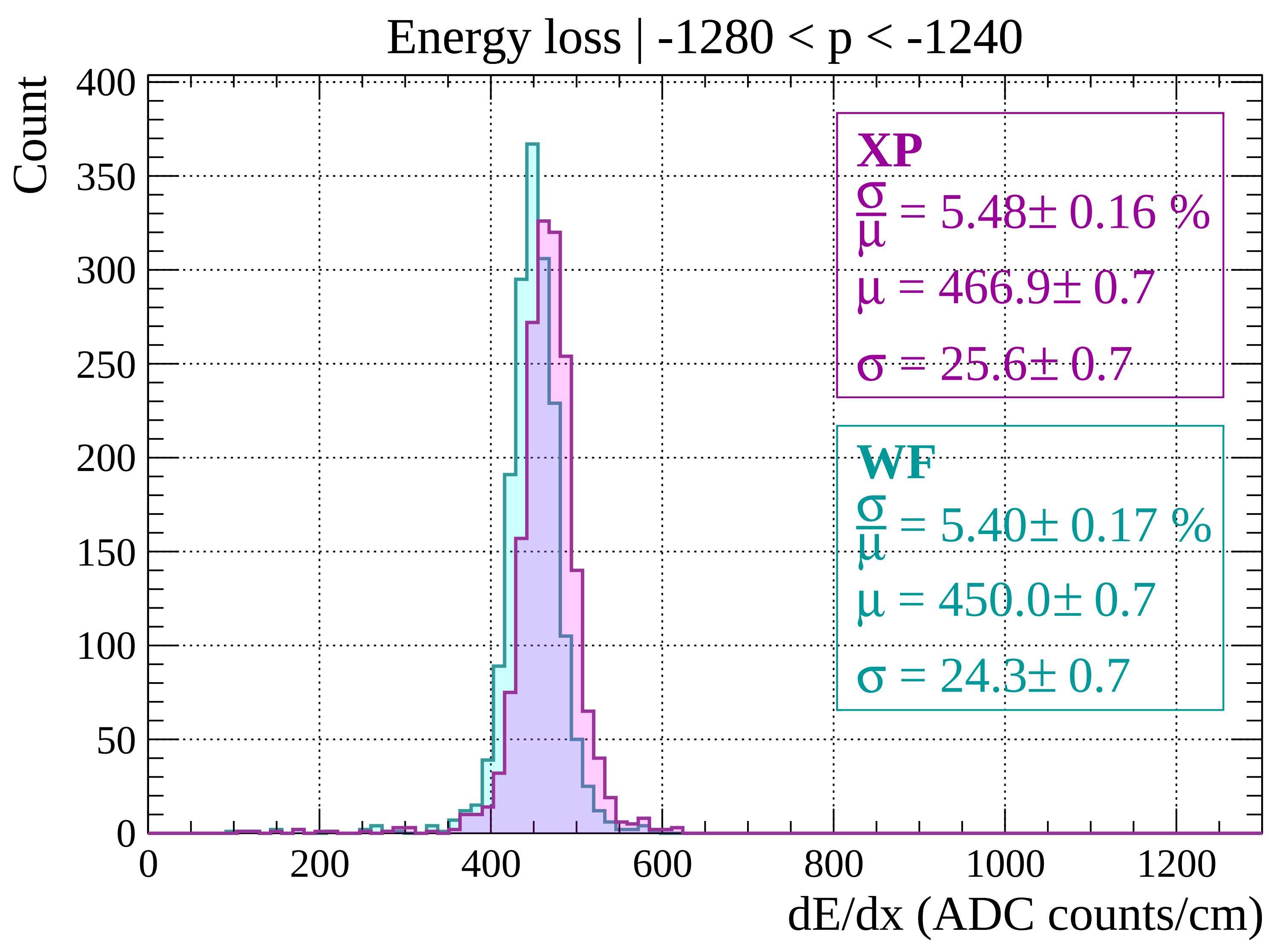


Energy loss | -1400 < p < -1360
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
σ/μ = 5.12± 0.17 %
μ = 471.4± 0.7
σ = 24.1± 0.7
WF
σ/μ = 4.89± 0.18 %
μ = 455.1± 0.7
σ = 22.3± 0.8

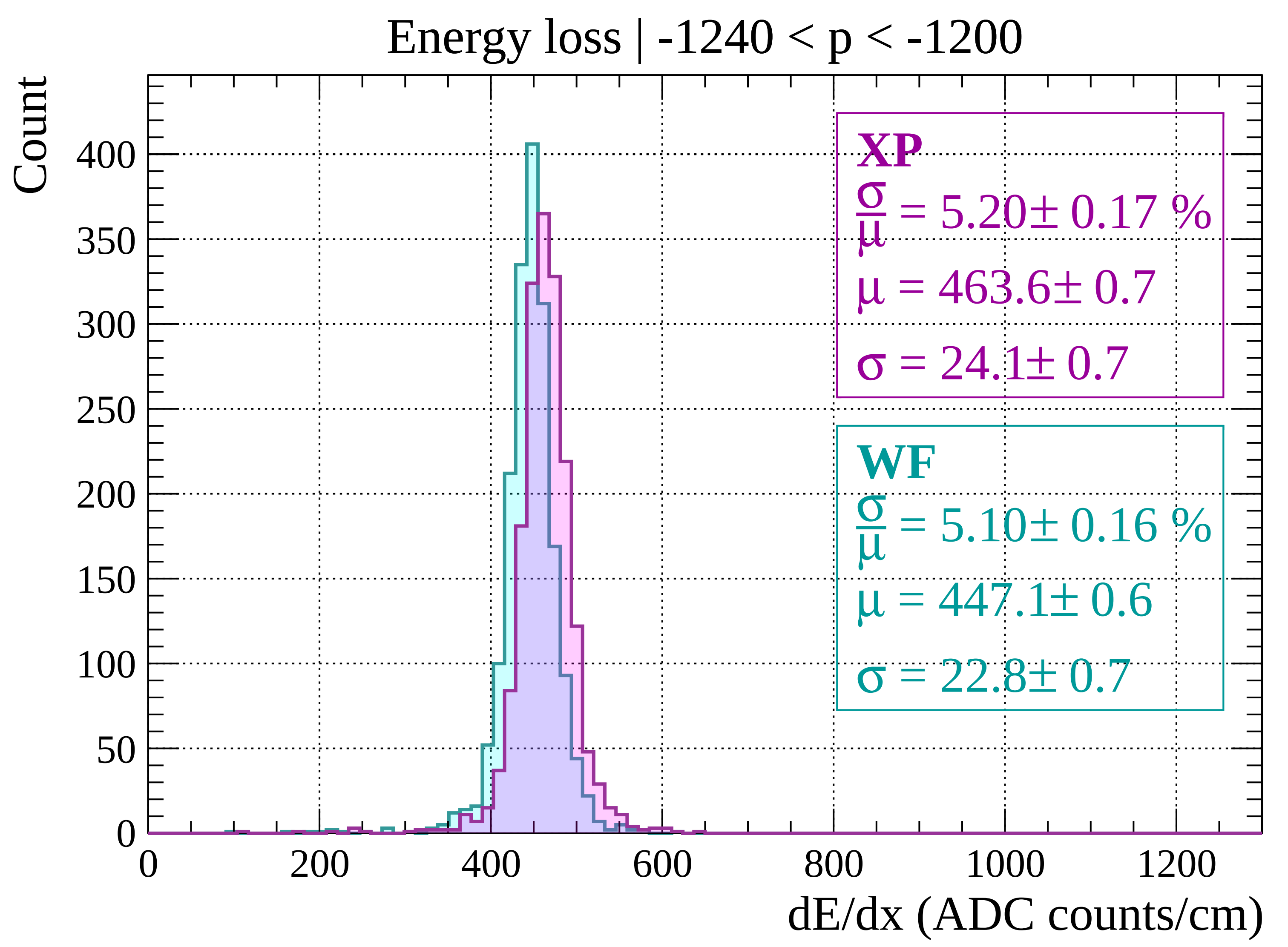


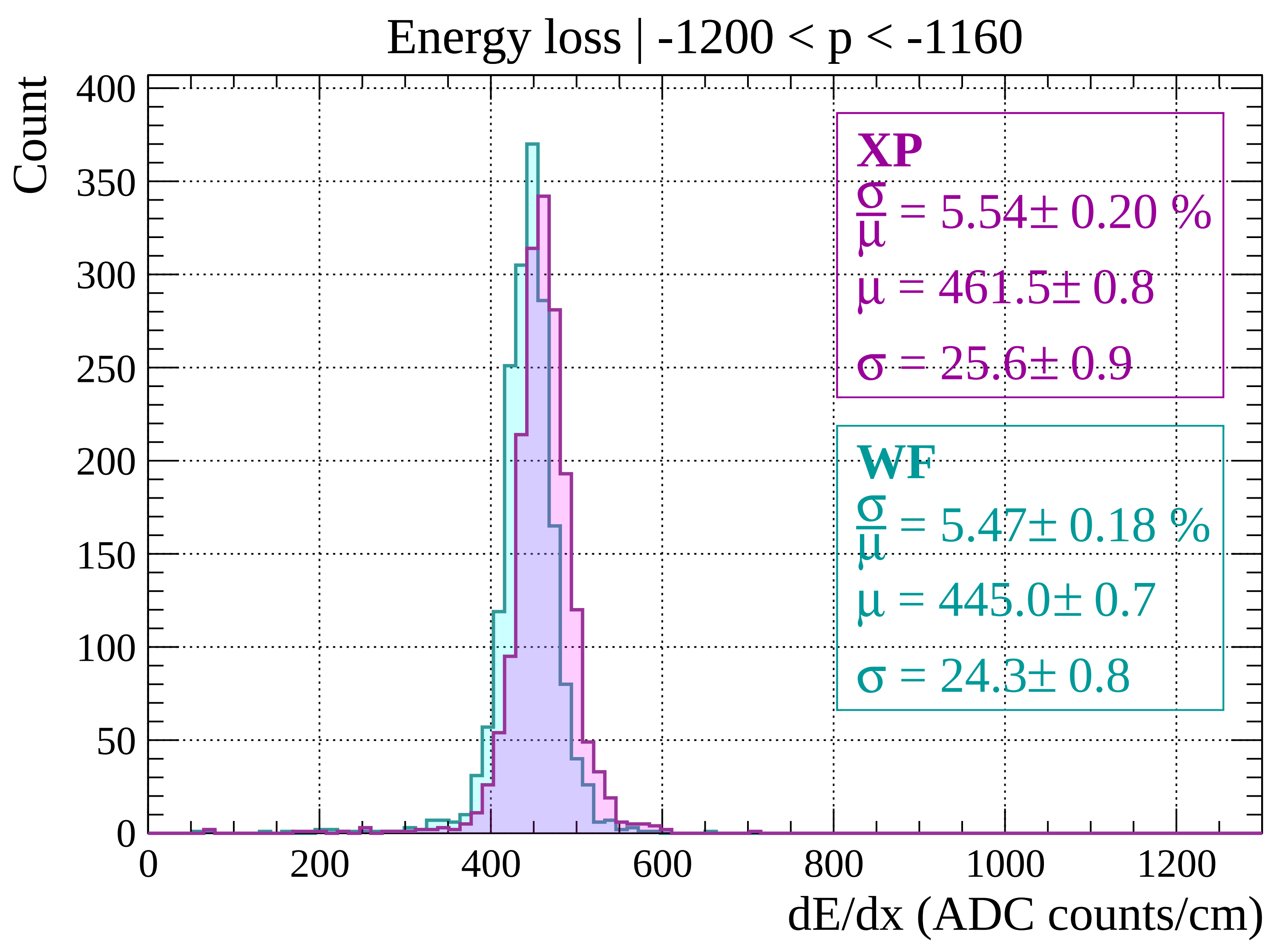
Energy loss | -1360 < p < -1320
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 5.40± 0.16 %
μ = 469.3± 0.7
σ = 25.3± 0.7
WF
$\frac{\sigma}{\mu}$ = 5.16± 0.17 %
μ = 452.3± 0.7
σ = 23.3± 0.7


Energy loss | -1320 < p < -1280
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
σ/μ = 5.72± 0.18 %
μ = 467.6± 0.8
σ = 26.8± 0.8
WF
σ/μ = 5.38± 0.18 %
μ = 450.8± 0.7
σ = 24.2± 0.8

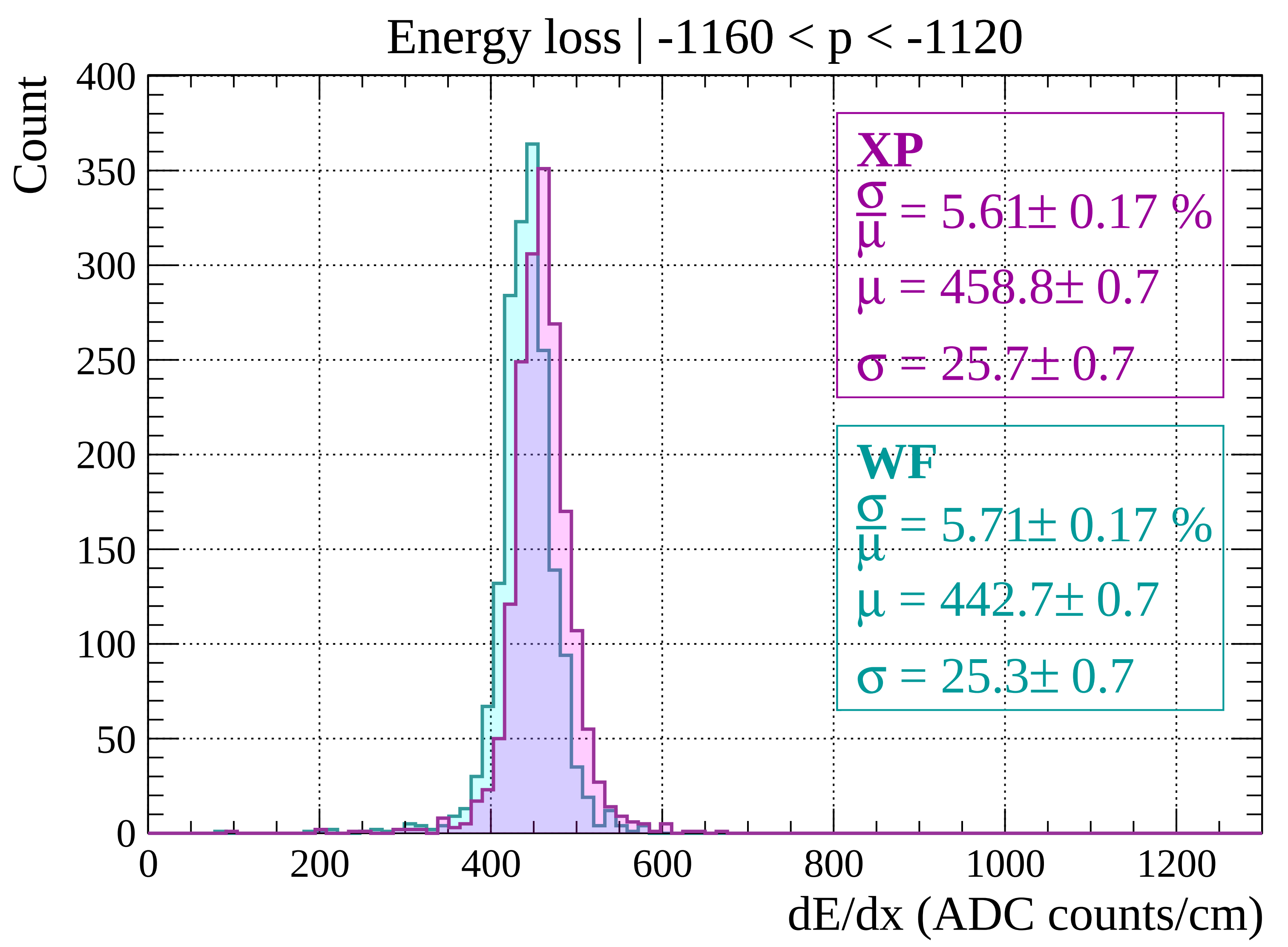


Energy loss | -1280 < p < -1240
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 5.48± 0.16 %
μ = 466.9± 0.7
σ = 25.6± 0.7
WF
$\frac{\sigma}{\mu}$ = 5.40± 0.17 %
μ = 450.0± 0.7
σ = 24.3± 0.7

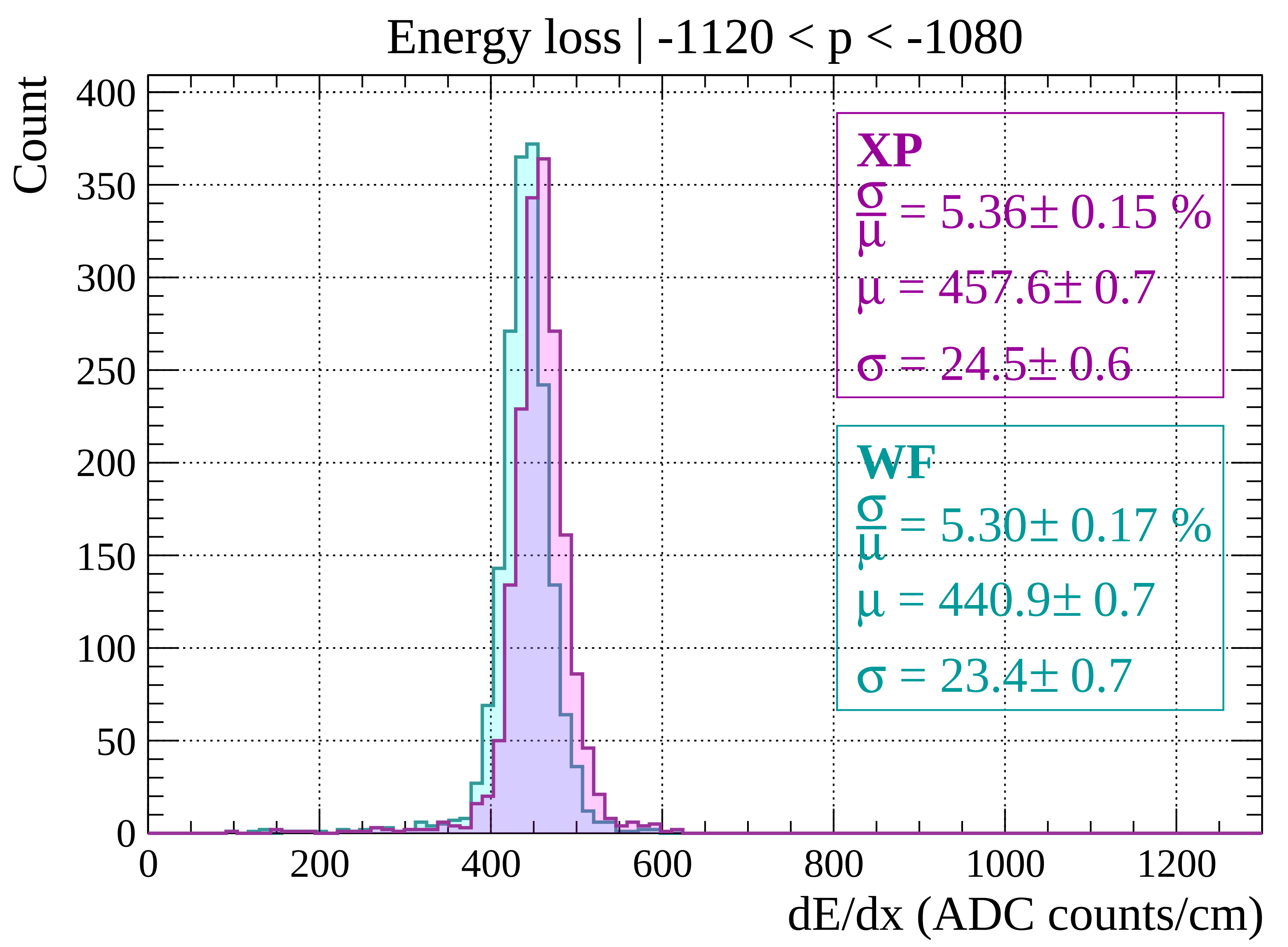

Energy loss | -1240 < p < -1200
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
400
XP
$\frac{\sigma}{\mu}$ = 5.20± 0.17 %
μ = 463.6± 0.7
σ = 24.1± 0.7
WF
$\frac{\sigma}{\mu}$ = 5.10± 0.16 %
μ = 447.1± 0.6
σ = 22.8± 0.7



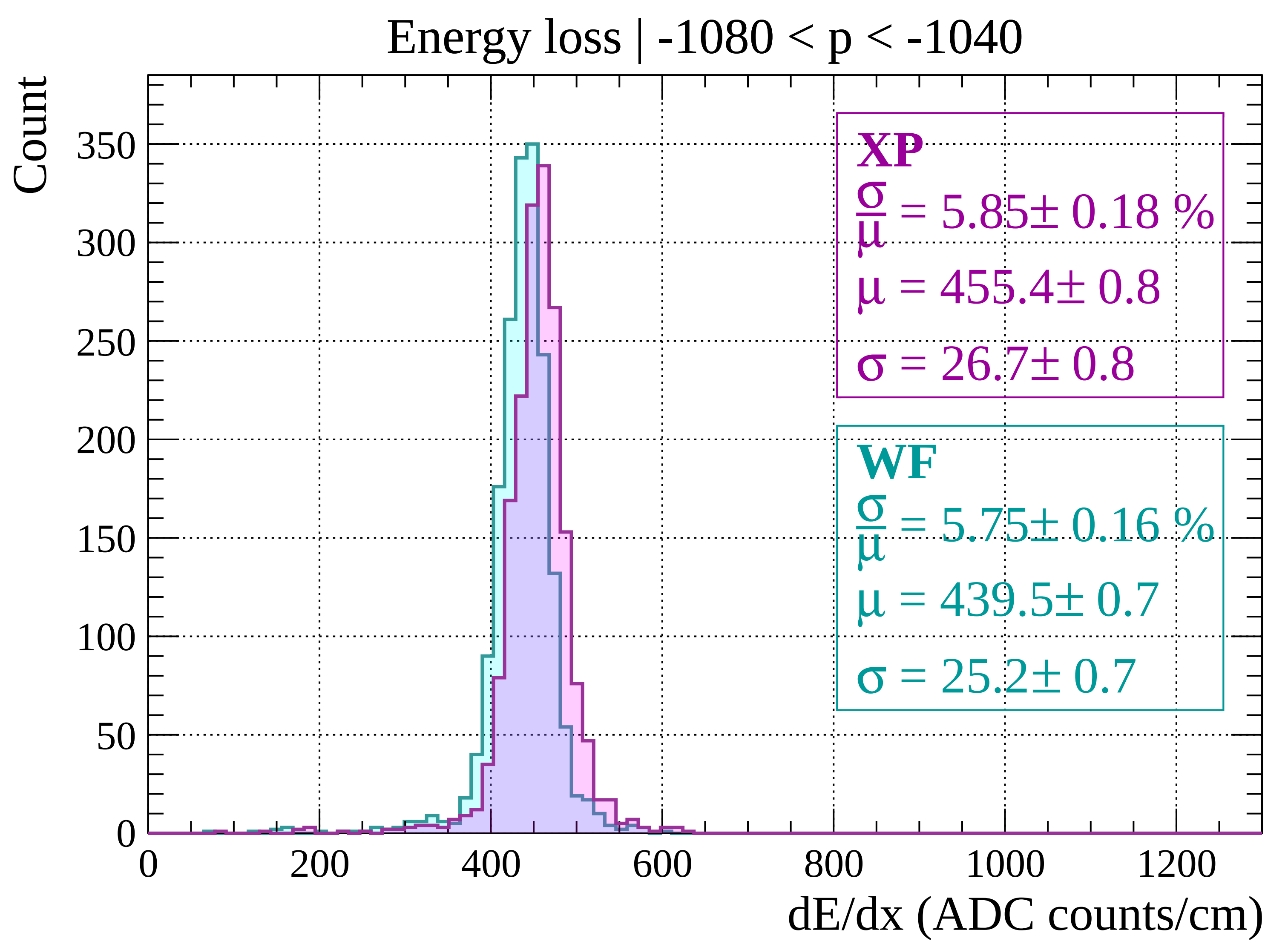
Energy loss | -1200 < p < -1160
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
400
XP
σ/μ = 5.54± 0.20 %
μ = 461.5± 0.8
σ = 25.6± 0.9
WF
σ/μ = 5.47± 0.18 %
μ = 445.0± 0.7
σ = 24.3± 0.8

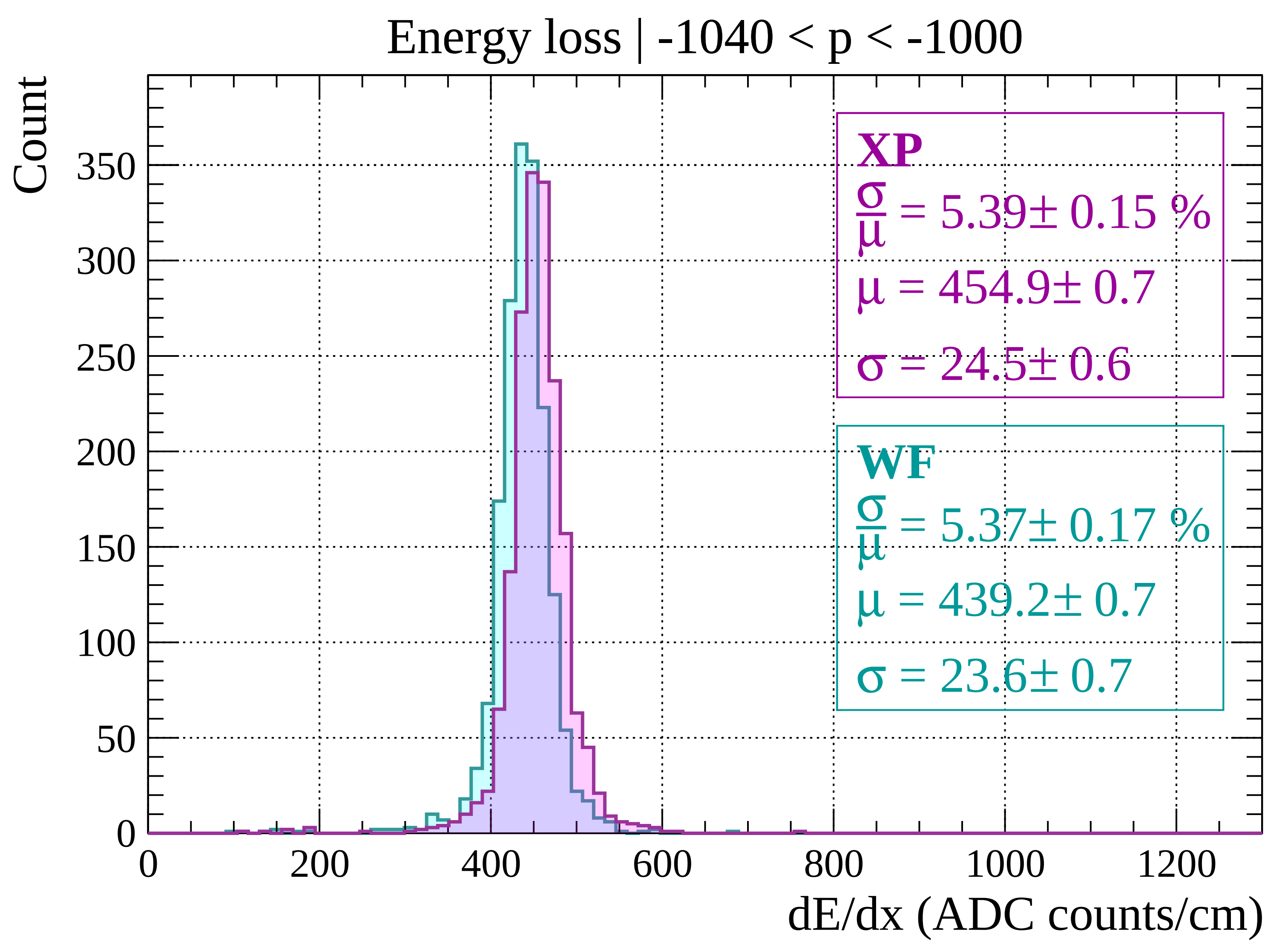


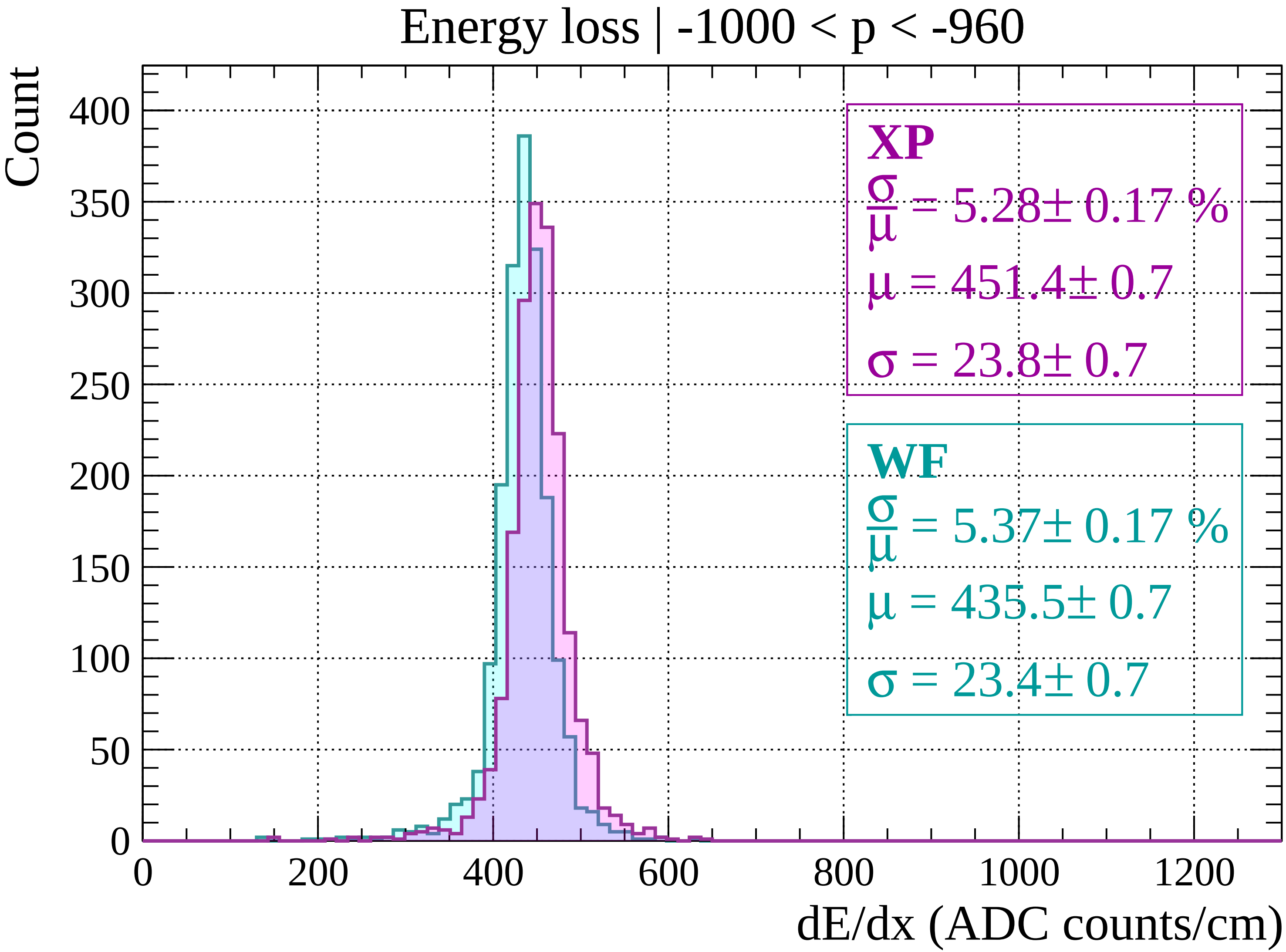
Energy loss | -1160 < p < -1120
Count
400
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 5.61± 0.17 %
μ = 458.8± 0.7
σ = 25.7± 0.7
WF
σ/μ = 5.71± 0.17 %
μ = 442.7± 0.7
σ = 25.3± 0.7

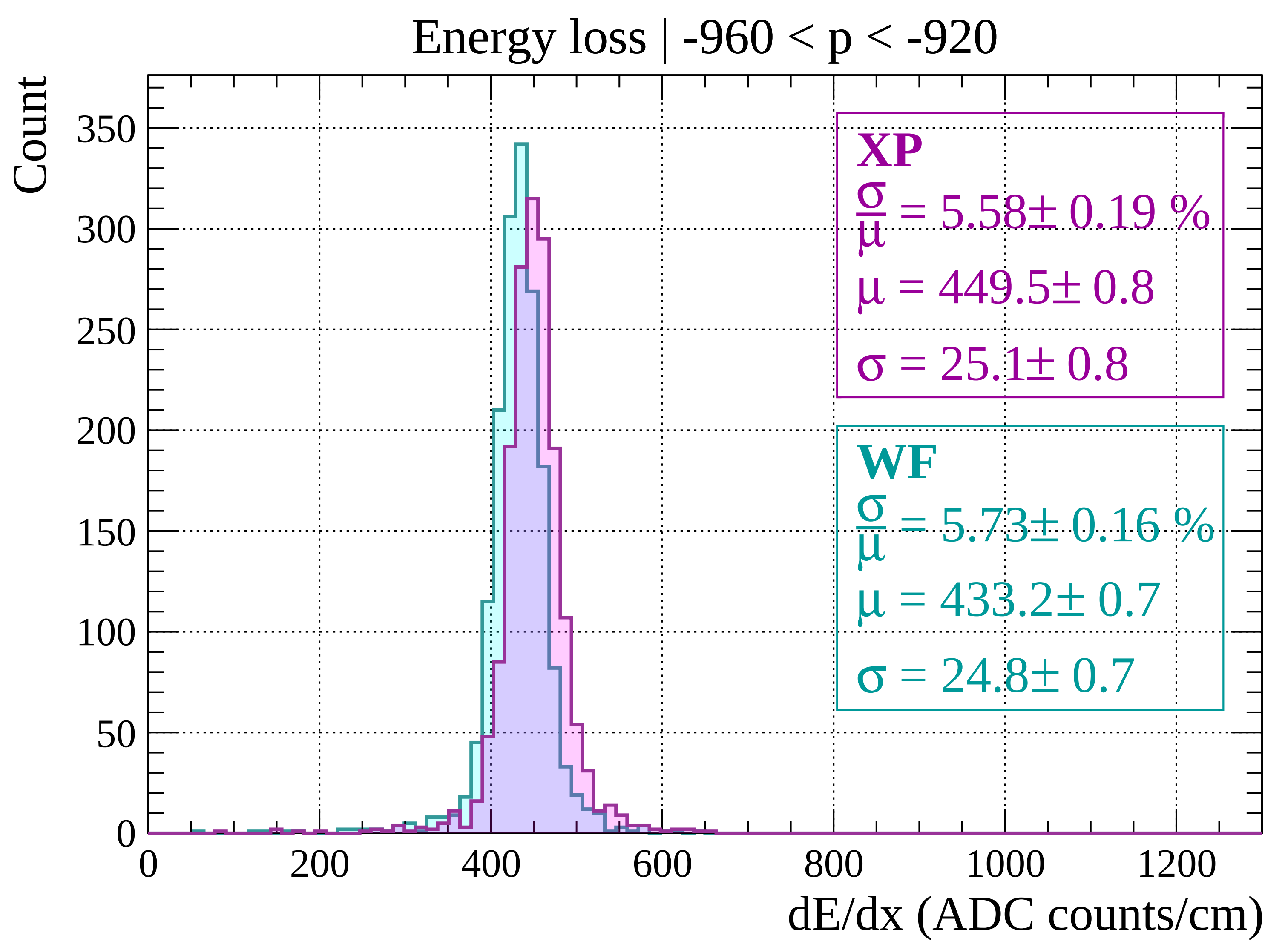


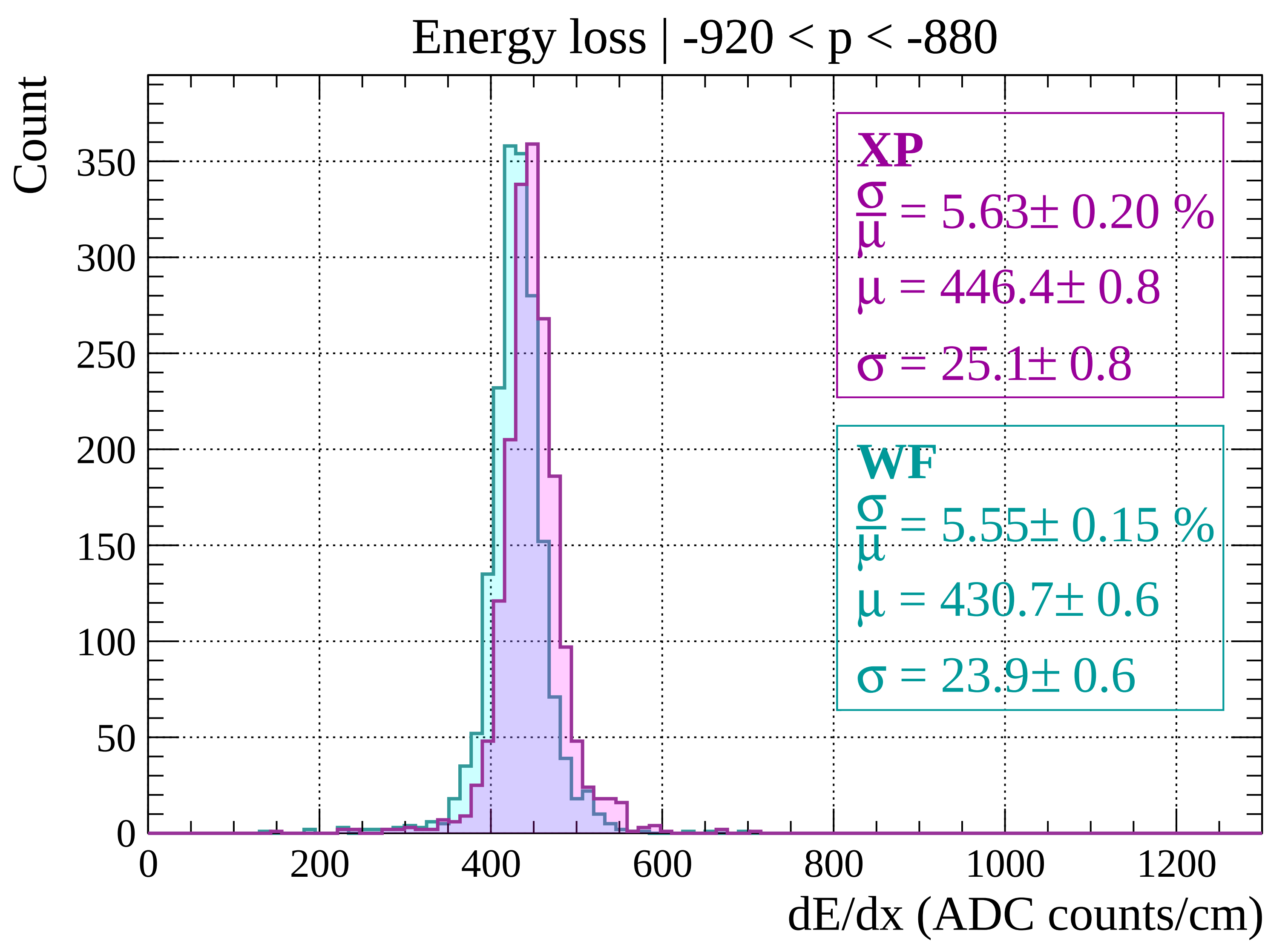
Energy loss | -1120 < p < -1080
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
σ/μ = 5.36± 0.15 %
μ = 457.6± 0.7
σ = 24.5± 0.6
WF
σ/μ = 5.30± 0.17 %
μ = 440.9± 0.7
σ = 23.4± 0.7

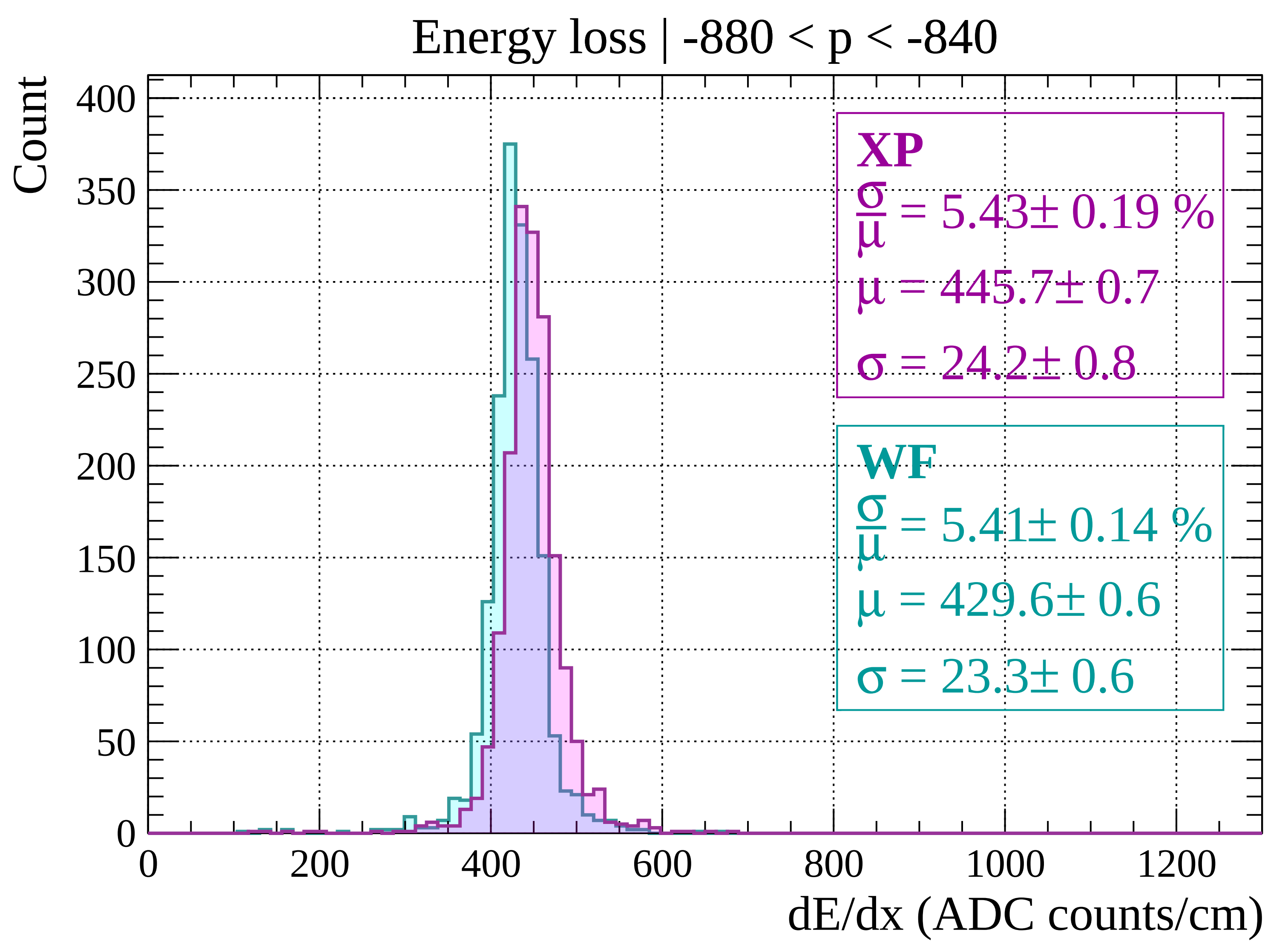

Energy loss | -1080 < p < -1040
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 5.85± 0.18 %
μ = 455.4± 0.8
σ = 26.7± 0.8
WF
$\frac{\sigma}{\mu}$ = 5.75± 0.16 %
μ = 439.5± 0.7
σ = 25.2± 0.7


Energy loss | -1040 < p < -1000
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 5.39± 0.15 %
μ = 454.9± 0.7
σ = 24.5± 0.6
WF
σ/μ = 5.37± 0.17 %
μ = 439.2± 0.7
σ = 23.6± 0.7

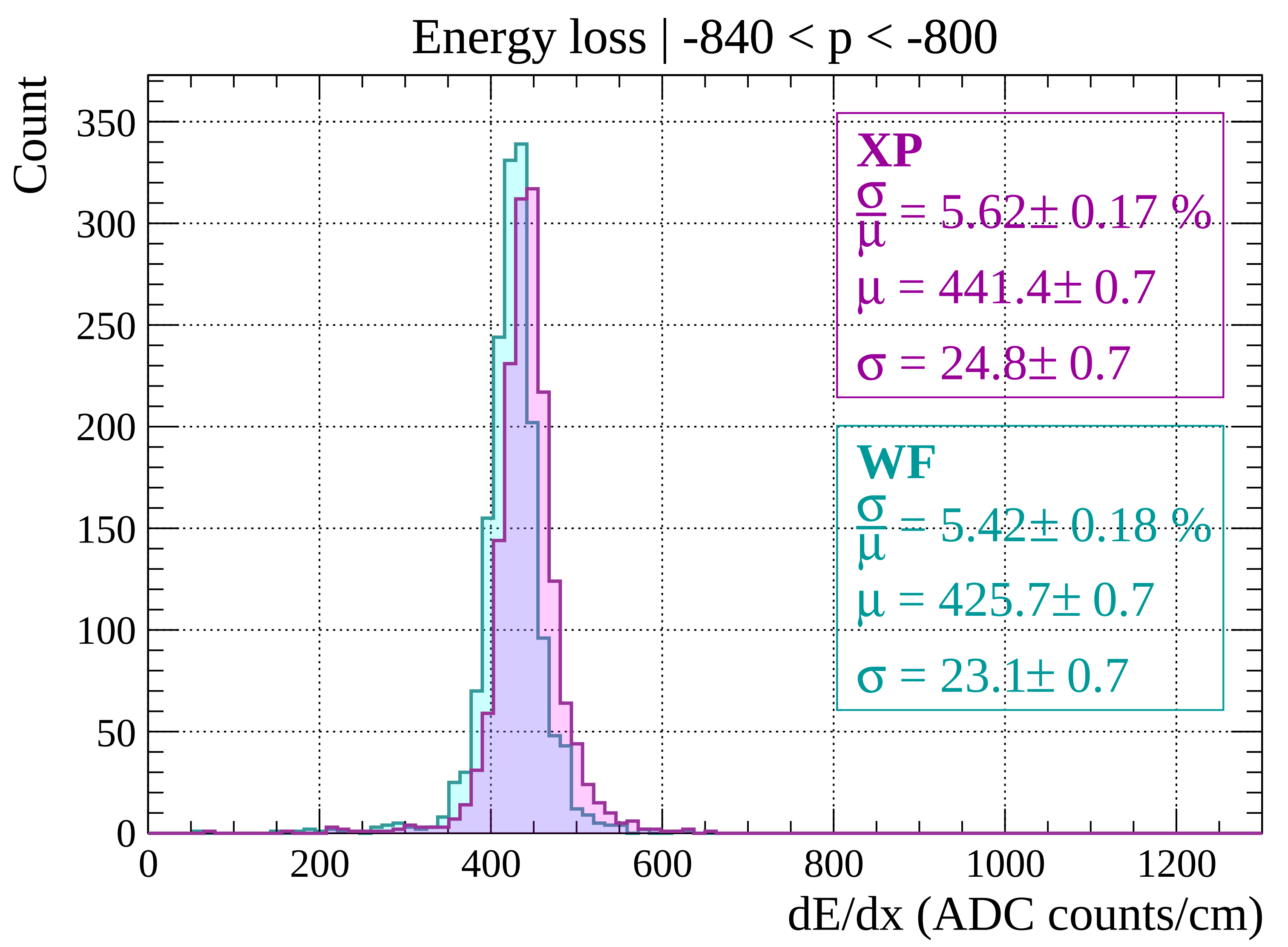

Energy loss | -1000 < p < -960
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
400
XP
σ/μ = 5.28± 0.17 %
μ = 451.4± 0.7
σ = 23.8± 0.7
WF
σ/μ = 5.37± 0.17 %
μ = 435.5± 0.7
σ = 23.4± 0.7


Energy loss | -960 < p < -920
Count
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 5.58± 0.19 %
μ = 449.5± 0.8
σ = 25.1± 0.8
WF
σ/μ = 5.73± 0.16 %
μ = 433.2± 0.7
σ = 24.8± 0.7

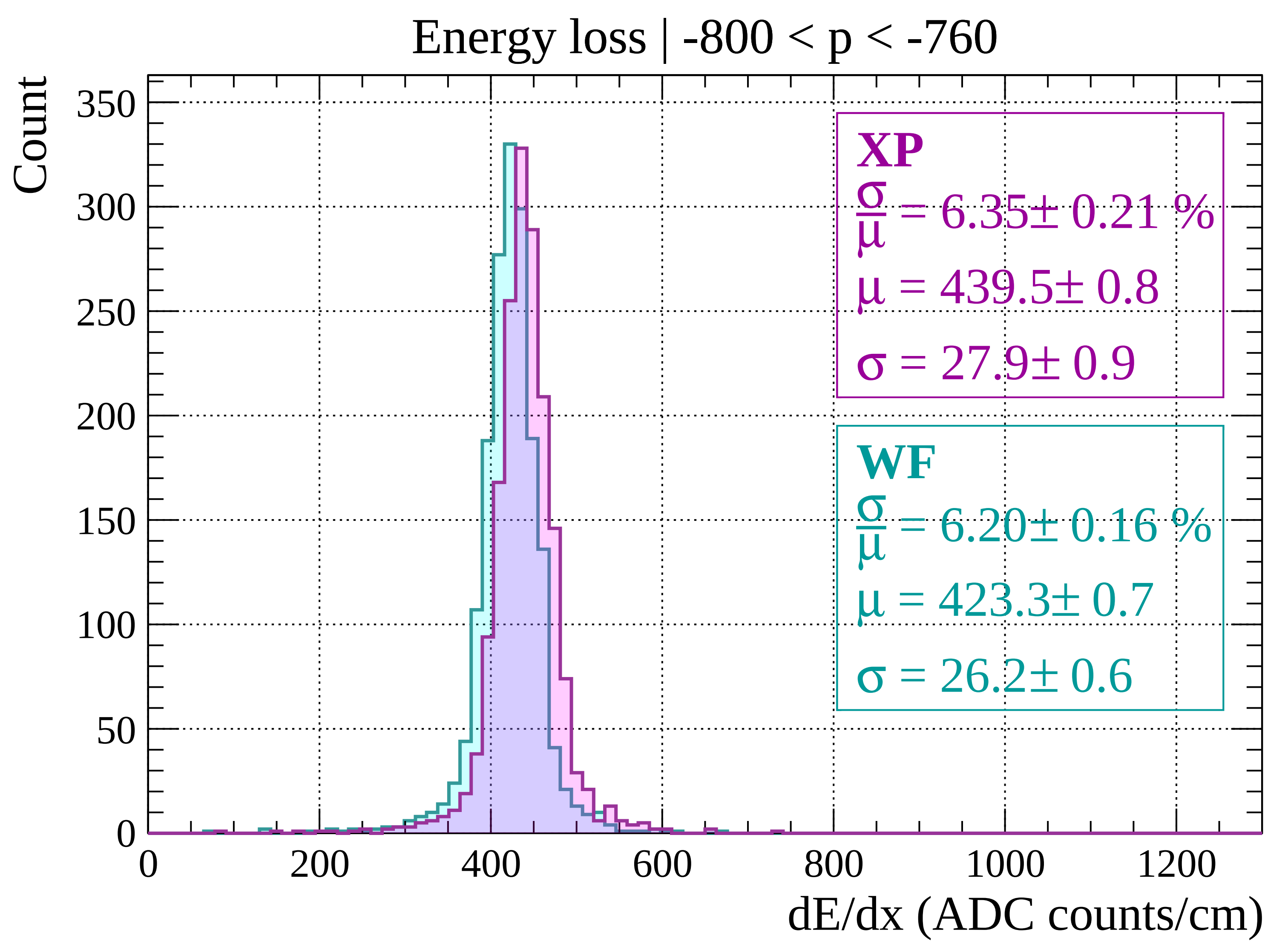

Energy loss | -920 < p < -880
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 5.63± 0.20 %
μ = 446.4± 0.8
σ = 25.1± 0.8
WF
σ/μ = 5.55± 0.15 %
μ = 430.7± 0.6
σ = 23.9± 0.6
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200

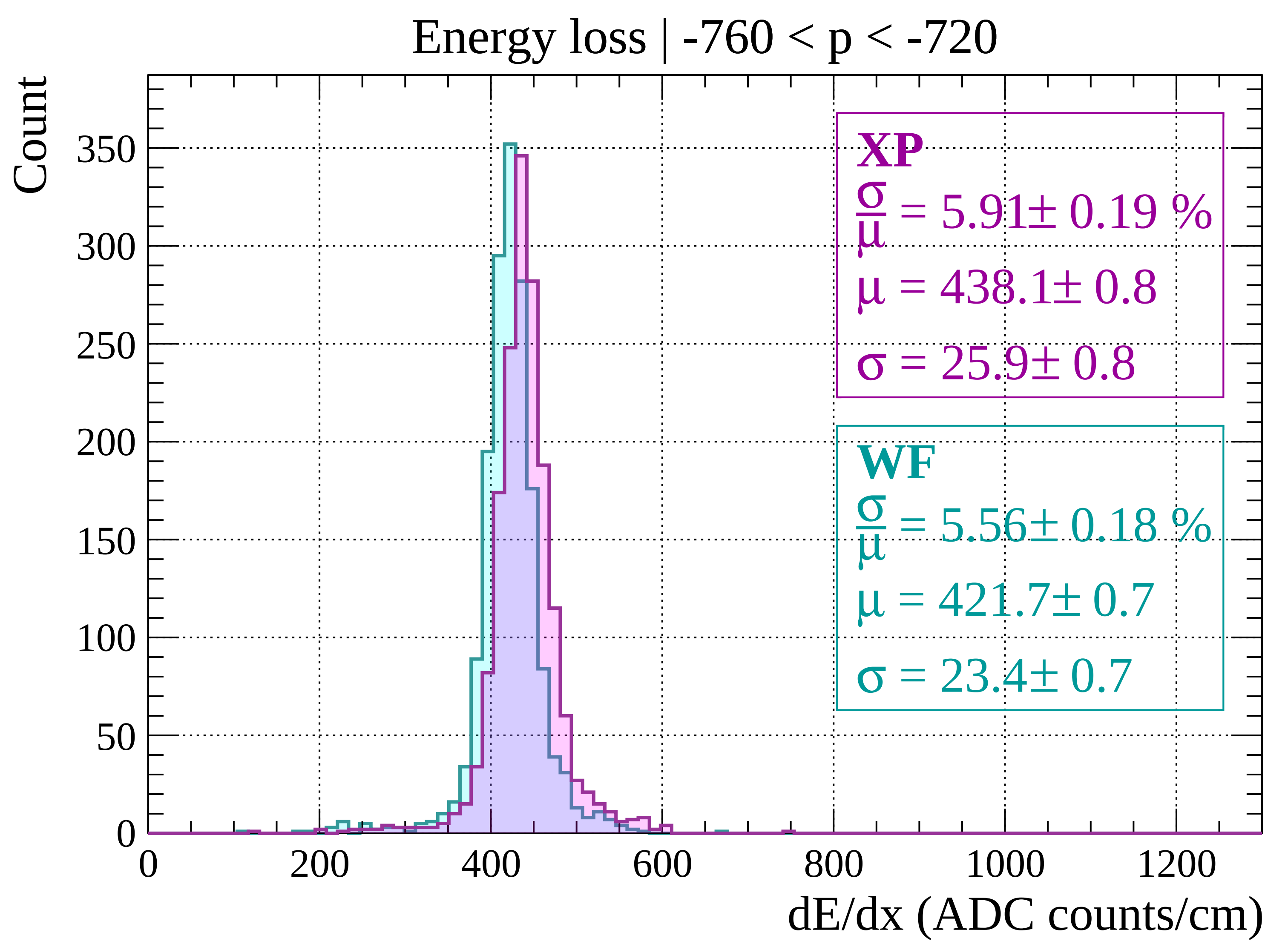


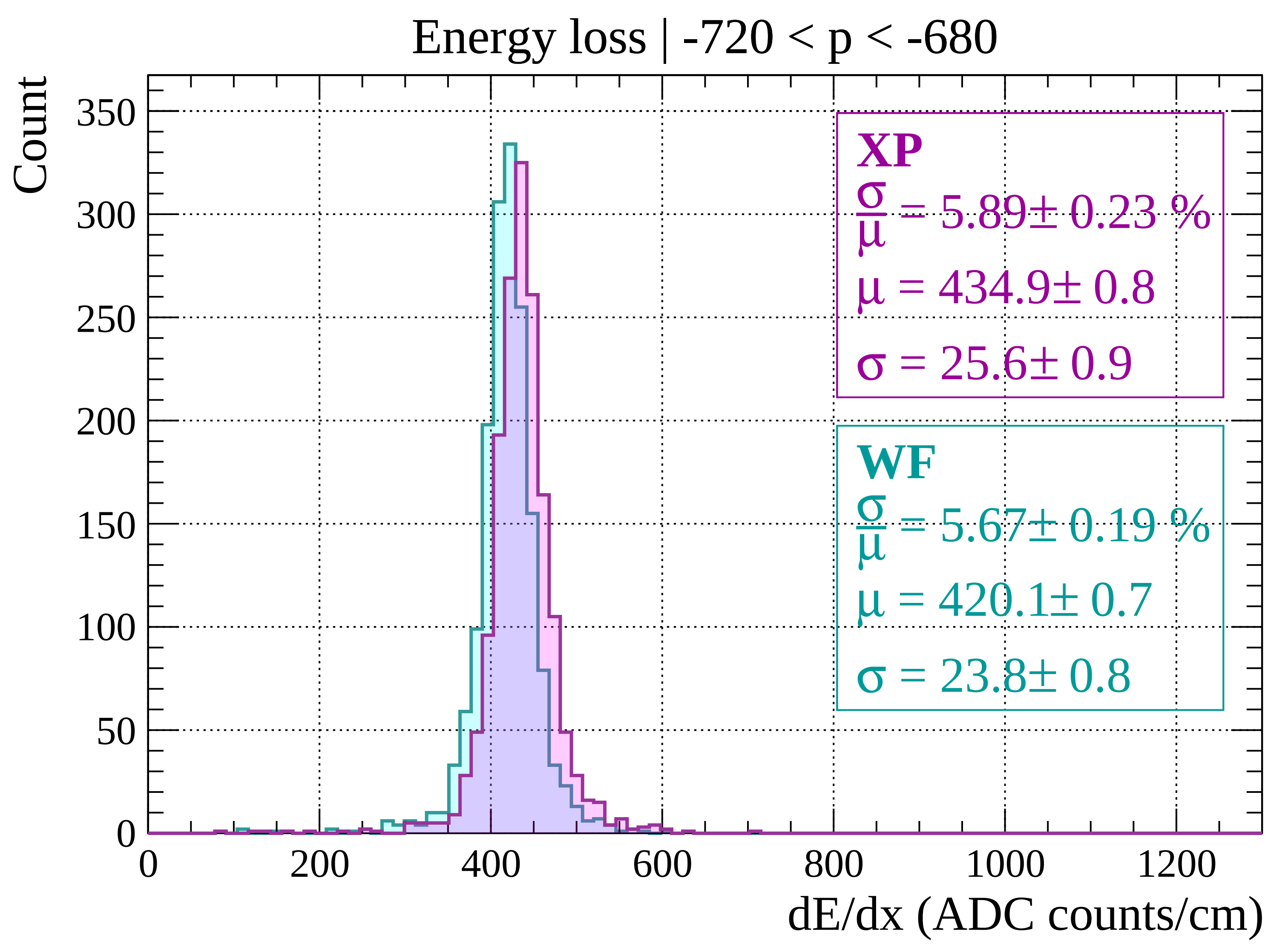
Energy loss | -880 < p < -840
Count
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 5.43± 0.19 %
μ = 445.7± 0.7
σ = 24.2± 0.8
WF
$\frac{\sigma}{\mu}$ = 5.41± 0.14 %
μ = 429.6± 0.6
σ = 23.3± 0.6

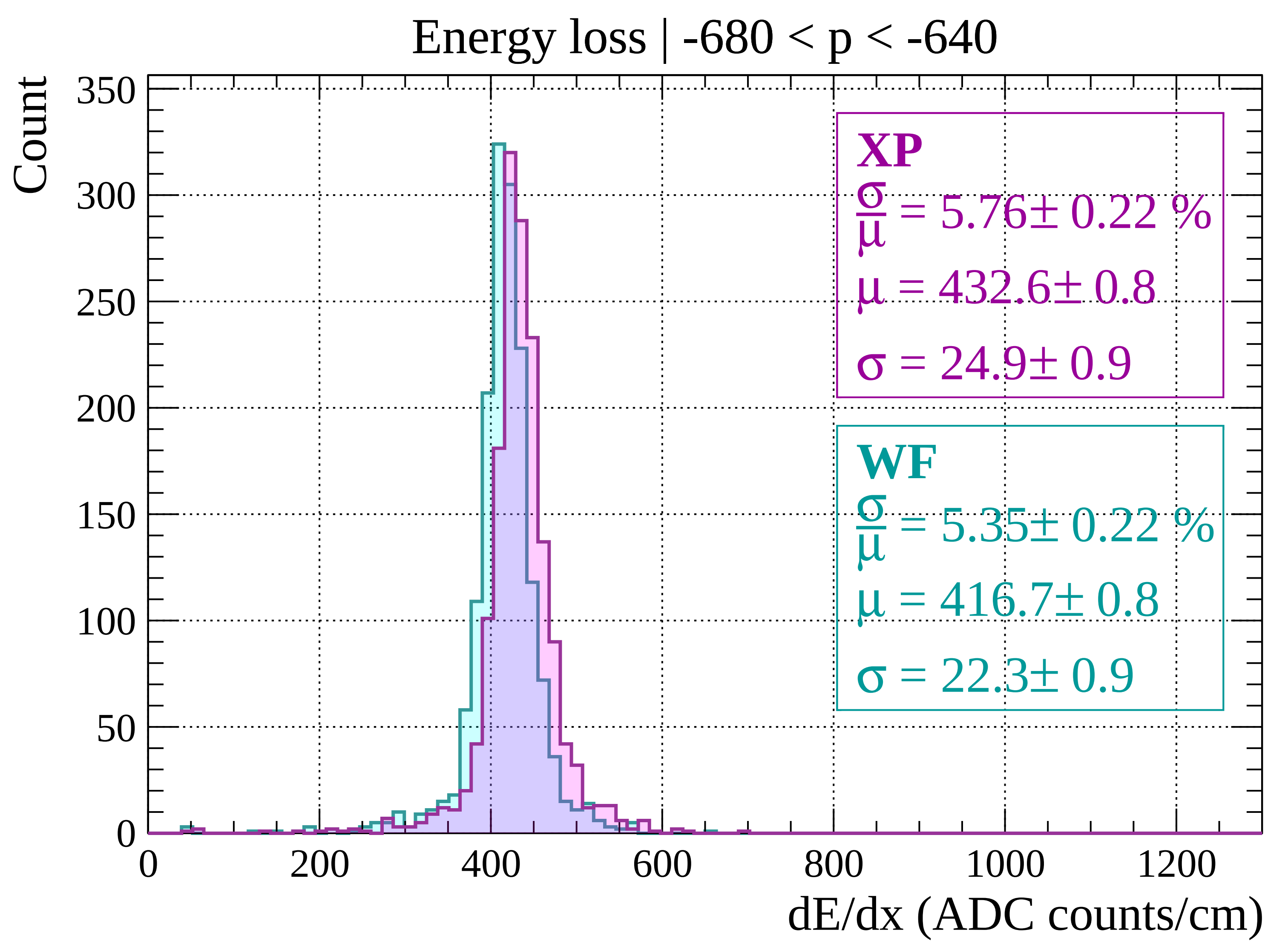


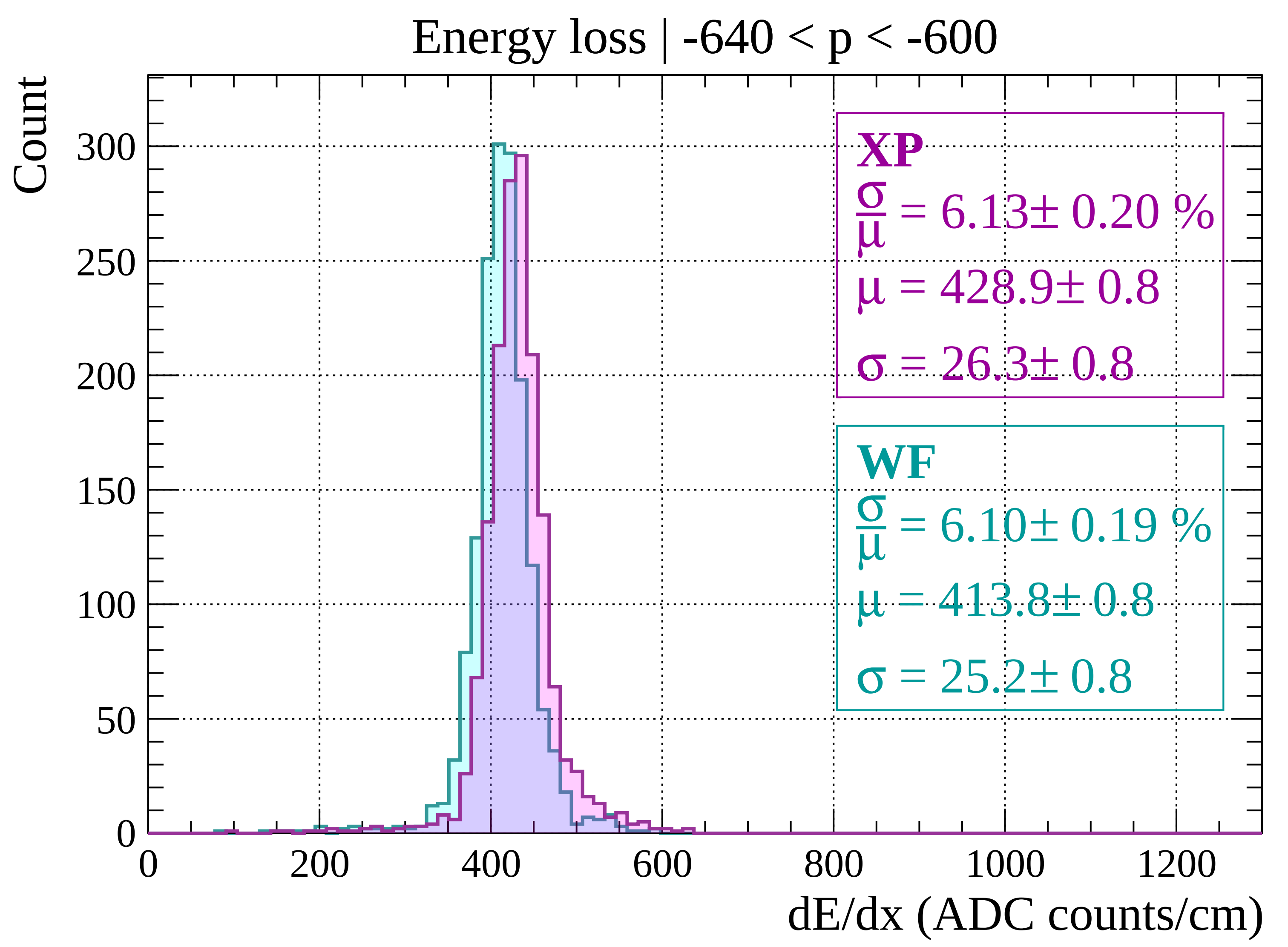
Energy loss | -840 < p < -800
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 5.62 \pm 0.17$ %
$\mu = 441.4 \pm 0.7$
$\sigma = 24.8 \pm 0.7$
WF
$\frac{\sigma}{\mu} = 5.42 \pm 0.18$ %
$\mu = 425.7 \pm 0.7$
$\sigma = 23.1 \pm 0.7$

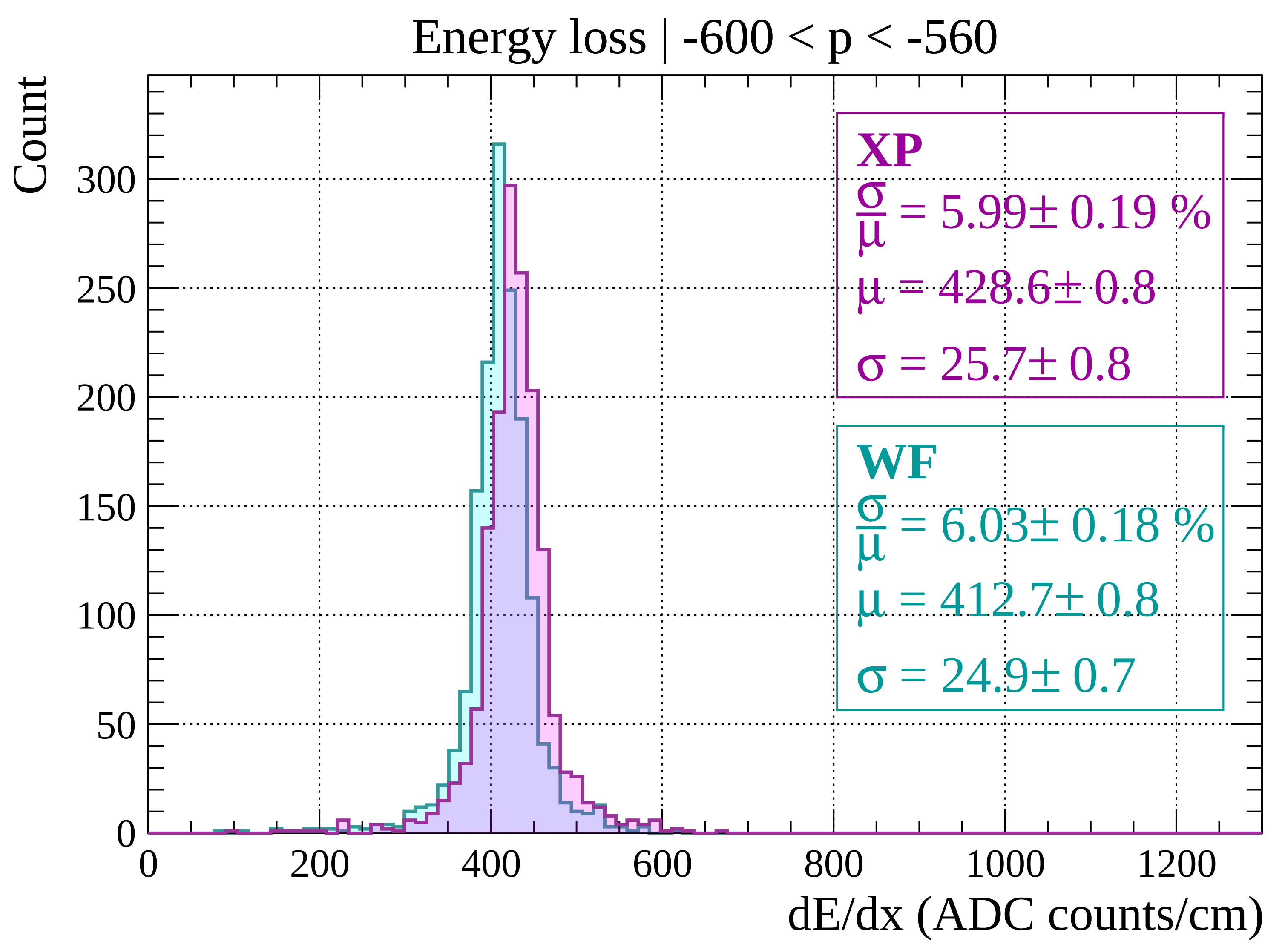

Energy loss | -800 < p < -760
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 6.35± 0.21 %
μ = 439.5± 0.8
σ = 27.9± 0.9
WF
σ/μ = 6.20± 0.16 %
μ = 423.3± 0.7
σ = 26.2± 0.6
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200

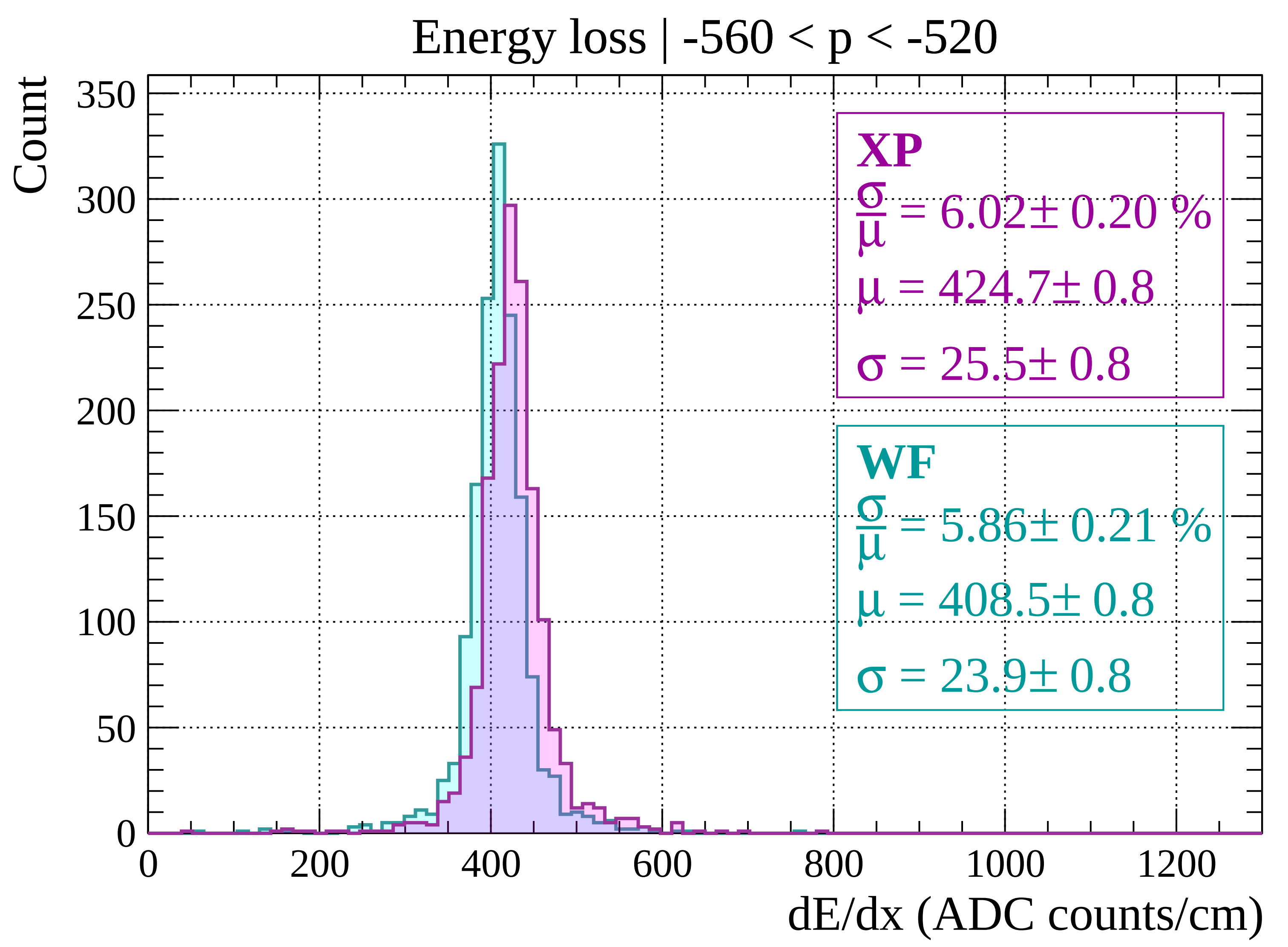

Energy loss | -760 < p < -720
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 5.91± 0.19 %
μ = 438.1± 0.8
σ = 25.9± 0.8
WF
σ/μ = 5.56± 0.18 %
μ = 421.7± 0.7
σ = 23.4± 0.7
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350


Energy loss | -720 < p < -680
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 5.89 \pm 0.23$ %
$\mu = 434.9 \pm 0.8$
$\sigma = 25.6 \pm 0.9$
WF
$\frac{\sigma}{\mu} = 5.67 \pm 0.19$ %
$\mu = 420.1 \pm 0.7$
$\sigma = 23.8 \pm 0.8$

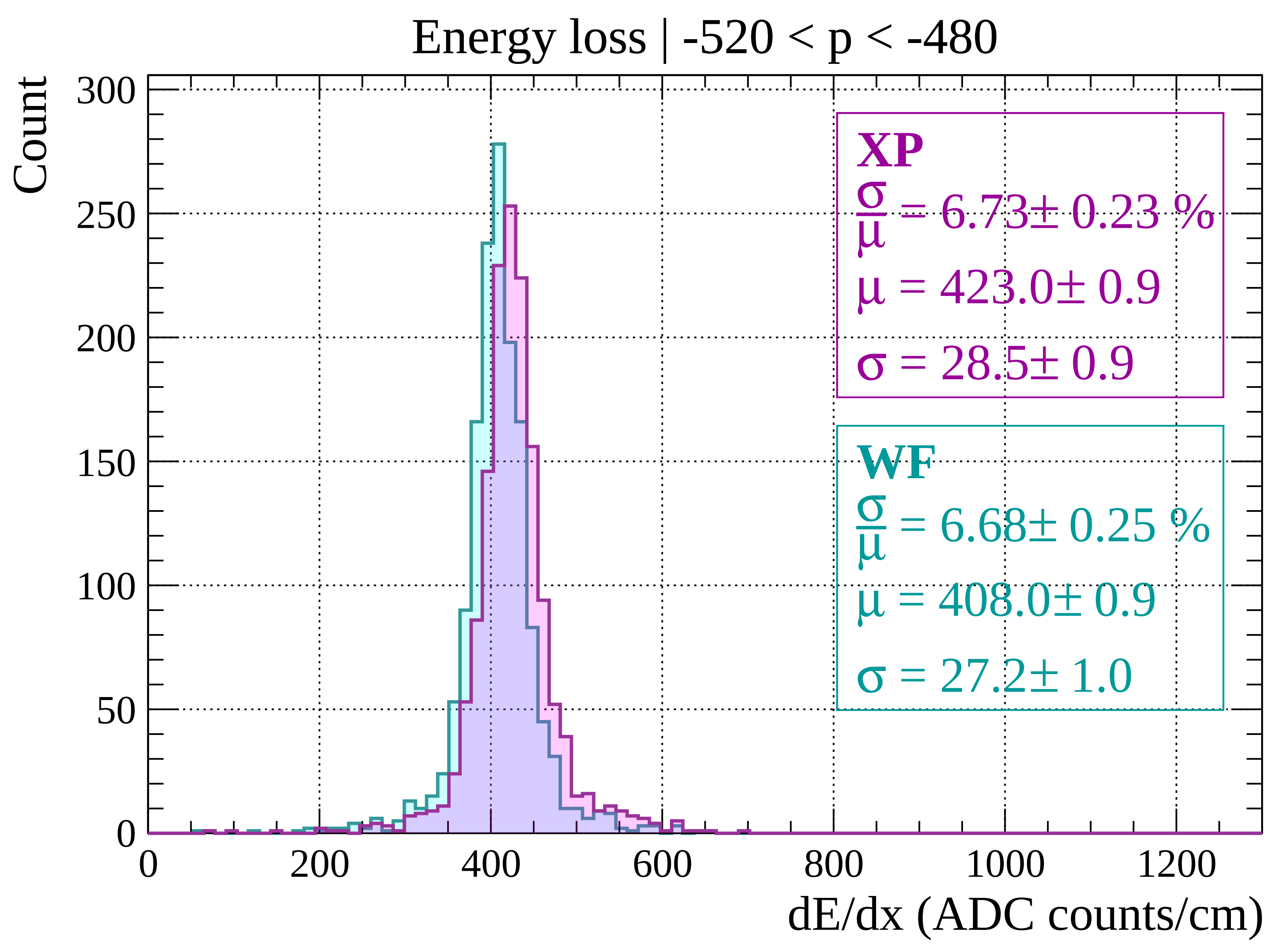

Energy loss | -680 < p < -640
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 5.76 \pm 0.22$ %
$\mu = 432.6 \pm 0.8$
$\sigma = 24.9 \pm 0.9$
WF
$\frac{\sigma}{\mu} = 5.35 \pm 0.22$ %
$\mu = 416.7 \pm 0.8$
$\sigma = 22.3 \pm 0.9$

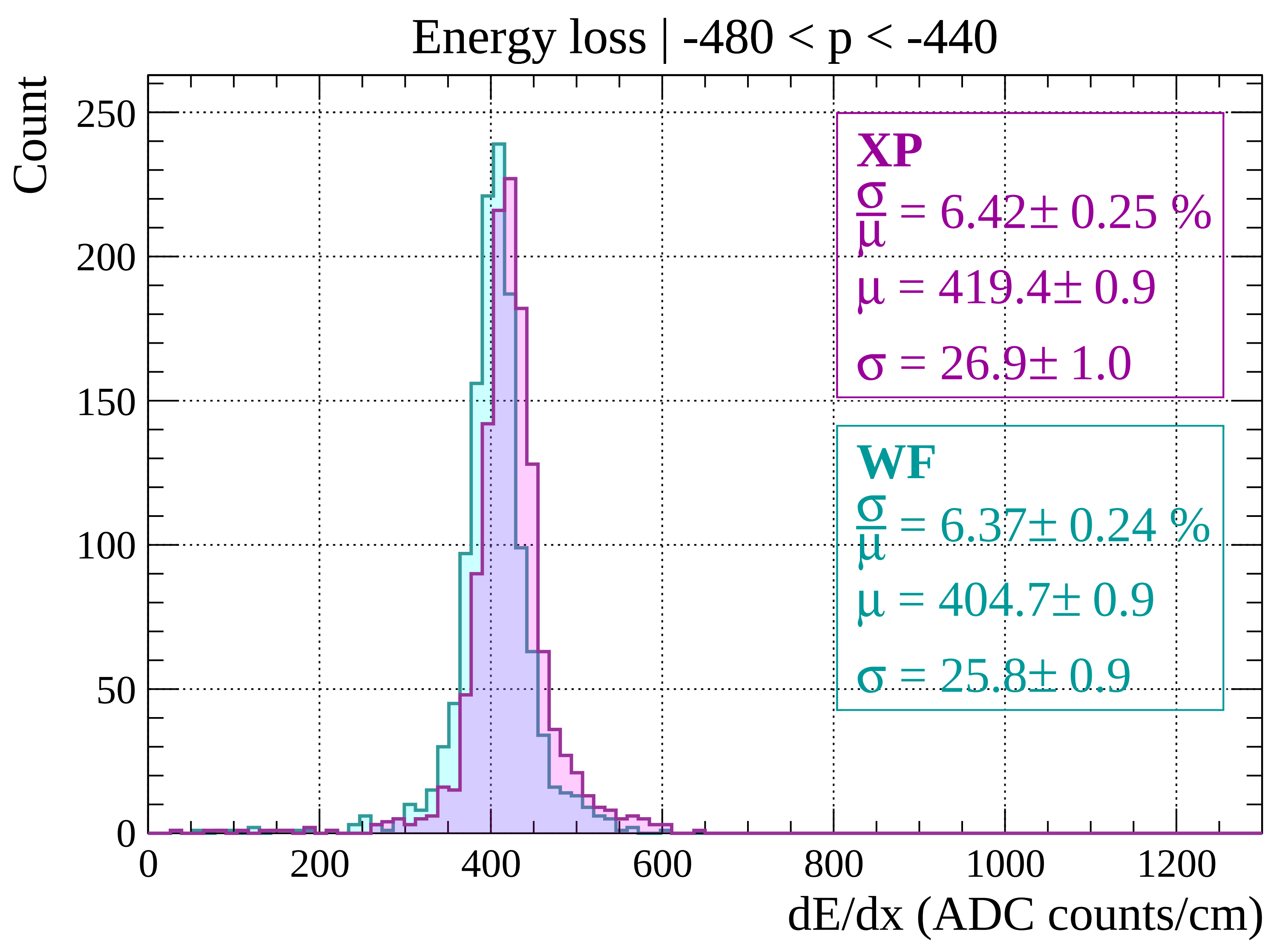

Energy loss | -640 < p < -600
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 6.13± 0.20 %
μ = 428.9± 0.8
σ = 26.3± 0.8
WF
$\frac{\sigma}{\mu}$ = 6.10± 0.19 %
μ = 413.8± 0.8
σ = 25.2± 0.8

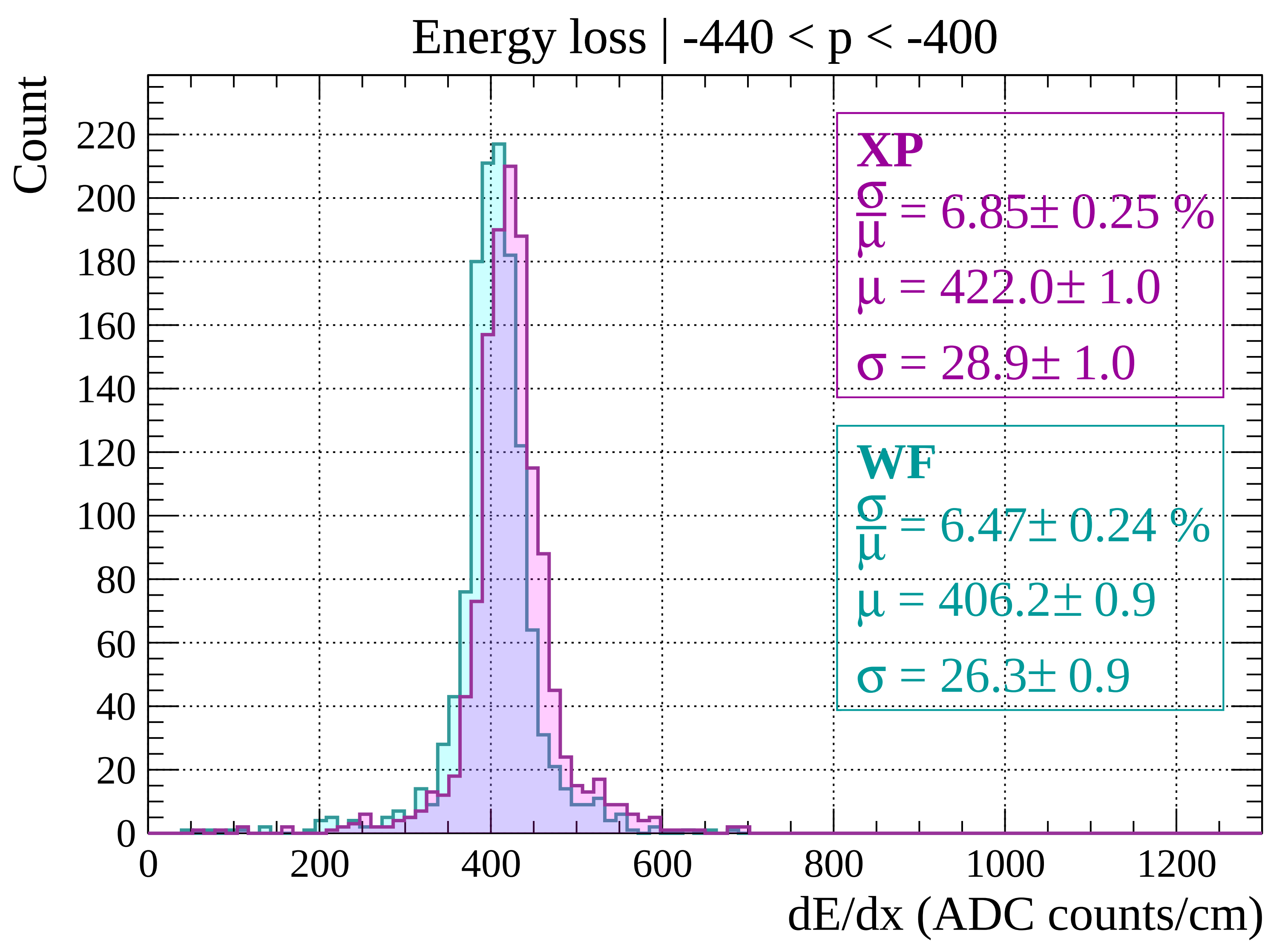

Energy loss | -600 < p < -560
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
XP
σ/μ = 5.99± 0.19 %
μ = 428.6± 0.8
σ = 25.7± 0.8
WF
σ/μ = 6.03± 0.18 %
μ = 412.7± 0.8
σ = 24.9± 0.7

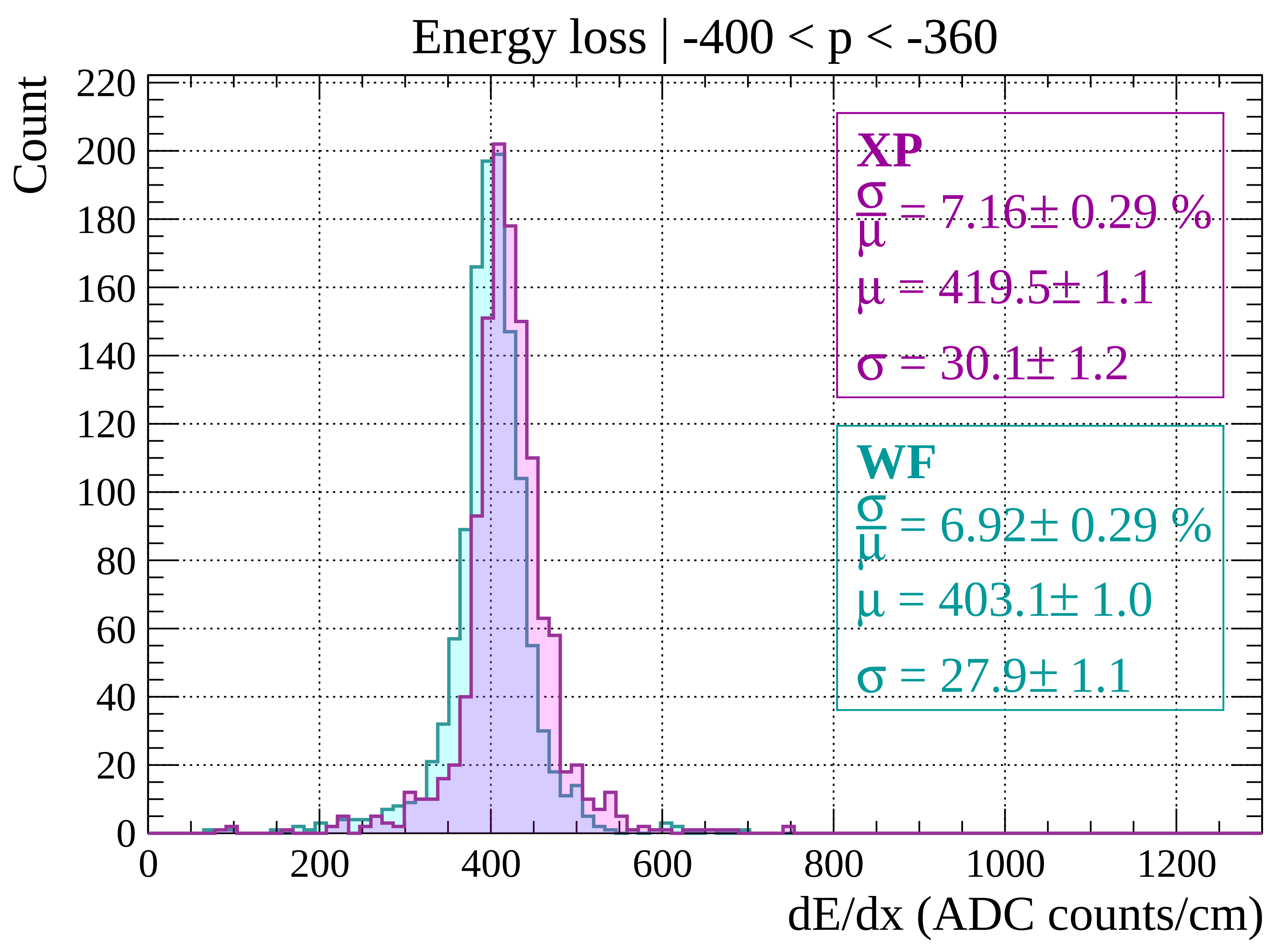


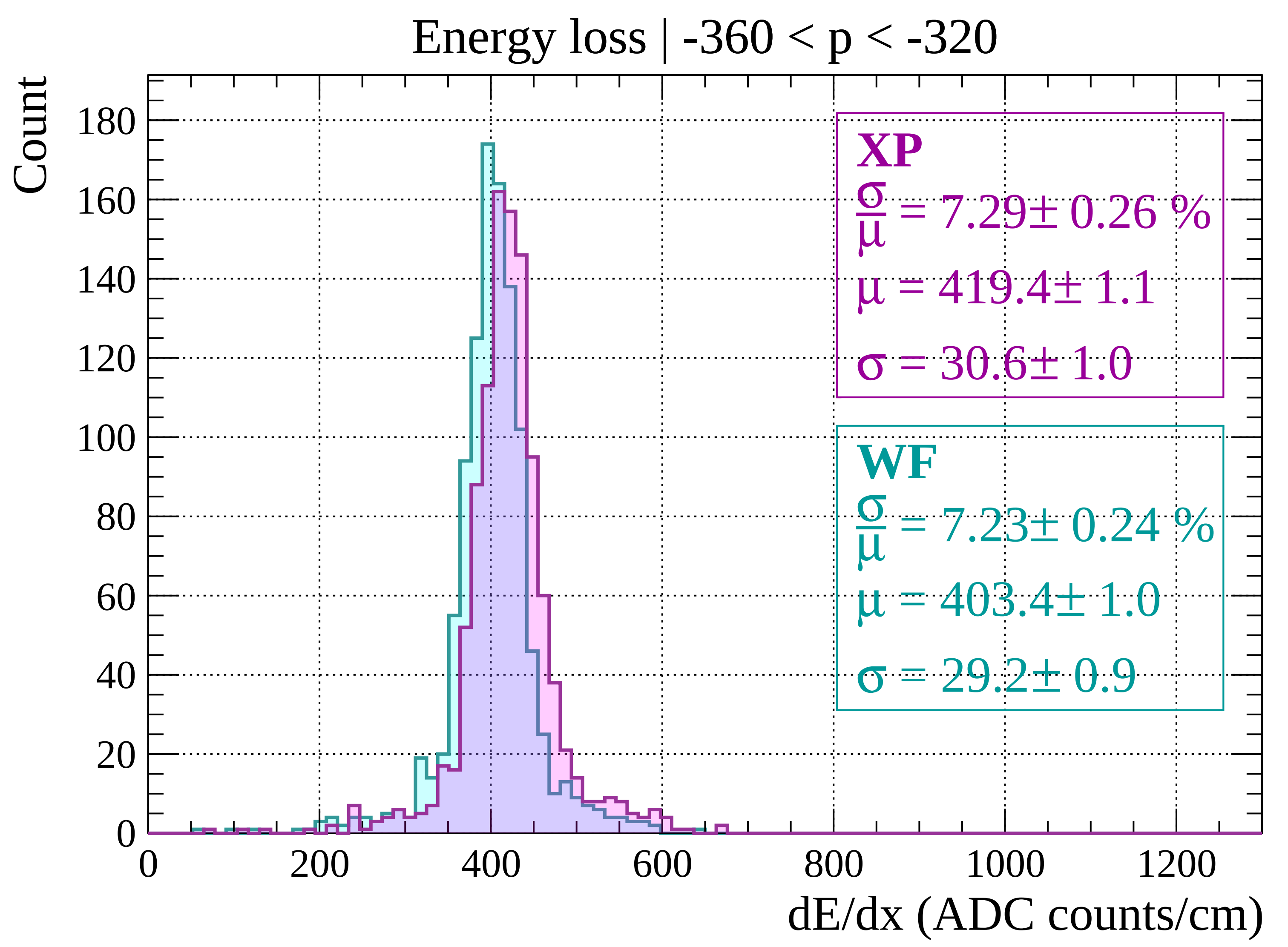
Energy loss | -560 < p < -520
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
σ/μ = 6.02± 0.20 %
μ = 424.7± 0.8
σ = 25.5± 0.8
WF
σ/μ = 5.86± 0.21 %
μ = 408.5± 0.8
σ = 23.9± 0.8

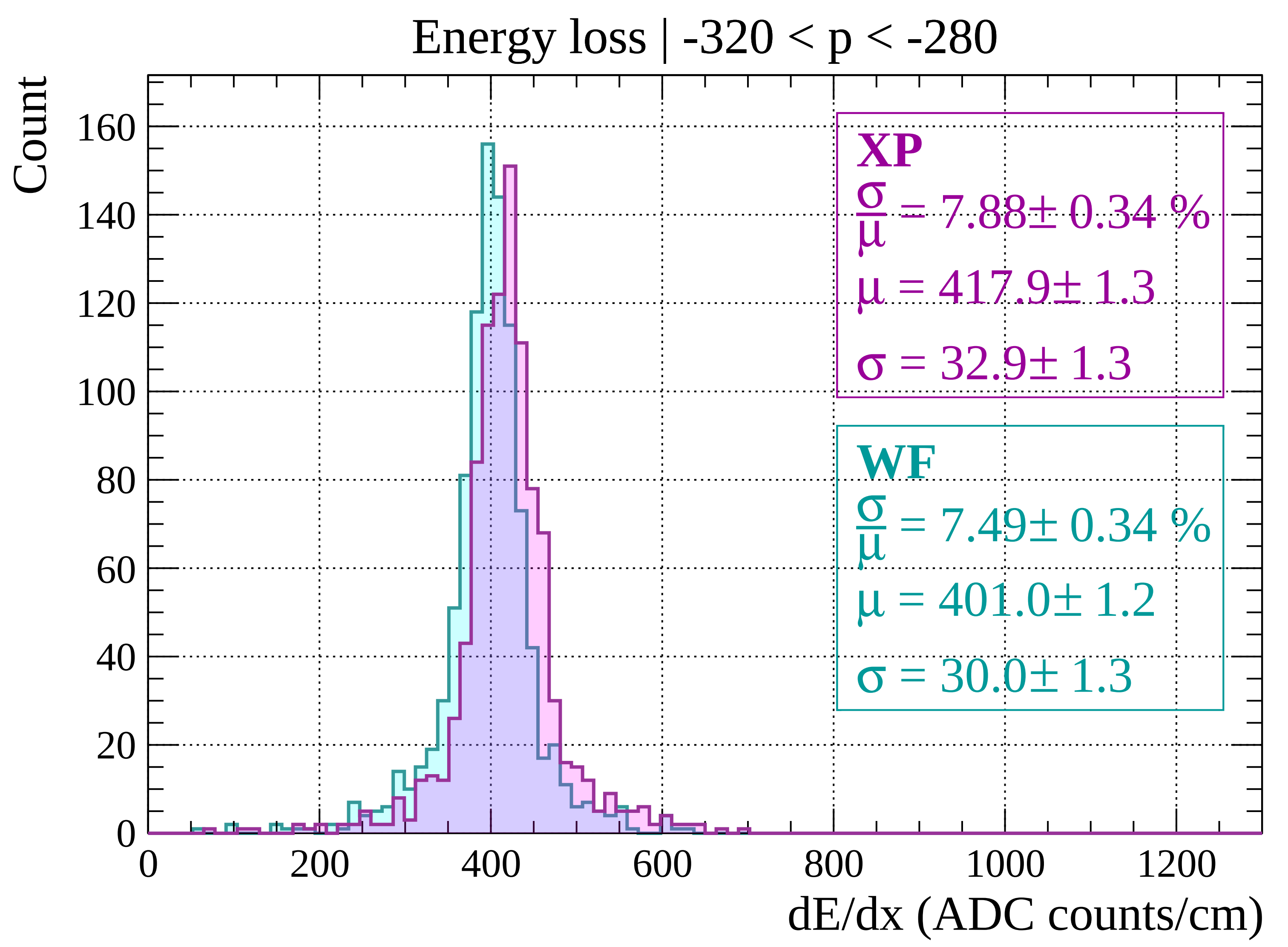

Energy loss | -520 < p < -480
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 6.73± 0.23 %
μ = 423.0± 0.9
σ = 28.5± 0.9
WF
σ/μ = 6.68± 0.25 %
μ = 408.0± 0.9
σ = 27.2± 1.0

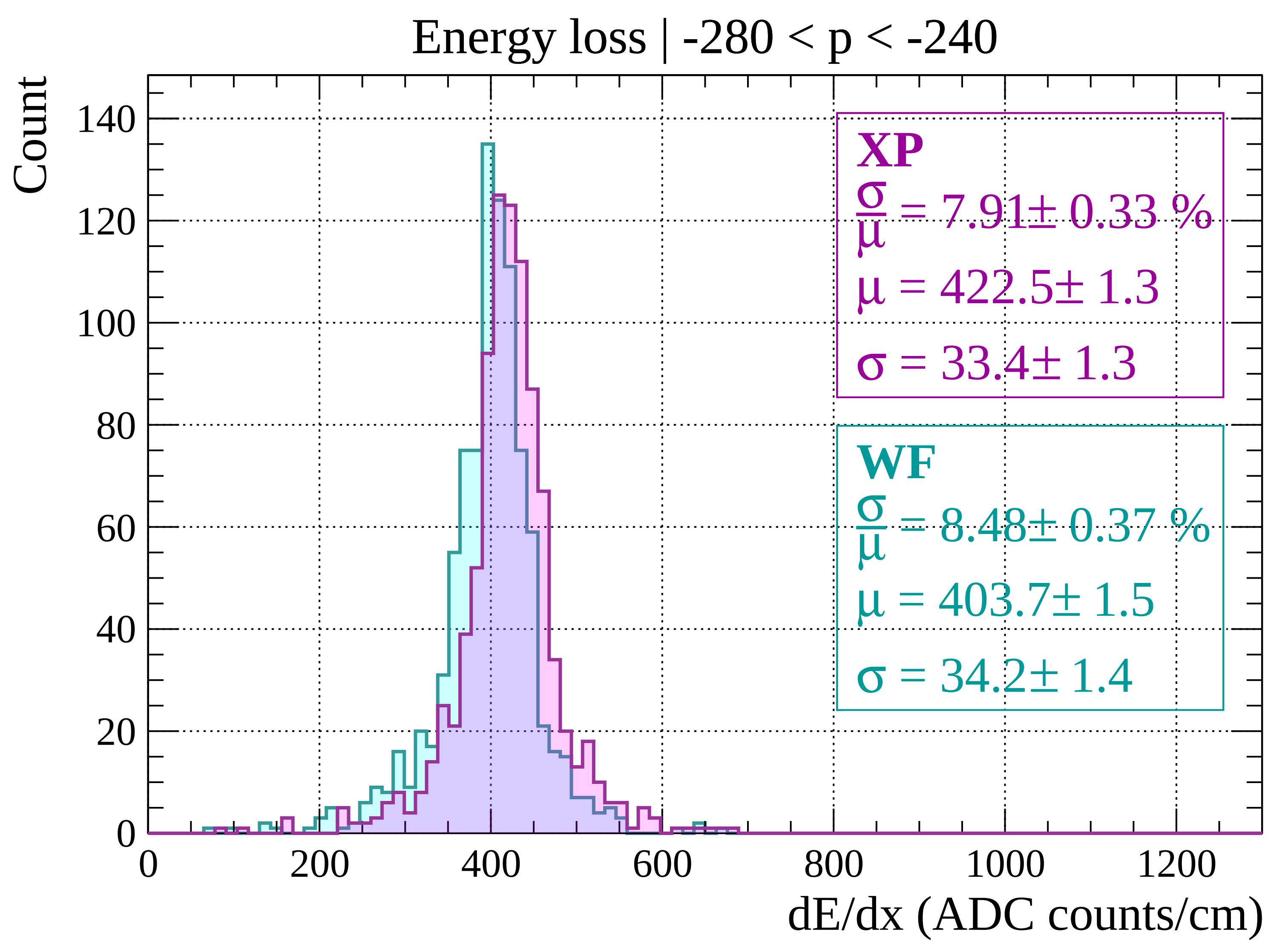

Energy loss | -480 < p < -440
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 6.42± 0.25 %
μ = 419.4± 0.9
σ = 26.9± 1.0
WF
σ/μ = 6.37± 0.24 %
μ = 404.7± 0.9
σ = 25.8± 0.9


Energy loss | -440 < p < -400
Count
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 6.85± 0.25 %
μ = 422.0± 1.0
σ = 28.9± 1.0
WF
σ/μ = 6.47± 0.24 %
μ = 406.2± 0.9
σ = 26.3± 0.9

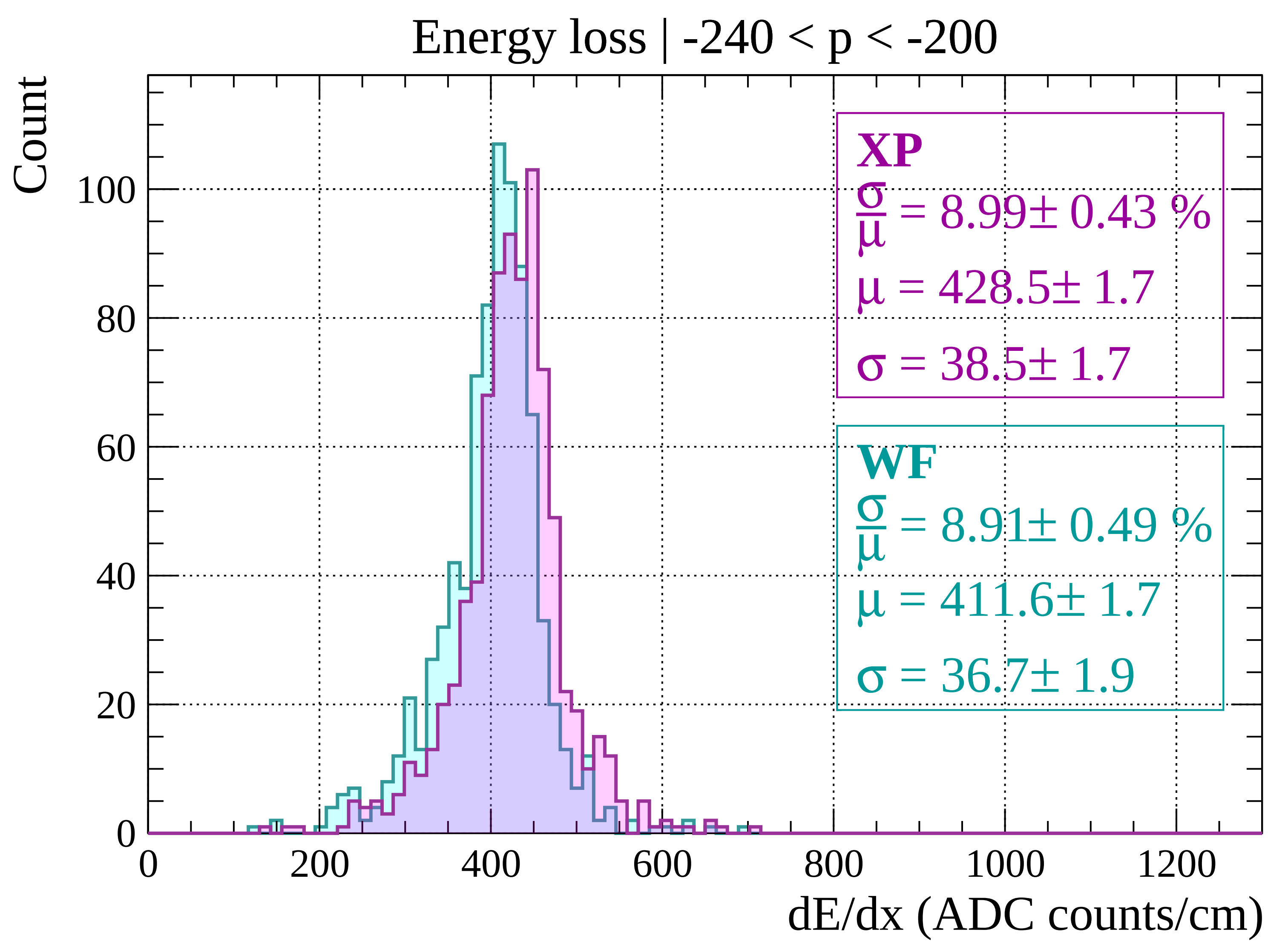

Energy loss | -400 < p < -360
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
220
XP
σ/μ = 7.16± 0.29 %
μ = 419.5± 1.1
σ = 30.1± 1.2
WF
σ/μ = 6.92± 0.29 %
μ = 403.1± 1.0
σ = 27.9± 1.1


Energy loss | -360 < p < -320
Count
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.29± 0.26 %
μ = 419.4± 1.1
σ = 30.6± 1.0
WF
$\frac{\sigma}{\mu}$ = 7.23± 0.24 %
μ = 403.4± 1.0
σ = 29.2± 0.9

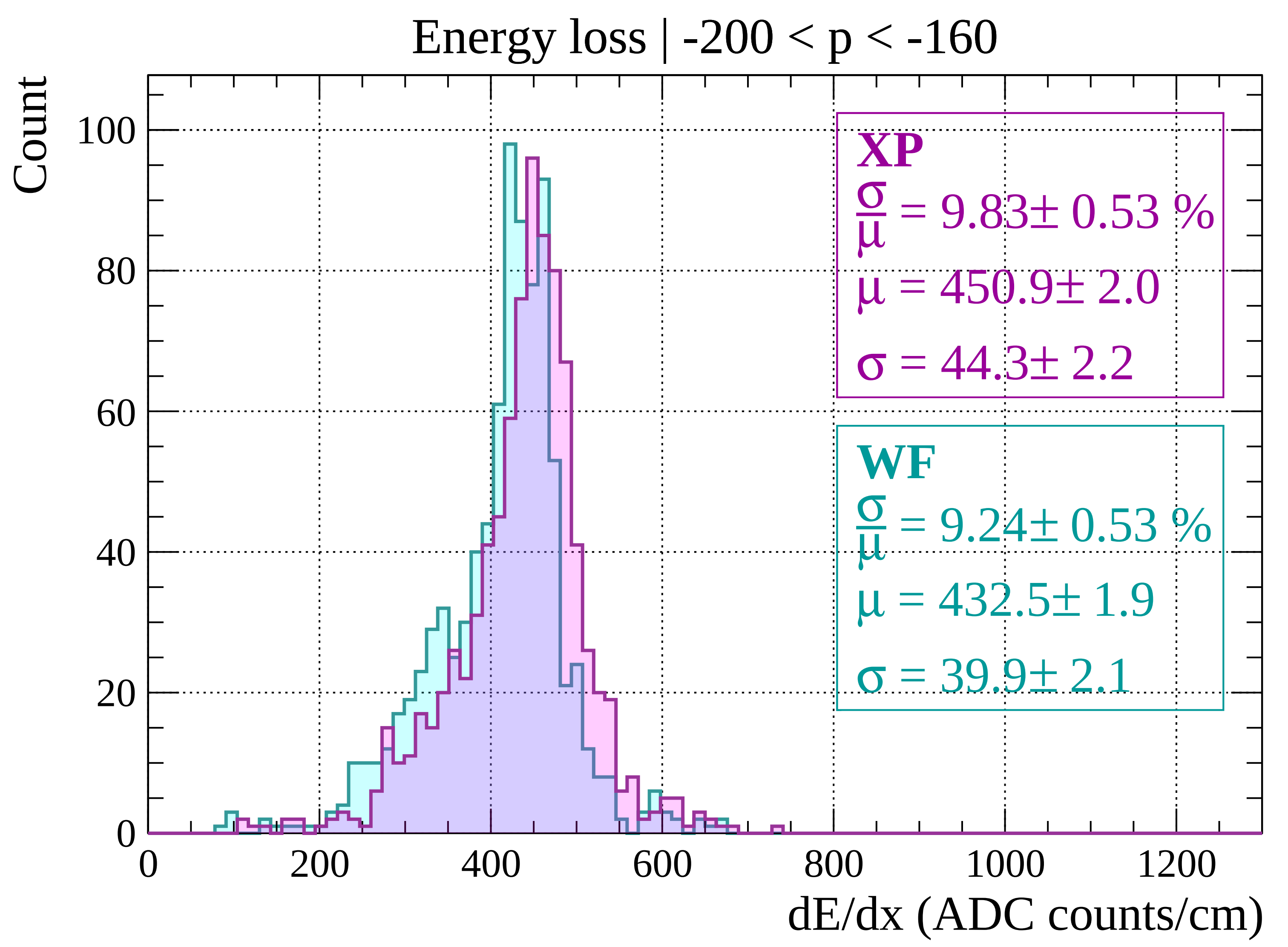

Energy loss | -320 < p < -280
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP
σ/μ = 7.88± 0.34 %
μ = 417.9± 1.3
σ = 32.9± 1.3
WF
σ/μ = 7.49± 0.34 %
μ = 401.0± 1.2
σ = 30.0± 1.3

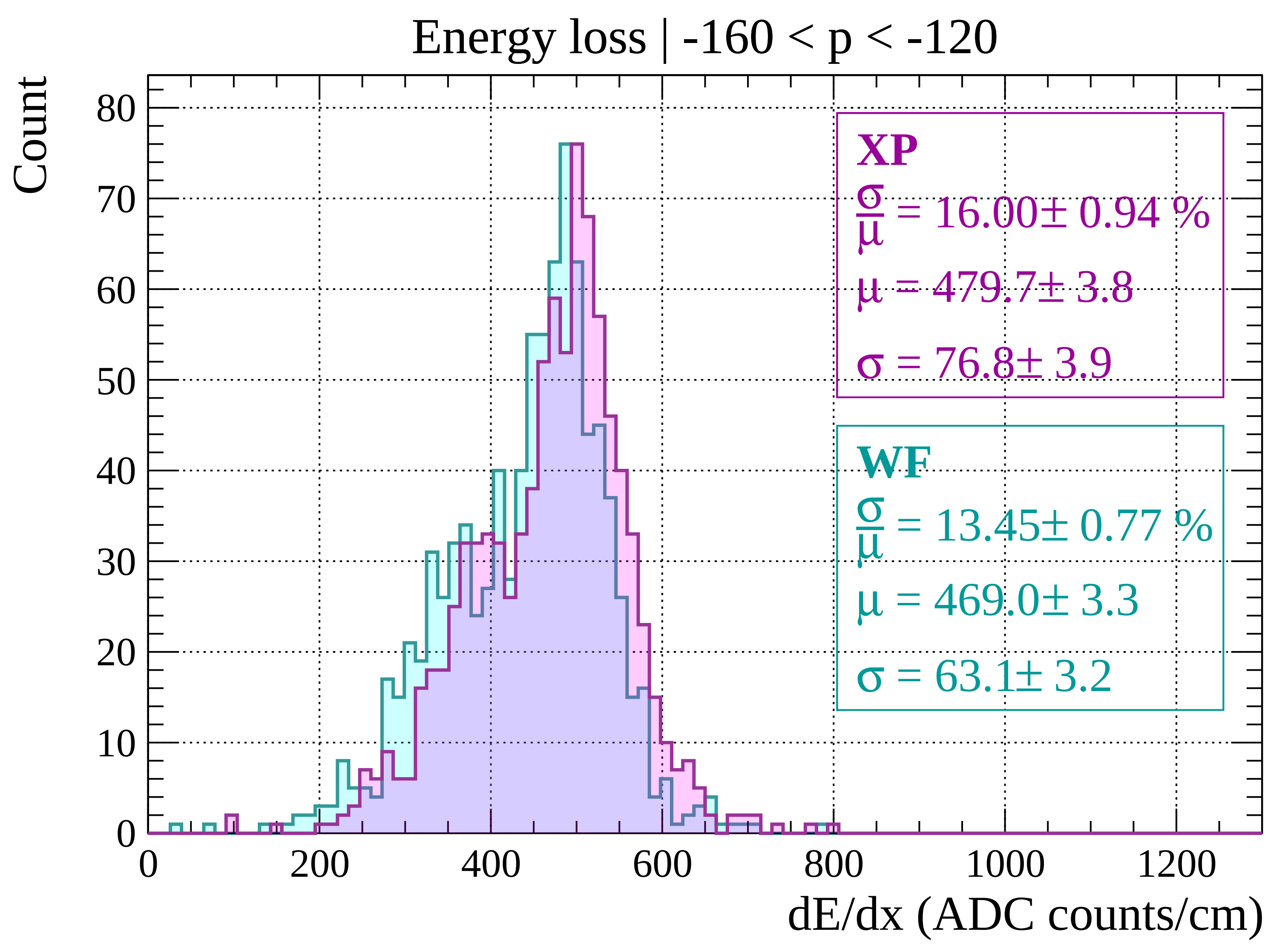


Energy loss | -280 < p < -240
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 7.91± 0.33 %
μ = 422.5± 1.3
σ = 33.4± 1.3
WF
$\frac{\sigma}{\mu}$ = 8.48± 0.37 %
μ = 403.7± 1.5
σ = 34.2± 1.4

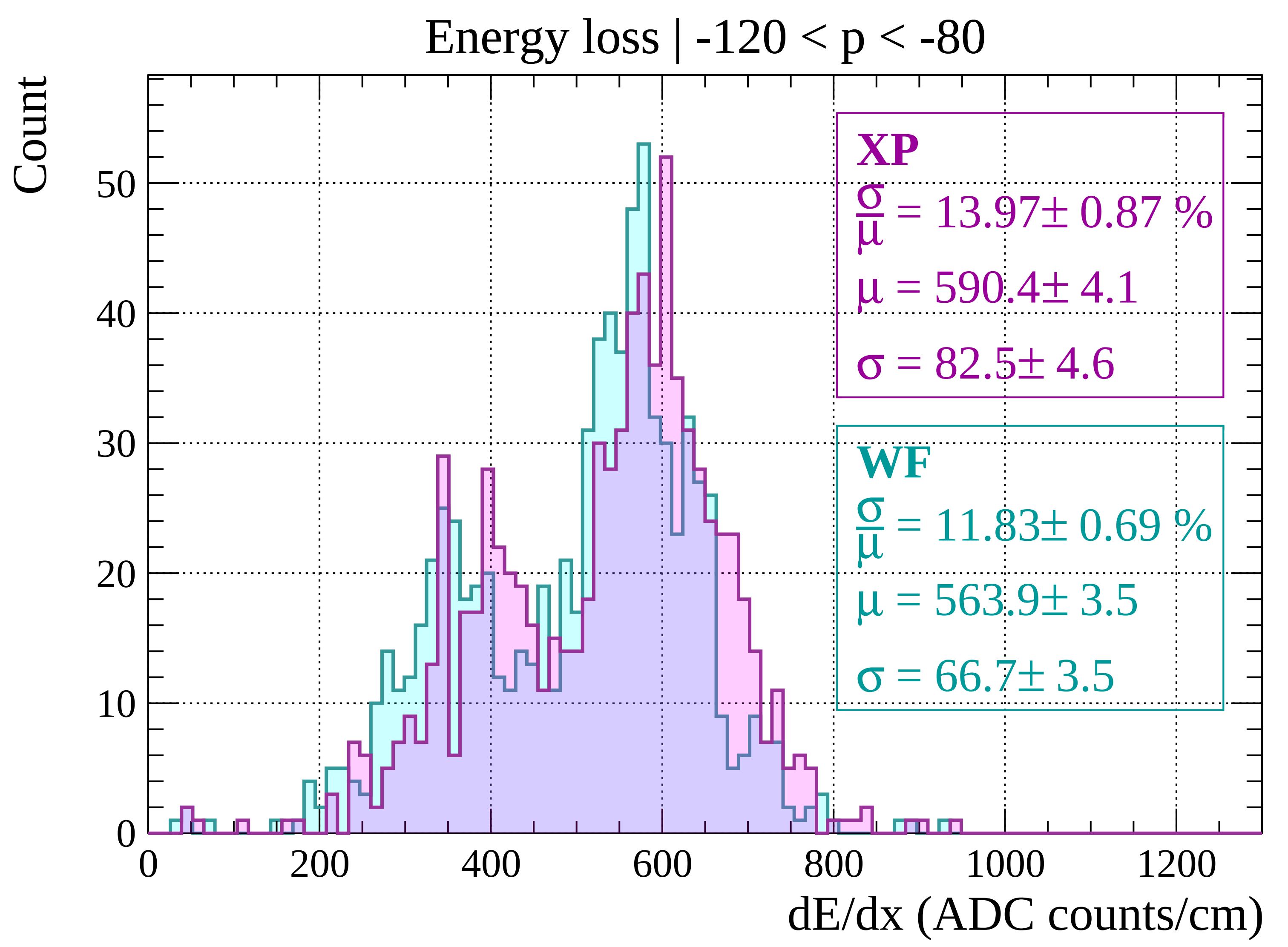


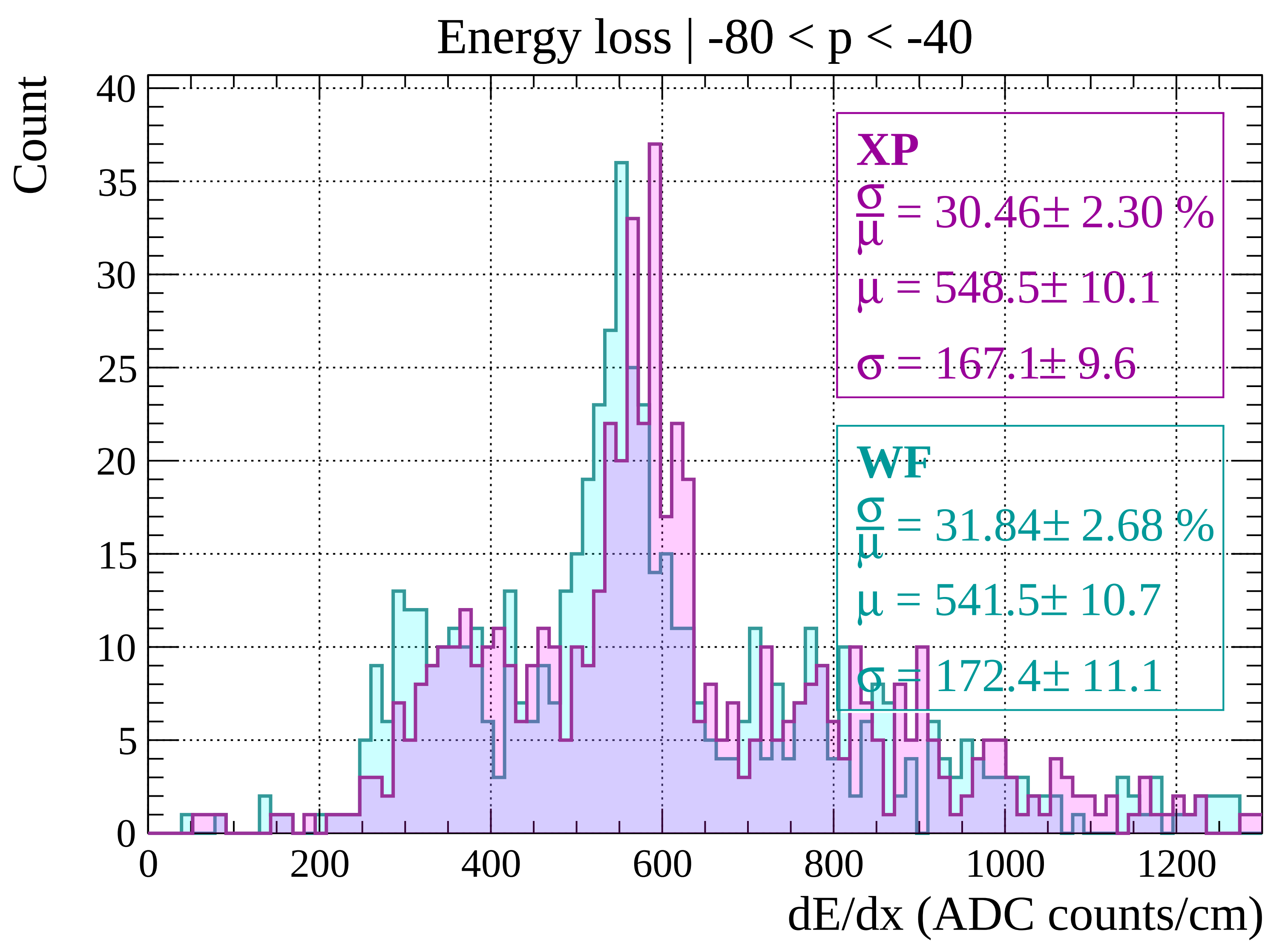
Energy loss | -240 < p < -200
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
XP
σ/μ = 8.99± 0.43 %
μ = 428.5± 1.7
σ = 38.5± 1.7
WF
σ/μ = 8.91± 0.49 %
μ = 411.6± 1.7
σ = 36.7± 1.9

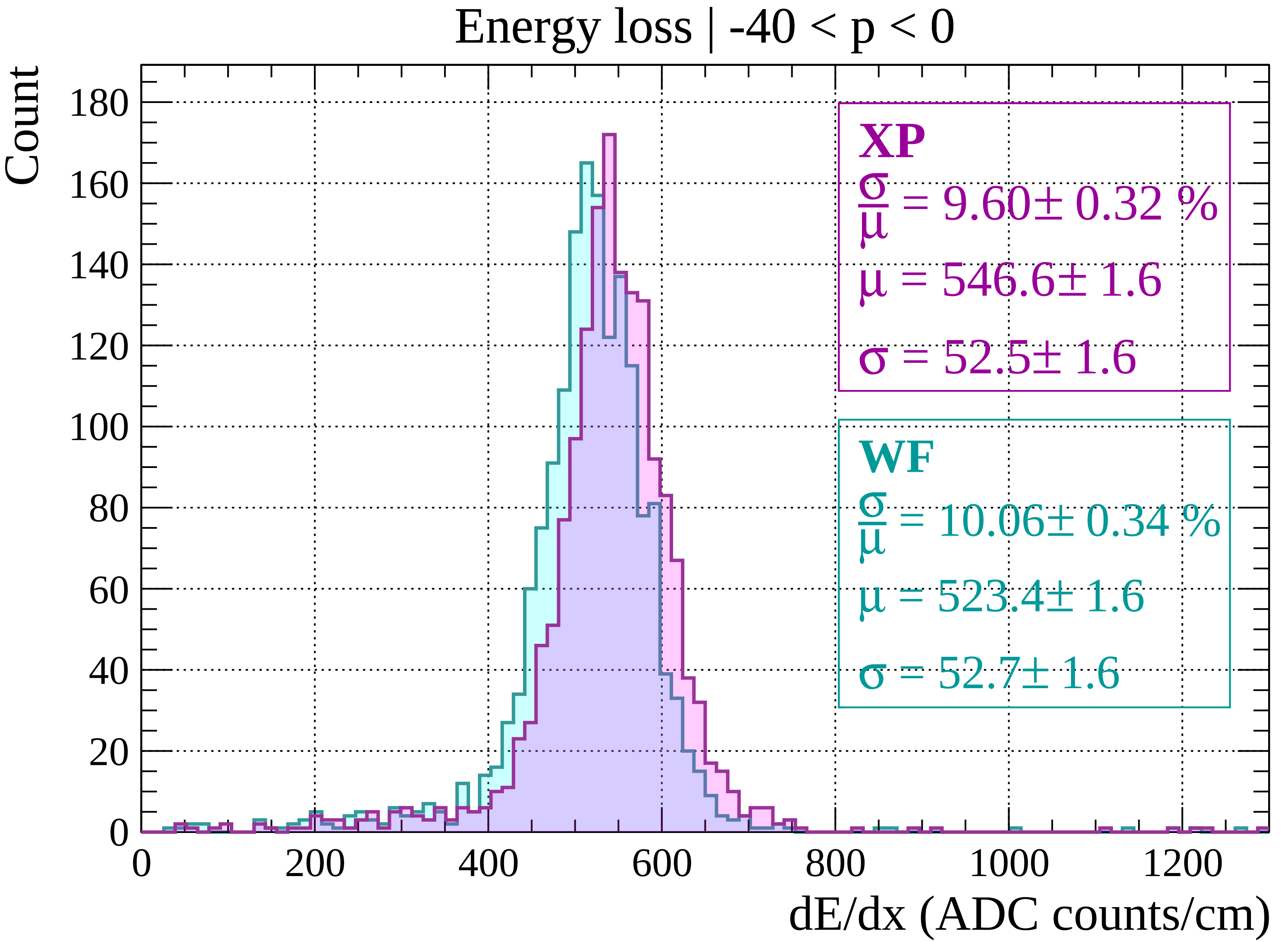

Energy loss | -200 < p < -160
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.83± 0.53 %
μ = 450.9± 2.0
σ = 44.3± 2.2
WF
$\frac{\sigma}{\mu}$ = 9.24± 0.53 %
μ = 432.5± 1.9
σ = 39.9± 2.1

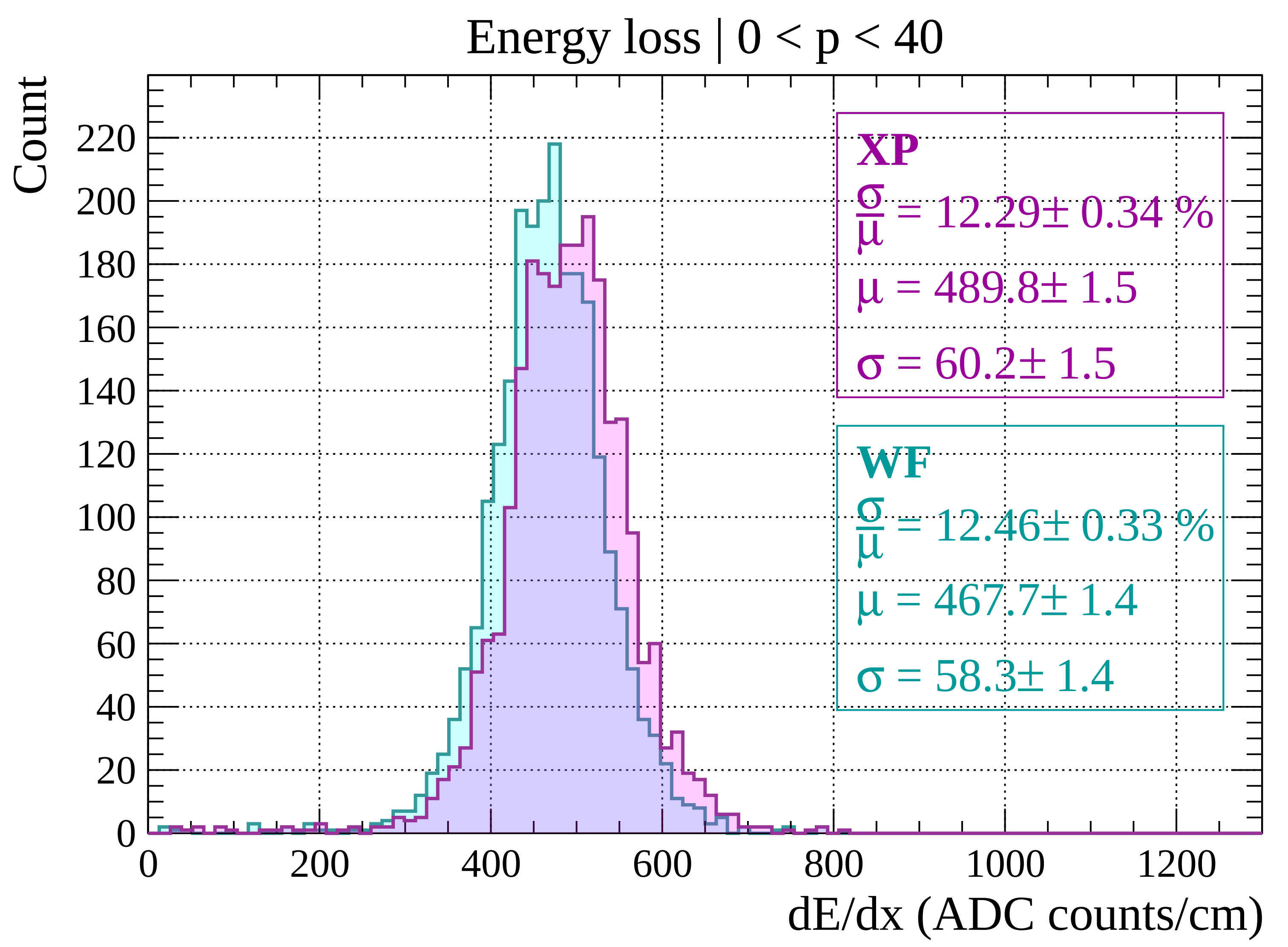


Energy loss | -160 < p < -120
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP
σ/μ = 16.00± 0.94 %
μ = 479.7± 3.8
σ = 76.8± 3.9
WF
σ/μ = 13.45± 0.77 %
μ = 469.0± 3.3
σ = 63.1± 3.2

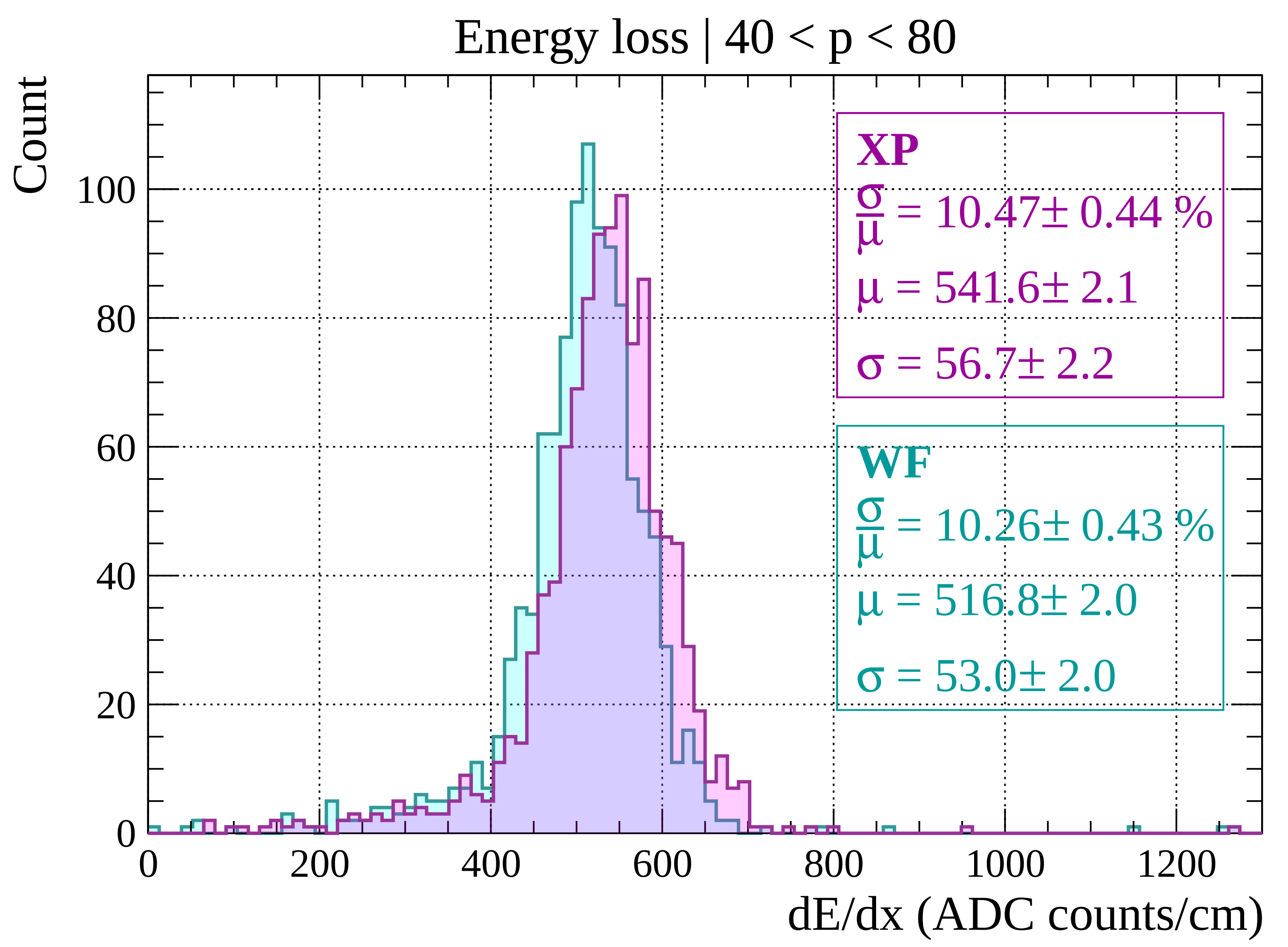

Energy loss | -120 < p < -80
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 13.97± 0.87 %
μ = 590.4± 4.1
σ = 82.5± 4.6
WF
σ/μ = 11.83± 0.69 %
μ = 563.9± 3.5
σ = 66.7± 3.5


Energy loss | -80 < p < -40
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 30.46± 2.30 %
μ = 548.5± 10.1
σ = 167.1± 9.6
WF
$\frac{\sigma}{\mu}$ = 31.84± 2.68 %
μ = 541.5± 10.7
σ = 172.4± 11.1
0
5
10
15
20
25
30
35
40
0
200
400
600
800
1000
1200

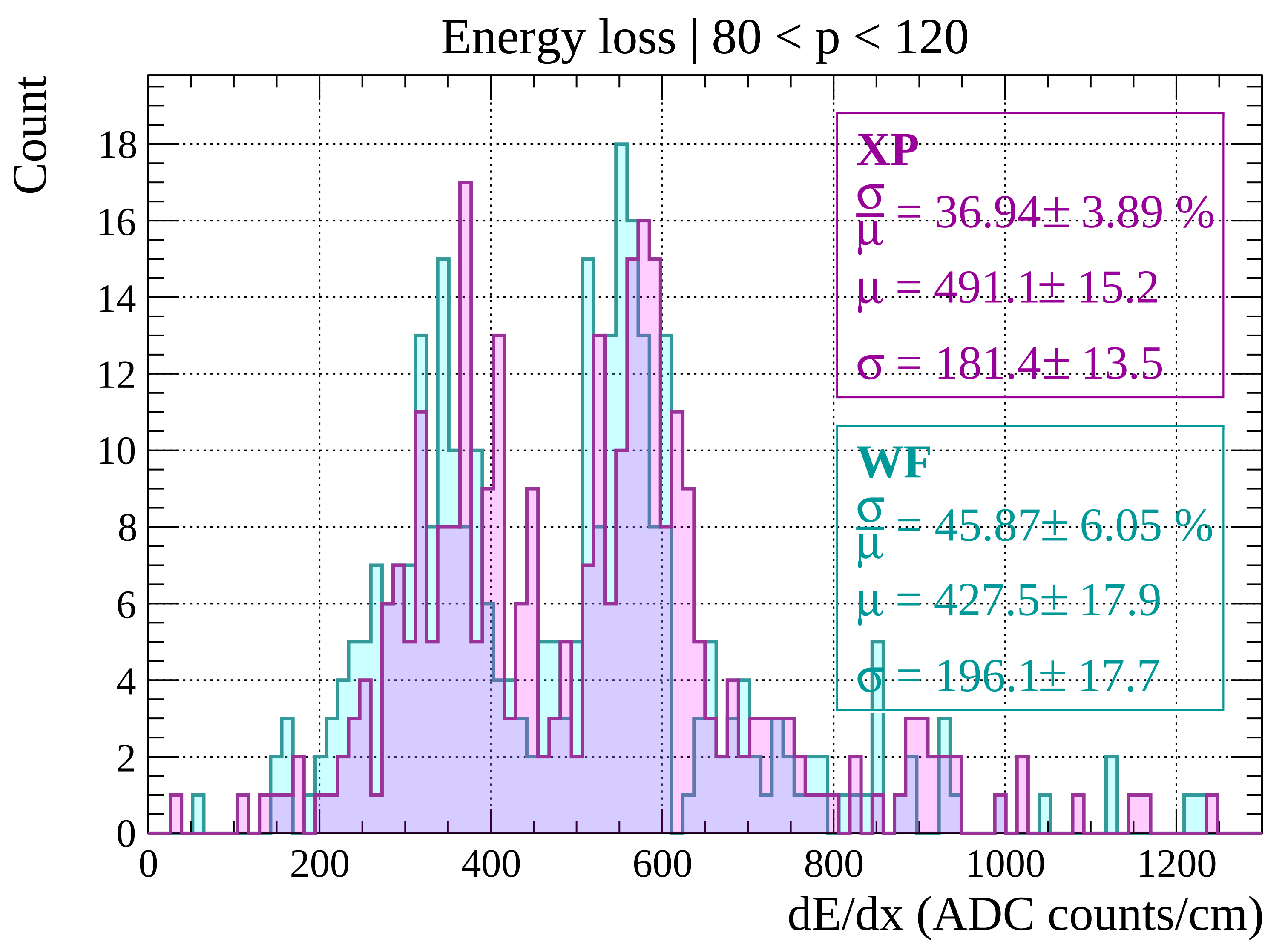

Energy loss | -40 < p < 0
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 9.60 ± 0.32 %
μ = 546.6 ± 1.6
σ = 52.5 ± 1.6
WF
σ/μ = 10.06 ± 0.34 %
μ = 523.4 ± 1.6
σ = 52.7 ± 1.6
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200

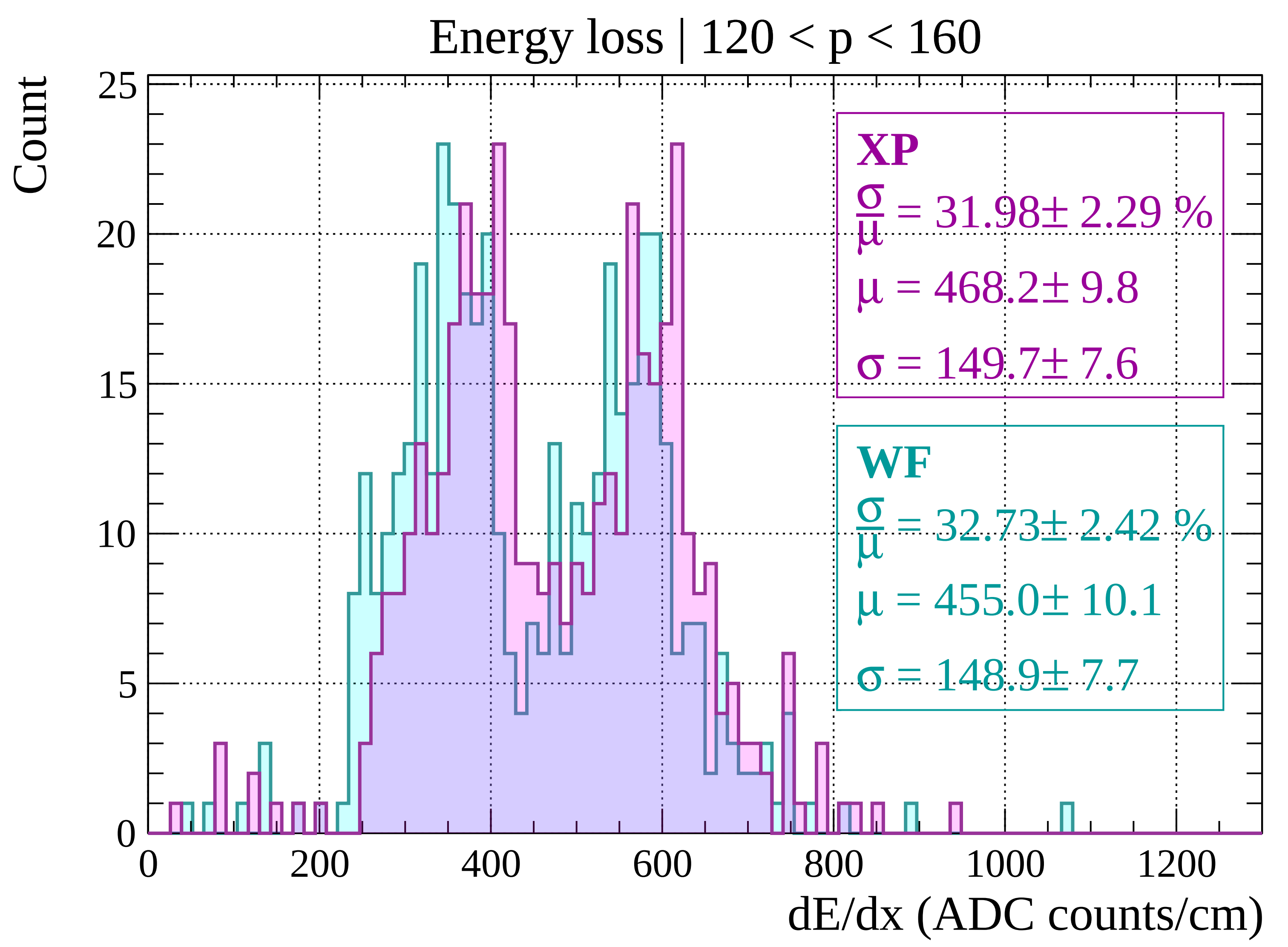


Energy loss | 0 < p < 40
Count
XP
$\frac{\sigma}{\mu}$ = 12.29± 0.34 %
μ = 489.8± 1.5
σ = 60.2± 1.5
WF
$\frac{\sigma}{\mu}$ = 12.46± 0.33 %
μ = 467.7± 1.4
σ = 58.3± 1.4
dE/dx (ADC counts/cm)

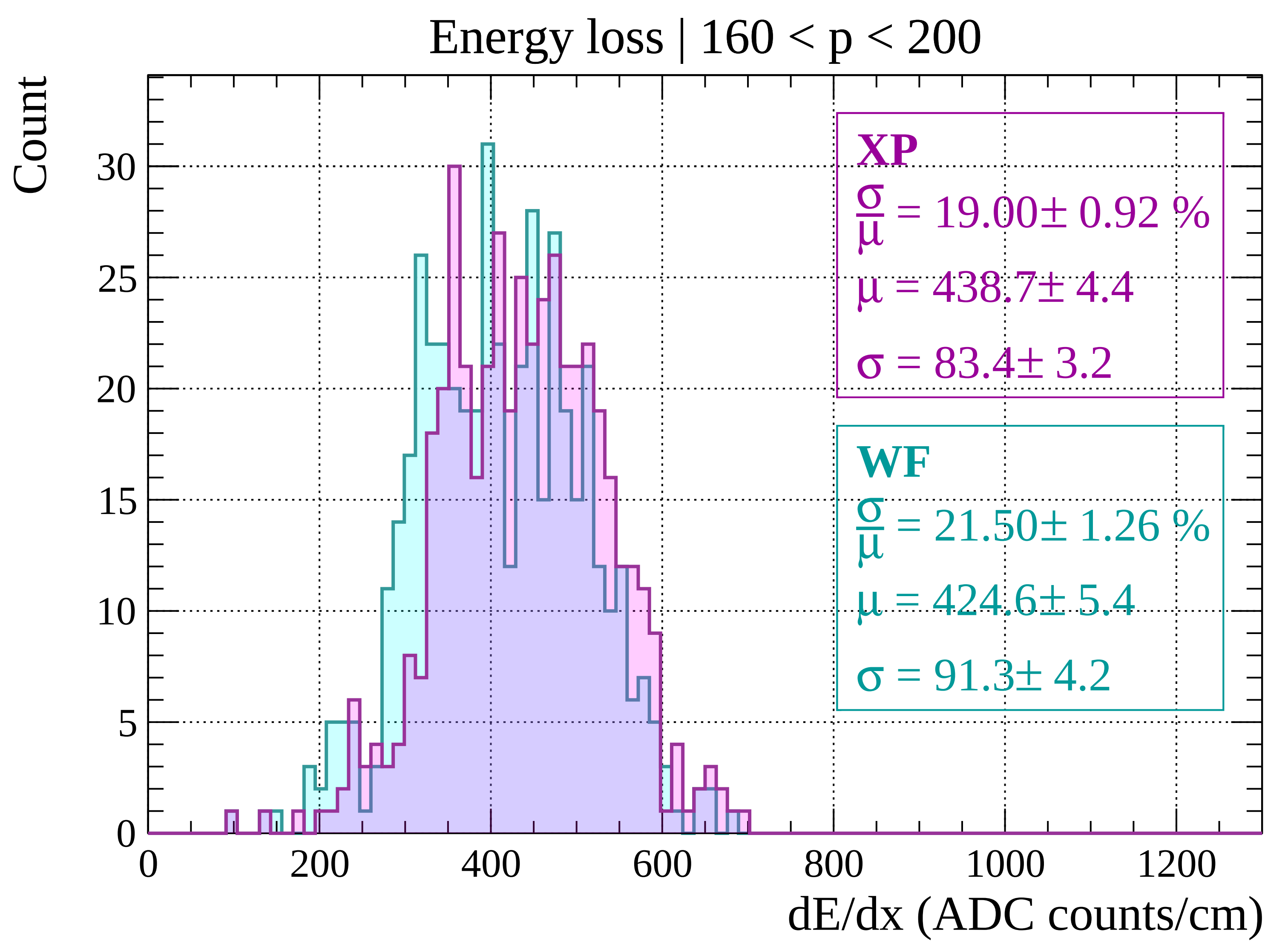


Energy loss | 40 < p < 80
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 10.47± 0.44 %
μ = 541.6± 2.1
σ = 56.7± 2.2
WF
$\frac{\sigma}{\mu}$ = 10.26± 0.43 %
μ = 516.8± 2.0
σ = 53.0± 2.0

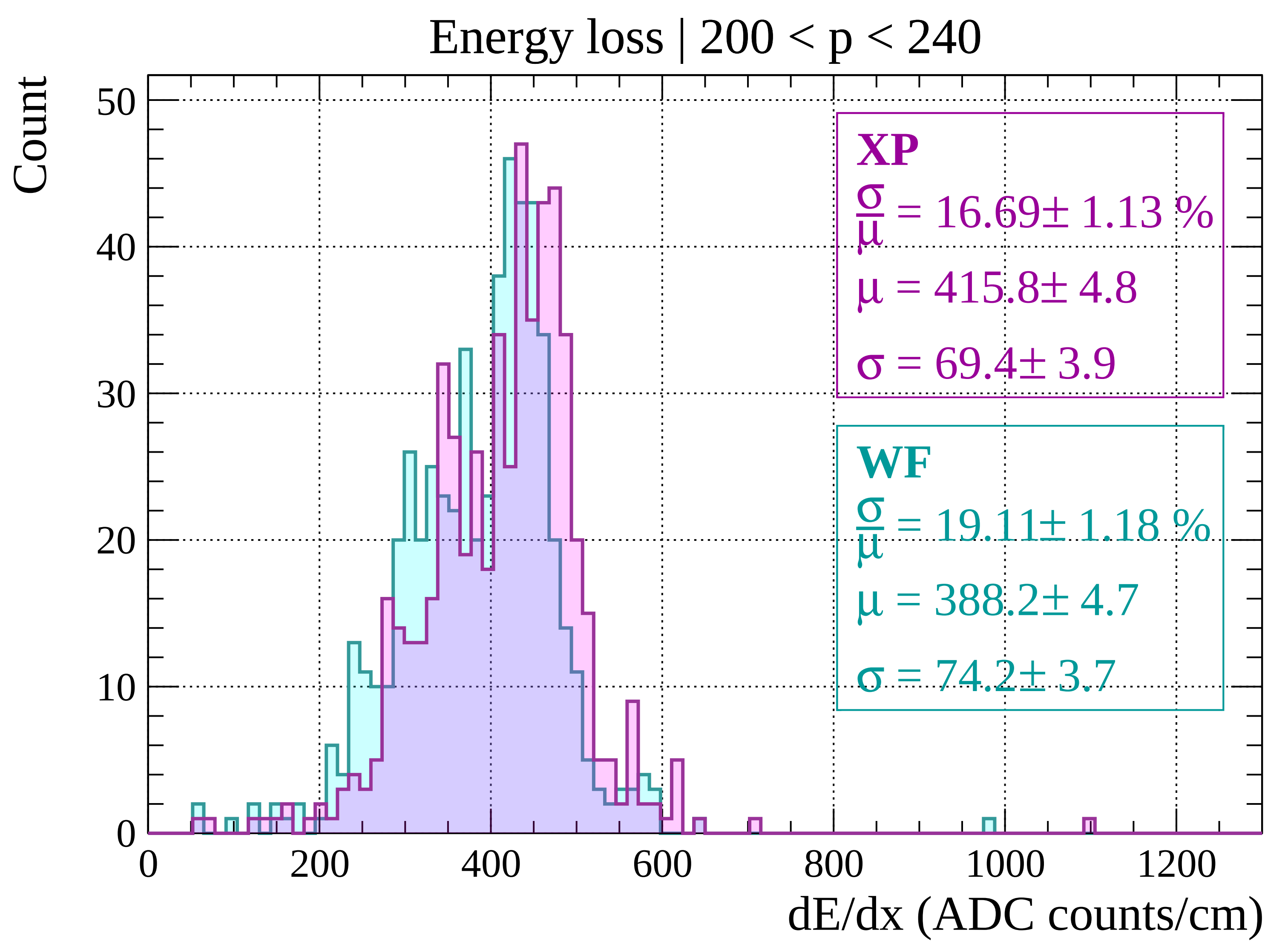

Energy loss | 80 < p < 120
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 36.94± 3.89 %
μ = 491.1± 15.2
σ = 181.4± 13.5
WF
σ/μ = 45.87± 6.05 %
μ = 427.5± 17.9
σ = 196.1± 17.7
0
2
4
6
8
10
12
14
16
18
0
200
400
600
800
1000
1200



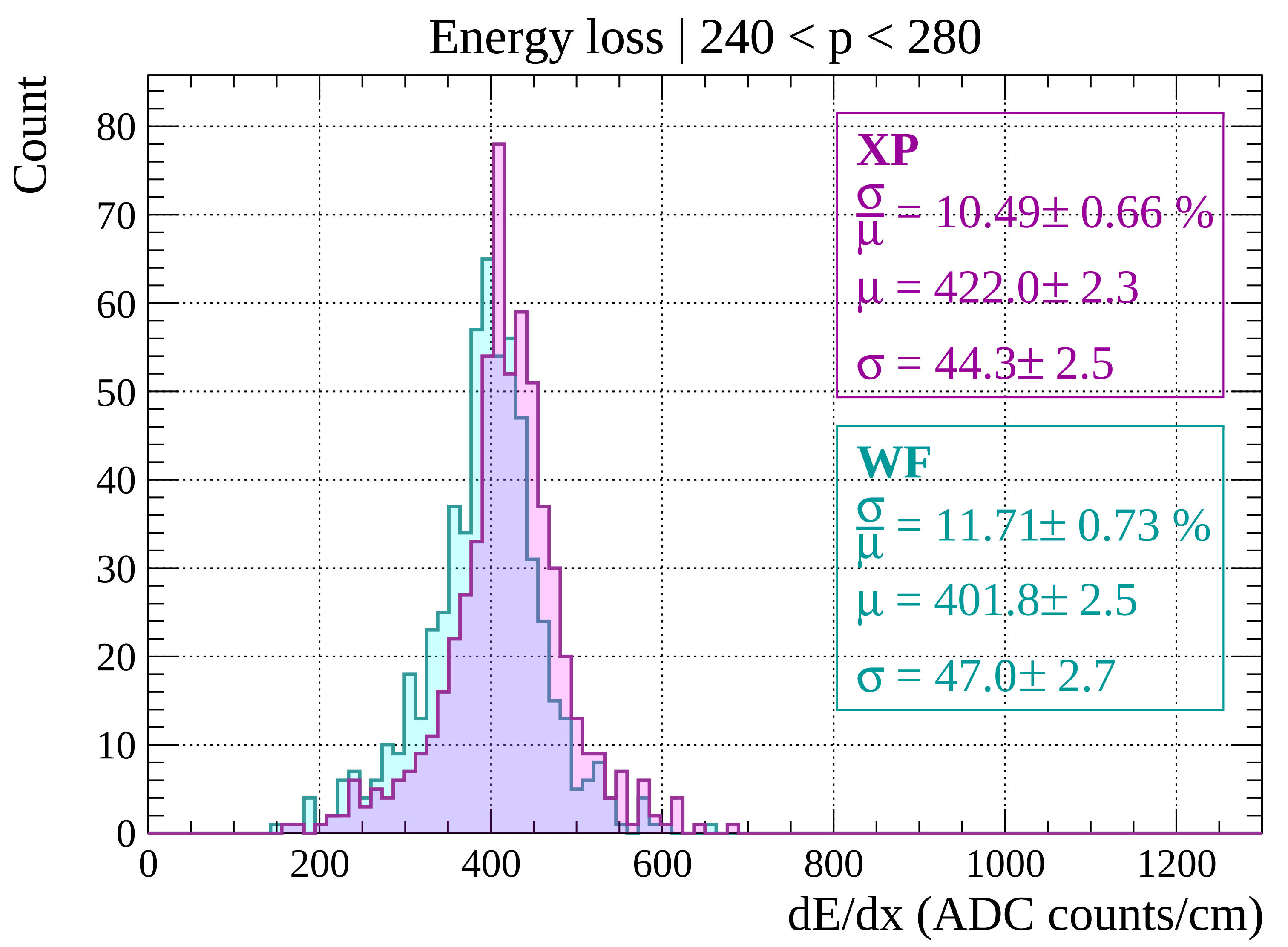
Energy loss | 120 < p < 160
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
XP
$\frac{\sigma}{\mu}$ = 31.98± 2.29 %
μ = 468.2± 9.8
σ = 149.7± 7.6
WF
$\frac{\sigma}{\mu}$ = 32.73± 2.42 %
μ = 455.0± 10.1
σ = 148.9± 7.7

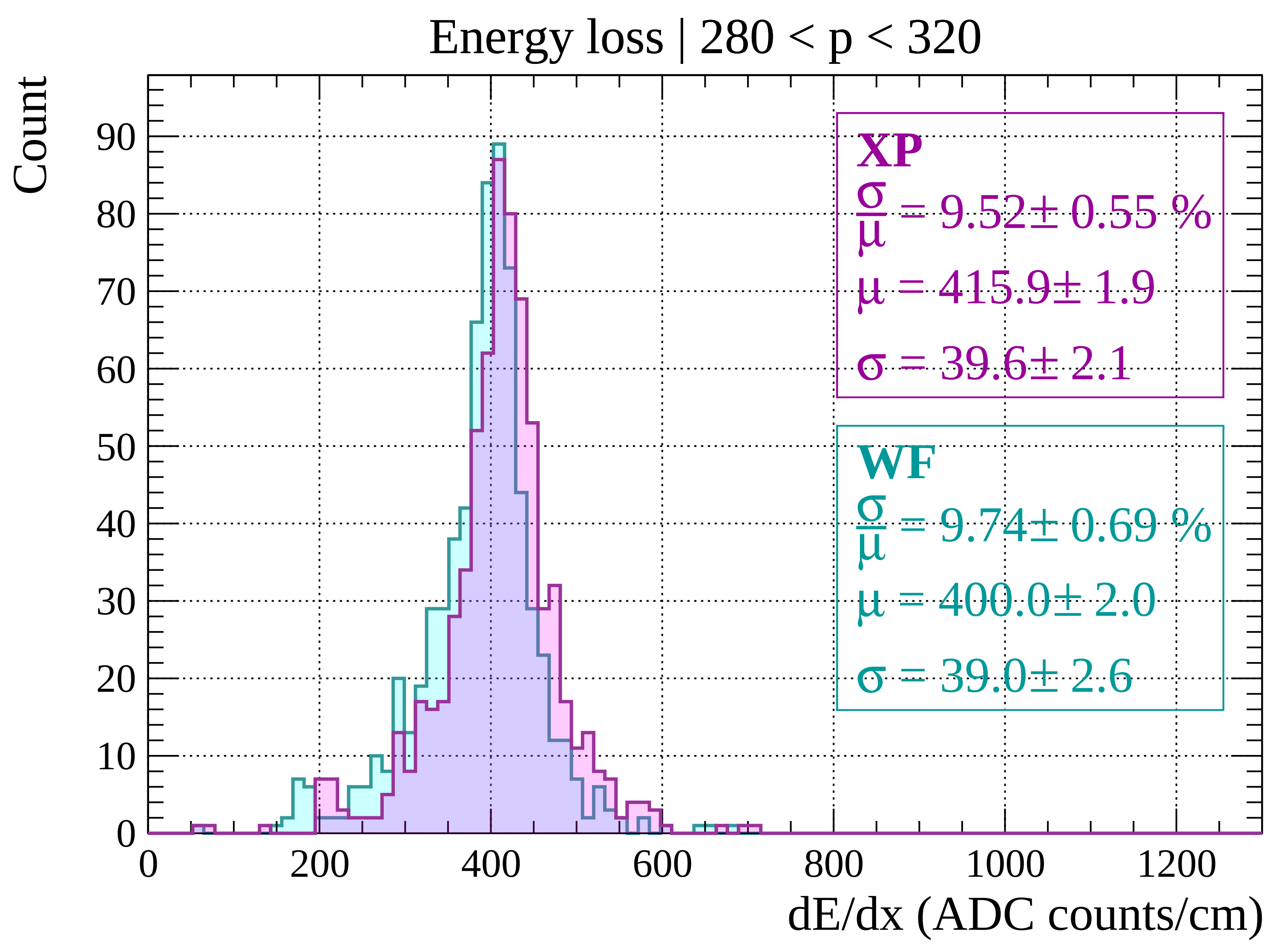


Energy loss | 160 < p < 200
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 19.00± 0.92 %
μ = 438.7± 4.4
σ = 83.4± 3.2
WF
σ/μ = 21.50± 1.26 %
μ = 424.6± 5.4
σ = 91.3± 4.2

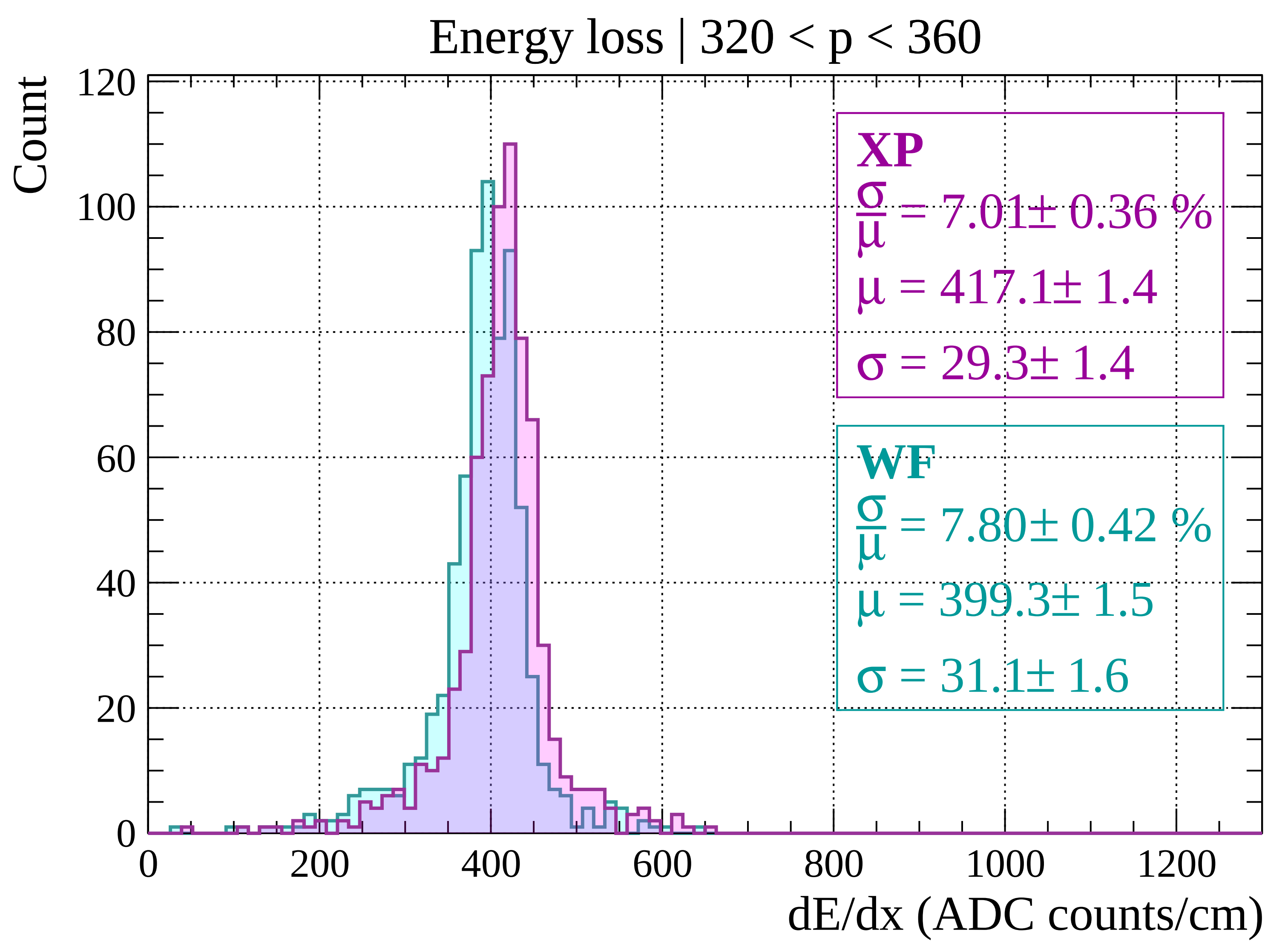


Energy loss | 200 < p < 240
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
σ/μ = 16.69± 1.13 %
μ = 415.8± 4.8
σ = 69.4± 3.9
WF
σ/μ = 19.11± 1.18 %
μ = 388.2± 4.7
σ = 74.2± 3.7

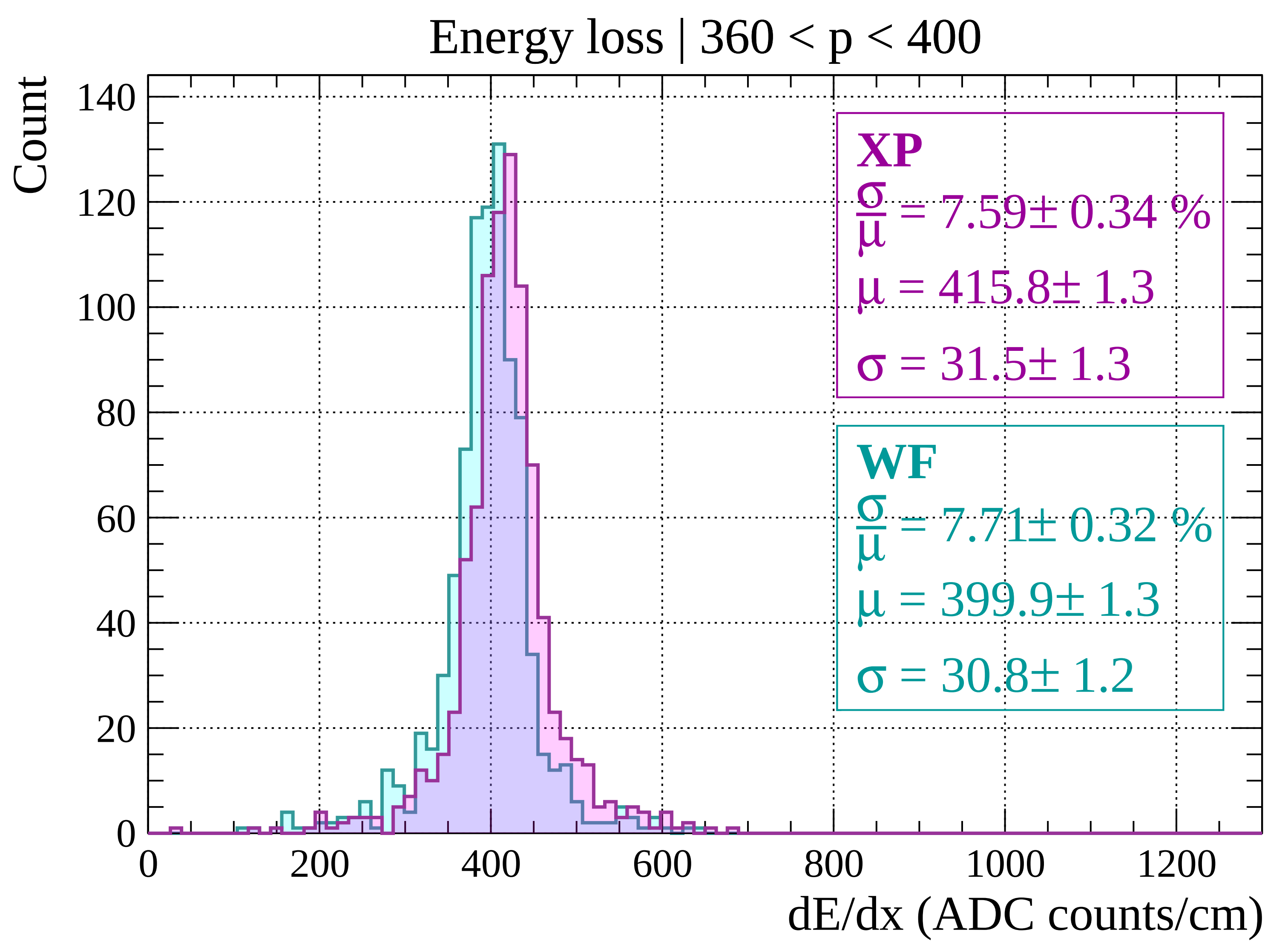

Energy loss | 240 < p < 280
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 10.49± 0.66 %
μ = 422.0± 2.3
σ = 44.3± 2.5
WF
$\frac{\sigma}{\mu}$ = 11.71± 0.73 %
μ = 401.8± 2.5
σ = 47.0± 2.7
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200



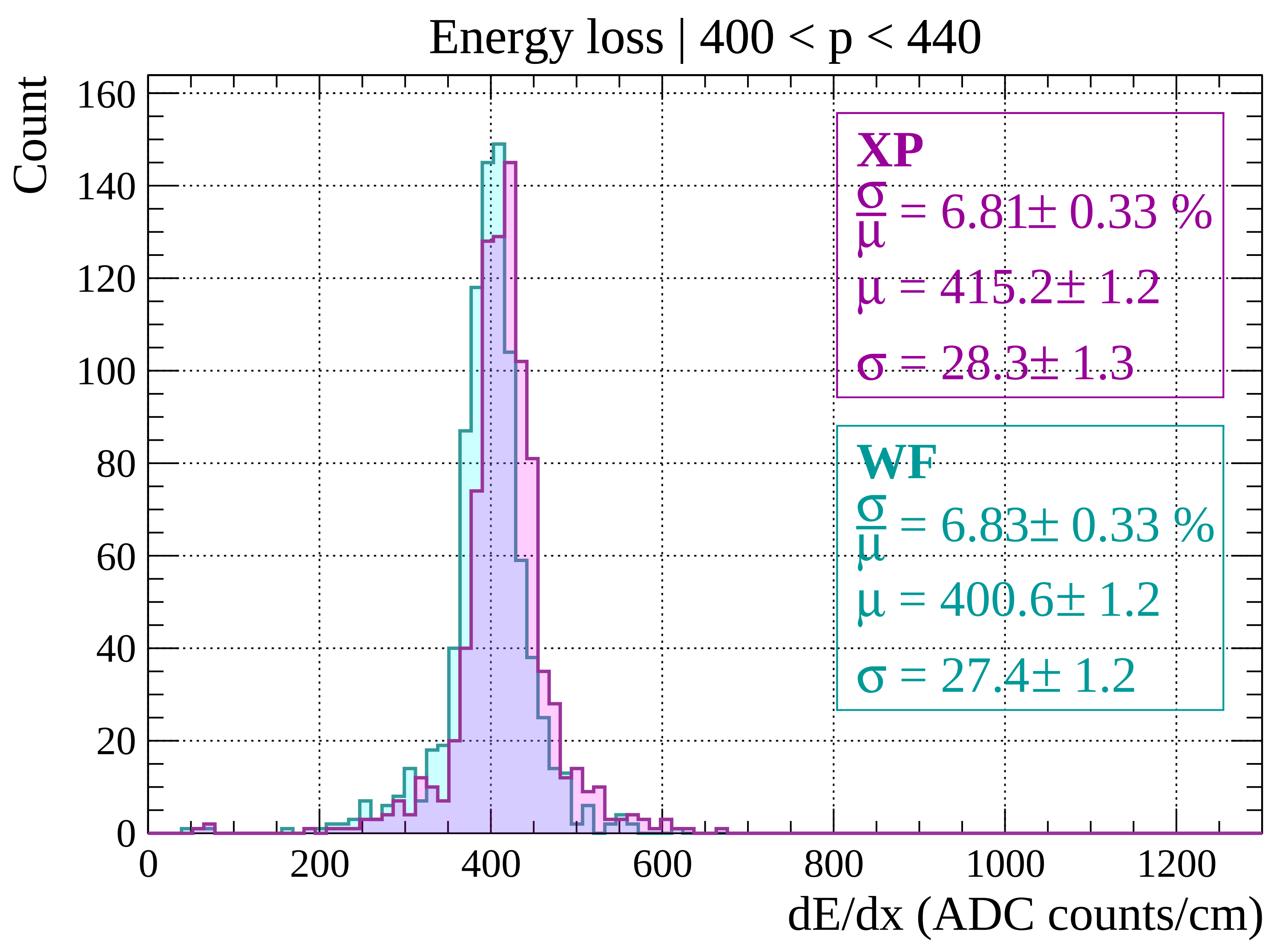
Energy loss | 280 < p < 320
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 9.52 \pm 0.55$ %
$\mu = 415.9 \pm 1.9$
$\sigma = 39.6 \pm 2.1$
WF
$\frac{\sigma}{\mu} = 9.74 \pm 0.69$ %
$\mu = 400.0 \pm 2.0$
$\sigma = 39.0 \pm 2.6$



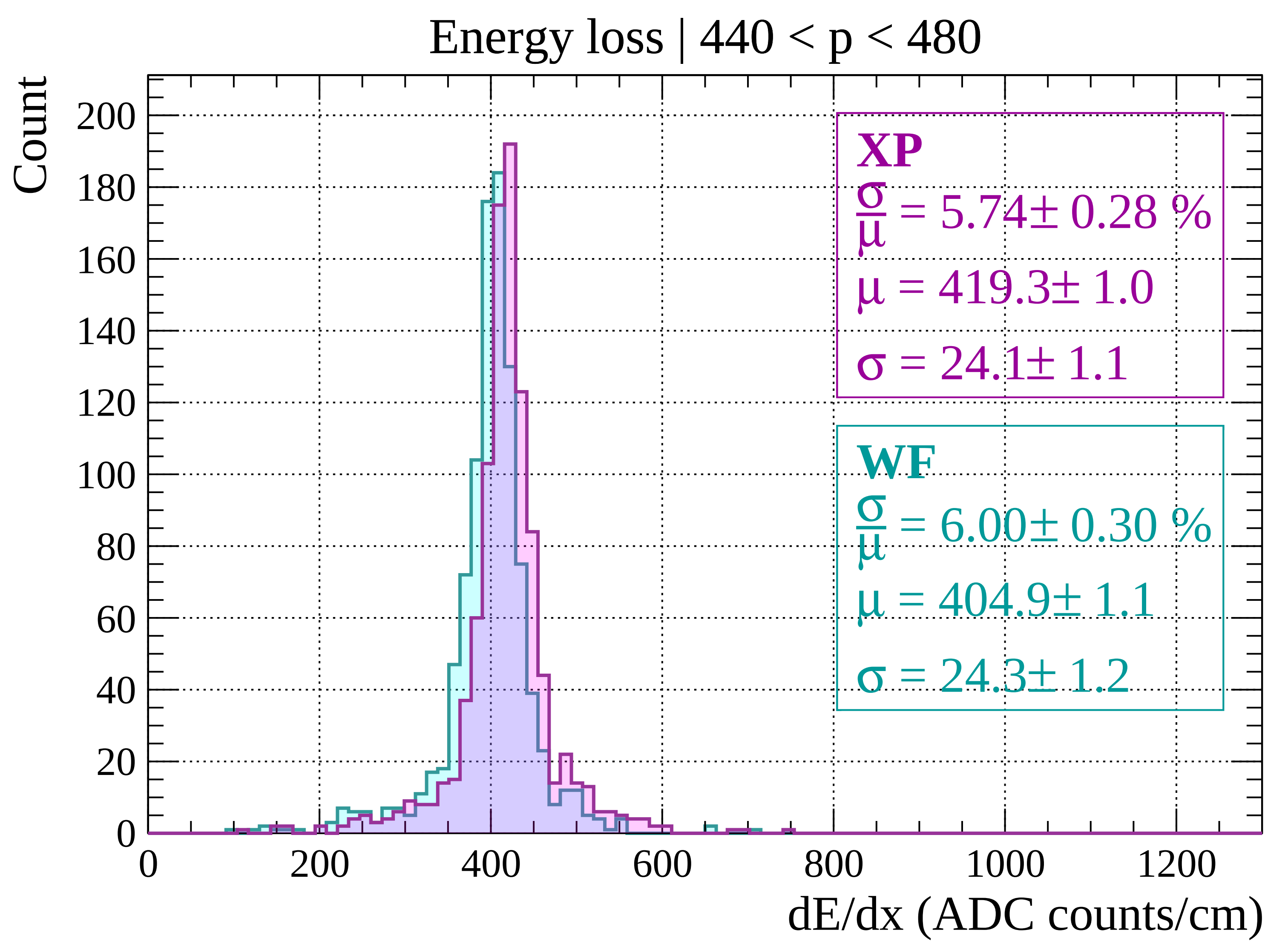
Energy loss | 320 < p < 360
Count
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.01± 0.36 %
μ = 417.1± 1.4
σ = 29.3± 1.4
WF
$\frac{\sigma}{\mu}$ = 7.80± 0.42 %
μ = 399.3± 1.5
σ = 31.1± 1.6

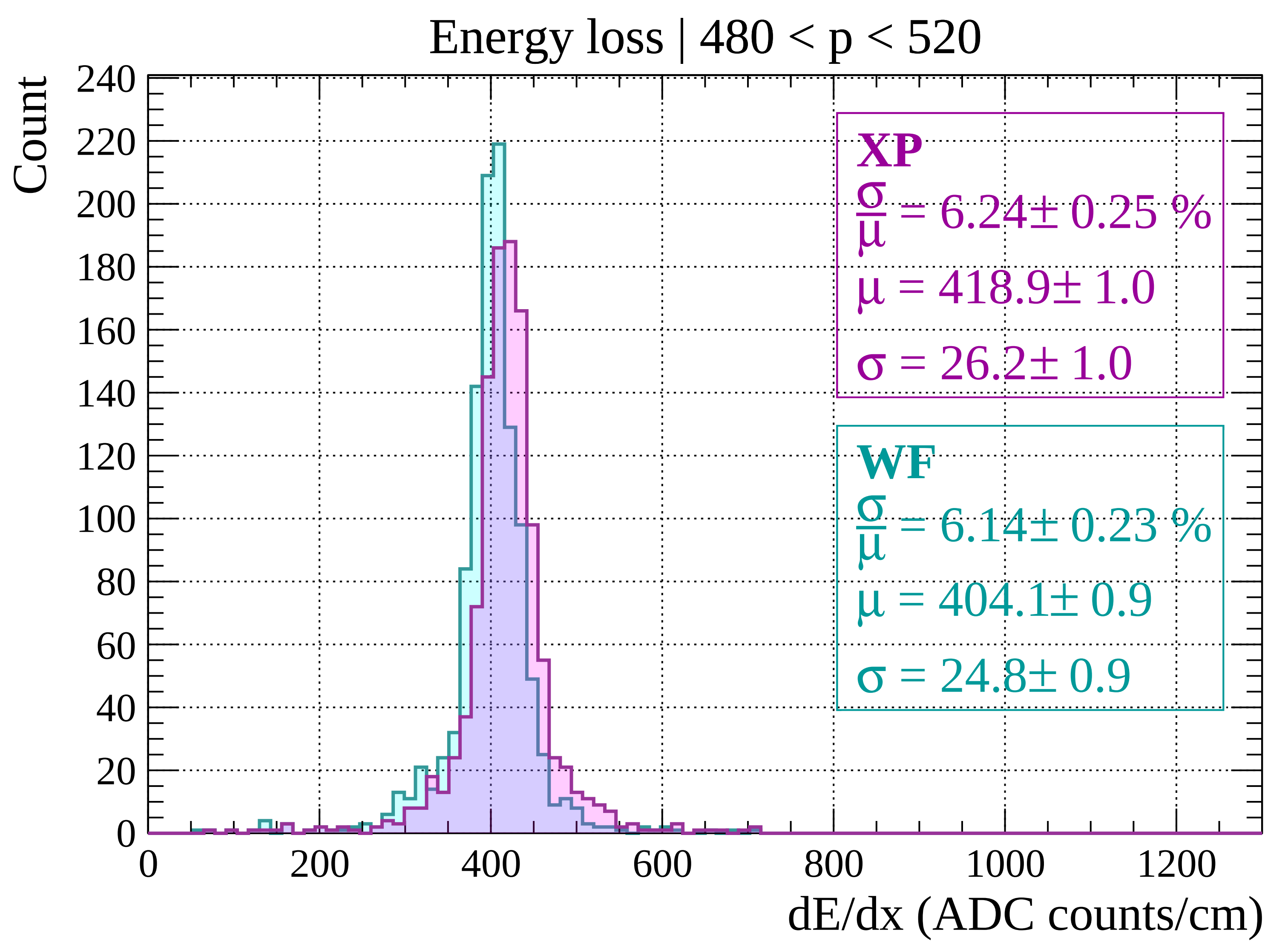


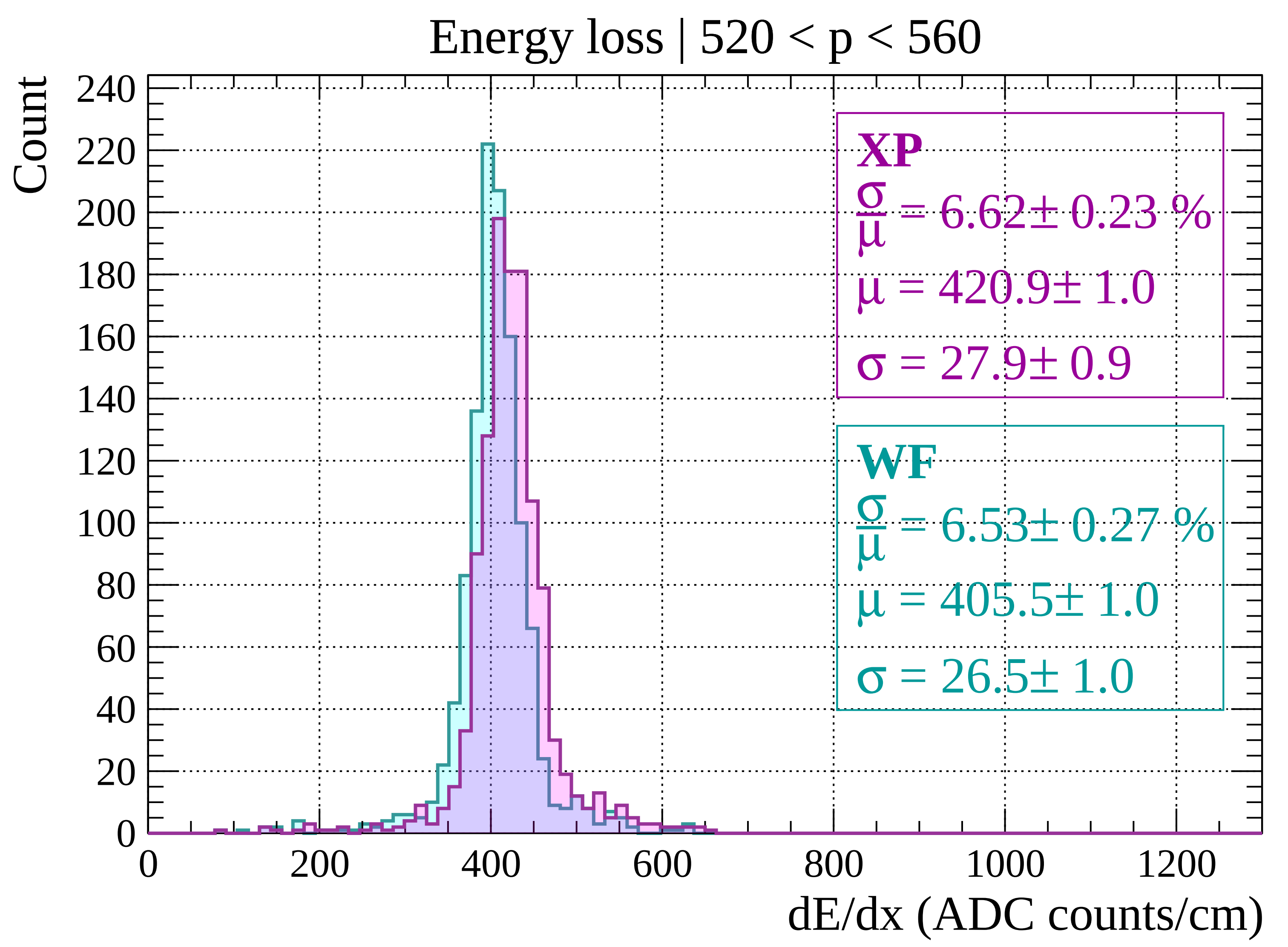
Energy loss | 360 < p < 400
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
σ/μ = 7.59± 0.34 %
μ = 415.8± 1.3
σ = 31.5± 1.3
WF
σ/μ = 7.71± 0.32 %
μ = 399.9± 1.3
σ = 30.8± 1.2

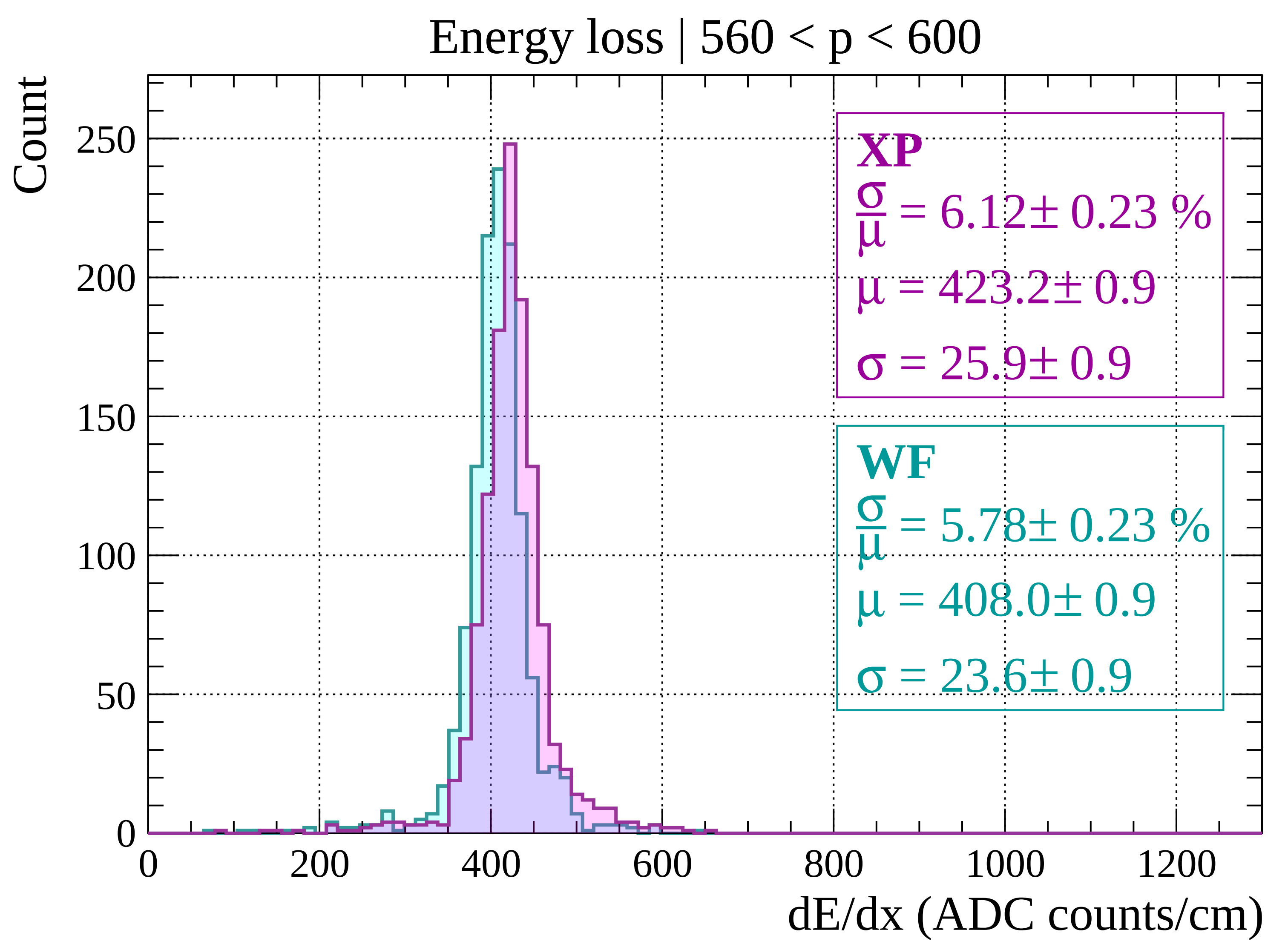


Energy loss | 400 < p < 440
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP
σ/μ = 6.81± 0.33 %
μ = 415.2± 1.2
σ = 28.3± 1.3
WF
σ/μ = 6.83± 0.33 %
μ = 400.6± 1.2
σ = 27.4± 1.2

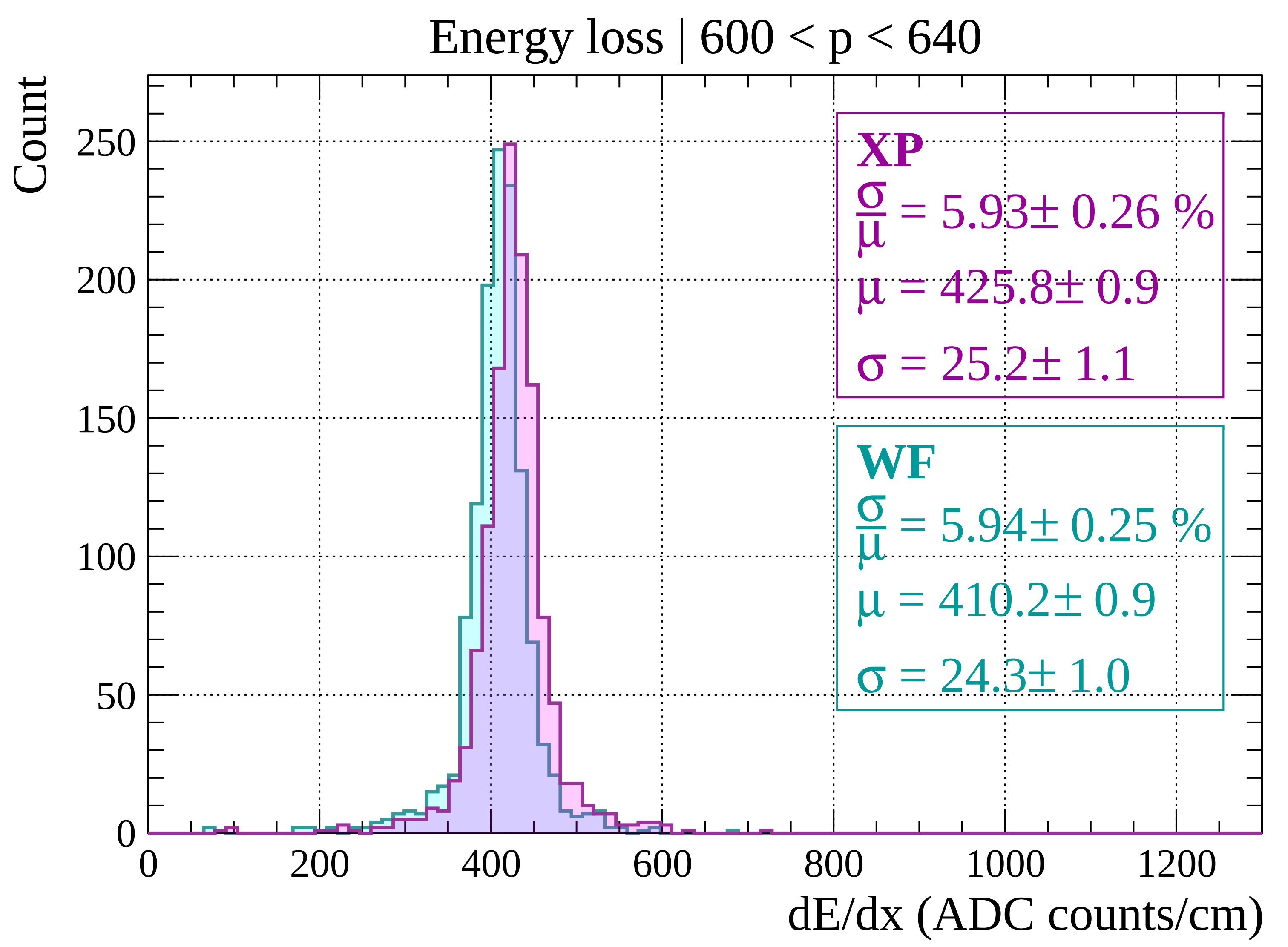

Energy loss | 440 < p < 480
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 5.74± 0.28 %
μ = 419.3± 1.0
σ = 24.1± 1.1
WF
$\frac{\sigma}{\mu}$ = 6.00± 0.30 %
μ = 404.9± 1.1
σ = 24.3± 1.2
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200

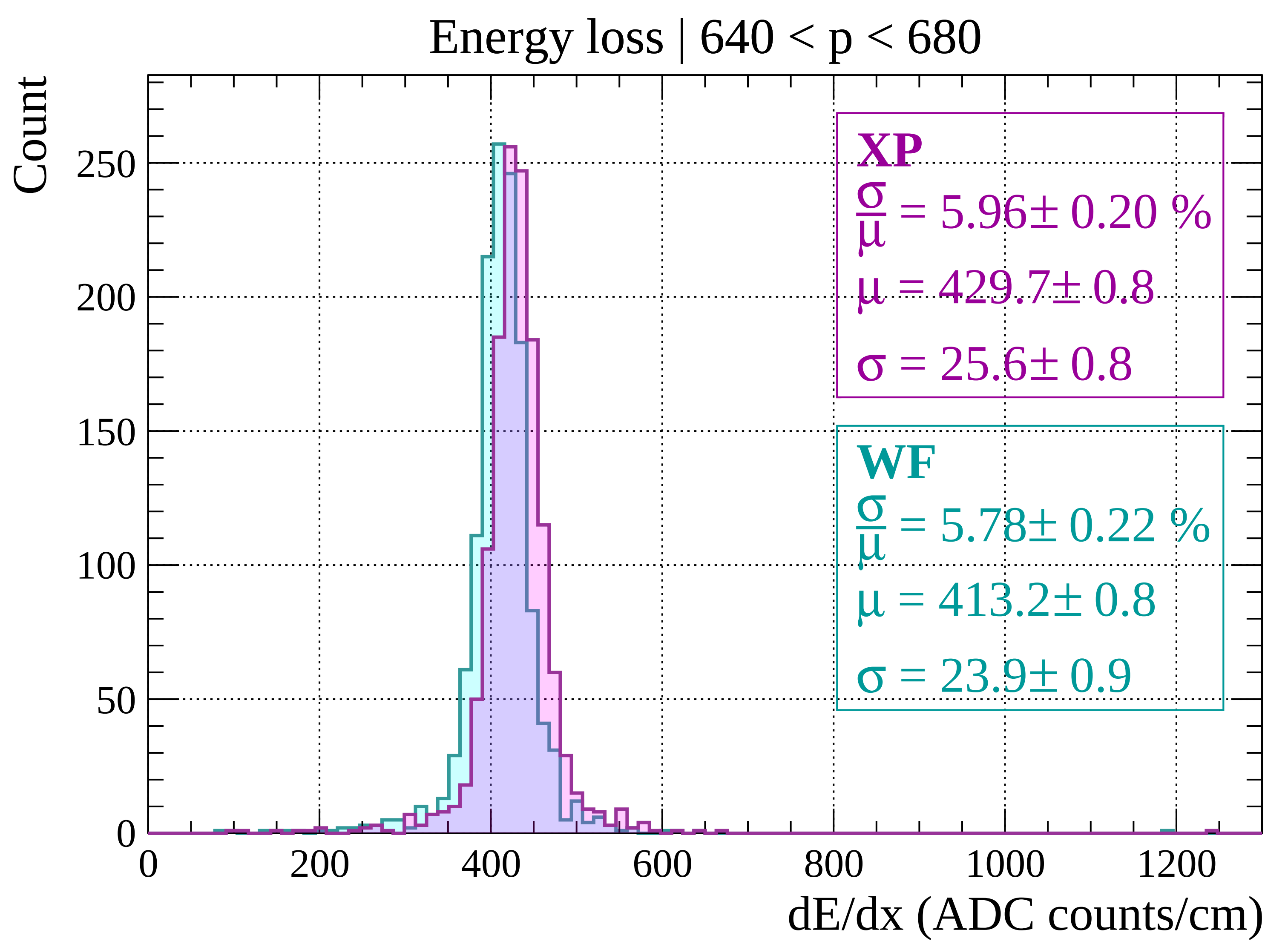

Energy loss | 480 < p < 520
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
220
240
XP
$\frac{\sigma}{\mu}$ = 6.24± 0.25 %
μ = 418.9± 1.0
σ = 26.2± 1.0
WF
$\frac{\sigma}{\mu}$ = 6.14± 0.23 %
μ = 404.1± 0.9
σ = 24.8± 0.9


Energy loss | 520 < p < 560
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 6.62± 0.23 %
μ = 420.9± 1.0
σ = 27.9± 0.9
WF
$\frac{\sigma}{\mu}$ = 6.53± 0.27 %
μ = 405.5± 1.0
σ = 26.5± 1.0
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
220
240

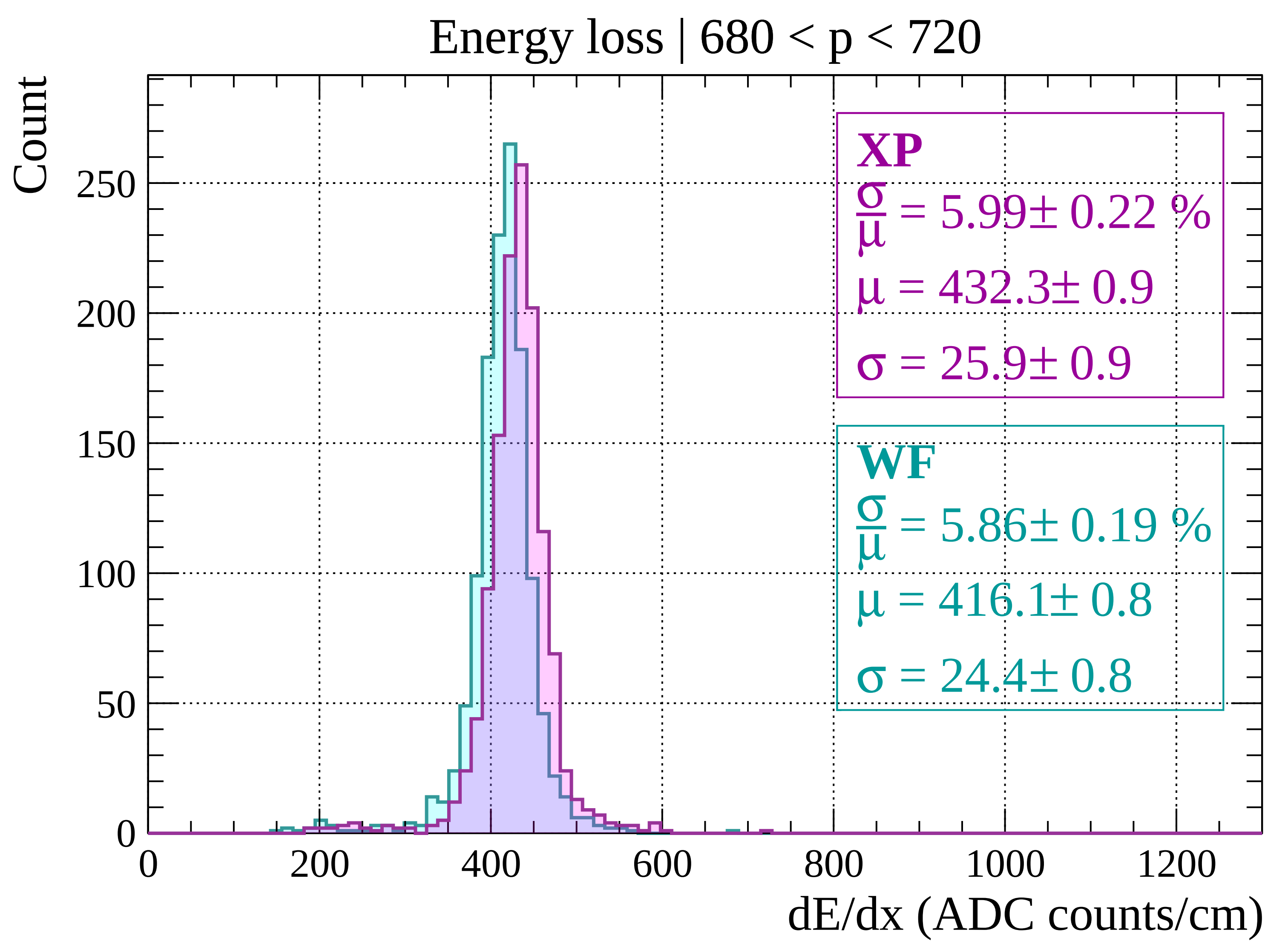


Energy loss | 560 < p < 600
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 6.12± 0.23 %
μ = 423.2± 0.9
σ = 25.9± 0.9
WF
σ/μ = 5.78± 0.23 %
μ = 408.0± 0.9
σ = 23.6± 0.9

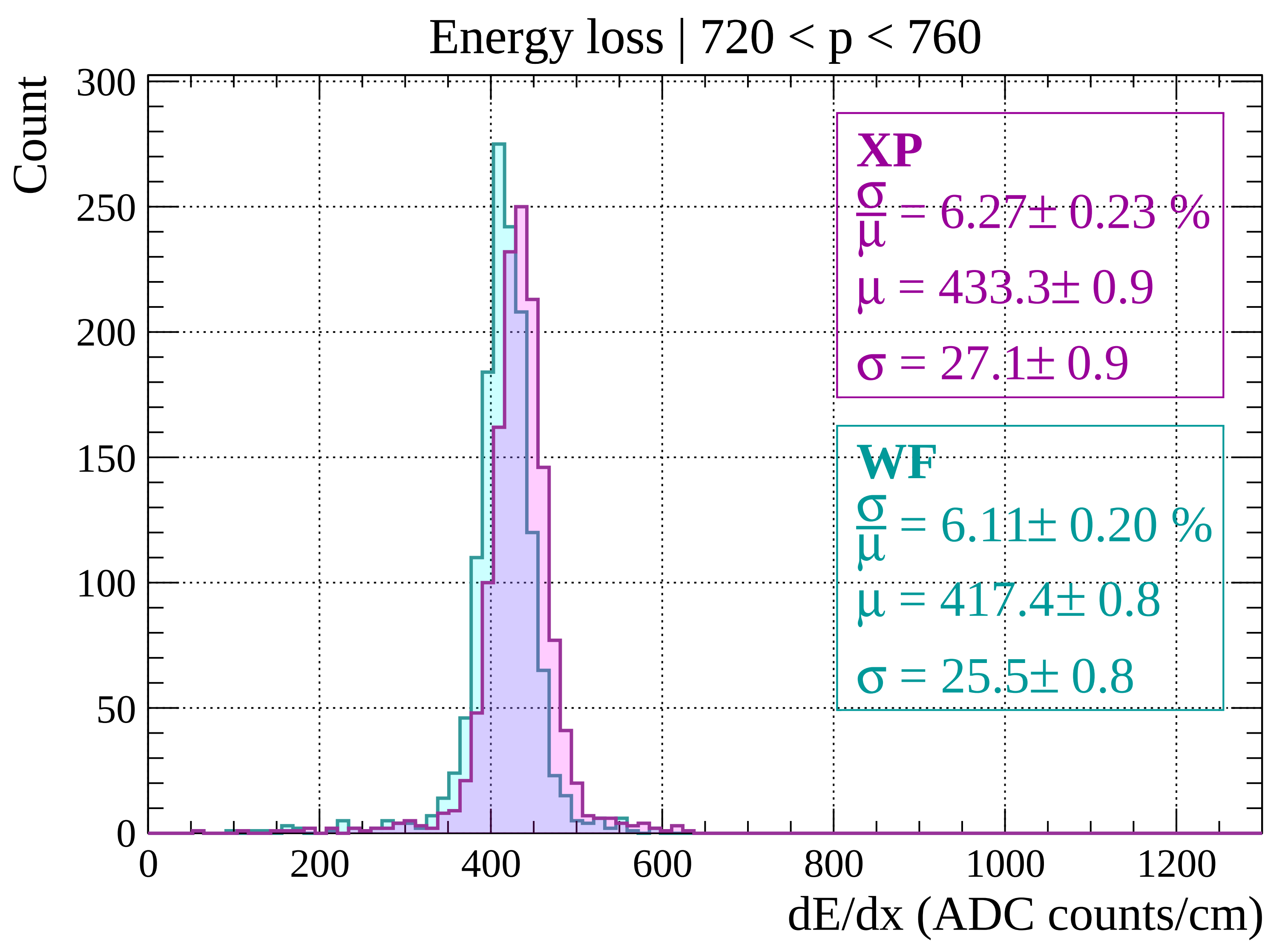

Energy loss | 600 < p < 640
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
XP
σ/μ = 5.93± 0.26 %
μ = 425.8± 0.9
σ = 25.2± 1.1
WF
σ/μ = 5.94± 0.25 %
μ = 410.2± 0.9
σ = 24.3± 1.0

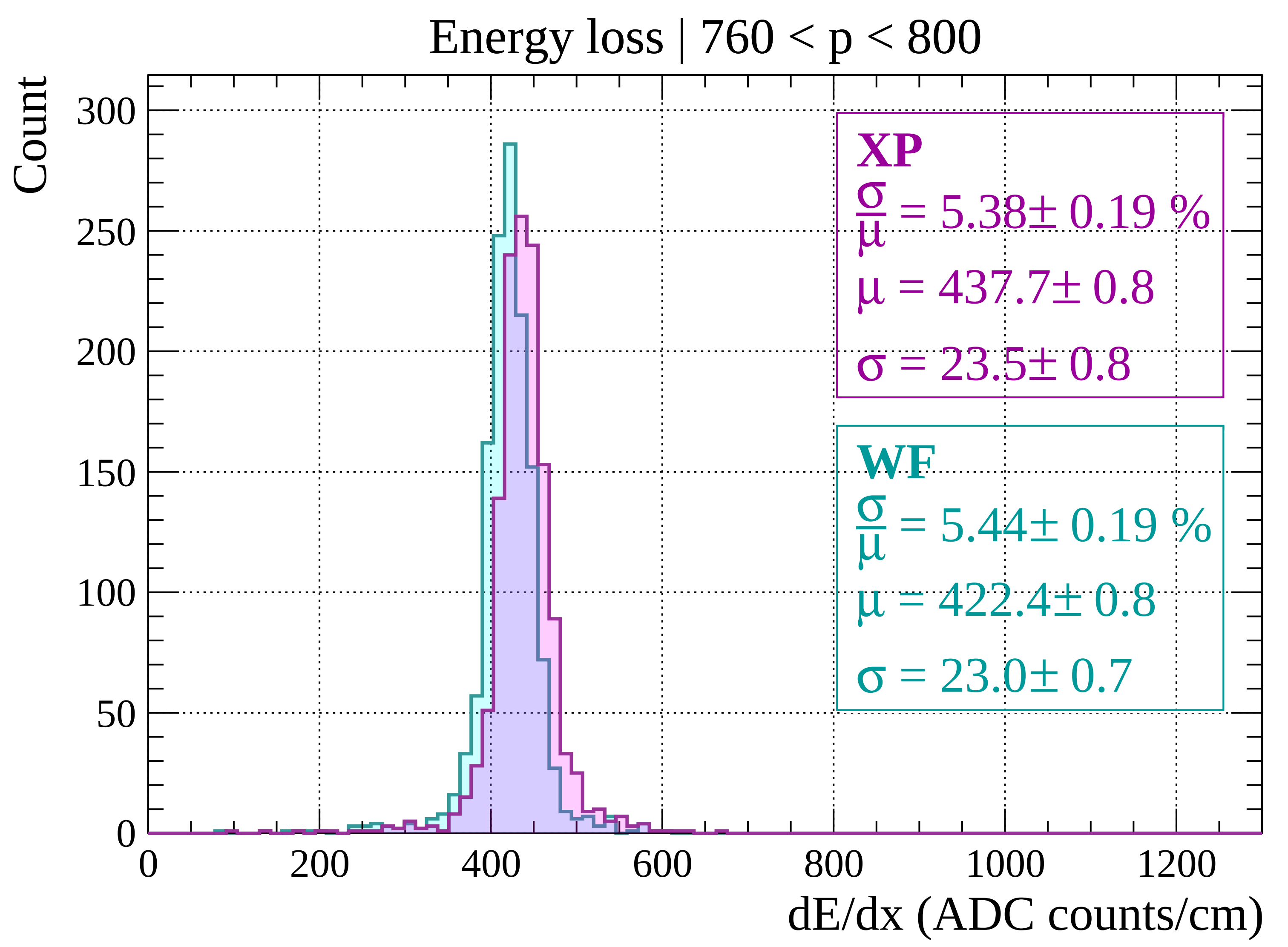


Energy loss | 640 < p < 680
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 5.96± 0.20 %
μ = 429.7± 0.8
σ = 25.6± 0.8
WF
$\frac{\sigma}{\mu}$ = 5.78± 0.22 %
μ = 413.2± 0.8
σ = 23.9± 0.9

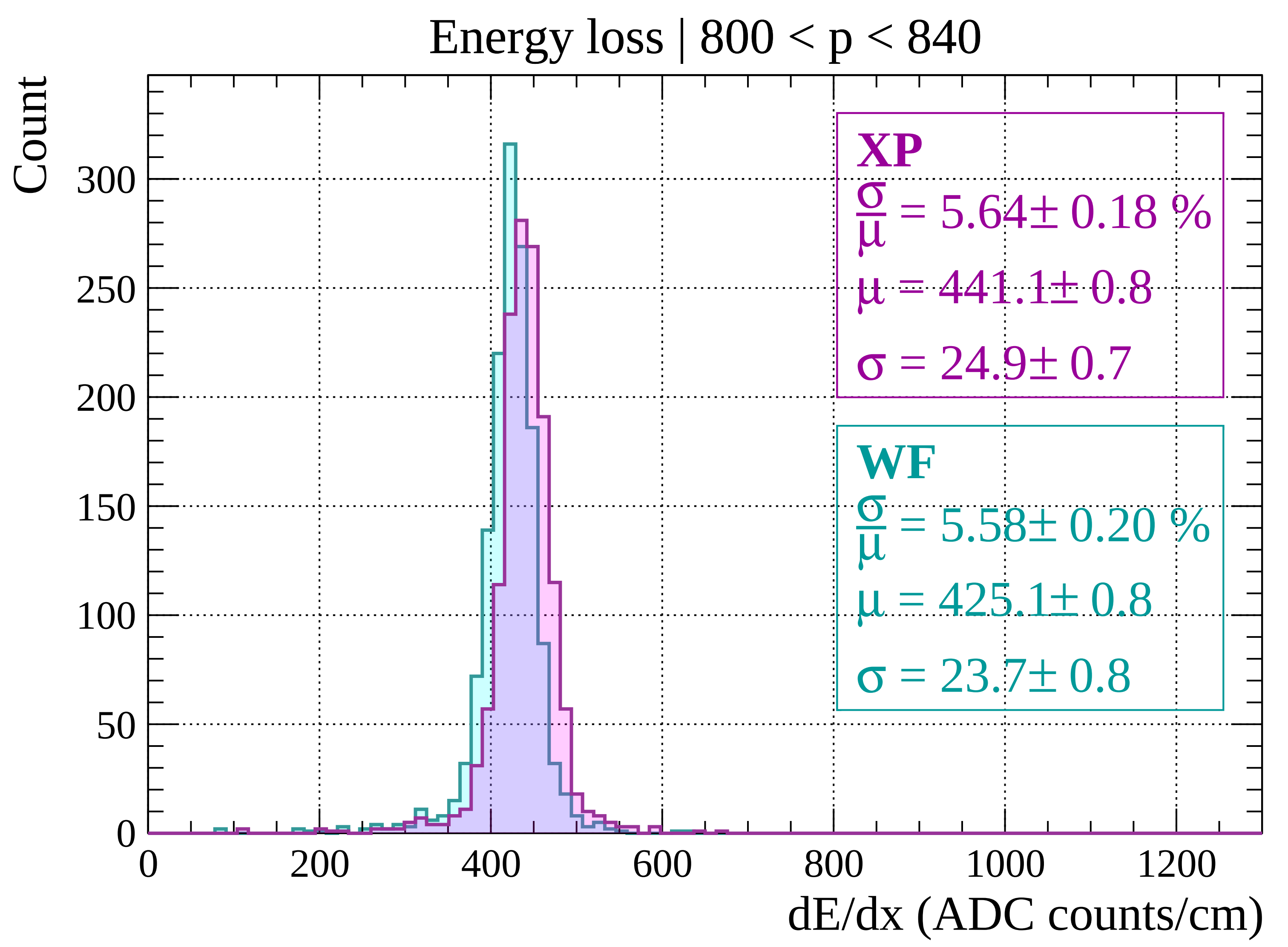


Energy loss | 680 < p < 720
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
XP
σ/μ = 5.99± 0.22 %
μ = 432.3± 0.9
σ = 25.9± 0.9
WF
σ/μ = 5.86± 0.19 %
μ = 416.1± 0.8
σ = 24.4± 0.8

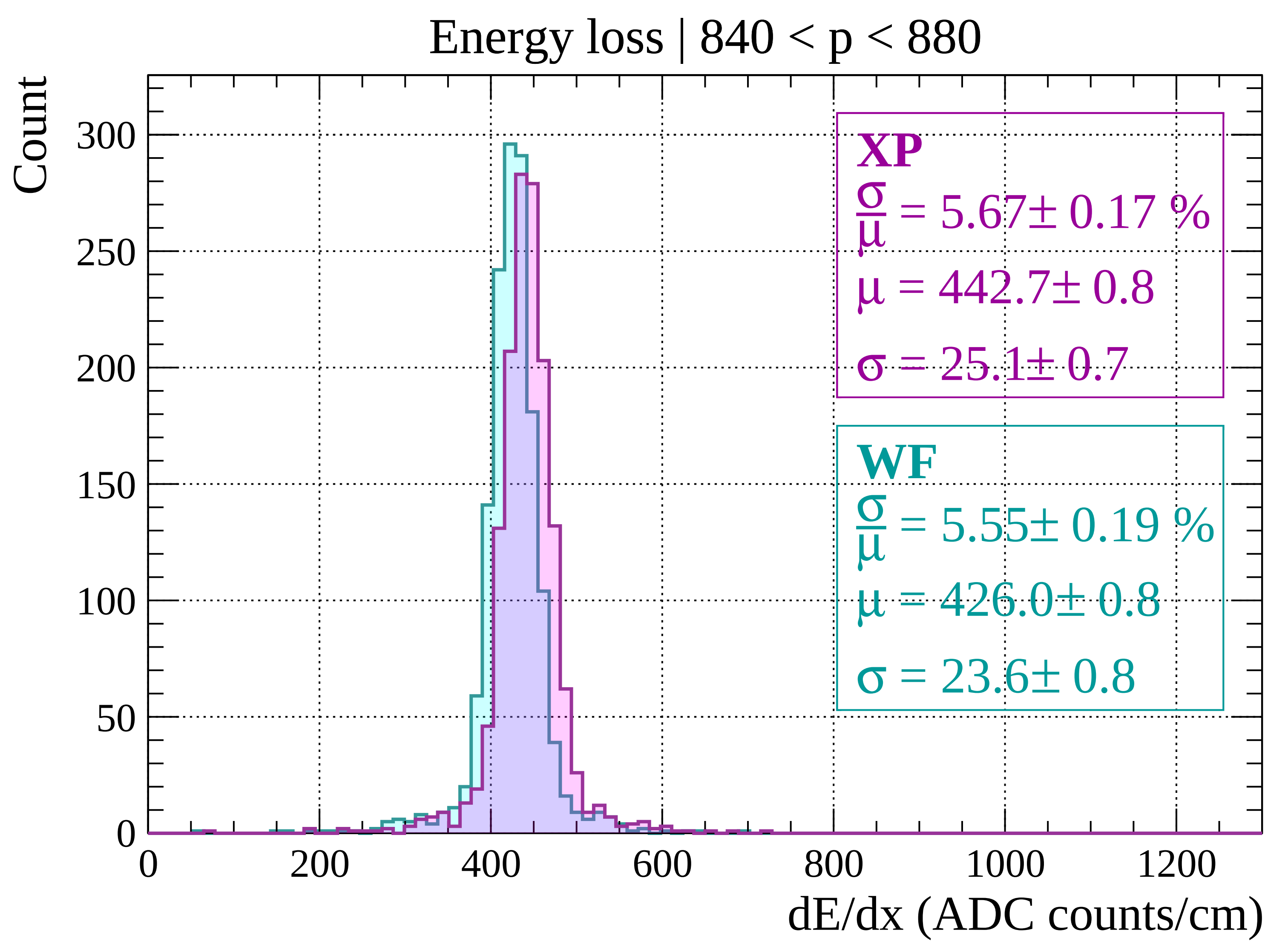

Energy loss | 720 < p < 760
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 6.27± 0.23 %
μ = 433.3± 0.9
σ = 27.1± 0.9
WF
$\frac{\sigma}{\mu}$ = 6.11± 0.20 %
μ = 417.4± 0.8
σ = 25.5± 0.8
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200



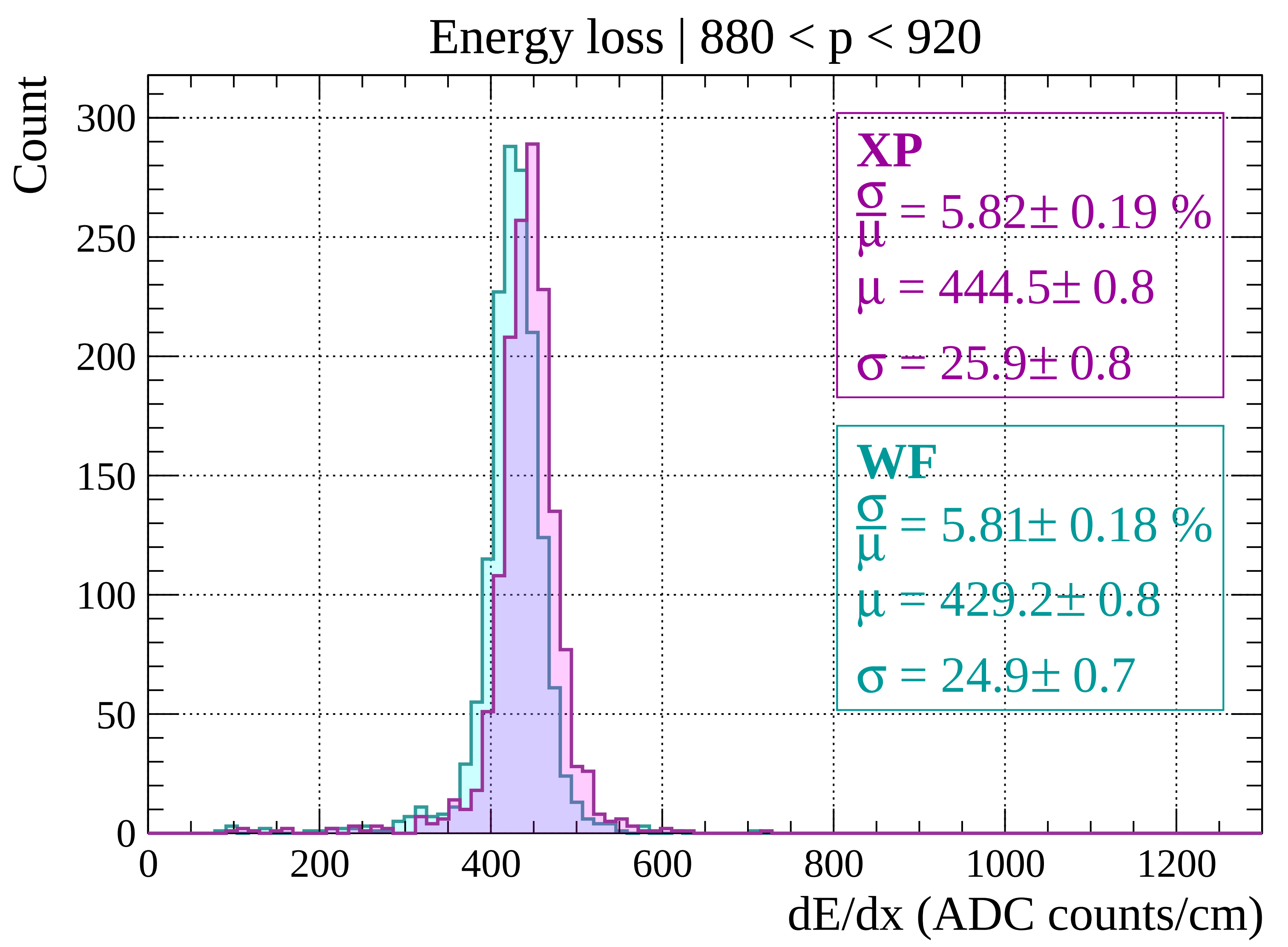
Energy loss | 760 < p < 800
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
σ/μ = 5.38± 0.19 %
μ = 437.7± 0.8
σ = 23.5± 0.8
WF
σ/μ = 5.44± 0.19 %
μ = 422.4± 0.8
σ = 23.0± 0.7



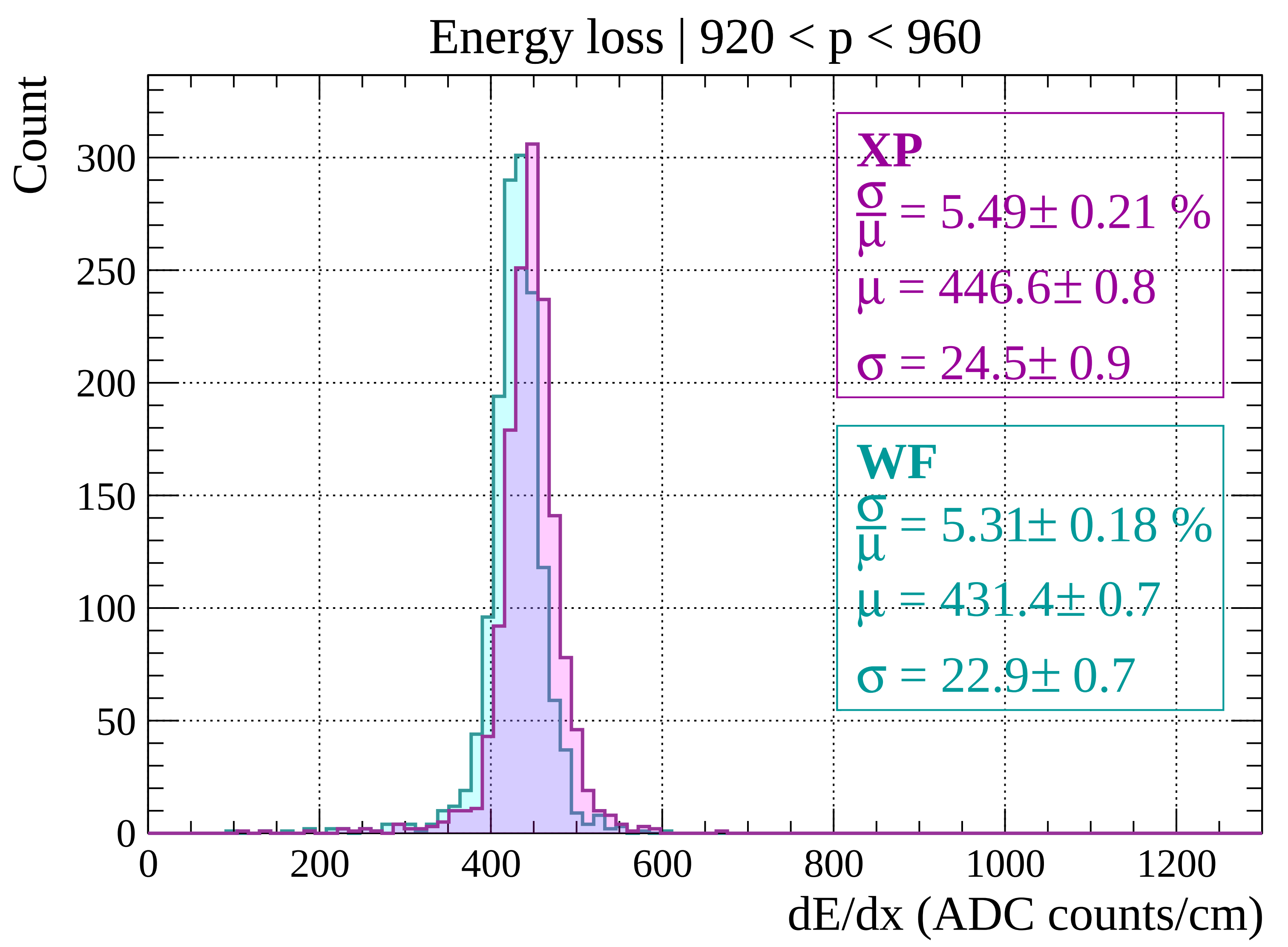
Energy loss | 800 < p < 840
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
σ/μ = 5.64± 0.18 %
μ = 441.1± 0.8
σ = 24.9± 0.7
WF
σ/μ = 5.58± 0.20 %
μ = 425.1± 0.8
σ = 23.7± 0.8



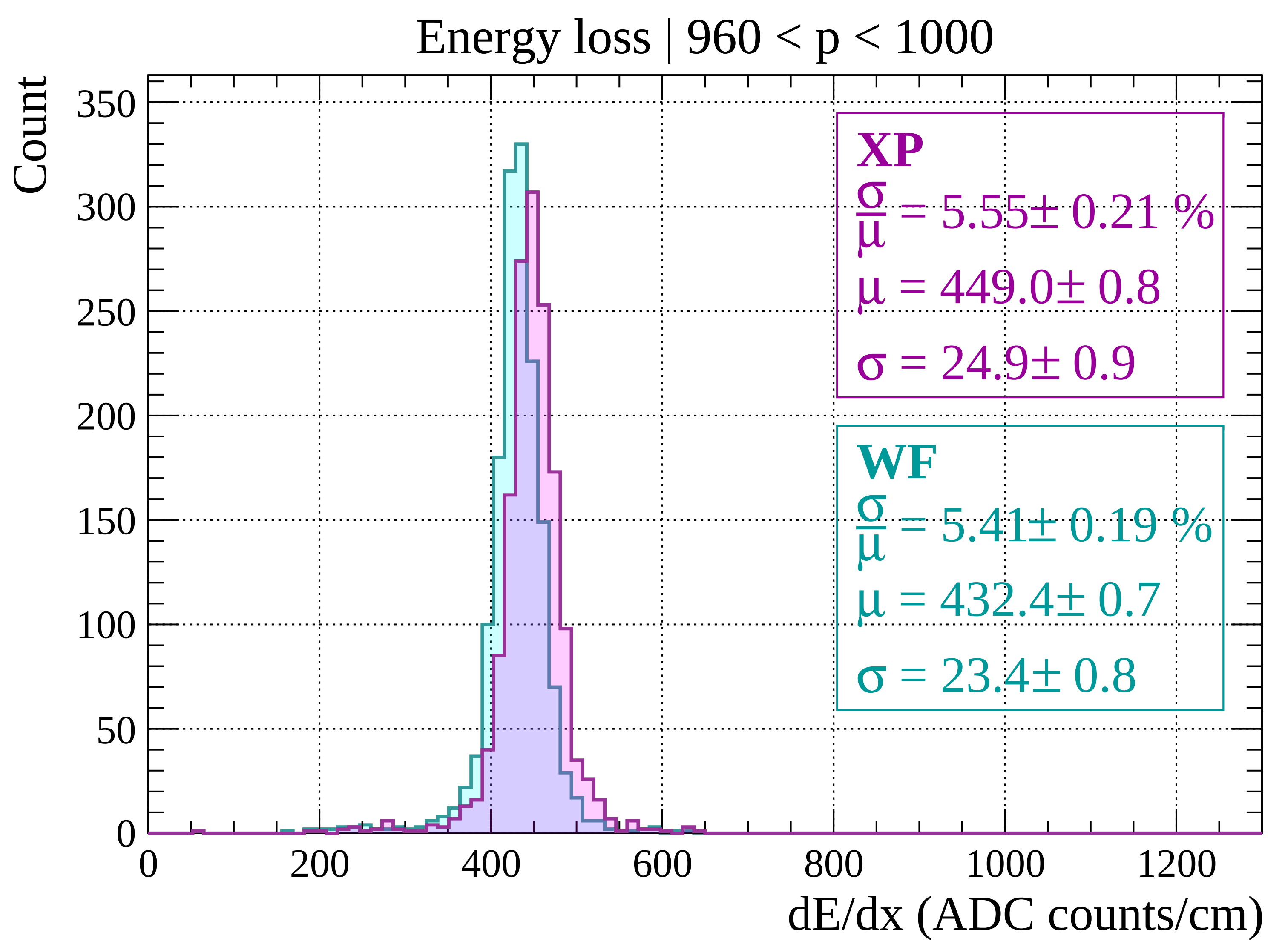
Energy loss | 840 < p < 880
Count
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 5.67 \pm 0.17$ %
$\mu = 442.7 \pm 0.8$
$\sigma = 25.1 \pm 0.7$
WF
$\frac{\sigma}{\mu} = 5.55 \pm 0.19$ %
$\mu = 426.0 \pm 0.8$
$\sigma = 23.6 \pm 0.8$



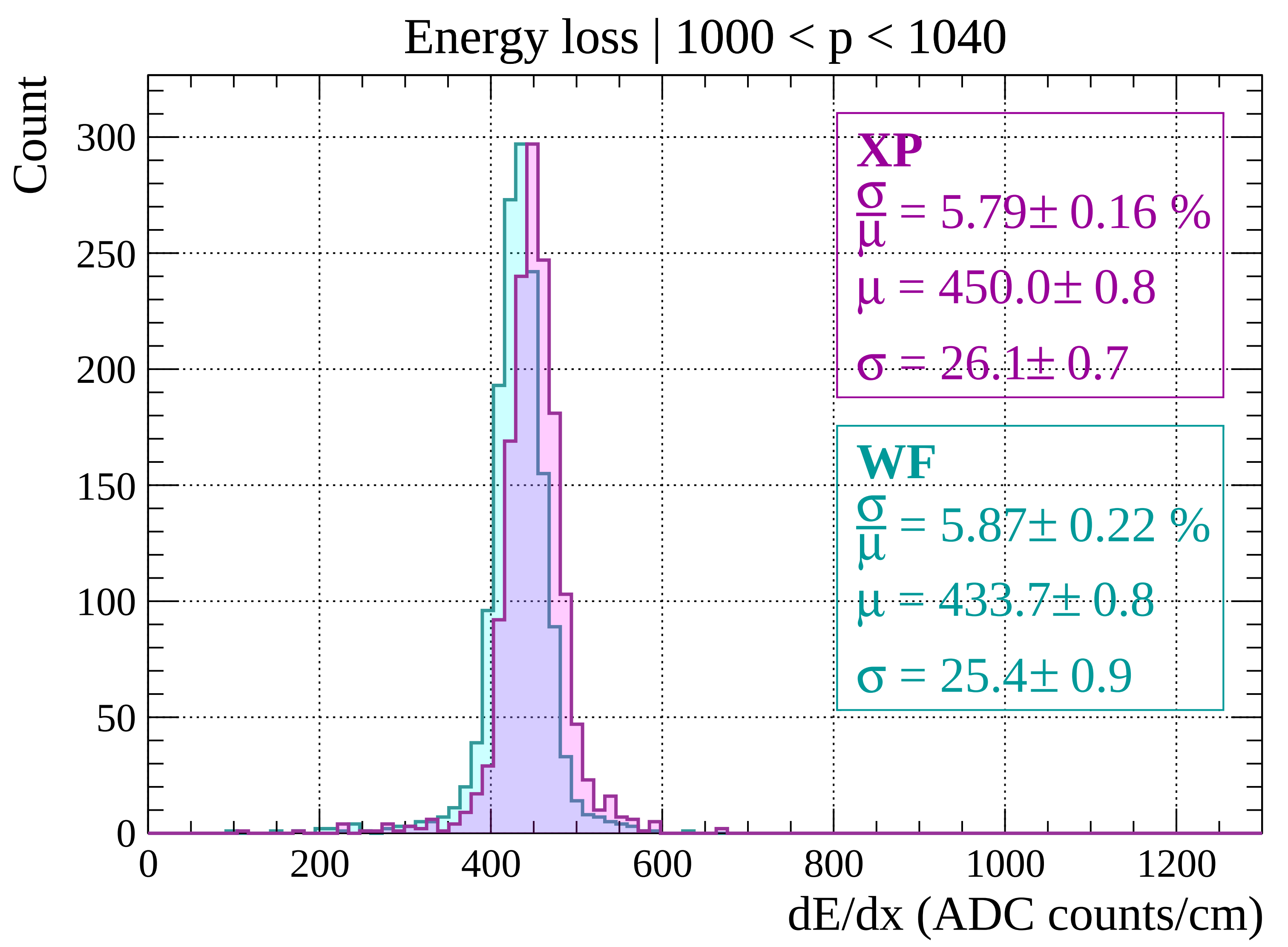
Energy loss | 880 < p < 920
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 5.82± 0.19 %
μ = 444.5± 0.8
σ = 25.9± 0.8
WF
σ/μ = 5.81± 0.18 %
μ = 429.2± 0.8
σ = 24.9± 0.7



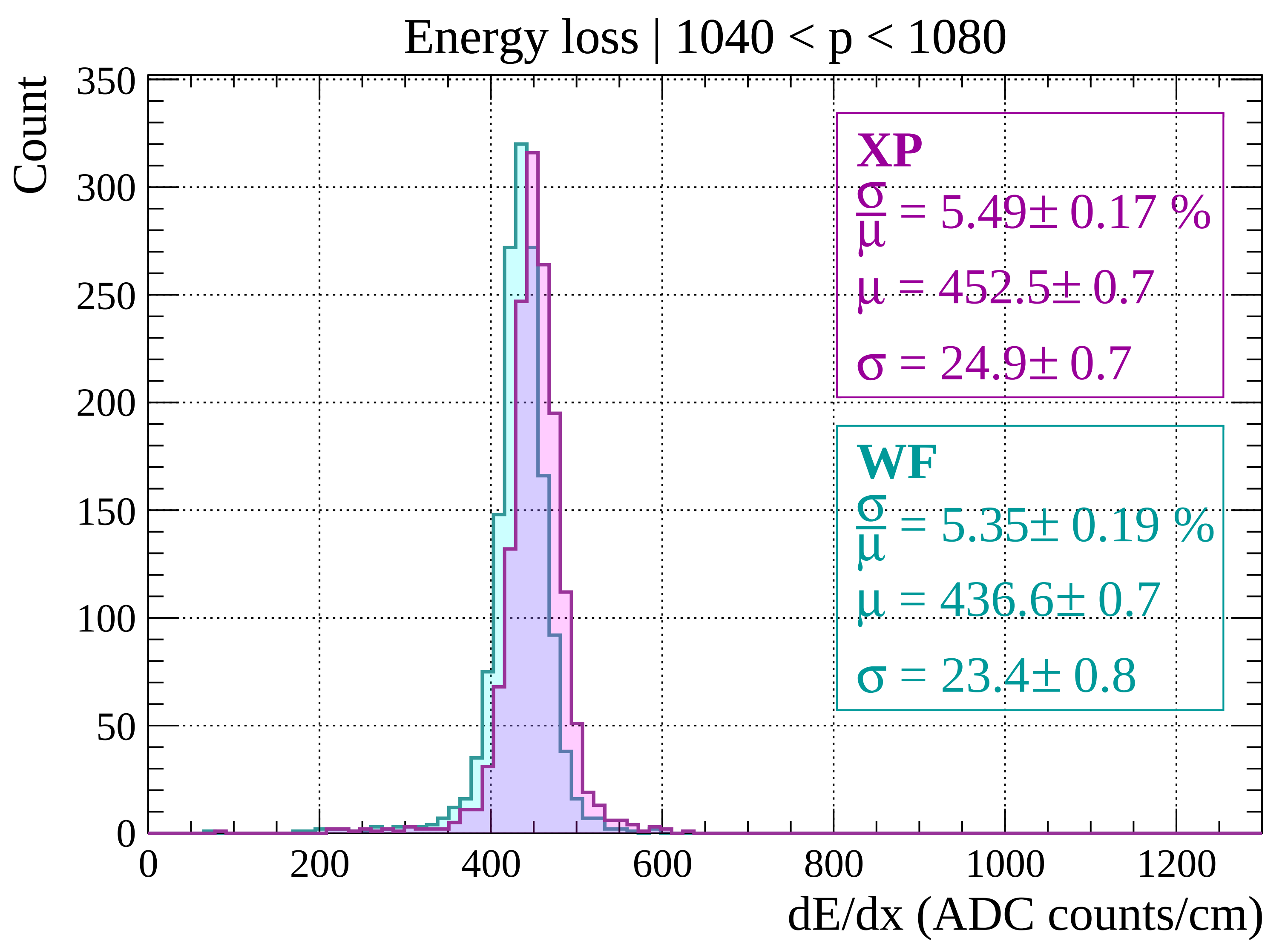
Energy loss | 920 < p < 960
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 5.49± 0.21 %
μ = 446.6± 0.8
σ = 24.5± 0.9
WF
$\frac{\sigma}{\mu}$ = 5.31± 0.18 %
μ = 431.4± 0.7
σ = 22.9± 0.7

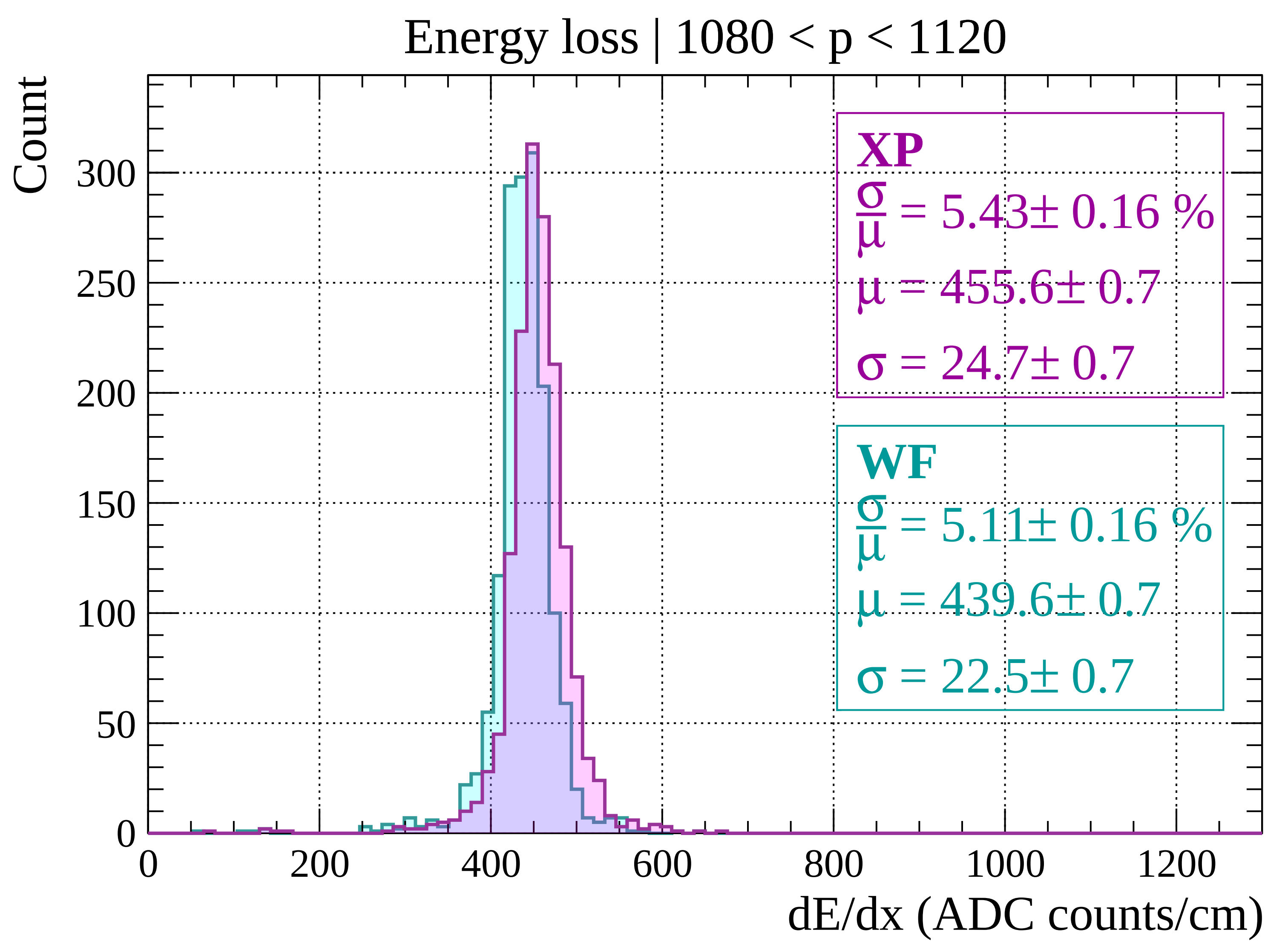


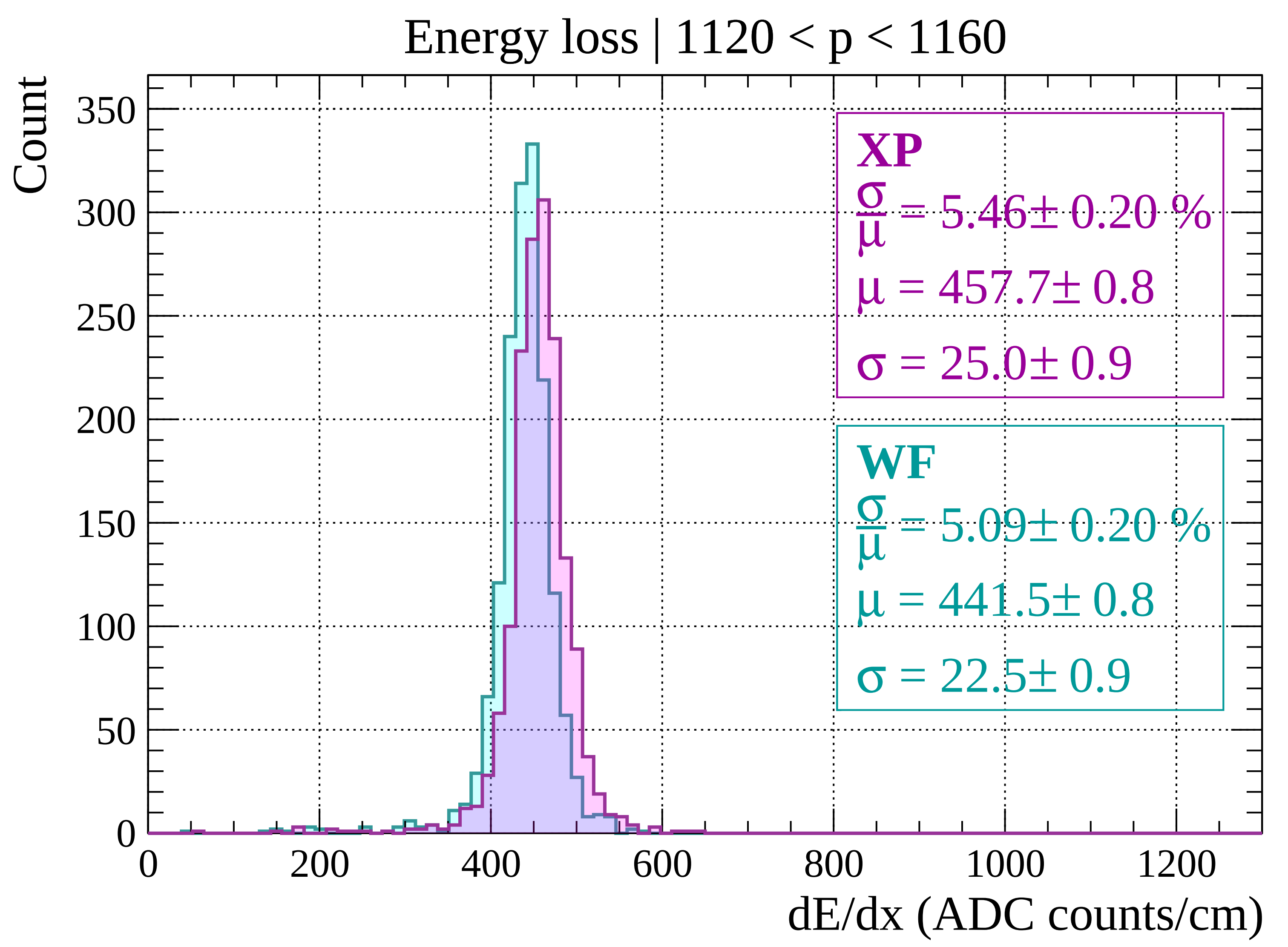
Energy loss | 960 < p < 1000
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
σ/μ = 5.55± 0.21 %
μ = 449.0± 0.8
σ = 24.9± 0.9
WF
σ/μ = 5.41± 0.19 %
μ = 432.4± 0.7
σ = 23.4± 0.8



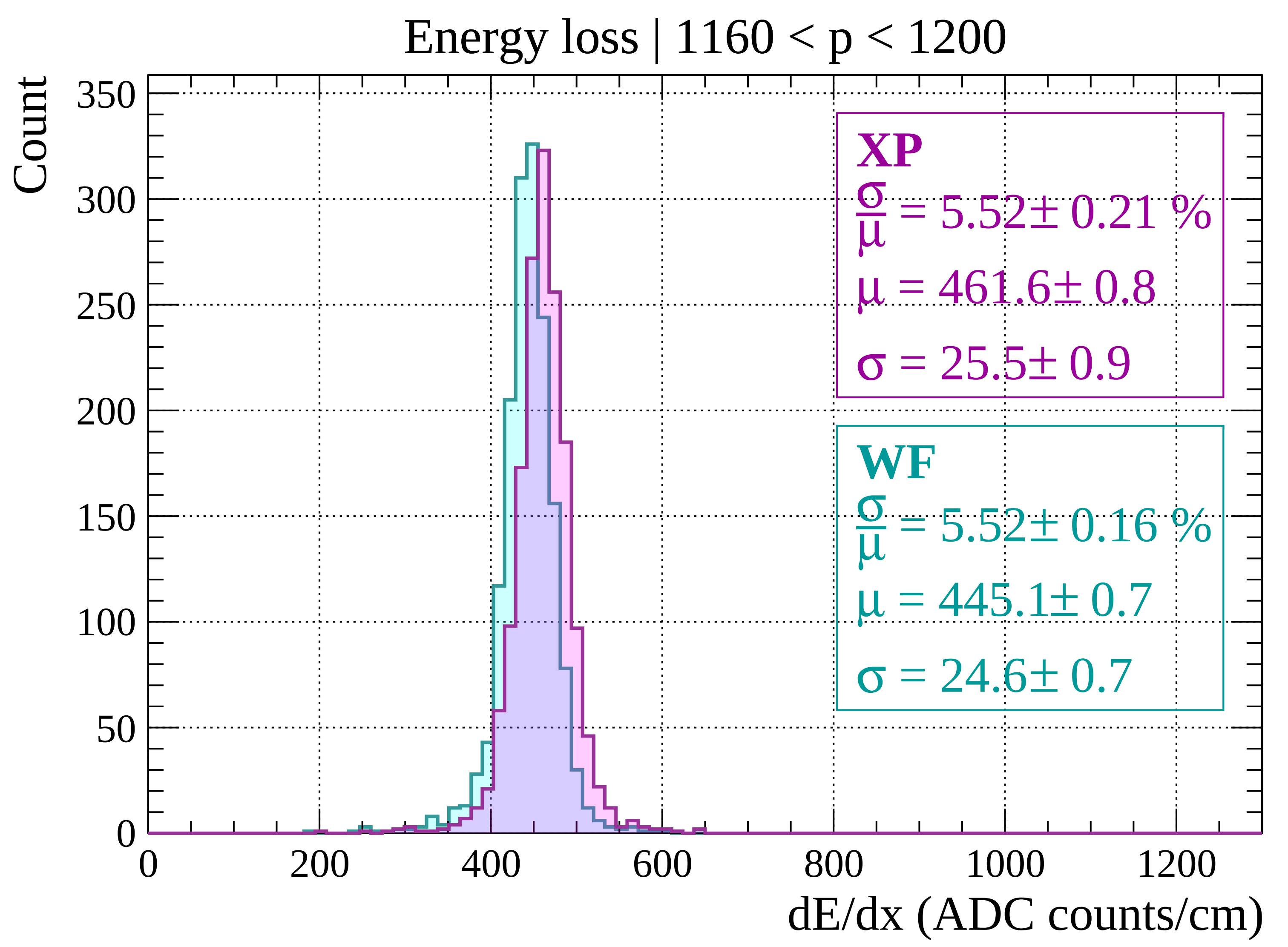
Energy loss | 1000 < p < 1040
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 5.79± 0.16 %
μ = 450.0± 0.8
σ = 26.1± 0.7
WF
$\frac{\sigma}{\mu}$ = 5.87± 0.22 %
μ = 433.7± 0.8
σ = 25.4± 0.9


Energy loss | 1040 < p < 1080
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
XP
σ/μ = 5.49± 0.17 %
μ = 452.5± 0.7
σ = 24.9± 0.7
WF
σ/μ = 5.35± 0.19 %
μ = 436.6± 0.7
σ = 23.4± 0.8

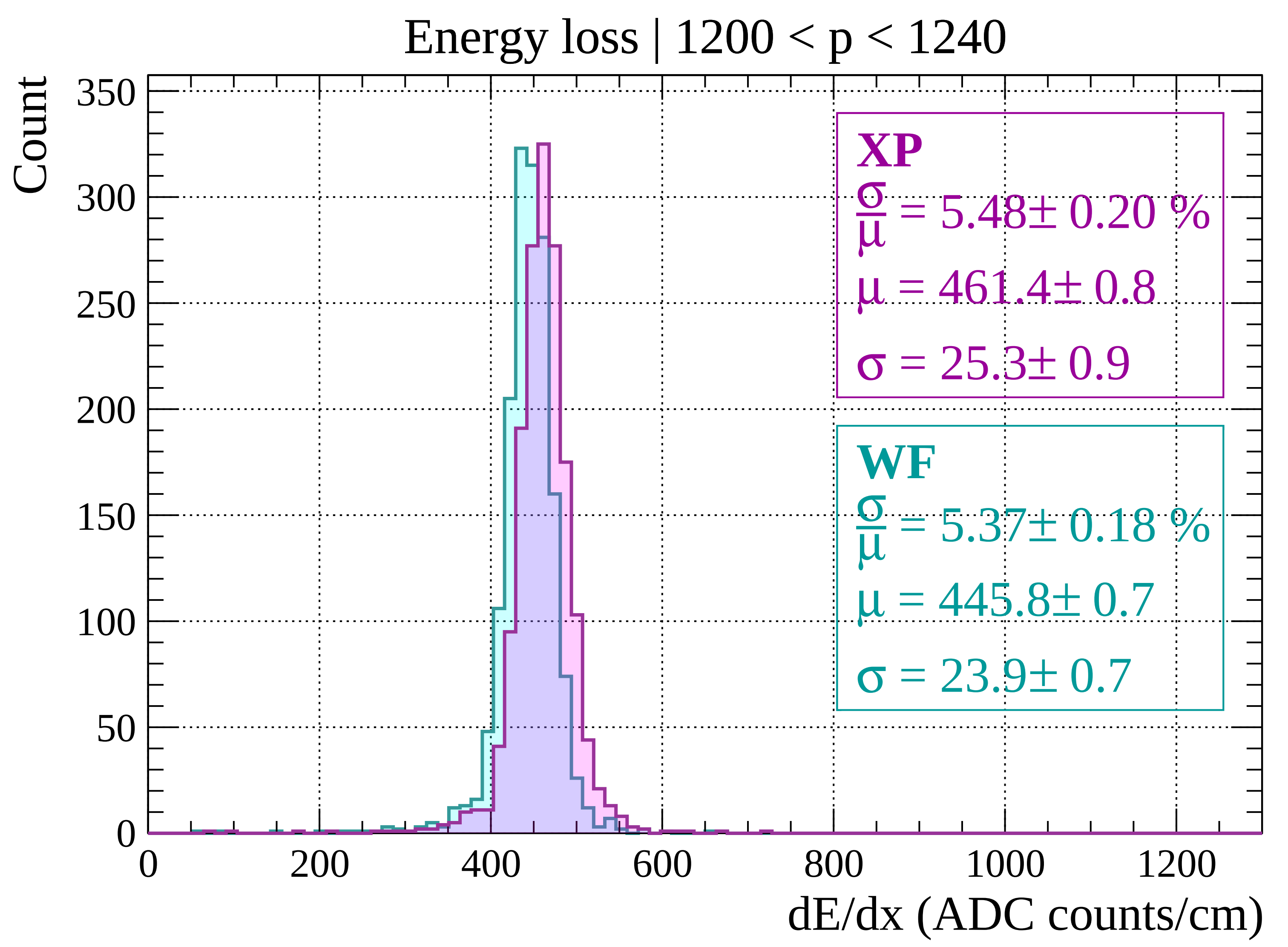

Energy loss | 1080 < p < 1120
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 5.43± 0.16 %
μ = 455.6± 0.7
σ = 24.7± 0.7
WF
$\frac{\sigma}{\mu}$ = 5.11± 0.16 %
μ = 439.6± 0.7
σ = 22.5± 0.7



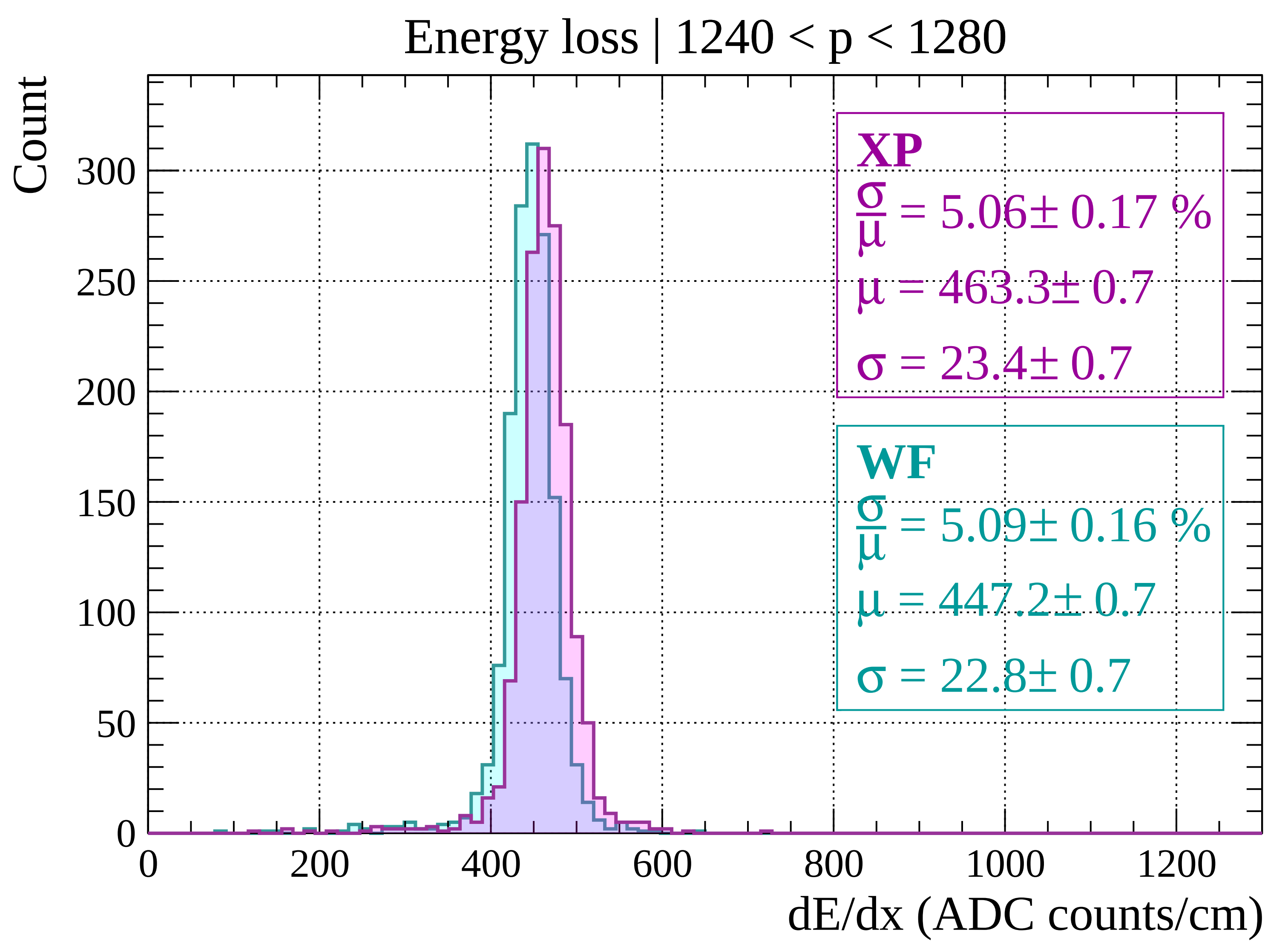
Energy loss | 1120 < p < 1160
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
σ/μ = 5.46± 0.20 %
μ = 457.7± 0.8
σ = 25.0± 0.9
WF
σ/μ = 5.09± 0.20 %
μ = 441.5± 0.8
σ = 22.5± 0.9



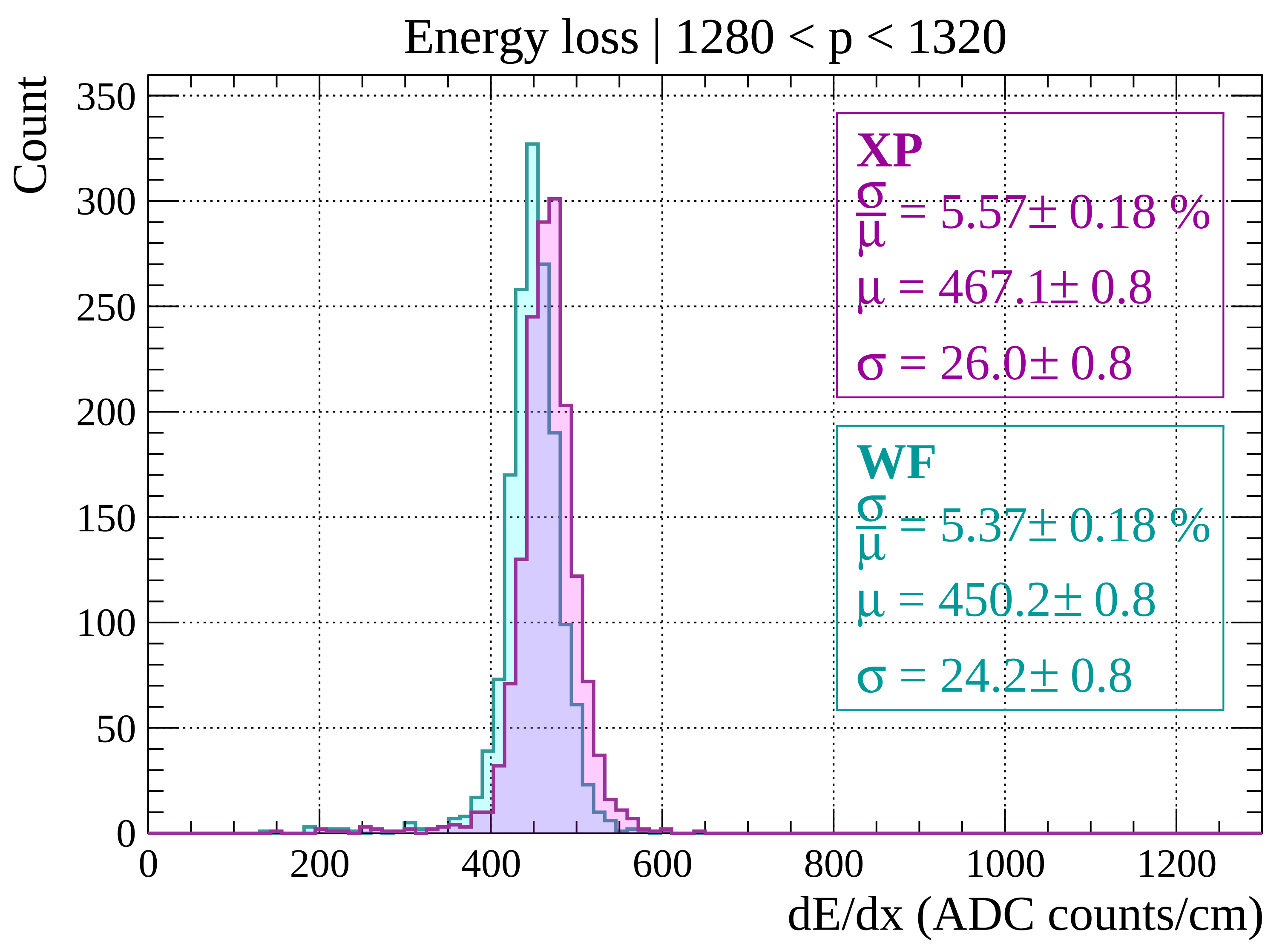
Energy loss | 1160 < p < 1200
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 5.52 ± 0.21 %
μ = 461.6 ± 0.8
σ = 25.5 ± 0.9
WF
σ/μ = 5.52 ± 0.16 %
μ = 445.1 ± 0.7
σ = 24.6 ± 0.7
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200

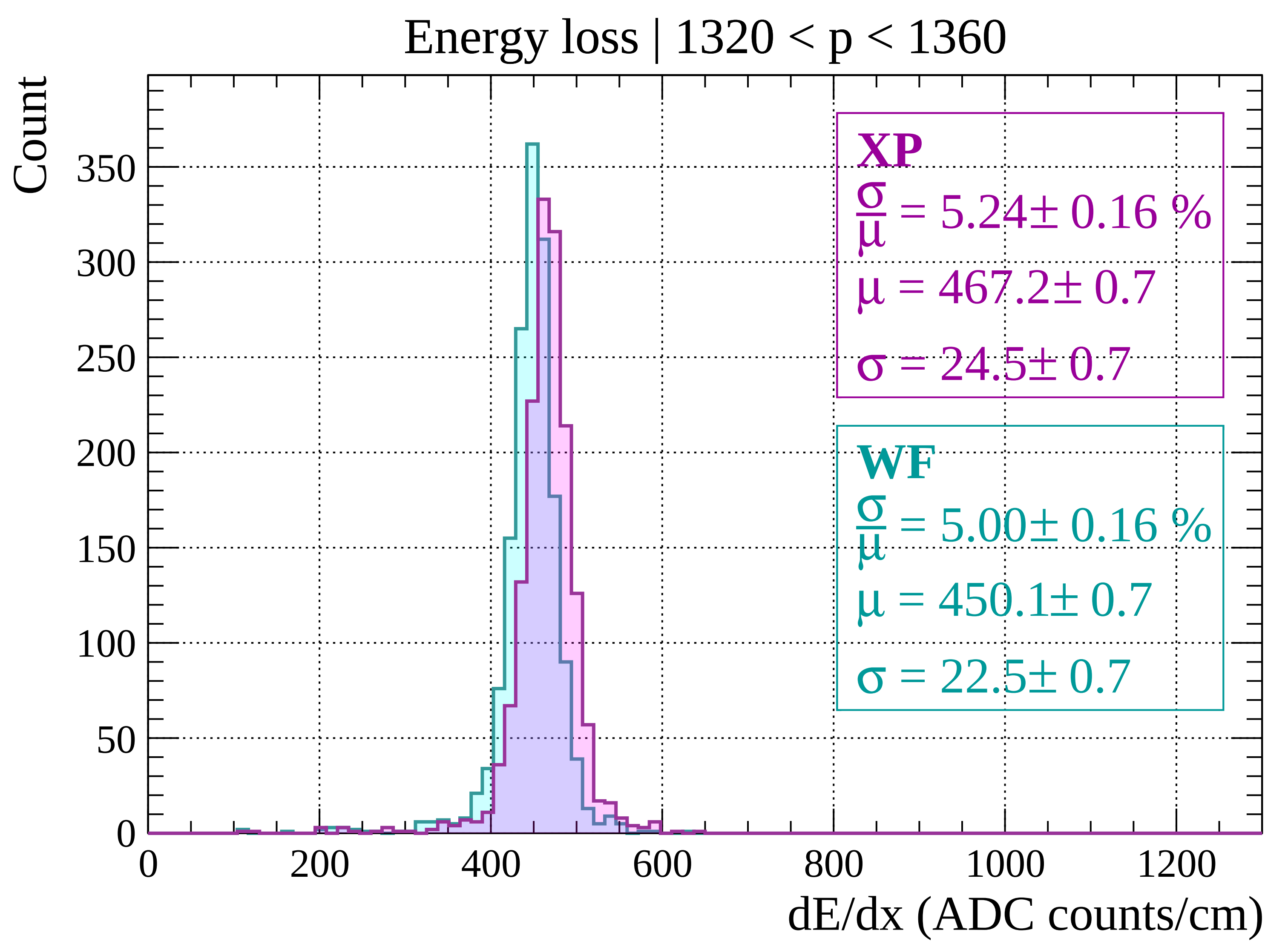


Energy loss | 1200 < p < 1240
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 5.48± 0.20 %
μ = 461.4± 0.8
σ = 25.3± 0.9
WF
$\frac{\sigma}{\mu}$ = 5.37± 0.18 %
μ = 445.8± 0.7
σ = 23.9± 0.7

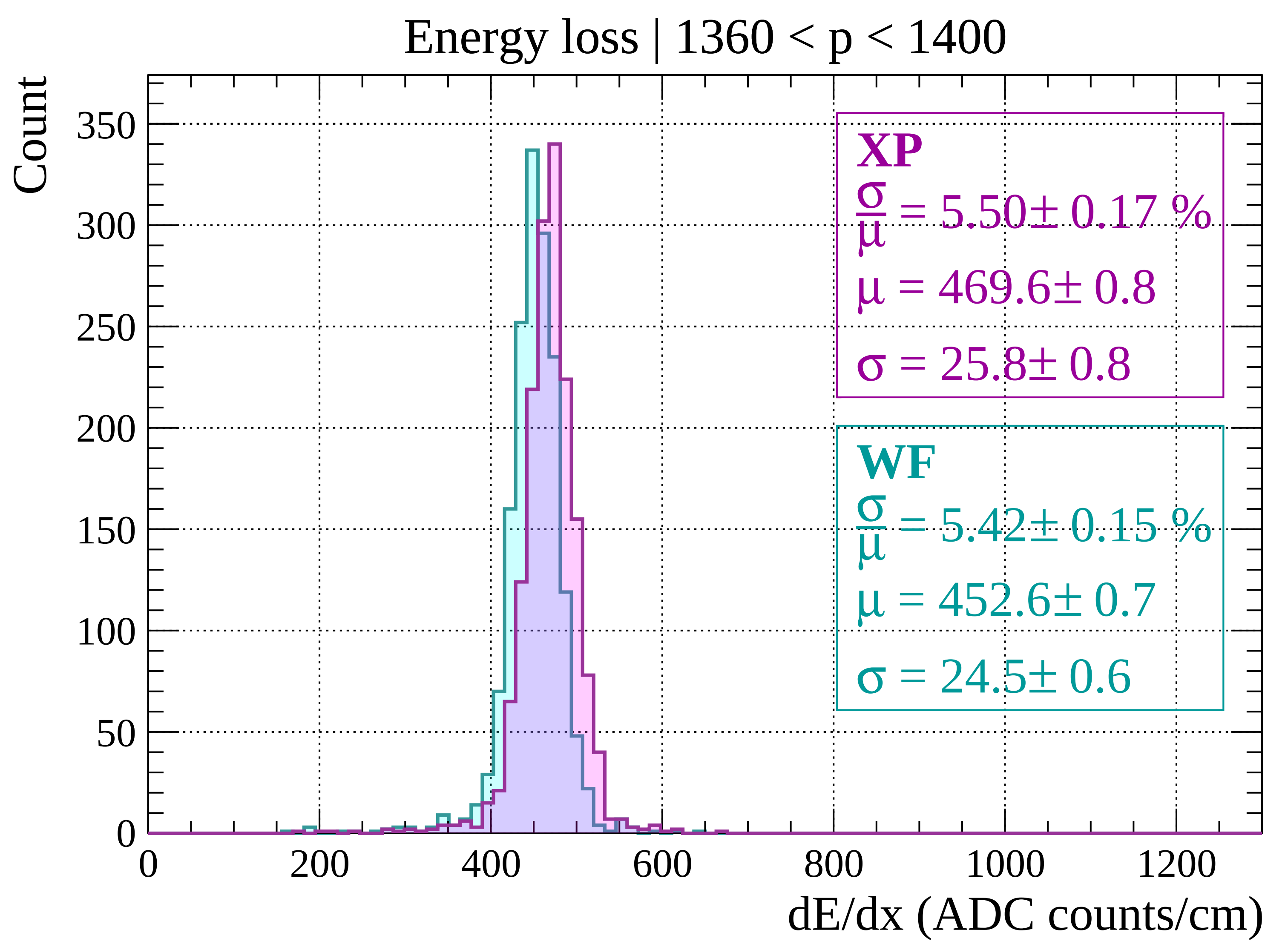


Energy loss | 1240 < p < 1280
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
σ/μ = 5.06± 0.17 %
μ = 463.3± 0.7
σ = 23.4± 0.7
WF
σ/μ = 5.09± 0.16 %
μ = 447.2± 0.7
σ = 22.8± 0.7

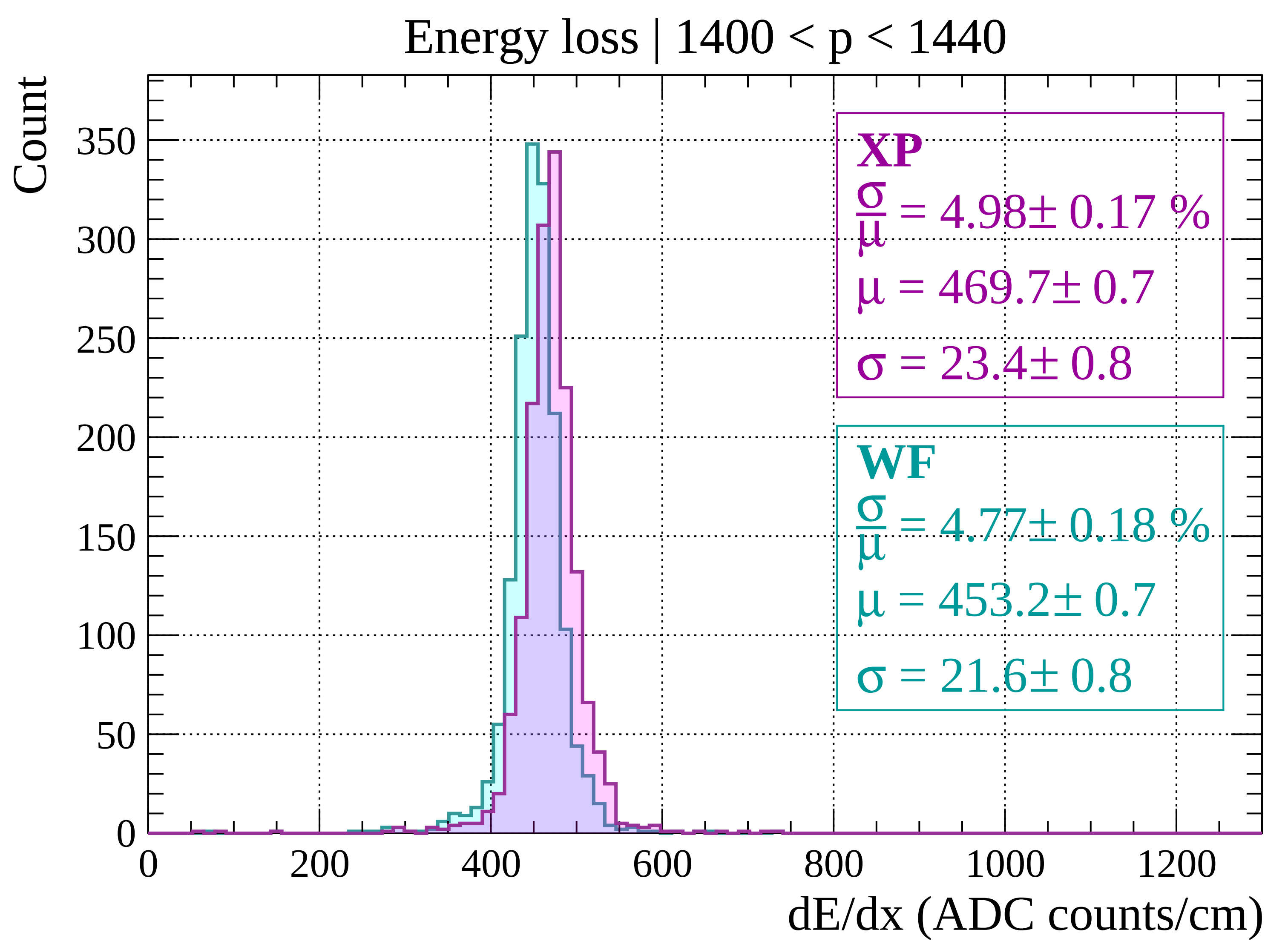

Energy loss | 1280 < p < 1320
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 5.57 ± 0.18 %
μ = 467.1 ± 0.8
σ = 26.0 ± 0.8
WF
σ/μ = 5.37 ± 0.18 %
μ = 450.2 ± 0.8
σ = 24.2 ± 0.8
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200



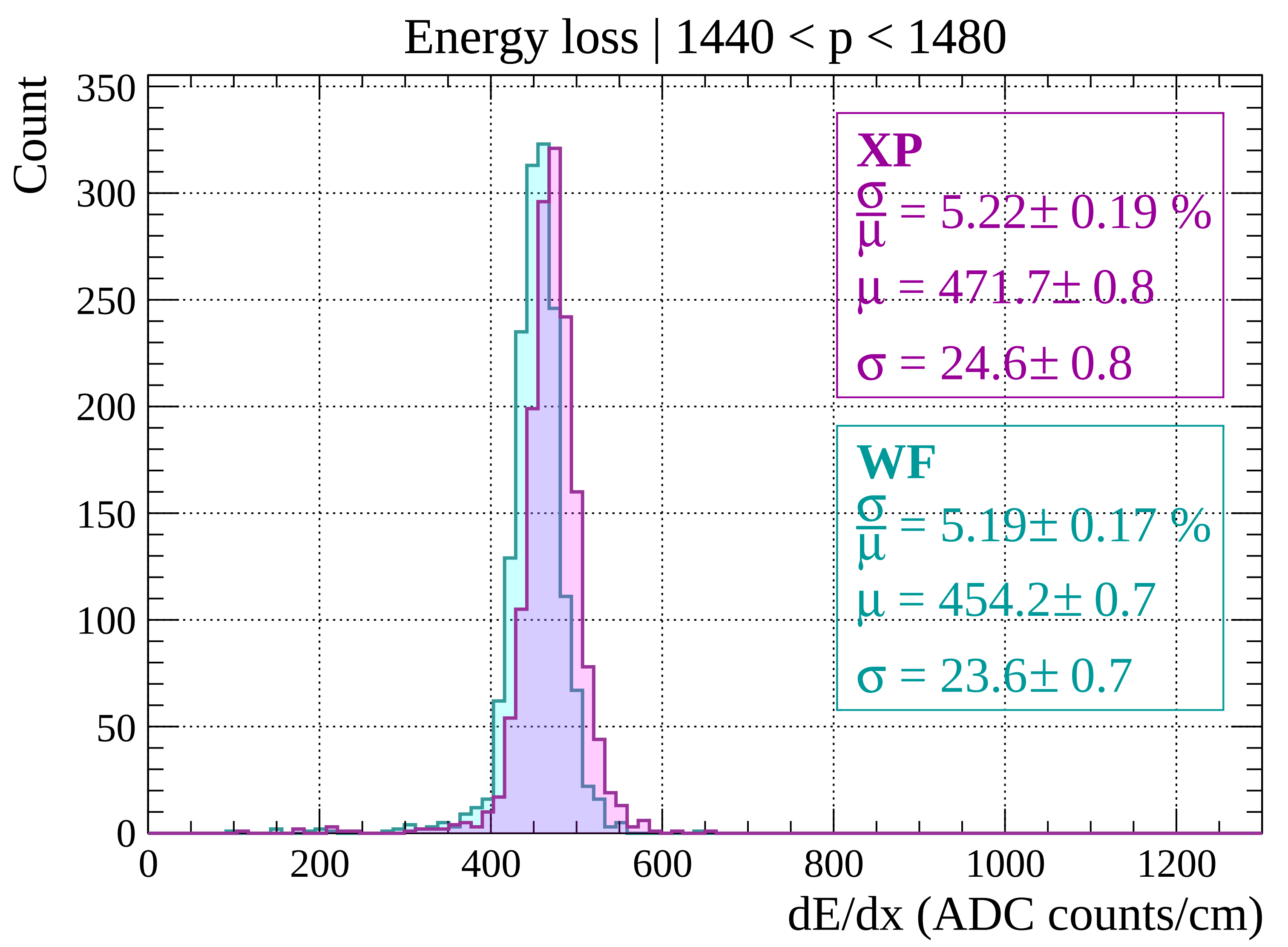
Energy loss | 1320 < p < 1360
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 5.24 \pm 0.16$ %
$\mu = 467.2 \pm 0.7$
$\sigma = 24.5 \pm 0.7$
WF
$\frac{\sigma}{\mu} = 5.00 \pm 0.16$ %
$\mu = 450.1 \pm 0.7$
$\sigma = 22.5 \pm 0.7$



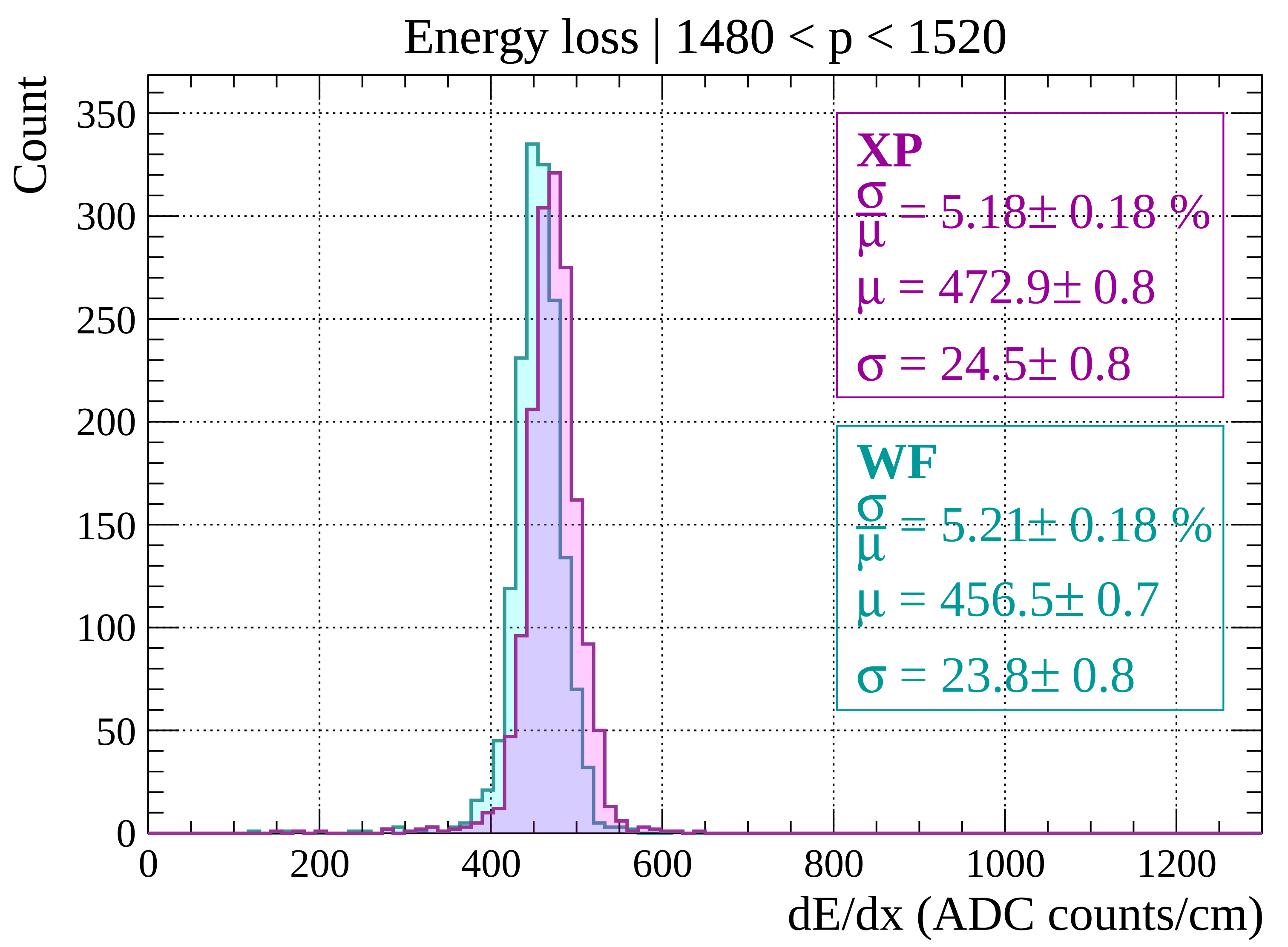
Energy loss | 1360 < p < 1400
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 5.50 \pm 0.17$ %
$\mu = 469.6 \pm 0.8$
$\sigma = 25.8 \pm 0.8$
WF
$\frac{\sigma}{\mu} = 5.42 \pm 0.15$ %
$\mu = 452.6 \pm 0.7$
$\sigma = 24.5 \pm 0.6$



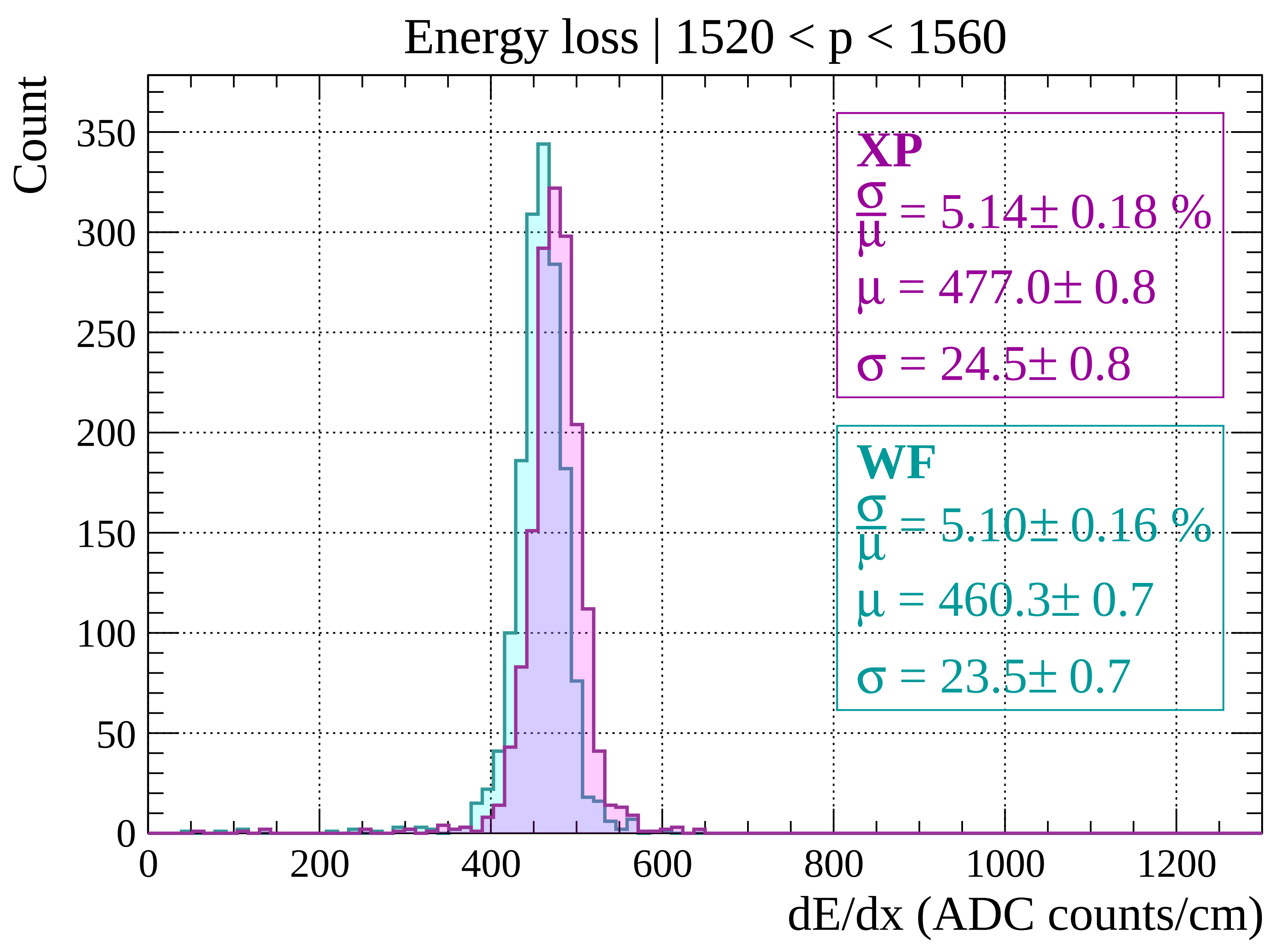
Energy loss | 1400 < p < 1440
Count
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 4.98± 0.17 %
μ = 469.7± 0.7
σ = 23.4± 0.8
WF
σ/μ = 4.77± 0.18 %
μ = 453.2± 0.7
σ = 21.6± 0.8



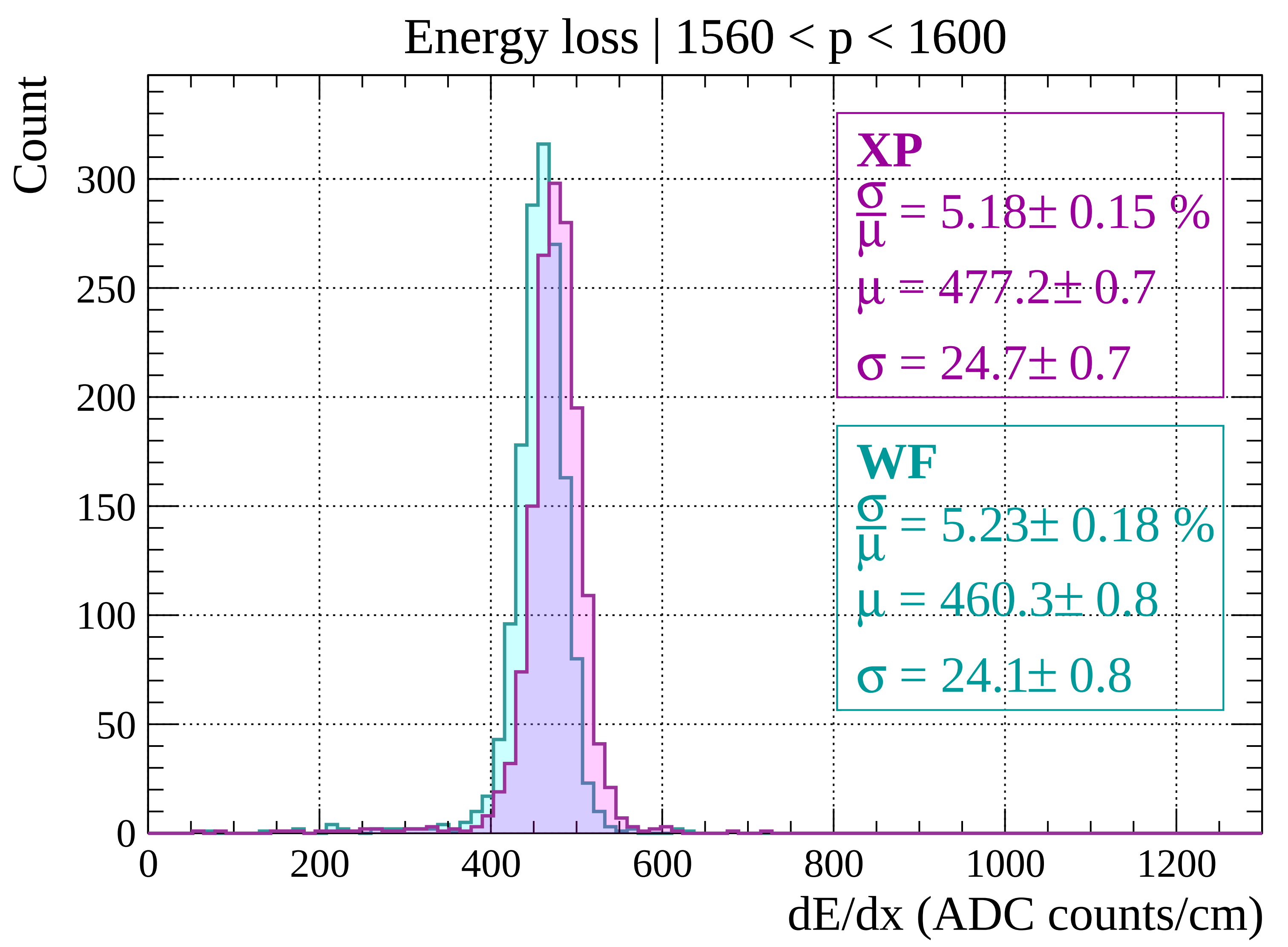
Energy loss | 1440 < p < 1480
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
XP
$\frac{\sigma}{\mu}$ = 5.22± 0.19 %
μ = 471.7± 0.8
σ = 24.6± 0.8
WF
$\frac{\sigma}{\mu}$ = 5.19± 0.17 %
μ = 454.2± 0.7
σ = 23.6± 0.7

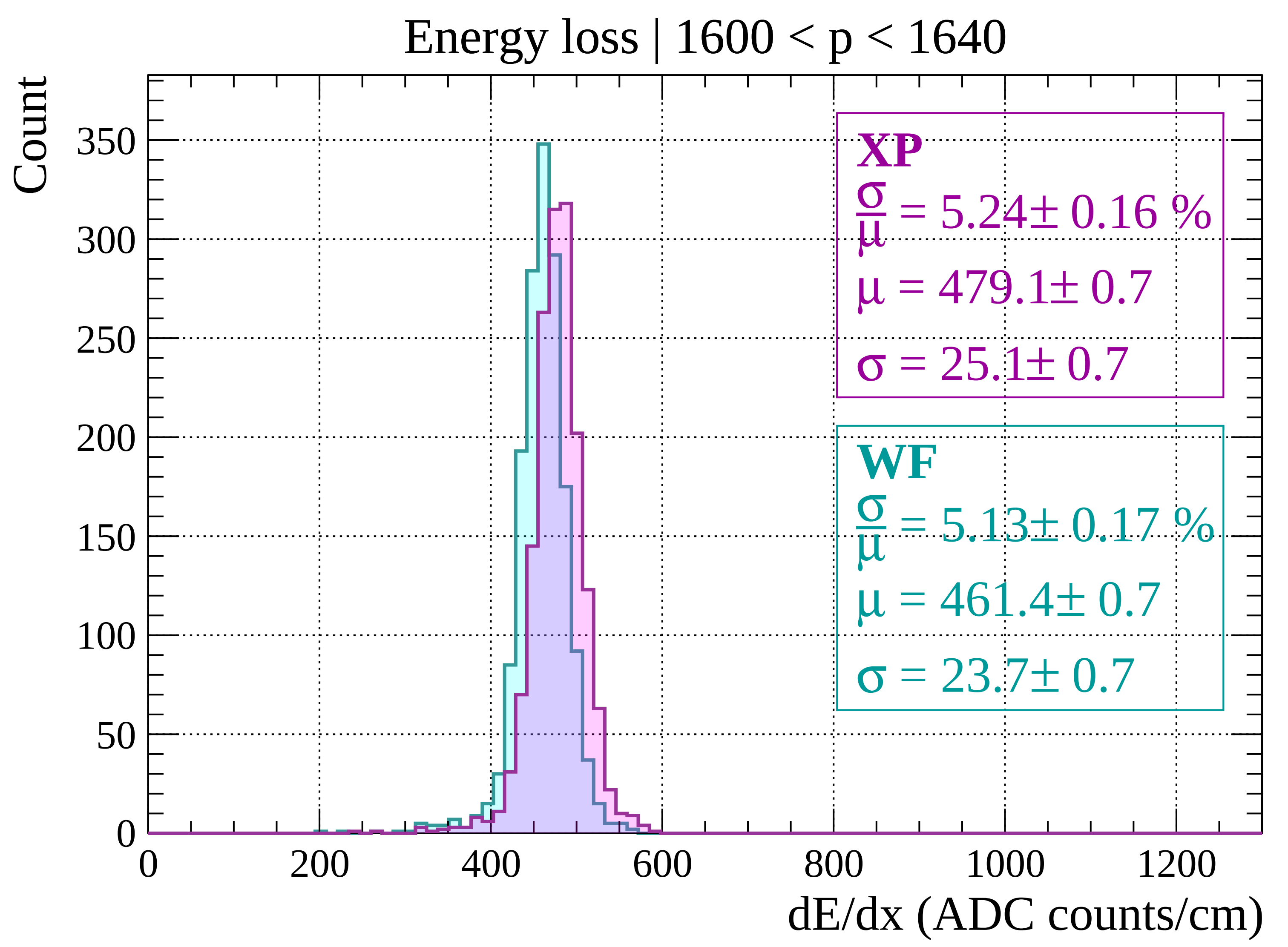


Energy loss | 1480 < p < 1520
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
σ/μ = 5.18± 0.18 %
μ = 472.9± 0.8
σ = 24.5± 0.8
WF
σ/μ = 5.21± 0.18 %
μ = 456.5± 0.7
σ = 23.8± 0.8

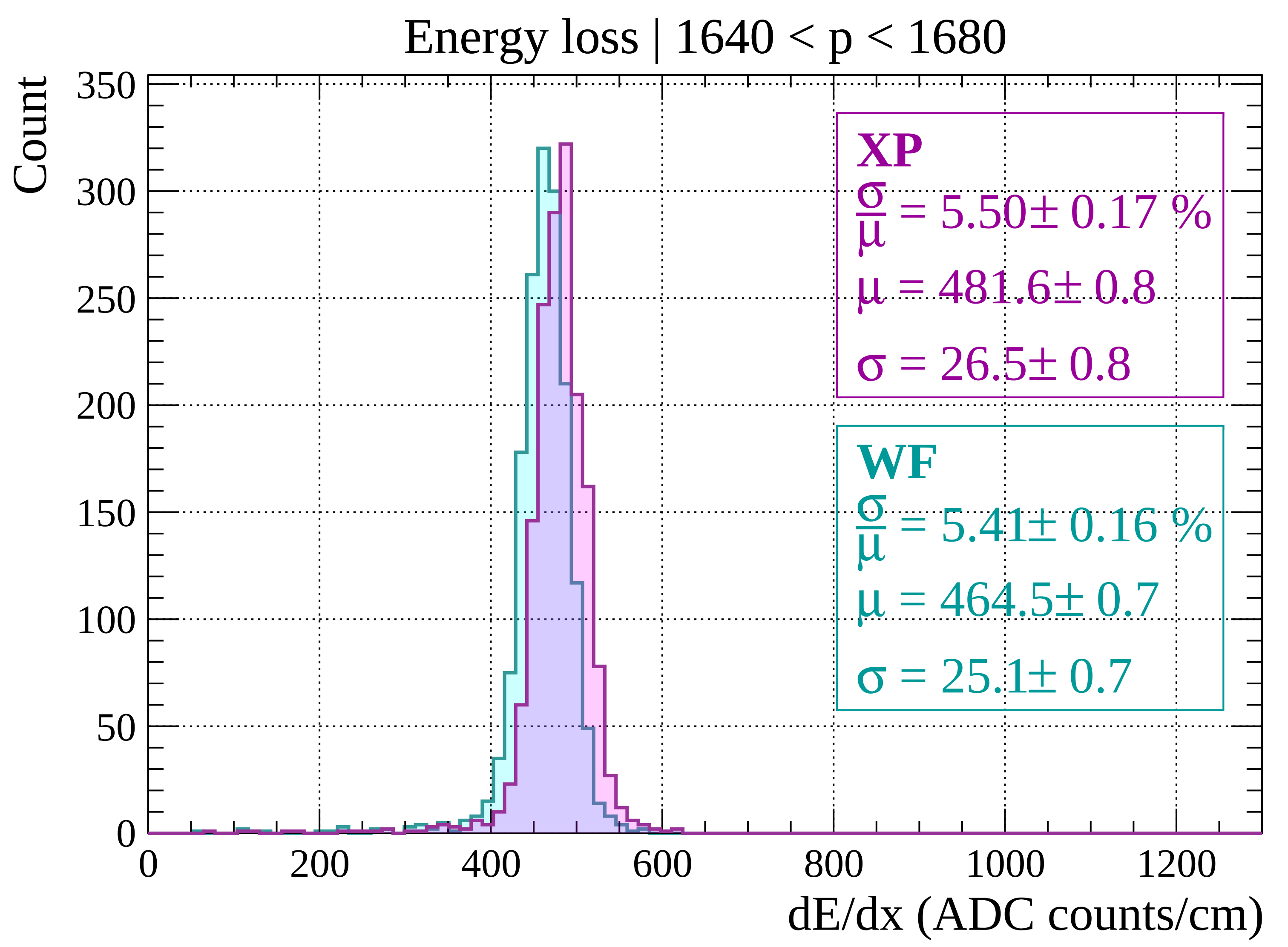


Energy loss | 1520 < p < 1560
Count
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 5.14± 0.18 %
μ = 477.0± 0.8
σ = 24.5± 0.8
WF
σ/μ = 5.10± 0.16 %
μ = 460.3± 0.7
σ = 23.5± 0.7

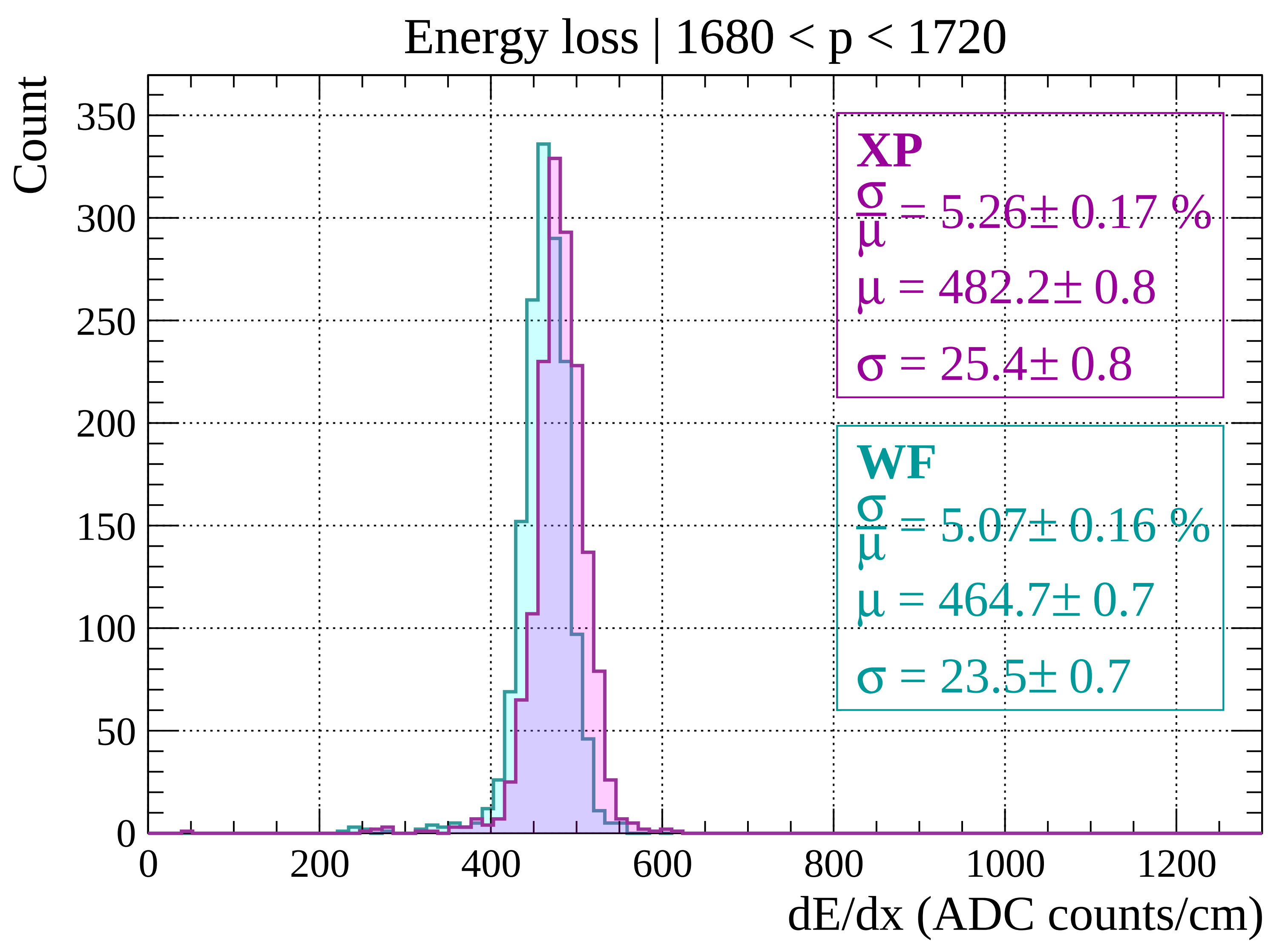

Energy loss | 1560 < p < 1600
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
XP
σ/μ = 5.18± 0.15 %
μ = 477.2± 0.7
σ = 24.7± 0.7
WF
σ/μ = 5.23± 0.18 %
μ = 460.3± 0.8
σ = 24.1± 0.8



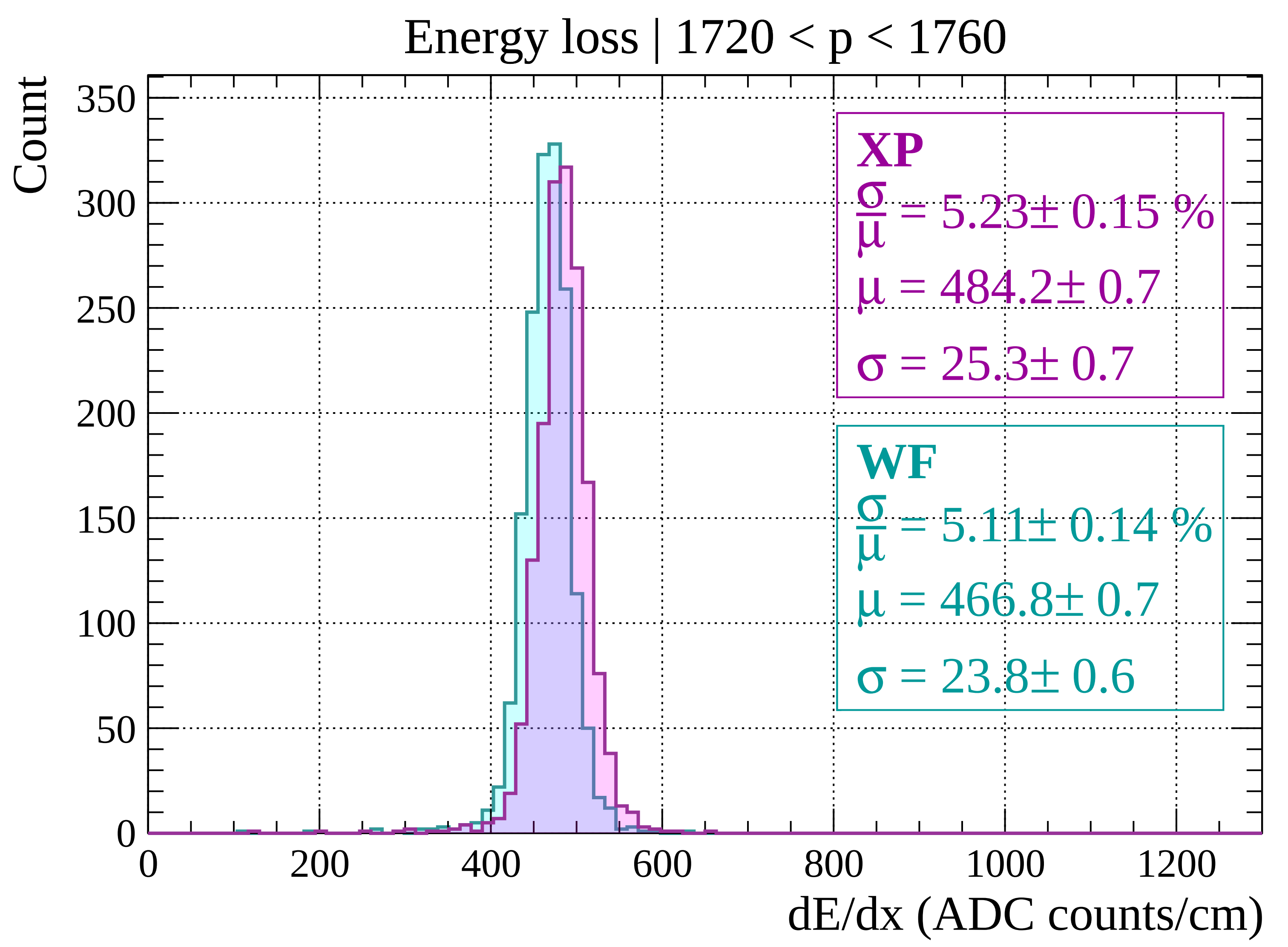
Energy loss | 1600 < p < 1640
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
σ/μ = 5.24± 0.16 %
μ = 479.1± 0.7
σ = 25.1± 0.7
WF
σ/μ = 5.13± 0.17 %
μ = 461.4± 0.7
σ = 23.7± 0.7

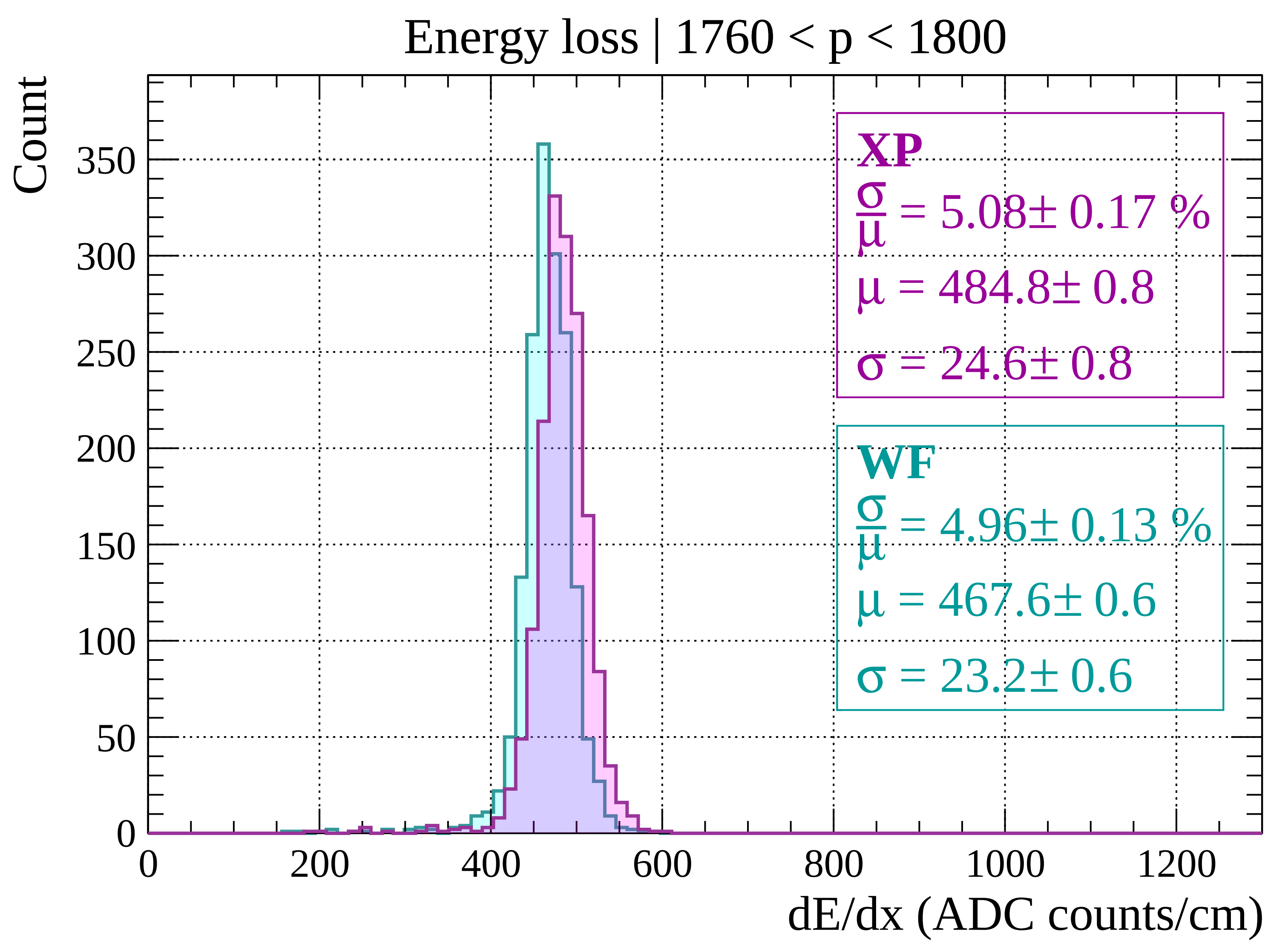


Energy loss | 1640 < p < 1680
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 5.50± 0.17 %
μ = 481.6± 0.8
σ = 26.5± 0.8
WF
$\frac{\sigma}{\mu}$ = 5.41± 0.16 %
μ = 464.5± 0.7
σ = 25.1± 0.7

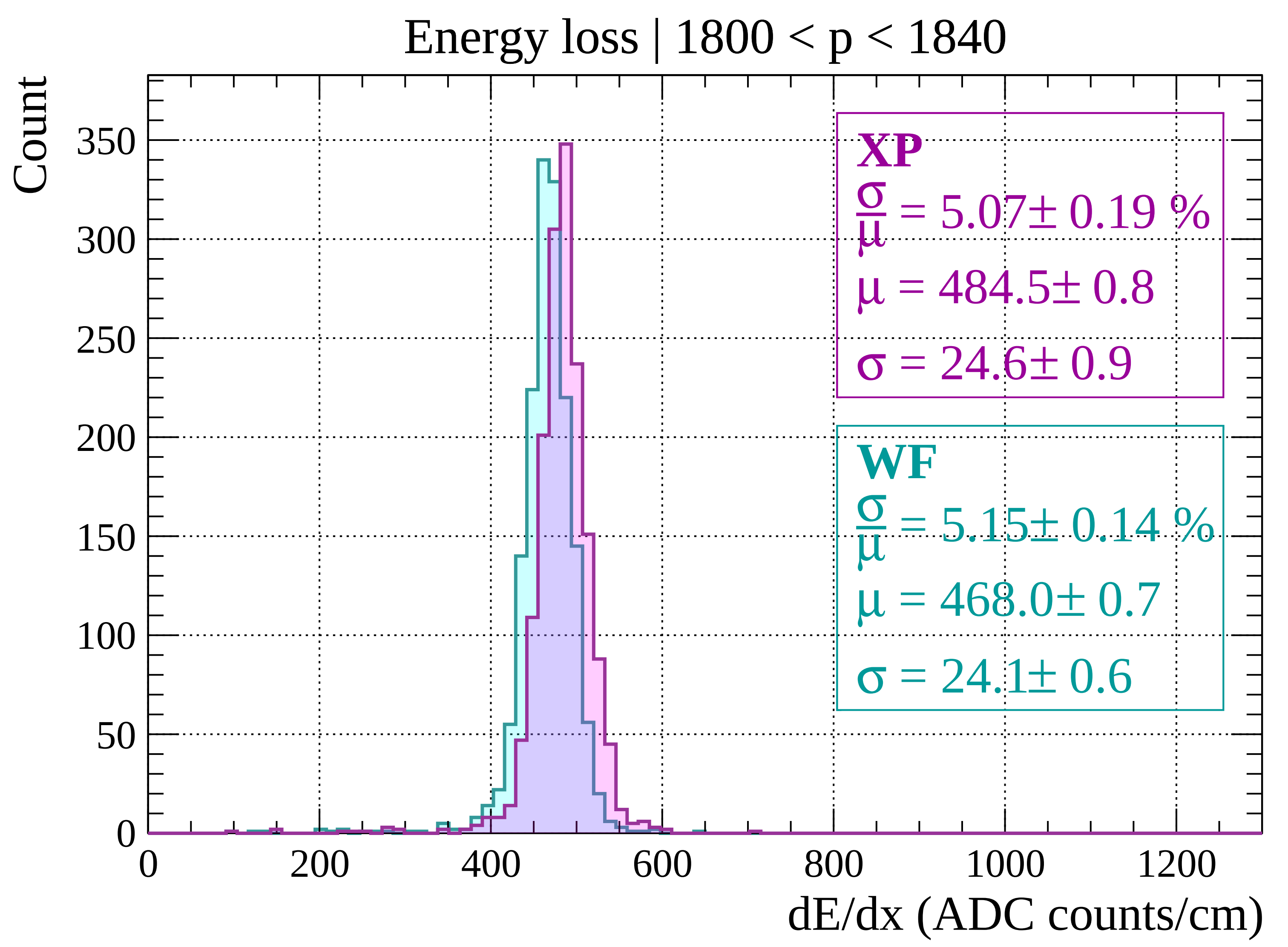


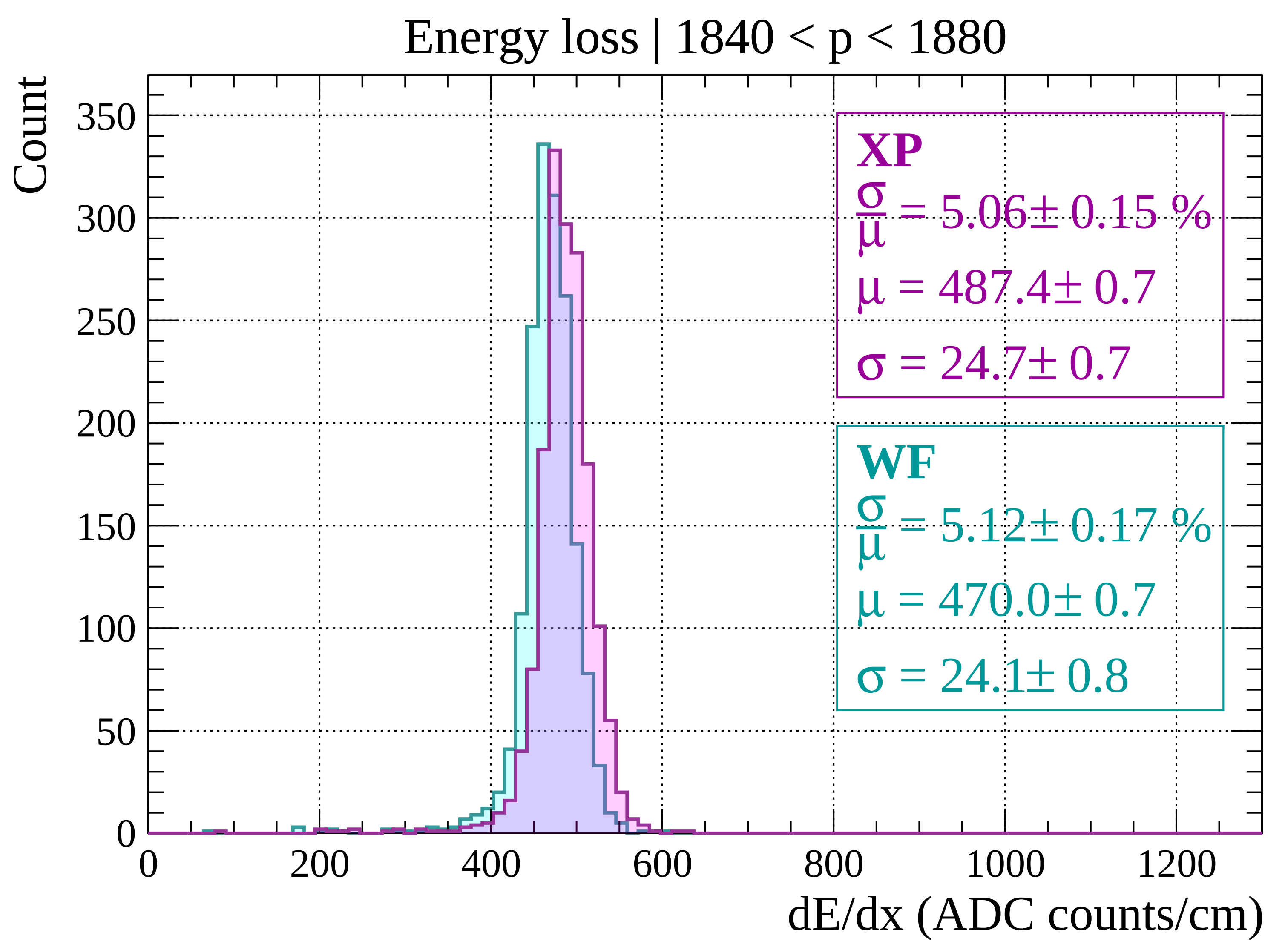
Energy loss | 1680 < p < 1720
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 5.26 \pm 0.17$ %
$\mu = 482.2 \pm 0.8$
$\sigma = 25.4 \pm 0.8$
WF
$\frac{\sigma}{\mu} = 5.07 \pm 0.16$ %
$\mu = 464.7 \pm 0.7$
$\sigma = 23.5 \pm 0.7$


Energy loss | 1720 < p < 1760
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 5.23± 0.15 %
μ = 484.2± 0.7
σ = 25.3± 0.7
WF
$\frac{\sigma}{\mu}$ = 5.11± 0.14 %
μ = 466.8± 0.7
σ = 23.8± 0.6

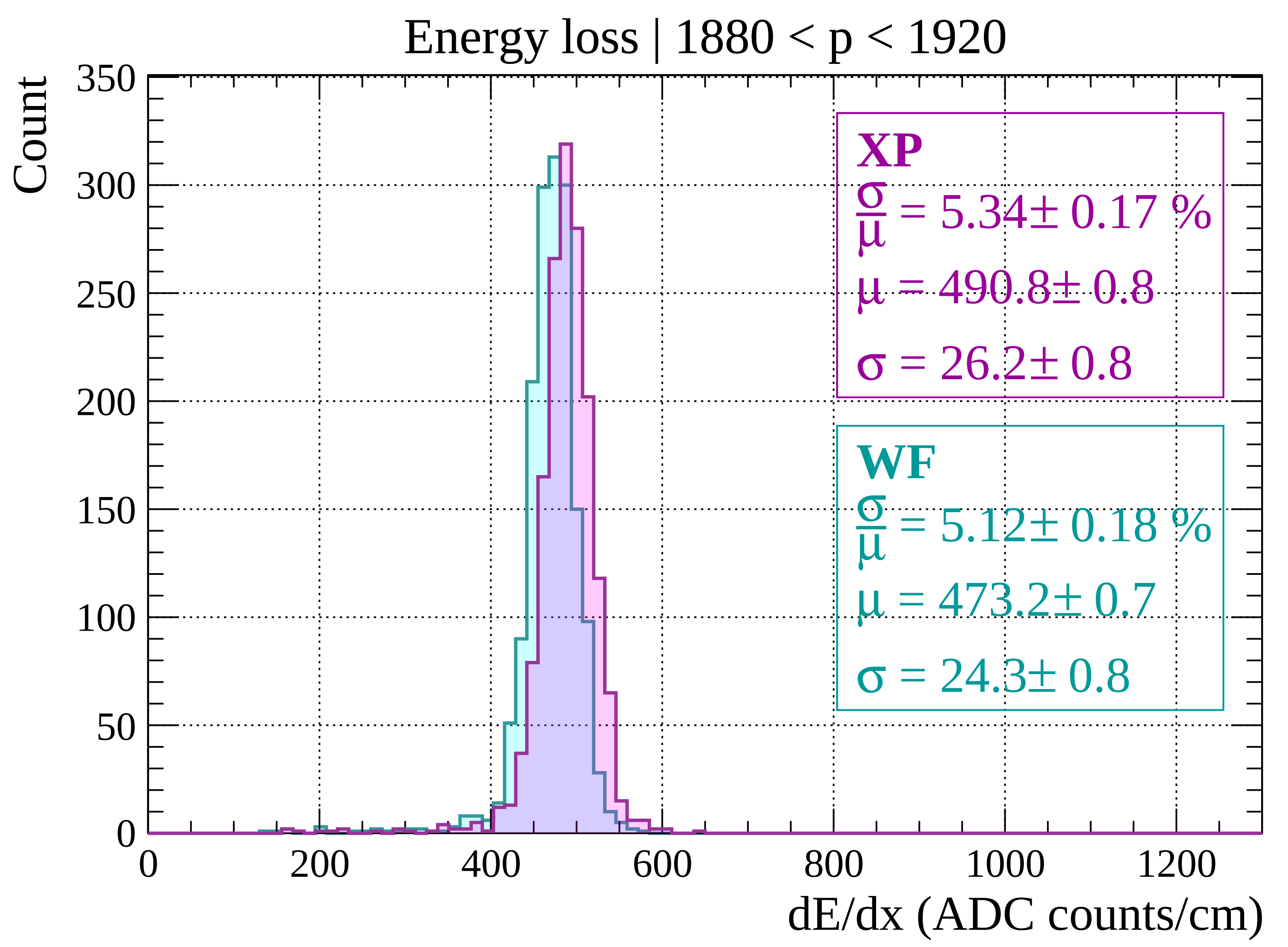


Energy loss | 1760 < p < 1800
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 5.08 \pm 0.17$ %
$\mu = 484.8 \pm 0.8$
$\sigma = 24.6 \pm 0.8$
WF
$\frac{\sigma}{\mu} = 4.96 \pm 0.13$ %
$\mu = 467.6 \pm 0.6$
$\sigma = 23.2 \pm 0.6$
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200

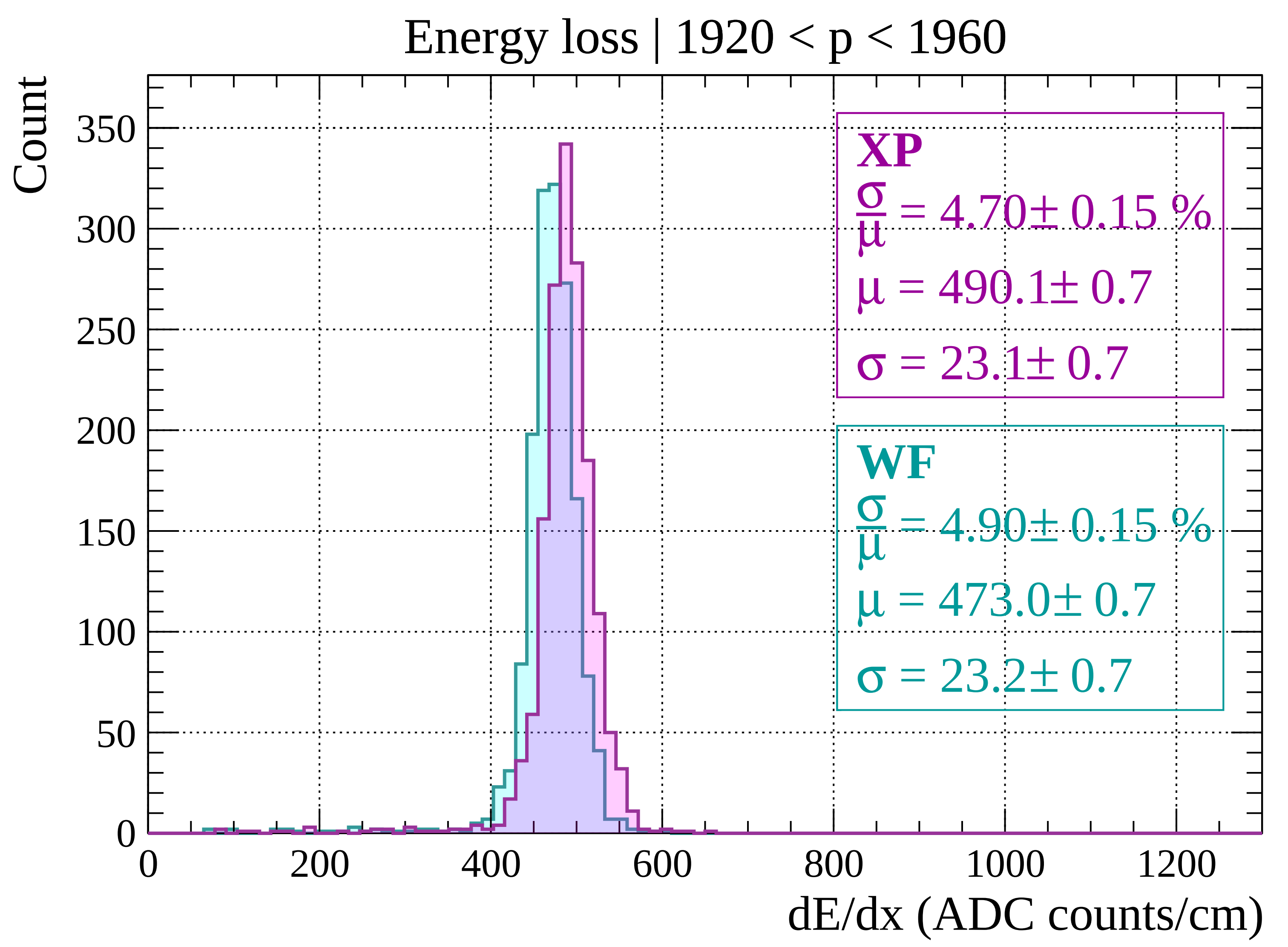

Energy loss | 1800 < p < 1840
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 5.07± 0.19 %
μ = 484.5± 0.8
σ = 24.6± 0.9
WF
$\frac{\sigma}{\mu}$ = 5.15± 0.14 %
μ = 468.0± 0.7
σ = 24.1± 0.6



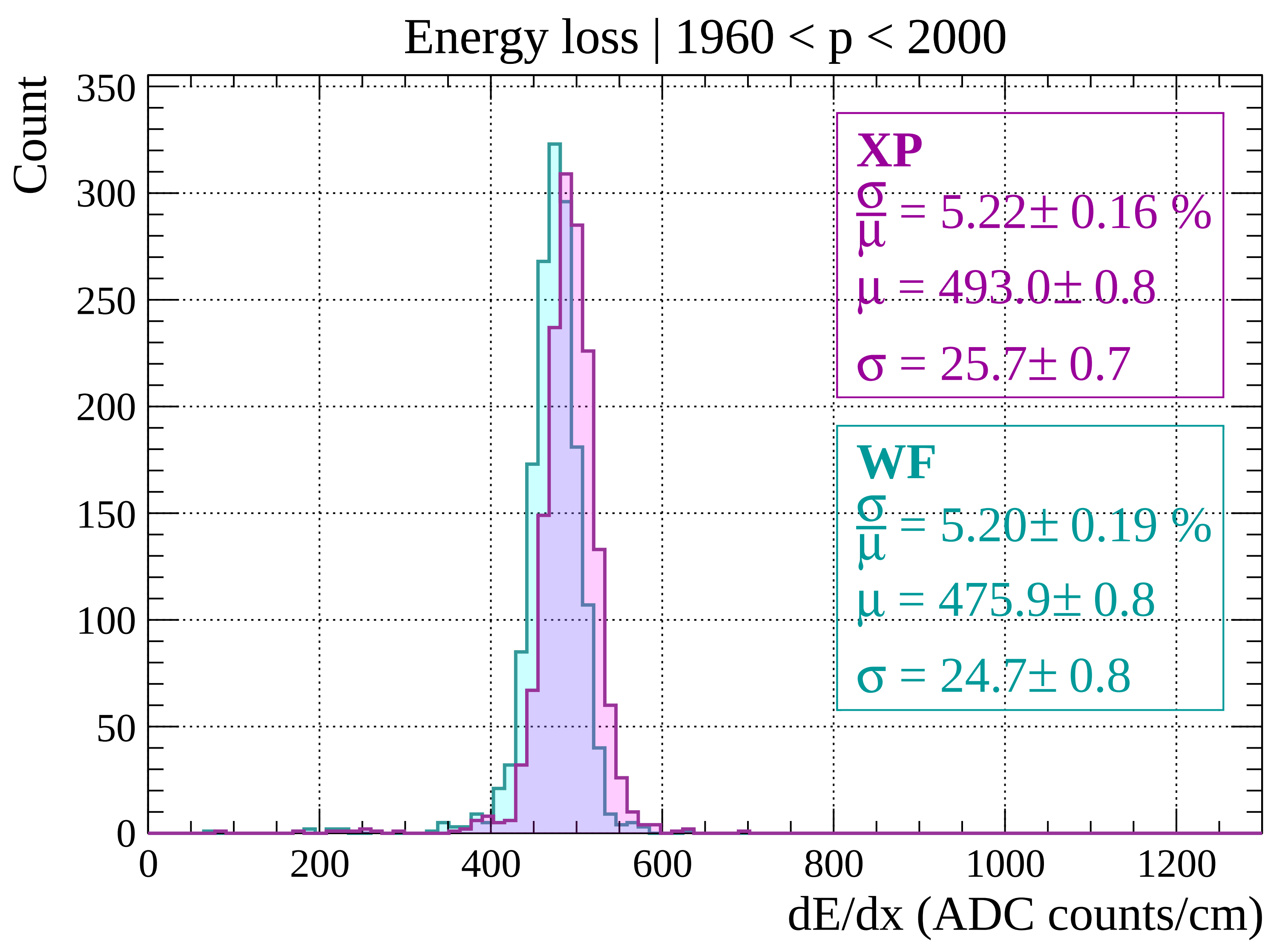
Energy loss | 1840 < p < 1880
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
σ/μ = 5.06± 0.15 %
μ = 487.4± 0.7
σ = 24.7± 0.7
WF
σ/μ = 5.12± 0.17 %
μ = 470.0± 0.7
σ = 24.1± 0.8

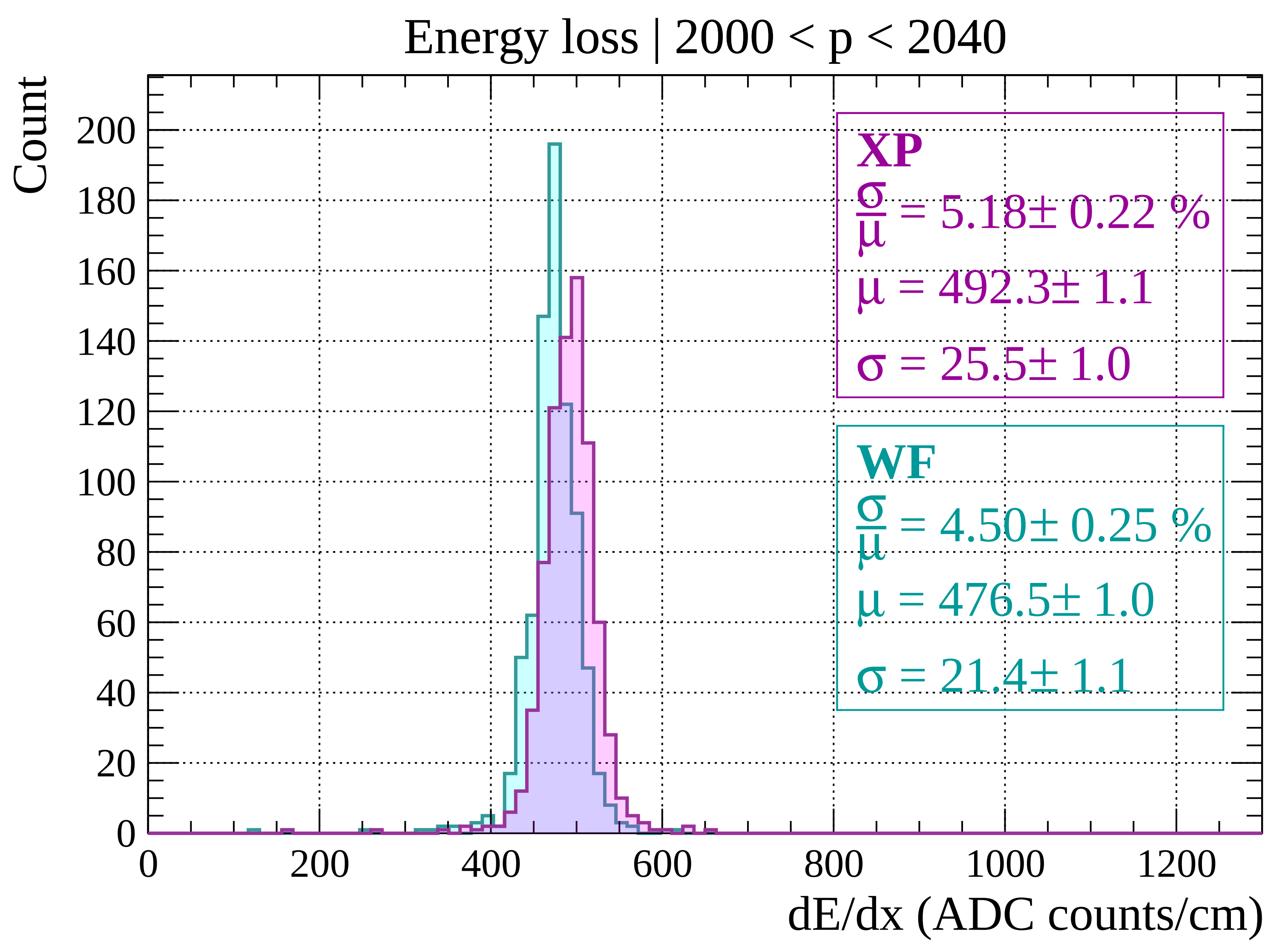


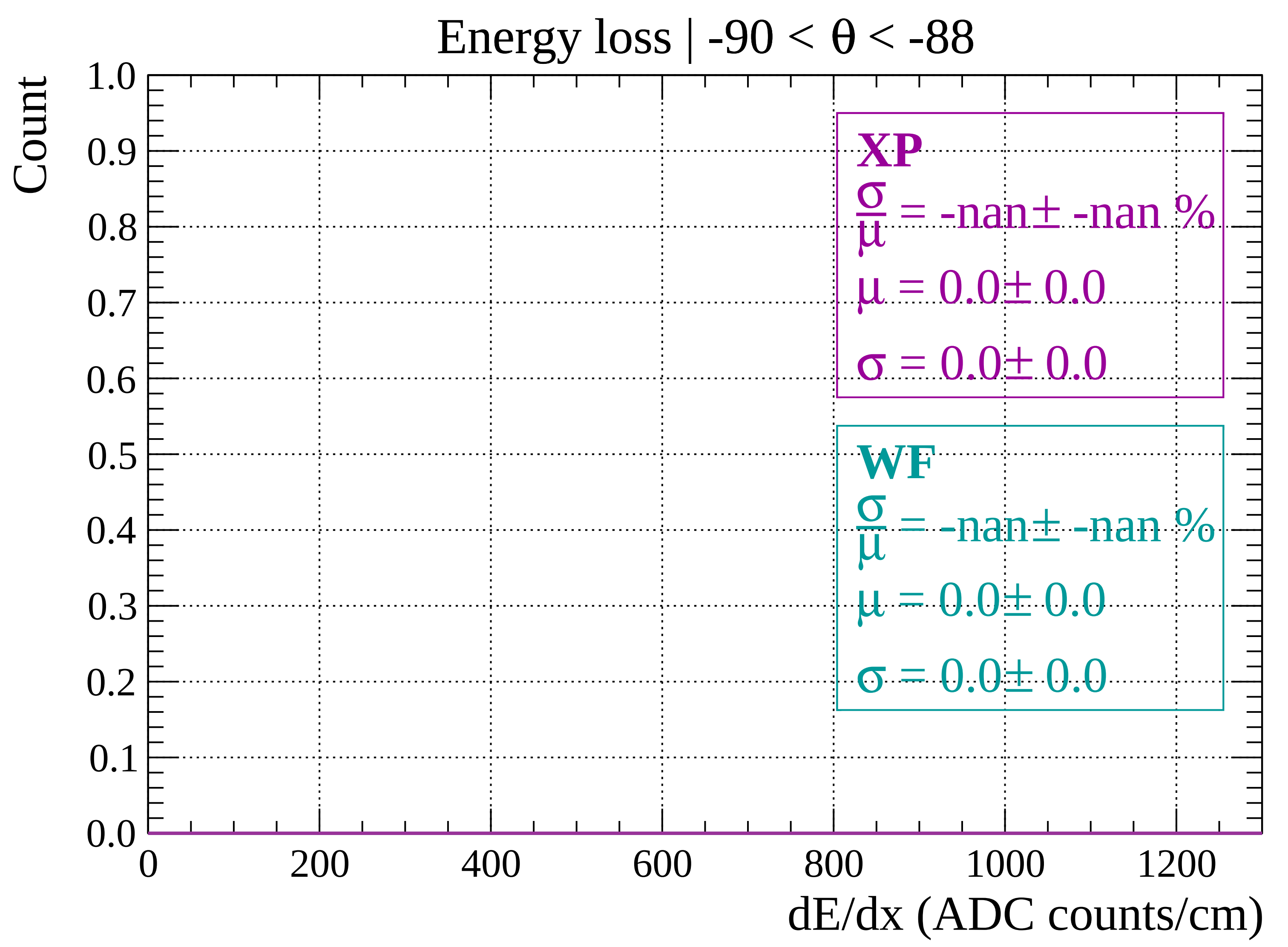
Energy loss | 1880 < p < 1920
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 5.34± 0.17 %
μ = 490.8± 0.8
σ = 26.2± 0.8
WF
$\frac{\sigma}{\mu}$ = 5.12± 0.18 %
μ = 473.2± 0.7
σ = 24.3± 0.8

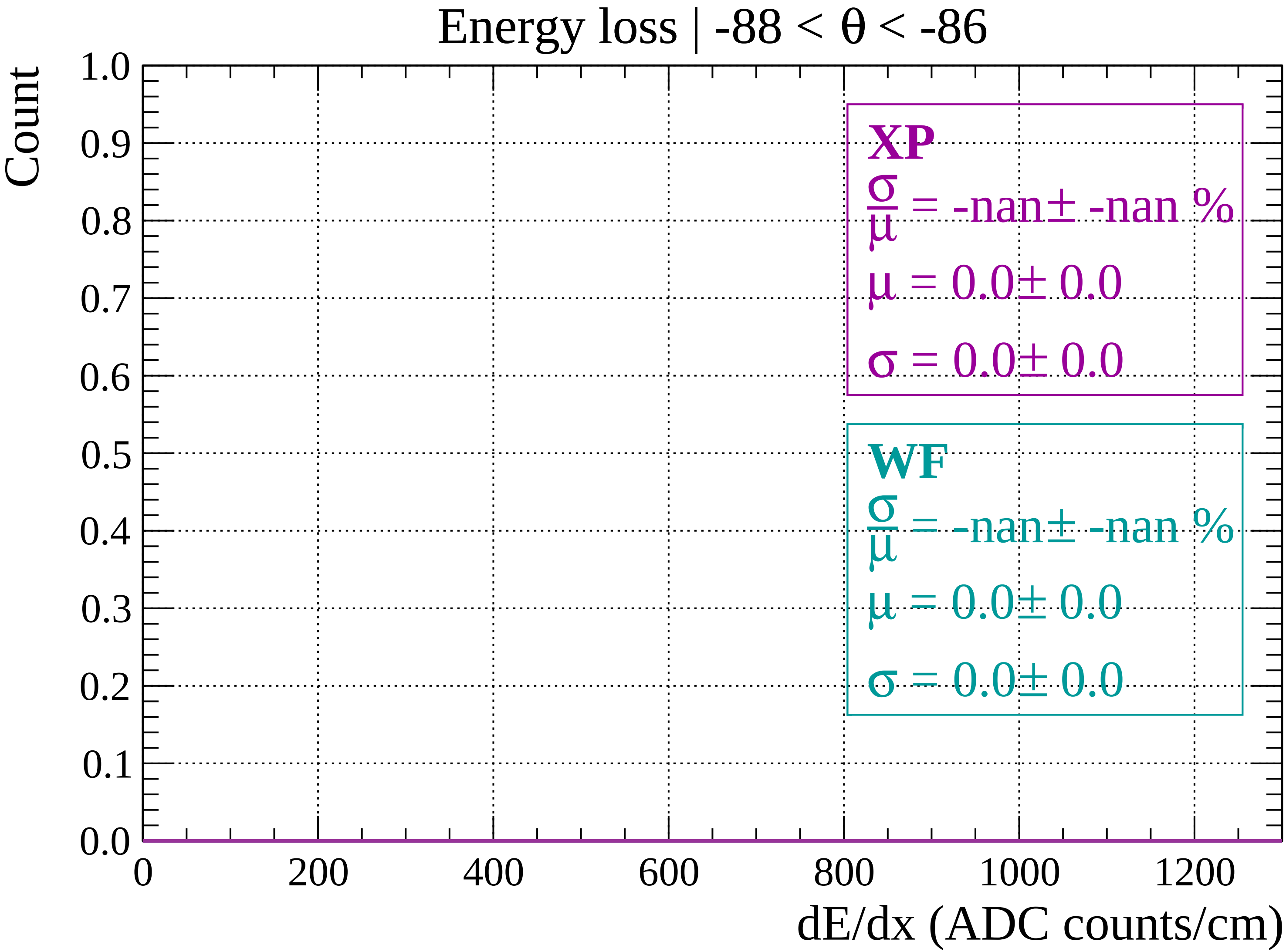


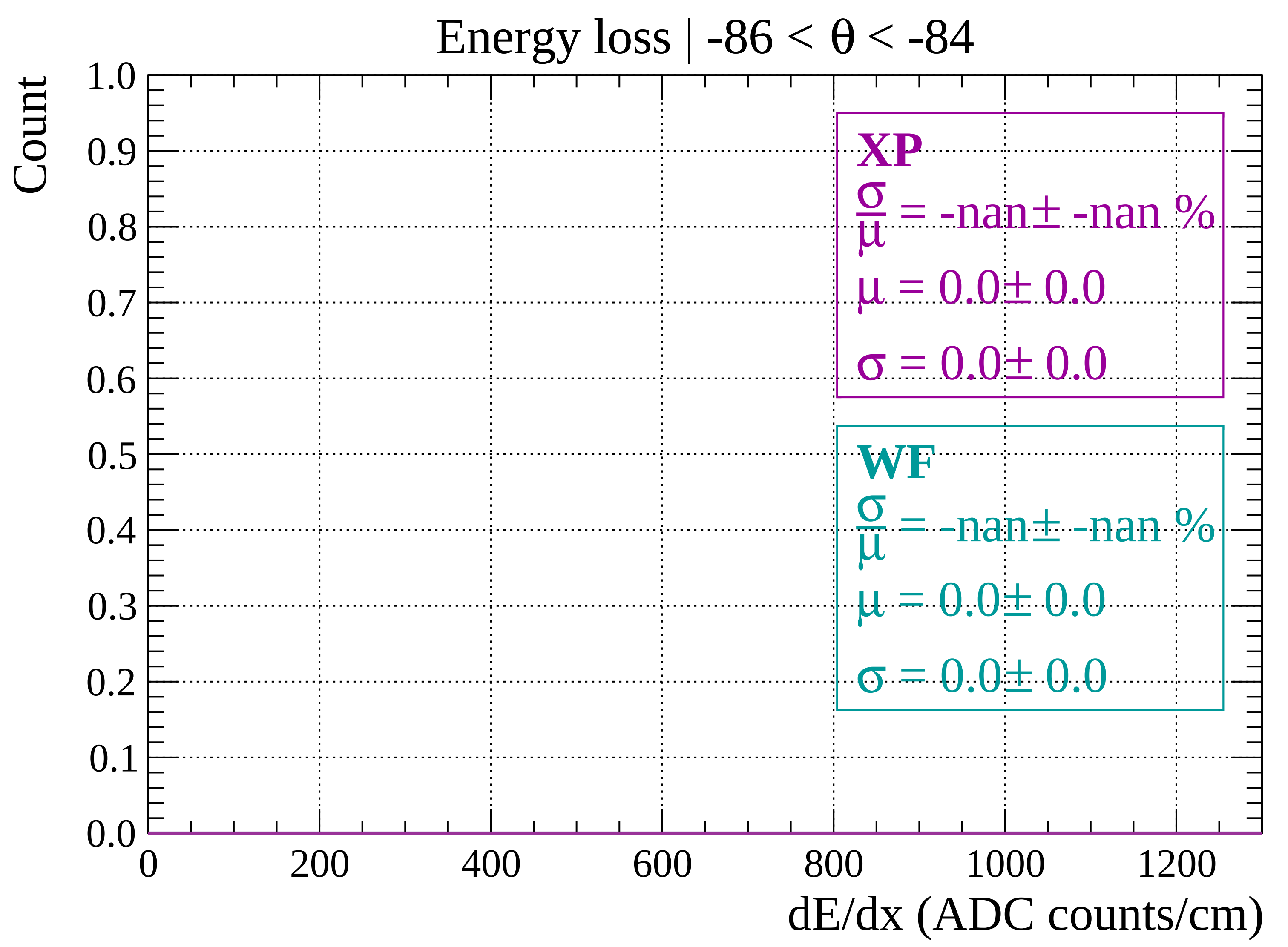
Energy loss | 1920 < p < 1960
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 4.70 \pm 0.15$ %
$\mu = 490.1 \pm 0.7$
$\sigma = 23.1 \pm 0.7$
WF
$\frac{\sigma}{\mu} = 4.90 \pm 0.15$ %
$\mu = 473.0 \pm 0.7$
$\sigma = 23.2 \pm 0.7$

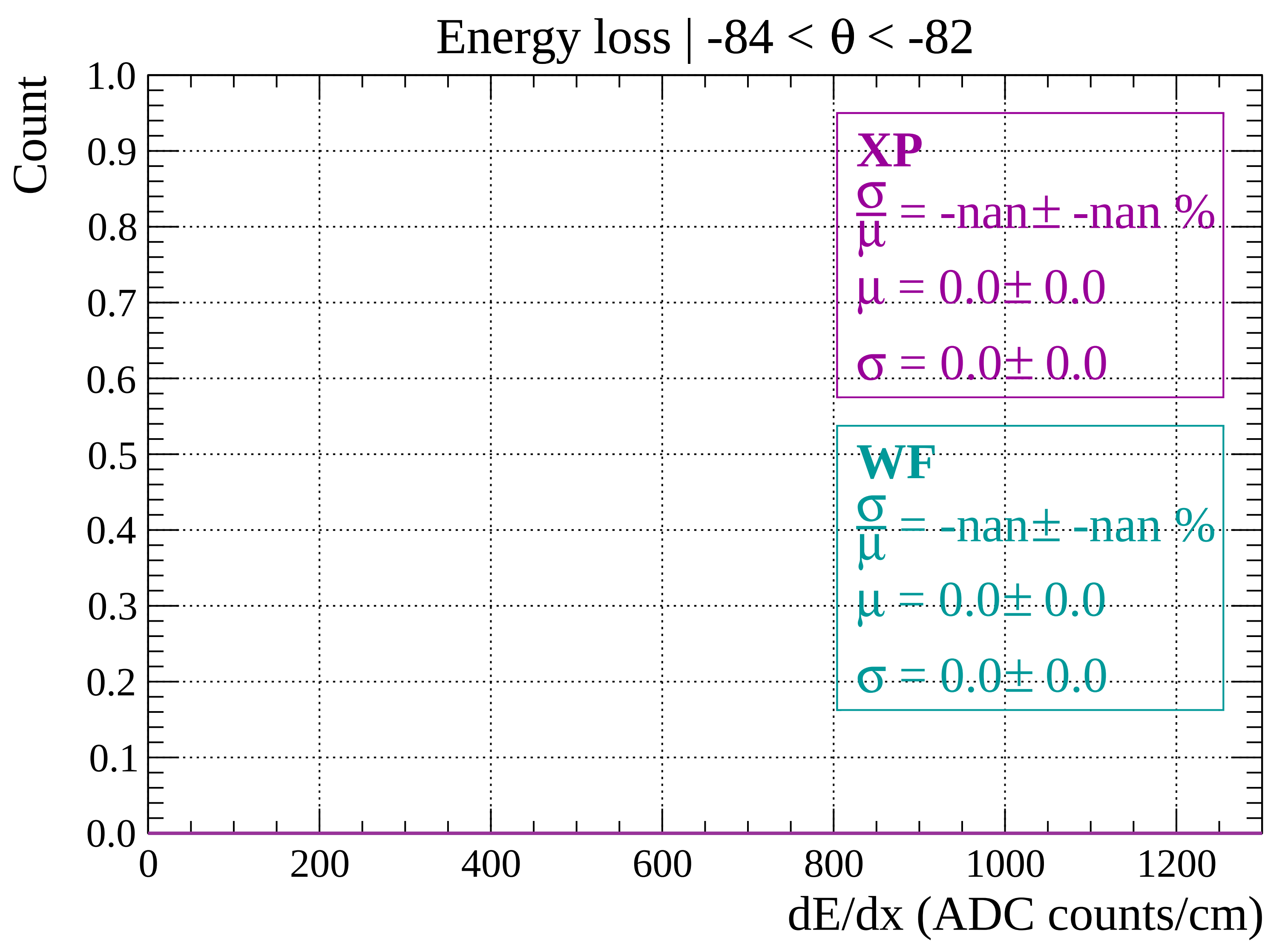

Energy loss | 1960 < p < 2000
Count
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 5.22 ± 0.16 %
μ = 493.0 ± 0.8
σ = 25.7 ± 0.7
WF
σ/μ = 5.20 ± 0.19 %
μ = 475.9 ± 0.8
σ = 24.7 ± 0.8

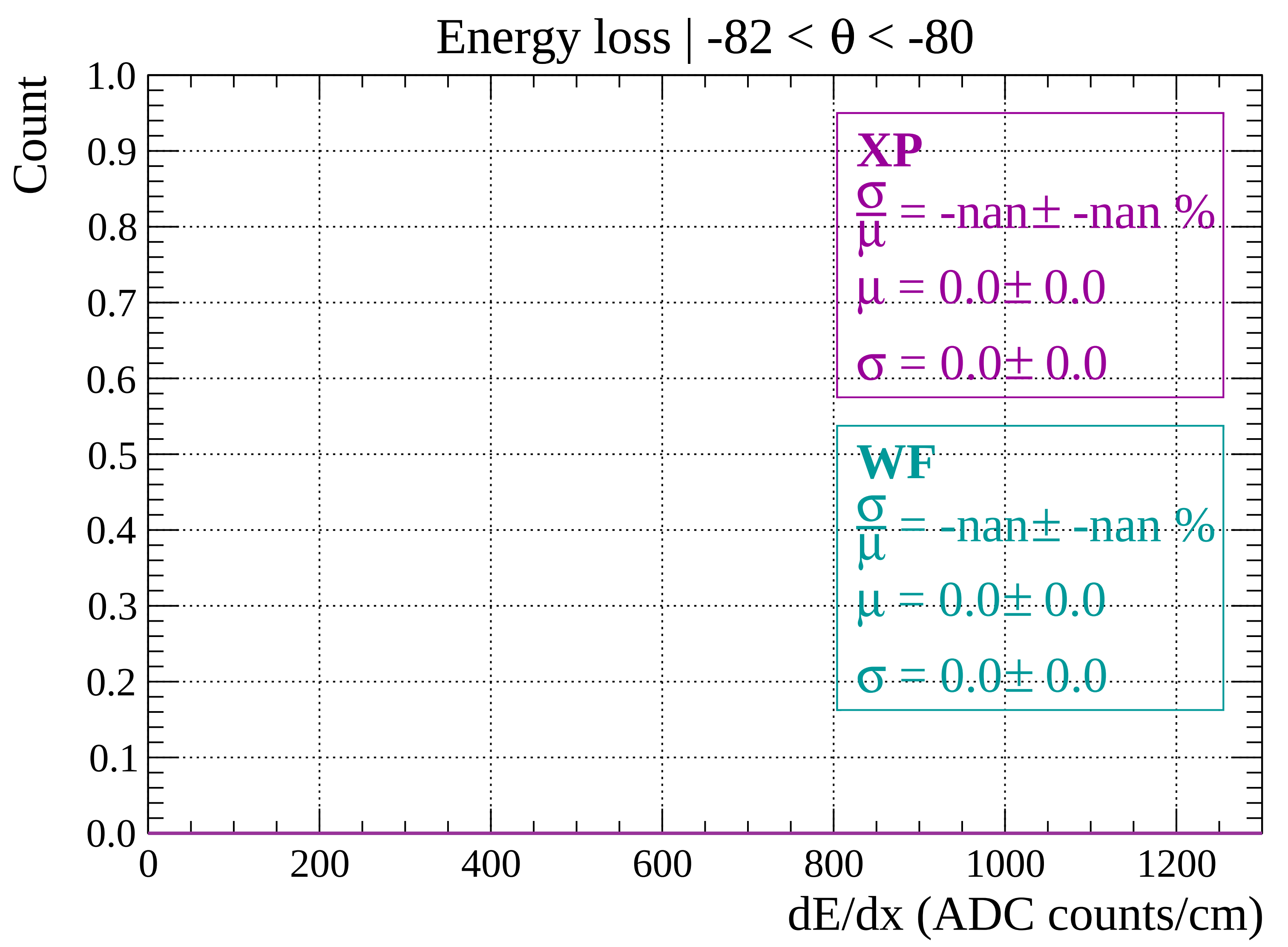

Energy loss | 2000 < p < 2040
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
XP
$\frac{\sigma}{\mu}$ = 5.18± 0.22 %
μ = 492.3± 1.1
σ = 25.5± 1.0
WF
$\frac{\sigma}{\mu}$ = 4.50± 0.25 %
μ = 476.5± 1.0
σ = 21.4± 1.1


Energy loss | -90 < θ < -88
Count
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

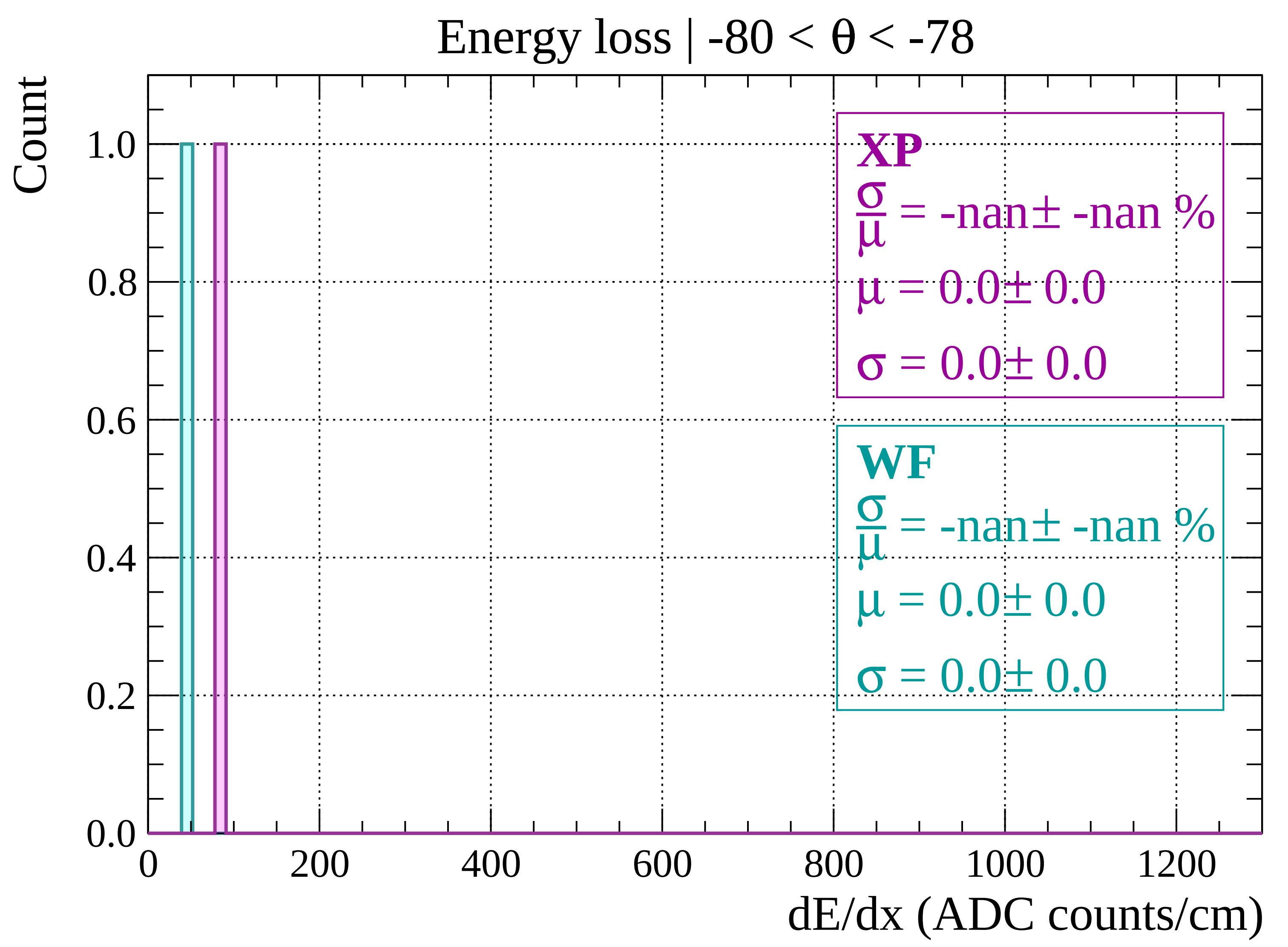

Energy loss | -88 < θ < -86
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

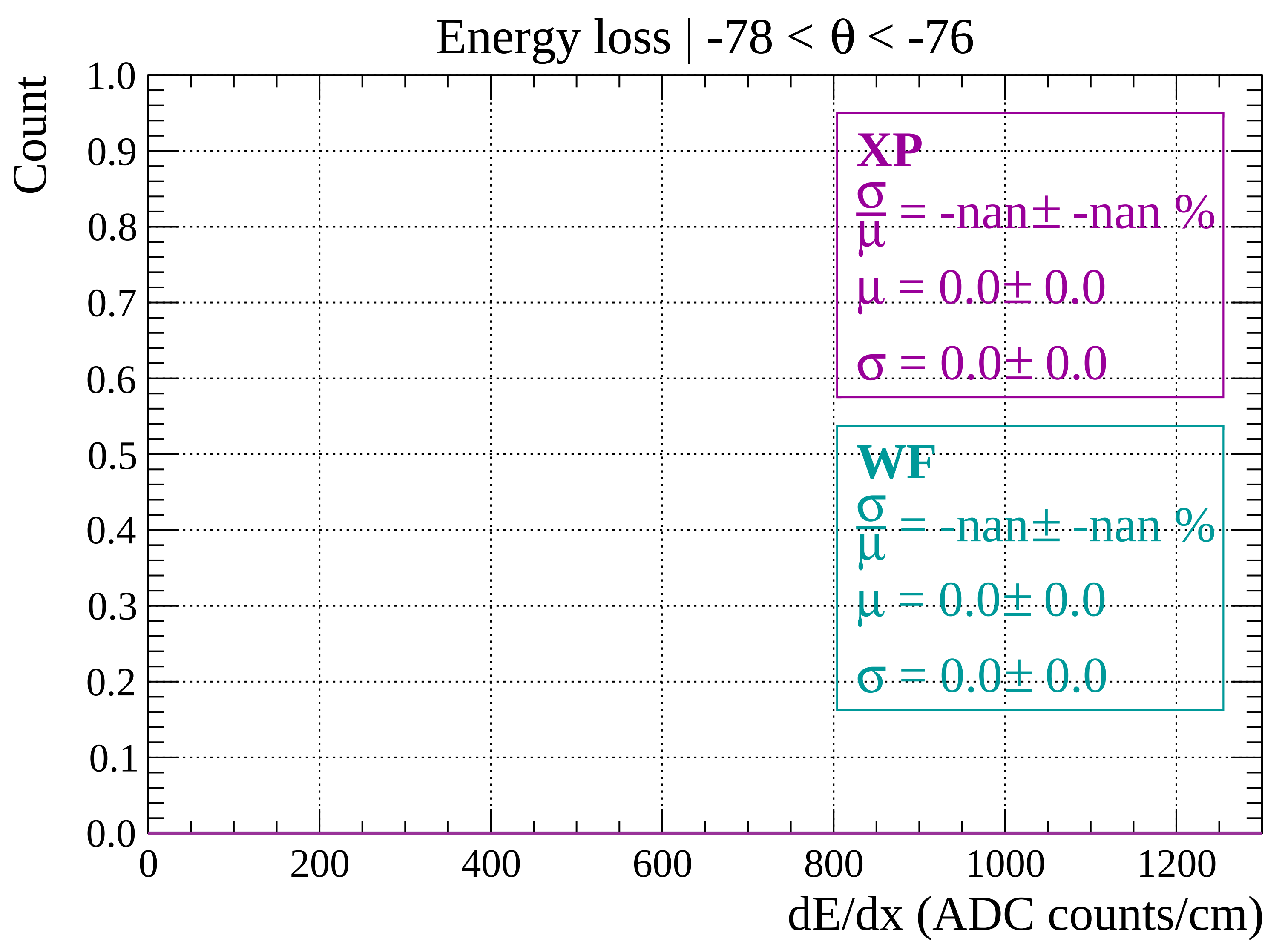

Energy loss | -86 < θ < -84
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | -84 < θ < -82
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

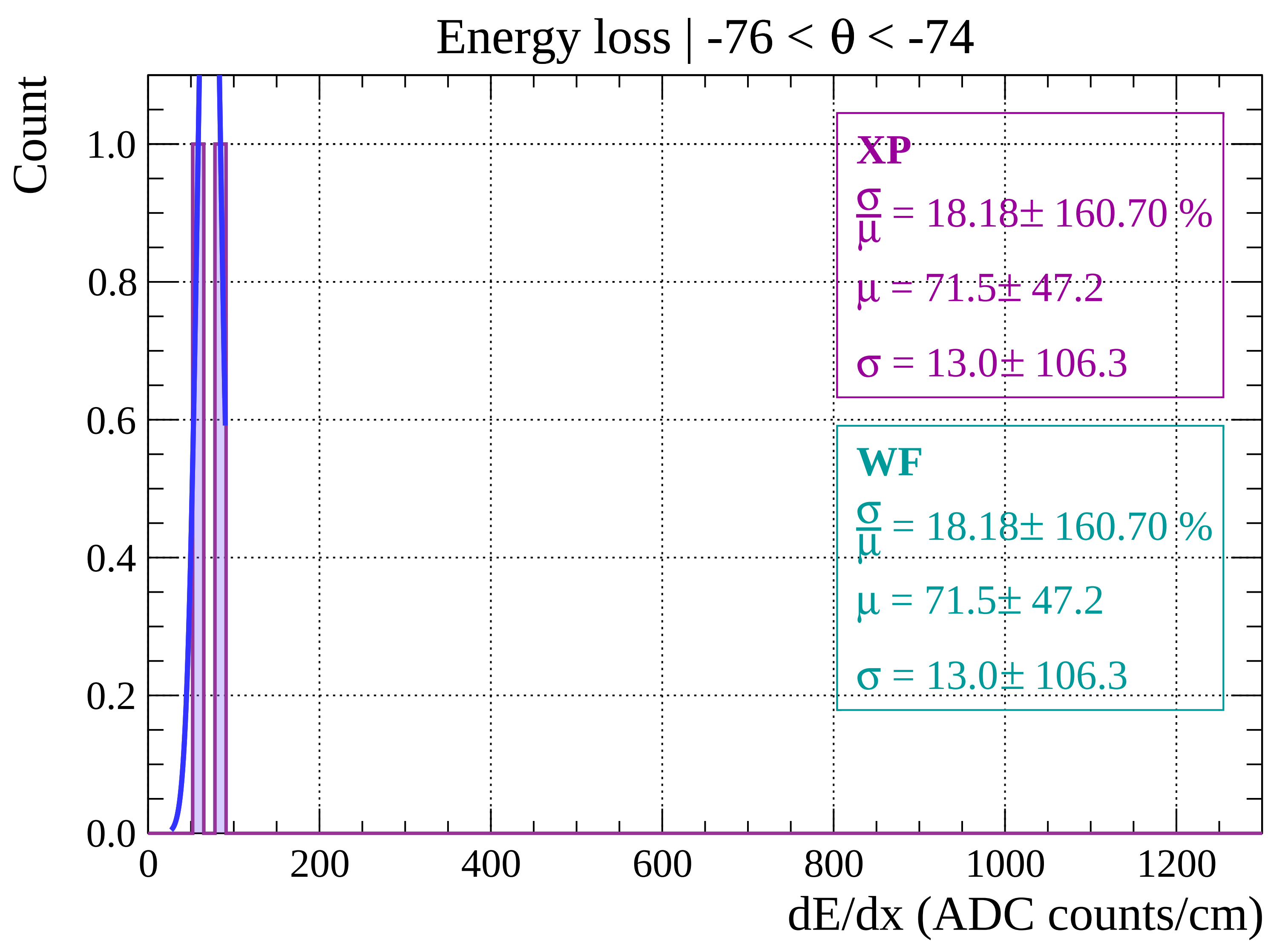


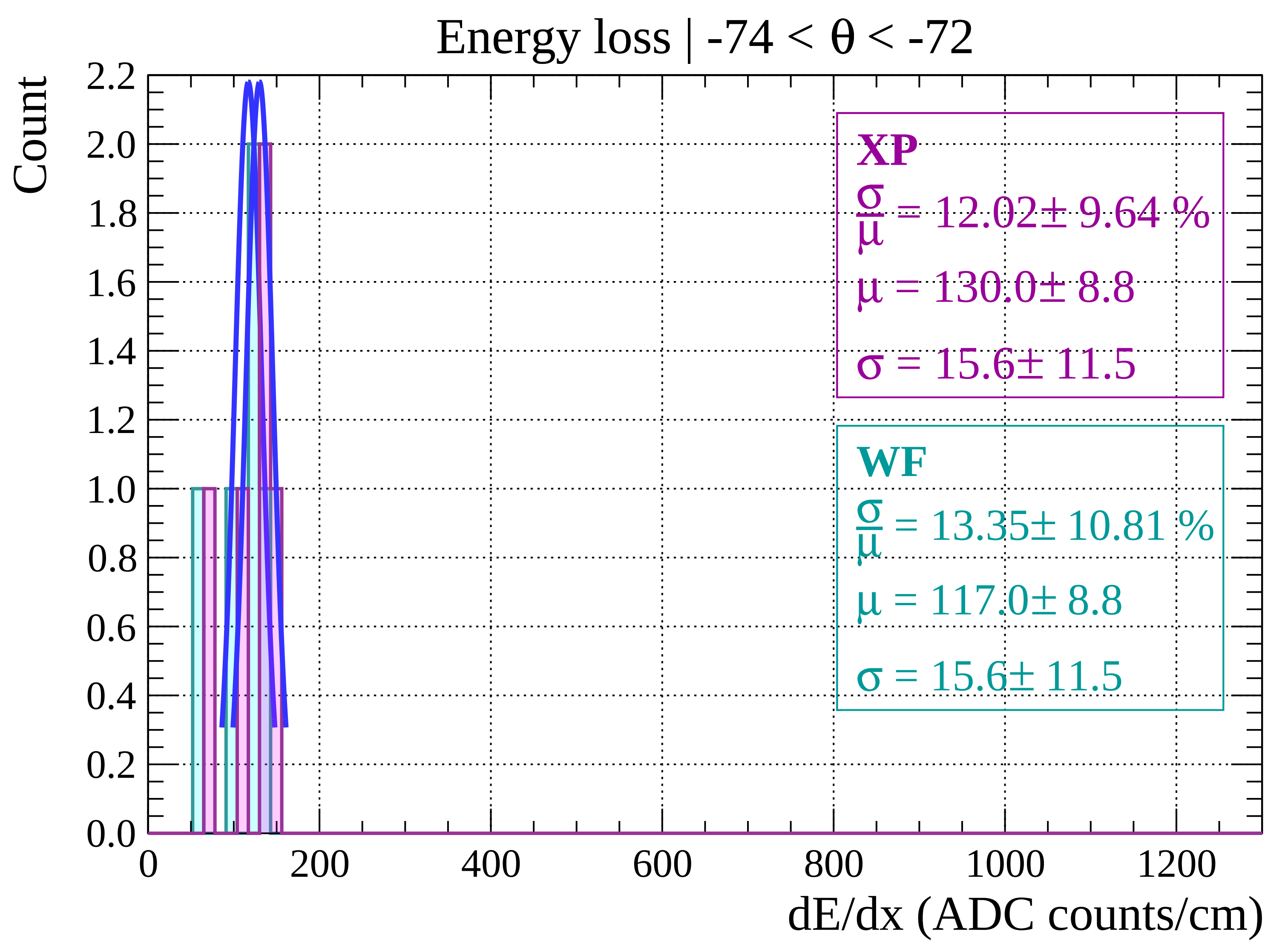
Energy loss | -82 < θ < -80
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | -80 < θ < -78
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

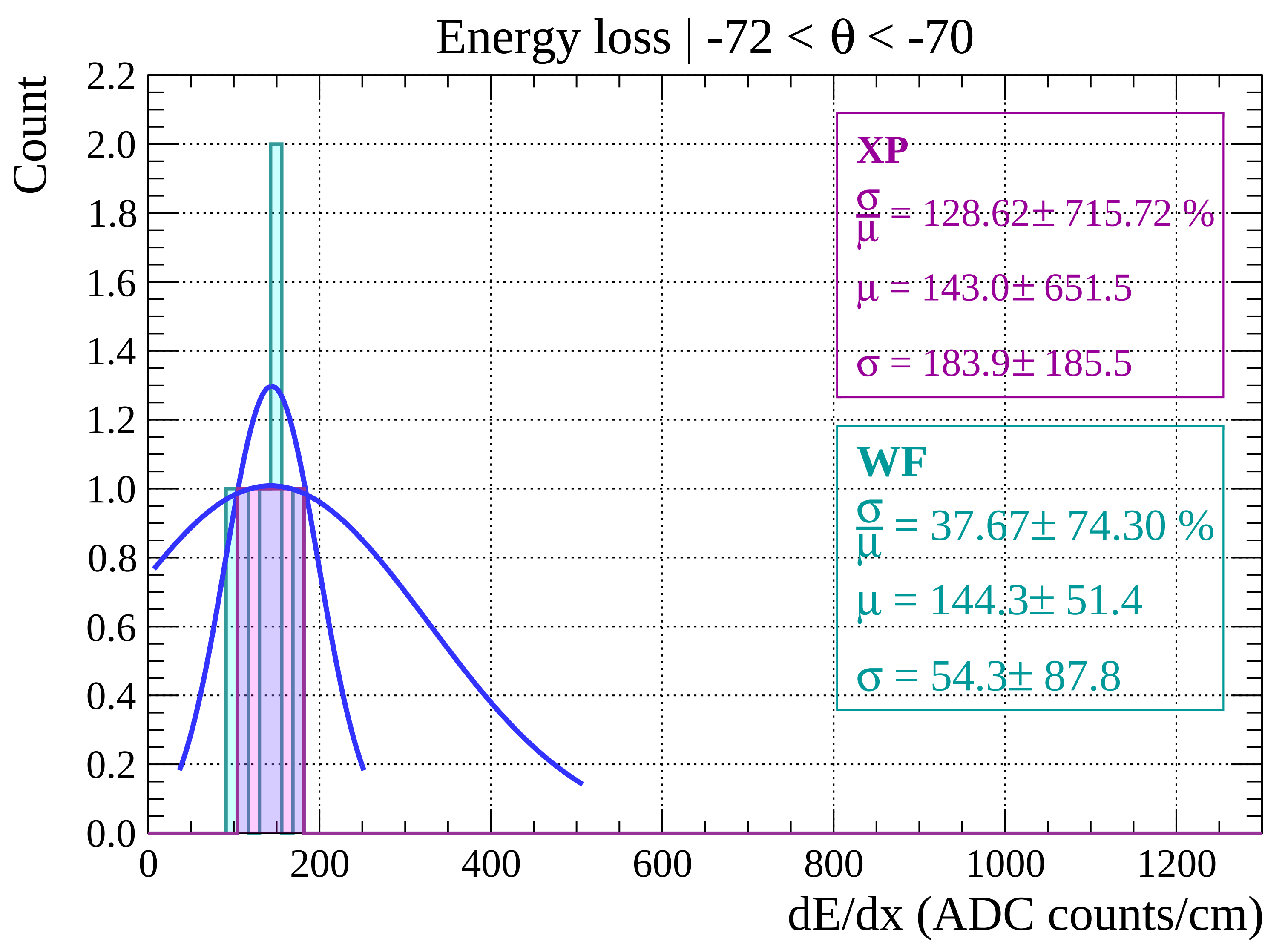


Energy loss | -78 < θ < -76
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

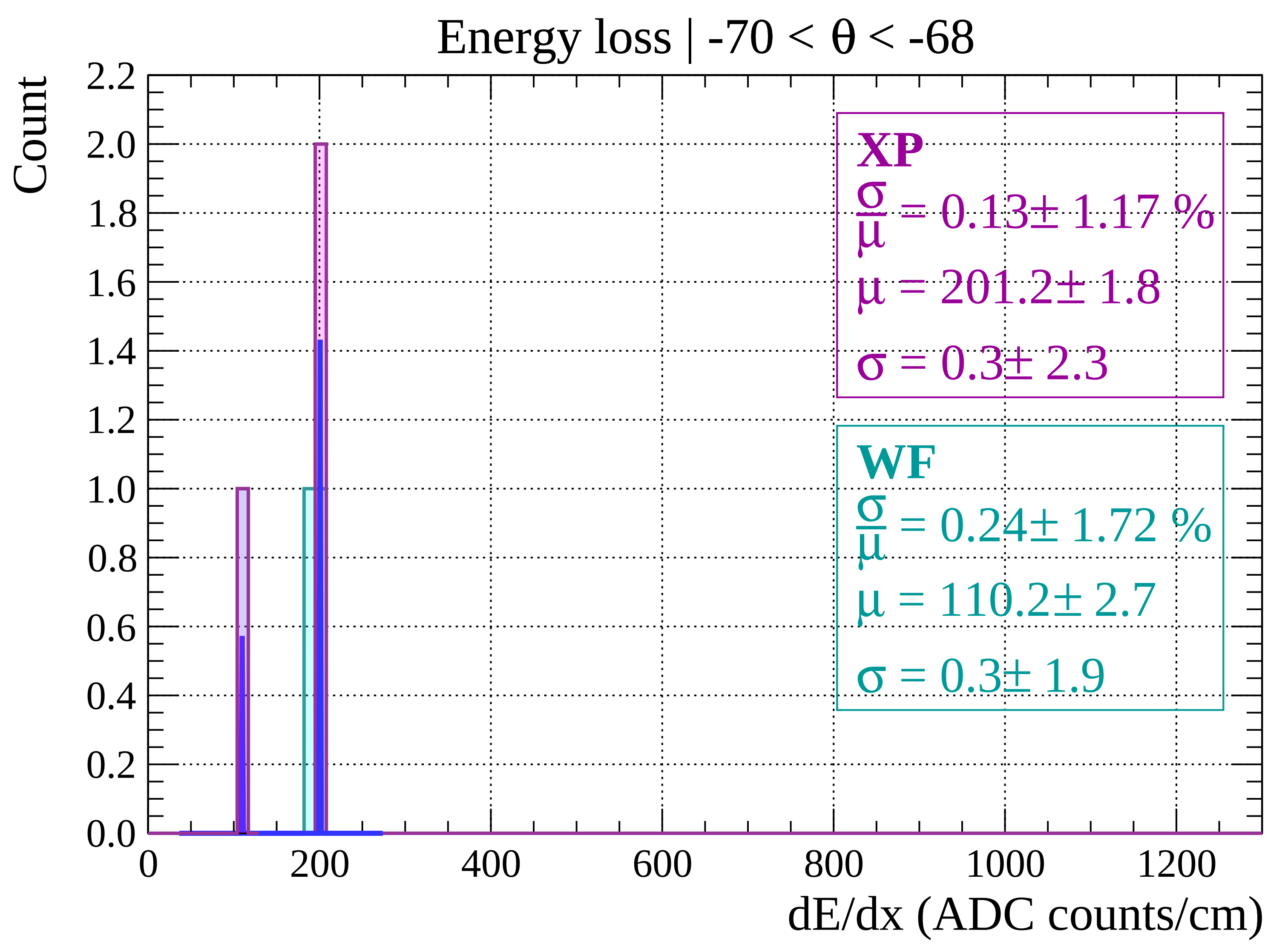

Energy loss | -76 < θ < -74
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 18.18± 160.70 %
μ = 71.5± 47.2
σ = 13.0± 106.3
WF
σ/μ = 18.18± 160.70 %
μ = 71.5± 47.2
σ = 13.0± 106.3

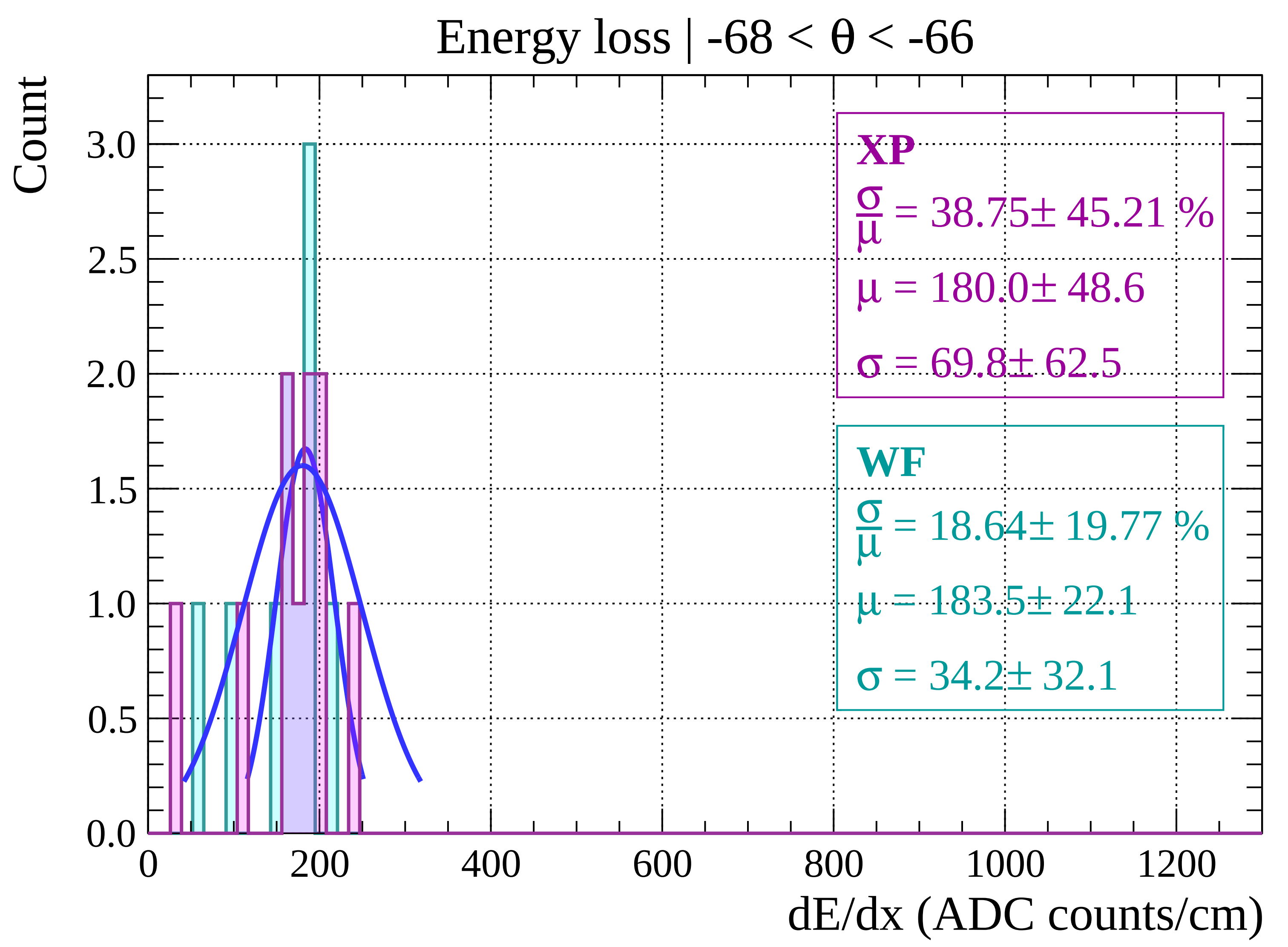


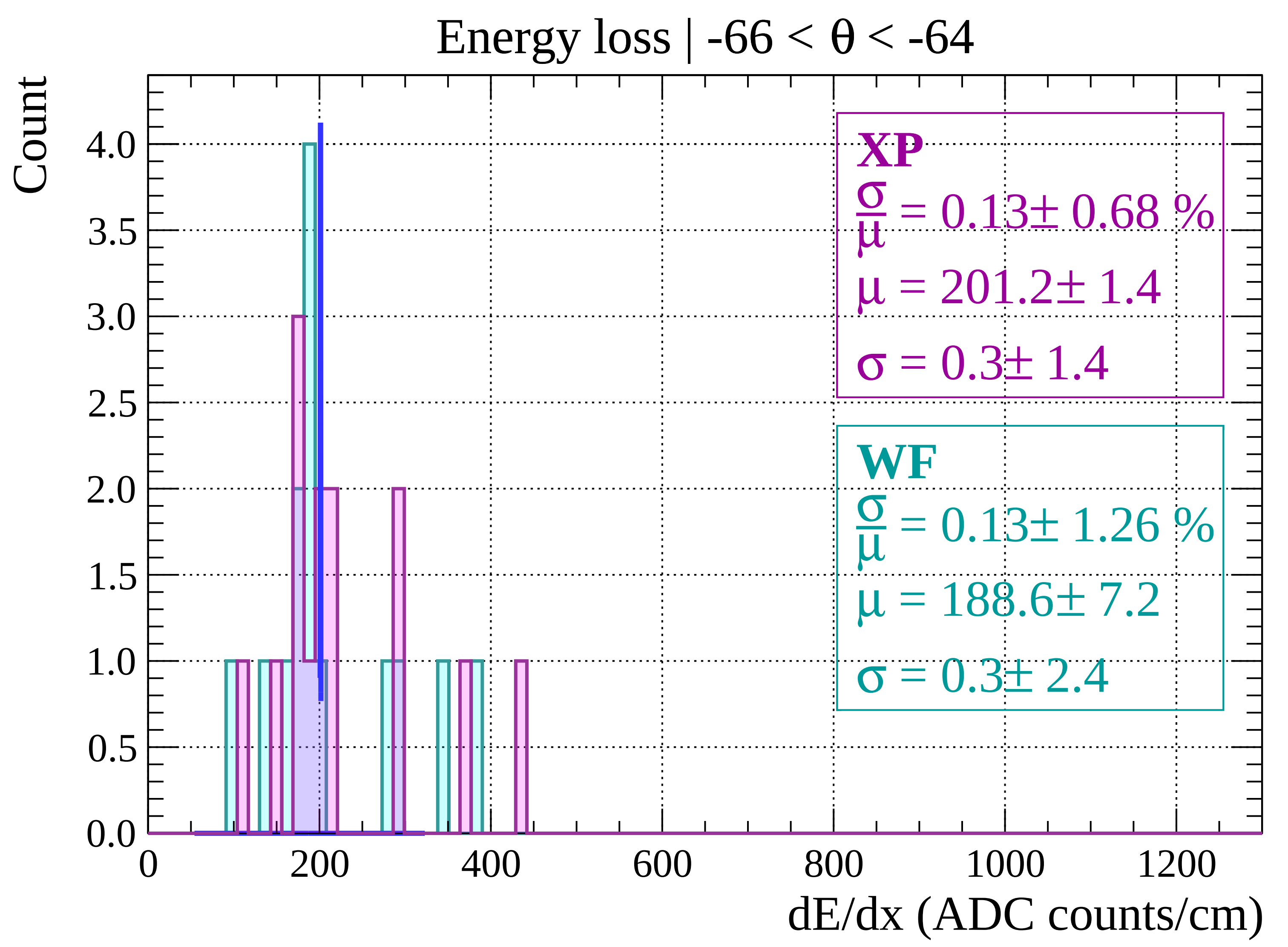
Energy loss | -74 < θ < -72
Count
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 12.02± 9.64 %
μ = 130.0± 8.8
σ = 15.6± 11.5
WF
σ/μ = 13.35± 10.81 %
μ = 117.0± 8.8
σ = 15.6± 11.5



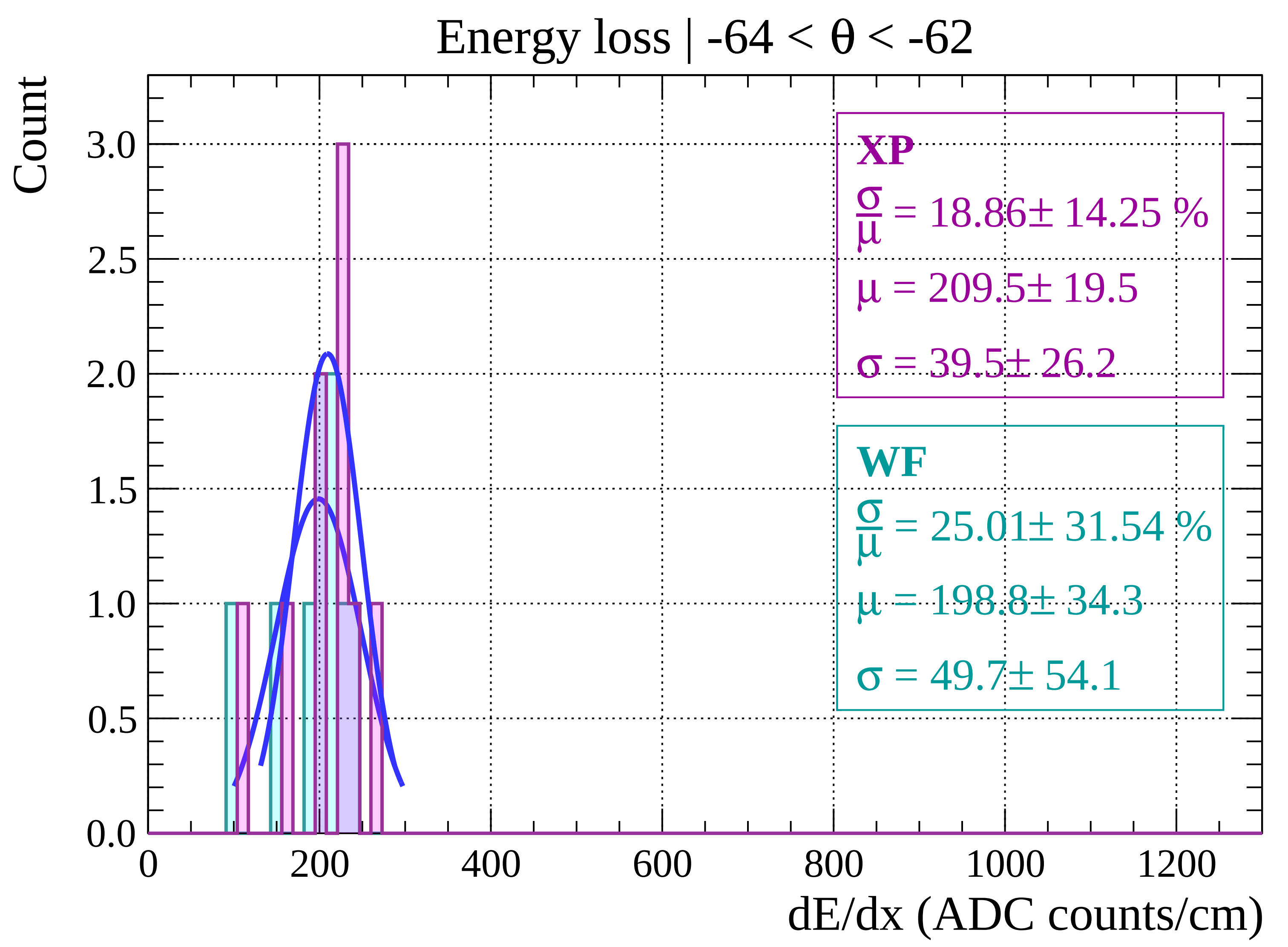
Energy loss | -72 < θ < -70
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 128.62± 715.72 %
μ = 143.0± 651.5
σ = 183.9± 185.5
WF
σ/μ = 37.67± 74.30 %
μ = 144.3± 51.4
σ = 54.3± 87.8


Energy loss | -70 < θ < -68
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 0.13± 1.17 %
μ = 201.2± 1.8
σ = 0.3± 2.3
WF
σ/μ = 0.24± 1.72 %
μ = 110.2± 2.7
σ = 0.3± 1.9

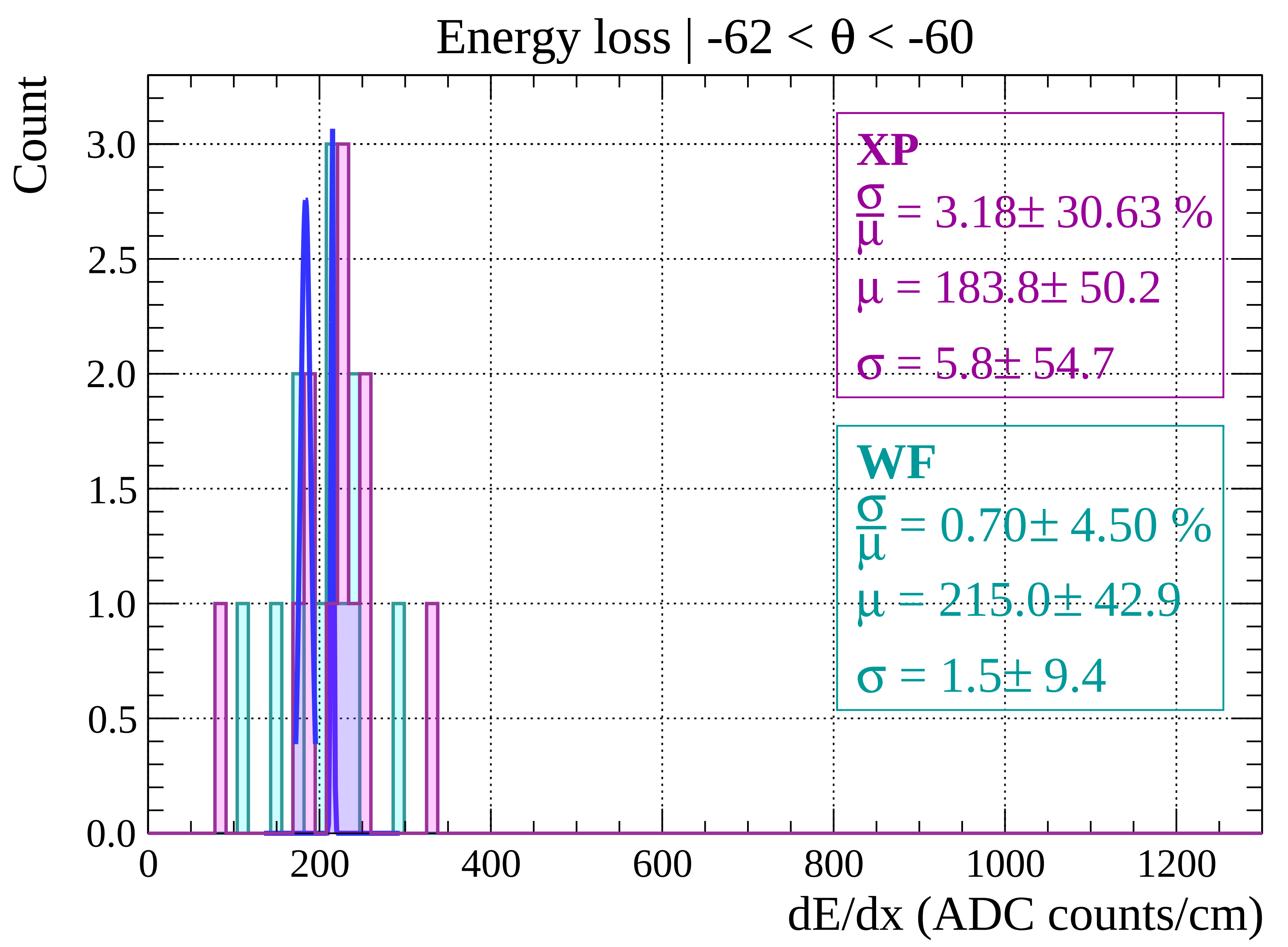

Energy loss | -68 < θ < -66
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 38.75± 45.21 %
μ = 180.0± 48.6
σ = 69.8± 62.5
WF
σ/μ = 18.64± 19.77 %
μ = 183.5± 22.1
σ = 34.2± 32.1



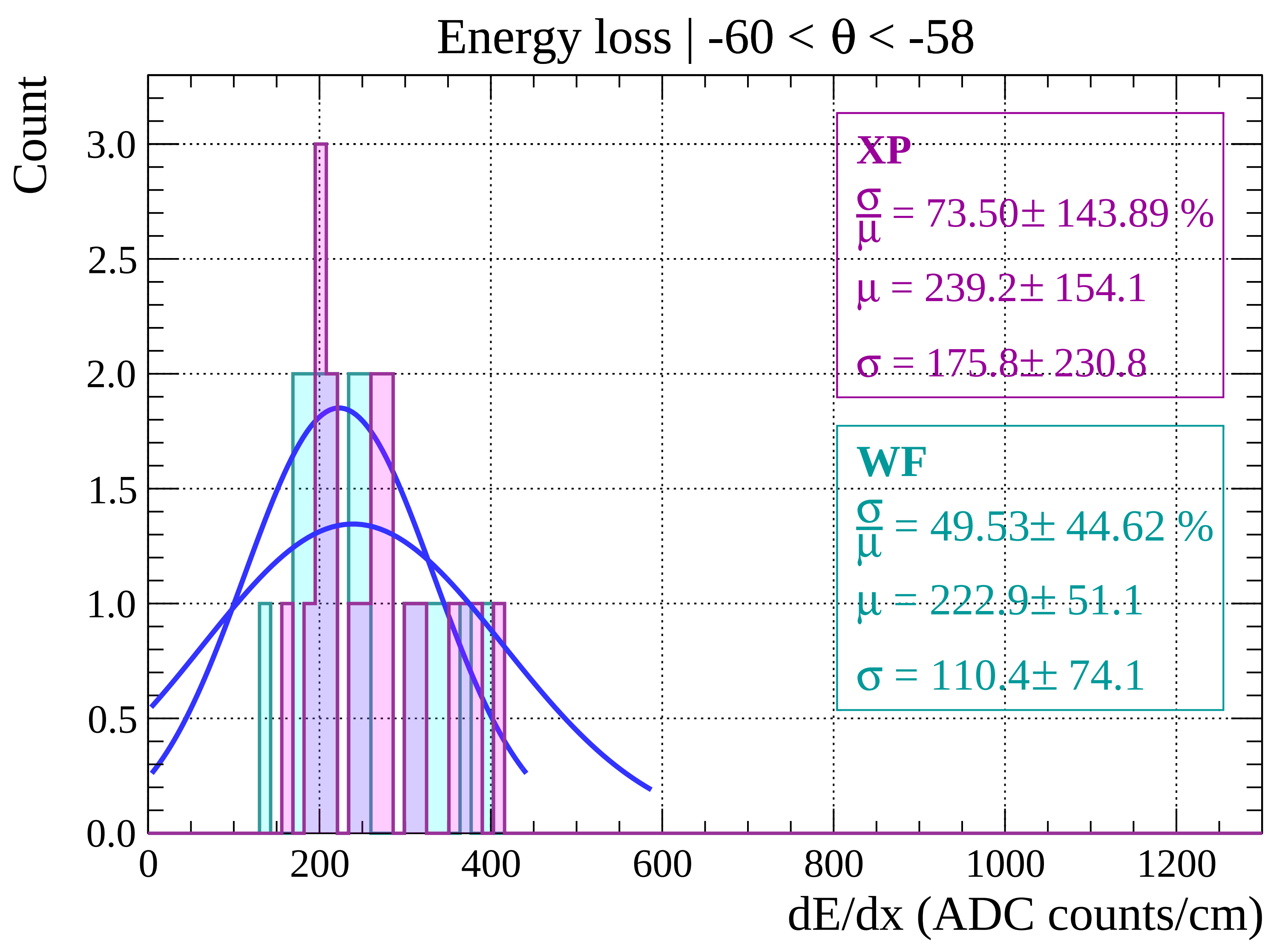
Energy loss | -66 < θ < -64
Count
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 0.13± 0.68 %
μ = 201.2± 1.4
σ = 0.3± 1.4
WF
σ/μ = 0.13± 1.26 %
μ = 188.6± 7.2
σ = 0.3± 2.4



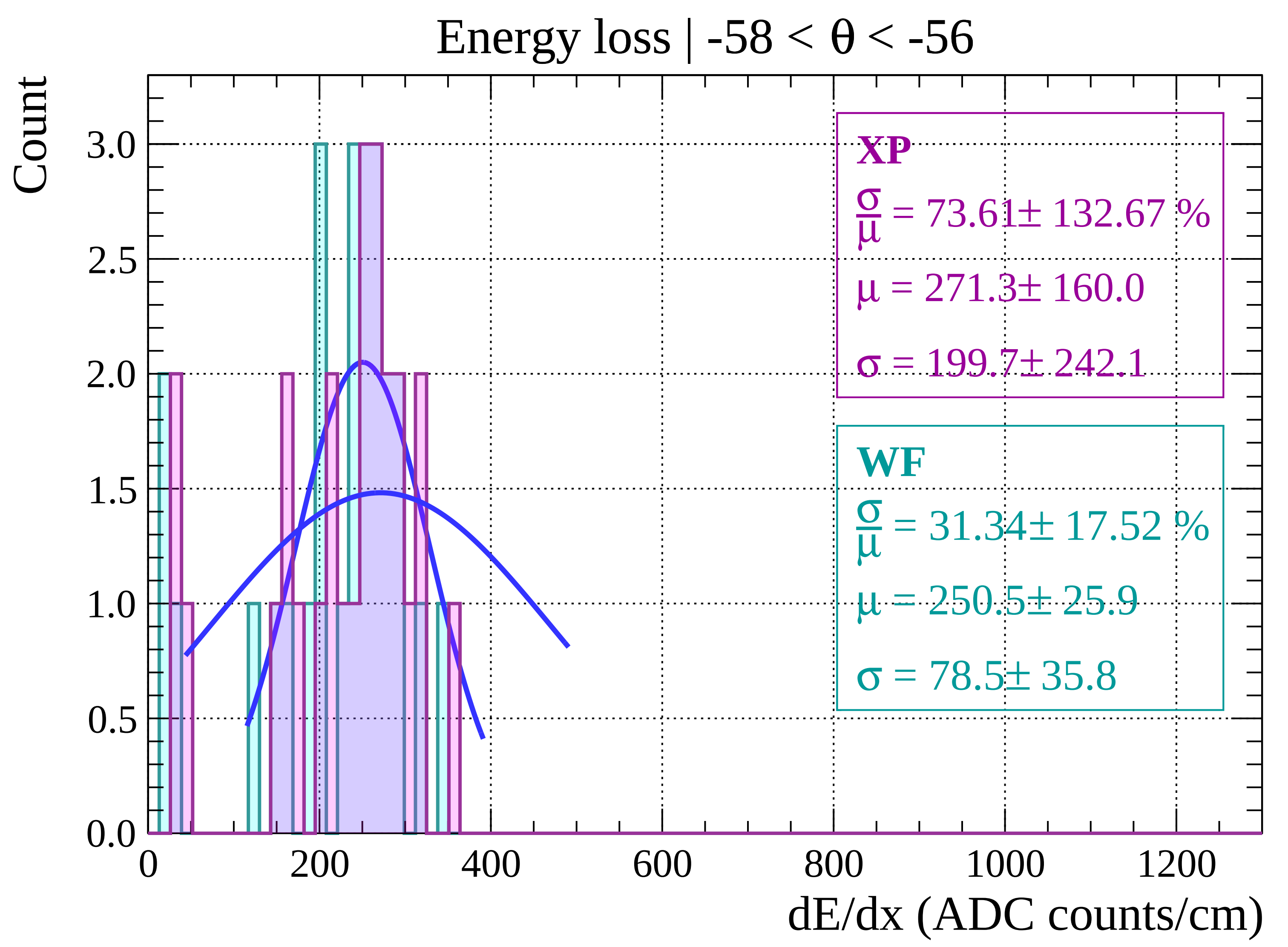
Energy loss | -64 < θ < -62
Count
dE/dx (ADC counts/cm)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 18.86± 14.25 %
μ = 209.5± 19.5
σ = 39.5± 26.2
WF
$\frac{\sigma}{\mu}$ = 25.01± 31.54 %
μ = 198.8± 34.3
σ = 49.7± 54.1



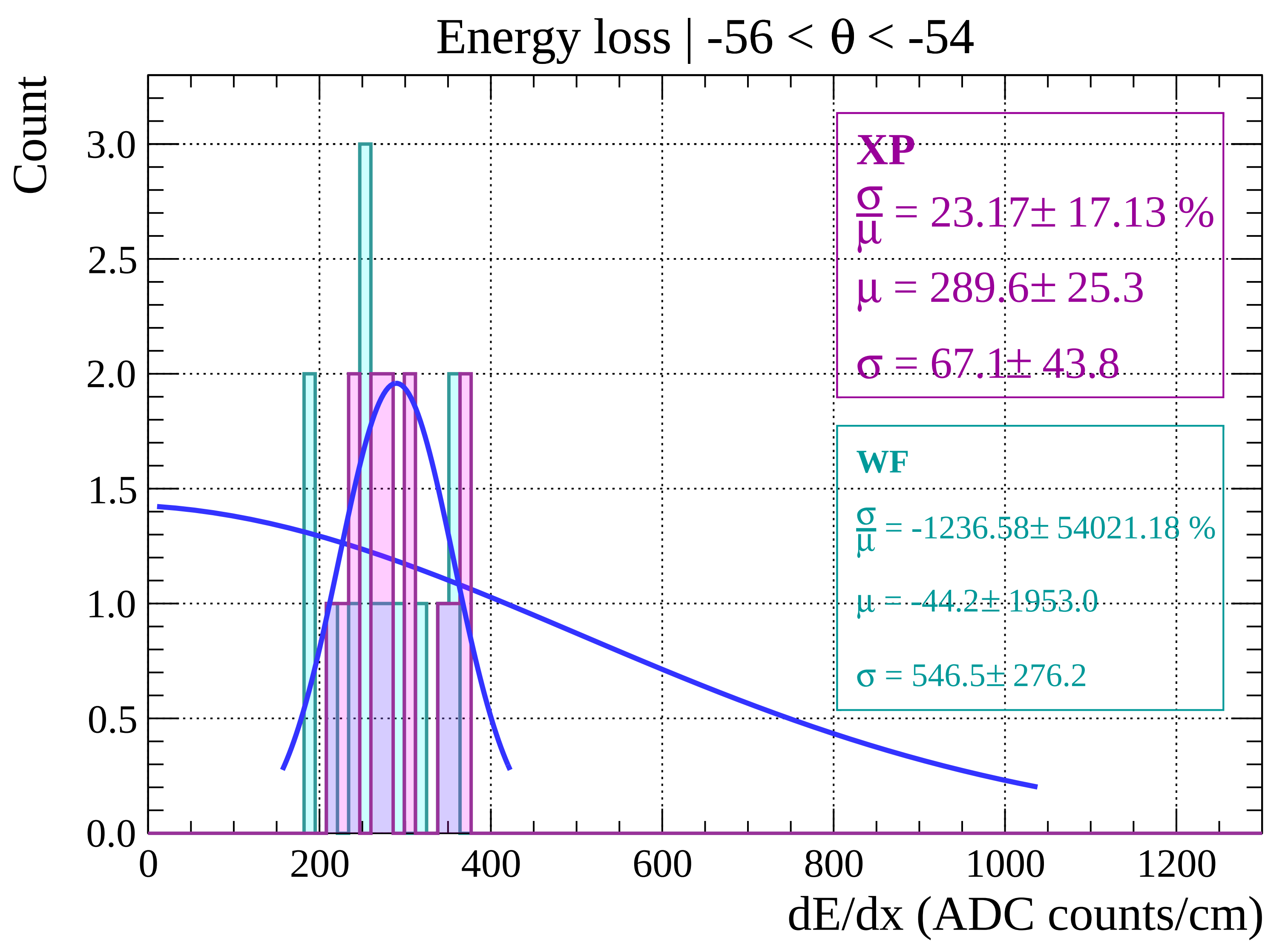
Energy loss | -62 < θ < -60
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 3.18± 30.63 %
μ = 183.8± 50.2
σ = 5.8± 54.7
WF
σ/μ = 0.70± 4.50 %
μ = 215.0± 42.9
σ = 1.5± 9.4

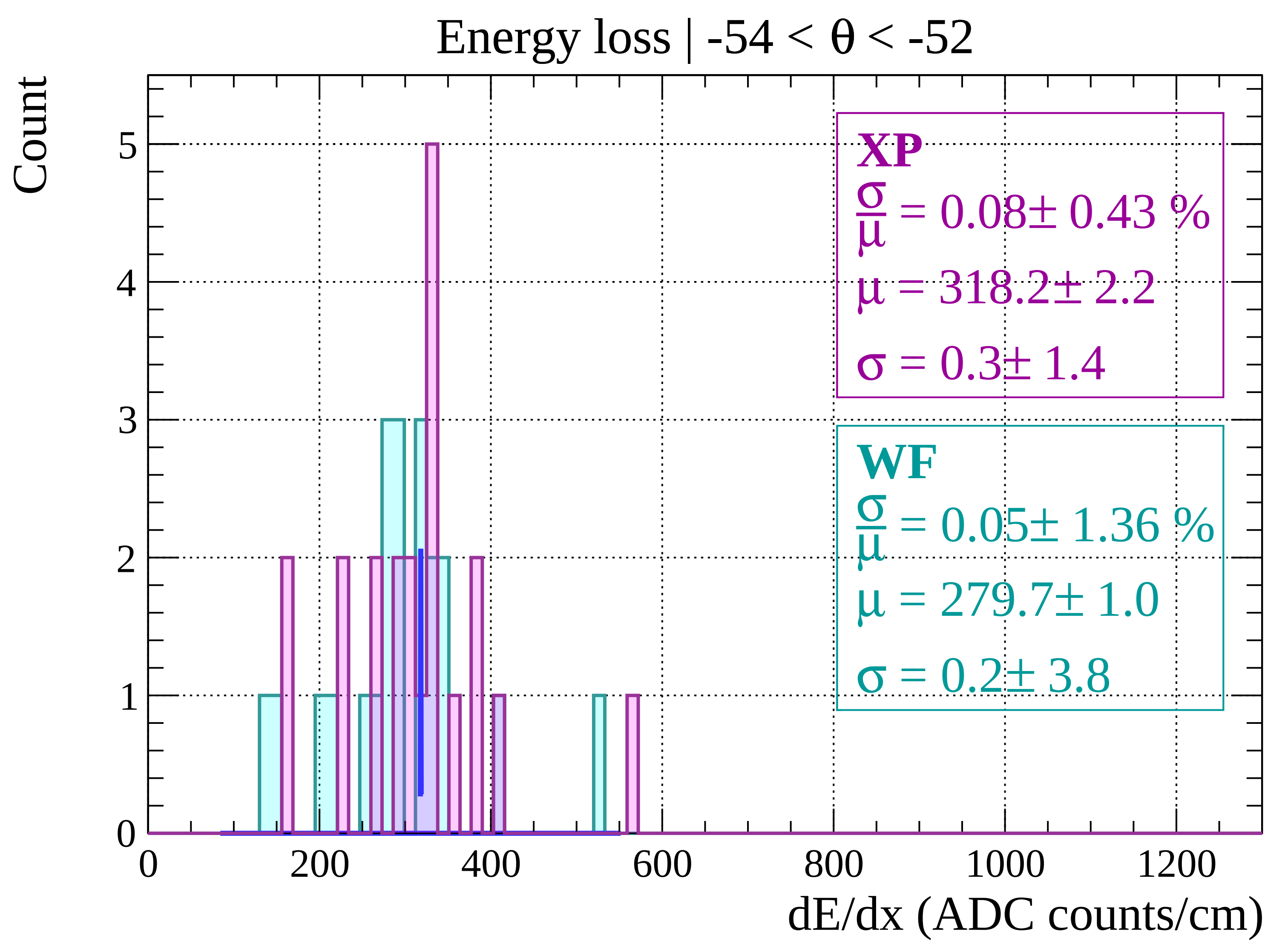


Energy loss | -60 < θ < -58
Count
dE/dx (ADC counts/cm)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
200
400
600
800
1000
1200
XP
σ/μ = 73.50± 143.89 %
μ = 239.2± 154.1
σ = 175.8± 230.8
WF
σ/μ = 49.53± 44.62 %
μ = 222.9± 51.1
σ = 110.4± 74.1

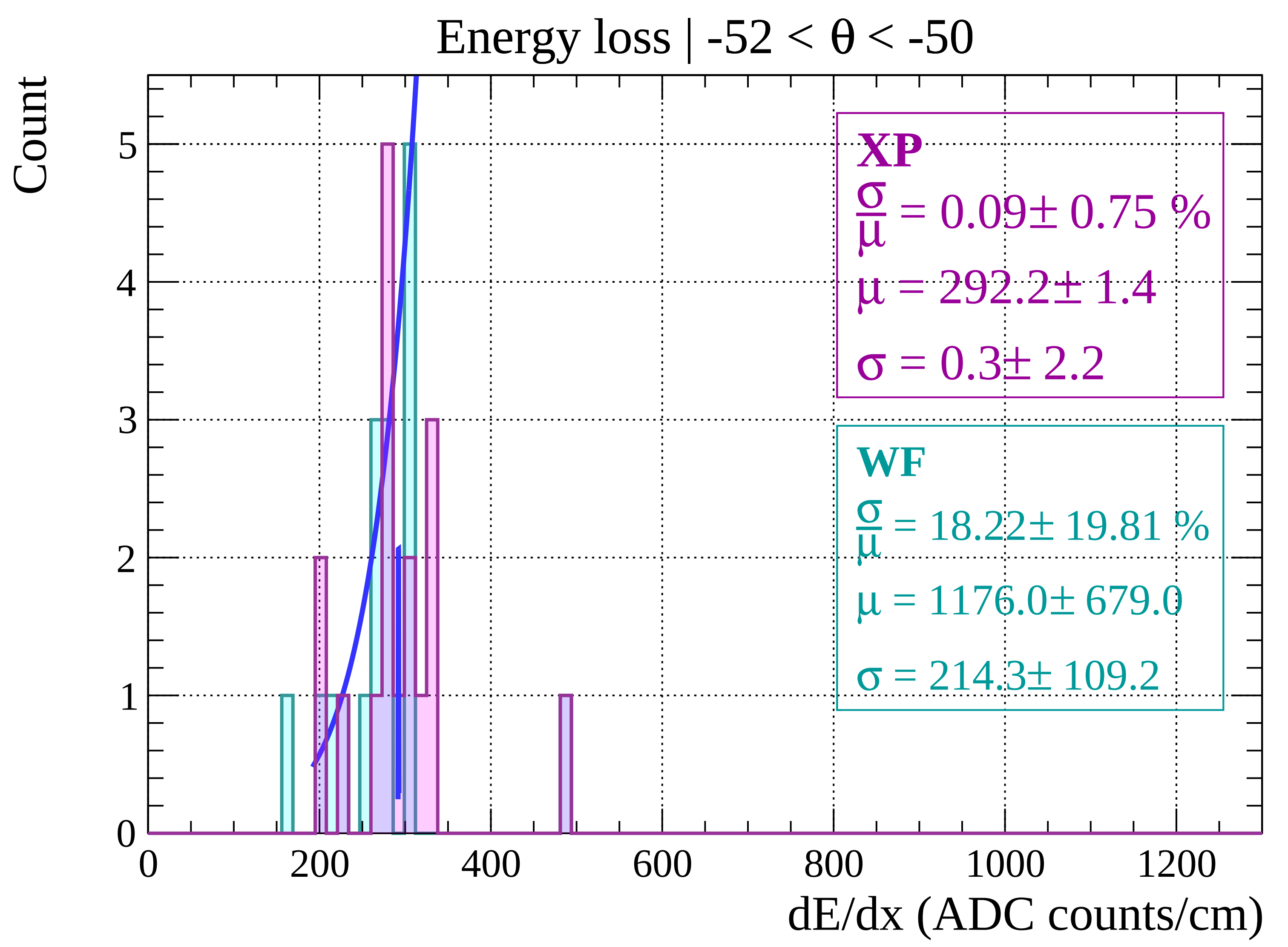


Energy loss | -58 < θ < -56
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 73.61± 132.67 %
μ = 271.3± 160.0
σ = 199.7± 242.1
WF
σ/μ = 31.34± 17.52 %
μ = 250.5± 25.9
σ = 78.5± 35.8

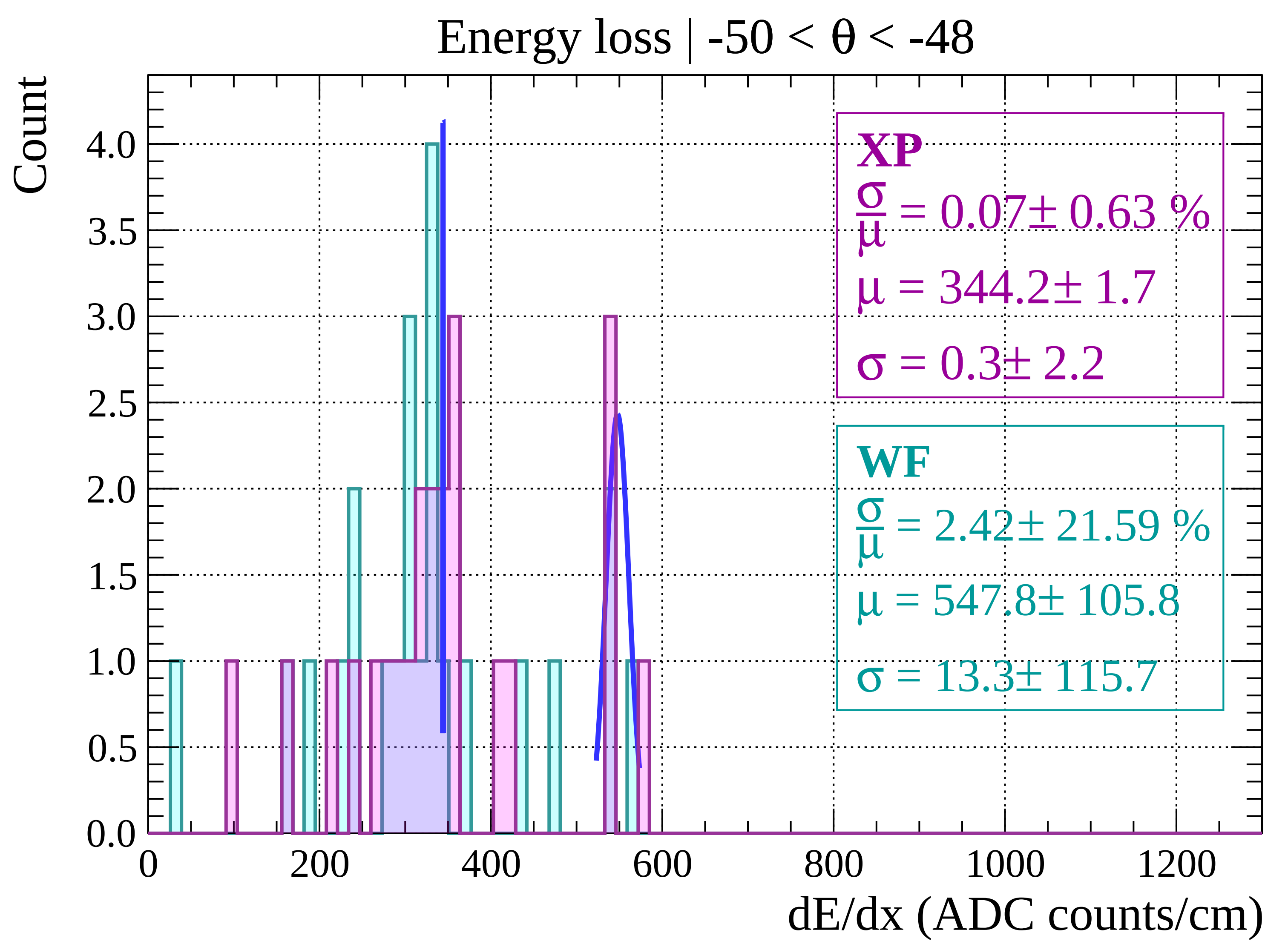

Energy loss | -56 < θ < -54
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 23.17± 17.13 %
μ = 289.6± 25.3
σ = 67.1± 43.8
WF
σ/μ = -1236.58± 54021.18 %
μ = -44.2± 1953.0
σ = 546.5± 276.2

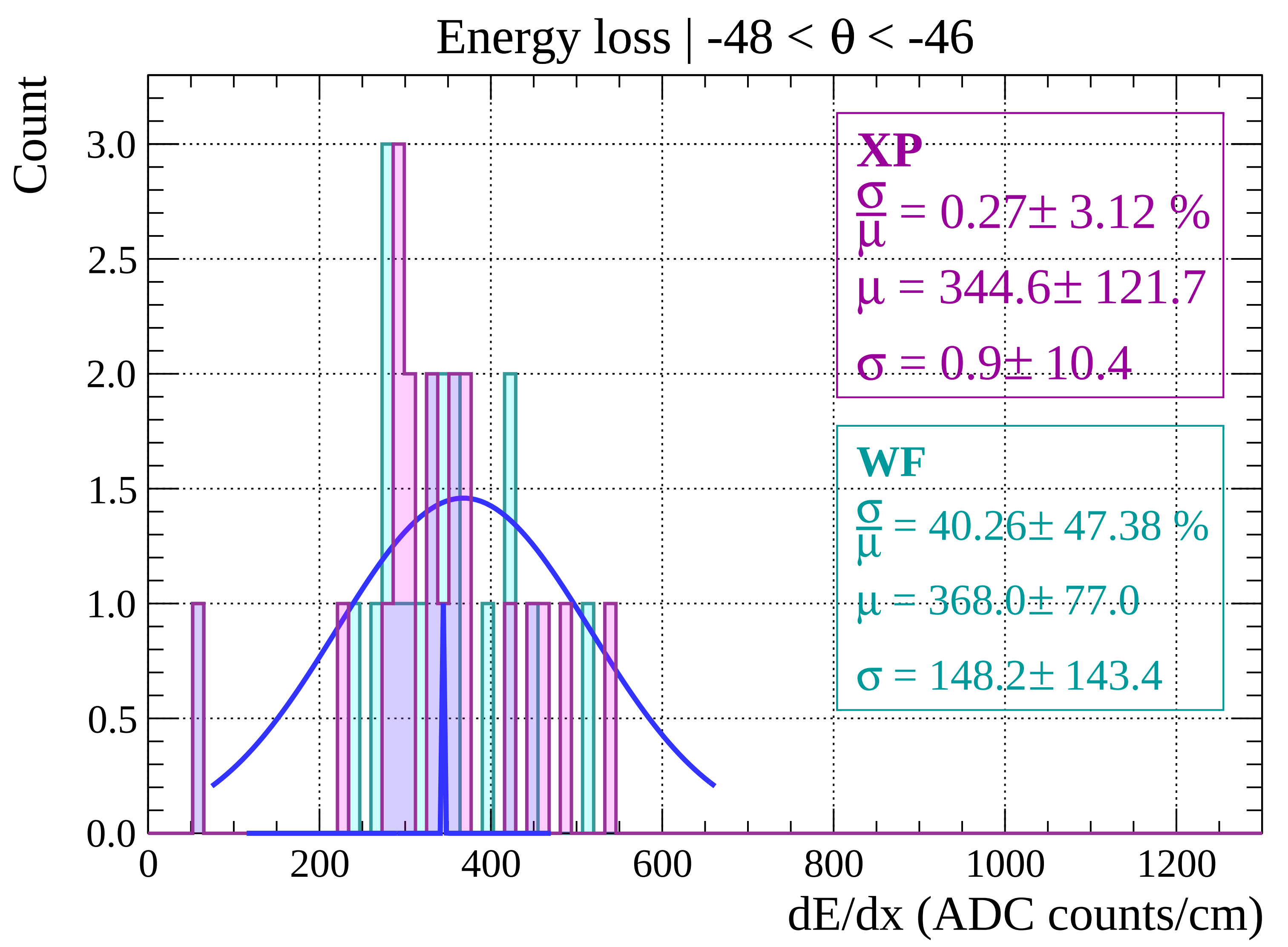


Energy loss | -54 < θ < -52
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1
2
3
4
5
XP
σ/μ = 0.08± 0.43 %
μ = 318.2± 2.2
σ = 0.3± 1.4
WF
σ/μ = 0.05± 1.36 %
μ = 279.7± 1.0
σ = 0.2± 3.8

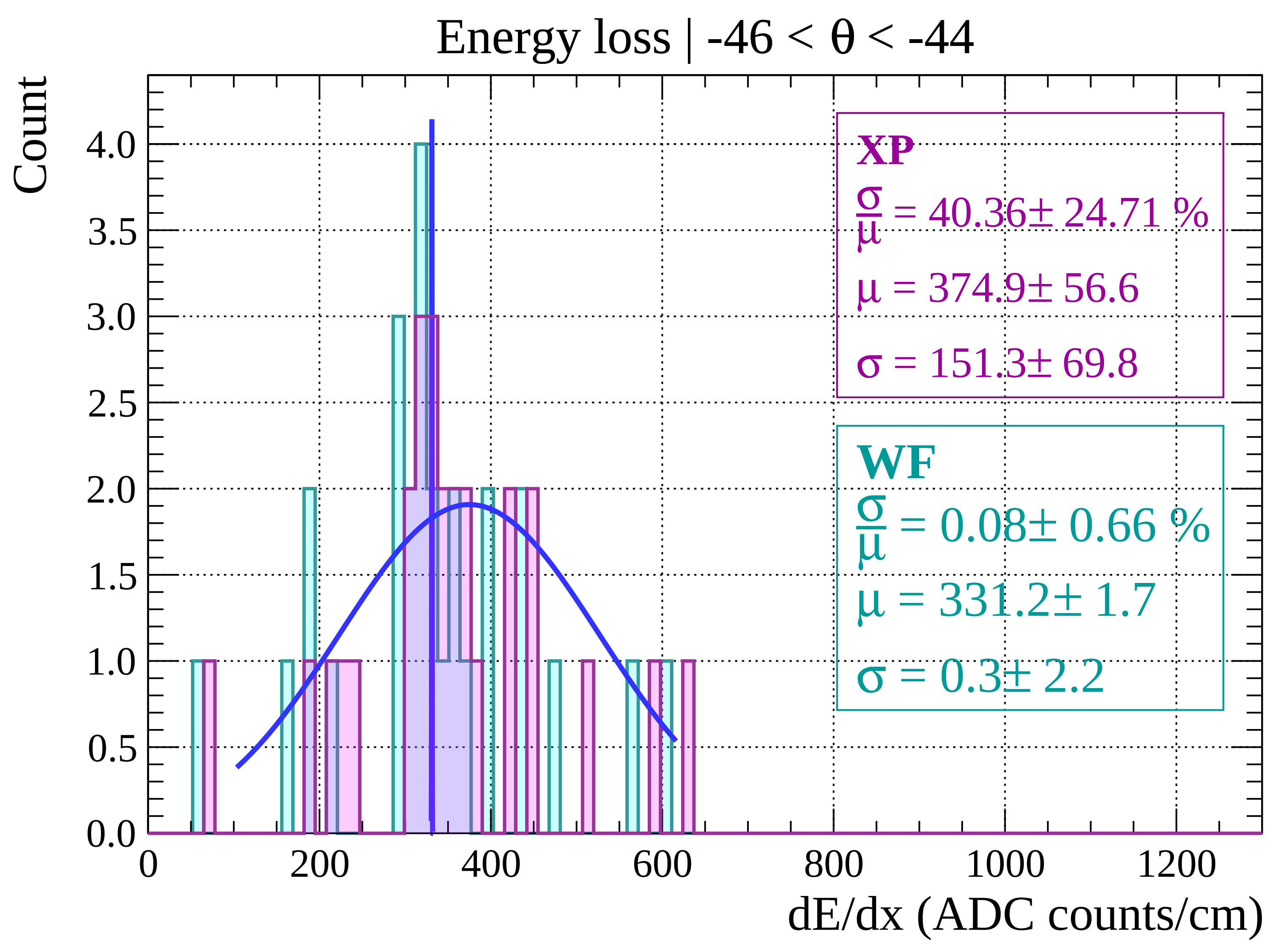


Energy loss | -52 < θ < -50
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 0.09± 0.75 %
μ = 292.2± 1.4
σ = 0.3± 2.2
WF
σ/μ = 18.22± 19.81 %
μ = 1176.0± 679.0
σ = 214.3± 109.2

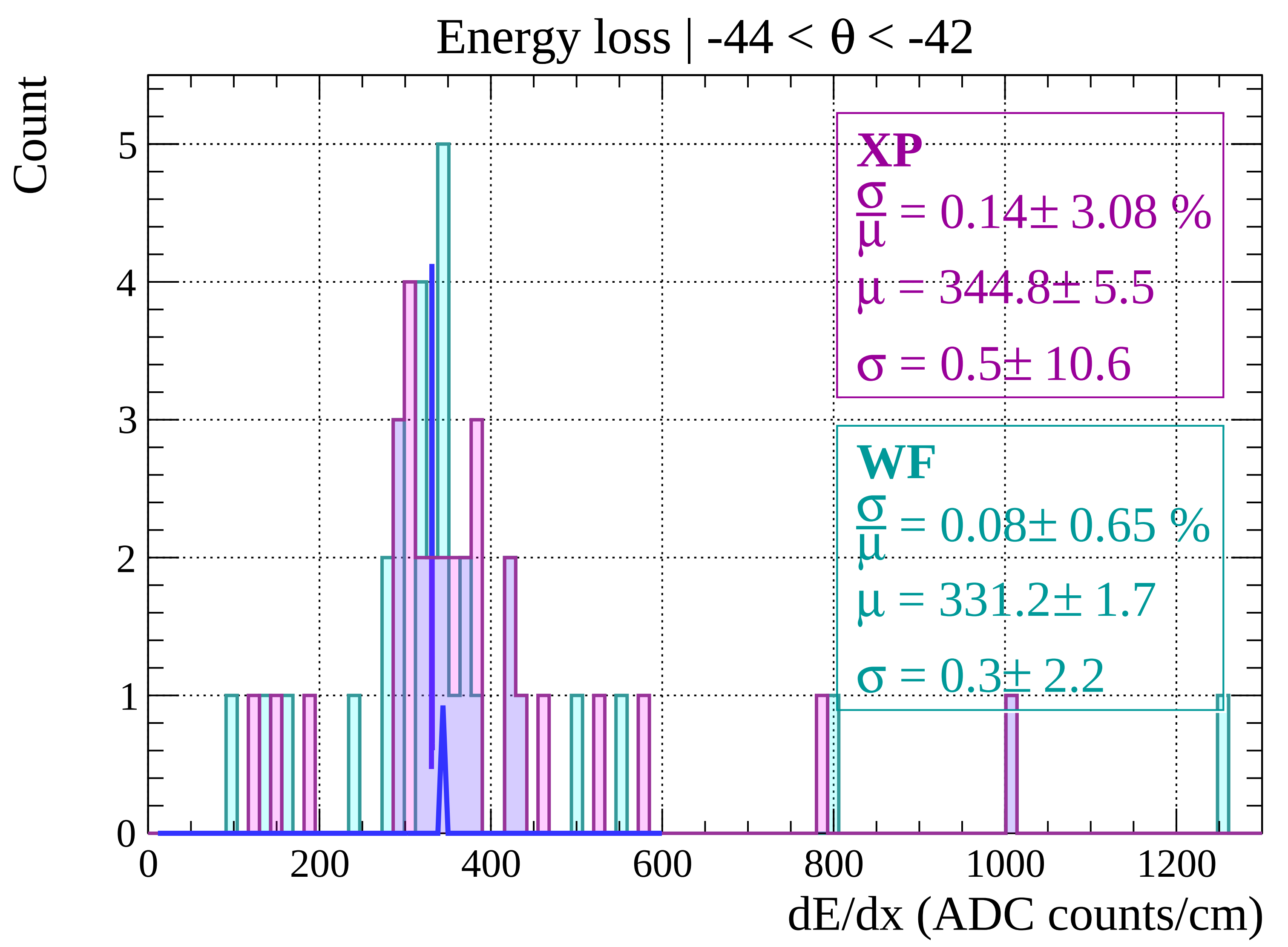

Energy loss | -50 < θ < -48
Count
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 0.07± 0.63 %
μ = 344.2± 1.7
σ = 0.3± 2.2
WF
σ/μ = 2.42± 21.59 %
μ = 547.8± 105.8
σ = 13.3± 115.7



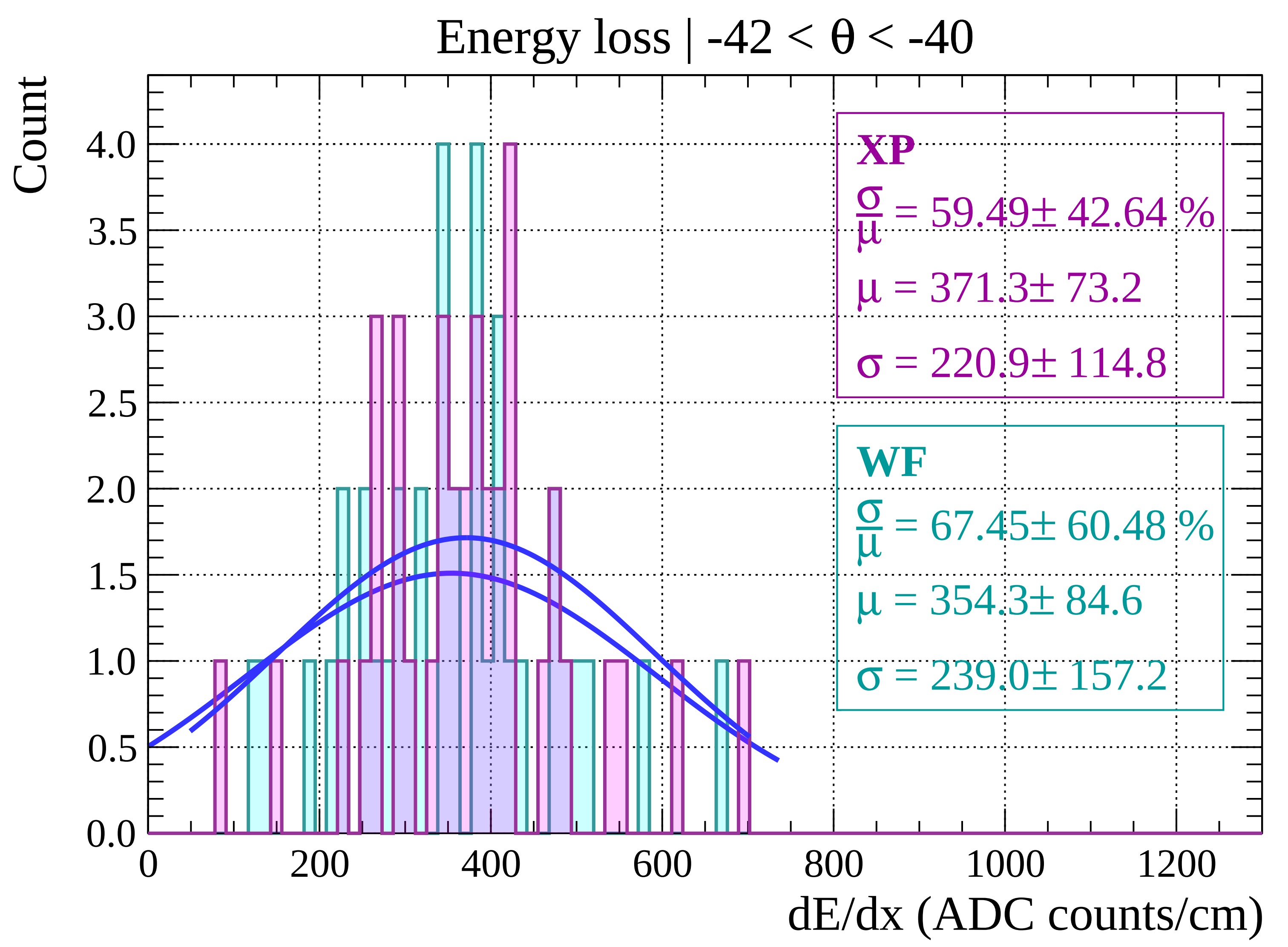
Energy loss | -48 < θ < -46
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 0.27± 3.12 %
μ = 344.6± 121.7
σ = 0.9± 10.4
WF
σ/μ = 40.26± 47.38 %
μ = 368.0± 77.0
σ = 148.2± 143.4

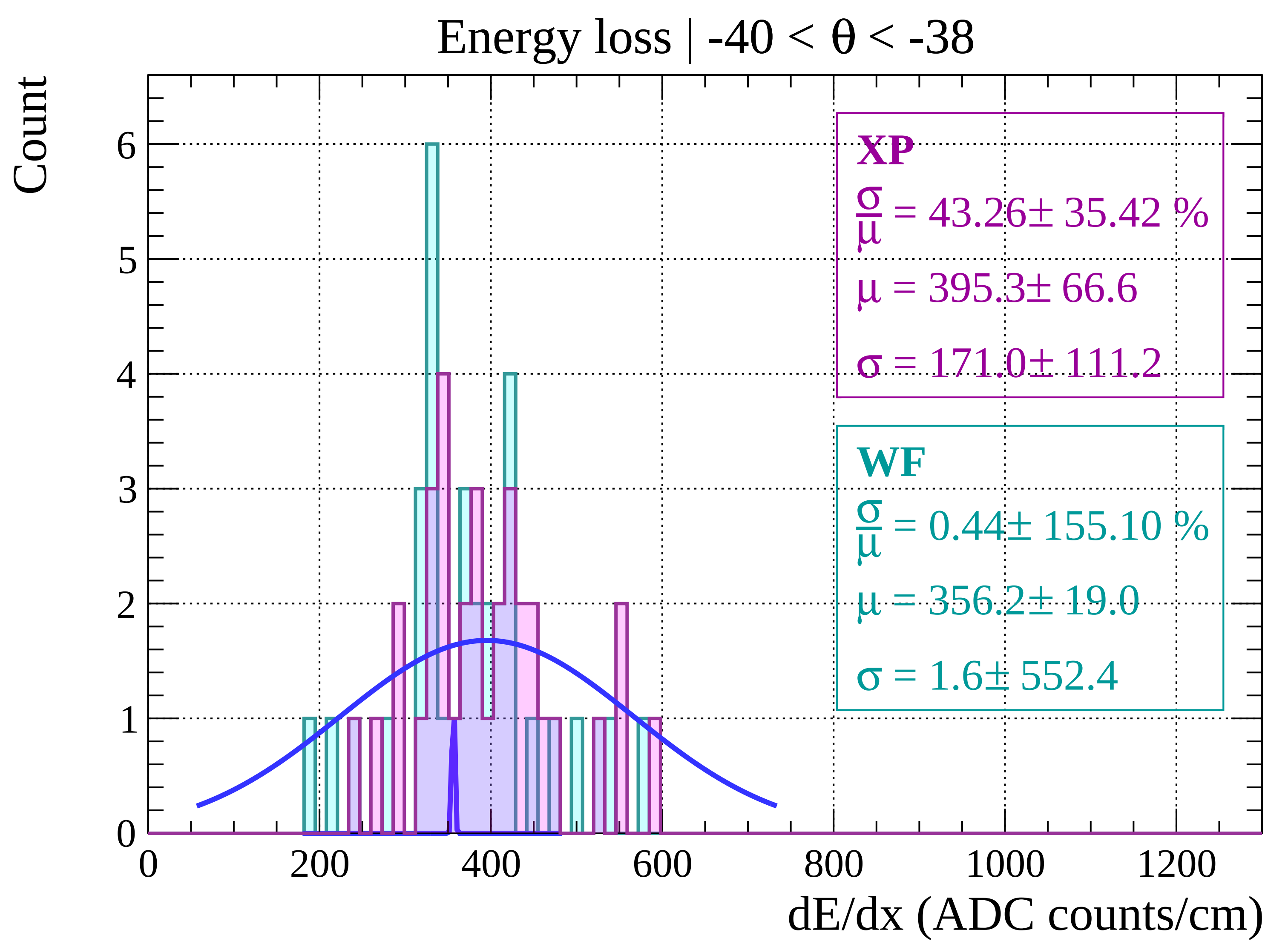


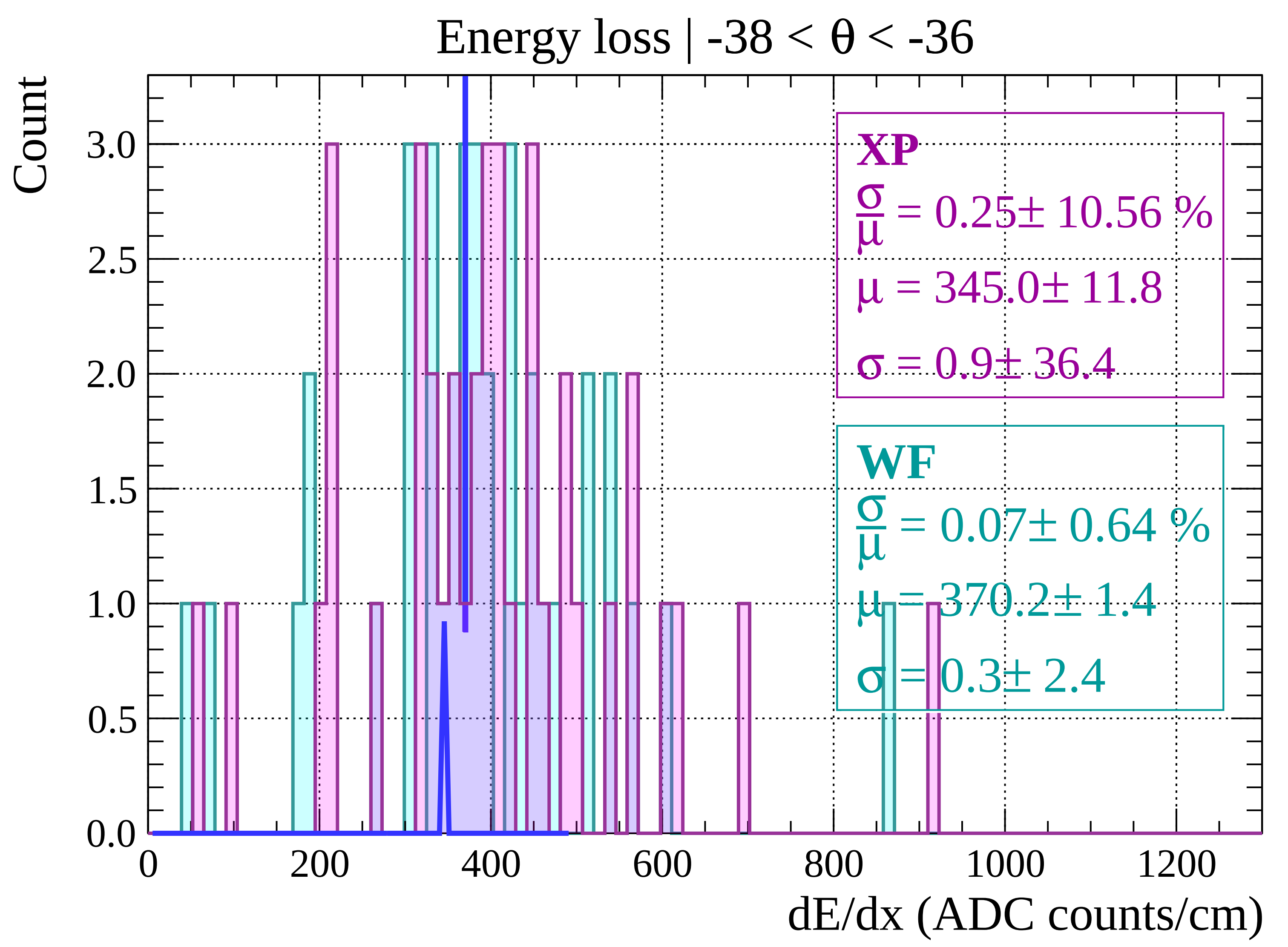
Energy loss | -46 < θ < -44
Count
XP
σ/μ = 40.36± 24.71 %
μ = 374.9± 56.6
σ = 151.3± 69.8
WF
σ/μ = 0.08± 0.66 %
μ = 331.2± 1.7
σ = 0.3± 2.2
dE/dx (ADC counts/cm)



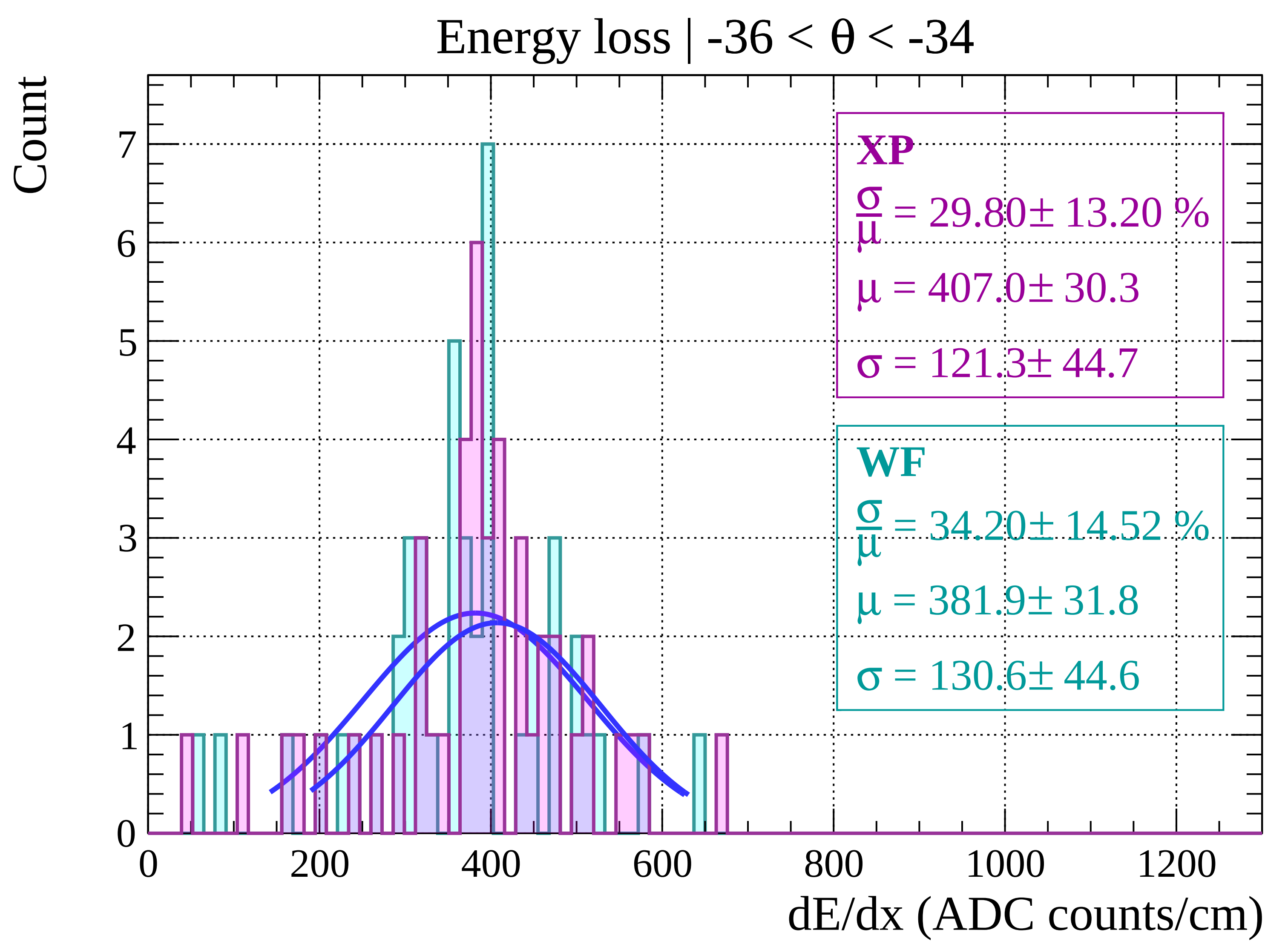
Energy loss | -44 < θ < -42
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1
2
3
4
5
XP
$\frac{\sigma}{\mu}$ = 0.14± 3.08 %
μ = 344.8± 5.5
σ = 0.5± 10.6
WF
$\frac{\sigma}{\mu}$ = 0.08± 0.65 %
μ = 331.2± 1.7
σ = 0.3± 2.2


Energy loss | -42 < θ < -40
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 59.49± 42.64 %
μ = 371.3± 73.2
σ = 220.9± 114.8
WF
σ/μ = 67.45± 60.48 %
μ = 354.3± 84.6
σ = 239.0± 157.2
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
200
400
600
800
1000
1200

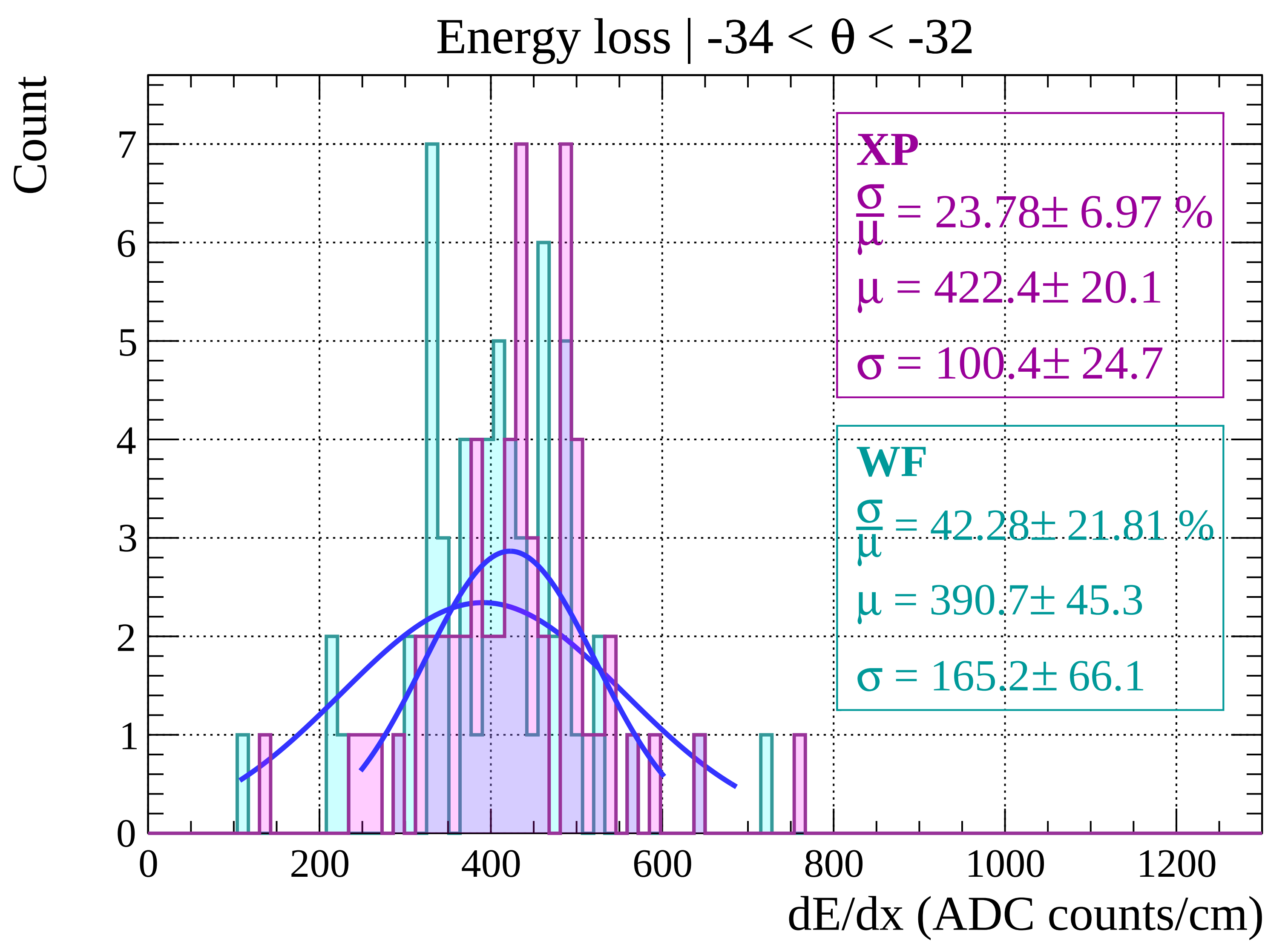

Energy loss | -40 < θ < -38
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
6
0
200
400
600
800
1000
1200
XP
σ/μ = 43.26± 35.42 %
μ = 395.3± 66.6
σ = 171.0± 111.2
WF
σ/μ = 0.44± 155.10 %
μ = 356.2± 19.0
σ = 1.6± 552.4



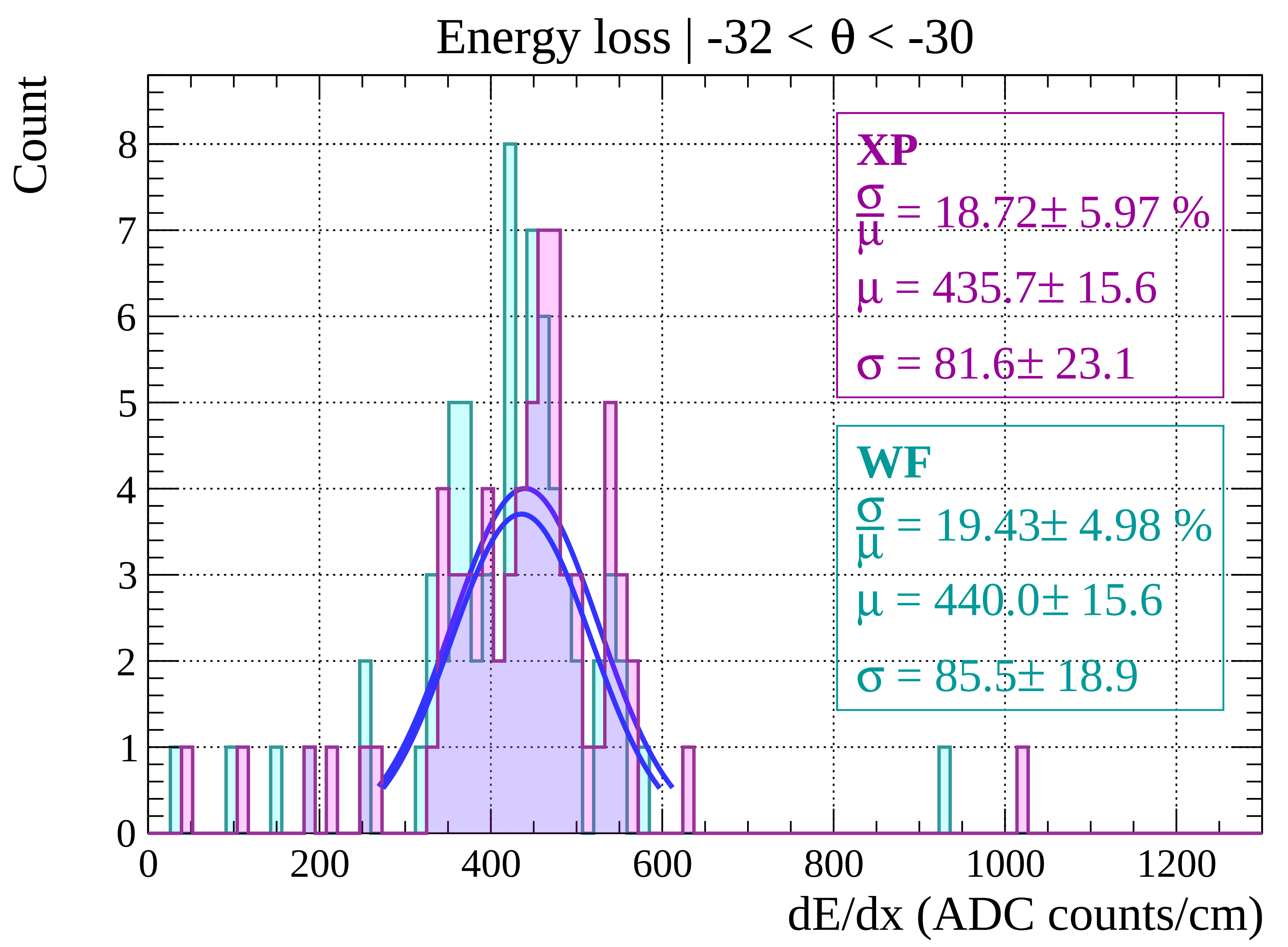
Energy loss | -38 < θ < -36
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 0.25± 10.56 %
μ = 345.0± 11.8
σ = 0.9± 36.4
WF
σ/μ = 0.07± 0.64 %
μ = 370.2± 1.4
σ = 0.3± 2.4



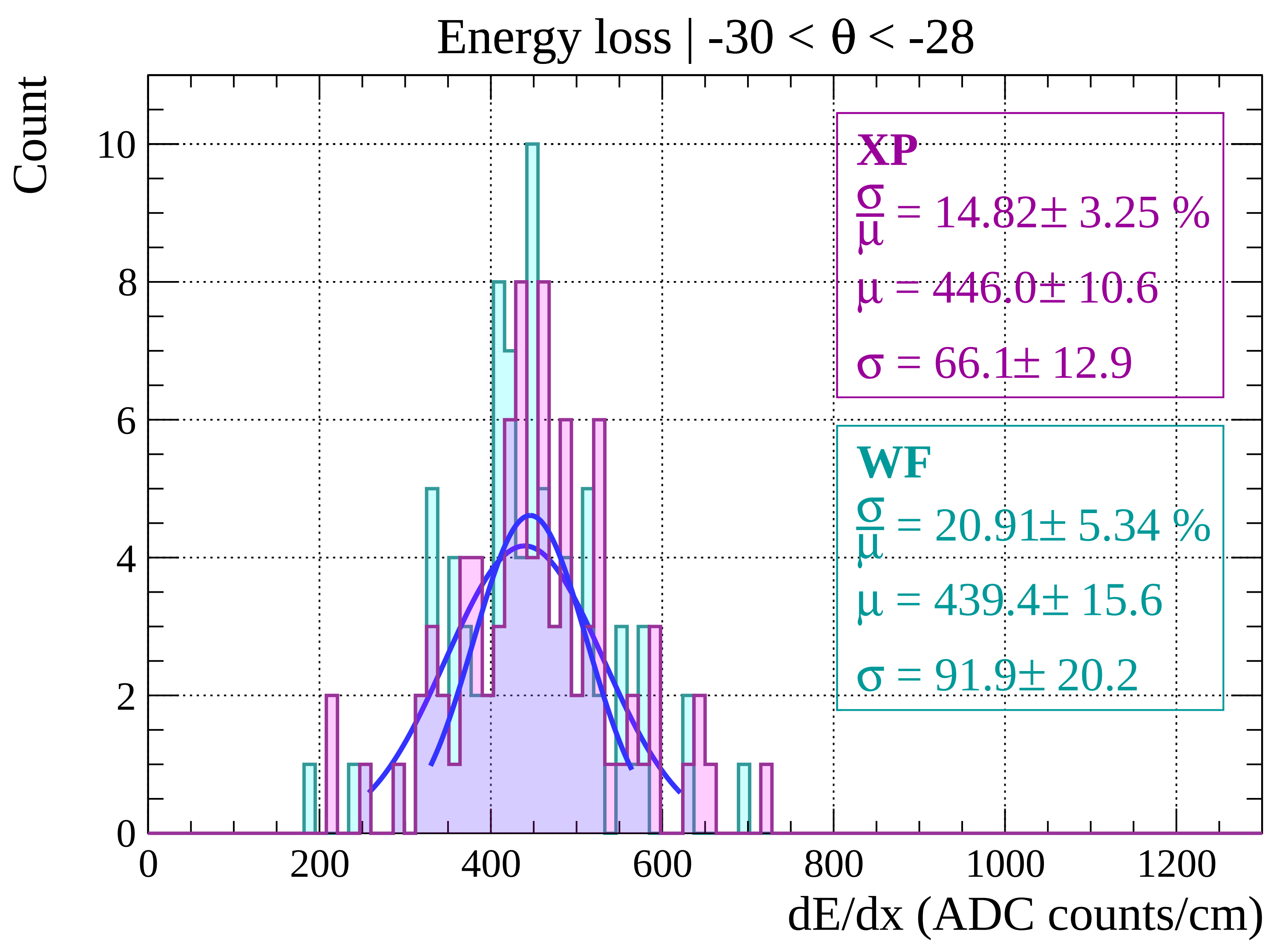
Energy loss | -36 < θ < -34
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1
2
3
4
5
6
7
XP
σ/μ = 29.80± 13.20 %
μ = 407.0± 30.3
σ = 121.3± 44.7
WF
σ/μ = 34.20± 14.52 %
μ = 381.9± 31.8
σ = 130.6± 44.6

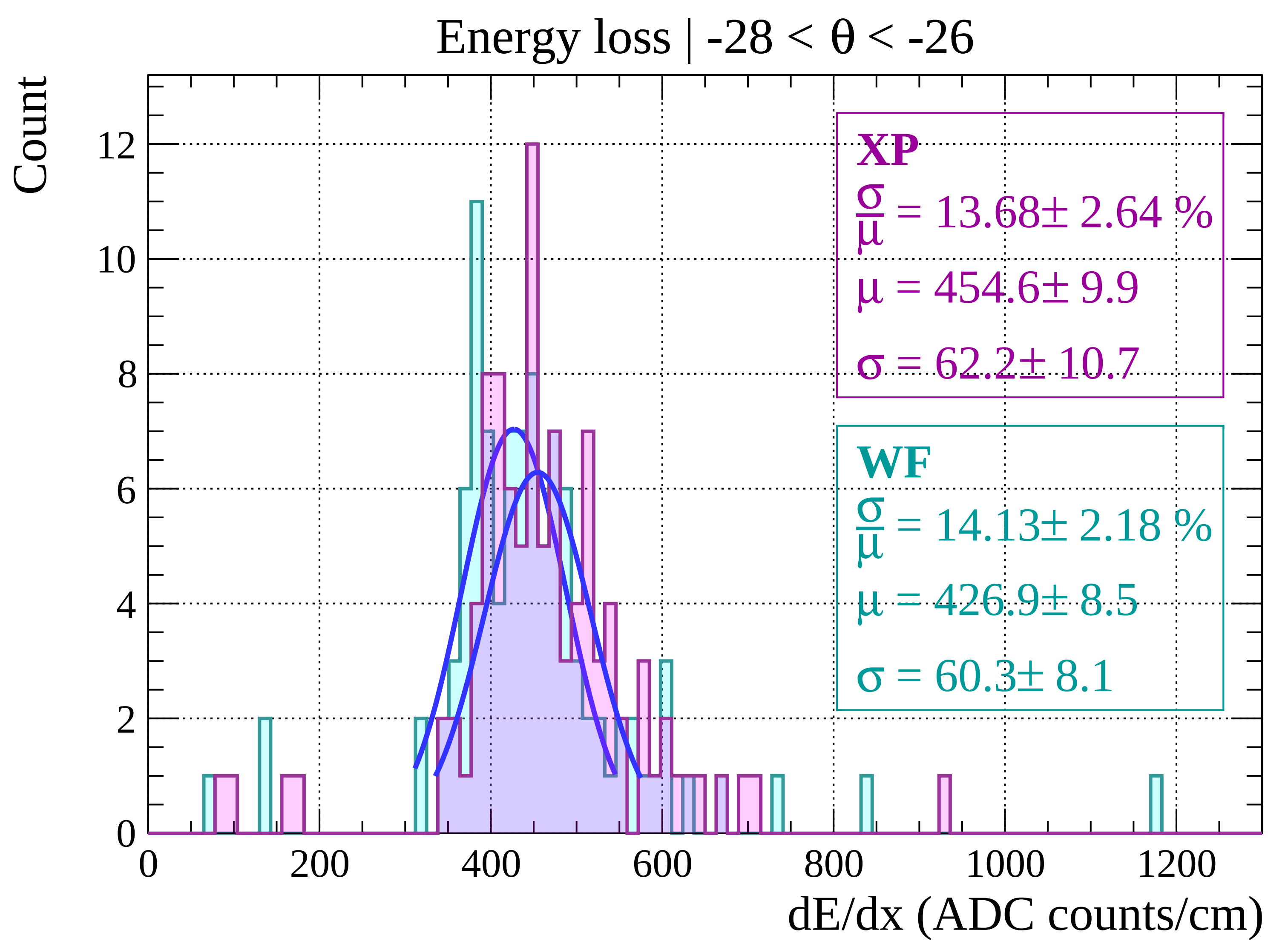


Energy loss | -34 < θ < -32
Count
0
1
2
3
4
5
6
7
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 23.78± 6.97 %
μ = 422.4± 20.1
σ = 100.4± 24.7
WF
σ/μ = 42.28± 21.81 %
μ = 390.7± 45.3
σ = 165.2± 66.1

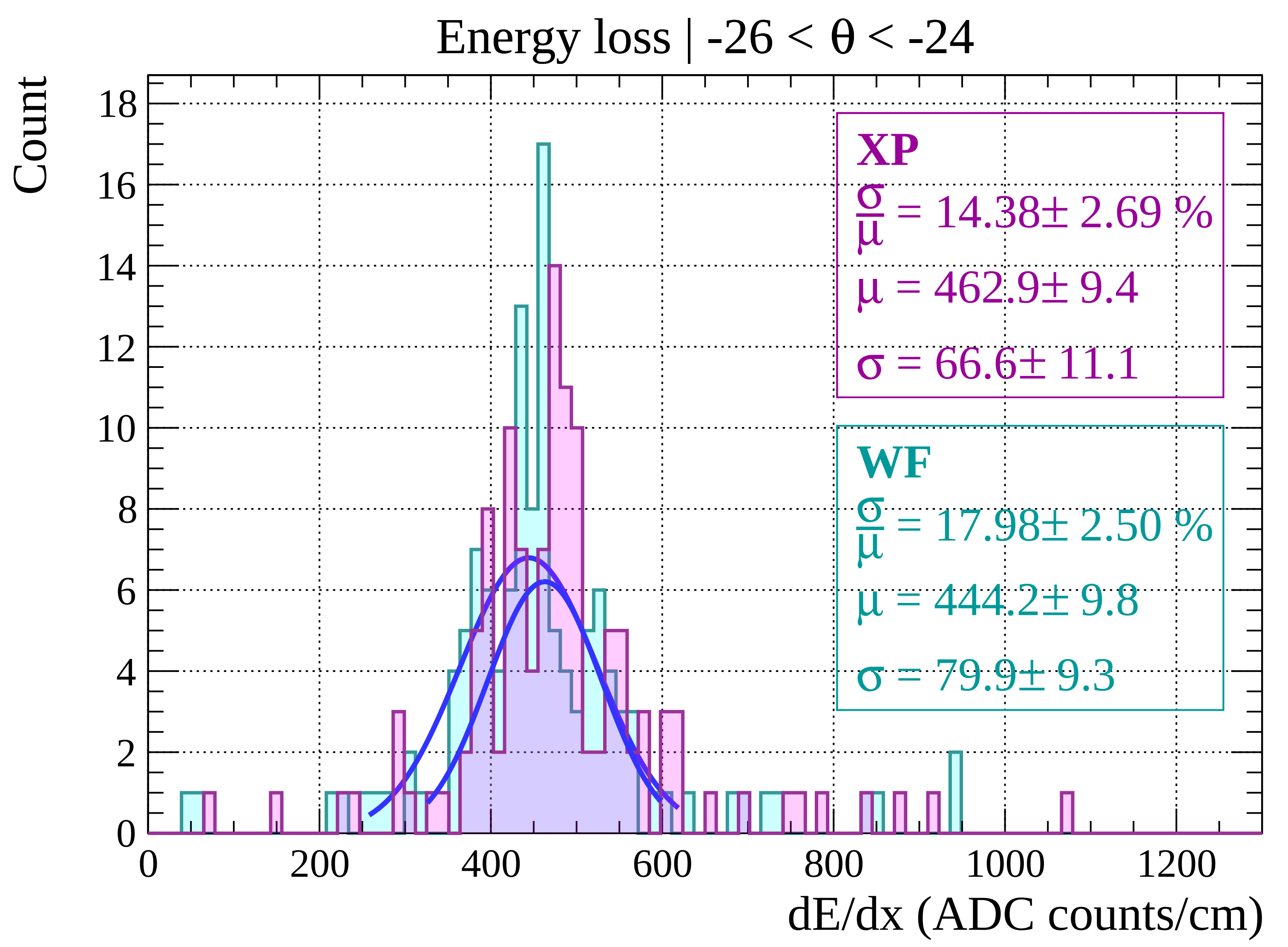


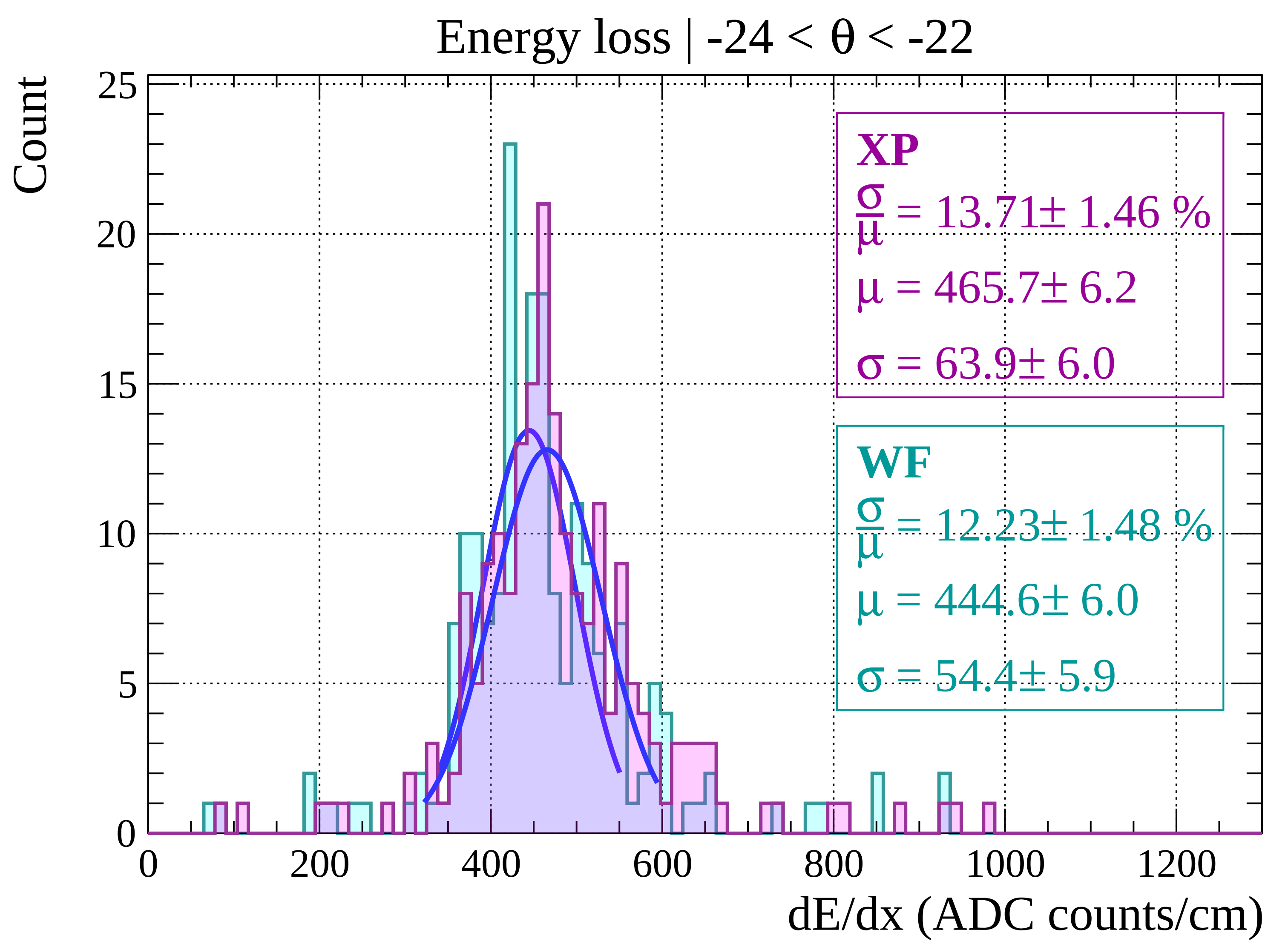
Energy loss | -32 < θ < -30
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 18.72± 5.97 %
μ = 435.7± 15.6
σ = 81.6± 23.1
WF
σ/μ = 19.43± 4.98 %
μ = 440.0± 15.6
σ = 85.5± 18.9
0
1
2
3
4
5
6
7
8
0
200
400
600
800
1000
1200


Energy loss | -30 < θ < -28
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
0
200
400
600
800
1000
1200
XP
σ/μ = 14.82± 3.25 %
μ = 446.0± 10.6
σ = 66.1± 12.9
WF
σ/μ = 20.91± 5.34 %
μ = 439.4± 15.6
σ = 91.9± 20.2

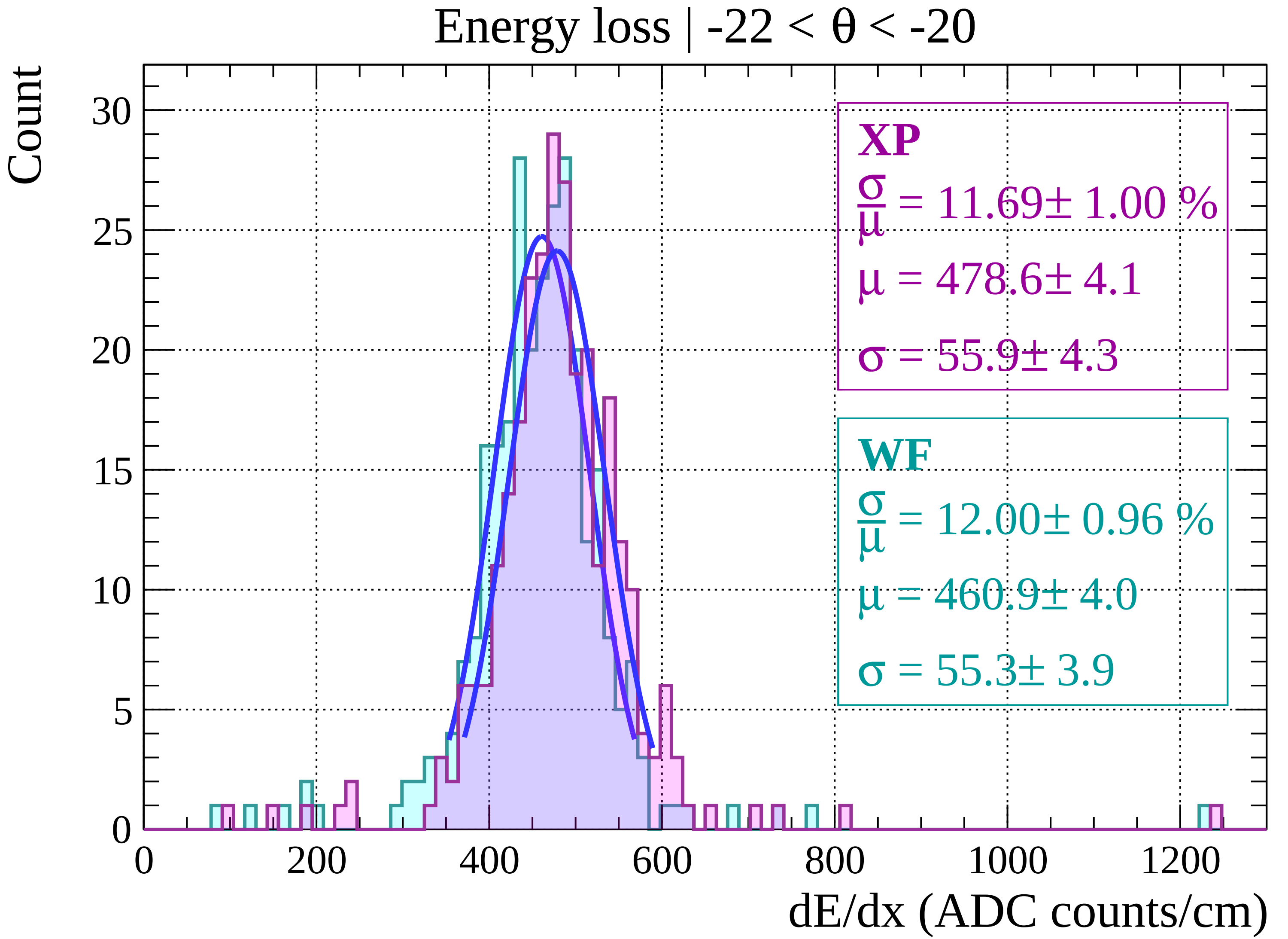


Energy loss | -28 < θ < -26
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
2
4
6
8
10
12
XP
σ/μ = 13.68± 2.64 %
μ = 454.6± 9.9
σ = 62.2± 10.7
WF
σ/μ = 14.13± 2.18 %
μ = 426.9± 8.5
σ = 60.3± 8.1

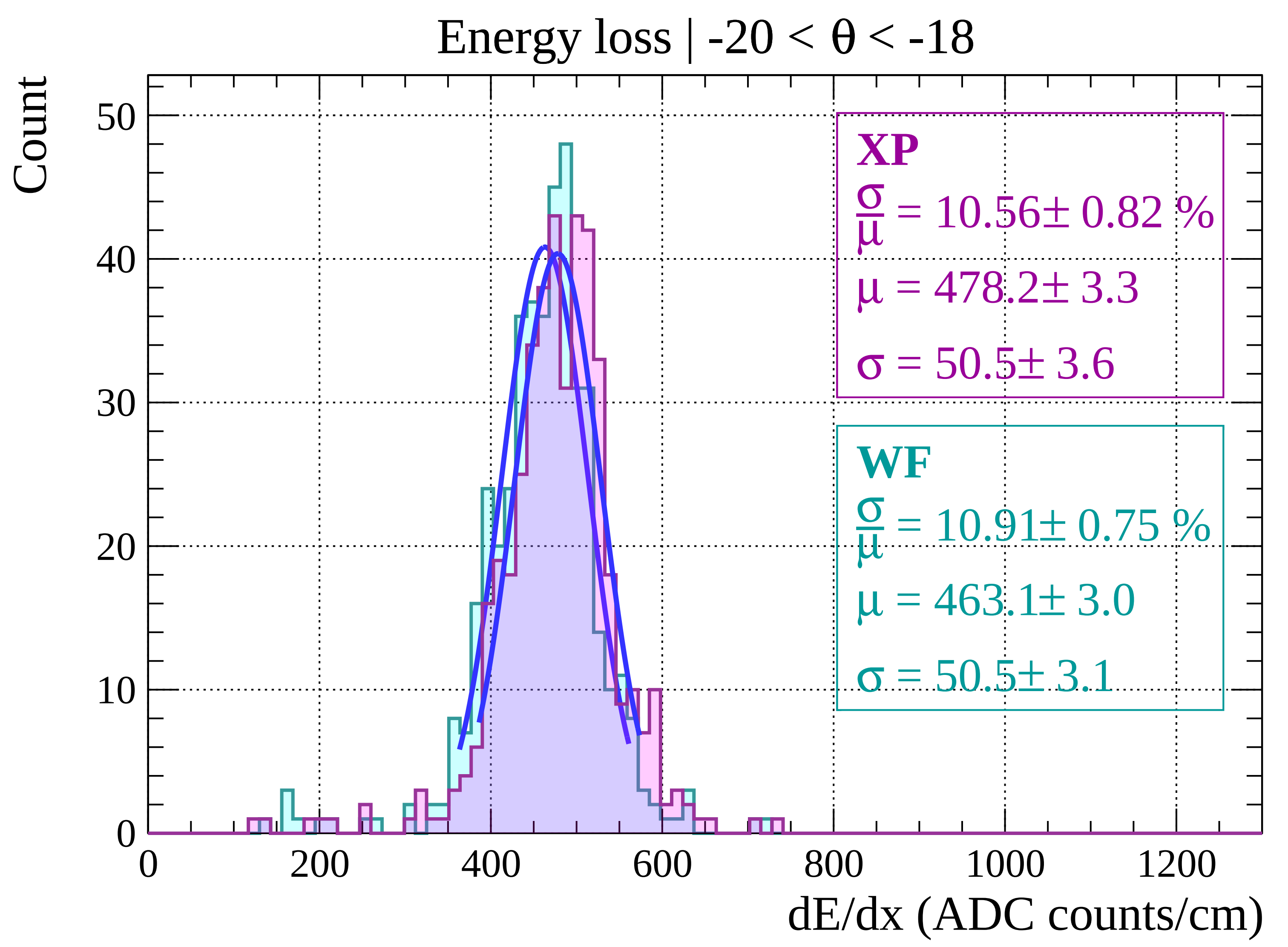

Energy loss | -26 < θ < -24
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
18
0
200
400
600
800
1000
1200
XP
σ/μ = 14.38± 2.69 %
μ = 462.9± 9.4
σ = 66.6± 11.1
WF
σ/μ = 17.98± 2.50 %
μ = 444.2± 9.8
σ = 79.9± 9.3



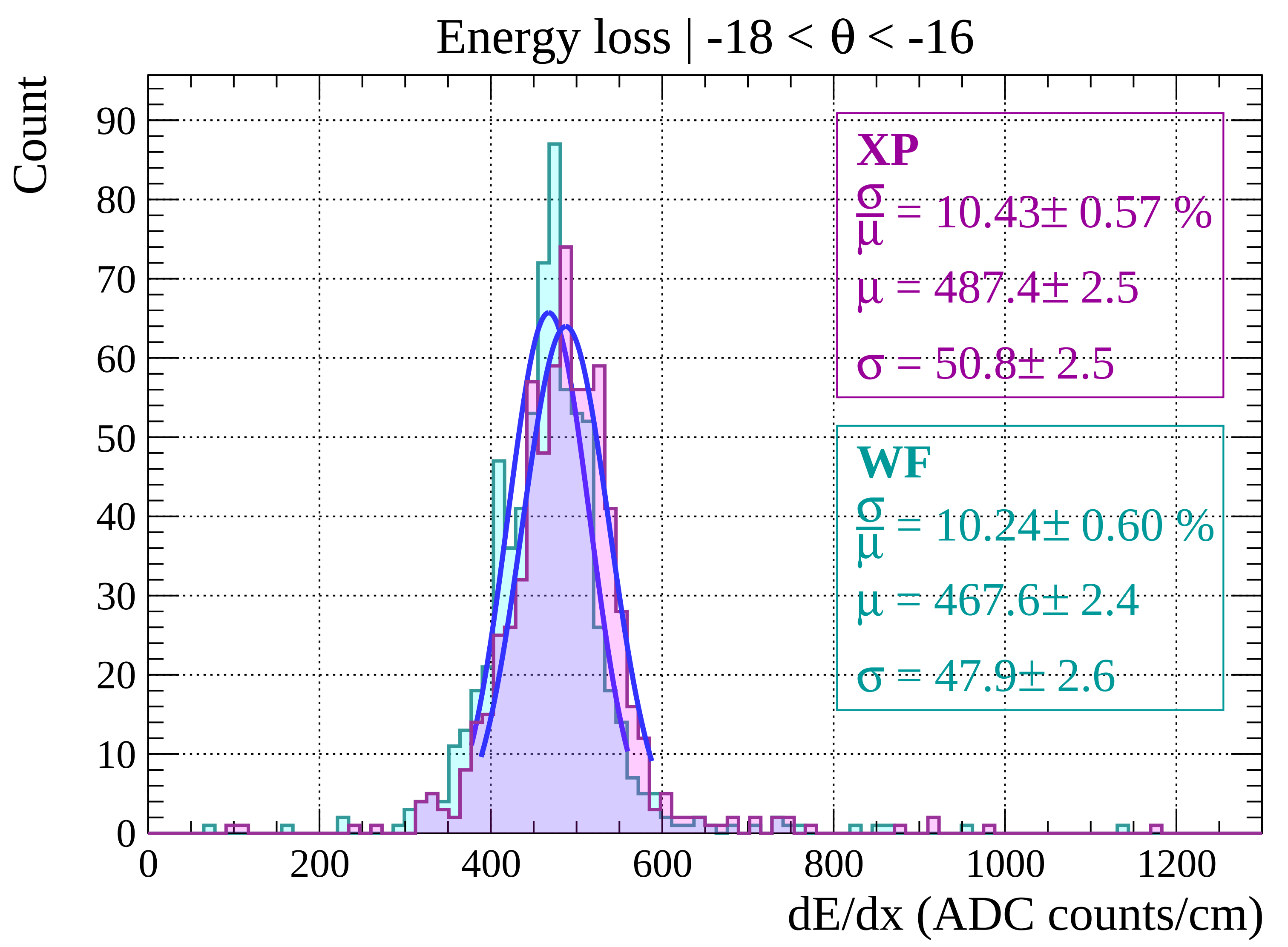
Energy loss | -24 < θ < -22
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 13.71± 1.46 %
μ = 465.7± 6.2
σ = 63.9± 6.0
WF
σ/μ = 12.23± 1.48 %
μ = 444.6± 6.0
σ = 54.4± 5.9
0
5
10
15
20
25
0
200
400
600
800
1000
1200



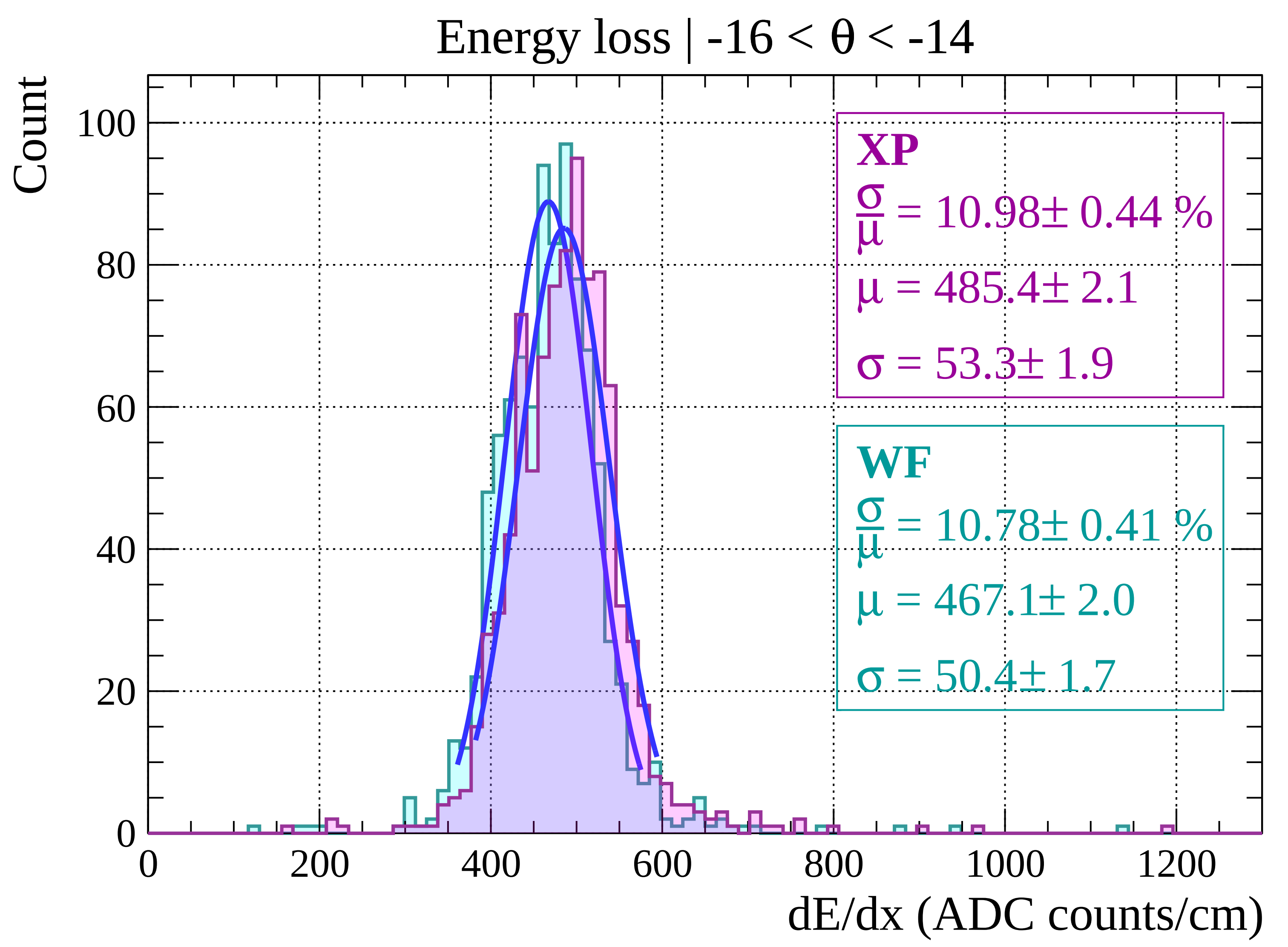
Energy loss | -22 < θ < -20
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200
XP
σ/μ = 11.69± 1.00 %
μ = 478.6± 4.1
σ = 55.9± 4.3
WF
σ/μ = 12.00± 0.96 %
μ = 460.9± 4.0
σ = 55.3± 3.9

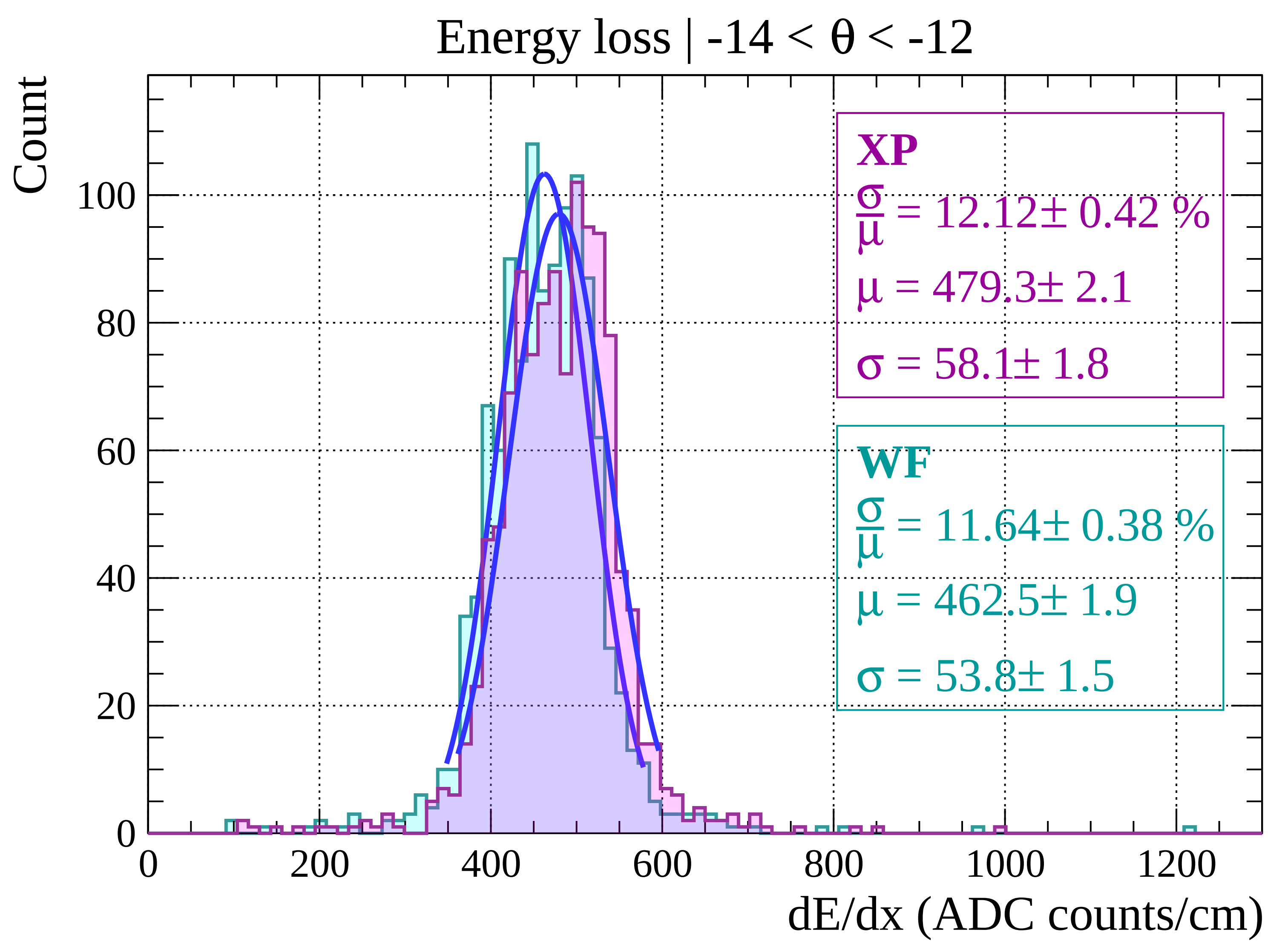


Energy loss | -20 < θ < -18
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
σ/μ = 10.56± 0.82 %
μ = 478.2± 3.3
σ = 50.5± 3.6
WF
σ/μ = 10.91± 0.75 %
μ = 463.1± 3.0
σ = 50.5± 3.1

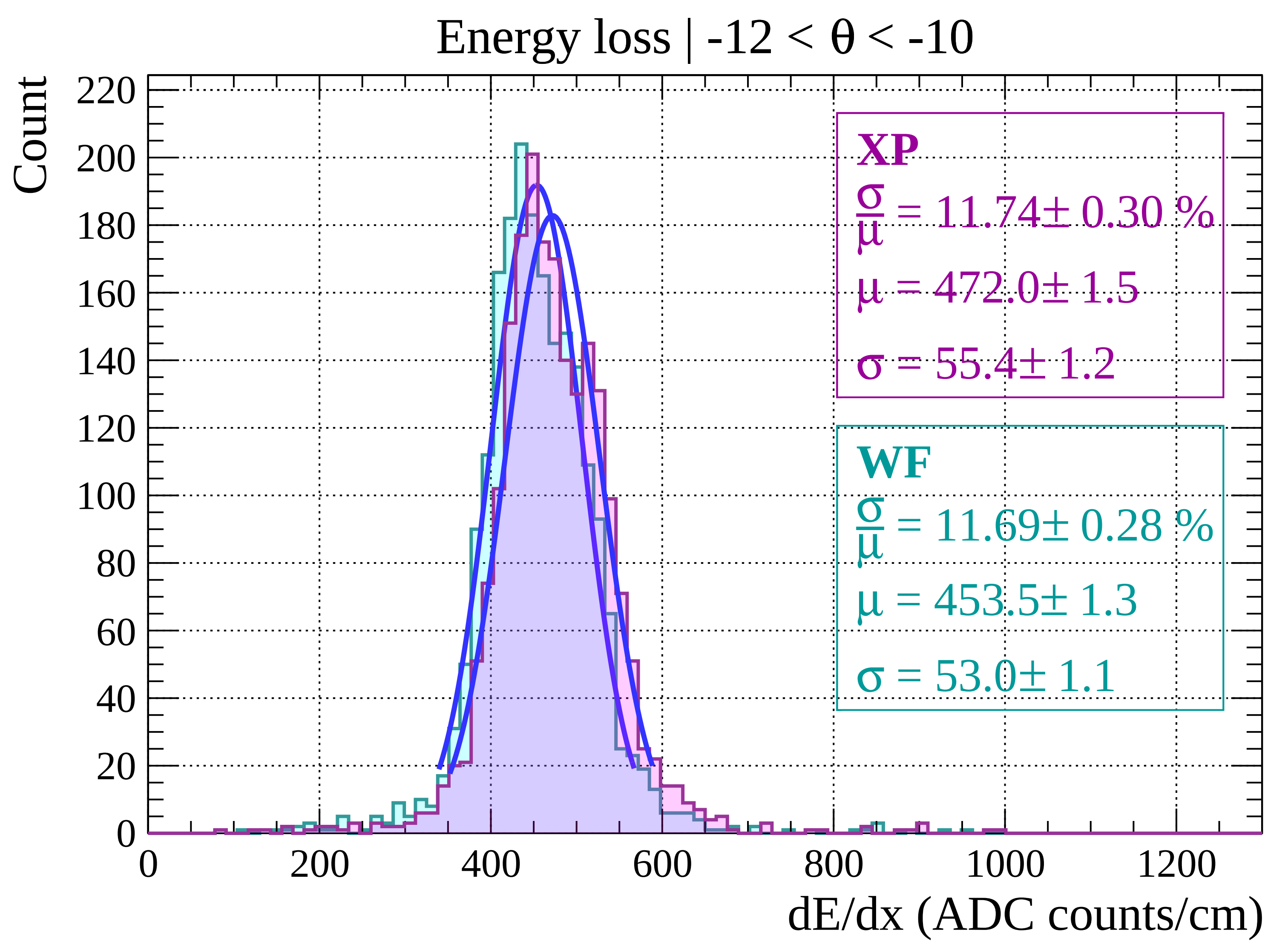


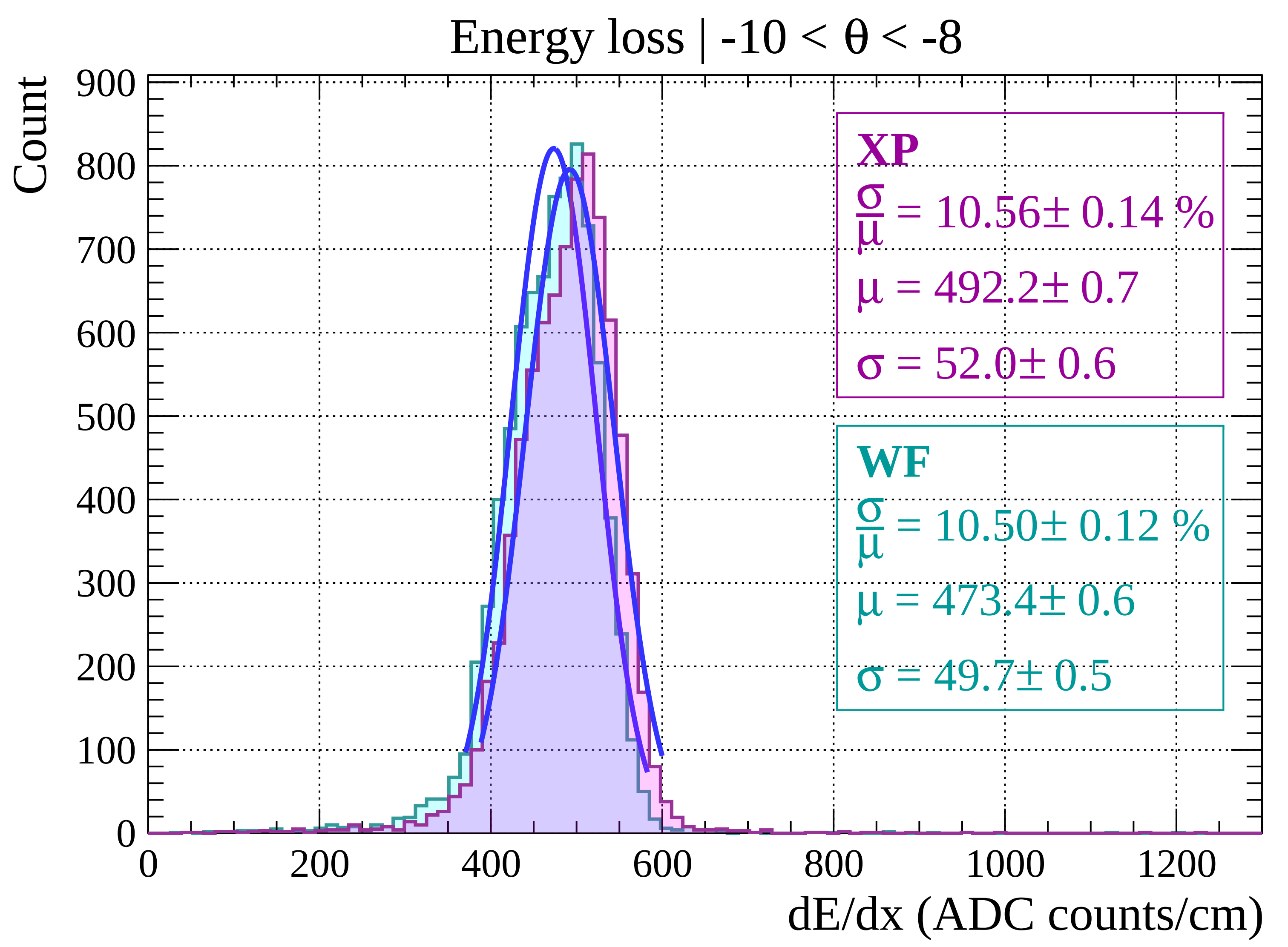
Energy loss | -18 < θ < -16
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP
σ/μ = 10.43± 0.57 %
μ = 487.4± 2.5
σ = 50.8± 2.5
WF
σ/μ = 10.24± 0.60 %
μ = 467.6± 2.4
σ = 47.9± 2.6


Energy loss | -16 < θ < -14
Count
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 10.98± 0.44 %
μ = 485.4± 2.1
σ = 53.3± 1.9
WF
σ/μ = 10.78± 0.41 %
μ = 467.1± 2.0
σ = 50.4± 1.7

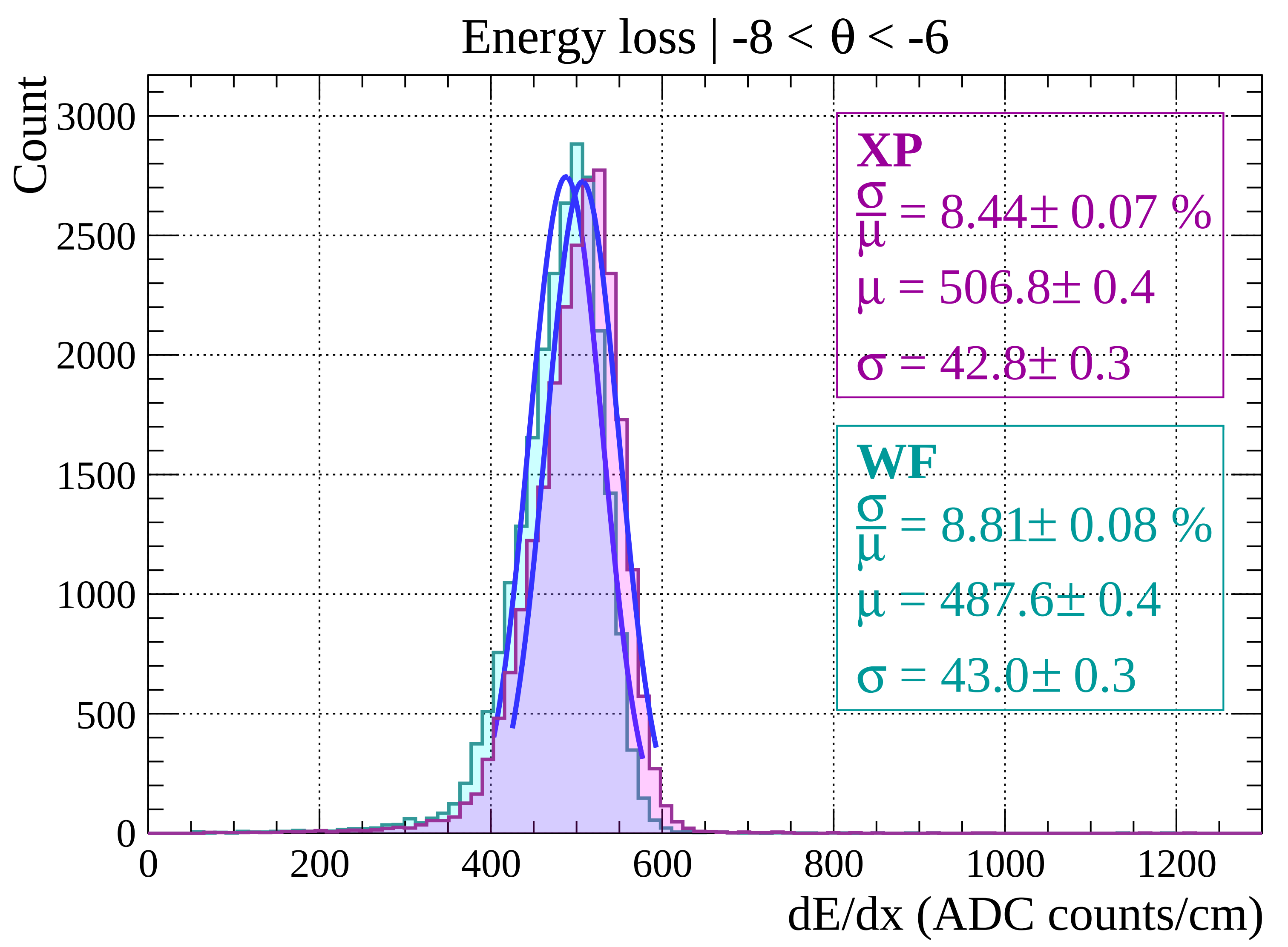


Energy loss | -14 < θ < -12
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 12.12± 0.42 %
μ = 479.3± 2.1
σ = 58.1± 1.8
WF
σ/μ = 11.64± 0.38 %
μ = 462.5± 1.9
σ = 53.8± 1.5
0
20
40
60
80
100
0
200
400
600
800
1000
1200

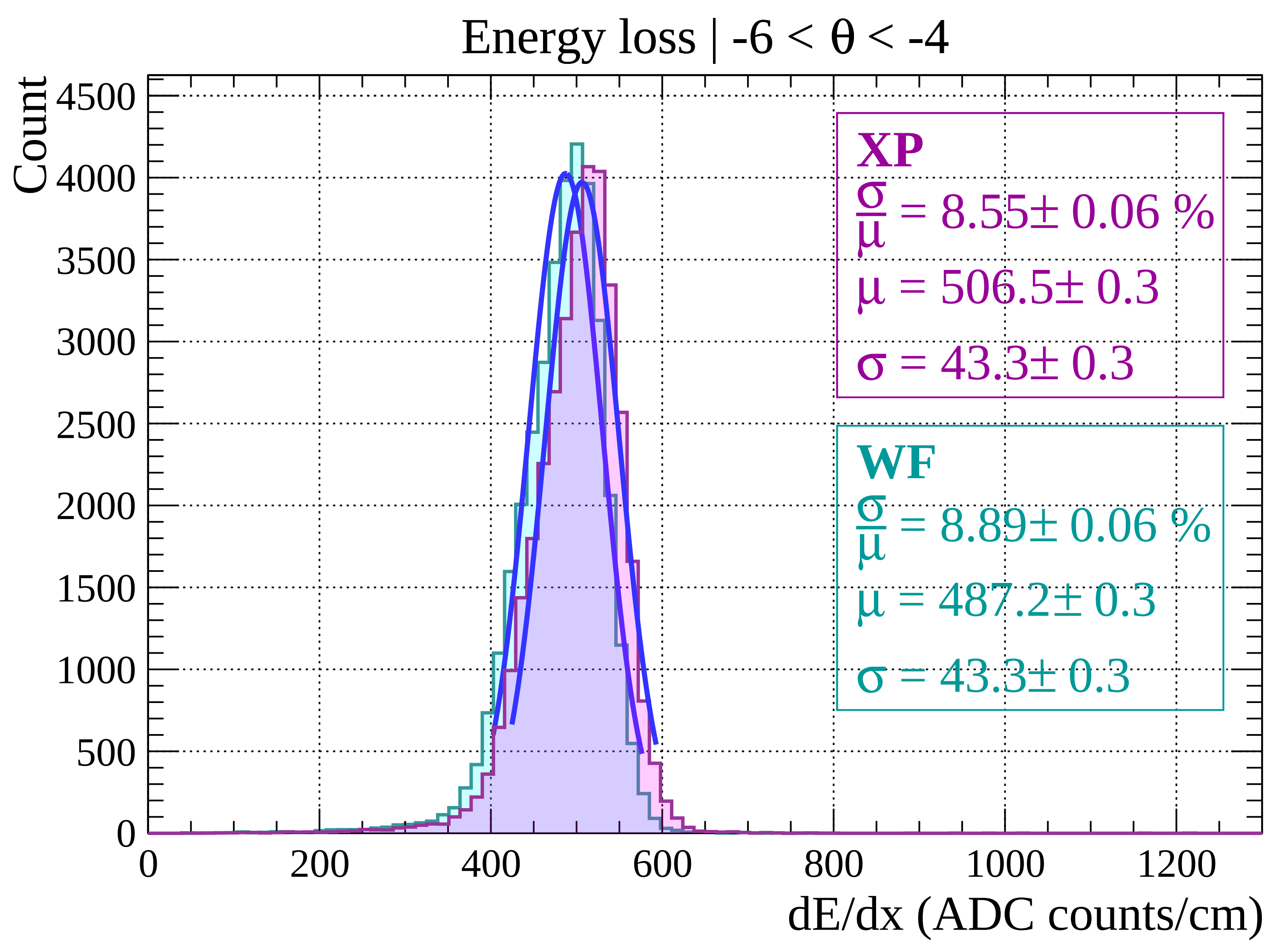

Energy loss | -12 < θ < -10
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.74± 0.30 %
μ = 472.0± 1.5
σ = 55.4± 1.2
WF
σ/μ = 11.69± 0.28 %
μ = 453.5± 1.3
σ = 53.0± 1.1

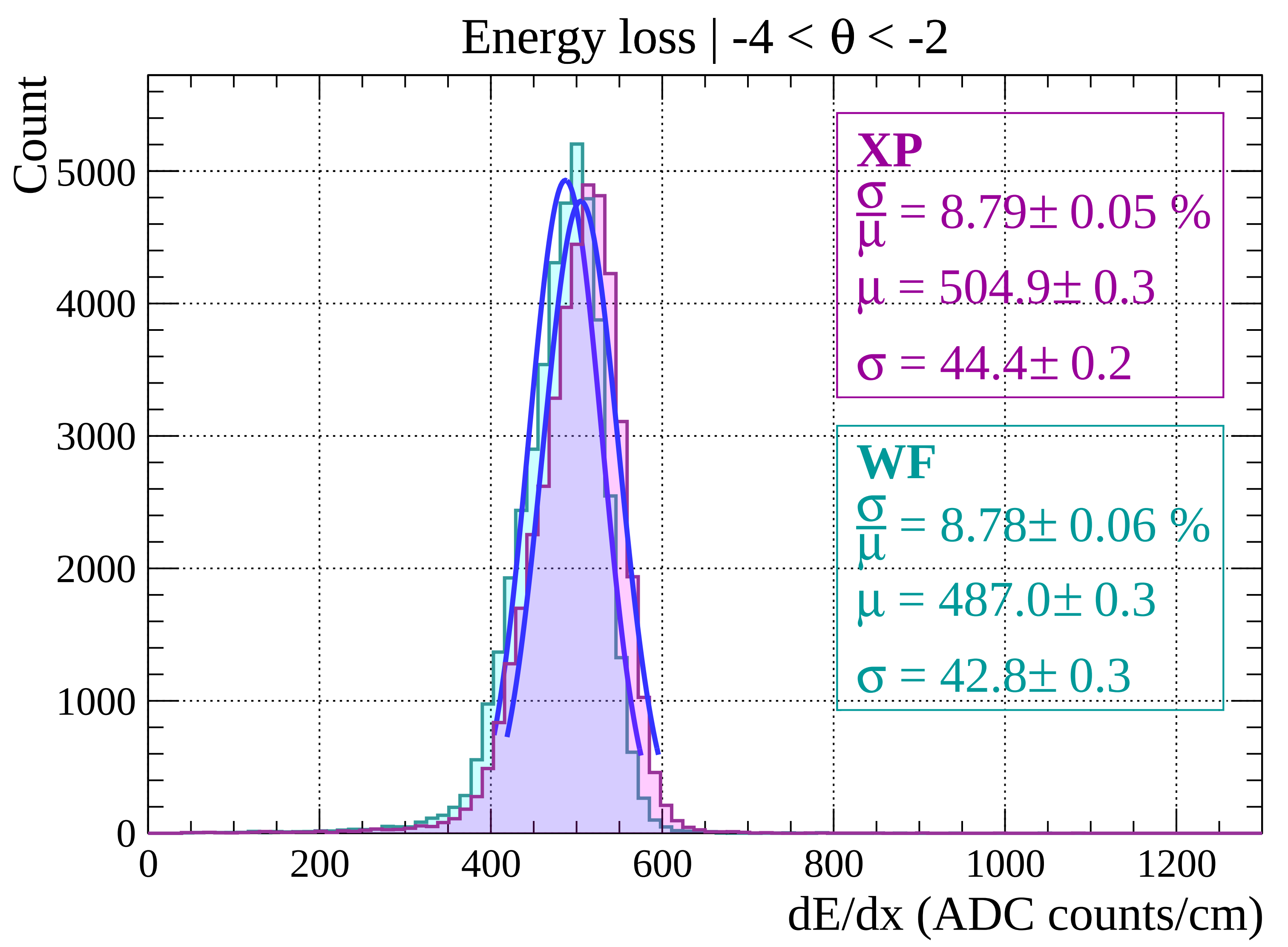


Energy loss | -10 < θ < -8
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
σ/μ = 10.56± 0.14 %
μ = 492.2± 0.7
σ = 52.0± 0.6
WF
σ/μ = 10.50± 0.12 %
μ = 473.4± 0.6
σ = 49.7± 0.5

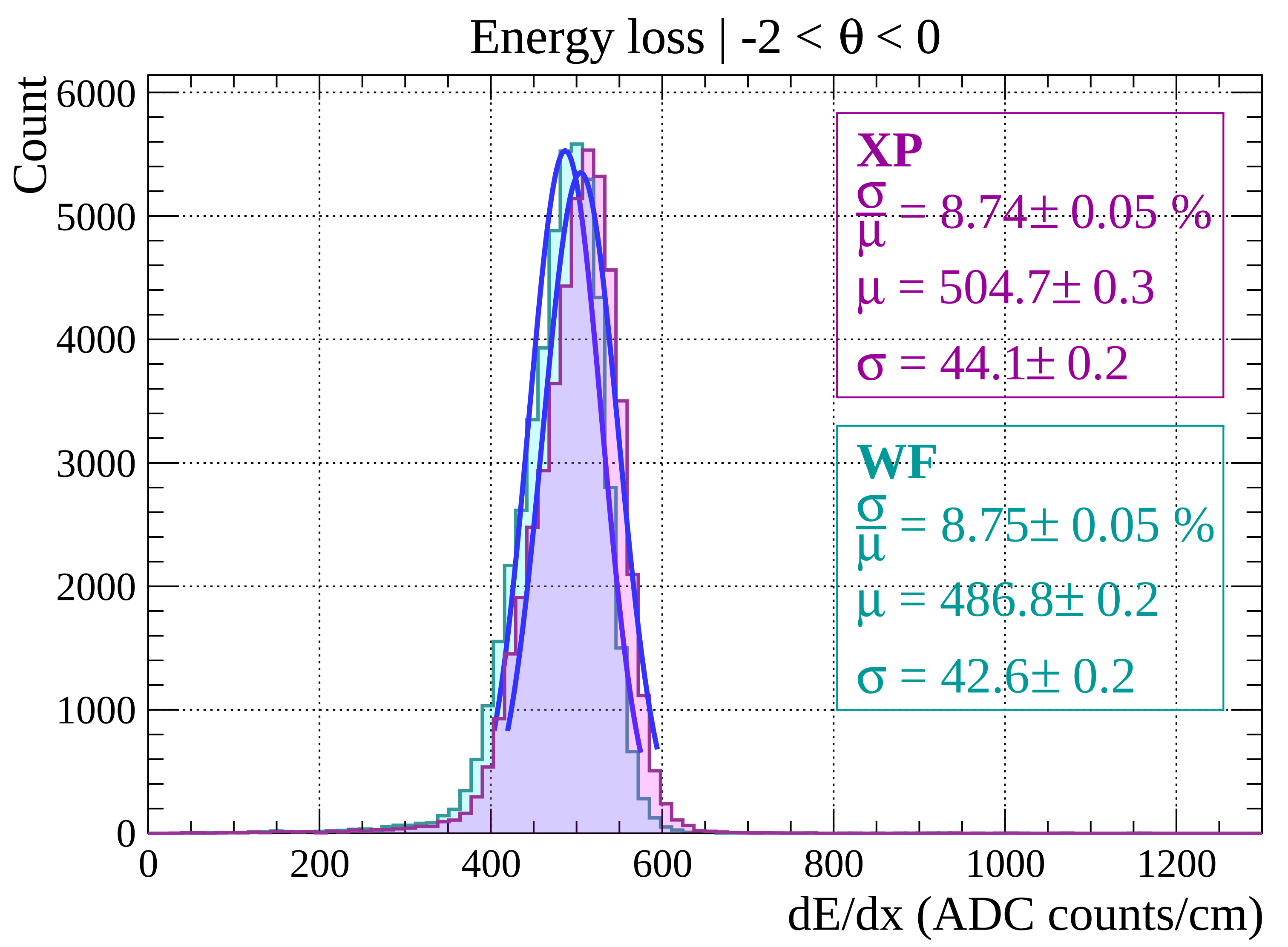


Energy loss | -8 < θ < -6
Count
3000
2500
2000
1500
1000
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.44± 0.07 %
μ = 506.8± 0.4
σ = 42.8± 0.3
WF
$\frac{\sigma}{\mu}$ = 8.81± 0.08 %
μ = 487.6± 0.4
σ = 43.0± 0.3

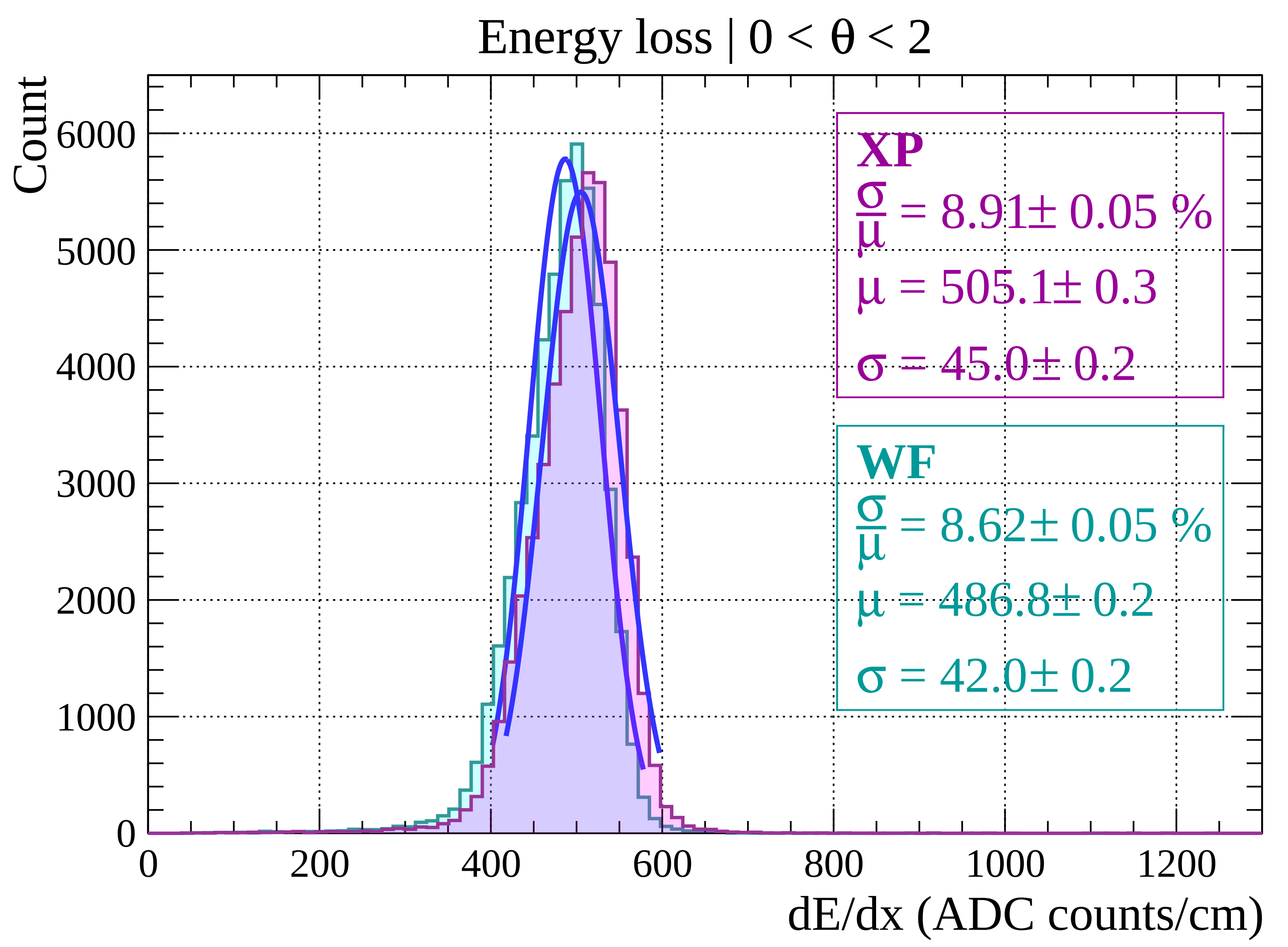

Energy loss | -6 < θ < -4
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
500
1000
1500
2000
2500
3000
3500
4000
4500
XP
σ/μ = 8.55± 0.06 %
μ = 506.5± 0.3
σ = 43.3± 0.3
WF
σ/μ = 8.89± 0.06 %
μ = 487.2± 0.3
σ = 43.3± 0.3

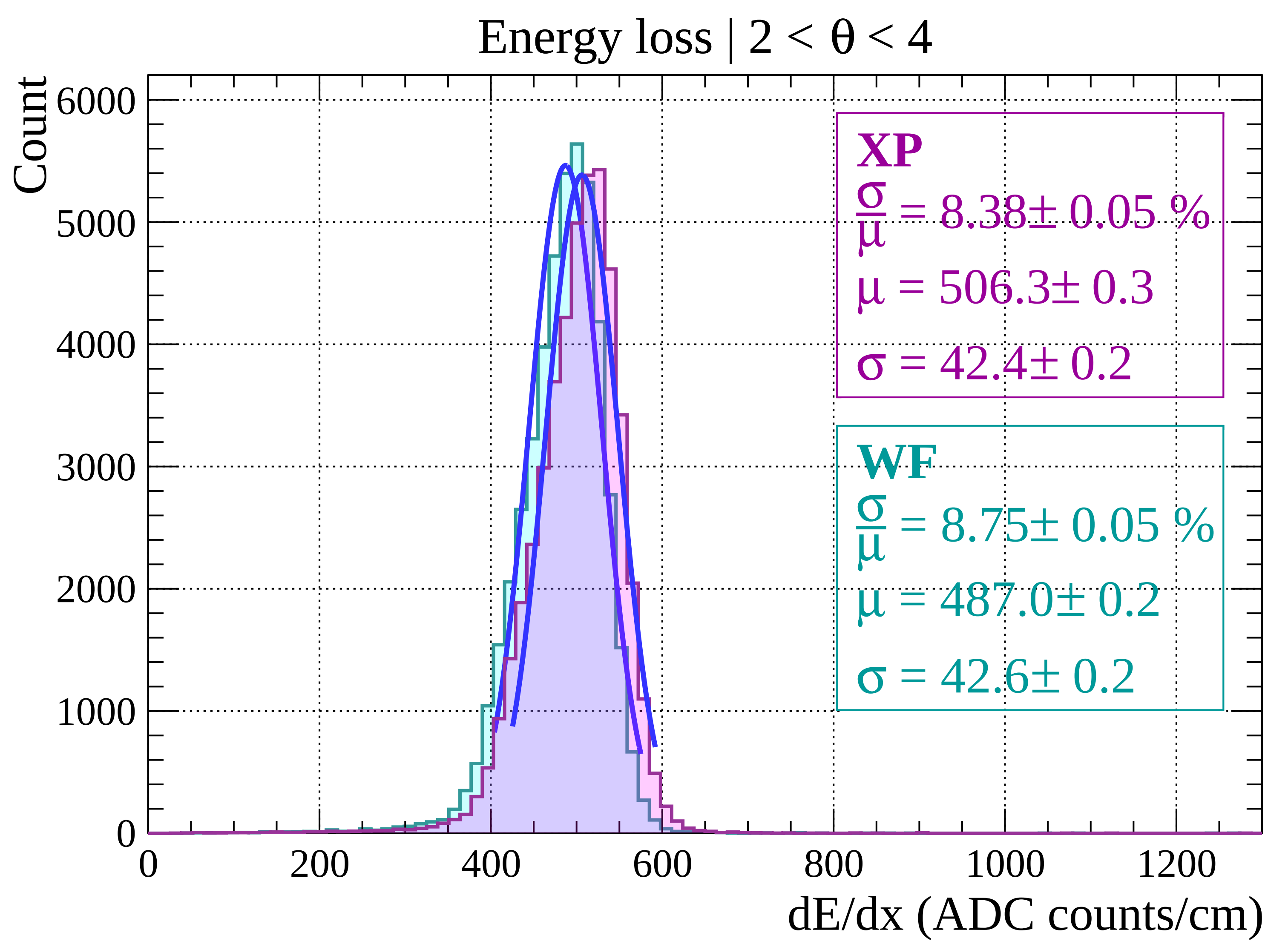


Energy loss | -4 < θ < -2
Count
0
1000
2000
3000
4000
5000
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.79± 0.05 %
μ = 504.9± 0.3
σ = 44.4± 0.2
WF
σ/μ = 8.78± 0.06 %
μ = 487.0± 0.3
σ = 42.8± 0.3

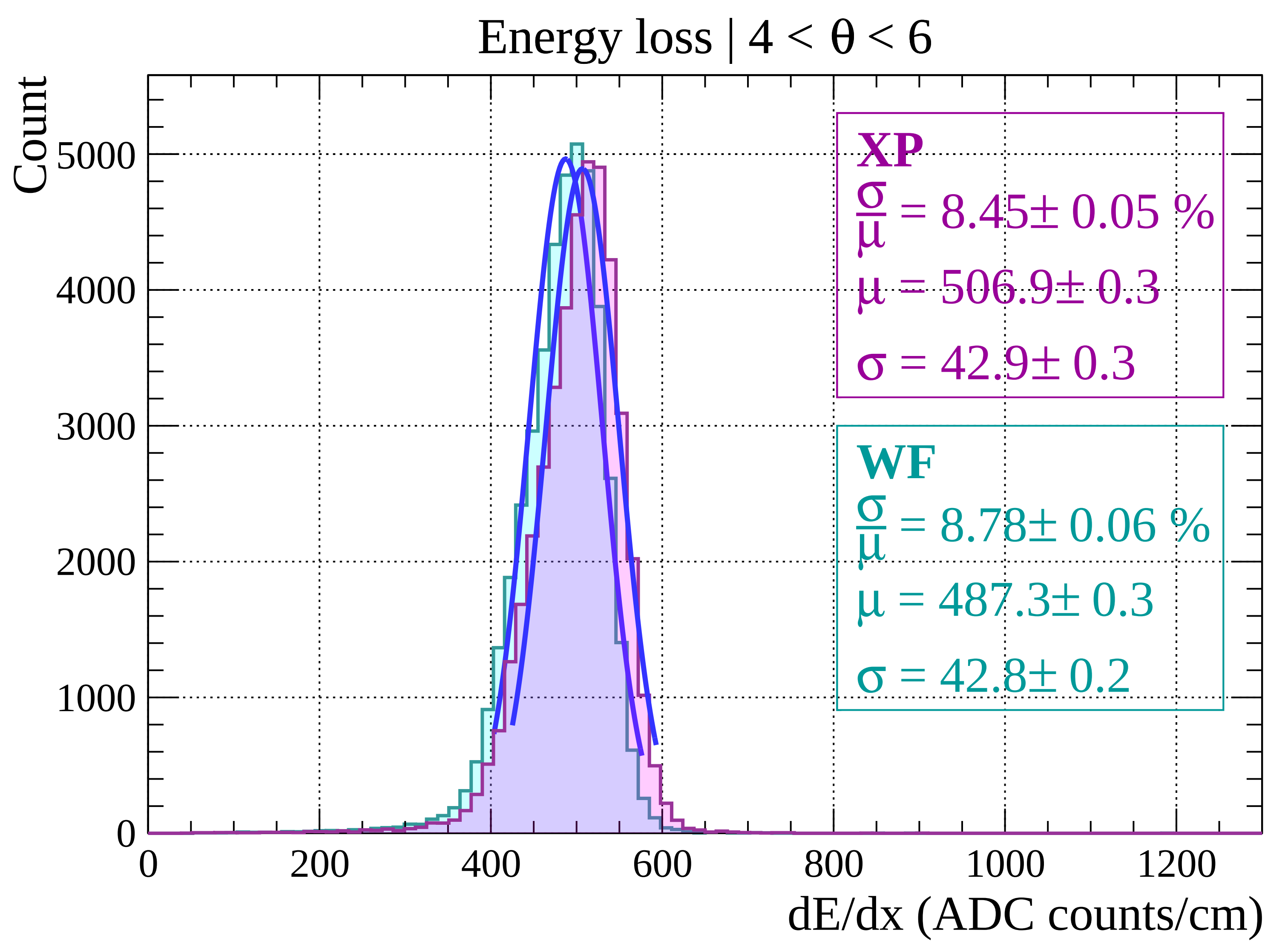


Energy loss | -2 < θ < 0
Count
6000
5000
4000
3000
2000
1000
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.74± 0.05 %
μ = 504.7± 0.3
σ = 44.1± 0.2
WF
σ/μ = 8.75± 0.05 %
μ = 486.8± 0.2
σ = 42.6± 0.2

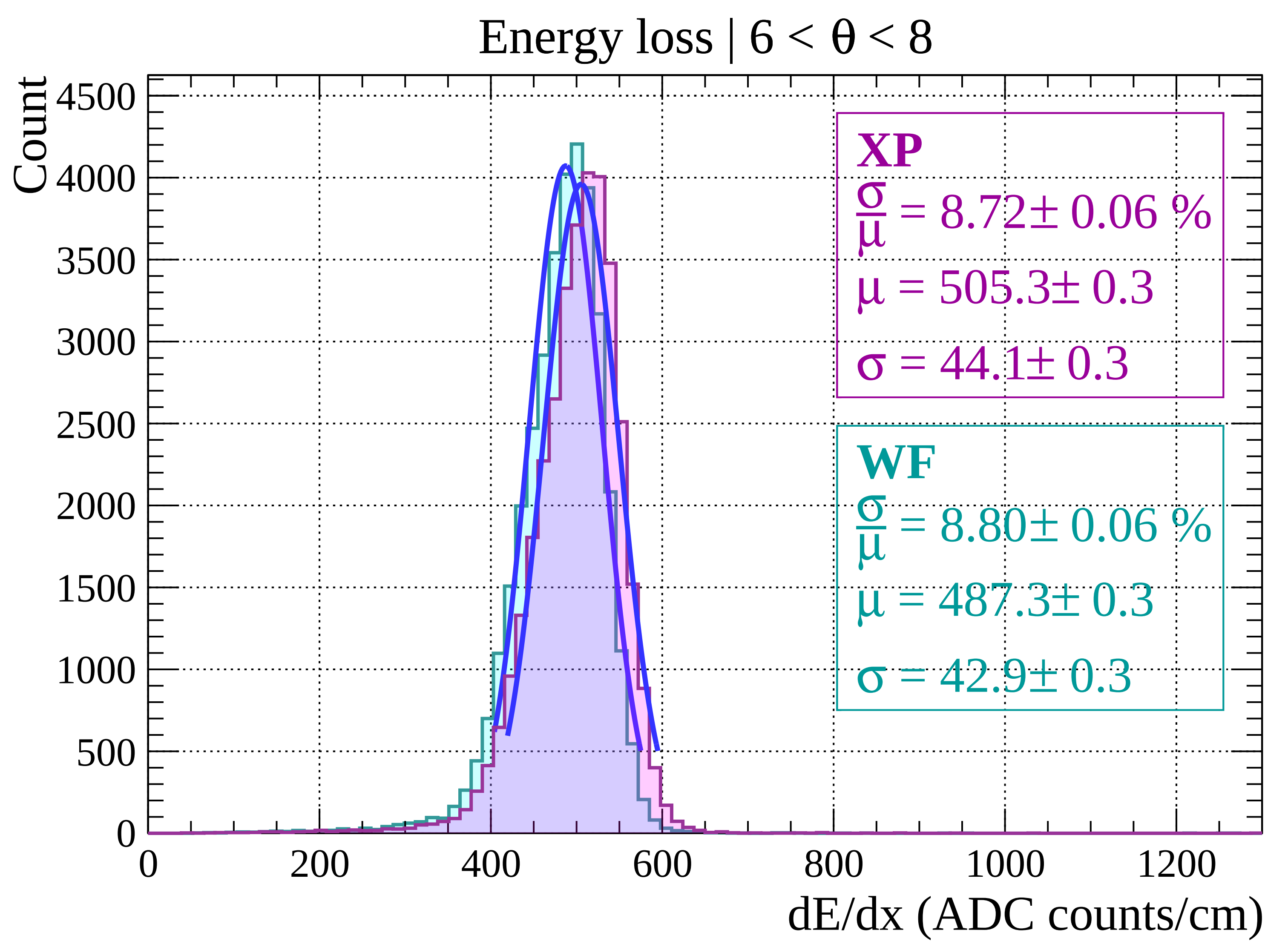

Energy loss | 0 < θ < 2
Count
dE/dx (ADC counts/cm)
0
1000
2000
3000
4000
5000
6000
0
200
400
600
800
1000
1200
XP
σ/μ = 8.91± 0.05 %
μ = 505.1± 0.3
σ = 45.0± 0.2
WF
σ/μ = 8.62± 0.05 %
μ = 486.8± 0.2
σ = 42.0± 0.2



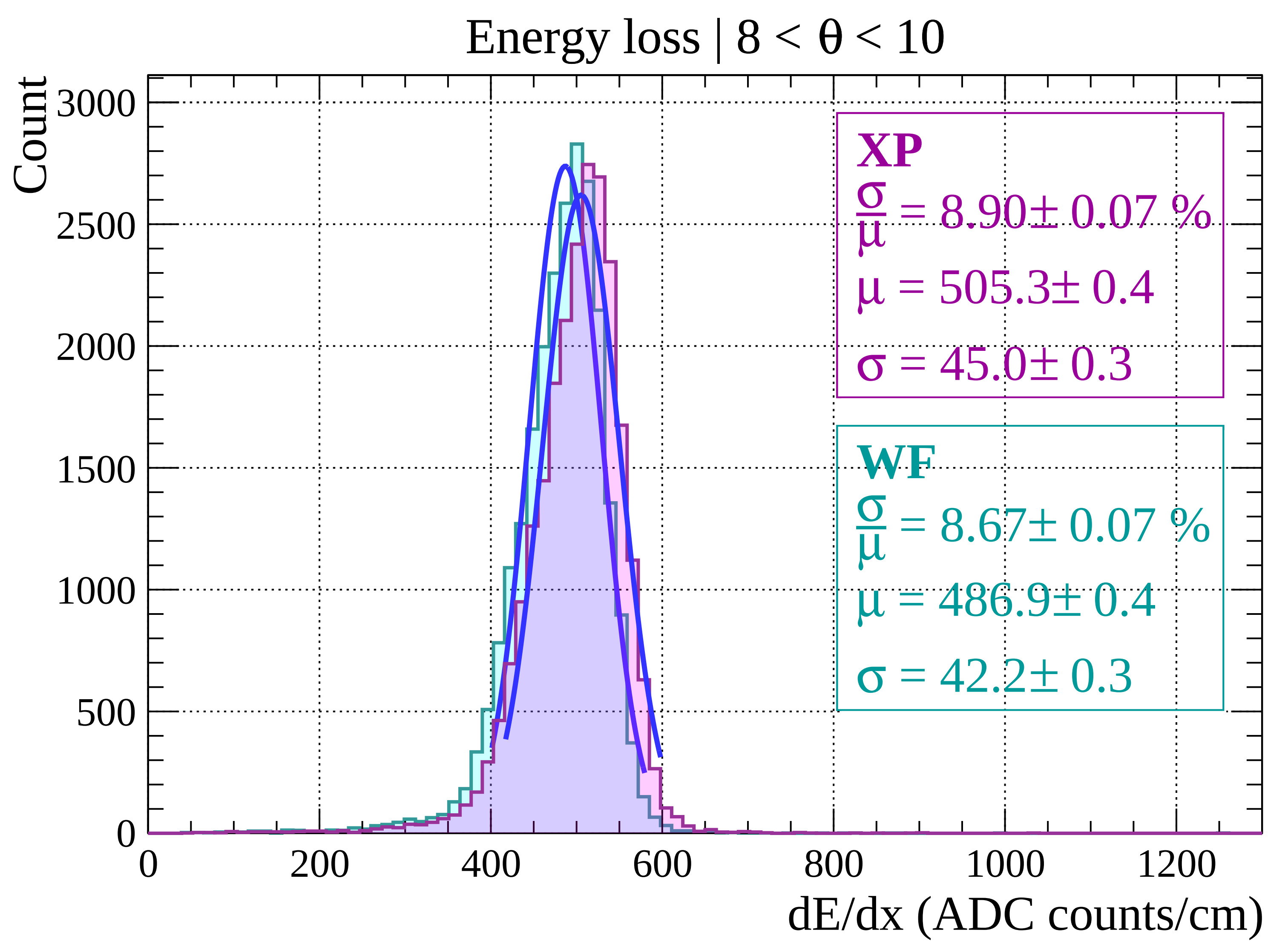
Energy loss | 2 < θ < 4
Count
6000
5000
4000
3000
2000
1000
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.38± 0.05 %
μ = 506.3± 0.3
σ = 42.4± 0.2
WF
σ/μ = 8.75± 0.05 %
μ = 487.0± 0.2
σ = 42.6± 0.2



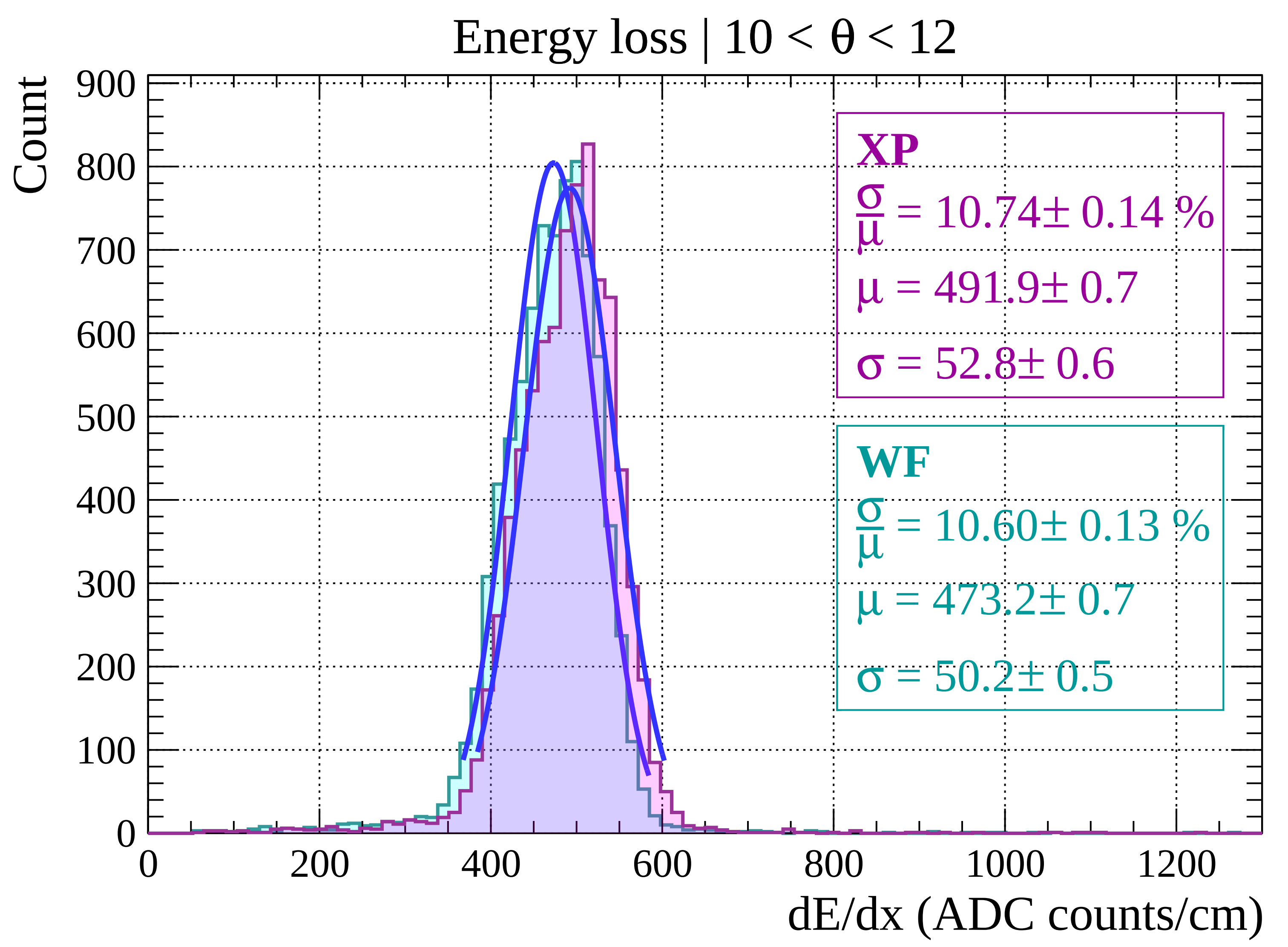
Energy loss | 4 < θ < 6
Count
5000
4000
3000
2000
1000
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.45± 0.05 %
μ = 506.9± 0.3
σ = 42.9± 0.3
WF
σ/μ = 8.78± 0.06 %
μ = 487.3± 0.3
σ = 42.8± 0.2



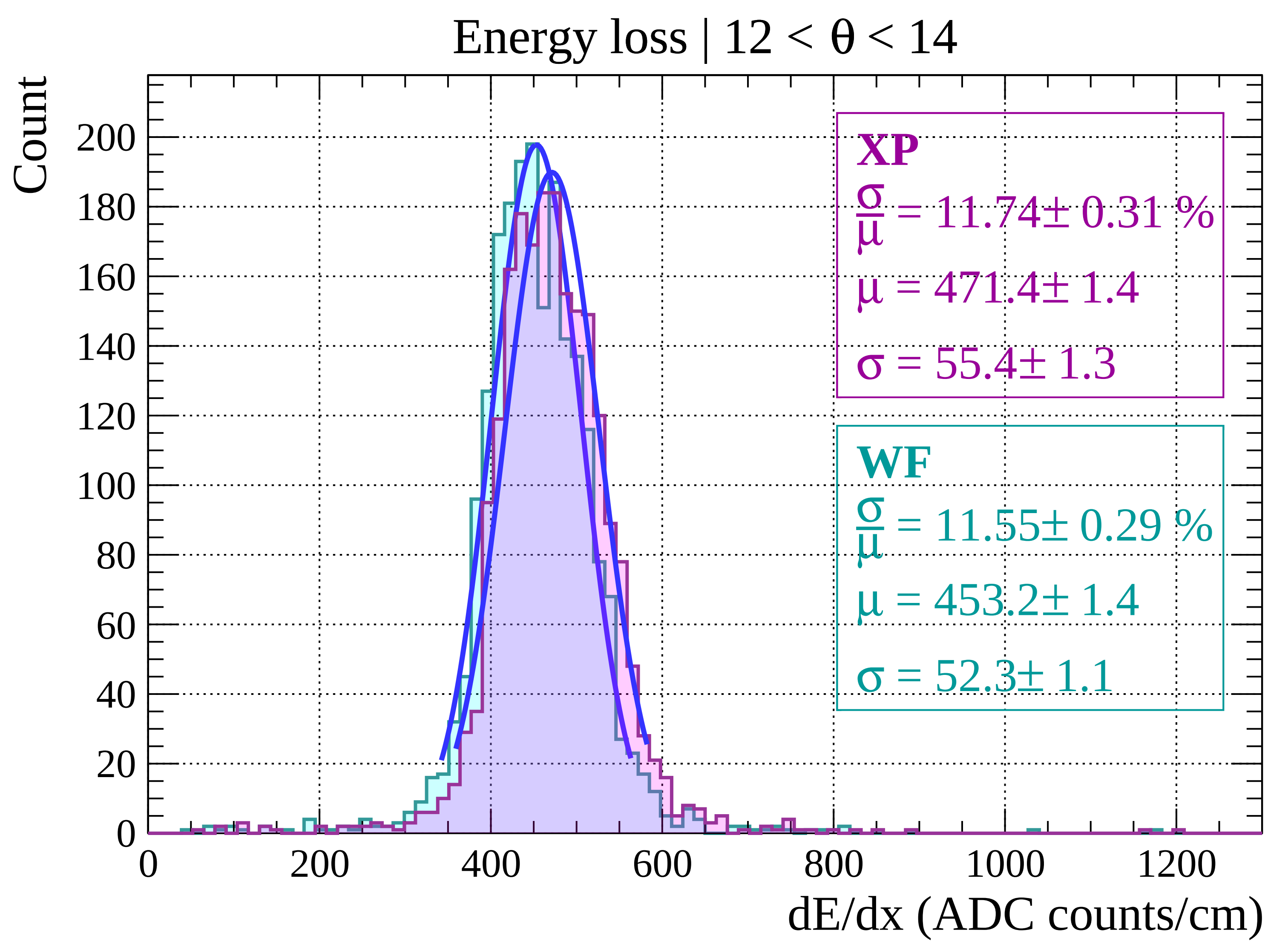
Energy loss | 6 < θ < 8
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
500
1000
1500
2000
2500
3000
3500
4000
4500
XP
σ/μ = 8.72± 0.06 %
μ = 505.3± 0.3
σ = 44.1± 0.3
WF
σ/μ = 8.80± 0.06 %
μ = 487.3± 0.3
σ = 42.9± 0.3

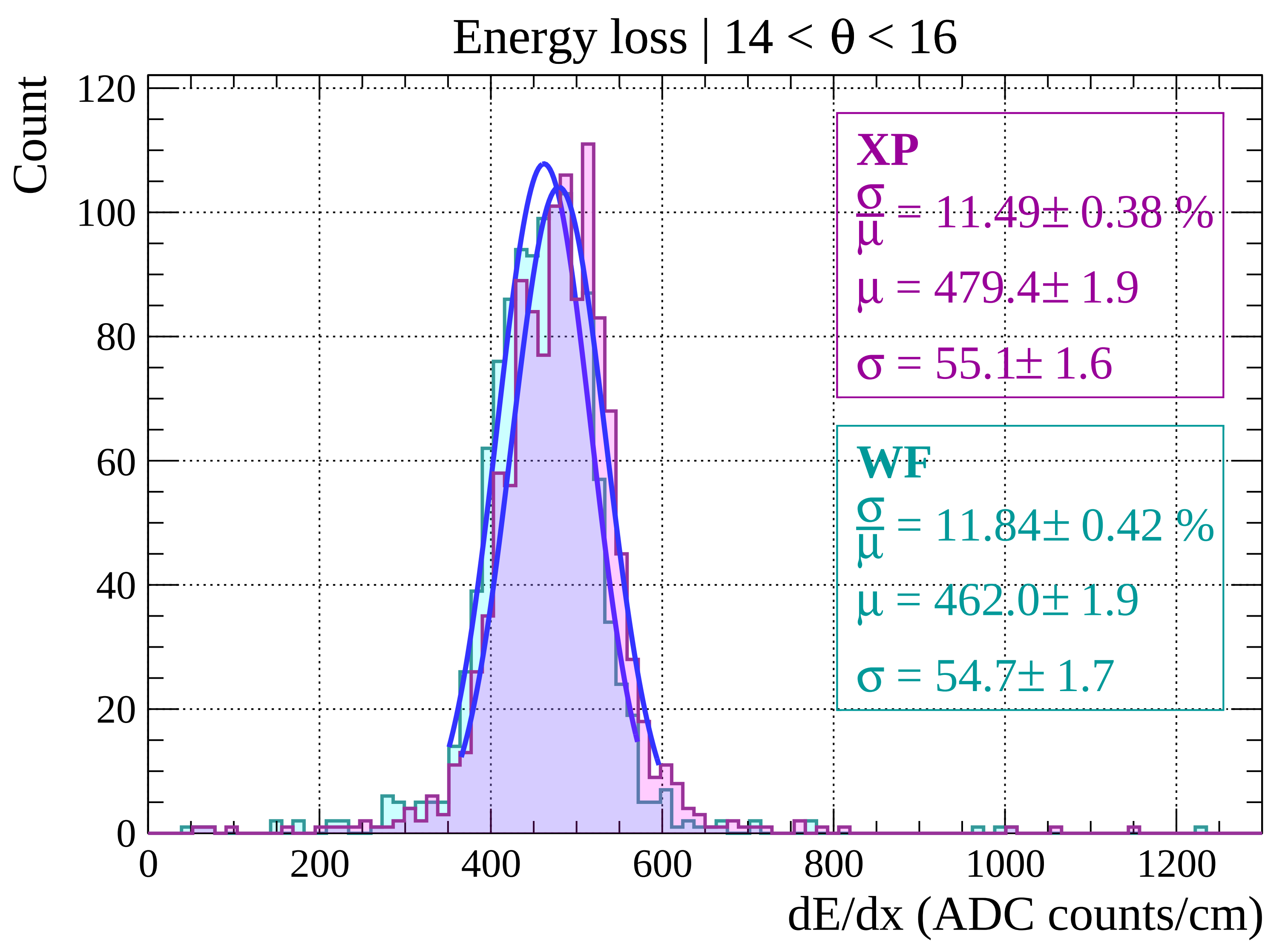


Energy loss | 8 < θ < 10
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
500
1000
1500
2000
2500
3000
XP
σ/μ = 8.90± 0.07 %
μ = 505.3± 0.4
σ = 45.0± 0.3
WF
σ/μ = 8.67± 0.07 %
μ = 486.9± 0.4
σ = 42.2± 0.3

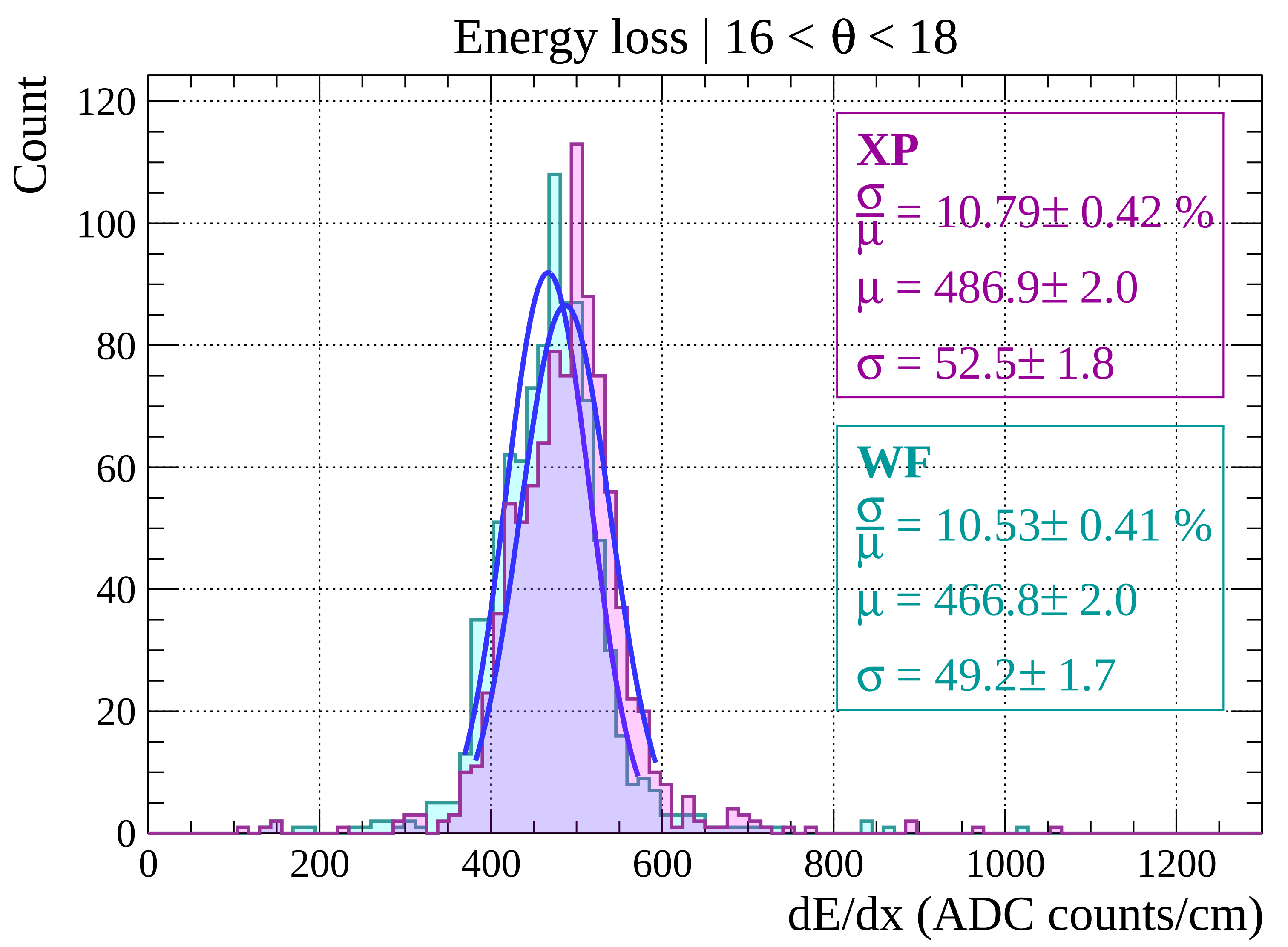


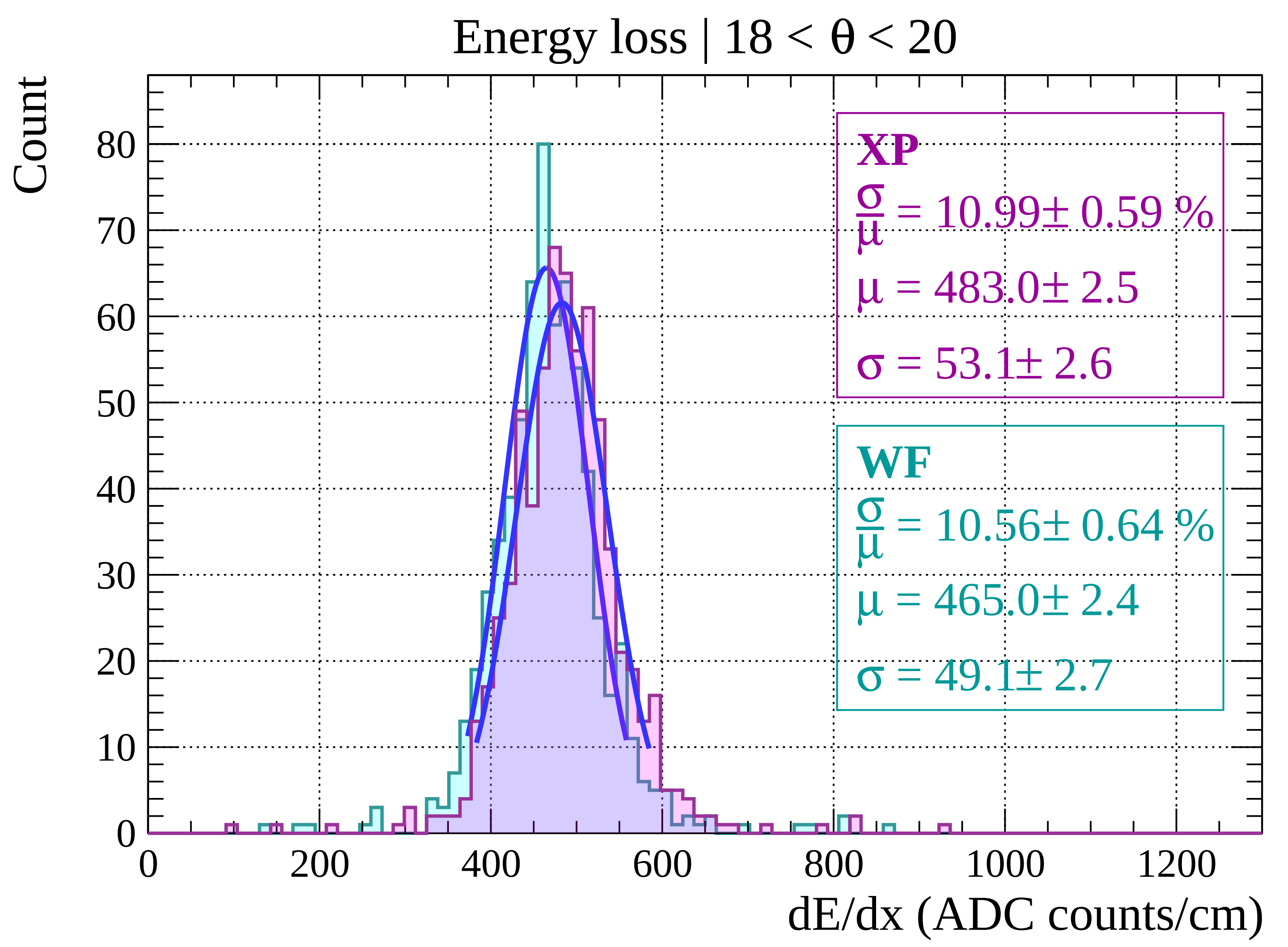
Energy loss | 10 < θ < 12
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
σ/μ = 10.74± 0.14 %
μ = 491.9± 0.7
σ = 52.8± 0.6
WF
σ/μ = 10.60± 0.13 %
μ = 473.2± 0.7
σ = 50.2± 0.5


Energy loss | 12 < θ < 14
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
XP
σ/μ = 11.74± 0.31 %
μ = 471.4± 1.4
σ = 55.4± 1.3
WF
σ/μ = 11.55± 0.29 %
μ = 453.2± 1.4
σ = 52.3± 1.1

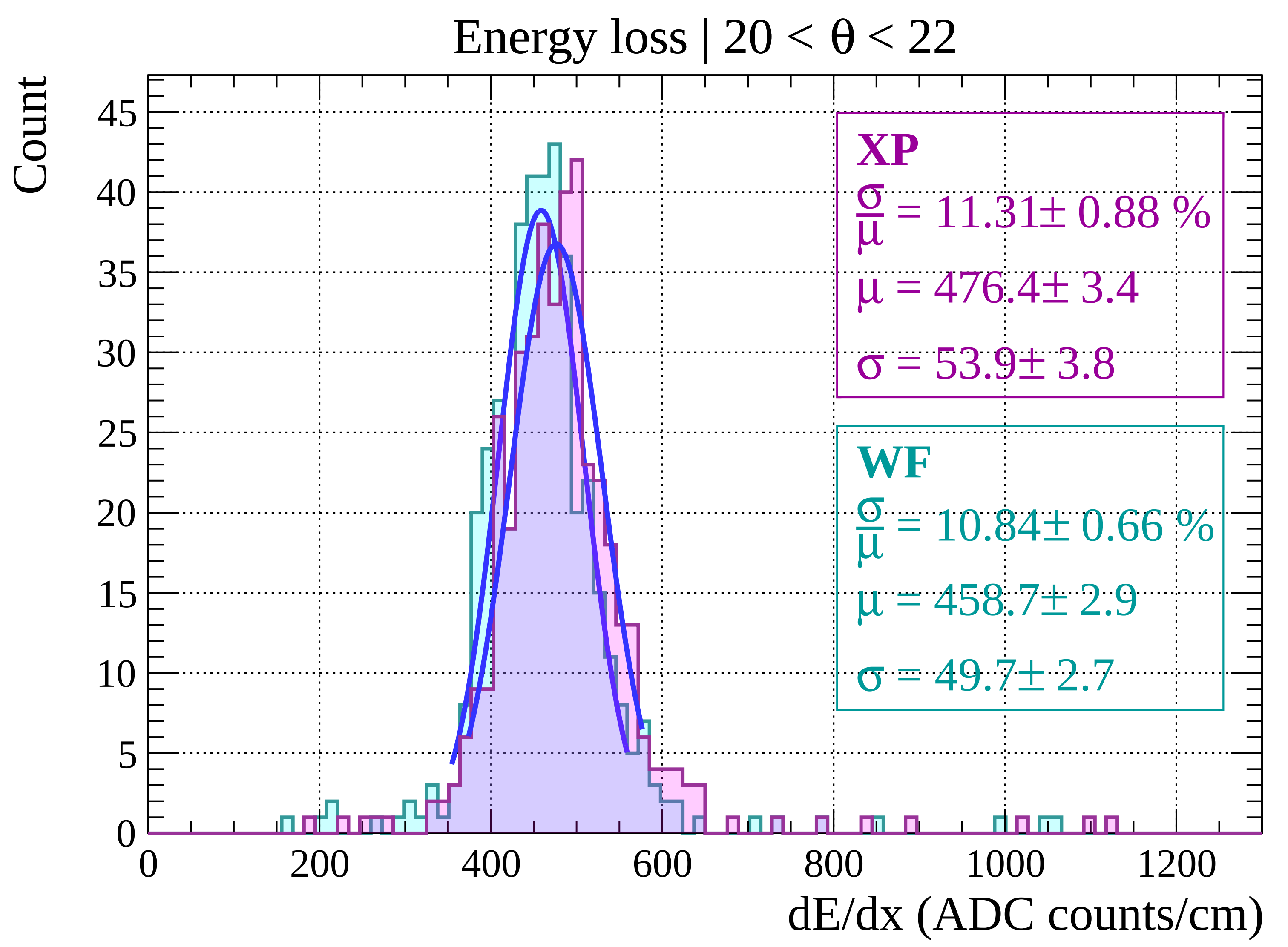


Energy loss | 14 < θ < 16
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
σ/μ = 11.49± 0.38 %
μ = 479.4± 1.9
σ = 55.1± 1.6
WF
σ/μ = 11.84± 0.42 %
μ = 462.0± 1.9
σ = 54.7± 1.7

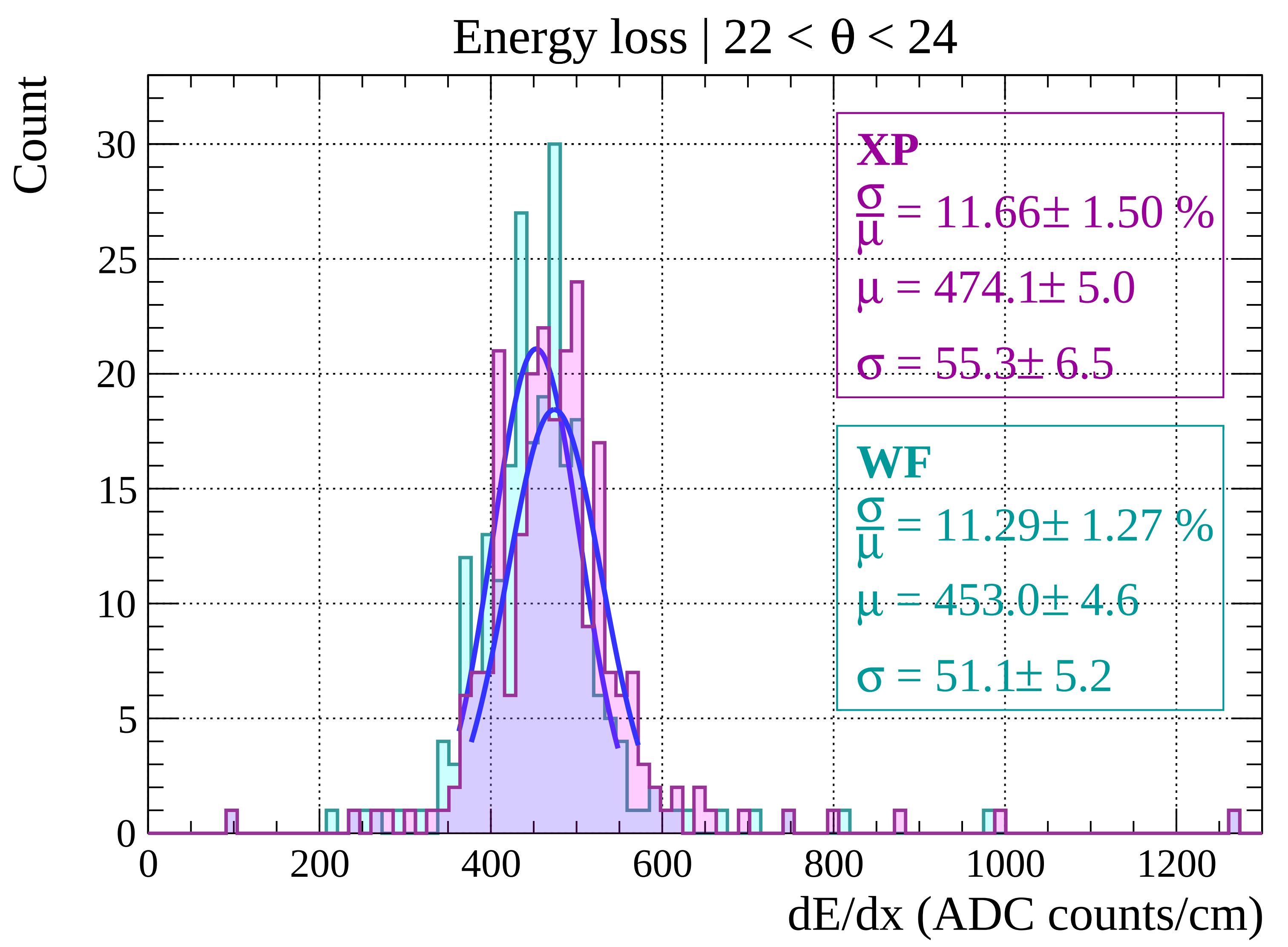

Energy loss | 16 < θ < 18
Count
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 10.79± 0.42 %
μ = 486.9± 2.0
σ = 52.5± 1.8
WF
$\frac{\sigma}{\mu}$ = 10.53± 0.41 %
μ = 466.8± 2.0
σ = 49.2± 1.7



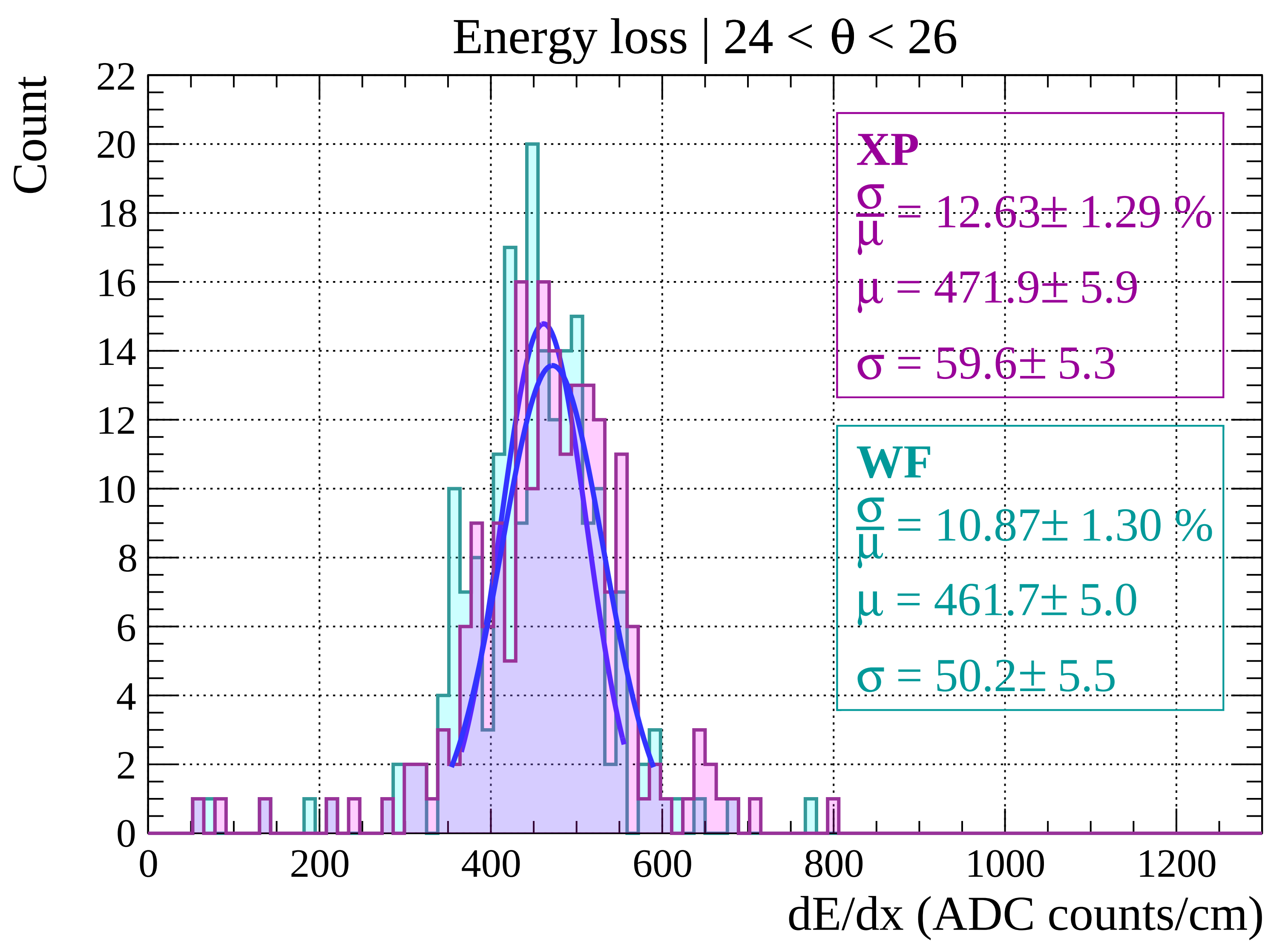
Energy loss | 18 < θ < 20
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 10.99 ± 0.59 %
μ = 483.0 ± 2.5
σ = 53.1 ± 2.6
WF
σ/μ = 10.56 ± 0.64 %
μ = 465.0 ± 2.4
σ = 49.1 ± 2.7
0
10
20
30
40
50
60
70
80
200
400
600
800
1000
1200

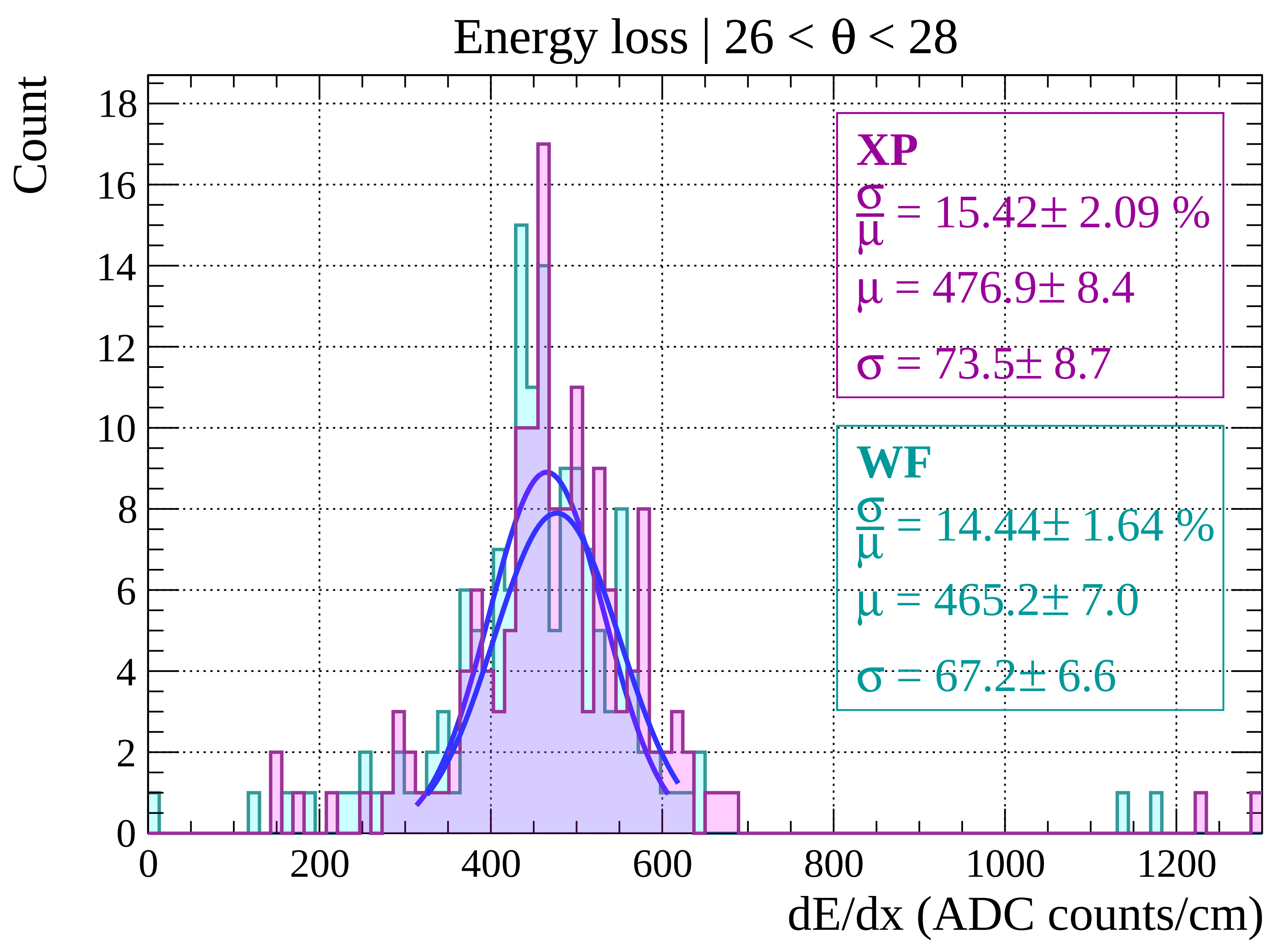


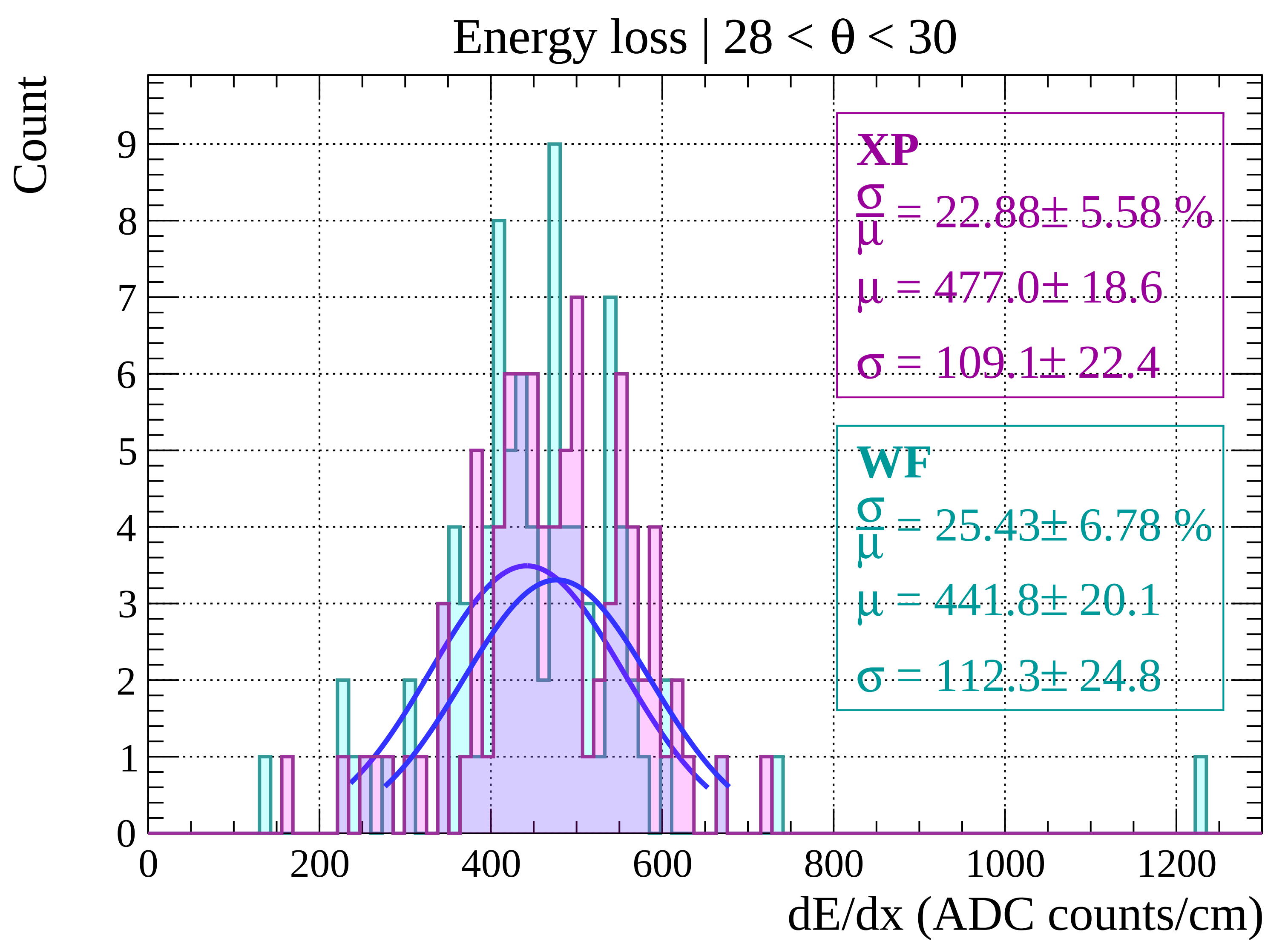
Energy loss | 20 < θ < 22
Count
0
5
10
15
20
25
30
35
40
45
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.31± 0.88 %
μ = 476.4± 3.4
σ = 53.9± 3.8
WF
σ/μ = 10.84± 0.66 %
μ = 458.7± 2.9
σ = 49.7± 2.7

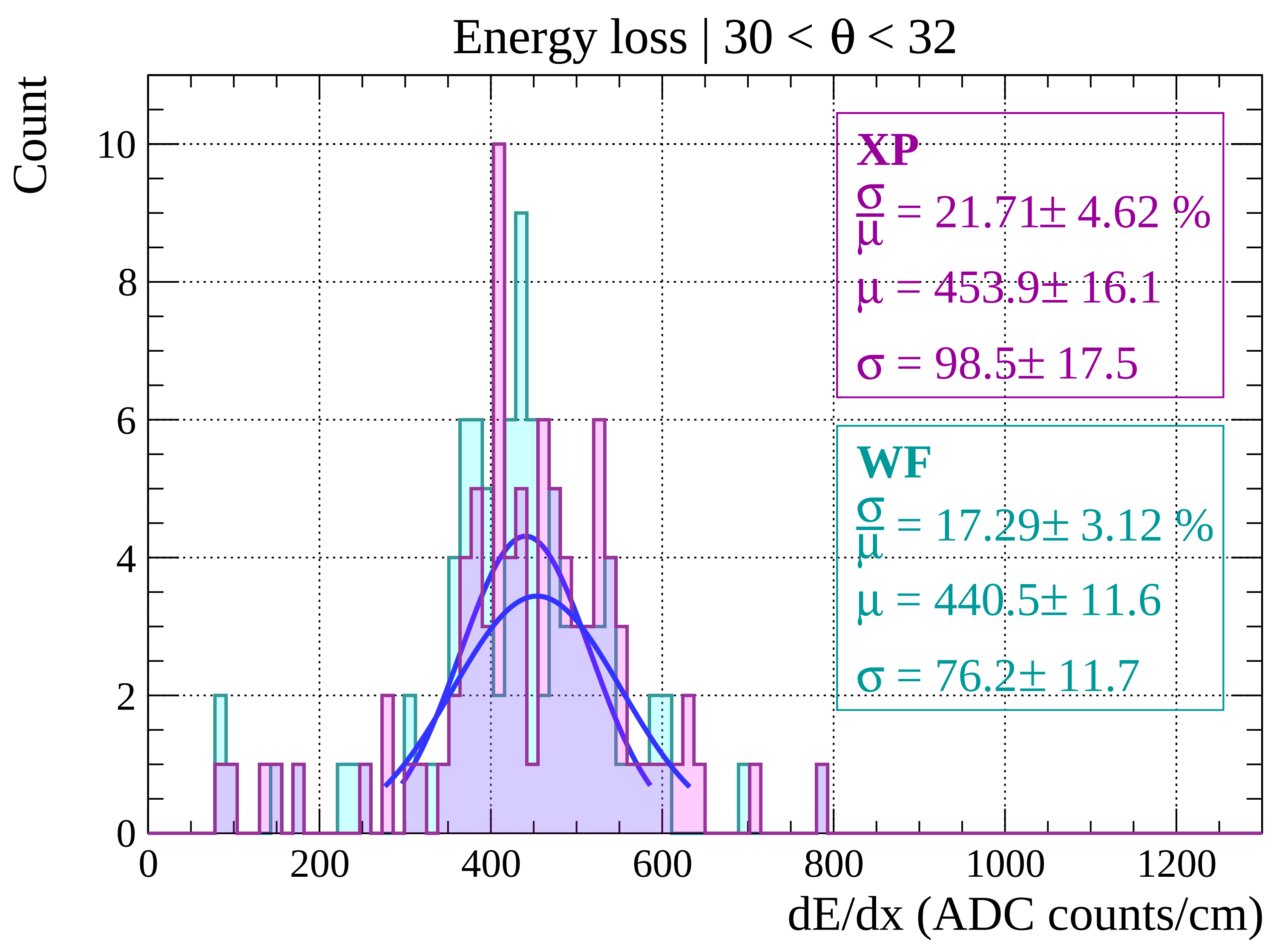


Energy loss | 22 < θ < 24
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
XP
σ/μ = 11.66± 1.50 %
μ = 474.1± 5.0
σ = 55.3± 6.5
WF
σ/μ = 11.29± 1.27 %
μ = 453.0± 4.6
σ = 51.1± 5.2

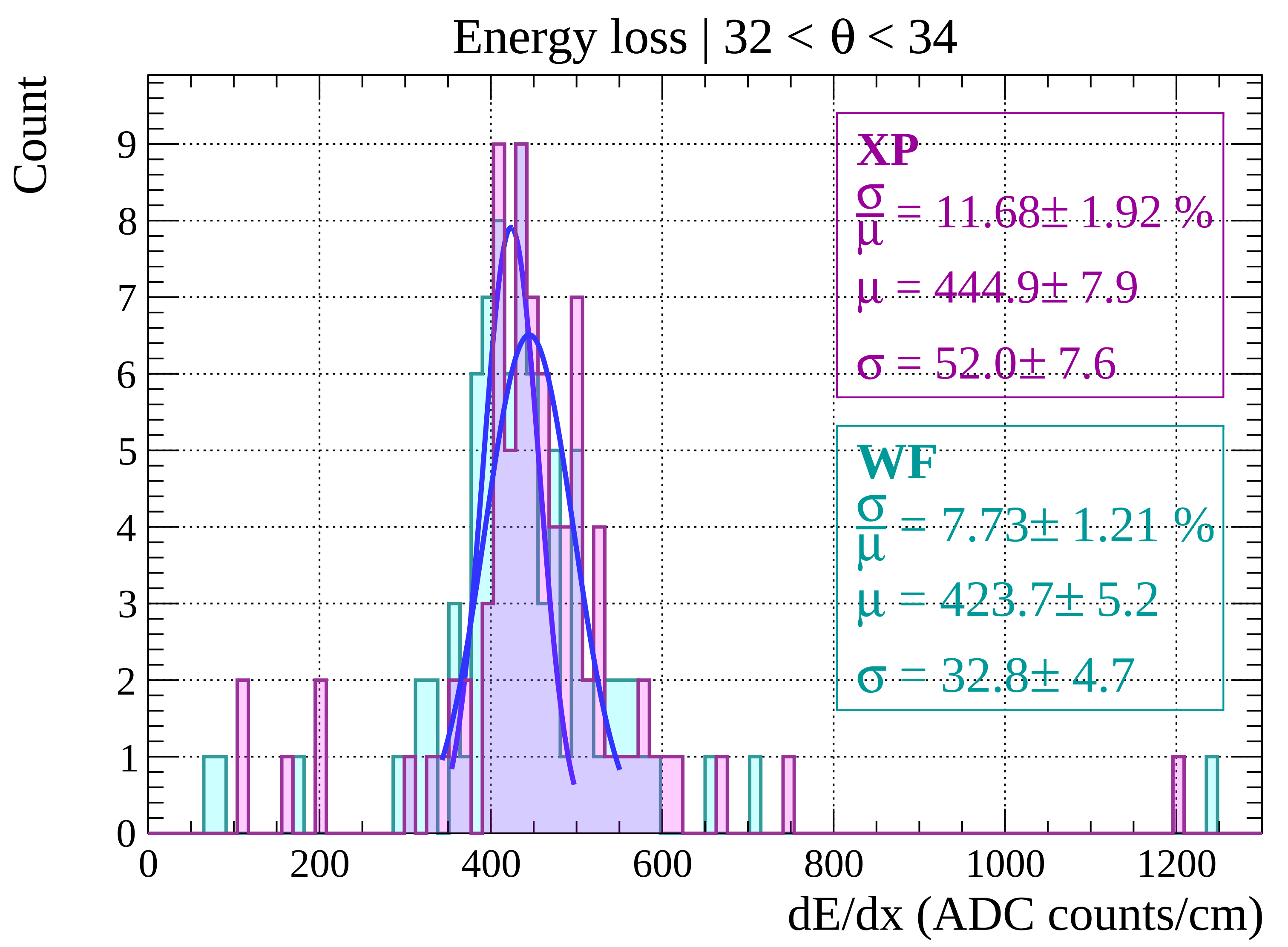

Energy loss | 24 < θ < 26
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
18
20
22
0
200
400
600
800
1000
1200
XP
σ/μ = 12.63± 1.29 %
μ = 471.9± 5.9
σ = 59.6± 5.3
WF
σ/μ = 10.87± 1.30 %
μ = 461.7± 5.0
σ = 50.2± 5.5


Energy loss | 26 < θ < 28
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.42± 2.09 %
μ = 476.9± 8.4
σ = 73.5± 8.7
WF
σ/μ = 14.44± 1.64 %
μ = 465.2± 7.0
σ = 67.2± 6.6
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14
16
18

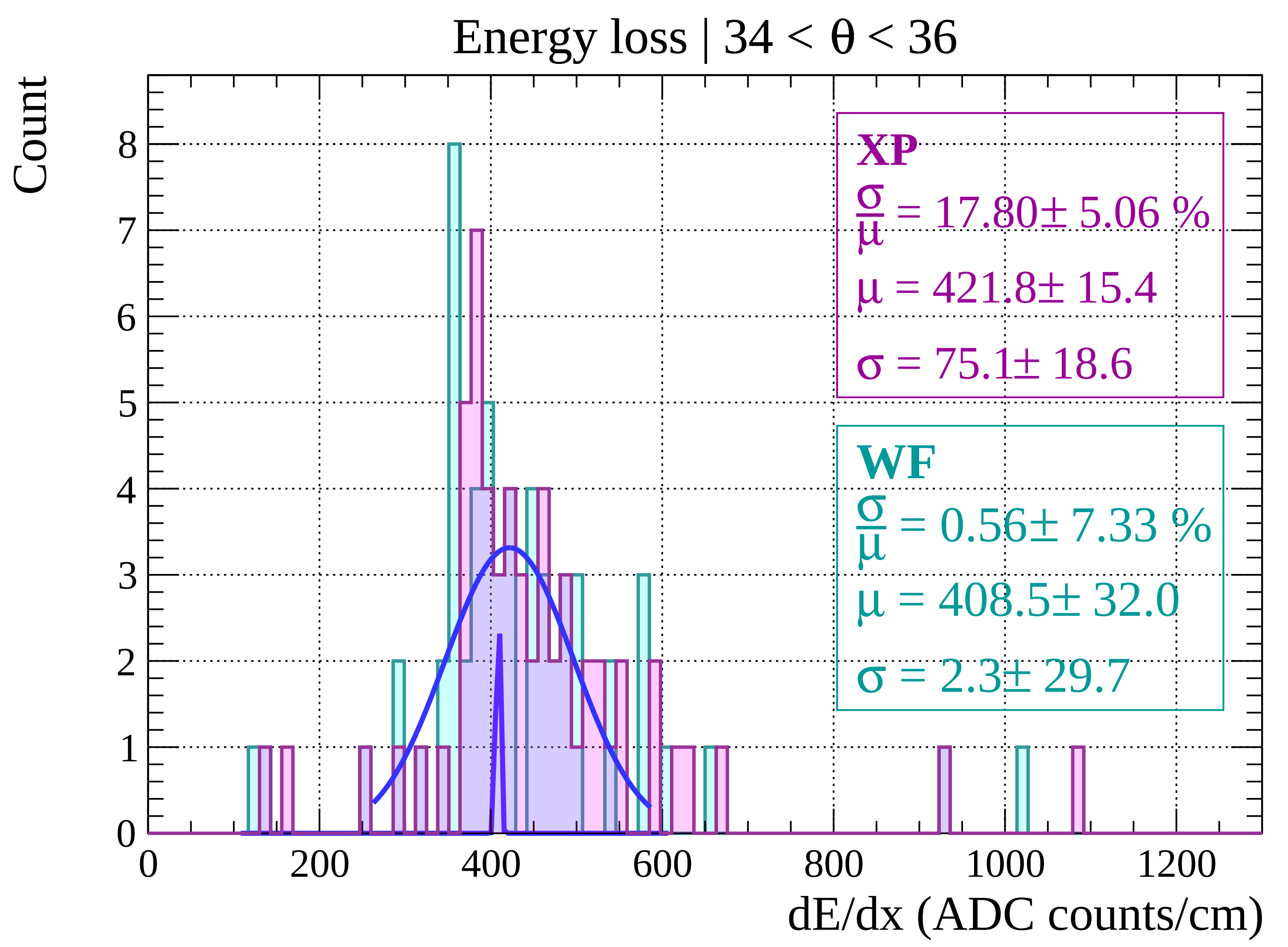

Energy loss | 28 < θ < 30
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
7
8
9
XP
σ/μ = 22.88± 5.58 %
μ = 477.0± 18.6
σ = 109.1± 22.4
WF
σ/μ = 25.43± 6.78 %
μ = 441.8± 20.1
σ = 112.3± 24.8



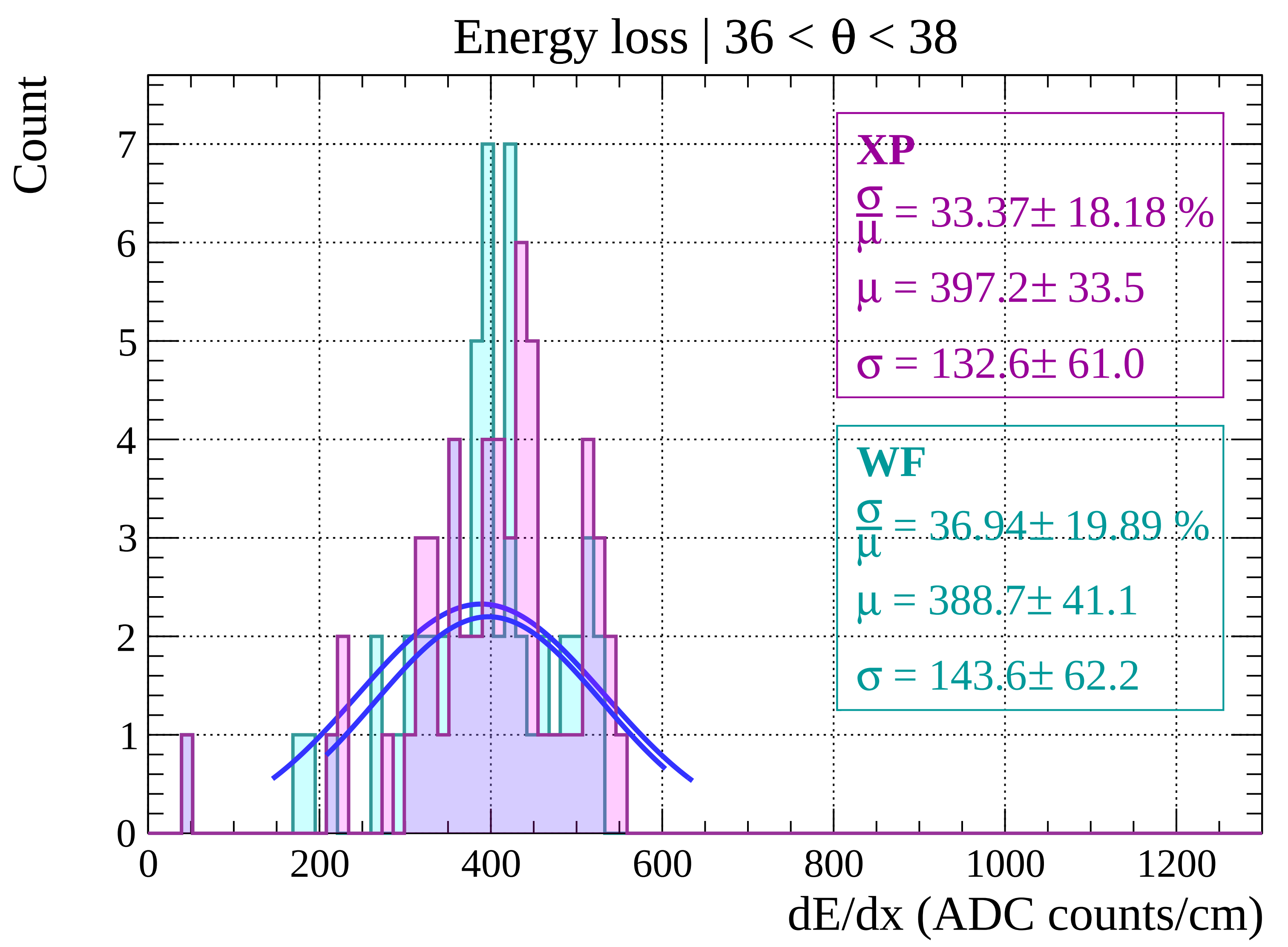
Energy loss | 30 < θ < 32
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
XP
$\frac{\sigma}{\mu}$ = 21.71± 4.62 %
μ = 453.9± 16.1
σ = 98.5± 17.5
WF
$\frac{\sigma}{\mu}$ = 17.29± 3.12 %
μ = 440.5± 11.6
σ = 76.2± 11.7

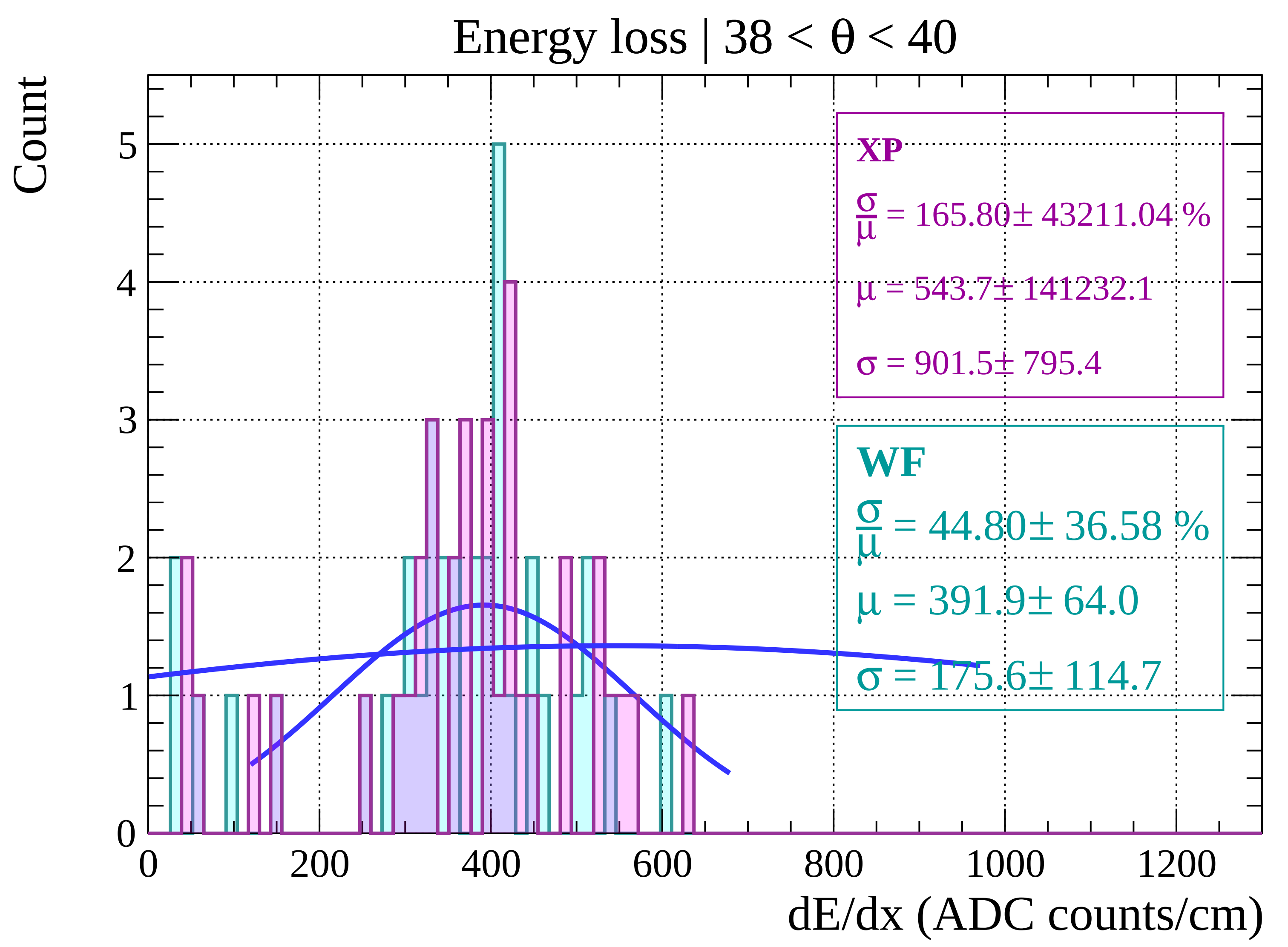


Energy loss | 32 < θ < 34
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1
2
3
4
5
6
7
8
9
XP
σ/μ = 11.68± 1.92 %
μ = 444.9± 7.9
σ = 52.0± 7.6
WF
σ/μ = 7.73± 1.21 %
μ = 423.7± 5.2
σ = 32.8± 4.7

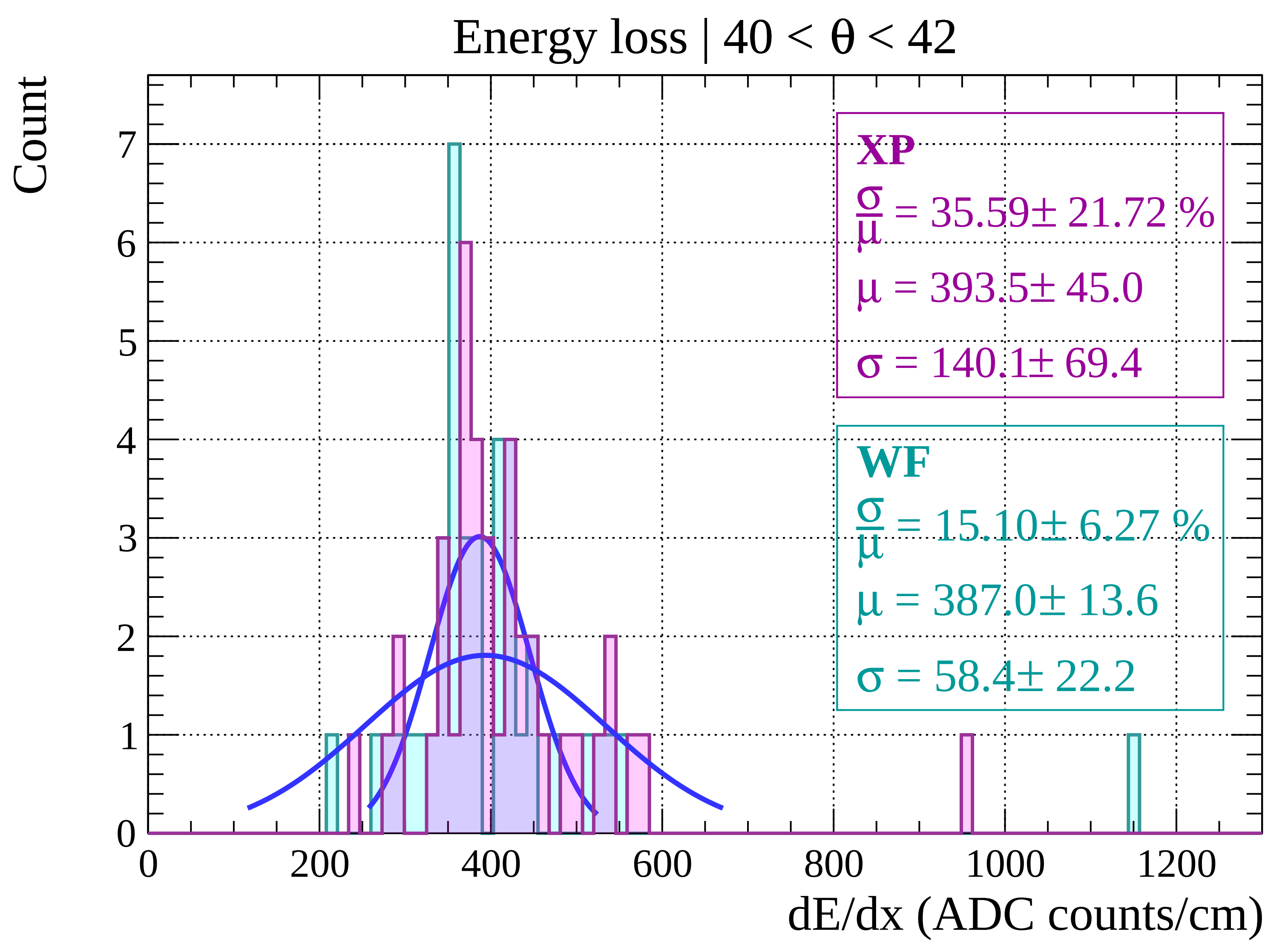


Energy loss | 34 < θ < 36
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 17.80± 5.06 %
μ = 421.8± 15.4
σ = 75.1± 18.6
WF
σ/μ = 0.56± 7.33 %
μ = 408.5± 32.0
σ = 2.3± 29.7

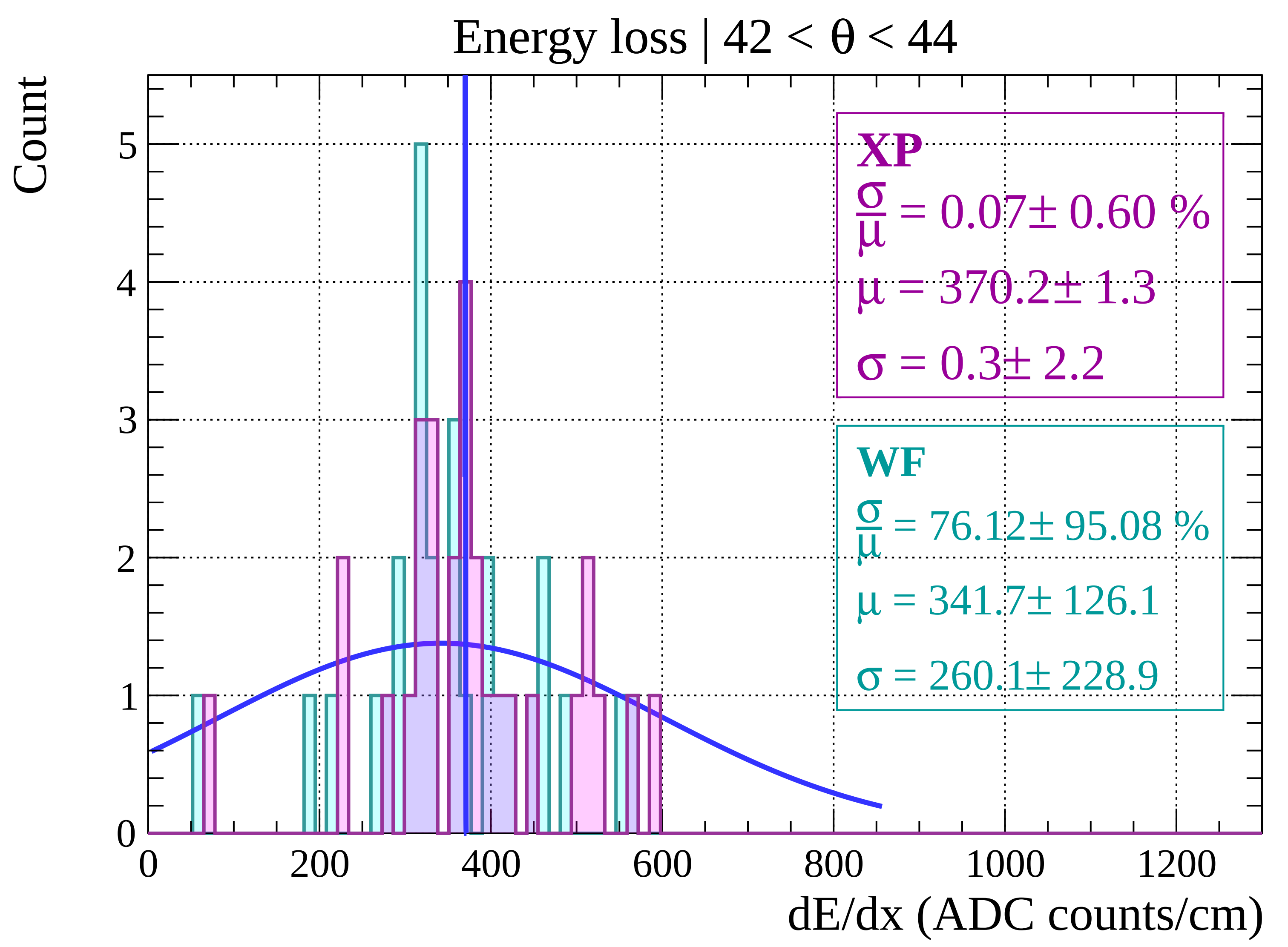

Energy loss | 36 < θ < 38
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 33.37± 18.18 %
μ = 397.2± 33.5
σ = 132.6± 61.0
WF
σ/μ = 36.94± 19.89 %
μ = 388.7± 41.1
σ = 143.6± 62.2
0
1
2
3
4
5
6
7
0
200
400
600
800
1000
1200



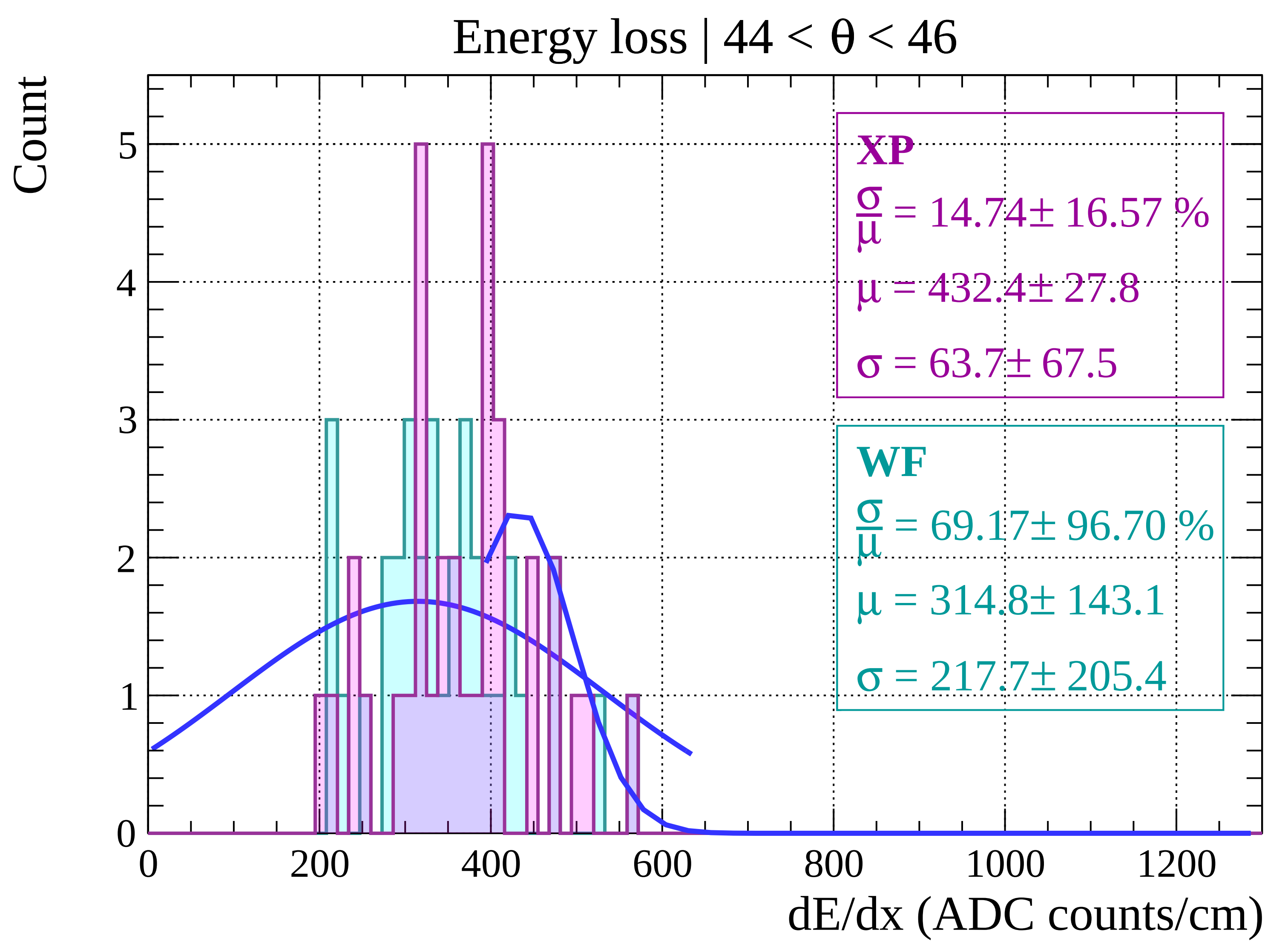
Energy loss | 38 < θ < 40
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 165.80± 43211.04 %
μ = 543.7± 141232.1
σ = 901.5± 795.4
WF
σ/μ = 44.80± 36.58 %
μ = 391.9± 64.0
σ = 175.6± 114.7
0
1
2
3
4
5
0
200
400
600
800
1000
1200



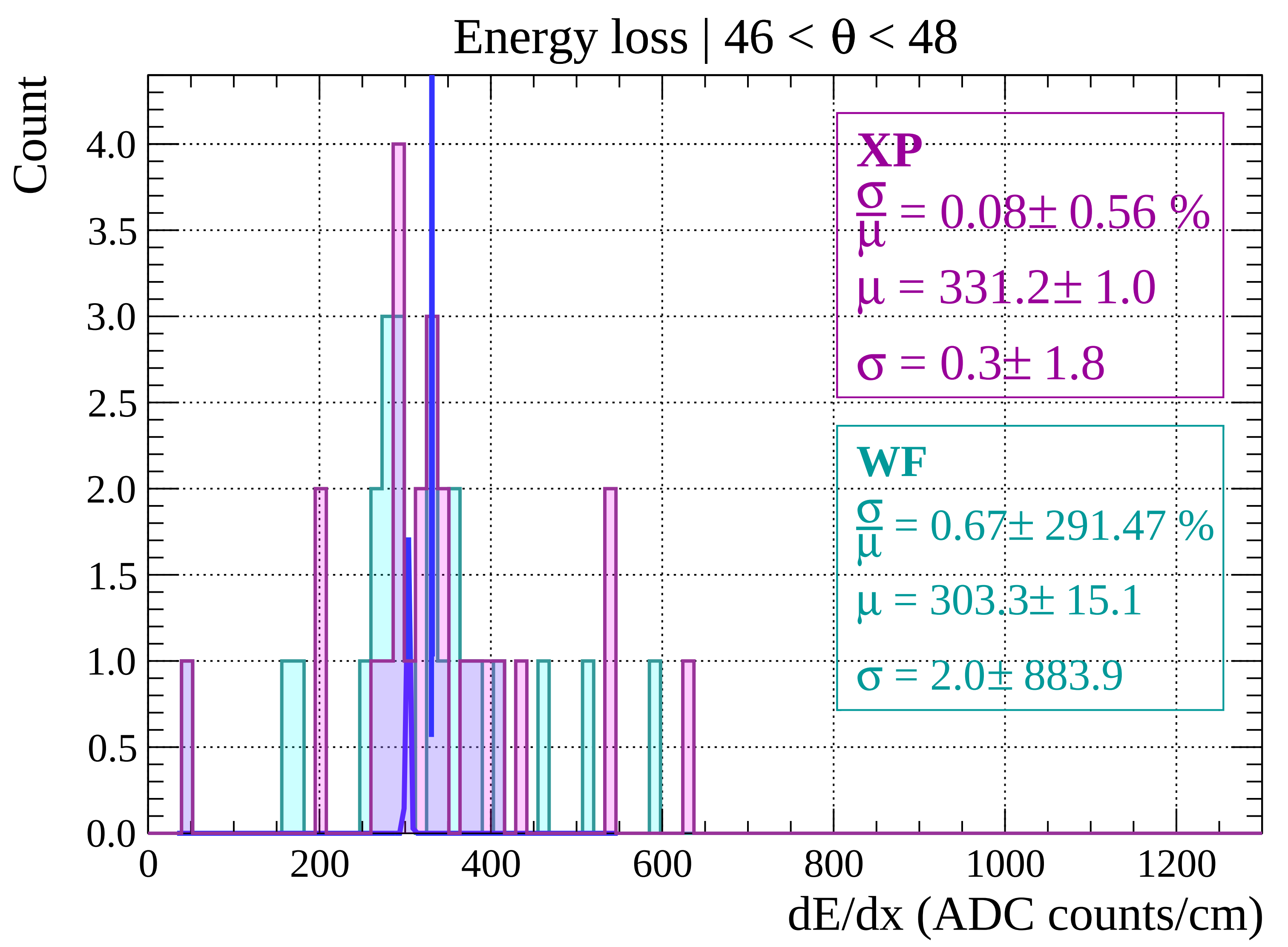
Energy loss | 40 < θ < 42
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
6
7
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 35.59± 21.72 %
μ = 393.5± 45.0
σ = 140.1± 69.4
WF
$\frac{\sigma}{\mu}$ = 15.10± 6.27 %
μ = 387.0± 13.6
σ = 58.4± 22.2



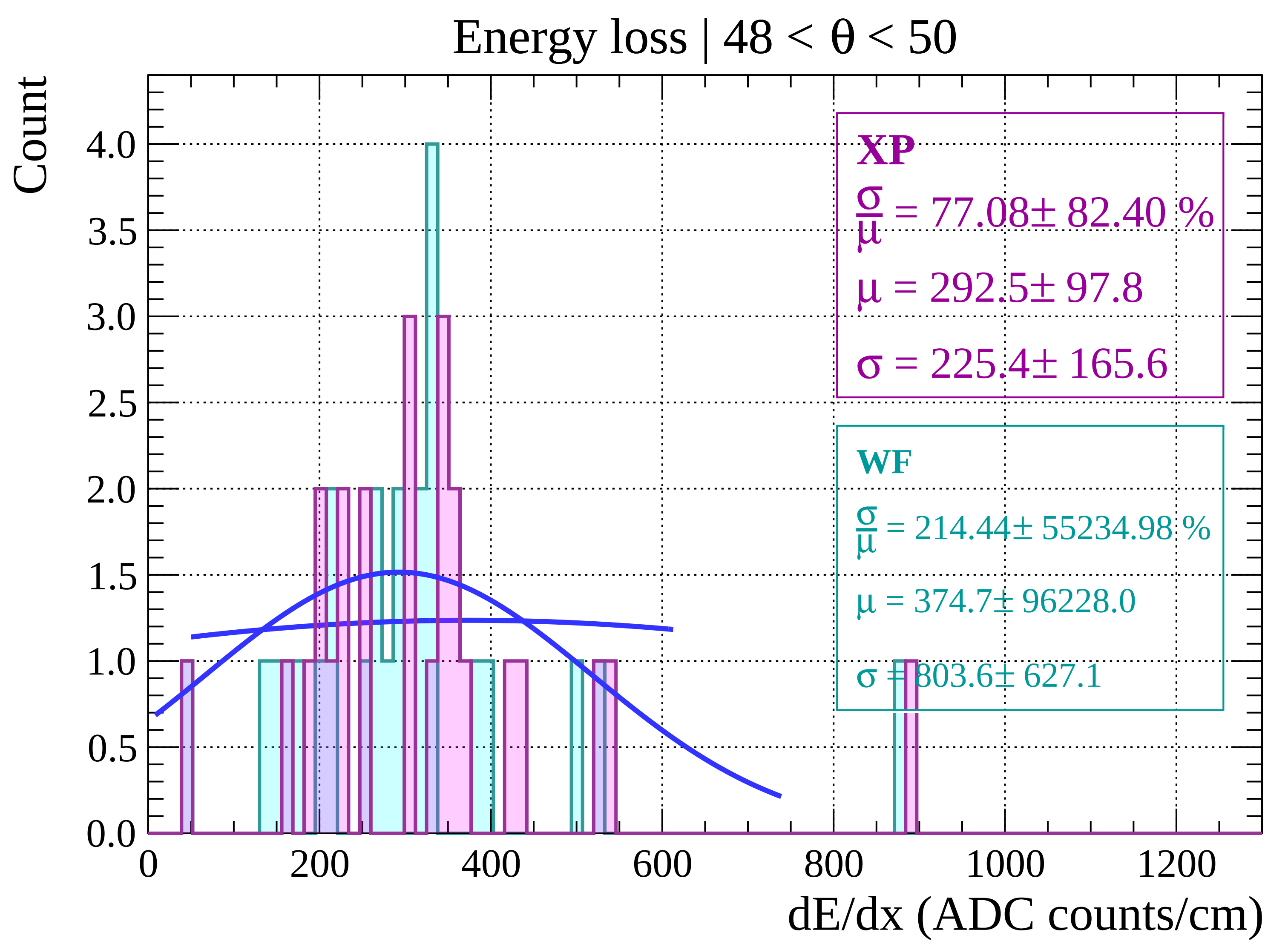
Energy loss | 42 < θ < 44
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
0
200
400
600
800
1000
1200
XP
σ/μ = 0.07± 0.60 %
μ = 370.2± 1.3
σ = 0.3± 2.2
WF
σ/μ = 76.12± 95.08 %
μ = 341.7± 126.1
σ = 260.1± 228.9



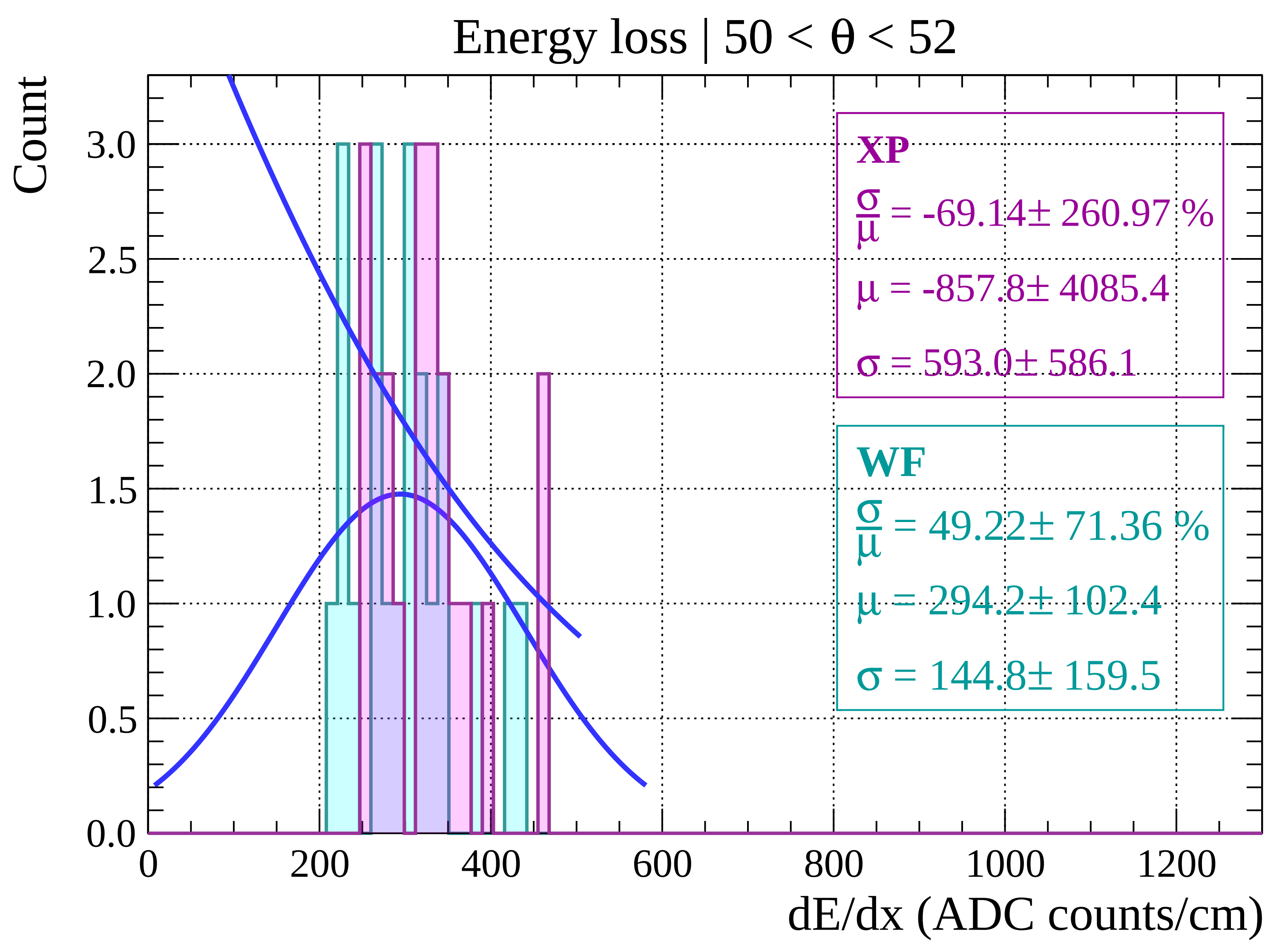
Energy loss | 44 < θ < 46
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
0
200
400
600
800
1000
1200
XP
σ/μ = 14.74± 16.57 %
μ = 432.4± 27.8
σ = 63.7± 67.5
WF
σ/μ = 69.17± 96.70 %
μ = 314.8± 143.1
σ = 217.7± 205.4



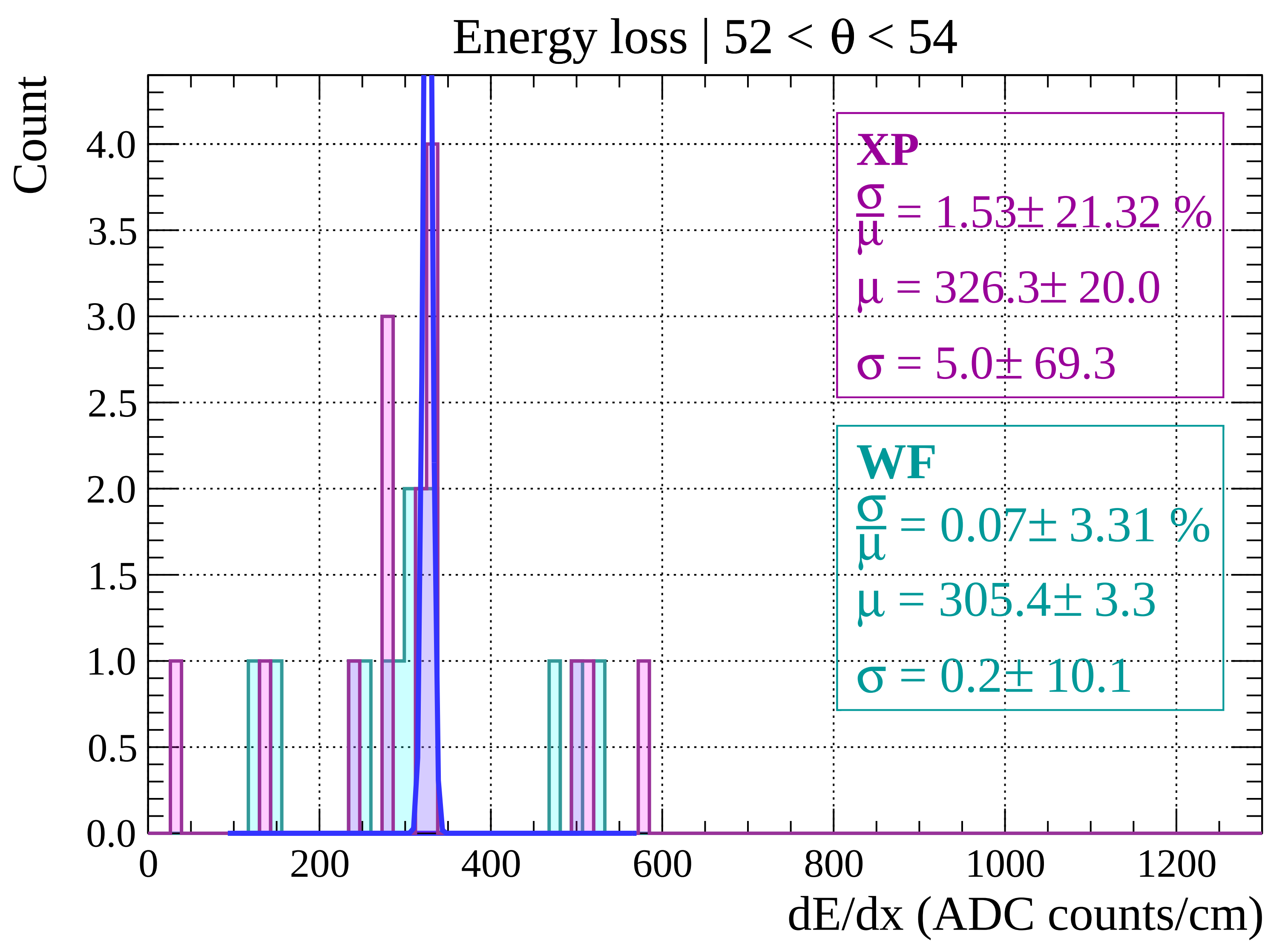
Energy loss | 46 < θ < 48
Count
dE/dx (ADC counts/cm)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
200
400
600
800
1000
1200
XP
σ/μ = 0.08± 0.56 %
μ = 331.2± 1.0
σ = 0.3± 1.8
WF
σ/μ = 0.67± 291.47 %
μ = 303.3± 15.1
σ = 2.0± 883.9



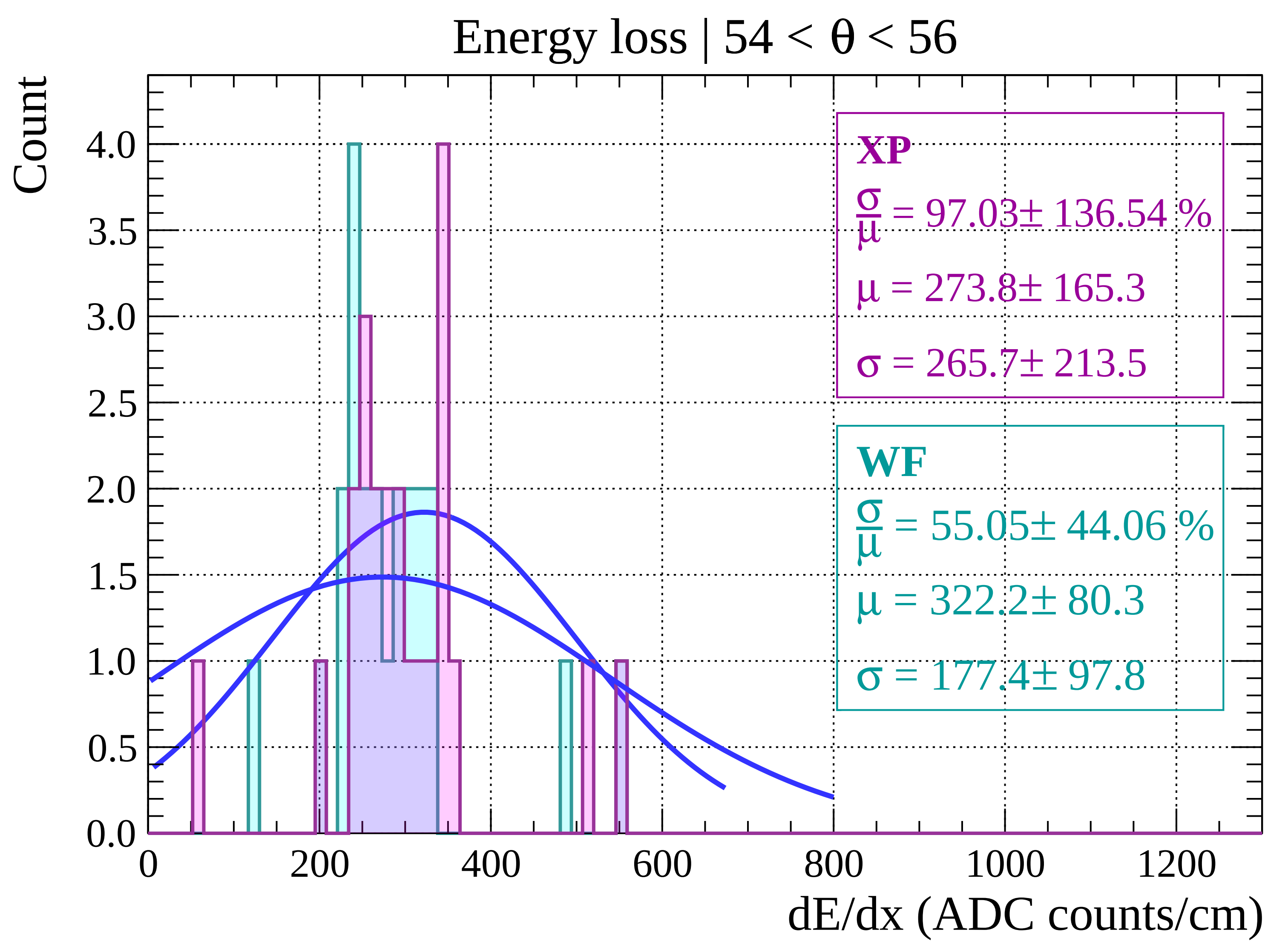
Energy loss | 48 < θ < 50
Count
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 77.08± 82.40 %
μ = 292.5± 97.8
σ = 225.4± 165.6
WF
σ/μ = 214.44± 55234.98 %
μ = 374.7± 96228.0
σ = 803.6± 627.1



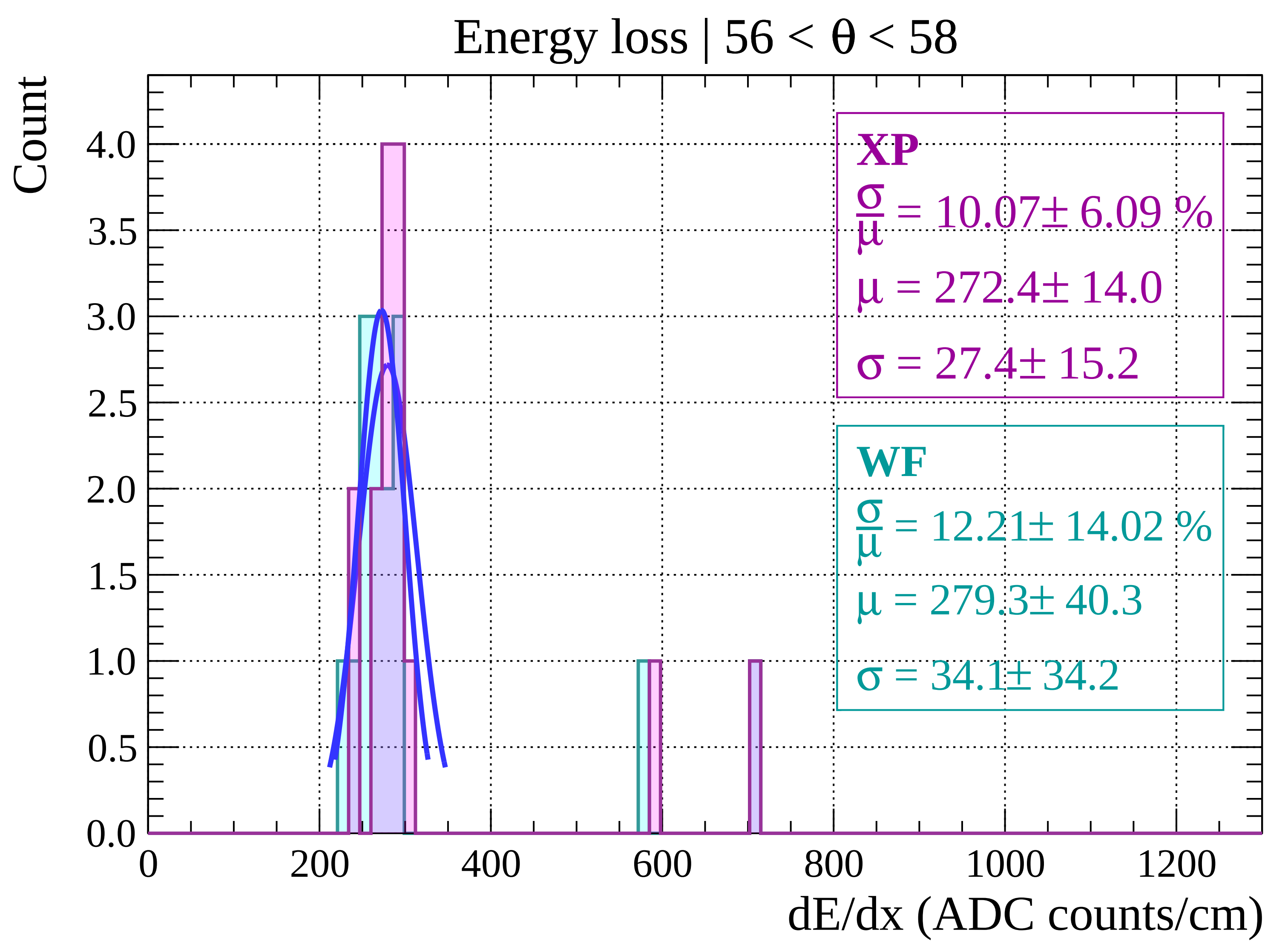
Energy loss | 50 < θ < 52
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -69.14± 260.97 %
μ = -857.8± 4085.4
σ = 593.0± 586.1
WF
σ/μ = 49.22± 71.36 %
μ = 294.2± 102.4
σ = 144.8± 159.5



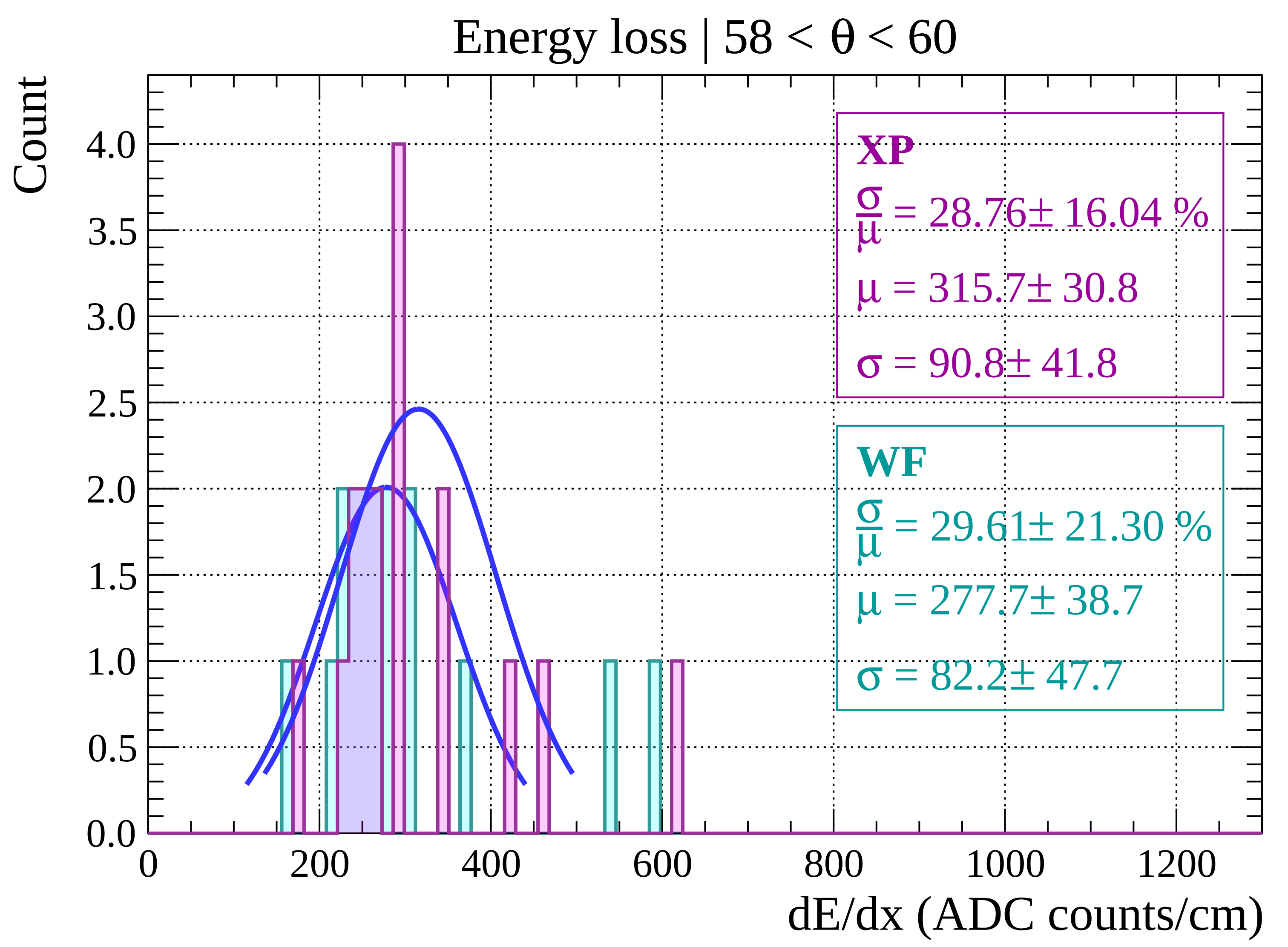
Energy loss | 52 < θ < 54
Count
dE/dx (ADC counts/cm)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 1.53± 21.32 %
μ = 326.3± 20.0
σ = 5.0± 69.3
WF
$\frac{\sigma}{\mu}$ = 0.07± 3.31 %
μ = 305.4± 3.3
σ = 0.2± 10.1



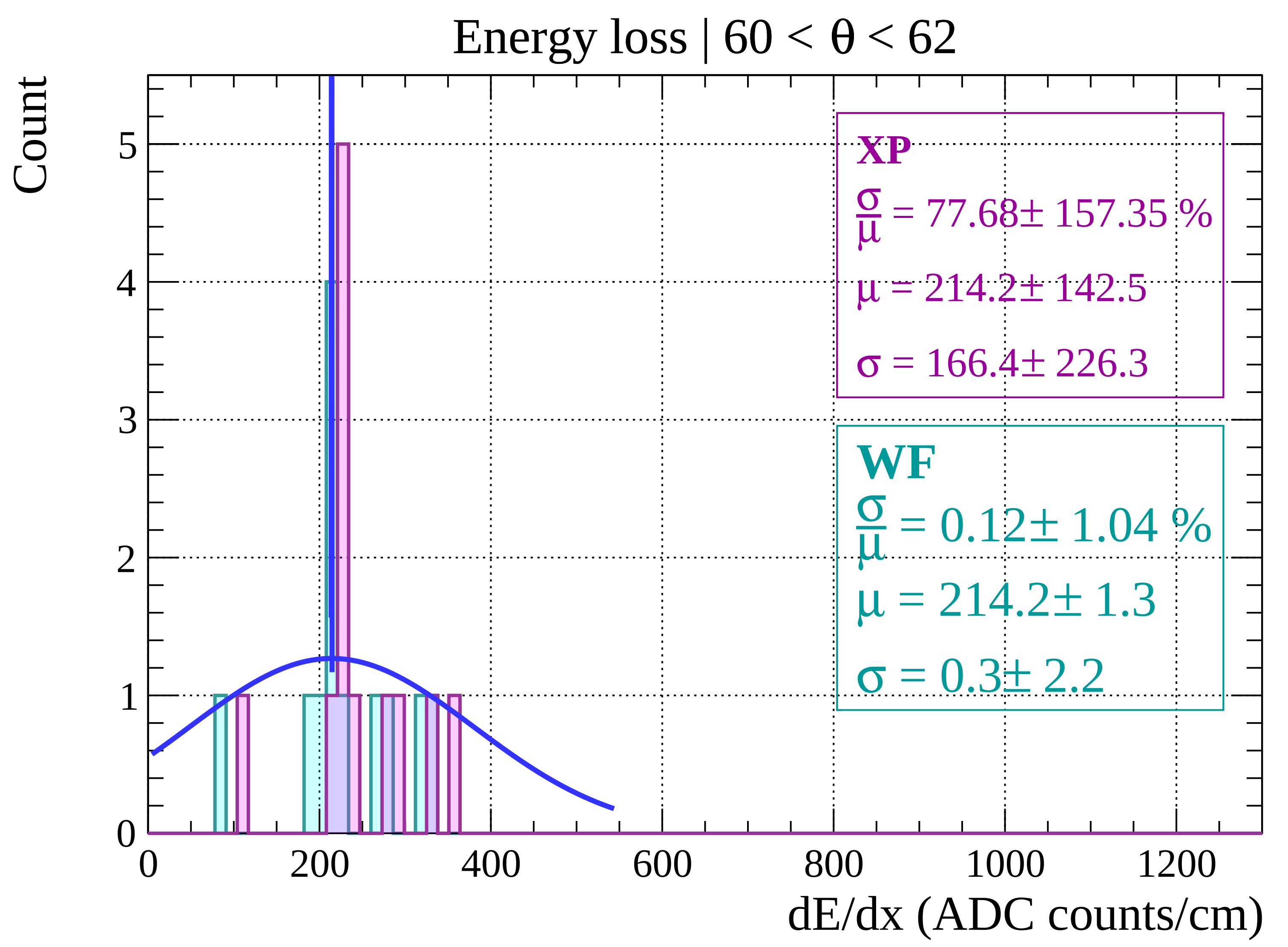
Energy loss | 54 < θ < 56
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 97.03± 136.54 %
μ = 273.8± 165.3
σ = 265.7± 213.5
WF
σ/μ = 55.05± 44.06 %
μ = 322.2± 80.3
σ = 177.4± 97.8



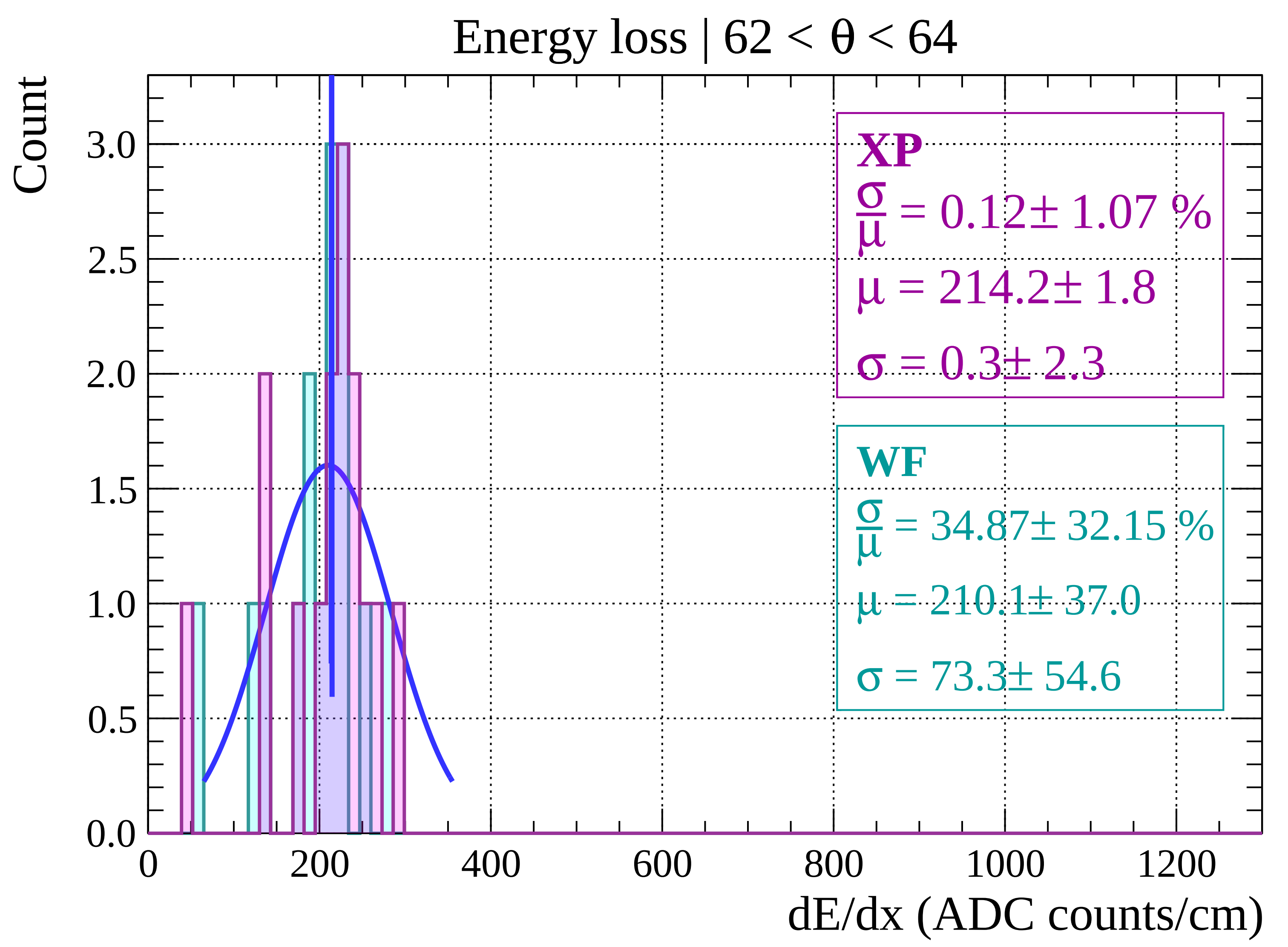
Energy loss | 56 < θ < 58
Count
dE/dx (ADC counts/cm)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 10.07± 6.09 %
μ = 272.4± 14.0
σ = 27.4± 15.2
WF
$\frac{\sigma}{\mu}$ = 12.21± 14.02 %
μ = 279.3± 40.3
σ = 34.1± 34.2



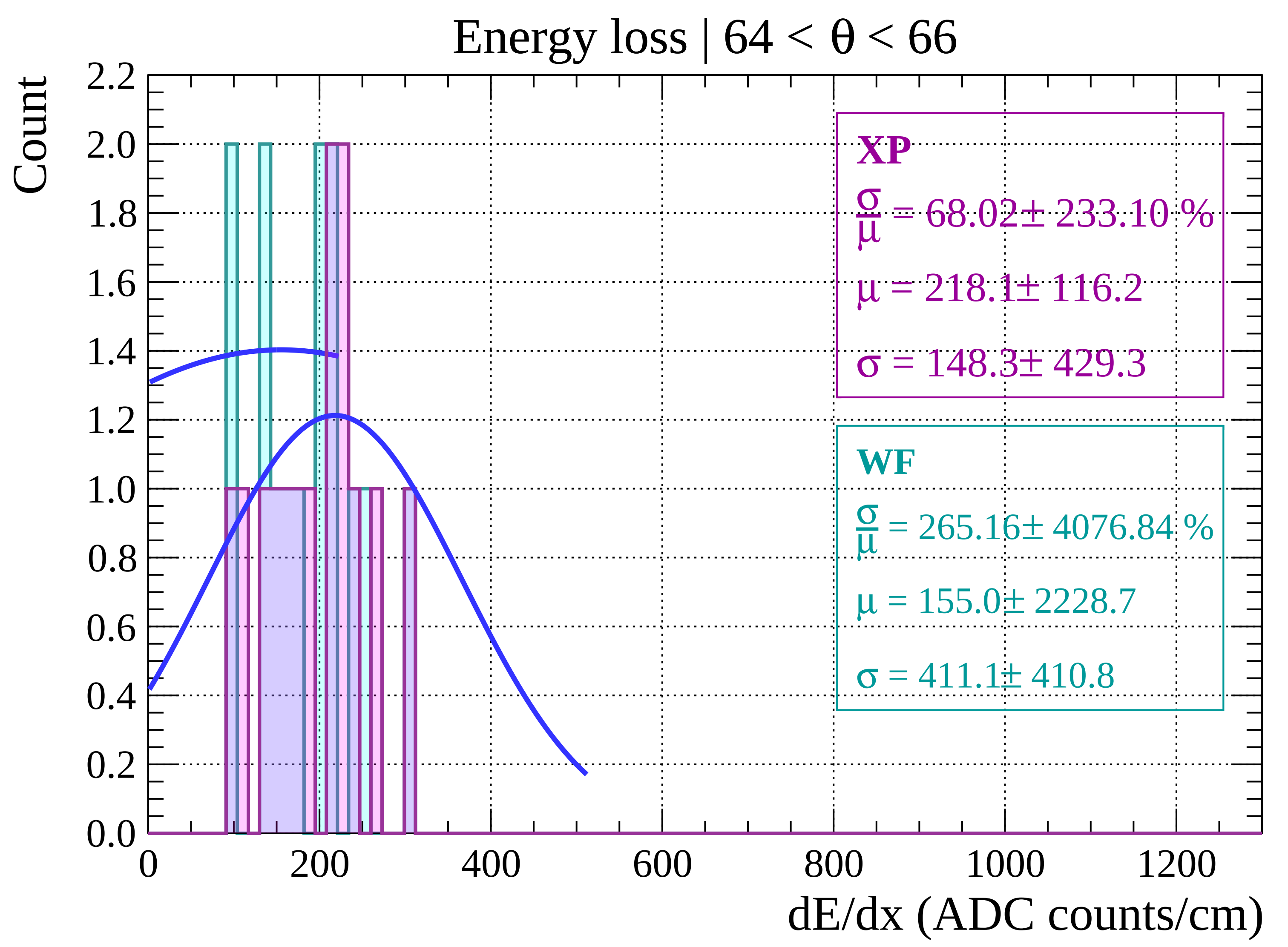
Energy loss | 58 < θ < 60
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 28.76± 16.04 %
μ = 315.7± 30.8
σ = 90.8± 41.8
WF
σ/μ = 29.61± 21.30 %
μ = 277.7± 38.7
σ = 82.2± 47.7

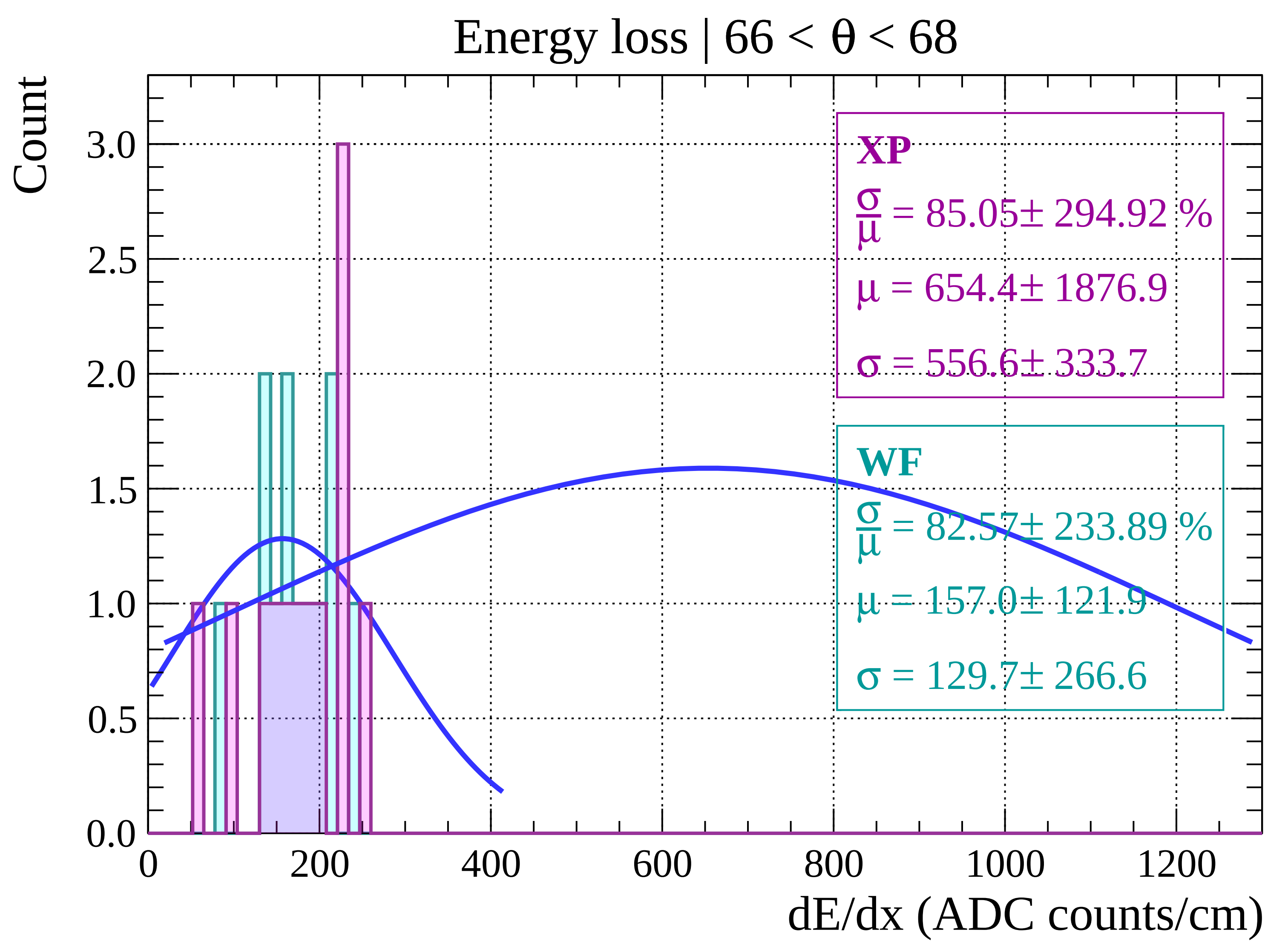


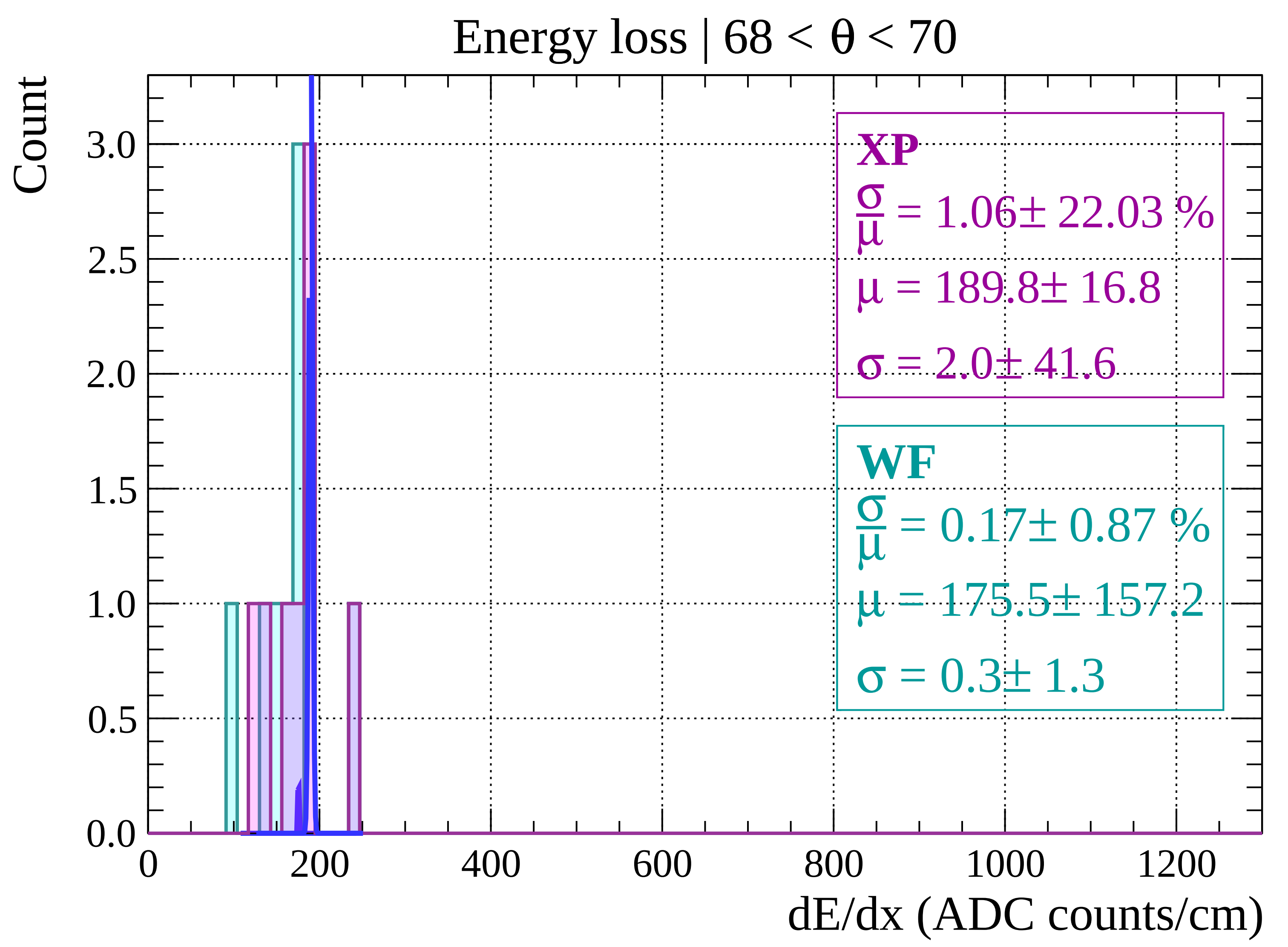
Energy loss | 60 < θ < 62
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
XP
σ/μ = 77.68± 157.35 %
μ = 214.2± 142.5
σ = 166.4± 226.3
WF
σ/μ = 0.12± 1.04 %
μ = 214.2± 1.3
σ = 0.3± 2.2



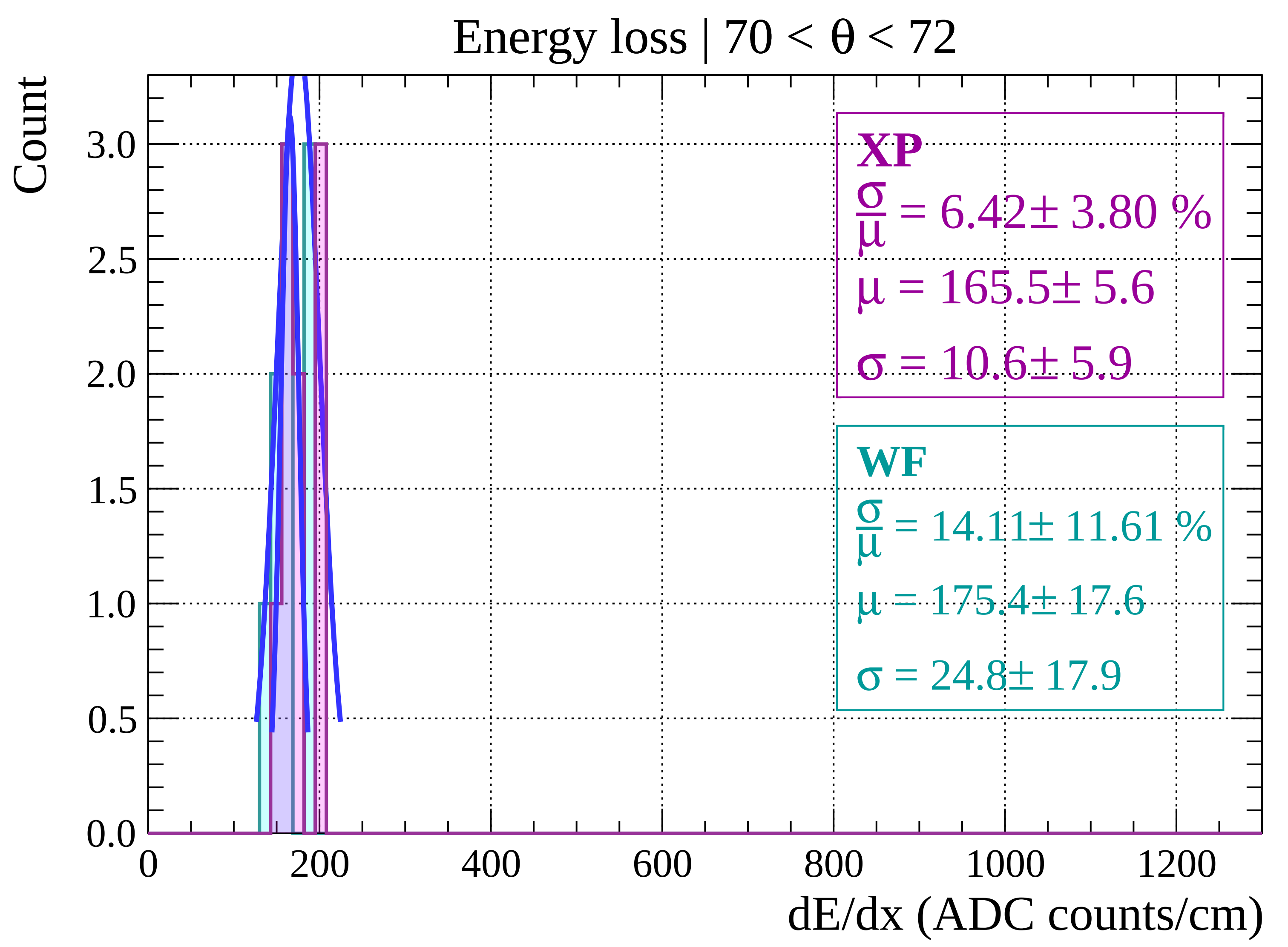
Energy loss | 62 < θ < 64
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 0.12± 1.07 %
μ = 214.2± 1.8
σ = 0.3± 2.3
WF
σ/μ = 34.87± 32.15 %
μ = 210.1± 37.0
σ = 73.3± 54.6
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
200
400
600
800
1000
1200

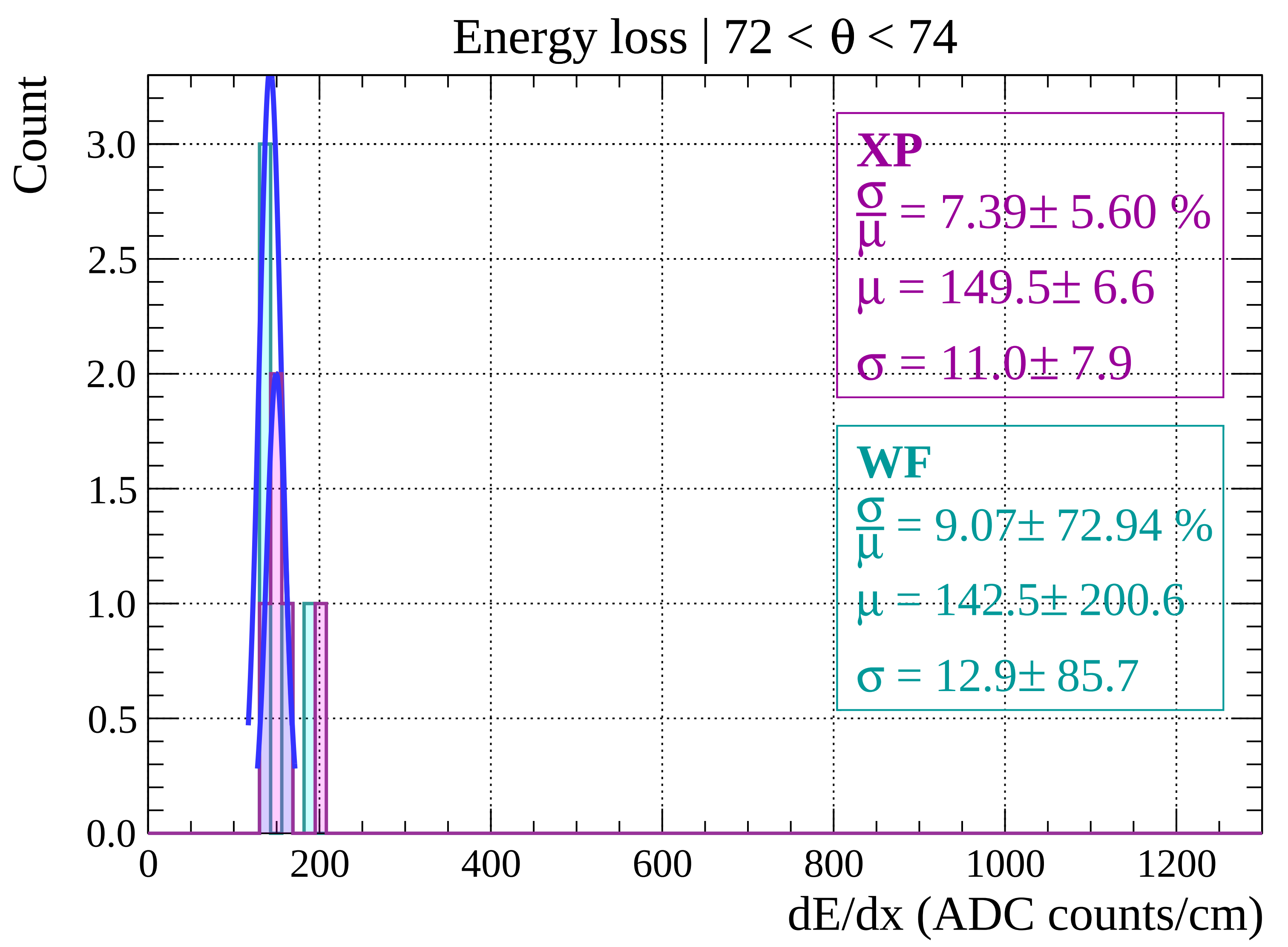

Energy loss | 64 < θ < 66
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
σ/μ = 68.02± 233.10 %
μ = 218.1± 116.2
σ = 148.3± 429.3
WF
σ/μ = 265.16± 4076.84 %
μ = 155.0± 2228.7
σ = 411.1± 410.8


Energy loss | 66 < θ < 68
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0.0
0.5
1.0
1.5
2.0
2.5
3.0
XP
σ/μ = 85.05± 294.92 %
μ = 654.4± 1876.9
σ = 556.6± 333.7
WF
σ/μ = 82.57± 233.89 %
μ = 157.0± 121.9
σ = 129.7± 266.6

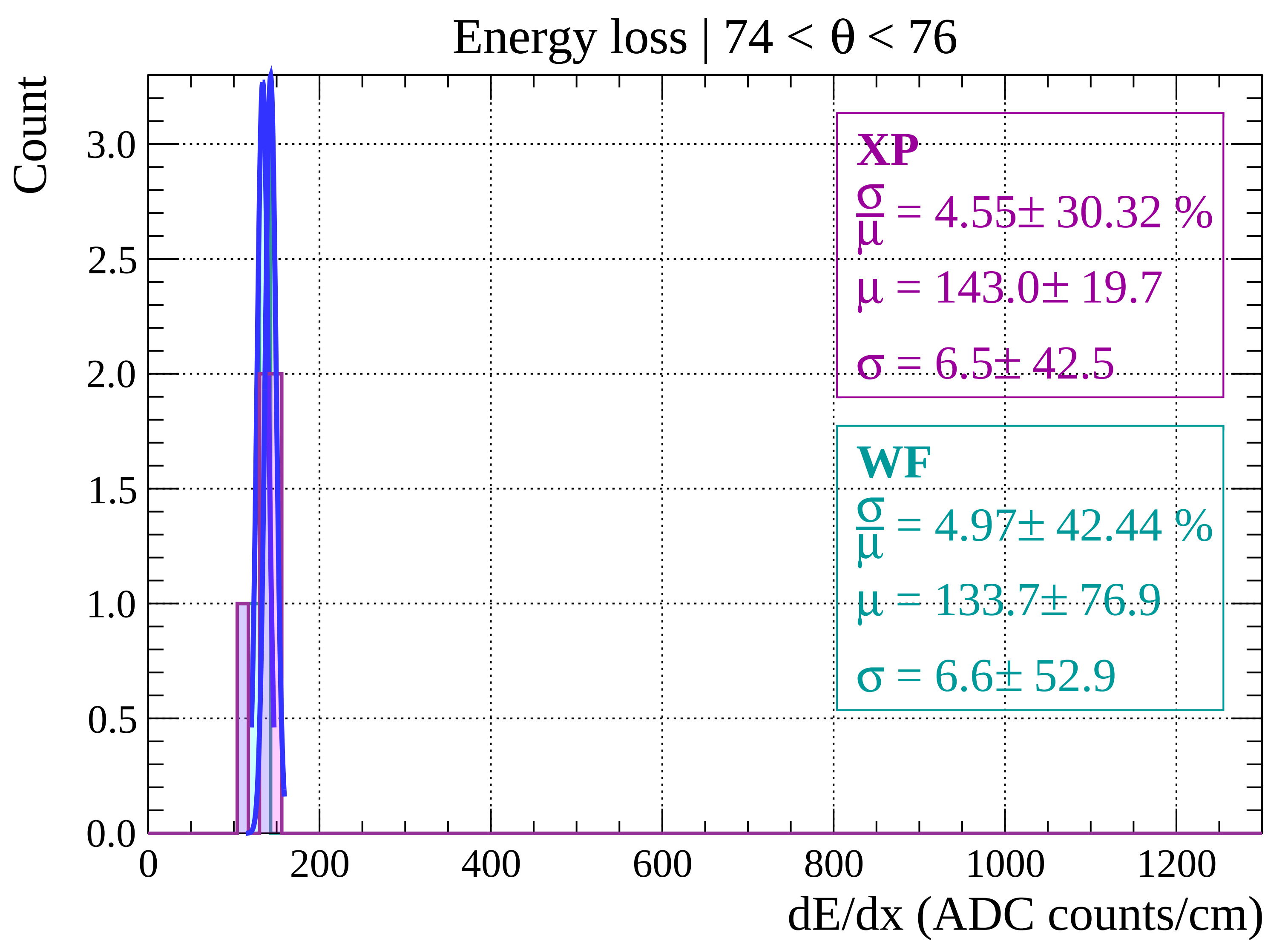


Energy loss | 68 < θ < 70
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 1.06± 22.03 %
μ = 189.8± 16.8
σ = 2.0± 41.6
WF
σ/μ = 0.17± 0.87 %
μ = 175.5± 157.2
σ = 0.3± 1.3

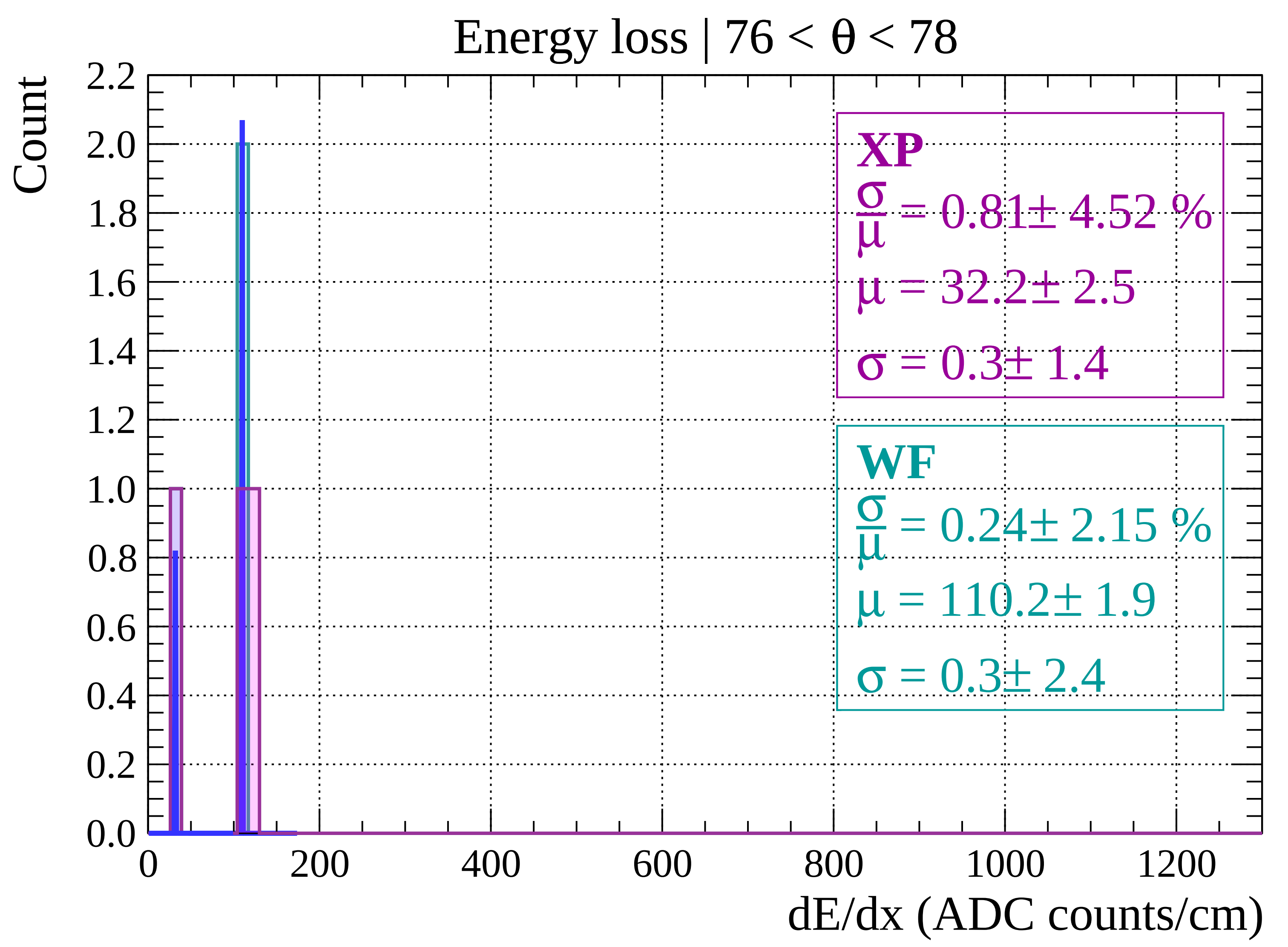

Energy loss | 70 < θ < 72
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0.0
0.5
1.0
1.5
2.0
2.5
3.0
XP
σ/μ = 6.42± 3.80 %
μ = 165.5± 5.6
σ = 10.6± 5.9
WF
σ/μ = 14.11± 11.61 %
μ = 175.4± 17.6
σ = 24.8± 17.9



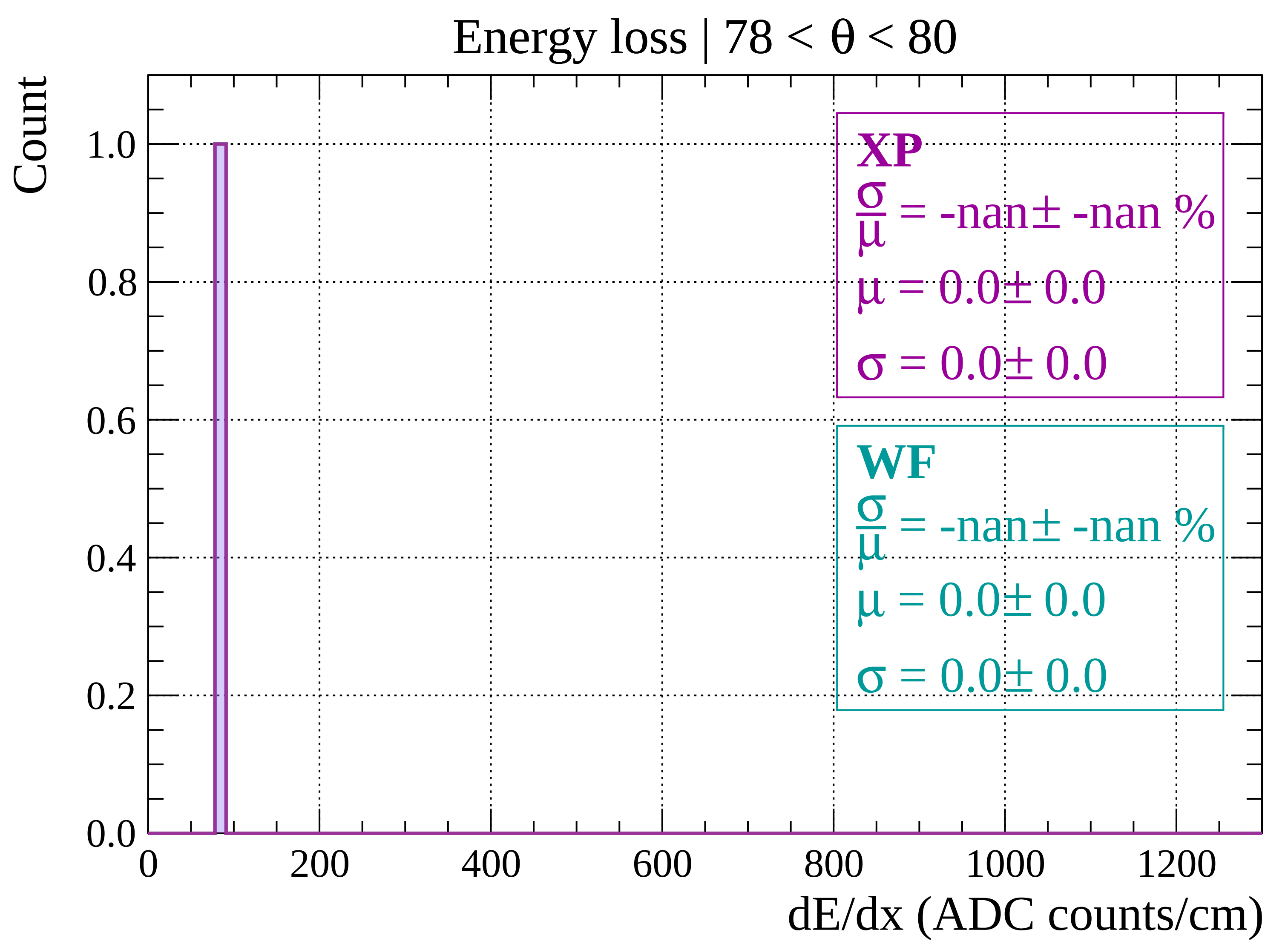
Energy loss | 72 < θ < 74
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 7.39± 5.60 %
μ = 149.5± 6.6
σ = 11.0± 7.9
WF
σ/μ = 9.07± 72.94 %
μ = 142.5± 200.6
σ = 12.9± 85.7



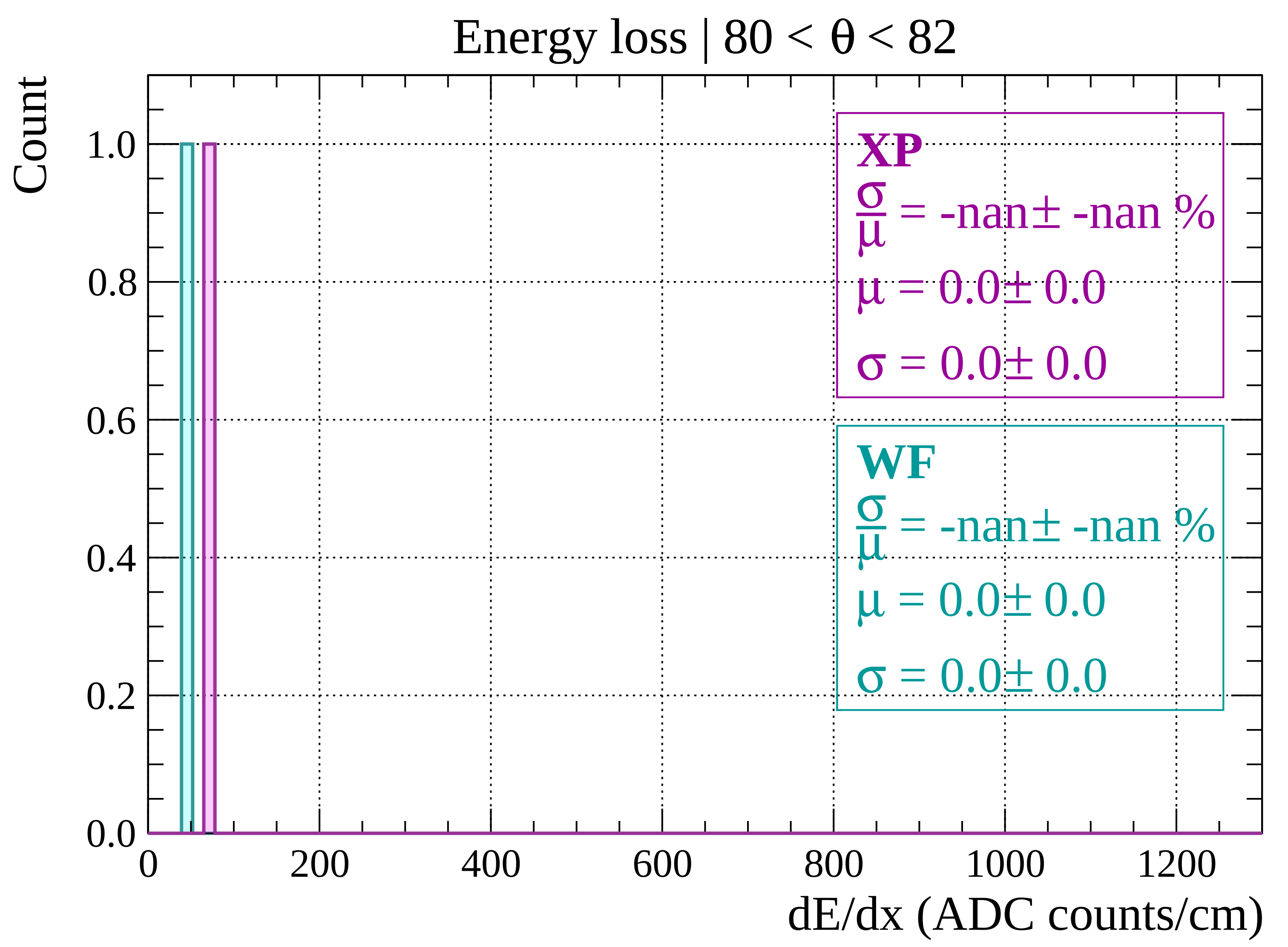
Energy loss | 74 < θ < 76
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 4.55± 30.32 %
μ = 143.0± 19.7
σ = 6.5± 42.5
WF
σ/μ = 4.97± 42.44 %
μ = 133.7± 76.9
σ = 6.6± 52.9

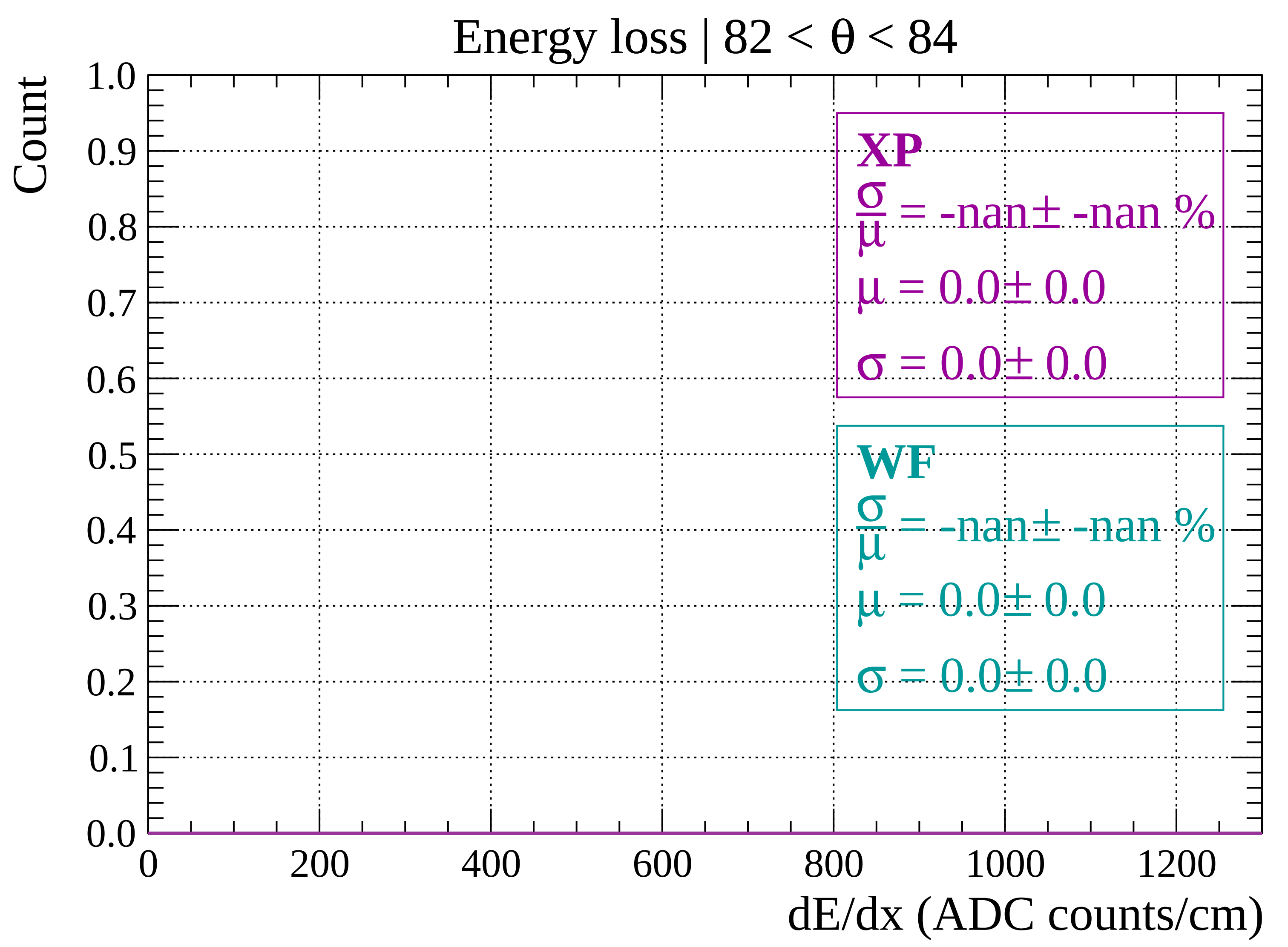


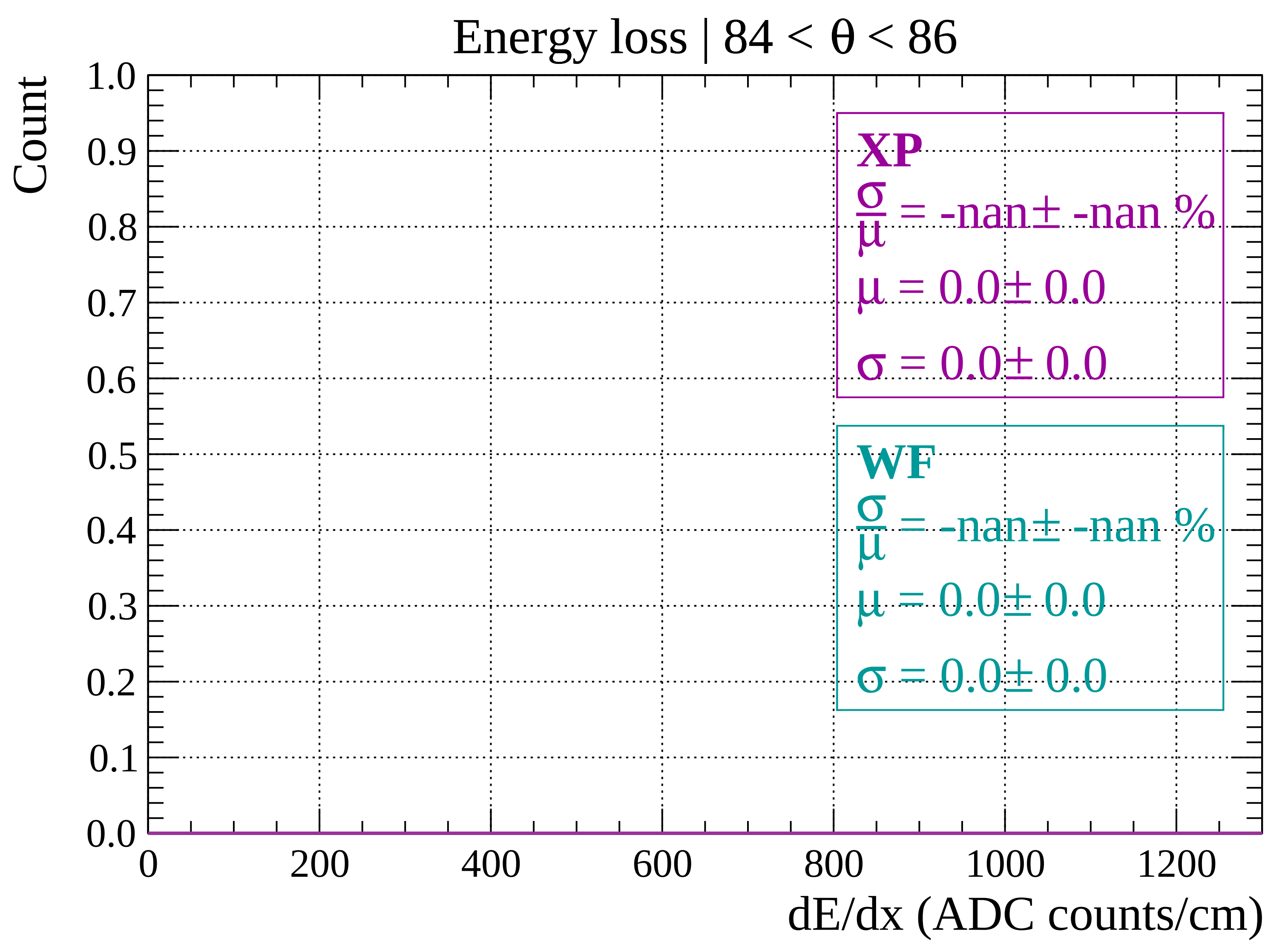
Energy loss | 76 < θ < 78
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
XP
σ/μ = 0.81± 4.52 %
μ = 32.2± 2.5
σ = 0.3± 1.4
WF
σ/μ = 0.24± 2.15 %
μ = 110.2± 1.9
σ = 0.3± 2.4

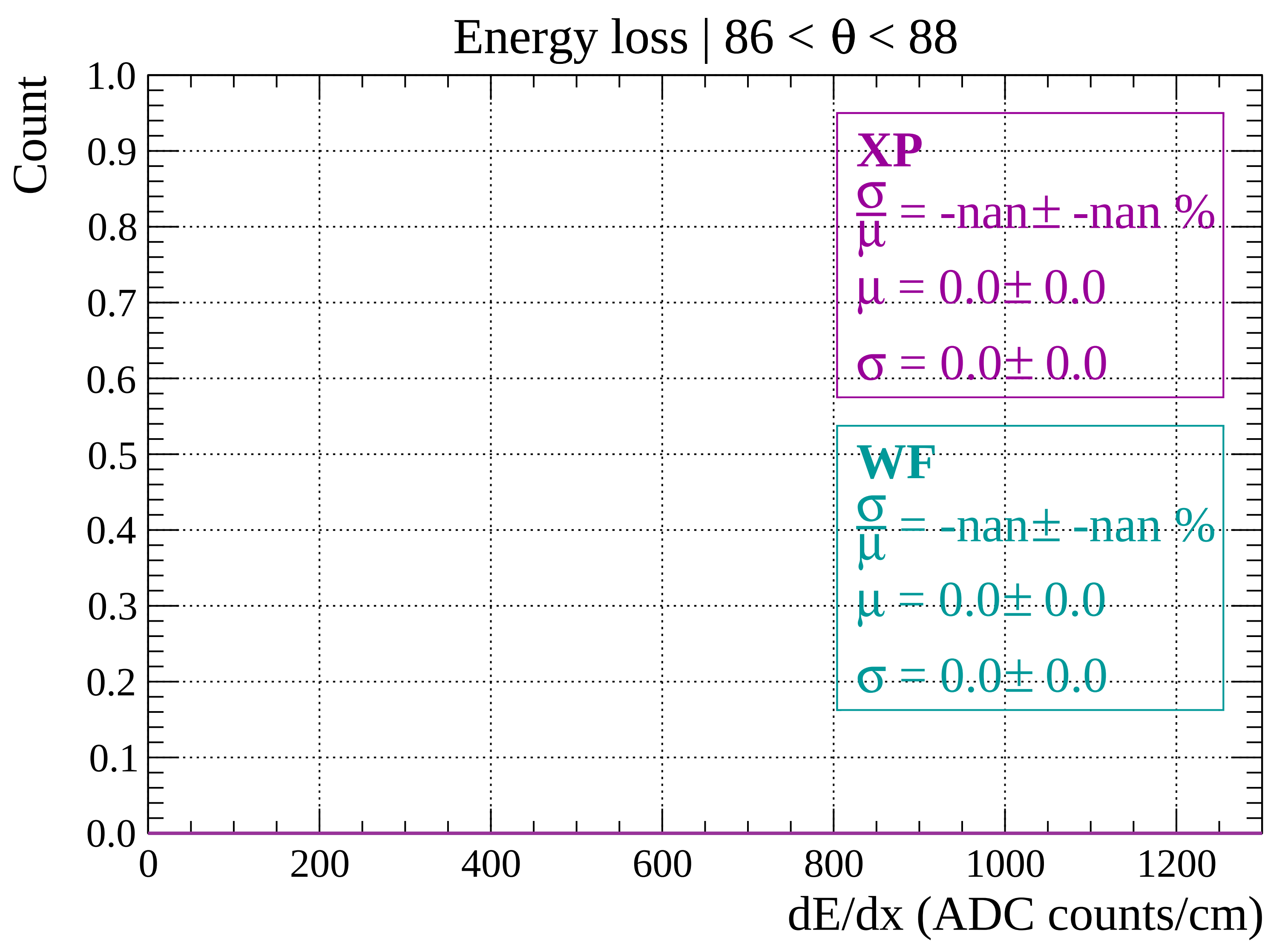


Energy loss | 78 < θ < 80
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

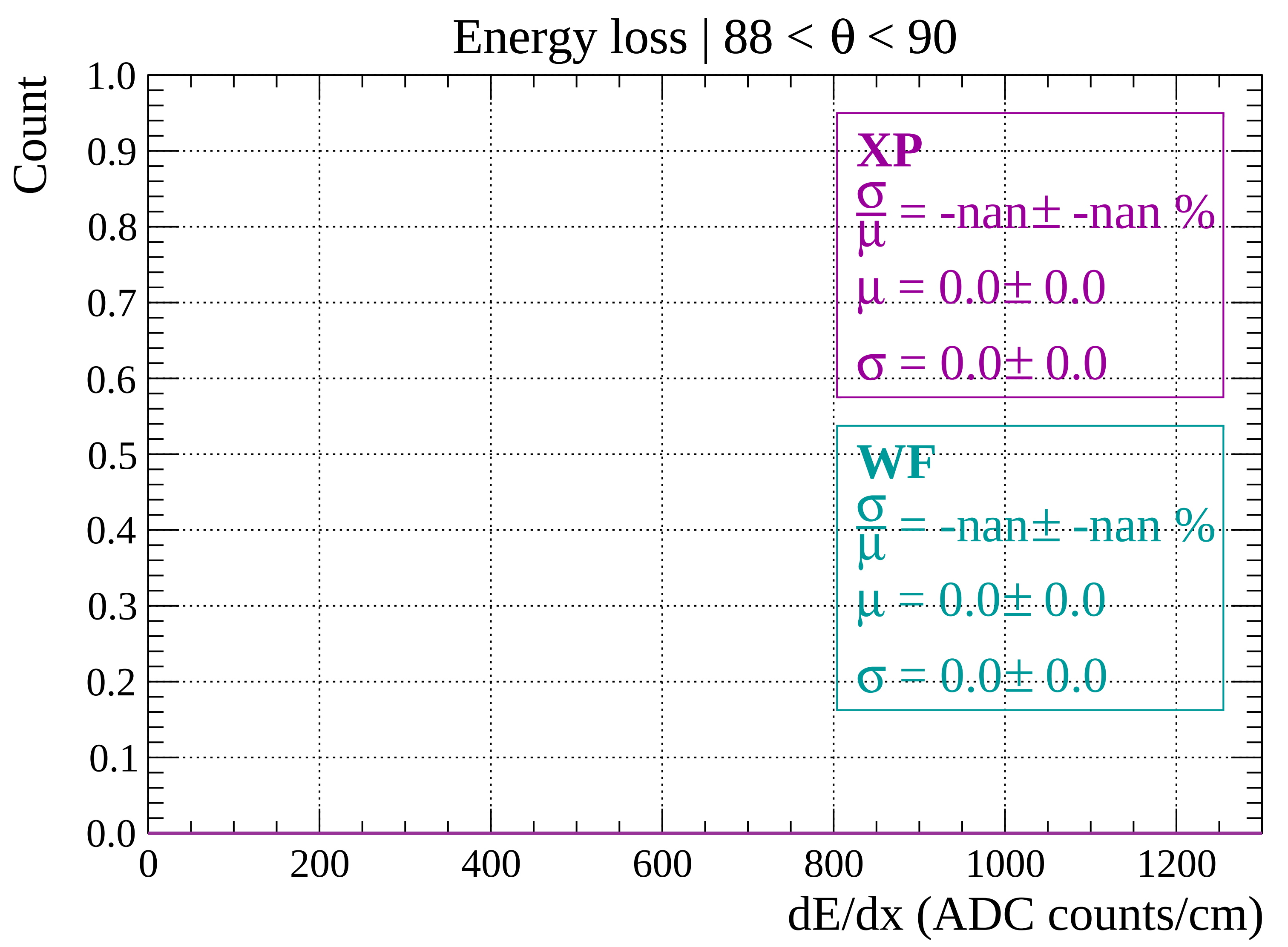

Energy loss | 80 < θ < 82
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

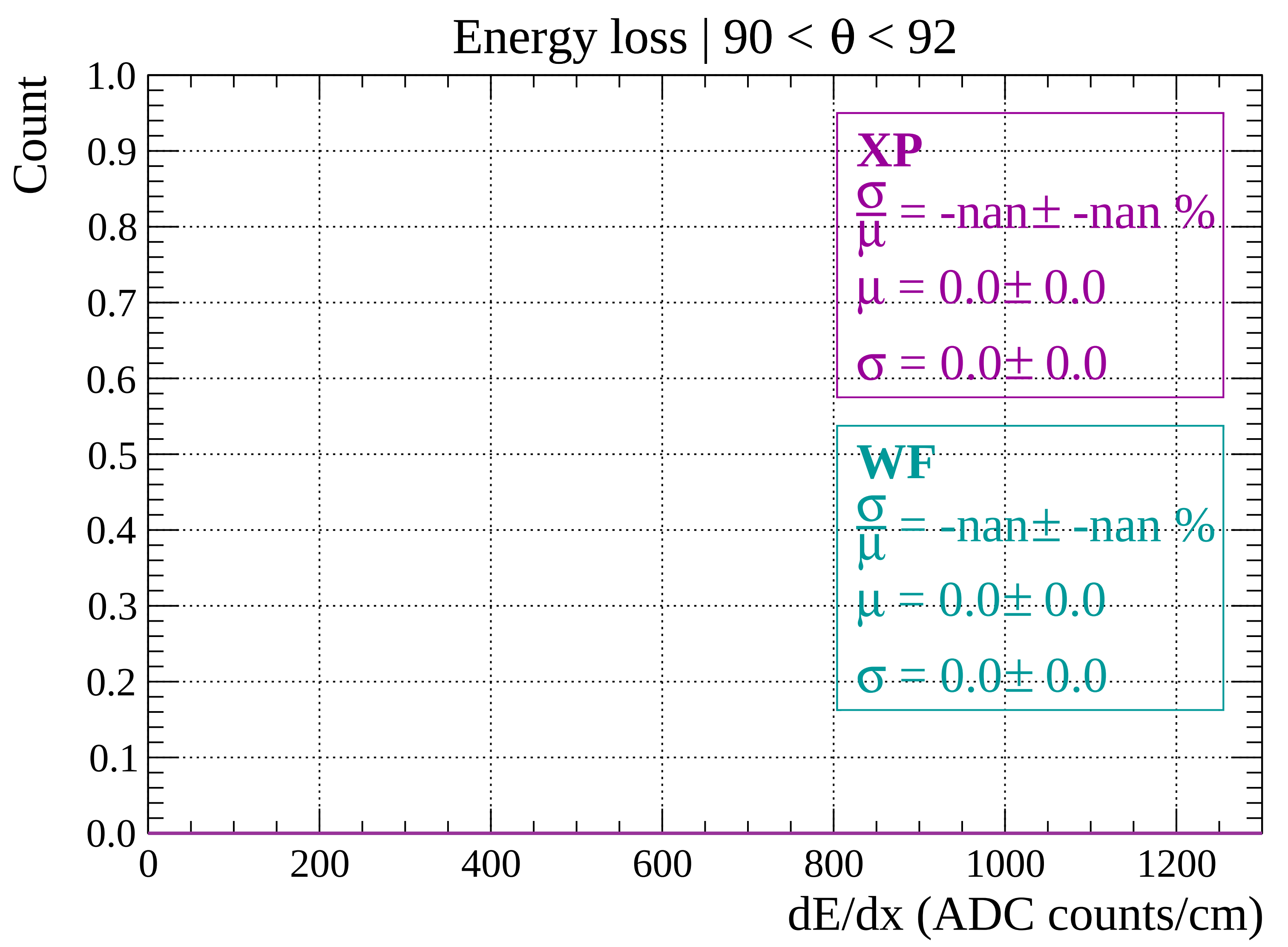

Energy loss | 82 < θ < 84
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 84 < θ < 86
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

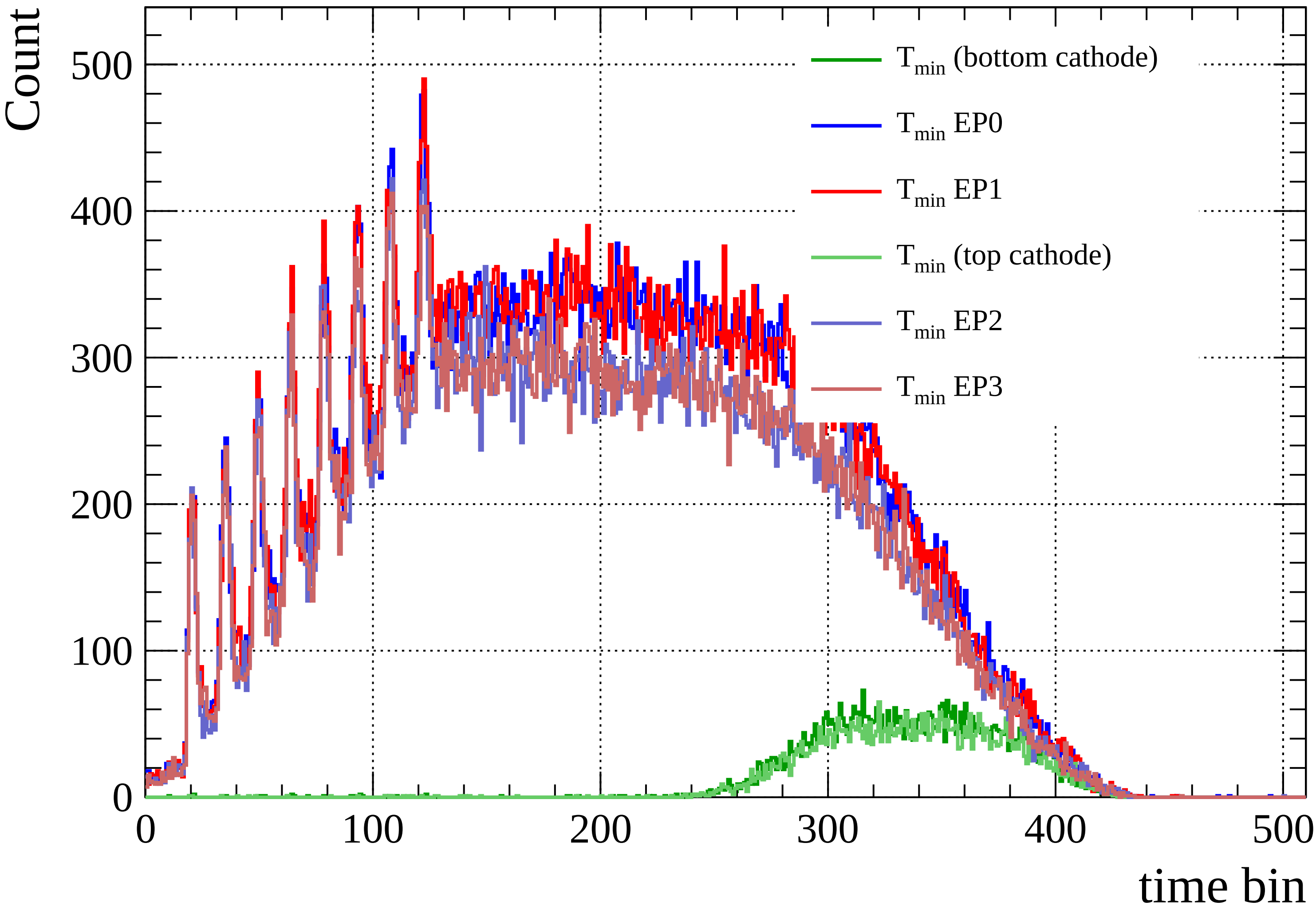


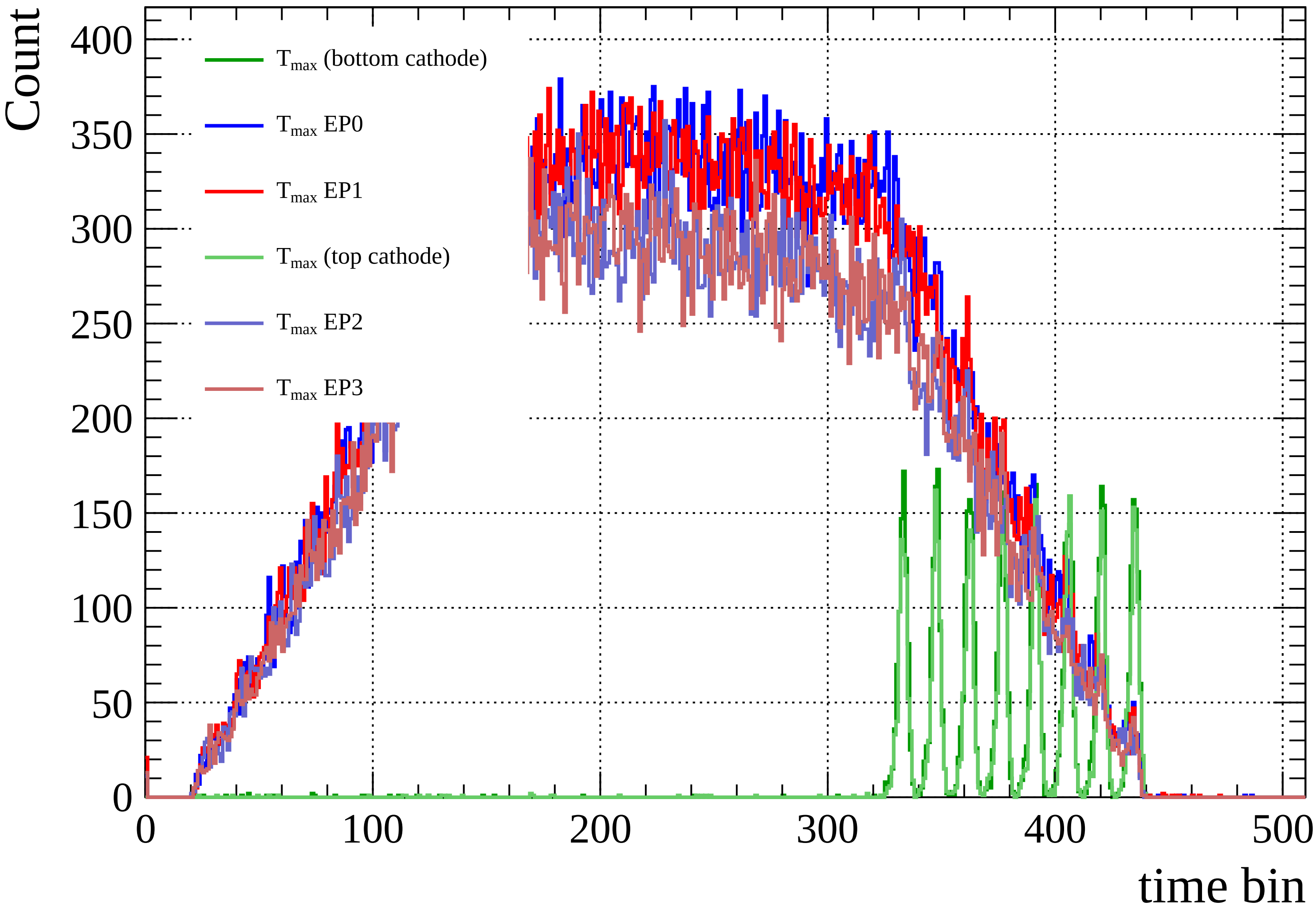
Energy loss | 86 < θ < 88
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 88 < θ < 90
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Energy loss | 90 < θ < 92
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Count
time bin
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3
0
100
200
300
400
500



Count
400
350
300
250
200
150
100
50
0
0
100
200
300
400
500
time bin
T_max (bottom cathode)
T_max EP0
T_max EP1
T_max (top cathode)
T_max EP2
T_max EP3